GNBF5020
Terminology Deep Dive
Every medical and biology term, explained

Introduction to Molecular Biology and Genetics · every medical and biology term, explained

English edition · Part 1: 12 topic chapters · Part 2: 905 dictionary entries
Dr. Tony Wong · September–October 2026

[How to use this book] This book flattens every name in the course into one book you can look things up in: each term gets a one-line summary, then the real mechanism, the consequences and the clinical use, and finally where it sits in the course and how much of it is examinable. A beginner can read it from end to end — when a technical term stops you, turn to Part 2 and find it in alphabetical order.

[Answer in English] Dr. Wong has stated this explicitly: **all assignments and the examination must be answered in English.** Gene names, signalling pathways, drug names and assay methods must all be written in **English** in your answers.

How this book is organised

Part 1	Topic deep dives (12 chapters, one domain each; each closes with a key-vocabulary table)
Part 2	A–Z dictionary (905 entries, sorted by the English term)

Part 1 is for understanding a domain, Part 2 is for looking up single terms; the two cross-reference each other, and the vocabulary tables and the dictionary lead to one another.

How to use this book

This is the terminology reference for GNBF5020. It does not repeat the lecture notes; it answers a more basic question: what does this term actually mean? Every medical and biology term the course touches gets a one-line summary first, and then a paragraph that explains the real mechanism, why it matters, and how the lecturer uses it in the exam.

The book has two parts

Part 1 — Topic deep dives. 12 chapters, one domain each. Each chapter runs as continuous prose that explains the domain from the ground up, follows it with a structure diagram for that domain, and closes with a key-vocabulary table that lists every term of the chapter in one place (Term, In one sentence, Source, Did he name it?). Read it front to back the first time you meet a domain.

Part 2 — A–Z dictionary. 905 entries, sorted by the English term. Every entry carries the full explanation; this is the part you open when you already know the word and want the detail behind it.

How an entry is built

▪ **Term** — the term itself.
In one sentence: — one line, to give you something you can hold on to.
Detailed explanation: — the heart of the book: what it actually does, the mechanism behind it, the consequences and the clinical use.
Why this course brings it up: — where the term sits in the course, and how firmly Dr. Wong said it would be examined.
Appears in: — the chapter and the lecture it appears in.

How to read the “Did he name it?” column

Examined — he named it explicitly and put it in scope. Memorise it.
Excluded — he said explicitly that it is not examined, or that only the direction of it needs to be understood.
Not examined — a background concept he never named on its own. Understanding it is enough.

How to find a term

1. You know the English term → go straight to Part 2 and use your viewer's search function; the dictionary is in alphabetical order.
2. You only remember the concept → check the key-vocabulary tables in Part 1 to find the English term, then return to Part 2 for the detailed entry.
3. You only know which chapter it is in → open that chapter's key-vocabulary table, or search the term in Part 2.
4. You want to know what to actually memorise → filter the “Did he name it?” column down to “Examined”; those are the entries that matter.
5. You want to see what a structure looks like → open the chapter for that domain in Part 1; its structure diagram is at the end of the chapter.

The conventions used throughout

· Technical terms are always left in English. Write your answers in English.
· “He” means Dr. Tony Wong.
· Two source notations, used the same way in both parts: “Lect N” is Dr. Wong's Nth lecture, and “Ch N” is Chapter N of this book. The Source column of the Part 1 key-vocabulary tables and the “Appears in” field of the Part 2 entries both use these two forms.
· Part 1 is for understanding a domain; Part 2 is the complete per-term reference.
· This book does not replace the lecture slides. It flattens the names scattered across them into something you can look up.

Cantonese audio edition Every entry is also read aloud in Cantonese, so you can listen and revise on the way in: **_cantonese/index.html**

Video guides

Twelve of the terms in this book carry the most exam weight, and each one has a short video guide: **English narration with Traditional Chinese subtitles**, about a minute each. The videos live in the project's **_video/assets/** folder, and **_video/index.html** plays all twelve on one page.

Video	Term	Chapter	MP4 file
V01	**Kaplan–Meier curve** A step chart of how many patients are still alive or free of progression at each moment.	Ch 10 · Cancer biology and targeted therapy	V01_Kaplan_Meier_curve.mp4
V02	**Contractile ring → binucleated cell** Mitosis and cytokinesis are two different jobs; block the pinching and you get one cell with two nuclei.	Ch 8 · Cell biology	V02_Contractile_ring_binucleated_cell.mp4
V03	**NHEJ defect → SCID** The cell borrows NHEJ to build antibody diversity, so a broken NHEJ means no antibody library at all.	Ch 6 · DNA damage and repair	V03_NHEJ_defect_SCID.mp4
V04	**Mitochondrial pedigree** Affected mothers pass it to every child, affected fathers pass it to none.	Ch 8 · Cell biology	V04_Mitochondrial_pedigree.mp4
V05	**Meiosis I vs Meiosis II (M1 vs M2)** M1 failure gives two normal and two abnormal; M2 failure gives one normal and three abnormal.	Ch 8 · Cell biology	V05_Meiosis_I_vs_Meiosis_II_M1_vs_M2.mp4
V06	**BER vs NER** BER repairs one damaged base; NER cuts out a bulky lesion that spans several nucleotides.	Ch 6 · DNA damage and repair	V06_BER_vs_NER.mp4
V07	**Hot tumour vs cold tumour** MMR deficiency makes a hot tumour with many neoantigens, and that is what checkpoint blockade can treat.	Ch 9 · Immunology and cancer immunity	V07_Hot_tumour_vs_cold_tumour.mp4
V08	**Driver vs passenger** A driver gives the cell a growth advantage; a passenger just hitched a ride.	Ch 10 · Cancer biology and targeted therapy	V08_Driver_vs_passenger.mp4
V09	**IgG1 vs IgG4** IgG1 triggers ADCC through the NK cell's Fc receptor; IgG4 cannot, but it stays in the blood longer.	Ch 9 · Immunology and cancer immunity	V09_IgG1_vs_IgG4.mp4
V10	**Predictive vs prognostic biomarker** One predicts how the patient will respond to a treatment; the other predicts how the disease itself will go.	Ch 11 · Assay techniques and omics	V10_Predictive_vs_prognostic_biomarker.mp4
V11	**cis vs trans** A cis element is DNA that regulates the gene on that same DNA; a trans factor is a protein made elsewhere.	Ch 7 · Control of gene expression	V11_cis_vs_trans.mp4
V12	**de novo vs maintenance methylation** De novo methyltransferases establish new marks; maintenance methyltransferase copies the old marks onto the daughter strand.	Ch 4 · Chromatin and epigenetics	V12_de_novo_vs_maintenance_methylation.mp4

Video	Term	Chapter	MP4 file
V13	**DNA replication** Semi-conservative, bidirectional and semi-discontinuous — the three adjectives that describe every replication fork.	Ch 5 · DNA replication	V13_DNA_replication.mp4
V14	**Transcription** RNA polymerase reads one template strand; eukaryotes add a cap, a tail and splicing before the message leaves the nucleus.	Ch 7 · Control of gene expression	V14_Transcription.mp4
V15	**Translation** The ribosome reads three bases at a time and builds one amino acid at a time, in one fixed direction.	Ch 2 · Proteins and enzymes	V15_Translation.mp4
V16	**DNA repair** Six pathways deal with the damage; when one of them fails, you get a disease.	Ch 6 · DNA damage and repair	V16_DNA_repair.mp4

Open the index page with all twelve videos →

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How to find a term in Part 2

905 entries, in alphabetical order of the English term (punctuation and case ignored; terms beginning with a number come first). The dictionary is far too long to list here, so find a term this way:

· **You know the English term** → go straight to Part 2 and use your viewer's search function.
· **You only remember the concept** → check the key-vocabulary tables in Part 1 for the English term, then return to Part 2 for the detailed entry.
· **You only know which chapter it is in** → open that chapter's key-vocabulary table, or search the term in Part 2.
· **You want to know what to memorise** → filter each chapter's “Did he name it?” column down to “Examined”.

PART 1

Topic deep dives

(One domain per chapter)

Each chapter runs as continuous prose that explains one domain from the ground up, and closes with that domain's key-vocabulary table.

This part contains

Chapter 1 Chemistry and biomolecules
Chapter 2 Proteins and enzymes
Chapter 3 DNA, genes and chromosomes
Chapter 4 Chromatin and epigenetics
Chapter 5 DNA replication
Chapter 6 DNA damage and repair
Chapter 7 Control of gene expression
Chapter 8 Cell biology
Chapter 9 Immunology and cancer immunity
Chapter 10 Cancer biology and targeted therapy
Chapter 11 Assay techniques and omics
Chapter 12 Drug development and clinical trials

Chapter 1

Chemistry and biomolecules

Start by getting to know the building materials of the factory. Everything is made of **Atom**s: in the centre of each atom is a dense nucleus holding positively charged **Proton**s and electrically neutral **Neutron**s, and outside the nucleus sit negatively charged **Electron**s arranged in layers called **Electron shell**s. What an element is is decided by its number of protons; the electrons do the bonding. The atoms inside you are not exotic — they are the same carbon, oxygen, hydrogen and nitrogen you find in the air and in your textbooks. What makes them special is only the order they sit in and what they are attached to. Dr. Wong never taught this section. All he said in Lecture 1 was: "Please don't memorize all these chemical structure… They would not be included in my exam." So atom, proton, neutron, electron and electron shell are all background: understand them, do not memorise them.

Atoms are held together by three kinds of attraction. A **Covalent bond** is a shared pair of electrons between two atoms, and it builds the skeleton of most biomolecules; an **Ionic bond** is the electrostatic attraction between oppositely charged ions, like table salt; a **Hydrogen bond** is the weakest of the three, a sideways attraction between a slightly positive hydrogen and a slightly negative oxygen or nitrogen. Hydrogen bonds are weak, but they can break and reform at any moment — which is exactly why a protein can fold and DNA can unzip. It is also one of the very few chemistry concepts on the exam list: **there are 3 hydrogen bonds between G–C and 2 between A–T**, so the more GC content a DNA has, the harder it is to pull apart and the higher its melting temperature goes; that direction is what decides the design conditions of a PCR primer.

When atoms combine they become a **Molecule**, and a molecule has a shape, and its shape decides what it is able to touch. The most special of the elements is **Carbon**: with four electrons in its outer shell that are never full, it always wants four more hands, so it can string itself into long chains, into rings, into branches and into cages. Sugars, lipids, proteins and DNA are all built on a carbon skeleton. The carbon chain is only the handle; what really decides a molecule's temperament is the **Functional group** hanging off that chain — hydroxyl, carboxyl, amino, phosphate. He states plainly that this one is not examined, because what would have to be memorised is the structure rather than the concept. What you really do need to remember is the phosphate functional group, because it is at the same time part of the DNA backbone and the cell's most frequently used switch.

The liquid inside a cell is water, and water is the **Solvent**. The water molecule is bent because oxygen pulls the shared electrons closer to itself, which gives it **Polarity**: one end slightly negative, one end slightly positive. Polarity attracts polarity, so water dissolves salts and sugars but refuses oil; and because of that polarity, a cell wall is even possible. Water has four jobs inside a cell: it is the solvent, it takes part in the reactions between proteins and ATP, it holds the three-dimensional shape of large molecules up with hydrogen bonds, and its high specific heat stops your body temperature from swinging wildly as cells work. Water's other role is **pH**: an acid-base value that changes a molecule's charge, and a change in charge changes the shape, and a change in shape changes the function. His Lecture 1 uses a chemical change to a therapeutic protein as the example — at high pH, asparagine deamidates into succinimide, which then takes up water and becomes aspartate or isoaspartate, and the mass goes from 114.04 Da to 97.02 Da and then to 115.03 Da. A chemical change to one side chain is enough to change how a drug binds its target.

Where does the cell's energy come from? **Energy** in a cell is not warmth; it is the capacity to do work. **Metabolism** is the name for all of the cell's chemical reactions: catabolic reactions (**Catabolic**) take big molecules apart and harvest the energy, while anabolic reactions (**Anabolic**) spend energy to build things. He states plainly that the details of the metabolic pathways are not examined, but the concept has to be understood, because his cancer-metabolism example is built on precisely this division of labour: a cancer cell sends most of its pyruvate to lactate even when oxygen is available, producing less, but leaving behind plenty of raw material for making nucleotides, lipids and amino acids. His Warburg slide carries three more numbers: a resting normal cell makes about 36 ATP aerobically and 2 anaerobically, while a proliferating cancer cell makes about 4 either way. Those three numbers are not on his list of numbers to learn; take them as background. The currency the cell pays with is **ATP** (adenosine triphosphate): a kinase takes the terminal phosphate off ATP and sticks it onto a protein, and a

phosphatase uses water to remove the phosphate again — turn it on, turn it off, and you have the cell's fastest switch. The Hippo example in Lecture 1 works exactly like that: MST1 activates LATS1, LATS1 phosphorylates YAP, and the phosphorylated YAP is held outside the nucleus, so the cell stops dividing. **Oxidation** is losing electrons and **Reduction** is gaining them, and the two always happen together; when the cell burns sugar, the sugar is oxidised and the oxygen is reduced, and the energy released is captured as ATP.

So we arrive at the real trunk of the course. The four great families in Dr. Wong's Lecture 1 are the **Biomolecule**s: **Carbohydrate**, **Lipid**, **Protein** and **Nucleic acid**, and he gives each of them a role — disease biomarker, energy source, structural component, signalling molecule. Those four roles are also the frame he sets for the whole course: between genotype and phenotype, the link is made by what biomolecules do, so measuring how much of a molecule is in blood, urine or tissue is reading out what is happening inside the cell. The entire clinical purpose of the course rests on that one sentence.

Carbohydrates are made of carbon, hydrogen and oxygen. A single sugar unit is a **Monosaccharide**, two joined together make a **Disaccharide**, and a long chain makes a **Polysaccharide**, with the units joined by **Glycosidic bond**s. The same chain of glucose does completely different work depending on its shape: **Glucose** is the cell's energy currency; **Glycogen** is the storage form in animals and fungi, and the branching lets the cell release sugar quickly from several places at once — his Lecture 1 clinical example is glucose-6-phosphatase deficiency, where the liver and the kidney are missing this enzyme, so liver cells cannot release free glucose and glycogen storage disease follows; **Starch** is the plant storage form; **Cellulose** is the structural form, and its glycosidic bonds run in the opposite direction to starch's, producing straight parallel fibres, which is why the bacteria in a cow's stomach can cut that bond and you cannot — so starch is digestible and cellulose ends up as faeces.

Lipids are the family that does not dissolve in water. The building unit is a **Fatty acid**: a carbon chain with a small acidic end. A fatty acid with a double bond in the chain is an **Unsaturated fatty acid**, and the chain bends and cannot pack tightly, so it is an oil at room temperature; one with no double bonds is a **Saturated fatty acid**, and the chain is straight and packs tightly, so it is solid at room temperature. Three fatty acids plus a glycerol make a **Triglyceride**, which is the body's fuel warehouse; the other main character is the **Phospholipid**, and it is **Amphipathic**: the head loves water and the tail avoids it, so in water it needs no energy and no builders to arrange itself into a **Bilayer**, heads out and tails in. **Cholesterol** sits inside the bilayer and tunes its fluidity; blood is made of water, lipids cannot cross it, so they are packaged into **Lipoprotein**s to be carried. Lipids also do signalling: **LPS** (lipopolysaccharide) is an outer-membrane component of Gram-negative bacteria, and it is recognised by TLR4, which switches on the IKK, IκB, NF-κB cascade and causes inflammation; in excess it causes septic shock, which he says is a common cause of death among surgical intensive-care patients.

The last two families. **Protein** is the folded machine, and all of Chapter 2 is about it; **Nucleic acid** is the family that carries information. Its building unit is the **Nucleotide** = base + sugar + phosphate, and take the phosphate away and all you have left is a **Nucleoside**; there are only five **Base**s — A, T, C, G and U — and on the central dogma diagram 4 × 4 × 4 = 64 codons correspond to 20 amino acids, the start codon is AUG, the stop codons are UAA, UAG and UGA, and when you move a gene into a CHO cell to be expressed you have to do codon optimisation. Bases lined up in a row, plus a **Sugar-phosphate backbone**, and only then do you have DNA. While we are here, the phosphate group has two jobs: it holds up the backbone, and it is ATP's energy wallet. So ATP is itself a nucleotide, and DNA and ATP are built from the same part — which is the single most worth-remembering coincidence in the whole course.

Chemistry and biomolecules: ATP and the four families

Know the building materials first, and every later chapter makes sense

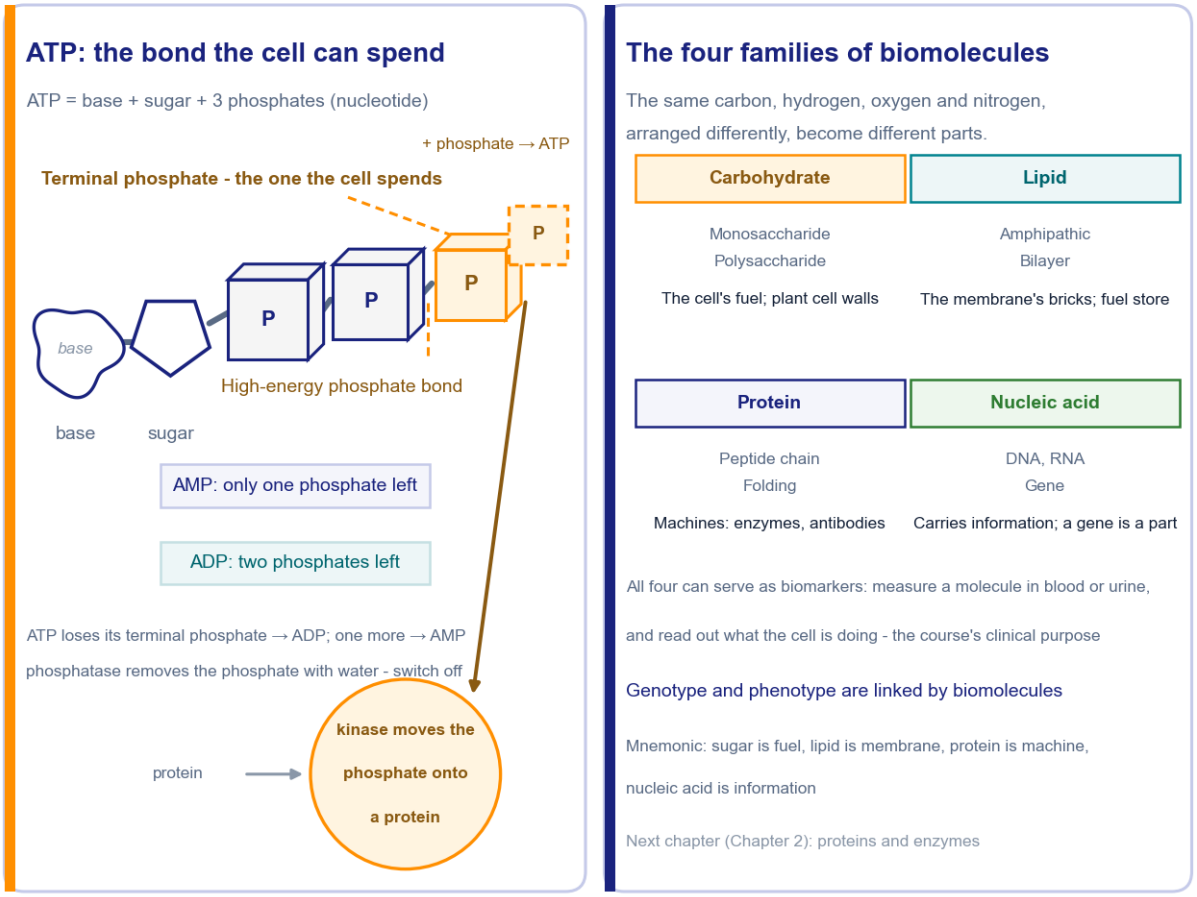


Figure 1: The three-dimensional structure of ATP and the four families of biomolecules. The only part the cell can actually spend is that last phosphate.

Key vocabulary for this chapter

Term	In one sentence	Source	Did he name it?
Atom	The smallest unit of an element; molecules are built from atoms joined together.	Ch 1	Not examined (he never taught it; he only said chemical structures are not examined)
Proton	A positively charged particle inside the nucleus; its number decides what the element is.	Ch 1	Not examined (same)
Neutron	An uncharged particle inside the nucleus; it adds weight without changing the chemistry.	Ch 1	Not examined (same)
Electron	A negatively charged particle orbiting the nucleus; it does all the bonding.	Ch 1	Not examined (same)
Electron shell	A fixed energy layer outside the nucleus; the outermost one decides who the atom bonds with.	Ch 1	Not examined (same)
Covalent bond	Two atoms sharing a pair of electrons — the most common way molecules are linked.	Ch 1	Not examined (background concept)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
Ionic bond	The attraction between oppositely charged ions, like table salt.	Ch 1	Not examined (background concept)
Hydrogen bond	A weak attraction between a slightly positive hydrogen and a nearby slightly negative atom.	Ch 1	Examined (§5 numbers: G–C = 3 bonds, A–T = 2 bonds; the more GC, the higher the Tm, which decides primer design)
Molecule	Two or more atoms joined in a fixed way; it has a shape and properties of its own.	Ch 1	Not examined (background concept)
Carbon	An element with four outer electrons that can bond to itself and to others; the backbone of biomolecules.	Ch 1	Not examined (background concept)
Functional group	A small group of atoms on a molecule that gives it its chemical personality and reactivity.	Ch 1	Excluded (he states plainly that "chemical structures" are not examined; you only need to understand the role of phosphate in the kinase/phosphatase switch)
Solvent	The liquid that other things dissolve in; the cell's solvent is water.	Ch 1	Not examined (background concept)
Polarity	One end of a molecule slightly positive and the other slightly negative, like a small magnet.	Ch 1	Not examined (background concept)
pH	An acid-base scale: 7 is neutral, below that is acidic and above it alkaline, and every step is a factor of ten.	Ch 1	Examined (his deamidation example happens at high pH: 114.04 → 97.02 → 115.03 Da)
Energy	The capacity to do work; in the cell it comes from breaking chemical bonds.	Ch 1	Not examined (background concept)
Metabolism	The collective name for all the cell's chemical reactions: taking things apart to harvest energy, building things up to spend it.	Ch 1, Lect 1	Not examined (you must understand the concept; he states plainly that the details of the metabolic pathways are not examined)
Catabolic	Breakdown metabolism: taking large molecules apart and harvesting the energy.	Ch 1, Lect 1	Excluded (he states plainly that the details of the metabolic pathways are not examined)
Anabolic	Building-up metabolism: spending energy to assemble large molecules.	Ch 1, Lect 1	Excluded (same)
ATP	The energy the cell can spend; breaking the terminal phosphate off releases it.	Ch 1	Examined (the phosphorylation switch: kinases add the phosphate using ATP, phosphatases remove it with water; the start of the Hippo–YAP chain)
Oxidation	Losing electrons (or gaining oxygen); it releases energy.	Ch 1	Not examined (background concept)
Reduction	Gaining electrons (or losing oxygen); it stores energy.	Ch 1	Not examined (background concept)
Biomolecule	A molecule that biology makes and uses: the four families of carbohydrate, lipid, protein and nucleic acid.	Ch 1, Lect 1	Examined (the frame of Lecture 1: the four families and their four roles — biomarker / energy / structure / signalling)
Carbohydrate	One of the four biomolecules; the main fuel and also a structural material.	Ch 1, Lect 1	Examined (the classification table on Lecture 1 slide 7; the example of glycogen storage disease)
Monosaccharide	A single sugar unit, such as glucose or fructose.	Ch 1, Lect 1	Not examined (a classification term; he never named it on its own)
Disaccharide	Two sugar units joined together, such as sucrose.	Ch 1, Lect 1	Not examined (a classification term)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
Polysaccharide	A long chain of sugar units: glycogen, starch or cellulose.	Ch 1, Lect 1	Not examined (a classification term)
Glycosidic bond	The chemical link that joins one sugar unit to the next in a sugar chain.	Ch 1	Not examined (background concept)
Glucose	The monosaccharide the cell burns for energy.	Ch 1, Lect 1	Examined (glucose-6-phosphatase deficiency causes glycogen storage disease — his Lecture 1 example)
Glycogen	The glucose storage form of animals and fungi: a branched sugar chain.	Ch 1, Lect 1	Examined (the α1,4 chains and α1,6 branch points on slide 11; storage liver disease)
Starch	The glucose storage form of plants.	Ch 1	Not examined (background concept)
Cellulose	The structural carbohydrate that makes up plant cell walls.	Ch 1	Not examined (background concept)
Lipid	Lipids: energy storage, membrane components, and signalling molecules.	Ch 1, Lect 1	Examined (LPS is a lipid; the phospholipid signalling pathway on slide 19)
Fatty acid	A carbon chain with a small acidic end; saturated or unsaturated depending on whether the chain has double bonds.	Ch 1, Lect 1	Not examined (background concept)
Saturated fatty acid	A fatty acid with no double bonds in its chain: straight and tightly packed.	Ch 1, Lect 1	Not examined (background concept)
Unsaturated fatty acid	A fatty acid with one or more double bonds in its chain: bent and loosely packed.	Ch 1	Not examined (background concept)
Triglyceride	Three fatty acids attached to a glycerol; the body's fuel warehouse.	Ch 1, Lect 1	Not examined (background concept)
Phospholipid	A molecule with one water-loving end and one water-avoiding end, which is why membranes can form themselves.	Ch 1	Examined (slide 19: PI → PI(4)P → PI(4,5)P → DAG + IP3)
Amphipathic	The property of having one water-loving end and one water-avoiding end; the key word for understanding the cell membrane.	Ch 1	Not examined (background concept)
Bilayer	Two layers of phospholipid, heads out and tails in — the wall of the cell.	Ch 1	Not examined (background concept)
Cholesterol	A lipid embedded in the membrane that keeps it fluid even at low temperature.	Ch 1	Not examined (background concept)
Lipoprotein	A particle with lipid wrapped in protein; it carries lipids through the blood.	Ch 1	Not examined (background concept)
LPS	A bacterial molecule that, once recognised by TLR4, starts the NF-κB inflammatory cascade.	Ch 1, Lect 1, Lect 5	Examined (the cascade stem listed in §8: LPS → TLR4 → IKK → IκB → NF-κB; excess endotoxin causes septic shock)
Protein	A machine made by stringing amino acids together and then folding them into shape.	Ch 1	Examined (one of the three main axes of Lecture 1: structural hierarchy, the phosphorylation switch, antibodies)
Nucleic acid	The family of biomolecules — DNA and RNA — that carries information.	Ch 1	Examined (the fourth category on Lecture 1 slide 7)
Nucleoside	A base plus a sugar, with no phosphate yet.	Ch 1, Ch 3, Lect 1	Examined (defined explicitly in Lecture 1: nucleoside = base + sugar)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
Nucleotide	A base plus a sugar plus a phosphate: the building unit of DNA and RNA.	Ch 1, Ch 3	Examined (the central dogma diagram: 4×4×4 = 64 codons, start AUG, stop UAA/UAG/UGA, and CHO cells need codon optimisation)
Base	One of the four letters of DNA; RNA swaps T for U.	Ch 1, Ch 3	Examined (A/T/C/G/U on his slide 38; G–C 3 bonds, A–T 2 bonds)
Sugar-phosphate backbone	The structural framework of one DNA strand, built of alternating sugar and phosphate.	Ch 1	Not examined (background concept; Chapter 3 covers it in detail)

Chapter 2

Proteins and enzymes

A protein is a folded machine. It is what actually does the work inside the factory: enzymes, receptors, antibodies and the cytoskeleton are all folded proteins. A machine needs parts, and parts only fit if their shapes match — and that sentence will keep coming back all the way through this course.

The unit of a part is the **Amino acid**. There are twenty of them, and they all share one skeleton: a central carbon carrying an amino group, a hydrogen, a carboxyl group, and a variable **R-group (side chain)**. The skeleton is the same, the hat is different, and so each amino acid has its own size, charge and temperament. The R-group decides how the whole chain folds: a bulky or charged R-group (tryptophan, arginine) makes that stretch of the chain stiff, while the smallest, glycine, is the most flexible. In Lecture 1 he puts the whole table of twenty amino acids on the screen and then immediately says: "don't memorize this… definitely not include in the exam." So the R-group table is explicitly not examined; what you need to remember is the idea: twenty of them, one shared skeleton, only the R-group changing.

Head to tail is joined by the **Peptide bond**: the carboxyl group of one amino acid reacts with the amino group of the next, a molecule of water leaves, and a **Polypeptide** chain forms. Because every amino acid has both a carboxyl group and an amino group, the chain always has one end and the other: the head keeps a free amino group and is the N-terminus; the tail keeps a free carboxyl group and is the C-terminus. Proteins are always written from N to C, which is what makes "position 114" mean anything — in his deamidation example, the residue that goes wrong is number 114.

Then come the four levels of folding. **Primary structure** is the sequence itself, that single line of one-letter code. **Secondary structure** is the regular local shape of the backbone: the **Alpha helix** is a spring-like coil, and the **Beta sheet** is a sheet folded back to lie alongside itself; both are held up by hydrogen bonds. A single hydrogen bond is too weak, but a whole helix or a whole sheet is packed with them, and enough weak bonds add up to fix the shape. **Tertiary structure** is the three-dimensional shape the whole chain folds into, and now the R-groups decide who hides inside the protein and who stays outside to meet the water; two cysteines can also form a covalent **Disulfide bond**, like a staple pinning the fold shut, and in Lecture 1 he states plainly that it stabilises the tertiary and quaternary structures. **Quaternary structure** is several folded chains assembled into one machine, each chain called a **Subunit**; the examples he names are haemoglobin and antibody.

There is one more layer of organisation inside the chain: a **Protein domain** is a stretch of sequence that can fold up and do a job on its own, which lets one protein do several jobs and lets the cell reuse the same module in different proteins. The nuclear receptor in Lecture 5 is the perfect example: each one has an NTD, a DBD and an LBD, and the DBD versus the LBD is one of his nineteen confusable pairs.

Some proteins are catalytic machines in themselves. An **Enzyme** makes a chemical reaction faster without being consumed itself; it seats the reactants in a small pocket called the **Active site**, the reactant is called the **Substrate**, and the energy hill the reaction has to climb is the **Activation energy**. He does not examine enzyme kinetics, but you must understand the concept, because every tool in the course — helicase, primase, polymerase, ligase, kinase, phosphatase, histone deacetylase, caspase — is a protein with a pocket, and blocking the pocket or changing its shape is the same thing as changing the chemistry.

How is a protein actually made? The gene is first transcribed into an mRNA copy, and the ribosome reads it three letters at a time; every three letters is a **Codon** specifying one amino acid. His central dogma diagram says it plainly: four letters, 4 × 4 × 4 = 64 codons for 20 amino acids, start codon AUG, stop codons UAA, UAG and UGA; and because the same amino acid has several possible spellings, and different species prefer different spellings, you get codon usage — so when you move a gene into a CHO cell to be expressed, codon optimisation means rewriting the codons to suit the new host's preferences. Once the chain is built, a **Chaperone** helps it fold and stops it tangling into a knot; once it is folded it can be chemically modified, and those modifications are collectively called a **Post-translational modification (PTM)**. The most important one is phosphorylation: a kinase adds the phosphate

using ATP, a phosphatase removes it with water, and the charged phosphates repel each other, pushing the protein open or letting it close — a switch that works in seconds. **Deamidation** in Lecture 1 is another example: at high pH asparagine deamidates into succinimide and then takes up water to become aspartate or isoaspartate, and the mass goes 114.04 Da → 97.02 Da → 115.03 Da; one side chain changes chemically and the drug's binding and potency can change with it. That is what he calls therapeutic protein liability.

One more modification decides where a protein is delivered. A **Signal peptide** is a short address label sitting on the N terminus that sends a newly made protein into the rough ER and hence into the secretory pathway. He teaches it because one of his postdocs once left it out and threw away four months of work: with no signal peptide, secretory and membrane proteins simply do not express. He puts it plainly: "signal peptide is really, really important for your study." Notice that it is nothing like the NTD of the nuclear receptor — which is one of his confusable pairs as well.

Antibodies are the most useful protein model there is. An **Antibody** is shaped like a Y, with two arms that grab the target; the target is called an **Antigen**, and the small patch the antibody actually touches is the **Epitope**. One antibody is built from two heavy chains and two light chains, and each arm is made of a **VH** and a **VL**; the elbow is the **Fab fragment**, about 50 kDa, and it does the grabbing; the stem is the **Fc fragment**, also about 50 kDa, and it calls in the rest of the immune system; cut the whole variable region off and you have the **Fv fragment**, about 25 kDa. The three hypervariable loops at the fingertips are **CDR1**, **CDR2** and **CDR3**, and they are the real gripping surface; they can take an astronomical number of shapes, which is why the immune system can build antibodies against countless enemies. His Lecture 1 slide lists these numbers and fragments, and Lecture 4 tells you the diversity is assembled by V(D)J recombination — and that when that machine breaks, you get SCID. Antibodies also come in versions, and this one is called **Isotype**: the Fc of **IgG1** grips the Fc receptor on an NK cell, which triggers **ADCC** (antibody-dependent cell-mediated cytotoxicity); **IgG4** cannot grip as tightly and so has no ADCC, but it stays in the blood longer. He pairs IgG1 with IgG4 as a confusable pair because "IgG1 kills, IgG4 does not kill but stays longer" is exactly the reason a clinician picks one antibody over another.

Last of all, how are proteins studied? Four methods, each answering one question. **Edman degradation** reads the sequence off one amino acid at a time from the N terminus, and his own words are "classical Edman degradation". **LC-MS/MS** is the modern workhorse: it cuts the protein into peptides, separates them by liquid chromatography, weighs them by mass spectrometry, and reads the sequence back out of the fragment masses against a database — and while it is at it, it tells you how much of the sample was there. **SDS-PAGE** uses SDS to give every protein the same negative charge and then separates them on the gel by size; four bands mean four sizes, not four identities, and when you want to separate first and identify afterwards you use **2D-PAGE**: the first dimension is isoelectric focusing, separating by charge (pH 4.0 to 9.0), the second is SDS-PAGE, separating by size (220 kDa down to 10 kDa), each protein lands in its own square, and you cut the spot out and send it to LC-MS/MS to be identified. The vimentin case in Lecture 1 goes exactly this way: 2D-PAGE separates the spots, LC-MS/MS identifies them, and finally ELISA measures the amount in serum, turning vimentin into a biomarker for liver cancer. The last method is **X-ray crystallography**, which solves the three-dimensional positions of the atoms; his Lecture 3 demonstrates it with the nucleosome, whose crystal structure he solves to 2.8 Å, and he calls anything below 2.0 Å "very excellent". Keep the four methods as one table and one sentence does it: Edman gives the sequence, LC-MS/MS gives the identity and the amount, SDS-PAGE gives the size, and X-ray crystallography gives the shape of the atoms. The whole **Proteome** of a cell is the sum of all its proteins, and every method in this chapter ends up serving the same thing: turning proteins into diagnostic markers you can measure in a blood sample.

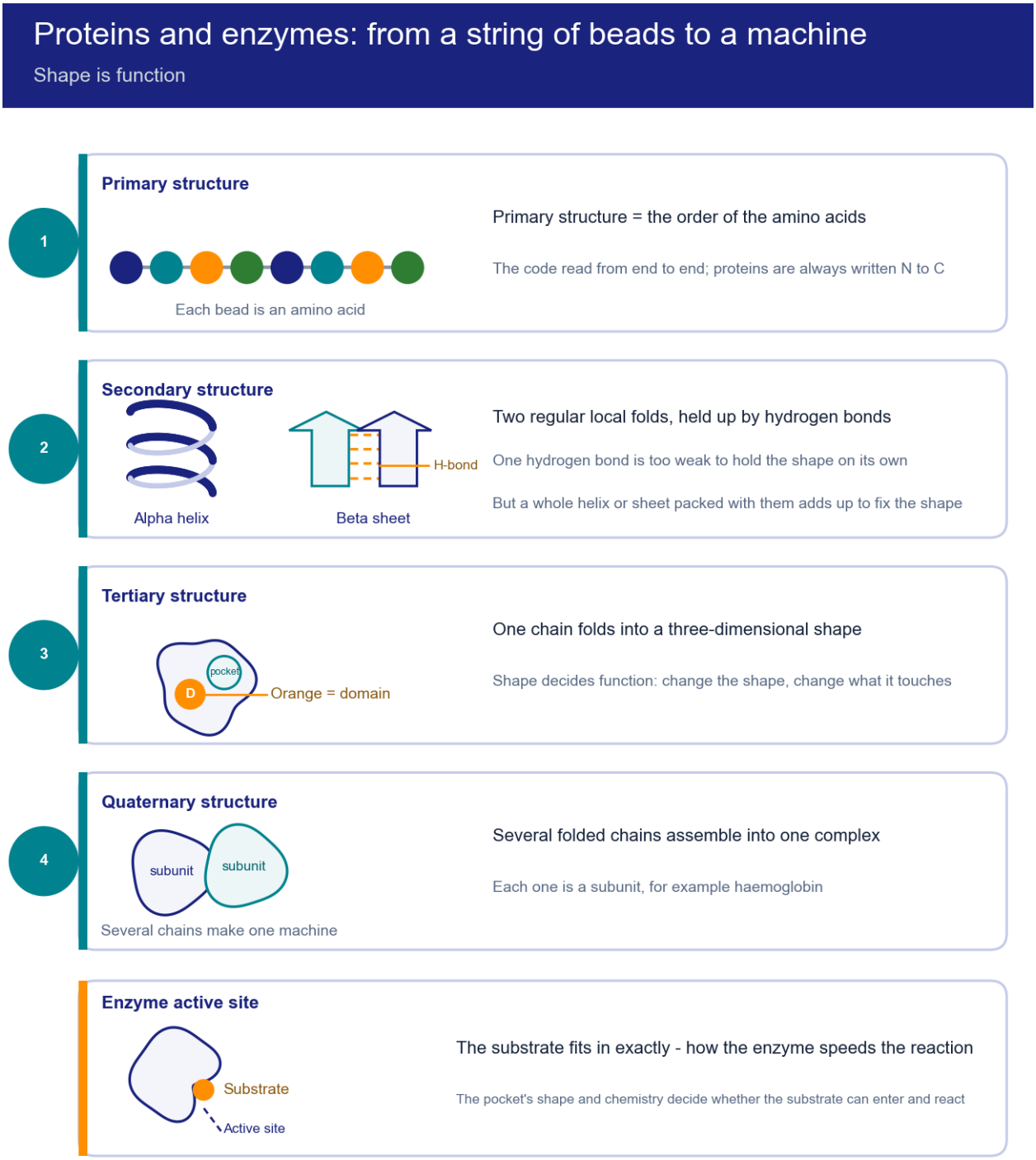


Figure 2: A protein folding from a string of beads into a machine, and how the active site lets the reaction happen.

Key vocabulary for this chapter

Term	In one sentence	Source	Did he name it?
Amino acid	One of the twenty small units that make up a protein, each with its own side chain.	Ch 2	Examined (the general amino acid structure on slide 23; also the origin of the N/C ends and of position numbering)
R-group (side chain)	The variable flap on an amino acid that decides how the whole chain folds.	Ch 2	Excluded (he states plainly "don't memorize this… definitely not include in the exam")
Peptide bond	The chemical link that joins one amino acid to the next.	Ch 2	Not examined (background concept)
Polypeptide	A long chain of amino acids joined by peptide bonds; folded, it is a protein.	Ch 2	Not examined (background concept)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
Primary structure	The amino acid sequence of a protein, read from one end to the other.	Ch 2	Examined (the four-level hierarchy on Lecture 1 slide 25)
Secondary structure	The regular local folding of a chain: the alpha helix and the beta sheet, held by hydrogen bonds.	Ch 2	Examined (same; he draws the hydrogen bond as C=O···H–N)
Tertiary structure	The overall three-dimensional fold of a single protein chain.	Ch 2	Examined (same)
Quaternary structure	Several protein chains assembled into one working complex.	Ch 2	Examined (he names haemoglobin and antibody as examples)
Alpha helix	The spring-like helix a protein chain often takes, held by hydrogen bonds.	Ch 2	Examined (slide 25; hydrogen bonds hold every turn)
Beta sheet	A sheet-like fold of several stretched chains lying side by side, held by hydrogen bonds.	Ch 2	Examined (slide 25; C=O···H–N hydrogen bonds)
Disulfide bond	A covalent link between two cysteines that helps fix a protein's shape.	Ch 2, Lect 1	Examined (he states plainly that it stabilises the tertiary and quaternary structures)
Protein domain	A stretch of protein that folds up and works on its own.	Ch 2, Lect 1	Examined (defined in Lect 1; the DBD vs LBD question in Lect 5 is its exam version)
Subunit	One of the chains in a protein made of several chains.	Ch 2	Not examined (a component of quaternary structure; he never lists it as an exam point)
Enzyme	A protein that makes a chemical reaction faster without consuming itself.	Ch 2	Not examined (background concept; he lists no enzyme-kinetics exam point)
Active site	The small pocket on an enzyme where the substrate sits and the reaction happens.	Ch 2	Not examined (background concept)
Substrate	The molecule an enzyme acts on.	Ch 2	Not examined (background concept)
Activation energy	The energy hill every reaction has to climb, which the enzyme lowers.	Ch 2	Not examined (background concept)
Codon	Three letters on the mRNA that specify one amino acid.	Ch 2, Lect 1	Examined (the central dogma diagram: 4×4×4 = 64, start AUG, stop UAA/UAG/UGA, and CHO cells need codon optimisation)
Chaperone	A protein that helps another protein fold correctly and stops it tangling.	Ch 2	Not examined (background concept)
Post-translational modification (PTM)	A chemical modification added after a protein has folded; phosphorylation is the most important kind.	Ch 2, Lect 1, Lect 5	Examined (both the phosphorylation switch and deamidation are PTMs)
Deamidation	A slow chemical change to a protein: asparagine turns into aspartate, which can change potency.	Ch 2, Lect 1	Examined (his therapeutic protein liability example: 114.04 → 97.02 → 115.03 Da)
Signal peptide	The address label at the front of a protein that delivers it to the ER and the secretory pathway.	Ch 2, Lect 2	Examined (Tier 2 §4.9: he says "really, really important for your study"; leaving it out blocks secretory protein expression completely)
Antibody	A Y-shaped protein made by B cells that grabs one antigen in particular.	Ch 2, Ch 9, Lect 1, Lect 2	Examined (Lect 1 and 2; the carrier of an ADC; antibody diversity)
Antigen	Something the immune system treats as foreign and therefore attacks.	Ch 2, Ch 9, Lect 1, Lect 2	Examined (the thing an antibody recognises)
Epitope	The small patch on the antigen that the antibody actually touches.	Ch 2, Ch 9, Lect 2	Not examined (background concept)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
Fab fragment	The arm of an antibody — the part that grabs the antigen.	Ch 2, Lect 1	Examined (slide 26: Fab about 50 kDa)
Fc fragment	The stem of an antibody — the part that calls in the rest of the immune system.	Ch 2, Lect 1	Examined (slide 26: Fc about 50 kDa; IgG1 uses it to trigger ADCC)
Fv fragment	The smallest antibody fragment, just the variable region, about 25 kDa.	Ch 2, Lect 1	Examined (slide 26: Fv ≈ 25 kDa = VH + VL)
VH	The variable region on an antibody heavy chain.	Ch 2, Lect 1, Lect 2	Examined (the VL/VH β-barrel on slide 26)
VL	The variable region on an antibody light chain.	Ch 2, Lect 1, Lect 2	Examined (the VL/VH β-barrel on slide 26)
CDR1	One of the three hypervariable loops in the antibody variable region; together they form the antigen-binding surface.	Ch 2, Lect 1	Examined (slide 26; the source of antibody diversity)
CDR2	One of the three hypervariable loops in the antibody variable region.	Ch 2, Lect 1	Examined (slide 26; the source of antibody diversity)
CDR3	One of the three hypervariable loops in the antibody variable region, and the most important of them.	Ch 2, Lect 1	Examined (slide 26; the source of antibody diversity)
ADCC	The antibody's stem grabs a killer cell such as an NK cell and orders it to kill the stuck-on cell.	Ch 2, Ch 9, Lect 1, Lect 2	Examined (the §7 confusable pair: IgG1 has ADCC, IgG4 does not)
Isotype	The version or class of an antibody; IgG1 kills with ADCC, IgG4 does not.	Ch 2, Ch 9, Lect 2	Examined (the §7 confusable pair; Lect 2 topic 12)
IgG1	The antibody class that kills its target cell with ADCC.	Ch 2, Ch 9, Lect 1, Lect 2	Examined (the §7 confusable pair: IgG1 vs IgG4)
IgG4	The antibody class with no ADCC, but a longer stay in the blood.	Ch 2, Ch 9, Lect 1, Lect 2	Examined (the §7 confusable pair: IgG1 vs IgG4)
Edman degradation	The standard method that reads a protein's sequence off one amino acid at a time from the N terminus.	Ch 2	Examined (his own phrase "classical Edman degradation", set against modern LC-MS/MS)
LC-MS/MS	A method that chops proteins up, weighs them and reads out the sequence; it tells you which proteins are present and how much of each.	Ch 2, Ch 11, Lect 1	Examined (his own words; the method behind every proteomics biomarker)
SDS-PAGE	The standard gel test that separates proteins by size; four bands are four sizes, not four identities.	Ch 2, Lect 1	Examined (the second dimension of 2D-PAGE on slide 34; the western blot loading control on slide 44)
X-ray crystallography	The method that solves the exact three-dimensional shape of a protein's atoms.	Ch 2	Examined (the nucleosome crystal at 2.8 Å; he calls below 2.0 Å "very excellent")
2D-PAGE	A two-dimensional gel that separates proteins by charge and then by mass; the spots are cut out and sent to LC-MS/MS.	Ch 2, Ch 11, Lect 1	Examined (slide 34: IEF pH 4.0–9.0 × SDS-PAGE 220→10 kDa; the vimentin case)
Proteome	All the proteins that a cell or a tissue makes.	Ch 2, Lect 1	Examined (the starting point of the proteomics workflow in Lect 1 topic 11)

Chapter 3

DNA, genes and chromosomes

First learn the alphabet. The building unit of DNA is the **Nucleotide** = base + sugar + phosphate; take the phosphate away and all you have left is a **Nucleoside**. The sugar in DNA is **Deoxyribose** and the sugar in RNA is **Ribose**, and the two differ by a single oxygen. There are only five **Base**s — A, T, C, G and U — with DNA using the first four and RNA swapping T for U. Bases lined up in a row, plus the two long rails of the **Sugar-phosphate backbone**, and only then does DNA have a shape.

Then comes the famous shape. Two strands wind around each other as a **Double helix**, with the bases pairing across the middle like the rungs of a ladder, and the pairing rule is called **Base pairing**: A always pairs with T, G always pairs with C. There are 2 hydrogen bonds between A–T and 3 between G–C — those are his own numbers, and they are the set of numbers most likely to turn into the stem of a multiple-choice question. The model came out of three questions answered by three groups. In 1928 **Griffith** found that material from dead pneumococcus bacteria could turn live harmless bacteria into virulent ones, but he could not say what it was, and he called it the transforming principle. In 1943 **Avery, MacLeod and McCarty** answered "what is it" with enzyme experiments: **DNase**, the enzyme that cuts DNA, destroyed the transforming activity, while the protease that cuts proteins did not — so the genetic material is DNA and not protein. **Watson, Crick and Franklin** answered "what does it look like", building the double-helix model from Rosalind Franklin's X-ray images. The model immediately became a working tool: heat pulls the two strands apart, and pulling them apart is exactly what PCR does. The two strands run in opposite directions, which is called **Antiparallel**: one is 5′→3′ and the other 3′→5′; the entire logic of the replication fork in Lecture 4 is built on that fact.

A **Gene** is a recipe: a stretch of DNA on a chromosome that is transcribed into RNA and then used to make a protein or a functional RNA. The direction of flow is set by the central dogma: DNA becomes mRNA by transcription, and mRNA becomes a polypeptide chain by translation. The gene name itself is never the clinical answer — K-Ras only means something once you know which slot it occupies in a signalling cascade: normal K-Ras only transmits when EGFR fires upstream, and the single-letter G12C change locks RAS in its on state, so the cell keeps dividing even when the upstream receptor is blocked by cetuximab.

A **Chromosome** is a printed manual: an extremely long piece of DNA plus the proteins that package it, compressed enough to fit inside a nucleus. A human cell has 46 of them, 23 pairs. It has anatomy: the **Centromere** is the waist, where the spindle fibre grabs during division; the waist divides the chromosome into a short **p-arm** and a long **q-arm**, which is why the band name q21.1 is read as q-two-one-point-one and not q-twenty-one-point-one. At each end there is a **Telomere**: the very tip is a unique sequence, moving inward are 100 to 300 kb of subtelomeric repeats, and moving further inward are 3 to 20 kb of the repeated TTAGGG telomere sequence. Because DNA polymerase cannot copy the very end of a linear chromosome, the telomere loses about 50 to 200 bp at every division; when it reaches the threshold you hit the **Hayflick limit** and the cell stops dividing. **Telomerase** can put the telomere back, and most cancers (about ninety per cent) switch it back on — and his other number is that the longest ten per cent of telomeres do worse, not better, for survival after recurrence. Chromosomes are big enough to see: stain them with Giemsa to make a **Karyotype**, and you get the banded pattern of light and dark stripes called **G-banding**; an **Ideogram** is the diagram that labels every band of every chromosome one by one. Its resolution is limited, though: it shows structural changes of roughly 0.5 to 10 Mb, and a change of a single base has to be found by sequencing.

The manual is too big, so it gets packed. The DNA in one cell is about a metre long and has to fit inside a nucleus far smaller than a millimetre, so it must be packaged. DNA winds around a protein spool to make a **Nucleosome** — one turn of 147 bp, about 10 nm across; nucleosomes linked together make **Chromatin**, and tightening that makes a chromosome. Loose, open chromatin that is being read is called **Euchromatin**, and tight, silent chromatin is called **Heterochromatin**: heterochromatin is not "a bit quieter", it physically keeps transcription factors and enhancers out so that nothing can get close. All of Chapter 4 is about this layer.

Next comes the most important word in the course: a **Mutation** is a typo in the recipe. Typos come in several kinds. A change of one letter is a **Point mutation**, and it may do nothing at all (**Silent mutation**, the protein is unchanged), or swap one amino acid (**Missense mutation**), or create an early stop codon that truncates the protein (**Nonsense mutation**). If the number of letters inserted or deleted is not a multiple of three, the reading frame shifts into a **Frameshift** and everything downstream is misread; if it is a multiple of three it is **In-frame**, the protein keeps its length but is missing a stretch. A frameshift also manufactures a **Neoantigen** — a fragment the immune system has never seen — and that is exactly why MSI-high tumours are called hot tumours and can take anti-PD-1. Bigger changes take a whole chunk: removing a stretch is a **Deletion**, an extra copy is a **Duplication**, a whole stretch turned back to front is an **Inversion**, and two chromosomes swapping pieces is a **Translocation**. He singles out inversion: it neither adds nor removes DNA and it does not change gene dosage, yet it can move a gene away from its enhancer or move an enhancer next to a different promoter, and the expression level changes dramatically — "you need to understand that some of these abnormalities can cause the loss of the genetic materials, but some does not." One step bigger is the number of whole chromosomes: a cell with the wrong number of chromosomes has **Aneuploidy**, caused by chromosomes failing to separate properly during meiosis, which is **Non-disjunction**; in M1 the homologous chromosomes fail to separate, giving 2 normal and 2 abnormal gametes, and in M2 the sister chromatids fail to separate, giving 1 normal and 3 abnormal gametes. He lists telling M1 and M2 apart as one of five questions. The results are **Trisomy 21** (Down syndrome) and **Turner syndrome (45,X)**, a woman with a single X. As for **Copy number variation (CNV)** — the same stretch of DNA being present in different numbers of copies in different people — he says explicitly that it will only "give you a taste" and that you should not be scared by all the mechanisms. What you really need is **Driver** (a mutation that gives the cell a growth advantage) and **Passenger** (one that merely hitched a ride), and the way to tell them apart is to combine the CNV profile with the mRNA expression: the one whose copy number went up and whose transcription went up with it is the driver, and the example he uses is NPM1 in leukaemia.

There is a clinical habit worth building here: ask the two questions separately. Is there a typo? Is that change actually there in the DNA sequence? And does the typo matter? Does it change the protein, does it give the cell a growth advantage? The first question is answered by sequencing, the second by functional and clinical data; a complete clinical interpretation needs an answer to both. And remember that a typo has to be fixed in place before it becomes a mutation: the cell has repair systems, and the overwhelming majority of errors are corrected on the spot during replication; only a change that was copied into a daughter cell and left unrepaired becomes a permanent letter in the DNA. A cancer genome and the origin of an inherited disease are both built up this way, one accumulated typo at a time.

Last of all, the tools that let you use DNA. **PCR** copies a chosen fragment into millions of copies by repeating three steps: **Denaturation** separates the two strands, **Annealing** lets the primer stick on, and **Extension** has the polymerase add the letters; after n cycles you have about 2^n copies. Designing a **Primer** means looking at the **Melting temperature (Tm)**: the more GC there is, the higher the Tm, because what you have to break is hydrogen bonds — and his slide gives the direction only, never the numbers. **DNA fingerprint** exploits differences between individuals in the length of microsatellite sequences to identify people, and Lecture 4 gives it a single sentence on a diagram. Beyond that there is the toolbox for copying and manipulating: a **Restriction enzyme** cuts out a fragment with **Sticky end**s, you join them into a **Plasmid** or another **Vector**, you carry out a **Ligation**, and the fragment can then be made in bulk in bacteria or sequenced.

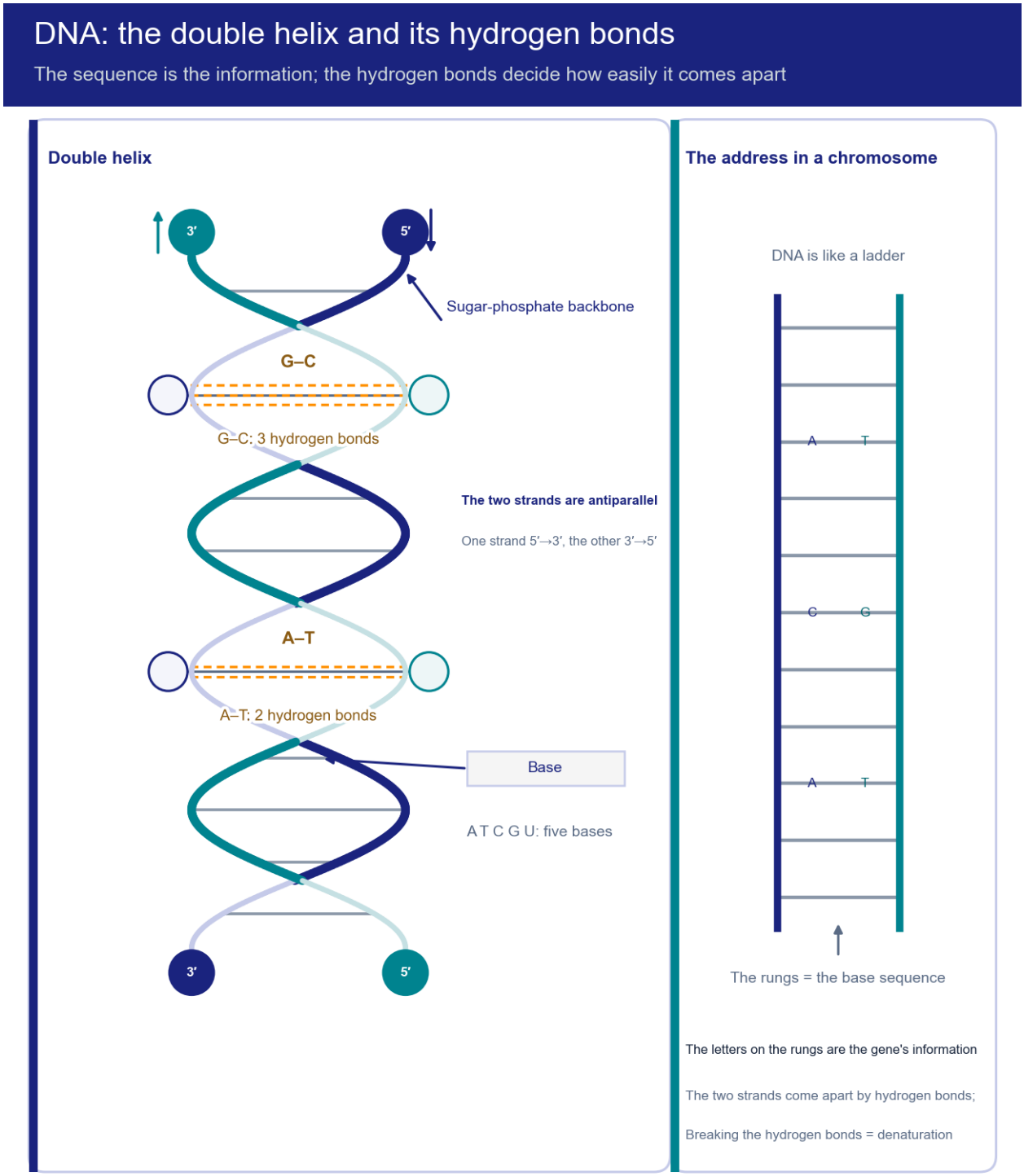


Figure 3: The three-dimensional structure of the DNA double helix, the number of hydrogen bonds between the bases, and the two strands running in opposite directions.

Key vocabulary for this chapter

Term	In one sentence	Source	Did he name it?
Nucleotide	A base plus a sugar plus a phosphate: the building unit of DNA and RNA.	Ch 1, Ch 3	Examined (his definition: nucleoside = base + sugar; slide 38)
Nucleoside	A base plus a sugar, with no phosphate yet.	Ch 1, Ch 3, Lect 1	Examined (defined explicitly in Lecture 1)
Deoxyribose	The sugar of DNA, one oxygen less than the sugar of RNA.	Ch 3	Not examined (background concept)
Ribose	The sugar of RNA.	Ch 3	Not examined (background concept)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
Base	One of the four letters of DNA; RNA swaps T for U.	Ch 1, Ch 3	Examined (A/T/C/G/U on his slide 38; G–C 3 bonds, A–T 2 bonds)
Double helix	Two DNA strands wound around each other, with the bases pairing across the middle like ladder rungs.	Ch 3	Examined (the Watson–Crick–Franklin model; he says this model is the basis of PCR)
Base pairing	The pairing rule for bases: A with T, G with C.	Ch 3, Lect 1	Examined (§5 numbers: G–C 3 bonds, A–T 2 bonds; the more GC, the higher the Tm, which decides primer design)
Antiparallel	The two DNA strands run in opposite directions, one 5′→3′ and one 3′→5′.	Ch 3	Examined (§4.5 and the §7 confusable pair: "elongation happens from 5 prime to 3 prime")
Gene	A recipe: a stretch of DNA specifying one product.	Ch 3	Examined (the central dogma; K-Ras as an example of a predictive biomarker)
Chromosome	A packaged copy of the genome — the printed manual.	Ch 3, Lect 3	Examined (§5: 23 pairs, the p/q arms, telomere structure)
Centromere	The waist of a chromosome, where the spindle fibre attaches.	Ch 3, Lect 3	Examined (§4.11)
Telomere	The cap of repeated DNA at the end of a chromosome; it protects the end and shortens with division.	Ch 3, Lect 3, Lect 4	Examined (§5: 3–20 kb TTAGGG, 100–300 kb subtelomeric; the longest ten per cent of telomeres do worse for survival after recurrence)
p-arm	The short arm of a chromosome, above the centromere.	Ch 3, Lect 3	Examined (§4.11)
q-arm	The long arm of a chromosome, below the centromere.	Ch 3, Lect 3	Examined (§4.11; q21.1 is read as q-two-one-point-one)
Nucleosome	The basic packaging unit: DNA wound around one turn of histones.	Ch 3, Ch 4, Lect 3	Examined (§5: 147 bp, about 10 nm)
Chromatin	DNA plus the proteins that package it — the manual and its wrapping.	Ch 3, Lect 3	Examined (§4.10)
Euchromatin	Loose, open chromatin that is being transcribed.	Ch 3, Lect 3	Examined (§4.10 and the §7 confusable pair)
Heterochromatin	Tight, silent chromatin that physically keeps transcription factors out of the promoter.	Ch 3, Lect 3	Examined (§4.10 and the §7 confusable pair)
Karyotype	A photograph of every chromosome in a cell, used to spot large structural abnormalities.	Ch 3, Lect 3	Examined (§4.11: G-banding resolution about 0.5–10 Mb; a single-base change needs sequencing)
G-banding	The pattern of light and dark stripes on a stained chromosome, which reveals large structural changes.	Ch 3, Lect 3	Examined (§4.11)
Ideogram	The diagram that labels every band of every chromosome one by one.	Ch 3, Lect 3	Examined (the chromosome 8 ideogram on slide 13 with q21.1 marked)
Mutation	A change in the DNA sequence — a typo in the recipe.	Ch 3, Lect 3	Examined (the driver/passenger distinction; K-Ras G12C)
Point mutation	A change of a single letter: it may be harmless, or it may be a driver.	Ch 3, Lect 3	Examined (K-Ras G12C is exactly a single-letter change)
Silent mutation	A single-letter mutation that leaves the protein completely unchanged.	Ch 3, Lect 3	Not examined (the name never appears in the five lectures; he never turns the three single-letter outcomes into a memory list)
Missense mutation	A single-letter mutation that swaps one amino acid.	Ch 3, Lect 3	Not examined (same; the name does not appear in the five lectures)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
Nonsense mutation	A single-letter mutation that creates an early stop codon and truncates the protein.	Ch 3, Lect 3	Not examined (same; the name does not appear in the five lectures)
In-frame	An insertion or deletion of a multiple of three letters, so the reading frame is unchanged.	Ch 3, Lect 1	Not examined (it only appears on the diagram on slide 47)
Frameshift	An insertion or deletion that is not a multiple of three, shifting the reading frame and usually truncating the protein.	Ch 3, Ch 9, Ch 10, Lect 1, Lect 4, Lect 5	Examined (§4.4: MSI-H → frameshift → neoantigen → hot tumour → anti-PD-1)
Deletion	The loss of a stretch of DNA.	Ch 3, Ch 10, Lect 3	Examined (one of the five structural abnormalities on slide 40)
Duplication	An extra copy of a stretch of DNA.	Ch 3, Lect 3	Examined (one of the five structural abnormalities on slide 40)
Inversion	A stretch of chromosome cut out and reinserted backwards; no DNA is added or lost, but expression can change a great deal.	Ch 3, Lect 3	Examined (§3.5 related: no gain or loss, yet the enhancer–promoter relationship changes)
Translocation	Two chromosomes exchange pieces.	Ch 3, Lect 3	Examined (one of the five structural abnormalities on slide 40)
Aneuploidy	A cell with the wrong number of chromosomes, caused by chromosomes not separating properly.	Ch 3, Lect 3	Examined (§3.5 and slide 37)
Non-disjunction	Chromosomes failing to separate properly during meiosis, producing an abnormal chromosome number.	Ch 3, Lect 3	Examined (§3.5 Tier 1: you must be able to tell M1 from M2)
Trisomy 21	Down syndrome: three copies of chromosome 21 instead of two.	Ch 3, Lect 3	Examined (§5 numbers and slide 37)
Turner syndrome (45,X)	A woman with a single X chromosome; the missing one was lost during meiosis.	Ch 3, Lect 3	Examined (§5 numbers and slide 37)
Copy number variation (CNV)	A stretch of DNA present in more or fewer copies than normal.	Ch 3, Lect 3	Excluded (he says plainly "I'm just give you a taste… don't be scared with all these mechanism")
Driver	A mutation that gives the cell a growth advantage — a typo in an important word.	Ch 3, Ch 10, Lect 3	Examined (§4.12 and the §7 confusable pair: combine the CNV profile with the mRNA expression, for example NPM1)
Passenger	A mutation that rides along with the driver and does no harm itself.	Ch 3, Ch 10, Lect 3	Examined (§4.12 and the §7 confusable pair)
PCR	The polymerase chain reaction, which makes large numbers of copies of a chosen DNA fragment.	Ch 3, Ch 11, Lect 1, Lect 3, Lect 4	Examined (slide 41: n cycles give about 2^n copies; the double-helix model is its foundation)
Primer	The PCR primer, which decides where copying starts.	Ch 3, Lect 1, Lect 3, Lect 4	Examined (GC content sets the Tm and hence primer design)
Denaturation	Separating double-stranded DNA into single strands; the first step of PCR.	Ch 3, Lect 1, Lect 3, Lect 4	Not examined (he excludes protocol parameters such as denaturation and temperature; the concept of the step must be understood)
Annealing	The primer binding to single-stranded DNA; the second step of PCR.	Ch 3, Lect 4	Not examined (same)
Extension	The polymerase adding new letters along the template; the third step of PCR.	Ch 3, Lect 1, Lect 4	Not examined (same)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
Melting temperature (Tm)	The temperature at which half of a double-stranded DNA has separated.	Ch 3, Ch 11, Lect 1, Lect 3	Examined (§5: the more GC, the higher the Tm; his slide gives the direction but never the numbers)
DNA fingerprint	Using differences in microsatellite length between individuals to identify them.	Ch 3, Lect 4	Not examined (a single sentence on Lecture 4 slide 28)
Plasmid	A small circular DNA that replicates inside bacteria; used as a vector.	Ch 3, Lect 4	Not examined (Lect 4 demonstrates the cloning workflow but does not list it as an exam point)
Vector	A vector: a carrier that takes foreign DNA into a cell and lets it replicate.	Ch 3, Lect 4	Not examined (same)
Restriction enzyme	An enzyme that recognises a specific sequence and cuts DNA there.	Ch 3, Lect 4	Not examined (same)
Sticky end	The single-stranded overhang a restriction enzyme leaves, which can join to a complementary overhang.	Ch 3, Lect 4	Not examined (same)
Ligation	The step in which an enzyme joins fragments into one complete molecule.	Ch 3, Lect 4	Not examined (same)

Chapter 4

Chromatin and epigenetics

Last chapter took the DNA packaging staircase about halfway up; this one starts at its finest step. A **Nucleosome** is a spool of DNA wound around one turn of **Histone** protein. The core is an **Octamer** — two copies each of H2A, H2B, H3 and H4; at the point where DNA goes in and comes out again there is a **Linker histone (H1)** keeping watch. One turn of DNA is about 147 bp and the nucleosome is about 10 nm across, so the "beads on a string" of Chapter 3 is simply these. Higher up, the string is folded into a **30 nm fibre**, then 300 nm, then 700 nm, and finally the 1,400 nm mitotic chromosome — the whole staircase is the series of numbers 2 nm, 11 nm, 30 nm, 300 nm, 700 nm and 1,400 nm. This structure was solved the hard way, by X-ray crystallography: the nucleosome core particle came out at 2.8 Å, and he calls anything below 2.0 Å "very excellent". It is hard because histones carry a heavy positive charge and crystallise badly. And precisely because histones are so highly conserved — the same from fruit fly to human — a result in a model organism can be extrapolated. One more thing to remember: histone and DNA are held together by hydrogen bonds and electrostatic forces, and **not** by covalent bonds, so chromatin can be opened or closed at any time.

Then come the switches. The loose, open, being-read form is called **Euchromatin**; the tight, silent form is called **Heterochromatin**. He uses hard language here: heterochromatin physically blocks transcription factors and enhancers so that they cannot get anywhere near the promoter; it is not "a bit quieter". The state is not fixed — the **Nucleosome** slides along the DNA, uncovering the promoter or covering it over, and two things push the sliding: the intrinsic affinity of a stretch of DNA for the histone itself, and the ATP-consuming **Chromatin remodelling complex**. On his Lecture 3 slide he lists four ATP-dependent families: SWI/SNF, INO80, ISWI and CHD; but he also says "don't memorize all these figures", so remembering "there are four families and they move nucleosomes using ATP" is enough, and you do not need the yeast/fly/human/mouse comparison table.

The second switch is on the DNA itself. **DNA methylation** is the addition of a **Methyl group** (-CH3) to the fifth carbon of cytosine, and the **CpG** position (a C followed by a G) is where it happens most often. The **Methyltransferase** enzymes that add the methyl come in two kinds, and this is his confusable pair: the **De novo methyltransferase** (DNMT3A, DNMT3B) **establishes** new marks, while the **Maintenance methyltransferase** (DNMT1) copies the marks on the old strand **onto the daughter strand** after DNA replication — so a mark does not fade away when the cell divides, and that is the precondition for epigenetics being able to "remember". The direction decides the outcome: when the promoter of a tumour suppressor gene becomes **Hypermethylated**, the gene is silenced and the brake is jammed, without a single letter of the DNA having changed; conversely, when pericentromeric repeat sequences become **Hypomethylated**, the chromosome starts to become unstable through mitotic recombination. Both routes his slide 20 draws as two arrows leading to cancer.

The third switch is on the histones, and that is the **Histone code**. You must be able to read the nomenclature: H3K9me3 means Histone 3, lysine at position 9, trimethylated; the same **Methylation** on a histone has a different effect depending on where it sits — **H3K9me3** marks "this patch of DNA stays closed", while **H3K4me3** marks "here is an active promoter". **Acetylation** is the other end of the same lever: adding an acetyl group neutralises the positive charge on the histone, the chromatin loosens and the gene is read out; taking the acetyl group away tightens it up and shuts the gene back down. The position of the acetylation carries information as well: **H3K18ac** (acetylation of lysine at position 18) being low predicts worse survival in pancreatic cancer, breast cancer, prostate cancer and lung cancer.

These three layers of switches are not independent — they pull on each other, which is what he calls **Cross-talk**: DNA methylation and histone marks strengthen each other, forming a **Positive feedback** loop that locks a state in place, so the same cell divides many times over and the gene stays on or off regardless — that is **Cellular memory**. The clinical meaning of epigenetics is right there: a gene can be fully present, with a perfectly normal sequence, and still be locked shut. The lock is rhythmic in development: his slide traces the methylation curve — dropping to its lowest in the morula and blastocyst stages, climbing steeply after the blastocyst, staying high in differentiating somatic cells and in germ cells (the primordial germ cell), with the PGCs going into the germ line being

demethylated once more. Another slide covers the reverse mechanism: the TET enzymes can remove the methyl from DNA, and H3K4me3 blocks DNMT from binding at a CpG, so the "on" histone mark and the "off" DNA methylation are actually mutually exclusive at the same site — that is the molecular version of cross-talk. While we are here, these marks can still be read after the tissue has left the body: because the methylation pattern survives in blood and in stool, stool tests and blood-based cell-free DNA can both be used for non-invasive screening, which is where Lecture 3 connects epigenetics to clinical testing.

Last of all, how it is measured and how it is used. **Bisulfite sequencing** is the gold standard: bisulfite converts unmethylated C into U while methylated C is left alone, so after sequencing you can read the methylation pattern at single-base resolution. Two clinical biomarkers you must know: **CIMP** (CpG Island Methylator Phenotype) is the widespread promoter hypermethylation pattern in colorectal cancer with a worse prognosis, and the number he gives in his multivariate analysis is OR 39.6 for CIMP-high against CIMP-0 (95% CI 20.6–76.1, $p < 0.0001$); **MGMT** promoter methylation is a predictive marker in glioblastoma — once MGMT is silenced, the tumour cell can no longer repair the DNA damage caused by alkylating drugs, so it is more sensitive to alkylating chemotherapy plus radiotherapy and survives better. On the treatment side there is the **HDAC inhibitor**: block deacetylation and acetylation accumulates, and the silenced tumour suppressor gene starts to be expressed again. His Lecture 3 also contains a lovely trick — an HDAC inhibitor can induce PD-L1 on the tumour cell, creating what he calls gene addiction, so the tumour becomes sensitive to anti-PD-1 checkpoint blockade and a cold tumour gets a chance to be turned into a hot one. Among the specific drugs on his slide 50, **MS-275** is a class I HDAC inhibitor that upregulates IFN-γR-STAT1, promotes M1 macrophage polarization and inhibits MDSC.

There is one more set of numbers and one more confusable pair to hold on to. He has run a complete methylation analysis on **CDKN2A** (that is, p16): the AUC is 0.847 for exon 1 and 0.924 for exon 2, and 0.938 for the two combined, so "exon 1 versus exon 2" is one of the pairs in his confusable set — exon 2 is the region with the higher diagnostic value. In his multivariate logistic regression the strongest predictor is CIMP-high (OR 39.6 against CIMP-0), and the factor predicting loss of p16 protein expression is CDKN2A methylation itself, OR 12.7 (7.84–20.7, $p < 0.0001$). There is also a diagram of the mechanism by which an HDAC inhibitor induces PD-L1: the HDAC3 inhibitor RGFP 966 acts through BRD4 on the PD-L1 promoter, pulls PD-L1 up, and the tumour becomes more sensitive to PD-L1 blockade. This "make the tumour open the switch itself, then close the switch" logic is the prettiest clinical deduction in Lecture 3.

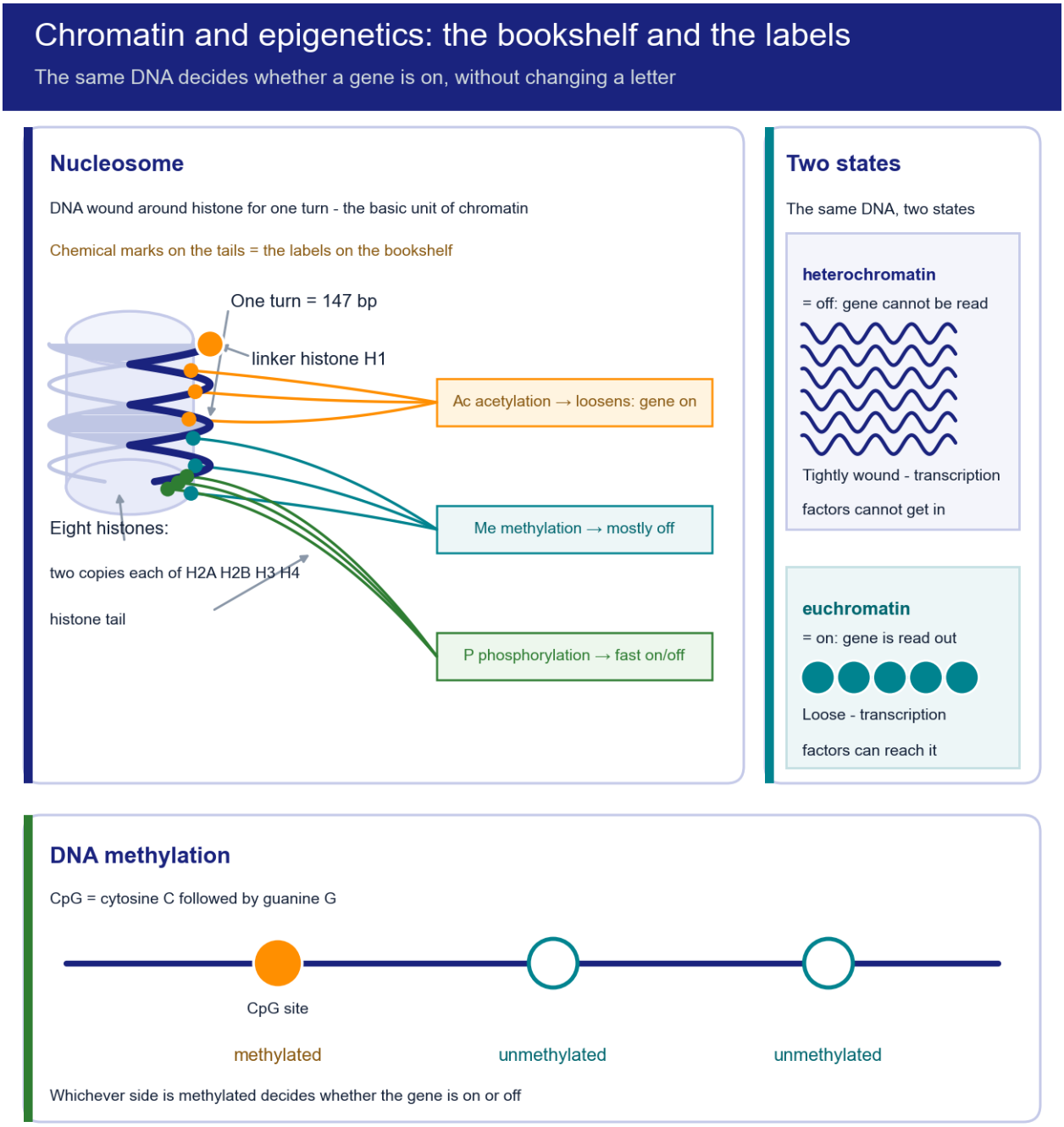


Figure 4: The nucleosome is a spool, and the chemical marks on the histone tails decide whether a gene is on or off.

Key vocabulary for this chapter

Term	In one sentence	Source	Did he name it?
Nucleosome	A spool of DNA wound around one turn of histone: the basic unit of chromatin packaging.	Ch 3, Ch 4, Lect 3	Examined (§5: one turn 147 bp, about 10 nm across)
Histone	The protein that DNA winds around to form a nucleosome.	Ch 4, Lect 3	Examined (§4.10: highly conserved, the binding is non-covalent, and the positive charge makes it hard to crystallise)
Octamer	The eight histones of the nucleosome core: two copies each of H2A, H2B, H3 and H4.	Ch 4, Lect 3	Examined (slide 9)
Linker histone (H1)	The histone that guards the point where DNA enters and leaves the nucleosome.	Ch 4, Lect 3	Examined (slides 8 and 9)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
147 bp	The length of DNA wrapped around the nucleosome core.	Ch 4, Lect 3	Examined (slide 9: core particle 147 bp, about 10 nm)
10 nm	The diameter of a nucleosome, which is why chromatin looks like beads on a string.	Ch 4, Lect 3	Examined (§5 numbers)
2.8 Å	The resolution of the crystal structure of the nucleosome core particle.	Ch 4, Lect 3	Examined (§5: below 2.0 Å he calls "very excellent")
30 nm fibre	The 30-nanometre chromatin fibre formed when nucleosomes pack together.	Ch 4, Lect 3	Examined (the 2 → 11 → 30 → 300 → 700 → 1,400 nm staircase on slide 4)
Chromatin remodelling complex	The machine that moves nucleosomes along the DNA using ATP, uncovering or covering a gene.	Ch 4, Lect 3	Examined (§4.10: one of the two drivers of sliding)
SWI/SNF	One of the four families of ATP-dependent chromatin remodelling.	Ch 4, Lect 3	Not examined (the family comparison table on slide 12; he says plainly that the details of the table need not be memorised — knowing there are four families is enough)
ISWI	One of the four families of ATP-dependent chromatin remodelling.	Ch 4, Lect 3	Not examined (same)
CHD	One of the four families of ATP-dependent chromatin remodelling.	Ch 4, Lect 3	Not examined (same)
DNA methylation	Adding a methyl group to the DNA, used to switch a gene off.	Ch 4, Lect 3, Lect 5	Examined (Tier 2 §4.1; the §7 confusable pair de novo vs maintenance)
CpG	The position where a C is followed by a G; the sequence that is methylated most often.	Ch 4, Lect 3	Examined (§4.1: promoter methylation at a CpG island silences a gene)
De novo methyltransferase	The enzymes that establish new methyl marks (DNMT3A, DNMT3B).	Ch 4, Lect 3	Examined (§4.1 and the §7 confusable pair: establishing vs copying)
Maintenance methyltransferase	The enzyme that copies the marks onto the daughter strand after replication (DNMT1).	Ch 4, Lect 3	Examined (§4.1 and the §7 confusable pair)
Hypermethylated	Methylated too much; promoter hypermethylation silences the gene.	Ch 4, Lect 3	Examined (§4.1: hypermethylation of a tumour suppressor promoter is a common cancer driver)
Hypomethylated	Methylated too little; it can bring about chromosomal rearrangement and genomic instability.	Ch 4, Lect 3	Examined (slide 20: the carcinogenic route through demethylation of the pericentromeric region)
Bisulfite sequencing	Bisulfite sequencing: reads out the methylation pattern at single-base resolution.	Ch 4, Ch 11, Lect 3	Examined (§4.1: the gold standard; unmethylated C → U, methylated C is kept)
Histone code	The system of marks on histones that tells the cell whether a patch of DNA is on or off.	Ch 4, Lect 3	Examined (§4.1; the nomenclature on slide 28)
H3K9me3	Trimethylation of lysine at position 9 of histone 3; it marks "this stays closed".	Ch 4, Lect 3	Examined (the nomenclature and its meaning, listed explicitly in §4.1)
Acetylation	Adding an acetyl group to a histone; the chromatin loosens and the gene is read out.	Ch 4, Lect 3, Lect 5	Examined (§4.1; slide 29: HAT up, HDAC down)
H3K18ac	Acetylation of lysine at position 18 of histone 3; a low level predicts worse survival.	Ch 4, Lect 3	Examined (slide 34: low H3K18ac predicts worse survival in pancreatic, breast, prostate and lung cancer)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
Methyltransferase	An enzyme that adds a methyl group to DNA or to a histone.	Ch 4, Lect 3, Lect 5	Examined (the DNMT family; adding the methyl means silencing)
HDAC inhibitor	A drug that blocks deacetylation, so chromatin opens and silenced genes are expressed again.	Ch 4, Lect 3, Lect 5	Examined (§4.4: cold tumour induces PD-L1 → gene addiction → checkpoint blockade; slide 34, restoring acetylation)
Cross-talk	The cross-talk by which DNA methylation and histone marks influence each other.	Ch 4, Lect 3	Examined (§4.1)
Positive feedback	A feedback loop in which two layers of marks strengthen each other and lock the state in.	Ch 4, Lect 3	Examined (§4.1: the mechanism that forms cellular memory)
Cellular memory	After the same cell has divided many times, the gene still stays on or off as it was.	Ch 4, Lect 3	Examined (§4.1)
MS-275	A class I HDAC inhibitor that upregulates IFN-γR-STAT1 and inhibits MDSC.	Ch 4, Lect 3	Examined (slide 50: promoting M1 macrophage polarization and tumour immunity)
CIMP	The widespread CpG island methylation phenotype of colorectal cancer, with a worse prognosis.	Ch 4, Lect 3	Examined (§4.1 and §5: CIMP-high vs CIMP-0 OR 39.6, 95% CI 20.6–76.1, $p < 0.0001$)
MGMT	Silenced by promoter methylation, which makes glioblastoma more sensitive to alkylating chemotherapy plus radiotherapy.	Ch 4, Lect 3	Examined (§4.1 and the §8 scenario question: the standard example of a predictive biomarker)

Chapter 5

DNA replication

The first thing a factory does when it becomes two factories is not split up — it copies the manual. Before a cell divides it has to manufacture the whole of its DNA once more, about three billion letters, so that every daughter cell receives a complete printed manual with no missing pages. The copy has to be almost perfect, because one typo in the manual is nothing much if it lands in an unimportant paragraph, and turns a normal cell into a cancer cell if it lands in an important one. In Lecture 4 Dr. Wong breaks the whole route into four stages: replication (copying faithfully), damage (copying wrongly, or being hurt), repair (the cell's response) and recombination (the cell deliberately cutting itself open and putting it back together). This chapter is about the first stage.

Copying a manual is harder than you would think, because DNA is not a flat line but a double helix like a zip. To copy it you have to open it, and the way to open it is to cut the hydrogen bonds that hold the two halves together. The machine that does this is **helicase**, which runs on ATP and walks along the DNA separating the two strands as it goes. Opening it creates tension ahead of the fork, like twisting a rubber band up, and that tension is relieved by **topoisomerase**, which briefly cuts the DNA, lets it rotate, and seals the cut back up. The two separated strands also need someone to hold them so they do not spring back together, and that handrail is the **single-strand binding protein (SSB)**.

The real trouble is that **DNA polymerase** cannot start a strand from nothing. So **primase** lays down a short stretch of RNA on the template first — the **RNA primer**, about 10 nucleotides long. Its only job is to provide a free 3′-OH so that DNA polymerase has something to build on. Each time a **dNTP** is added, the polymerase attaches it to the 3′ end and releases pyrophosphate.

Next comes the central structural problem of the whole theory of replication: the two strands of the double helix are antiparallel, while DNA can only grow in one direction, from 5′ to 3′. So of the two templates, one can be copied straight through in one go, and the other has to be copied backwards, in short blocks laid end to end. The continuous one is the **leading strand**; the blocked one is the **lagging strand**, and its short blocks are **Okazaki fragment**s. In plain terms: two people pulling the same rope in opposite directions, each holding a pen that can only write left to right. The one on the left writes smoothly; the one on the right has to write a short bit, stop, and start again somewhere else.

The order of the steps behind the replication fork is fixed: primase lays a new RNA primer; DNA polymerase extends it, making a new Okazaki fragment; the old RNA primer is removed by an exonuclease — his slide names **RNase H** specifically for this job; DNA polymerase fills the gap; and finally a **nick** is left in the backbone, which **DNA ligase** seals. The seal is made by forming a **phosphodiester bond** between a 5′ phosphate and a 3′ hydroxyl. The polymerase making the leading strand needs only one primer for the whole strand; the polymerase making the lagging strand gives every short block its own primer, and the primers are about 200 nucleotides apart.

Remember the direction rules as two of them, one about adding and one about removing: nucleotide elongation goes 5′→3′; proofreading and excision go 3′→5′. This is the pair of directions in Lecture 4 most likely to be turned into a multiple-choice question.

Proofreading is the quality control built into the copying machine. As well as synthesising, DNA polymerase has a **3′→5′ exonuclease** activity: it cuts off the one mismatched letter at the end of the strand it has just made and puts the right one back. A base that has paired with the wrong partner — A against G, for instance — is a **mismatch**. The order is: add the wrong letter → the 3′ end bulges and the polymerase stops → the exonuclease cuts the wrong letter off → the correct one is added and copying continues. On his slide the leading strand uses Pol ε and the lagging strand uses Pol δ, each with one POL domain and one EXO domain; and the mismatch-repair heterodimers MSH2/MSH6 and MLH1/PMS2 sit on standby as a second safety net.

Here there is a clinical turn. A **Nucleoside analogue** is a drug designed to look like an ordinary nucleotide; once it is mixed into DNA it jams the extension of the chain, and that is how it kills a dividing cell. But if the 3′→5′ exonuclease activity cuts the drug itself away, the polymerase simply puts an ordinary base back and the cell carries on living.

That is **drug resistance**. His slide splits the replication fork into two roads: if p53's exonuclease removes the nucleoside analogue, the result is drug resistance; if further DNA synthesis is blocked, the result is cell death signalling. And **p53** has its own 3′→5′ exonuclease activity, playing the part of an external proofreader.

And the same defect, in a different situation, is good news. A mutation that disables proofreading lets the cell accumulate a very high mutation load and produce a great many **neoantigen**s — protein fragments the immune system has never seen, so it has no tolerance for them, and a cytotoxic T lymphocyte can present and recognise them through MHC class I and kill the cell. On his slide, a proofreading mutation in Pol ε turns the cell into an ultramutator: more neoantigens, more tumour-infiltrating lymphocytes, and the result is a good prognosis — the Kaplan–Meier curve of POLE-mutant patients sits above that of POLE wild-type patients. A broken proofreading machine builds a tumour that is easier to see and easier to kill.

So how does replication actually carry on? Two models once competed: the conservative model said the old double helix stays exactly as it is and a brand-new one is built alongside, and the dispersive model said the old strands are broken up and scattered into the new molecules. What settled it was the **Meselson–Stahl** type of isotope-labelling experiment: DNA made with heavy nitrogen does not settle in the middle of the tube when it is centrifuged. After one round of division, if semi-conservative replication is right, every molecule is half heavy and half light, and only the middle band appears in the tube. The result matched **semi-conservative replication**: every daughter helix is made of one old strand and one newly synthesised strand. Every DNA strand you are carrying today has a continuous ancestral line that traces back to your parents.

Eukaryotic chromosomes are far too big for replication to start at a single point, so instead it starts at **origin of replication**s in thousands of places at once. The line on his second-level list is this: **thousands of origins, each opening a replication bubble, each bubble with two replication forks running in opposite directions (bidirectional).** This arrangement is called multiple replicons, and on his electron micrograph you can see several bubbles on the same fibre, with a 5 kb scale bar. The reason is purely one of volume: a human chromosome carries billions of bases, and if only one origin fired, replication would take far too long and the cell could never divide.

Last of all there is the problem at the end of the chromosome — the **end-replication problem**. The lagging strand is built backwards in blocks, and at the very end of the chromosome there is not enough room for the last primer to sit down, so a gap is left and the chromosome gets a little shorter with every round. Both ends of the chromosome therefore carry a tail of repeated DNA, the **telomere**, whose human repeat unit is TTAGGG. And **telomerase** can rebuild that tail. It is unusual: it is a reverse transcriptase, an enzyme that makes DNA from an RNA template, and it carries the template inside itself — the complex has two named parts, **TERC** (the RNA template) and **TERT** (the catalytic enzyme). His slide draws four steps: telomerase pairs with its RNA template and the 3′ overhang → extends it with GGTTAG repeat units → the enzyme moves along and repeats the whole thing → primase and DNA polymerase finish the complementary strand.

When the telomere runs out, the cell stops dividing, and that ceiling is called the **Hayflick limit**: the number of times an ordinary human cell can divide before it becomes senescent. Cancer cells' way out is to switch telomerase back on. His liver cancer story runs like this: a foetal liver has long telomeres and high TERT expression; a healthy adult liver has short telomeres and TERT switched off; chronic liver injury drives hepatocyte turnover and telomere shortening; cirrhosis reaches the Hayflick limit, which starts the DNA damage response and apoptosis; and then a single mutation in the TERT gene pushes telomerase activity up and restores telomere length, so the tumour can keep dividing. The clinical direction of that telomere marker has to be read the right way round: in R0-resected hepatocellular carcinoma (n = 578), the tumours whose telomere length fell into the top 10 per cent group had worse early-recurrence-free survival (early-recurrence-free survival P = 0.006), with a hazard ratio of 1.66 (1.05–2.61) in the multivariate analysis, p = 0.029. In other words, the tumours that managed to keep their long telomeres are exactly the ones that recur sooner. The treatment version of this is an **antisense oligonucleotide (ASO)** against TERT: in a flank xenograft experiment in immunodeficient mice, tumours implanted with no prior treatment took at 9/12, with two weeks of anti-TERT pre-treatment at 4/12 (Fisher's p = 0.1), and with three weeks at 2/12 (Fisher's p = 0.01).

One last clinical application, and this is the step where Lecture 4 walks from molecules to drugs. A **Topoisomerase inhibitor** kills every dividing cell, so on its own it is almost useless; but hang the same poison on an antibody that recognises only the tumour's receptor and it becomes an **ADC** (antibody–drug conjugate), with the payloads he names being SN-38 and DXd and the clinical targets being HER2 and TROP2. The antibody is the postcode and the poison is the letter. And keep his exclusion list in mind: he will not examine the structure of an ADC or the chemistry

of a dual-payload ADC; what will be examined is the reason for putting a topoisomerase inhibitor inside an antibody — selective delivery.

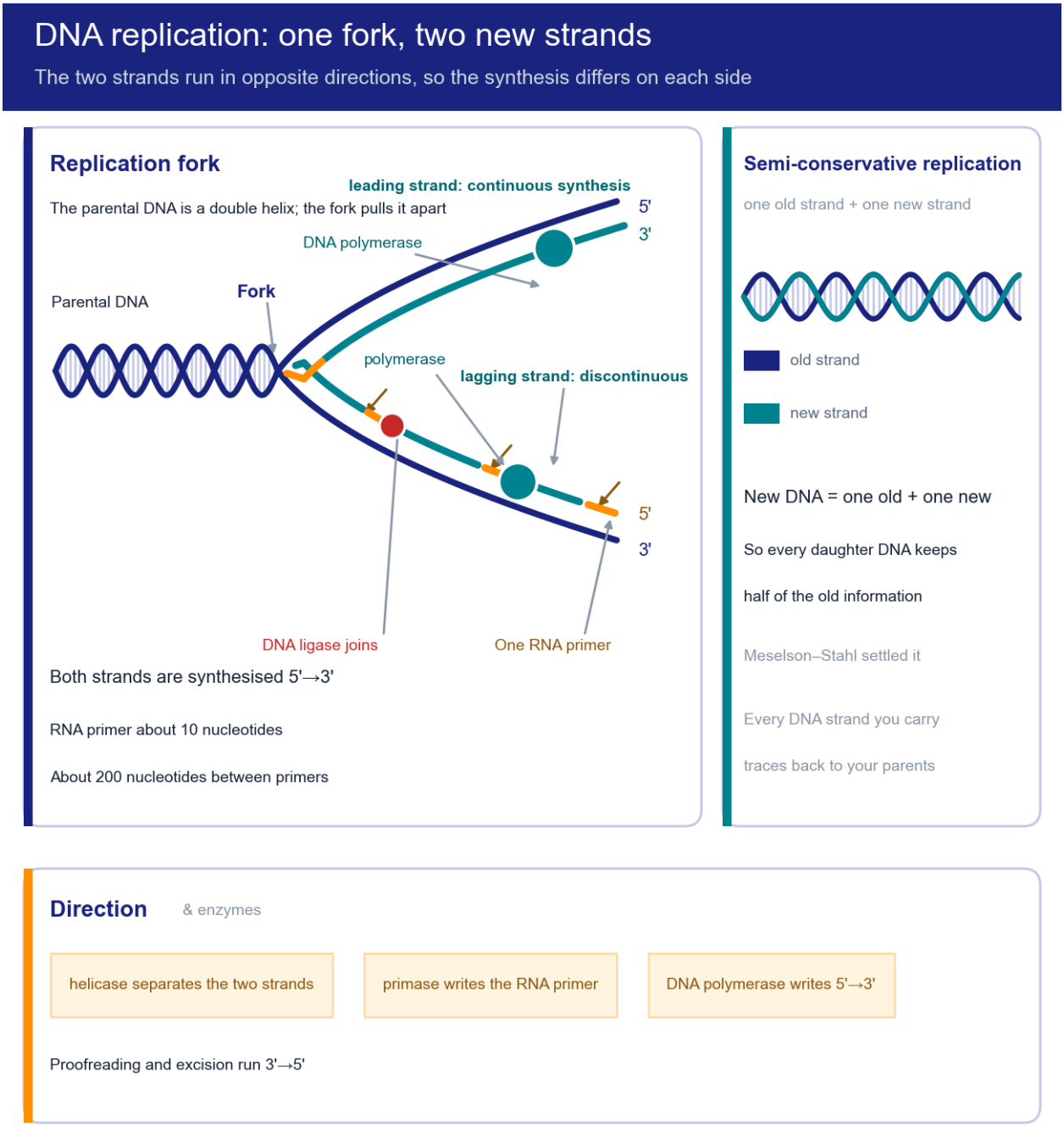


Figure 5: The replication fork. The leading strand is made continuously, while the lagging strand is made in Okazaki fragments.

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
origin of replication	The site where replication begins; a eukaryotic cell has thousands of them.	Ch 5, Lect 4	Examined (second-level)
replication bubble	The replication bubble opened up by one origin.	Ch 5, Lect 4	Examined (second-level)
replication fork	The Y-shaped opening at either end of a replication bubble, running in opposite directions.	Ch 5, Lect 4	Examined (second-level)
bidirectional	Two replication forks leaving the same origin in opposite directions.	Ch 5, Lect 4	Examined (second-level)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
helicase	The machine that uses ATP to cut the hydrogen bonds and unwind the double helix.	Ch 5, Lect 4	Examined (he covered it)
topoisomerase	Cuts DNA and seals it again, relieving the tension ahead of the replication fork.	Ch 5, Lect 4	Examined (he covered it)
topoisomerase inhibitor	The poison that kills the machine that untangles DNA; it is an ADC payload.	Ch 5, Lect 4	Examined (he covered it)
primase	The small RNA polymerase that lays down a short RNA primer.	Ch 5, Lect 4	Examined (he covered it)
RNA primer	A short RNA that provides a free 3′-OH; about 10 nucleotides long.	Ch 5, Lect 4	Examined (he covered it)
DNA polymerase	The copying machine that adds the dNTPs one by one.	Ch 5, Lect 4	Examined (he covered it)
dNTP	The triphosphate deoxynucleotide added to the new strand, releasing pyrophosphate as it goes.	Ch 5, Lect 4	Examined (he covered it)
leading strand	The strand that can be copied straight through continuously, needing only one primer.	Ch 5, Lect 4	Examined (second-level)
lagging strand	The strand that is copied in short blocks.	Ch 5, Lect 4	Examined (second-level)
Okazaki fragment	A short stretch of new DNA on the lagging strand, each with its own primer.	Ch 5, Lect 4	Examined (second-level)
nick	A break in one strand, sealed at the end by ligase.	Ch 5, Lect 4	Examined (slide)
DNA ligase	Seals the nick and joins the pieces into one complete strand.	Ch 5, Lect 4	Examined (he covered it)
phosphodiester bond	The bond ligase forms between a 5′ phosphate and a 3′ hydroxyl.	Ch 5, Lect 4	Examined (the second-level direction)
proofreading	The quality control built into the copying machine.	Ch 5, Lect 4	Examined (second-level)
3′→5′ exonuclease	The direction of proofreading and excision; it can also cut away a nucleoside analogue drug.	Ch 5, Lect 4	Examined (second-level)
semi-conservative replication	Every daughter helix = one old strand + one new strand.	Ch 5, Lect 4	Examined (he covered it)
Meselson–Stahl	The heavy-nitrogen labelling experiment that settled semi-conservative replication.	Ch 5, Lect 4	Examined (he covered it)
single-strand binding protein (SSB)	Holds the separated strands apart so they do not spring back together.	Ch 5, Lect 4	Examined (slide)
end-replication problem	There is no room at the end for the last primer, so the chromosome shortens every round.	Ch 5, Lect 3, Lect 4	Examined (he covered it)
telomerase	A reverse transcriptase that carries its own RNA template and rebuilds the telomere.	Ch 5, Ch 12, Lect 3, Lect 4	Examined (he covered it)
TERC	The RNA template part of the telomerase complex.	Ch 5, Lect 4	Examined (he covered it)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
TERT	The catalytic enzyme part of the telomerase complex; the target in liver cancer.	Ch 5, Lect 4	Examined (he covered it)
Hayflick limit	The number of times an ordinary human cell can divide before it stops dividing.	Ch 5, Lect 3, Lect 4	Examined (he covered it)

Chapter 6

DNA damage and repair

The last chapter ended at the moment the cell finished copying the manual. But the copy is only the starting point: from that moment on, the manual is under attack — all day and all night. When he taught Lecture 4, Dr. Wong called this the darkest problem in the whole course, and he split it in two: first "what is the damage", then "how does the cell fix it". The order of the lecture follows that — describe, classify, alarm system, five repair shops, and finally the two things he says in so many words will be examined: how a defect in NHEJ becomes SCID, and how the same machinery can be turned into a drug.

DNA damage can be asked about in two directions: how severe is the damage? and where did the injury come from? The source question divides in two. **Endogenous** damage is caused by the cell itself: **replication errors**, where the polymerase simply gets it wrong; strand breaks from an **abortive topoisomerase**, where the enzyme cuts the DNA and cannot seal it back; and metabolic by-products, especially **ROS** (reactive oxygen species) attacking the telomere. **Exogenous** damage comes from outside: **Ultraviolet light** causes a **pyrimidine dimer**, two adjacent bases gluing together chemically; ionising radiation causes double-strand breaks; an **intercalating agent** slides between the bases and twists the helix out of shape; and viruses, which can integrate into the genome, as HBV does in liver cancer. The chemical category is **alkylation** and **adduct** — a chemical group is attached to a base, so the letter is still there but it has changed shape.

On the severity question, the scale runs from a single base carrying a chemical tag all the way to both strands of the helix being cut at the same position. His slide lists the lesion types: single-strand break, double-strand break, bulky adduct, mismatch, insertion and deletion, and base modification (for example alkylation). These are not arbitrary labels — each one is the job description of one of the repair shops that follows.

The cell's first response is not to repair; it is to **raise the alarm**. The **DNA Damage Response (DDR)** is a four-stage cascade: signals (the damage itself and the replication stress) → sensors (the things that sit on the damaged DNA and notice) → transducers (which pass the message down) → effectors (which do the work, and he names **p53** and **BRCA**). From the effectors there are four possible outcomes: stop the cell cycle, turn transcription on, carry out DNA repair, or trigger apoptosis (the cell's controlled suicide). His slide marks "stop the cell cycle" with a road sign and apoptosis with a gravestone. The two key sensor kinases are **ATM** and **ATR**, and one of the first things the cell does is inhibit CDK so that the cycle slows down and buys repair time. If the damage is already beyond repair, the caspase cascade starts. The link to cancer is therefore direct: **cancer cells very often switch this braking system off**, and a cell that is broken but keeps dividing and keeps accumulating damage grows into a tumour.

Five repair shops. **Direct reversal** is the simplest: no nucleotide removal, no resynthesis, no ligation — his example is **MGMT**, which removes the alkyl group from O6-methylguanine. **BER** (base-excision repair) handles a single damaged base: a **DNA glycosylase** removes the damaged base and leaves an **abasic site**, an **AP site** (the sugar is still there, the letter has gone), and the gap is then excised, filled by a polymerase and sealed by ligase. **NER** (nucleotide-excision repair) handles bulky lesions: it comes in two flavours, **GG-NER** uses **UV-DDB** (DDB1 and DDB2, the DNA damage-binding proteins) plus **XPC** to find the lesion anywhere in the genome, and **TC-NER** uses CSA and CSB to spot a lesion stalled on RNA polymerase II; both routes finally funnel into **TFIIH** (XPB and XPD), and then **XPA**, RPA and two endonucleases (**XPF** and **XPG**) cut out a stretch of 22–32 nucleotides. **MMR** (mismatch repair) fixes replication mismatches and small insertions and deletions: a single-base mismatch is spotted by **MutSα** (the mismatch-recognition complex **MSH2**/MSH6), a mismatch in a repeat sequence is spotted by MutSβ (MSH2/MSH3), and both recruit **MutLα** (the mismatch-excision complex **MLH1**/PMS2), which excises, and then polymerase and ligase put it back. The fifth category is the broken double strand (**DSB**, double-strand break), handled by **HR** and **NHEJ**.

These two contrasts are the discriminating points you must learn, and they are the distractors multiple-choice questions love: **BER repairs a single damaged base; NER repairs a bulky lesion that spans several nucleotides**, such as a UV dimer. **MMR repairs replication mismatches and small indels; HR is the**

high-fidelity repair of broken double strands. And keep his exclusion list in mind: he will not examine the chemistry of DNA glycosylase and the AP site ("I can ensure that they will not be included in the coming midterm"), but you must know that BER removes a single damaged base and NER cuts out a stretch of nucleotides around a bulky lesion.

When the third repair shop breaks, the story runs all the way to the clinic. A **Microsatellite** is a short tandem repeat scattered around the genome — a dinucleotide or trinucleotide repeated over and over, of a length that differs from person to person, and that difference is the basis of DNA fingerprinting. When the polymerase copies such a repeat it can **slip**, leaving an unpaired CA loop on the new strand. When mismatch repair is intact, the loop is found and corrected, and the result is a stable microsatellite, that is **MSS**; when mismatch repair is defective (**MMR deficiency**, the synonym dMMR), the slip is not corrected, the repeat gets longer or shorter, and the result is **MSI** (microsatellite instability), and at the highest level of instability it is called **MSI-H**. The consequences run on from there: a slipped repeat often sits inside a gene, so the cell accumulates **frameshift mutations** and produces a great many truncated proteins and, more importantly, a great many mutation-associated neoantigens — fragments that MHC class I presents to a cytotoxic T lymphocyte. To escape, the tumour puts PD-L1 on its surface so that it binds PD-1 on the T cell and switches the T cell off; that is called T-cell anergy, and an anti-PD-1 antibody that blocks this lets the T cell wake up. So the whole chain is: MMR deficiency → MSI-H → frameshift → neoantigen → hot tumour → response to immune checkpoint inhibitors. And the number to remember: **MSI arises from mismatch repair deficiency and occurs in 15 % of colorectal cancer.** He also reminds you that one common cause of losing MMR is **hypermethylation of the MLH1 promoter** — the methylation switch from Lecture 3 switching off the repair shop from Lecture 4.

The fourth and fifth shops handle broken double strands, and the contrast between them is the pair he calls "the one most likely to be used as a multiple-choice distractor". **Homologous recombination (HR)** uses a homologous template — the sister chromatid — to put the break back, so it is exact, but sister chromatids exist only in S and G2, which means **HR happens only in S/G2**. Its machinery is RAD51, Rad52 and RPA carrying out the strand invasion into the paired duplex, and **BRCA1** and **BRCA2** are the genes that command this invasion. **Non-homologous end joining (NHEJ)** needs no template at all: it simply grabs the two loose ends and joins them — the quick fix, but error-prone (the result is often a deletion, or a deletion plus an addition), and it works at every stage of the cell cycle, including G1, including in cells that will never divide again. On his slide the broken ends are first protected by **Ku**, and then DNA-PKcs and ligase complete the joining; the factors that push the cell towards the NHEJ side are **53BP1** and **Shieldin**. There are other doors in between too: **alternative end joining (a-EJ)** needs only a few base pairs of homology, while single-strand annealing needs more than about 20 base pairs. Five things are enough to remember: HR uses a template, is exact, and happens only in S and G2; NHEJ uses no template, works in every phase and is error-prone; HR is what meiotic recombination uses and NHEJ is what the immune system uses; and in CRISPR, HR gives you a knock-in and NHEJ gives you a knock-out.

Synthetic lethality describes two errors: the cell can survive one of them on its own, but the two together it cannot bear. The pharmacological use is this: if the tumour has already lost a gene by itself, you use a drug to block the second one. A **BRCA1/2** mutation shuts down the tumour's HR, so giving it a **PARP inhibitor** has a special effect — the PARP inhibitor blocks PARylation and traps PARP on single-strand breaks, the replication forks stall and accumulate double-strand breaks, and the HR-deficient tumour cell dies, while a normal cell can still repair those breaks with its intact HR and survive. That difference opens up a wide **therapeutic window** — the margin between the minimum effective dose and the dose at which toxicity begins.

And the most surprising part is this: the repair work is not only repair. The cell deliberately cuts its own DNA open and puts it back together, in order to manufacture the millions of different antibodies the immune system needs. **V(D)J recombination** is the process by which, during the development of B cells and T cells, the antibody genes and the T cell receptor genes are deliberately reassembled. V stands for variable, D for diversity and J for joining. On his slide 40 the germline DNA of the immunoglobulin heavy chain carries the segments **V 39, D 27, J 6** and C 6 (his numbers are V ≈ 39, D = 27, J = 6); the cell first does D-J and then V-DJ, two rounds of somatic recombination, and the product is transcribed, spliced into a message and translated into the heavy chain. The light chain is simpler: V 36, J 5, recombined as V-J. Two details make the whole mechanism work: each segment has a **recombination-signal sequence** on both sides, and the complex made of **RAG1** (recombination-activating protein) and **RAG2** (recombination-activating protein) recognises these and does the cutting; and the machinery that processes and joins the ends is **NHEJ**. T cells do the same thing: V-J for the α chain, V-D-J for the β chain,

assembled on the cell surface into an αβ heterodimer with CD3ε, CD3δ and the zeta chain beside it, on a CD4 T cell that is reading an antigen presented by MHC class II.

Now the failure case, which he has labelled in his own words as a first-level exam question. His words are: "if the people showing a defective in this non-homologous end joining, what's the outcome? Definitely the outcome is the severe immune deficiency... Because he cannot generate a sufficient amount of the antibody pool to recognize the thousand copies of the foreign antigen." The causal chain you must be able to draw is this: **NHEJ is borrowed by V(D)J recombination to join the V, D and J segments → producing the diversity of antibodies (B cells) and TCRs (T cells) → an NHEJ defect means the diversity library cannot be built → SCID (severe combined immunodeficiency).** The clinical consequence is a child with no usable pool of antibodies.

The same machinery can also be used as a drug. **CRISPR/Cas9** is the bacterial immune system, re-engineered by engineers into a pair of molecular scissors with an address label: **gRNA** (guide RNA) specifies where to cut, and once Cas9 has recognised the short necessary sequence next to it (the **PAM**), the two nuclease domains **RuvC** and **HNH** cut out a double-strand break. What happens next is the choice you made deliberately: if the cell repairs it with NHEJ, small insertions or deletions are made, and since the coding sequence is read in threes, those small changes usually destroy the gene — that is a **knock-out**, and it works at every stage of the cell cycle. If the cell uses **HDR** (homology-directed repair) and a **donor template** is supplied, the new sequence is written in accurately — that is a **knock-in**, and it works only in S and G2, because that is when sister chromatids are available. The difference between the two routes is not a preference but a limitation of cell biology — and it is exactly the HR-versus-NHEJ difference above. ZFN is an earlier generation of the same tool and is now only a backup option.

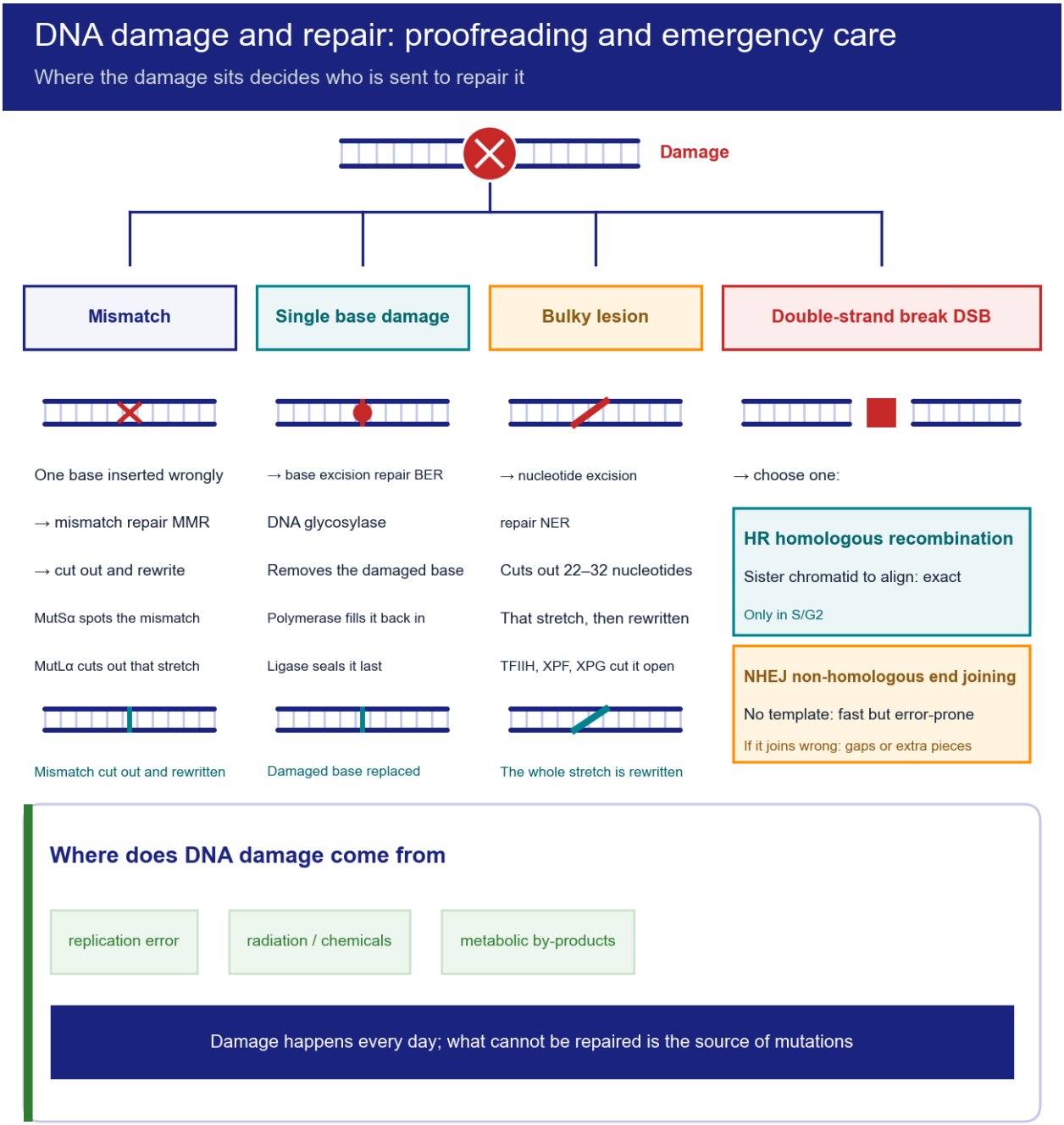


Figure 6: Where the damage sits decides who repairs it; get the pairing wrong and chromosomal rearrangements accumulate.

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
DNA damage	A chemical or structural injury to the DNA.	Ch 6, Lect 4	Examined (he covered it)
endogenous	A source of damage caused by the cell itself.	Ch 6, Lect 4	Examined (slide)
exogenous	A source of damage caused from outside.	Ch 6, Lect 4	Examined (slide)
adduct	An adduct formed when a chemical group is attached to a base.	Ch 6, Lect 4	Examined (slide)
pyrimidine dimer	Two adjacent bases glued together by ultraviolet light.	Ch 6, Lect 4	Examined (he covered it)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
ROS	Reactive oxygen species; an important source of endogenous damage.	Ch 6, Lect 4	Examined (slide)
alkylation	Chemical damage in which an alkyl group is attached to a base.	Ch 6, Lect 3, Lect 4	Examined (slide)
intercalating agent	A drug that slides between the bases and twists the helix out of shape.	Ch 6, Lect 4	Examined (he covered it)
DNA Damage Response (DDR)	The alarm system by which the cell detects damage and decides whether to repair or die.	Ch 6, Lect 4	Examined (he covered it)
ATM	A sensor kinase of the DDR; it can also serve as an ADC payload partner.	Ch 6, Lect 4	Examined (he covered it)
ATR	The other sensor kinase of the DDR.	Ch 6, Lect 4	Examined (slide)
p53	An effector of the DDR; it also has its own 3′→5′ exonuclease activity.	Ch 6, Ch 10, Lect 2, Lect 3, Lect 4	Examined (he covered it)
BER	The pathway that repairs a single damaged base.	Ch 6, Lect 4	Examined (second-level)
DNA glycosylase	The enzyme that cuts out the damaged base and does the first step of BER.	Ch 6, Lect 4	Excluded (he says the chemistry of glycosylase and the AP site is not examined)
abasic site / AP site	The base-less site where the sugar remains but the letter has gone.	Ch 6, Lect 4	Excluded (he says the chemistry of glycosylase and the AP site is not examined)
NER	The pathway that repairs bulky lesions.	Ch 6, Lect 4	Examined (second-level)
GG-NER	Global genome repair; finds the lesion with UV-DDB and XPC.	Ch 6, Lect 4	Examined (slide)
TC-NER	Transcription-coupled repair; uses CSA and CSB to find the stalled RNA Pol II.	Ch 6, Lect 4	Examined (slide)
TFIIH	The point where GG-NER and TC-NER join up; it contains XPB and XPD.	Ch 6, Lect 4, Lect 5	Examined (slide)
XPC	The protein that does the genome-wide search for the lesion in GG-NER.	Ch 6, Lect 4	Examined (slide)
XPA	The protein that positions the cut for the endonuclease in NER.	Ch 6, Lect 4	Examined (slide)
MMR	The pathway that fixes replication mismatches and small insertions and deletions.	Ch 6, Lect 4	Examined (second-level)
MutSα	The recognition heterodimer of MMR (MSH2/MSH6).	Ch 6, Lect 4	Examined (slide)
MutLα	The downstream heterodimer of MMR (MLH1/PMS2).	Ch 6, Lect 4	Examined (slide)
MSH2	The recognition protein present in both MutSα and MutSβ.	Ch 6, Lect 4	Examined (slide)
MLH1	One of the components of MutLα; methylation of its promoter switches MMR off.	Ch 6, Lect 4	Examined (slide)
microsatellite	A short tandem repeat in the genome; the basis of DNA fingerprinting.	Ch 6, Lect 4	Examined (he covered it)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
MSI	Microsatellite lengths start to change once mismatch repair is broken.	Ch 6, Ch 10, Lect 4	Examined (he covered it)
MSI-H	Microsatellite high instability; it comes with a great many neoantigens.	Ch 6, Ch 9, Lect 4	Examined (the key number: 15 % of colorectal cancers)
MSS	The microsatellite stays stable, that is, MMR is intact.	Ch 6, Lect 4	Examined (slide)
MMR deficiency	Loss of mismatch repair function; the synonym dMMR.	Ch 6, Lect 4	Examined (he covered it)
DSB	A double-strand break; the most severe structural injury.	Ch 6, Lect 4	Examined (he covered it)
HR	High-fidelity DSB repair using the sister chromatid as a template; S/G2 only.	Ch 6, Lect 4	Examined (second-level)
NHEJ	Template-free, fast but error-prone DSB repair; it works in every phase.	Ch 6, Ch 9, Lect 4	Examined (first-level)
alternative end joining (a-EJ)	The DSB repair shortcut that needs only a few base pairs of homology.	Ch 6, Lect 4	Examined (slide)
synthetic lethality	Losing one gene alone is survivable; losing both is fatal.	Ch 6, Ch 10, Lect 4	Examined (second-level)
therapeutic window	The margin between the minimum effective dose and the toxic dose.	Ch 6, Ch 10, Lect 4	Examined (second-level)
PARP inhibitor	The synthetic-lethal weapon against BRCA-mutant tumours.	Ch 6, Ch 10, Lect 4	Examined (second-level)
BRCA1	Commands the strand invasion in HR; a mutation means HR deficiency.	Ch 6, Lect 4	Examined (second-level)
BRCA2	Commands HR together with BRCA1; a mutation means HR deficiency.	Ch 6, Lect 4	Examined (second-level)
Ku	The protein that protects the broken ends first in NHEJ.	Ch 6, Lect 4	Examined (slide)
DNA-PKcs	The kinase that works with Ku in NHEJ.	Ch 6, Lect 4	Examined (slide)
RAD51	The protein that carries out the strand invasion in HR.	Ch 6, Lect 4	Examined (slide)
53BP1	The factor that pushes the cell towards the NHEJ side.	Ch 6, Lect 4	Examined (slide)
Shieldin	The factor that pushes the broken ends towards NHEJ together with 53BP1.	Ch 6, Lect 4	Examined (slide)
V(D)J recombination	Assembles the V, D and J segments to produce antibody and TCR diversity.	Ch 6, Ch 9, Lect 2, Lect 4	Examined (first-level)
RAG1	The protein that recognises the recombination-signal sequence and cuts the DNA.	Ch 6, Lect 4	Examined (slide)
RAG2	Forms the recognition-and-cutting complex together with RAG1.	Ch 6, Lect 4	Examined (slide)
recombination-signal sequence	The recognition tag on both sides of every V, D and J segment.	Ch 6, Lect 4	Examined (slide)
SCID	Severe combined immunodeficiency; the clinical result of an NHEJ defect.	Ch 6, Ch 9, Lect 2, Lect 4	Examined (first-level)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
CRISPR/Cas9	Gene-editing scissors re-engineered from a bacterial immune system.	Ch 6, Lect 2, Lect 4	Examined (second-level)
gRNA	The RNA that carries the address label and guides Cas9 to the target DNA.	Ch 6, Lect 2, Lect 4	Examined (slide)
PAM	The short DNA sequence Cas9 must find beside the target before it can cut.	Ch 6, Lect 2, Lect 4	Examined (slide)
RuvC	One of Cas9's nuclease domains; it cuts one side.	Ch 6, Lect 2, Lect 4	Examined (slide)
HNH	Cas9's other nuclease domain; it cuts the other side.	Ch 6, Lect 2, Lect 4	Examined (slide)
knock-out	Switching a gene off with the small indels made by NHEJ; it works in every phase.	Ch 6, Lect 4	Examined (second-level)
knock-in	Writing a sequence in precisely with HDR plus a donor template.	Ch 6, Lect 4	Examined (second-level)
HDR	Homology-directed repair; a branch of HR that works only in S/G2.	Ch 6, Lect 4	Examined (slide)
donor template	The template DNA that supplies the correct sequence during a knock-in.	Ch 6, Lect 4	Examined (slide)
ZFN	An earlier generation of gene-editing tool, now a backup.	Ch 6, Lect 2, Lect 4	Examined (slide)

Chapter 7

Control of gene expression

A factory has one master manual but makes hundreds of different products. The manual itself never changes. What changes is which page you photocopy, how many copies you make, how long each copy lasts, and where the finished goods get shipped. Biology calls the manual DNA, calls the page a gene, calls the copy messenger RNA, and calls "choosing which page to photocopy" gene expression. Dr. Wong calls Lecture 5 the place where the whole course comes together, because almost every drug and every biomarker he has shown works by changing gene expression somewhere along this chain.

The skeleton of the whole chain is the **central dogma**: the message in DNA becomes RNA (**transcription**), and the message in RNA becomes protein (**translation**). And regulation can happen at every layer: the same cell carries the same DNA, yet a liver cell and a nerve cell do completely different things, because they are reading different recipes. Your **genotype** is the recipe you inherited, your **phenotype** is what those recipes actually make, and the sum of all the regulation is what turns genotype into phenotype. That explains why the same mutation — a BRCA mutation, say — does not guarantee that you will get sick: the same typo in the same manual can end up producing completely different diseases.

The first distinction is prokaryote versus eukaryote. A **Prokaryote** has no nucleus, transcription and translation happen in the same compartment, and a ribosome can start reading a message while it is still being copied. A **Eukaryote** separates things: transcription happens in the nucleus, and the message is edited before it is sent out. Those separate compartments are exactly why eukaryotes need splicing.

The second distinction is **cis** versus **trans**. A **Cis** element (promoter, enhancer, silencer, insulator) is a stretch of DNA that regulates the gene sitting on that very same piece of DNA. A **Trans** factor (transcription factor, co-activator) is a diffusible protein, encoded somewhere else, floating freely in the cell until it binds a cis element. In the factory metaphor: a cis element is a panel wired to one particular machine, while a trans factor is a handheld remote control — one remote can open many different machines, even in different rooms. His example is a repressor encoded on chromosome 1 acting on a target gene on chromosome 13; he also points out that trans regulation is more common in eukaryotic systems. And remember: **not every enhancer is upstream** — a regulatory sequence can equally sit at the 3′ end of its target gene. This one he says you must understand.

The third distinction is the best-studied control system in prokaryotes: the **lac operon**. In the bacterium E. coli the three genes that use lactose (lacZ, lacY, lacA) sit side by side on the DNA, preceded by a promoter and an operator, and another regulatory gene makes the blocking protein **Lac repressor**. The rule has two conditions, not one. First, **lactose** must be present: the enzyme made by lacZ converts lactose into **allolactose**, and it is allolactose, not lactose itself, that knocks the Lac repressor off the operator. Second, **glucose** must be low: at that point the second signal makes **cAMP** rise, cAMP binds the protein **CAP**, and the CAP–cAMP complex sits on the promoter and pulls RNA polymerase in, turning transcription from slow into strong. The **trp operon** is the exact mirror: tryptophan is the end product, and when it is plentiful it acts as a **corepressor**, binding the Trp repressor so that the repressor sits on the operator and switches off tryptophan synthesis. The laboratory version uses **IPTG** as an artificial inducer: it looks enough like allolactose to switch off the Lac repressor, but it cannot be metabolised, so you can add it, remove it, and the switch stays wherever you put it; his slide puts the best performance at 100 μM IPTG, and the target gene on that pET plasmid carries an **ORF** (open reading frame). The other way round, mammalian **CHO cells** are used for proteins that need glycosylation, and they require codon optimisation — "So remember that when you use the CHO cell, you need to do the codon optimization."

The fourth distinction is the structure of a eukaryotic gene. **Gene structure** in eukaryotes is short exons separated by long introns. An **Exon** is kept in the mature message, an **intron** is not; both are copied into the first draft, the **pre-mRNA**, and then the introns are cut out by the **spliceosome** and the exons joined. The spliceosome is made of **snRNP**s (small nuclear ribonucleoproteins), and his slide draws three steps: five snRNP bind the intron → the intron loops and pulls the exons together → the intron is released and the snRNP recycled. **Alternative splicing** joins the

same set of exons in different combinations, so one gene makes several isoforms. In cancer cells, aberrant or cancer-specific splicing creates **neoantigen**s, fragments the immune system has never seen; and splicing out the stretch of protein that a drug binds on creates drug resistance. His worked example is a single-nucleotide substitution in a liver cancer gene that creates a new splice site, giving a shorter isoform; patients carrying the variant had markedly worse overall survival ($P < 0.001$), and on the gel there are two bands, 425 bp and 250 bp.

The message is not finished when it leaves the nucleus. **mRNA stability** is itself a layer of regulation: a message destroyed quickly makes very little protein, and one that hangs around for several hours makes far more. Three features decide its lifetime — the 5′ cap, the **poly-A tail**, and the structure of the 3′ UTR. His rule is simple: the longer the poly-A tail, the more translation; the shorter it is, the faster the message decays.

The fifth distinction is how transcription itself starts. In front of a eukaryotic gene's **promoter** there is a short **core promoter**, and the transcription machinery assembles there; the **TATA box** is a six-letter word sitting about −30 upstream of the transcription start site. The order of assembly is: TFIID (which contains **TBP**, the TATA-box-binding protein) arrives first → TFIIA (a general transcription factor), TFIIB (a general transcription factor) → RNA polymerase II together with TFIIE (a general transcription factor), TFIIF (a general transcription factor) and TFIIH. **Enhancer**s and **silencer**s can sit thousands of bases away from the gene they control, and DNA-bending proteins loop them back to the promoter. To find out which stretch is really doing the work you do **promoter mapping**: a series of deletion constructs of a candidate promoter are each put in front of a light-emitting **reporter gene** (luciferase, for example) and their activity measured, and this **reporter assay** points out which TATA variant works. And **cell-type specificity** is specific in time as well — different cell types express different sets of transcription factors, which produce different gene expression programmes; in his words, "This is the concept that I always emphasize."

The sixth distinction is pharmacology. A **Nuclear receptor** is a protein with three working parts: an **NTD** (the variable regulatory region, containing **AF-1**), a **DBD** (conserved, containing a **zinc finger**, which binds the DNA response element) and an **LBD** (which changes conformation on ligand binding and recruits a co-activator or a co-repressor, and contains **AF-2**). Take the oestrogen receptor as the example: **HSP90** holds the receptor in the cytoplasm; once the ligand binds, HSP90 lets go, the receptor dimerises, enters the nucleus and binds the ERE. And **dimerisation** is itself a regulatory switch: **monomers bind DNA very poorly, and dimerisation raises the affinity dramatically.** His example is **HER2**: it cannot homodimerise effectively and instead forms heterodimers with HER1, HER3 or HER4, and **Herceptin** (trastuzumab) blocks exactly that dimerisation. The same drug can play a different role in a different tissue: **Tamoxifen** is a **SERM** (selective estrogen receptor modulator), an antagonist in the breast (anti-tumour) but a partial agonist in the endometrium, which is why endometrial cancer risk goes up slightly. **Receptor occupancy (RO)** is a clinical parameter: how much of the receptor has to be occupied before you see an effect. His thresholds are — a typical antibody **1 %**, Herceptin in breast cancer **2 %**, in gastric cancer **10 %**; the same drug with a different dosing strategy, because the RO needed is tissue-specific.

The seventh distinction is the inflammatory clock. **NF-κB** is normally held in the cytoplasm by **I-κB**. Once bacterial **LPS** is recognised by a cell-surface receptor, it activates **IKK** (IκB kinase), which phosphorylates IκB-alpha; the phosphorylated IκB is then destroyed, NF-κB is released, moves into the nucleus and binds the promoter of an inflammatory gene. What he emphasises is the timing. The order of this **cytokine cascade** is: **TNF-α** comes first, peaking at **4 to 6 hours**, then IL-1β and IFN-γ, then IL-2 and IL-15; and the cytokines that are secreted feed back and push NF-κB harder, and that positive feedback loop is why the response is so fast. **But do not memorise that exact order** — he puts it on the "not examined" list, in his own words: "don't worry about all these cytokines. I'm just want you to show you that different responses can be observed in different times." What you need to be able to explain is the shape of it.

The eighth distinction is the oxygen sensor. **HIF-1α** is made continuously and, when oxygen is normal, is quickly degraded: **PHD** and **FIH** hydroxylate it, **VHL** recognises it and promotes polyubiquitination and sends it to the proteasome. Under hypoxia — the core of a solid tumour, for instance — HIF-1α stabilises, moves into the nucleus, joins HIF-1β, recruits p300/CBP and then binds the **HRE** (hypoxia-responsive element), switching on adaptive genes such as **VEGF**, GLUT1 and **EPO**. His application is to put CAR expression under the control of an HRE, so that the CAR-T only fires in the hypoxic tumour microenvironment — and about half of solid tumours contain hypoxic tissue.

The ninth distinction is the **Hippo** pathway, which decides whether a cell divides or differentiates. **MST1** phosphorylates and activates **LATS1**, and LATS1 phosphorylates **YAP**; phosphorylated YAP binds **14-3-3** in the

cytoplasm, is held there and is then degraded, and the cell stops proliferating. When the cascade fails, YAP enters the nucleus, binds **TEAD** and turns on the growth genes — his fruit-fly hpo clone and his mouse Mst1-/- Mst2-/- liver both grow a great deal of tissue that way.

The tenth distinction is the conversation between cells. The ligand of **Notch** sits on a neighbouring cell, and when it binds it triggers a proteolytic cleavage of the receptor; the intracellular domain that is cut off moves into the nucleus and switches transcription on. **synNotch** is the synthetic-biology version: CAR expression is placed under an AND gate of "tumour antigen plus a second context ligand", so it only fires when both are present.

Last is the newest layer: **lncRNA** (long non-coding RNA) is more than 200 nucleotides long, does not code for protein, and yet acts as a scaffold and a guide for chromatin remodelers and transcription factors. The examples on his slide include APOLO, which regulates genes through an RNA-DNA hybrid; PNCTR, which binds PTBP1 (a splicing factor) and affects splicing; **XIST**, which binds SPEN (an RNA-binding protein) and PRC2 (a chromatin-repressive complex) in epigenetic silencing; and **TINCR**, which influences mRNA stability through base pairing. And **miRNA** is a short RNA of about 22 nucleotides: the primary transcript is first processed by **Drosha** into a precursor — the Microprocessor (the pri-miRNA splicing complex) made with Pasha — and after export from the nucleus **Dicer** cuts it into a duplex, which is loaded into AGO to form **RISC**; RISC pairs with the target mRNA's 3′ **UTR** through the **seed** sequence, causing degradation of the mRNA or inhibition of translation. Because miRNA is so stable in serum it has become a popular non-invasive biomarker. The same layer also has **siRNA** and **ASO** (antisense oligonucleotide); in Lecture 4 he ran a preclinical liver-cancer trial with an anti-TERT ASO, and he also reminds you that **CRISPR/Cas9 is now the standard way to do gene knock-out and knock-in, and the older siRNA and shRNA knockdown techniques have essentially been superseded.**

Two names he never uses in the whole lecture, so that you do not read them elsewhere and fail to match them up. A **Second messenger** is the general name for the small molecules that carry a signal inside the cell; in this lecture he describes the specific example of cAMP and never uses the general term. A **Splicing factor** is the general name for proteins that help with splicing; here he uses the more precise **snRNP** and talks about what the spliceosome is made of rather than unpacking the individual splicing factors. Neither is an exam point for him, but recognising them will help you read material from elsewhere.

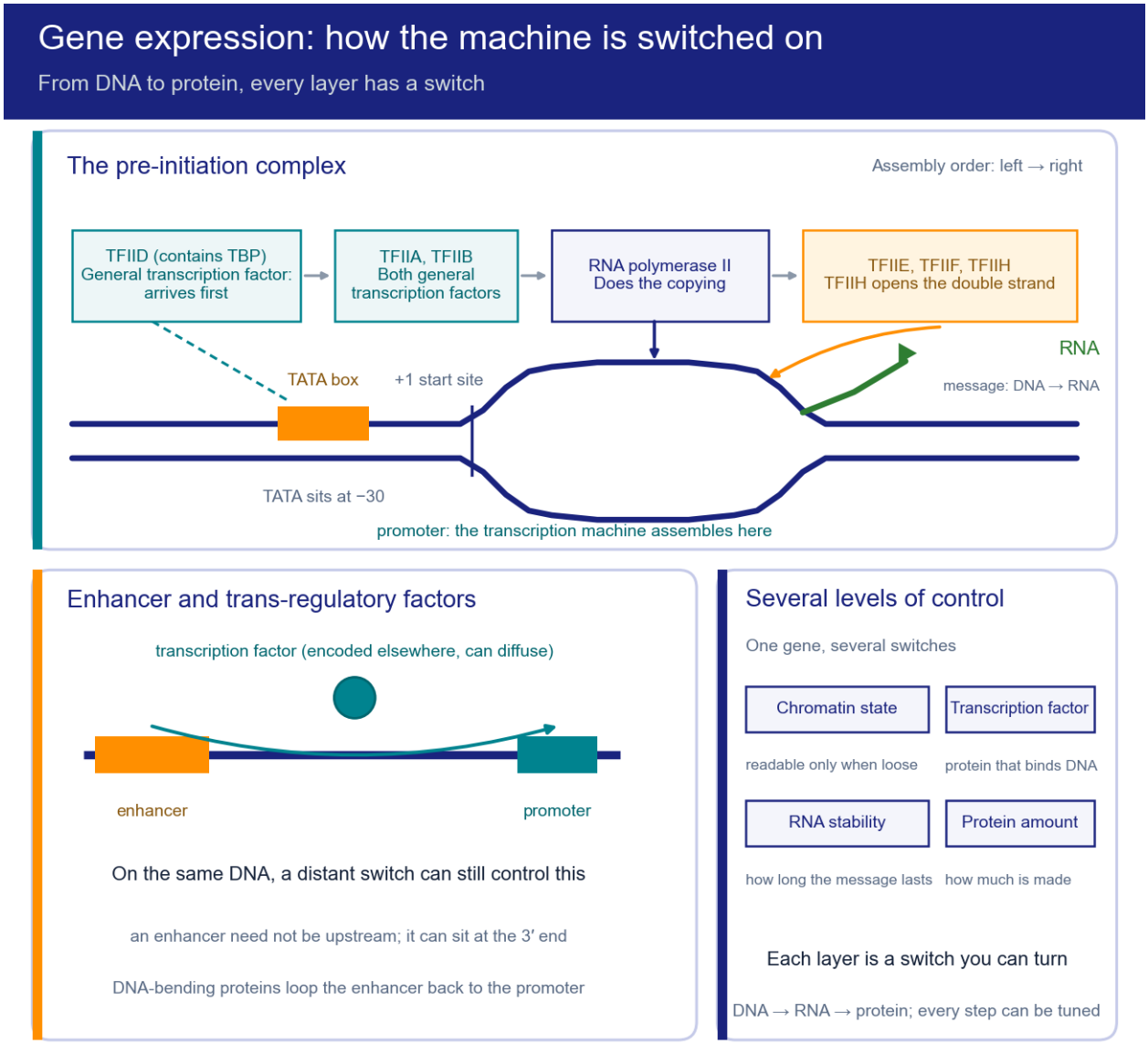


Figure 7: The transcription machinery assembling on the promoter, and the way an enhancer pulls it in from far away.

Key vocabulary for this chapter

Term	In one sentence	Source	Did he name it?
central dogma	The flow of information DNA → RNA → protein.	Ch 7, Lect 1, Lect 5	Examined (he covered it)
transcription	The layer that makes RNA from DNA.	Ch 7, Lect 5	Examined (he covered it)
translation	The layer that makes protein from RNA.	Ch 7, Lect 1, Lect 5	Examined (he covered it)
gene structure	The structure of a gene, made of exons and introns alternating.	Ch 7, Lect 5	Examined (he covered it)
exon	The segment that is kept in the mature message.	Ch 7, Lect 5	Examined (he covered it)
intron	The segment that gets cut out.	Ch 7, Lect 5	Examined (he covered it)
spliceosome	The machine made of snRNP that cuts out the introns.	Ch 7, Lect 5	Examined (slide)
snRNP	Small nuclear ribonucleoprotein; the building block of the spliceosome.	Ch 7, Lect 5	Examined (slide)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
alternative splicing	Joining the same exons in different combinations, so one gene makes several isoforms.	Ch 7, Lect 5	Examined (he covered it)
promoter	The sequence on the DNA that tells the copying machine "the gene starts here".	Ch 7, Lect 5	Examined (second-level)
core promoter	The stretch of promoter where the transcription machinery assembles.	Ch 7, Lect 5	Examined (he covered it)
TATA box	Six letters sitting about −30 upstream of the transcription start site.	Ch 7, Lect 5	Examined (he covered it)
TBP	The TATA-box-binding protein; a component of TFIID.	Ch 7, Lect 5	Examined (slide)
TFIID	The transcription factor complex that arrives first; it contains TBP.	Ch 7, Lect 5	Examined (slide)
enhancer	A cis element that turns a gene up; it can be a long way away.	Ch 7, Lect 5	Examined (second-level)
silencer	A cis element that turns a gene down.	Ch 7, Lect 5	Examined (slide)
insulator	A cis element that stops an enhancer's action from reaching past its target.	Ch 7, Lect 5	Examined (slide)
cis	A regulatory element that acts on the very DNA molecule it sits on.	Ch 7, Lect 5	Examined (second-level)
trans	Proteins such as transcription factors that diffuse and can act across chromosomes.	Ch 7, Lect 5	Examined (second-level)
lac operon	The operon in which three lactose-using genes are regulated together.	Ch 7, Lect 5	Examined (second-level)
Lac repressor	The blocking protein that sits on the operator and shuts the lac operon.	Ch 7, Lect 5	Examined (second-level)
allolactose	The derivative of lactose; it is the molecule that actually knocks off the Lac repressor.	Ch 7, Lect 5	Examined (second-level)
IPTG	An artificial inducer that switches off the Lac repressor but is not metabolised.	Ch 7, Lect 5	Examined (second-level)
cAMP	The second messenger that rises when glucose is low.	Ch 7, Lect 5	Examined (second-level)
CAP	The protein that binds cAMP and then sits on the promoter.	Ch 7, Lect 5	Examined (second-level)
trp operon	The operon switched off by its own end product, tryptophan.	Ch 7, Lect 5	Examined (he covered it)
corepressor	The end product that helps the repressor bind the operator.	Ch 7, Lect 5	Examined (he covered it)
mRNA stability	How long a message lives; a regulatory layer in its own right.	Ch 7, Lect 5	Examined (he covered it)
poly-A tail	The poly(A) tail at the 3′ end; the longer it is, the more translation.	Ch 7, Lect 5	Examined (he covered it)
UTR	The stretch of message that is not translated into protein but carries binding sites.	Ch 7, Lect 5	Examined (he covered it)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
ORF	Open reading frame; the position of the target gene on the expression plasmid.	Ch 7, Lect 5	Examined (slide)
reporter assay	Using a light-emitting reporter gene to measure the strength of a promoter.	Ch 7, Lect 5	Examined (he covered it)
promoter mapping	Using a deletion series to find out which promoter sequence is really the active one.	Ch 7, Lect 5	Examined (slide)
cell-type specificity	Different cell types express different sets of genes.	Ch 7, Lect 5	Examined (second-level)
nuclear receptor	A receptor that is itself a transcription factor, built from NTD, DBD and LBD.	Ch 7, Lect 5	Examined (second-level)
NTD	The N-terminal regulatory region of a nuclear receptor; it contains AF-1.	Ch 7, Lect 5	Examined (slide)
DBD	The DNA-binding domain of a nuclear receptor; it contains the zinc finger.	Ch 7, Lect 5	Examined (second-level)
LBD	The ligand-binding domain of a nuclear receptor; it contains AF-2.	Ch 7, Lect 5	Examined (second-level)
zinc finger	The structural motif in the DNA-binding domain that grabs the DNA.	Ch 7, Lect 5	Examined (slide)
AF-1	The transcription-activation region inside the NTD.	Ch 7, Lect 5	Examined (slide)
AF-2	The transcription-activation region inside the LBD.	Ch 7, Lect 5	Examined (slide)
HSP90	The chaperone that holds the nuclear receptor in the cytoplasm.	Ch 7, Lect 5	Examined (slide)
dimerisation	Monomers bind DNA badly; dimerisation raises the affinity a great deal.	Ch 7, Lect 5	Examined (second-level)
HER2	The receptor that cannot homodimerise effectively and must heterodimerise with HER1/3/4.	Ch 7, Ch 10, Lect 2, Lect 4, Lect 5	Examined (second-level)
Herceptin	The antibody drug that blocks HER2 dimerisation.	Ch 7, Ch 10, Lect 2, Lect 5	Examined (second-level)
tamoxifen	A SERM; an antagonist in the breast, a partial agonist in the endometrium.	Ch 7, Ch 10, Lect 5	Examined (second-level)
SERM	Selective estrogen receptor modulator.	Ch 7, Ch 10, Lect 5	Examined (second-level)
receptor occupancy	The proportion of receptor that must be occupied for a clinical effect.	Ch 7, Ch 10, Lect 2, Lect 5	Examined (the key numbers: 1 % / 2 % / 10 %)
NF-κB	The transcription factor held by I-κB that enters the nucleus once it is stimulated.	Ch 7, Lect 1, Lect 5	Examined (he covered it)
IKK	IκB kinase; it phosphorylates I-κB and brings about its destruction.	Ch 7, Lect 5	Examined (slide)
I-κB	The inhibitory protein that normally holds NF-κB in the cytoplasm.	Ch 7, Lect 1, Lect 5	Examined (he covered it)
TNF-α	The cytokine that peaks first in the inflammatory cascade, at 4–6 hours.	Ch 7, Lect 1, Lect 5	Examined (key number)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
cytokine cascade (exact order)	TNF-α → IL-1β → IFN-γ → IL-2 / IL-15.	Ch 7, Lect 5	Excluded (he says: don't worry about these cytokines, I just want you to show that different responses can be observed at different times)
HIF-1α	The oxygen sensor that stabilises under hypoxia and switches on HRE-controlled genes.	Ch 7, Lect 5	Examined (he covered it)
PHD	The enzyme that hydroxylates HIF-1α when oxygen is normal.	Ch 7, Lect 5	Examined (slide)
FIH	The other enzyme that hydroxylates HIF-1α when oxygen is normal.	Ch 7, Lect 5	Examined (slide)
VHL	The protein that recognises hydroxylated HIF-1α and brings about its degradation.	Ch 7, Lect 5	Examined (slide)
HRE	The hypoxia-responsive element; the DNA sequence the HIF complex binds.	Ch 7, Lect 5	Examined (slide)
VEGF	The angiogenesis factor controlled by HIF.	Ch 7, Lect 5	Examined (he covered it)
EPO	The erythropoietin controlled by HIF.	Ch 7, Lect 5	Examined (slide)
Hippo	The pathway that decides whether a cell divides or differentiates.	Ch 7, Lect 5	Examined (he covered it)
MST1	The upstream kinase of the Hippo pathway; it activates LATS1.	Ch 7, Lect 1, Lect 5	Examined (slide)
LATS1	The kinase that phosphorylates YAP.	Ch 7, Lect 1, Lect 5	Examined (slide)
YAP	The downstream effector of Hippo; when the pathway is off it enters the nucleus and binds TEAD.	Ch 7, Lect 1, Lect 5	Examined (he covered it)
14-3-3	The protein that holds phosphorylated YAP in the cytoplasm.	Ch 7, Lect 1, Lect 5	Examined (slide)
TEAD	The transcription factor YAP binds in the nucleus to switch on the growth genes.	Ch 7, Lect 5	Examined (slide)
Notch	The receptor that signals through a neighbour cell's ligand and a proteolytic cut.	Ch 7, Lect 5	Examined (he covered it)
synNotch	The engineered AND-gate CAR that raises tumour specificity.	Ch 7, Lect 5	Examined (he covered it)
lncRNA	A long RNA of more than 200 nt that does not code for protein.	Ch 7, Lect 5	Examined (he covered it)
XIST	The lncRNA responsible for X chromosome silencing; it binds SPEN and PRC2.	Ch 7, Lect 5	Examined (slide)
TINCR	The lncRNA that binds mRNA by base pairing and controls its stability.	Ch 7, Lect 5	Examined (slide)
miRNA	A short RNA of about 22 nt that silences its target mRNA through the seed sequence.	Ch 7, Lect 5	Examined (he covered it)
seed	Bases 2 to 8 of the miRNA-target mRNA pairing.	Ch 7, Lect 5	Examined (slide)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
RISC	The complex that carries the miRNA and carries out mRNA degradation or blocks translation.	Ch 7, Lect 1, Lect 5	Examined (slide)
Dicer	The enzyme that cuts the precursor miRNA into a duplex.	Ch 7, Lect 1, Lect 5	Examined (slide)
Drosha	The enzyme that processes the primary miRNA into a precursor inside the nucleus.	Ch 7, Lect 1, Lect 5	Examined (slide)
Pasha	The protein that forms the Microprocessor together with Drosha.	Ch 7, Lect 1, Lect 5	Examined (slide)
siRNA	The older gene-knockdown tool, now superseded by CRISPR.	Ch 7, Lect 1, Lect 5	Examined (he covered it)
ASO	Antisense oligonucleotide; he ran a preclinical liver-cancer trial with an anti-TERT ASO.	Ch 7, Lect 5	Examined (he covered it)
Second messenger	A small molecule that carries a signal inside the cell; the course does not use this name.	Ch 7, Lect 5	Not examined
Splicing factor	A protein that helps with splicing; the course does not use this name.	Ch 7, Lect 5	Not examined

Chapter 8

Cell biology

A factory is a **cell**: it takes in raw materials, photocopies its instruction manual, builds machines, and ships the finished machines out the door. Dr. Wong built the whole of Lecture 2 on a single idea — before you can measure, diagnose or treat anything, you have to be able to pick the right cell out of a mixed-up pot of soup. And this lecture takes a step back to look at the cell itself: which room does which job, how the cell divides, and why one inherited disease travels with the mother and not with the father.

The first distinction is the rooms of the cell. A eukaryotic cell is a set of rooms, each doing one job. **Organelle** is the collective name for those rooms: the **nucleus** is the control room and stores the genomic DNA; the **nucleolus** is where ribosomal subunits are made inside the nucleus. The **ER** (endoplasmic reticulum) is the workshop: the **rough ER** carries **ribosome**s that make proteins, which a chaperone then folds and which glycosylation then decorates with sugar groups; the **smooth ER** makes lipids such as phospholipids. The **Golgi** is the packing room: transport **vesicle**s deliver the protein to the **cis face**, it passes through the stacked cisternae and comes out of the **trans face** into a secretory vesicle, and finally the vesicle fuses with the **plasma membrane** and releases the protein outside the cell. This route is called the secretory pathway. The **lysosome** is the recycling and breakdown room. The **cytoskeleton** is both the skeleton and the freight system: **microtubule**s hold the shape up and form the spindle, and **actin** with **myosin** do the contracting.

The **mitochondrion** is the power plant, with its inner membrane folded into **cristae** where the energy machinery sits; it runs the TCA cycle and oxidative phosphorylation to make most of the ATP, and it is also the control centre of apoptosis. It is the one that does not fit the pattern, because it keeps its own little circle of **mtDNA**: a double-stranded small ring of about 17 kilobases carrying 37 genes. Why does it? The **Endosymbiotic theory** says that a long time ago an ancestral cell swallowed a bacterium and did not digest it, and the two later fused into one organism — which explains its bacteria-like circular genome and its two membranes.

That history created an inheritance pattern, and it is why this topic is a first-level exam question. After fertilisation the paternal mitochondria are tagged and cleared, so the embryo's mitochondria come from the egg, and this is **maternal inheritance** (**maternal inheritance**): every living human can trace their mitochondrial DNA back through the maternal line to one and the same woman, **Mitochondrial Eve**. Her family tree has three features: an affected **mother** passes the disease to **all** of her children; an affected **father** passes it to **none** of them; and the disease appears in **every** generation. But carrying a mutation is not the same as being ill. A cell can hold normal and mutant mtDNA at the same time, and that mixture is called **heteroplasmy**; when every copy is normal it is called **homoplasmy**. Disease appears only once the proportion of mutant mtDNA passes a **threshold** (**threshold effect**), and the threshold differs from tissue to tissue: the high-energy skeletal muscle has a **low** threshold and fails first, while hearing has a **high** one. But there is a turn that trips people up: **mitochondrial disease is not necessarily mtDNA.** Many of the proteins that keep mitochondria working are encoded by **nuclear genes**, and a mutation in a nuclear gene produces exactly the same clinical picture with a completely different inheritance pattern — ordinary Mendelian inheritance. So a pedigree that "looks mitochondrial" may actually be nuclear, and the answer has to mention both. The second mechanism appears in the next paragraph: at cytokinesis the organelles are distributed at random, so siblings who inherited the same mutant mtDNA can differ enormously in severity.

The second distinction is prokaryote versus eukaryote. A **Prokaryote** has no membrane-bound nucleus; its DNA sits concentrated in a region with no membrane around it, and that region is called the **nucleoid**; a **eukaryote** shuts its DNA inside a nucleus. This is the contrast used by both Lecture 2 and Lecture 5: bacterial systems are cheap and fast but cannot do complex glycosylation, so when glycosylation is needed you have to use a mammalian CHO cell.

The third distinction is the cell cycle. A **cell cycle** is a timetable: **G1** for growth, **S phase** for copying the DNA, **G2** for checking the copy, **M phase** for separating the chromosomes, and then cytokinesis cuts the cell in half; cells that do not divide sit resting on the **G0** branch. The timetable is driven by two things that lock together like a key in a lock: a

cyclin is the key and a **cyclin-dependent kinase (CDK)** is the machine, and every time a cyclin rises and slots into its matching CDK, that pair pushes the cycle to the next step. The brakes matter just as much: two families of inhibitors, **INK4** and **Cip/Kip**; and the release mechanism runs mitogen signal → cyclin D with CDK4/6 → **pRb** phosphorylation → E2F released → the S-phase genes are switched on, including cyclin E, which starts a positive feedback loop. **Palbociclib** blocks CDK4/6 and the cell cannot get out of G1. In research, to line the cells up at the same starting point you use **serum starvation**: drop the **FBS** (fetal bovine serum) in the medium from 10 % down to **0.5–1 %** and the cells leave the cycle and sit in G1. **But he explicitly excludes the details:** "you don't need to memorize all this cyclin or cyclin-dependent kinase, CDK." So do you need to memorise the cyclin–CDK pairing table? No. What you need is the lock-and-key principle, and the fact that "a drug can freeze the cell in one phase". His slide also points out that **p53** mutations appear in about 40 % of cancers — two out of every five tumours — so the cell's own brakes are broken.

The fourth distinction, and the core of the whole lecture, is that **mitosis** and **cytokinesis** are two separate jobs done by two separate machines. Mitosis is separating the contents: the replicated chromosomes line up on the microtubule **mitotic spindle** and are pulled apart, and at **anaphase** the sister chromatids separate, each new nucleus receiving a complete set. Cytokinesis is cutting the cell in half: a **contractile ring** of actin and myosin pinches the cytoplasm in the middle until the cell splits in two. You can break one and let the other carry on: block the contractile ring with an inhibitor and mitosis still happens perfectly — the DNA is copied, the chromosomes separate, two nuclei form — but the "pinch" never happens. The result is **a cell with two nuclei** (a binucleated cell); if the block lasts longer, or the cell divides again, you get a multinucleated cell. On a flow cytometry DNA-content plot this shows up as a single **4N** peak instead of the usual separate **2N** and 4N peaks. **This is his first-level exam question**, and he says only half the students got it right last year.

The fifth distinction is meiosis. **Meiosis** is the specialised division that makes gametes, and it does two rounds in a row. In **meiosis I** what separates is the **homologues**: the one you got from your mother goes one way and the one from your father goes the other; in **meiosis II** what separates is the **sister chromatids**, exactly as in mitosis. The difference comes from **crossing over**, also called **recombination**: when the homologues pair up in meiosis I they swap pieces, so information from mother and father is spliced together. Add random segregation and random fertilisation and you end with four haploid gametes that are all different from one another. And the M1 and M2 non-disjunction is his other first-level question: when M1 fails, one meiosis produces **2 normal + 2 abnormal** (the abnormal ones all of the same type); when M2 fails it produces **1 normal + 3 abnormal** (two of one type and one of the other).

The sixth distinction is the cell's two ways of dying. **Apoptosis** is programmed cell death: the cell takes itself apart, because it is damaged, because it is infected, or because it is no longer needed. There are two routes. The **intrinsic** route starts with stress inside the cell: the mitochondrion releases **cytochrome c**, which together with APAF-1 and procaspase-9 (the precursor form of caspase-9) forms the **apoptosome**, which then activates caspase-9. The **extrinsic** route starts with an outside signal: the ligand TRAIL binds a death receptor on the cell surface, which switches on caspase-8, which activates caspase-3 and at the same time cuts bid into tbid to push the signal into the mitochondrion. Either way the cell ends up carrying out its own death in an orderly fashion, without spilling its contents everywhere and causing inflammation. The **caspase** family is the heart of this executioner, and the **inhibitor of apoptosis (IAP)** proteins that hold them back are the cell's way of buying itself a little more time. When this goes wrong in the wrong direction the result is cancer: a cell that should have died stays alive and goes on dividing, so dysregulated apoptosis is one of the hallmarks of cancer. On the treatment side, **CDK4/6 inhibitors** do not kill the cell, they only arrest it; a **bispecific antibody** has one binding head aimed at the death receptor and another at the tumour-specific antigen, and only fires when both are present — his example is **BI 905711** against TRAILR2 and CDH17: in normal tissue one condition is missing from each side and nothing happens, while in colorectal cancer both are present and the tumour's apoptosis is lit. **Autophagy** is the opposite gesture — the cell does not die, it swallows and digests the parts of itself it does not need in order to get nutrients; his example is the hypoxic core of a solid tumour, where there is no blood supply and the cells survive by eating their own furniture.

Last of all, stem cells. A **stem cell** can both copy itself and turn into more specialised cells, and there are three kinds: **ESC** (embryonic stem cell) comes from the blastocyst, is pluripotent and can renew itself indefinitely; an **adult stem cell** is multipotent with a limited capacity; and an **iPSC** (induced pluripotent stem cell) is an ordinary somatic cell pushed back into a stem-cell-like state. This reprogramming was done in 2006 by Takahashi and Yamanaka, and the recipe is four **Yamanaka factors**: **Oct3/4**, **Sox2**, **c-Myc** and **Klf4**, fed to an ordinary fibroblast, which within

a few weeks grows into a tight round colony. Claiming that a cell is an iPSC is easy; proving it is hard, so there is a validation ladder: morphology, then markers (PCR and flow cytometry), and then the gold standard, the **teratoma assay**: inject the candidate cells into an immunodeficient mouse and see what they grow into; if you find tissue derived from all three **germ layers** — ectoderm, mesoderm and endoderm — then those cells really are pluripotent. The clinical importance is that iPSC solves the ethical problem with embryos, and it brings disease models and drug screening: see the cardiac toxicity first in iPSC-derived cardiomyocytes.

While we are here, three easily confused names, put side by side for comparison. **Necrosis** is the death in which an injured cell bursts, spills its contents and causes inflammation, the opposite of apoptosis; this course taught only the two programmed routes, apoptosis and autophagy. **RNA interference (RNAi)** is the older idea of gene knockdown — using a small RNA to dim down the output of a particular gene — and Dr. Wong points out that the siRNA and shRNA techniques have essentially been superseded by **CRISPR/Cas9**.

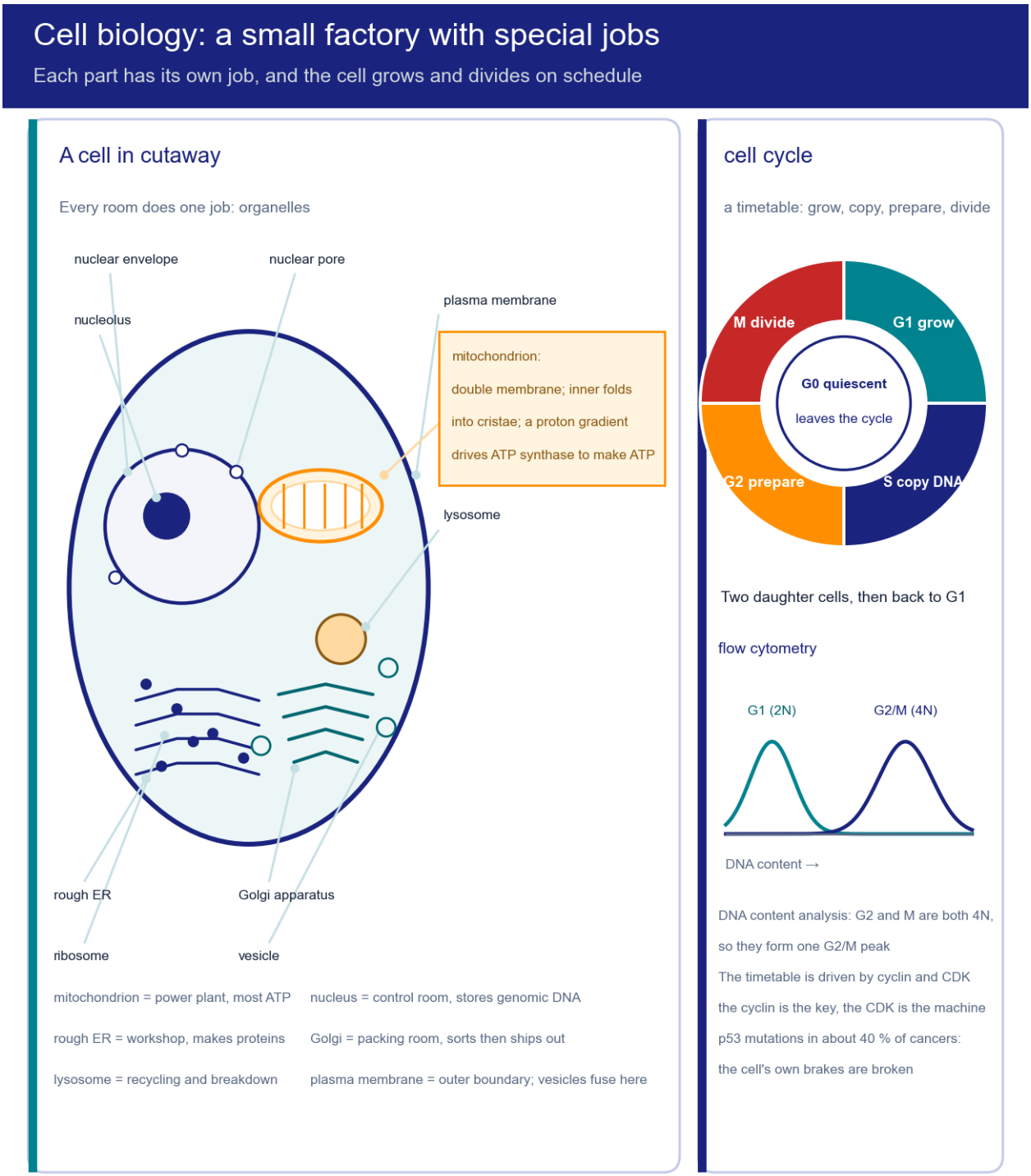


Figure 8: A cutaway of one cell, and the cycle in which it grows, copies itself and divides.

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
cell	A small factory that eats raw materials, copies the manual and builds machines.	Ch 8, Lect 2	Examined (he covered it)
prokaryote	A cell with no membrane-bound nucleus.	Ch 8, Lect 2, Lect 5	Examined (he covered it)
eukaryote	A cell with a nucleus.	Ch 8, Lect 2, Lect 5	Examined (he covered it)
plasma membrane	The outer boundary of the cell; the secretory vesicle finally fuses with it.	Ch 8, Lect 2	Examined (slide)
nucleus	The control room; it stores the genomic DNA.	Ch 8, Lect 2	Examined (he covered it)
nucleolus	The place inside the nucleus where ribosomal subunits are made.	Ch 8, Lect 2	Examined (slide)
organelle	The rooms inside the cell, each with its own job.	Ch 8, Lect 2	Examined (he covered it)
mitochondrion	The power plant; TCA cycle and oxidative phosphorylation.	Ch 8, Lect 2	Examined (he covered it)
mtDNA	A double-stranded small ring of about 17 kb carrying 37 genes.	Ch 8, Lect 2	Examined (he covered it)
cristae	The folds of the inner mitochondrial membrane.	Ch 8, Lect 2	Examined (slide)
Endosymbiotic theory	The mitochondrion descends from a bacterium that was swallowed and not digested.	Ch 8, Lect 2	Examined (he covered it)
ribosome	The machine that makes protein; it sits on the rough ER.	Ch 8, Lect 2	Examined (slide)
ER	The workshop where protein is synthesised and lipid is made.	Ch 8, Lect 2	Examined (he covered it)
rough ER	The ER that carries ribosomes and makes and folds proteins.	Ch 8, Lect 2	Examined (he covered it)
smooth ER	The ER that makes lipids, for example phospholipids.	Ch 8, Lect 2	Examined (slide)
Golgi	The packing room; it takes things in at the cis face and sends them out at the trans face.	Ch 8, Lect 2	Examined (slide)
cis face	The side of the Golgi that receives vesicles from the ER.	Ch 8, Lect 2	Examined (slide)
trans face	The side of the Golgi that sends out the secretory vesicles.	Ch 8, Lect 2	Examined (slide)
vesicle	The small membrane bubble that carries goods between organelles.	Ch 8, Lect 2	Examined (slide)
lysosome	The organelle for recycling and breakdown.	Ch 8, Lect 2	Examined (slide)
cytoskeleton	Both skeleton and freight system; actin and myosin do the contracting.	Ch 8, Lect 2	Examined (he covered it)
microtubule	The fibre that holds the shape up and forms the spindle.	Ch 8, Lect 2	Examined (slide)
actin	One of the two proteins that make up the contractile ring.	Ch 8, Lect 2	Examined (slide)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
myosin	One of the two proteins that make up the contractile ring.	Ch 8, Lect 2	Examined (slide)
contractile ring	The ring of actin and myosin that cuts the cell in two.	Ch 8, Lect 2	Examined (first-level)
mitotic spindle	The microtubule machine that pulls the replicated chromosomes apart.	Ch 8, Lect 2	Examined (slide)
cell cycle	The timetable G1 → S → G2 → M → cytokinesis.	Ch 8, Lect 2	Examined (he covered it)
G1	The growth phase; serum-starved cells stop here.	Ch 8, Lect 2	Examined (he covered it)
S phase	The phase in which the DNA is copied.	Ch 8, Lect 2	Examined (he covered it)
G2	The phase that checks the copy before division.	Ch 8, Lect 2	Examined (slide)
M phase	The phase in which the chromosomes are separated.	Ch 8, Lect 2	Examined (slide)
cyclin	The key of the cell cycle; it rises cyclically.	Ch 8, Lect 2	Examined (he covered it)
CDK	The machine that pairs with a cyclin and drives the cycle.	Ch 8, Lect 2	Examined (he covered it)
cyclin–CDK pairing	The table of which cyclin pairs with which CDK.	Ch 8, Lect 2	Excluded (he says: you do not need to memorise all the cyclin and CDK pairings)
pRb	Once phosphorylated it releases E2F and lets the cell enter S phase.	Ch 8, Lect 2	Examined (slide)
serum starvation	Dropping FBS to 0.5–1 % to synchronise the cells in G1.	Ch 8, Ch 11, Lect 2, Lect 4	Examined (key number)
FBS	Fetal bovine serum; the source of growth factors that feed cells in the medium.	Ch 8, Lect 2	Examined (key number)
G0	The branch where cells that do not divide sit.	Ch 8, Lect 2	Examined (slide)
mitosis	Separating the replicated chromosomes into two sets.	Ch 8, Lect 2	Examined (first-level)
cytokinesis	Pinching the cytoplasm into two separate cells.	Ch 8, Lect 2	Examined (first-level)
anaphase	The stage of mitosis in which the sister chromatids separate.	Ch 8, Lect 2, Lect 3	Examined (slide)
2N	The DNA content of a G1 cell (one genome).	Ch 8, Lect 2	Examined (first-level)
4N	The DNA content of a G2/M cell; also the peak of a binucleated cell.	Ch 8, Lect 2	Examined (first-level)
meiosis	The two consecutive divisions that make gametes.	Ch 8, Lect 2, Lect 3	Examined (he covered it)
homologues	A pair of chromosomes, one from your mother and one from your father.	Ch 8, Lect 2	Examined (first-level)
sister chromatids	The two identical copies produced when one chromosome is duplicated.	Ch 8, Lect 2	Examined (first-level)
crossing over	The swap of pieces between homologues in meiosis I.	Ch 8, Lect 2, Lect 3	Examined (he covered it)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
recombination	The rearrangement between homologous chromosomes; also another name for HR.	Ch 8, Lect 2	Examined (he covered it)
apoptosis	Programmed cell death; its dysregulation is a hallmark of cancer.	Ch 8, Lect 2	Examined (he covered it)
apoptosome	The complex made of cytochrome c, APAF-1 and procaspase-9.	Ch 8, Lect 2	Examined (slide)
caspase	The protease cascade that executes apoptosis.	Ch 8, Lect 2	Examined (slide)
cytochrome c	The protein released by the mitochondrion that starts the intrinsic apoptosis route.	Ch 8, Lect 2	Examined (slide)
autophagy	The cell digesting its own non-essential organelles to get nutrients.	Ch 8, Lect 2	Examined (he covered it)
stem cell	A cell that can copy itself and differentiate into specialised cells.	Ch 8, Lect 2	Examined (he covered it)
iPSC	A somatic cell pushed back into a pluripotent state.	Ch 8, Lect 2	Examined (he covered it)
Yamanaka factors	The four reprogramming factors Oct3/4, Sox2, c-Myc and Klf4.	Ch 8, Lect 2	Examined (he covered it)
Oct3/4	One of the Yamanaka factors.	Ch 8, Lect 2	Examined (he covered it)
Sox2	One of the Yamanaka factors.	Ch 8, Lect 2	Examined (he covered it)
c-Myc	One of the Yamanaka factors.	Ch 8, Lect 2	Examined (he covered it)
Klf4	One of the Yamanaka factors.	Ch 8, Lect 2	Examined (he covered it)
teratoma assay	The gold-standard validation: inject candidate cells into an immunodeficient mouse.	Ch 8, Lect 2	Examined (he covered it)
germ layers	The three layers ectoderm, mesoderm and endoderm.	Ch 8, Lect 2	Examined (he covered it)
Mitochondrial Eve	The maternal common ancestor of the mtDNA of all living people.	Ch 8, Lect 2	Examined (he covered it)
maternal inheritance	A mother passes it to all of her children; a father passes it to none.	Ch 8, Lect 2	Examined (first-level)
heteroplasmy	The state in which a cell holds a mixture of normal and mutant mtDNA.	Ch 8, Lect 2	Examined (first-level)
homoplasmy	The state in which the cell's mtDNA is all normal or all mutant.	Ch 8, Lect 2	Examined (slide)
threshold effect	Disease appears only once the proportion of mutant mtDNA crosses a tissue-specific threshold.	Ch 8, Lect 2	Examined (first-level)
Nucleoid	The region in a prokaryotic cell where DNA sits, with no membrane around it.	Ch 8, Lect 5	Not examined
RNA interference (RNAi)	The older small-RNA knockdown idea, now superseded by CRISPR.	Ch 8, Lect 2	Not examined
Inhibitor of apoptosis (IAP)	The protein that inhibits caspases and lets the cell live a little longer.	Ch 8, Lect 2	Not examined
Necrosis	The death in which an injured cell bursts and spills its contents, causing inflammation.	Ch 8, Lect 2	Not examined

Chapter 9

Immunology and cancer immunity

The immune system is like a security team: it patrols constantly, and anything it does not recognise, it takes apart. What makes a tumour frightening is not that it runs fast or is strong, but that it is still "one of us" — it has merely got a few letters of itself wrong, and that is enough for the security team not to recognise it. This chapter deals with two threads: how the immune system kills cells, and how the tumour switches the immune system off. Dr. Wong taught the two threads interleaved across Lectures 2 and 4, and added in Lecture 5 the dividing line that separates "hot tumour" from "cold tumour".

Antibodies: the Y shape, and two different ends

What the immune system recognises is an **antigen**, and an antibody recognises only a small piece of that antigen; that small piece is the **epitope**. An **Antibody** is a Y-shaped protein: the two arms at the top are the **Fab** (which grabs the antigen) and the stem below is the **Fc** (which calls the immune system over). What actually decides the clinical behaviour is which class the Fc end belongs to, and that class is called the **isotype**.

His discriminating point is very crisp: **IgG1** binds the Fc receptor on an NK cell and therefore triggers **ADCC** (antibody-dependent cellular cytotoxicity); **IgG4** binds the Fc receptor poorly and therefore has **no** ADCC, but in exchange it has a longer serum half-life. This contrast is on his list of the things "most likely to be used as a multiple-choice distractor", so learn it well enough to say the difference in one second. On the same list is **CDC** (complement-dependent cytotoxicity): the complement is a set of plasma proteins that can kill a cell with no cell involved. The way to remember it is simple — **ADCC needs a cell, CDC relies on complement; IgG1 gives you ADCC, IgG4 does not.**

The sheer number of different antibodies also depends on a thread running through Lecture 4. An antibody's recognition region is not made by one gene; it is assembled from different segments by **V(D)J recombination**: the B cell's heavy chain has **V of about 39, D 27 and J 6** gene segments, joined first D-J and then V; the light chain uses the V-J joining. This assembly borrows **NHEJ** (non-homologous end joining) to seal the joints, so once NHEJ breaks, the antibody diversity cannot be built, and the clinical result is **SCID** (severe combined immunodeficiency). His own words are: **Lecture 4:** "if the people showing a defective in this non-homologous end joining, what's the outcome? Definitely the outcome is the severe immune deficiency… Because he cannot generate a sufficient amount of the antibody pool to recognize the thousand copies of the foreign antigen."

First you have to be able to see the cells

To talk about immunity, the first step is to tell "among this pile of cells, which ones are the ones I want". **Flow cytometry** sweeps a laser across the cells one at a time: **FSC** (forward scatter) reflects cell **size**, **SSC** (side scatter) reflects **granularity / internal complexity**; **gating** is drawing a gate on the scatter plot around the population you want, for example picking the CD3+CD4+ helper T cells out of PBMCs; and **FACS** goes a step further and physically **sorts the living cells** into tubes or wells so that you can culture them or analyse them single-cell.

Tissue specimens are far more troublesome, because a tumour sample has cancer cells, fibroblasts and immune cells all mixed together. **LCM** (laser capture microdissection) uses a laser to cut out a homogeneous piece of tissue under the microscope, so the signal is not diluted by stromal and immune cells. To see where a protein is, you stain: **IHC** uses an enzyme substrate that makes colour, and **IF** (immunofluorescence) uses a fluorescent dye; **multiplex IF** stains several proteins at once, and **DAPI** stains the nucleus. **Confocal microscopy** uses a **pinhole** to block out-of-focus light, which is what lets it make optical sections and three-dimensional reconstructions — something widefield cannot do. These are his analysis tools; the concepts must be understood, but in Lecture 5 he pulls the emphasis back to gene expression.

Where the brakes are: the immune checkpoints

The immune system has brakes. An **Immune checkpoint** is that molecular brake. The most important one is **PD-1**: it sits on the T cell, and **PD-L1** sits on the tumour cell; the moment the two bind, the T cell is switched off and enters **anergy**, and the tumour has completed its **immune evasion**. His Lecture 2 slide also draws the other brakes — **CTLA-4**, **LAG3**, **TIM3** and **VISTA** on the effector cell, and **PD-L2** on the target cell — but he does not want you to memorise that string.

What you do have to memorise is this figure: in colorectal cancer patients with **MSI-H / dMMR** (mismatch repair deficiency) who receive anti-PD-1, **progression-free survival** is clearly better than in the **pMMR (MMR-proficient)** group, **P<0.001 by log-rank test**. His comment on the figure is: **Lecture 4:** "I would say this figure is one of the most important figures in modern medicine, because when you attend any seminars about the immune checkpoint inhibitors or immunotherapies or colorectal cancer, I can for sure that the presenter will include this figure in his presentation deck."

The mechanism chain behind that figure has to be explainable in one breath: **MMR deficiency → frameshift mutations → a great many neoantigens (new peptides the immune system has never seen) → the tumour becomes a hot tumour → anti-PD-1 works.** He puts it like this: **Lecture 5:** "So guys, you definitely remember that for the tumor that present many neoantigen and can be treated with the immunotherapy, we call this is a hot tumor, yeah. And not all the solid tumor is a hot tumor." The other way round, a tumour with few neoantigens is a **cold tumour**, which responds poorly to checkpoint inhibitors — but it can be "converted into hot": an HDAC inhibitor induces PD-L1 on the tumour cell, producing **gene addiction**, and then checkpoint blockade works again.

Neoantigens do not only come from MMR deficiency. He gives two other sources: a **proofreading**-defective Pol ε turns the cell into an **ultramutator**, so mutations explode → more neoantigens → more **TIL** → and the prognosis is actually better; and aberrant alternative splicing that creates new splice sites likewise produces peptides the immune system has never seen. There is also the Lecture 3 point about the HDAC inhibitor making a silenced tumour suppressor re-express itself, which belongs to the same strategy of "getting the immune system's attention back".

A tumour is not a lump of cancer cells

The term **Tumour microenvironment (TME)** is there to remind you: a tumour = cancer cells + immune cells + **CAF** (cancer-associated fibroblast) + ECM. The CAFs lay down a stroma-rich ECM; the **TAM** (tumour-associated macrophage), **Treg** (regulatory T cell) and **MDSC** (myeloid-derived suppressor cell) are all immune cells on the tumour's side, and they suppress T cells through TGFβ and IL-10; and with hypoxia, low pH, adenosine and IDO on top, the inside of a tumour is a genuinely immunosuppressive environment. The number of **TIL**s (tumour-infiltrating lymphocytes) that stay inside the tumour often rises together with the neoantigen burden and with a better prognosis.

Before immune cells can enter the tumour they also have to climb out of the blood. This **extravasation** process is drawn on his slide in four steps: **chemotaxis** is guided by **MCP-1**; **transmigration** across the endothelium is handled by **PECAM-1**; adhesion is via **ICAM-1** and **VCAM-1**; and the first step, rolling, depends on **P-selectin** and E-selectin.

CAR-T: attaching the antibody's grip to the T cell's pedal

A **CAR** (chimeric antigen receptor) is a receptor assembled out of borrowed parts: **scFv** (single-chain variable fragment, which joins VH and VL with a linker and is the antibody's two arms) → **hinge** → **transmembrane domain** (which anchors the receptor in the cell membrane) → the co-stimulation region (**CD28** or **4-1BB**) → **CD3ζ** (which carries the ITAM, the immunoreceptor tyrosine-based activation motif, and passes the signal on to **ZAP70**). When the signal arrives, the T cell transcribes IL-2, starts to proliferate, and releases perforin and granzyme to punch holes in the target and kill it.

He has also drawn the manufacturing process step by step: blood is taken from the patient and the T cells are collected (leukapheresis) → they are activated with anti-CD3/CD28 magnetic beads plus **IL-2** and **IL-15** → the CAR is inserted → they are expanded to millions → **lymphodepletion** comes first (chemotherapy that clears part of the patient's own lymphocytes so the CAR-T has room to expand) → and then the infusion. Three sources have to be kept apart: **autologous** (the patient's own cells), **allogeneic** (from a healthy donor's cord blood or from iPSCs, which can be edited with ZFN or CRISPR at the TRAC locus to reduce rejection), and **in vivo CAR-T** (mRNA is

packed into a **lipid nanoparticle**, delivered into the patient's body by an anti-CD5-decorated LNP, which turns the patient's own T cells into cells carrying a **FAP** CAR in vivo).

CAR-T has one price that he gave explicitly: **USD 370,000 to 475,000** (for one course of cell therapy). Keep that number apart from the **US$200,000 per patient** he quotes in Lecture 4 when he talks about the cost of clinical trials — the latter is the cost of running one trial for one person, and the two must not be mixed up.

Seeing the tumour without going to surgery: liquid biopsy

A **Liquid biopsy** looks for the tumour in the blood instead of cutting out a piece of tissue. A **CTC** (circulating tumour cell) drops off a solid tumour into the blood and can be isolated by screening on **EpCAM+ / CD45-** (EpCAM catches cancer cells of epithelial origin, CD45 excludes the white blood cells), which allows metastatic disease to be monitored non-invasively. But keep one eye open: **EMT** (epithelial-mesenchymal transition) changes the surface markers of a CTC, so marker expression **on a CTC is not the same thing as** marker expression in the tumour in situ.

How this chapter will be examined

The three most likely directions: reading that dMMR-versus-pMMR progression-free survival curve (the curve steps down from 1.0; **steep and early = early involvement and a poor prognosis; a late drop = late involvement; the two curves separate = the gene can tell the two groups of patients apart; they cross or overlap = it cannot**); answering contrast questions such as IgG1 versus IgG4 and ADCC versus CDC; and answering source questions on autologous versus allogeneic versus in vivo CAR-T. As for the string CTLA-4, LAG3, TIM3 and VISTA, and the name of every CAR domain, knowing them is background and memorising them is not necessary.

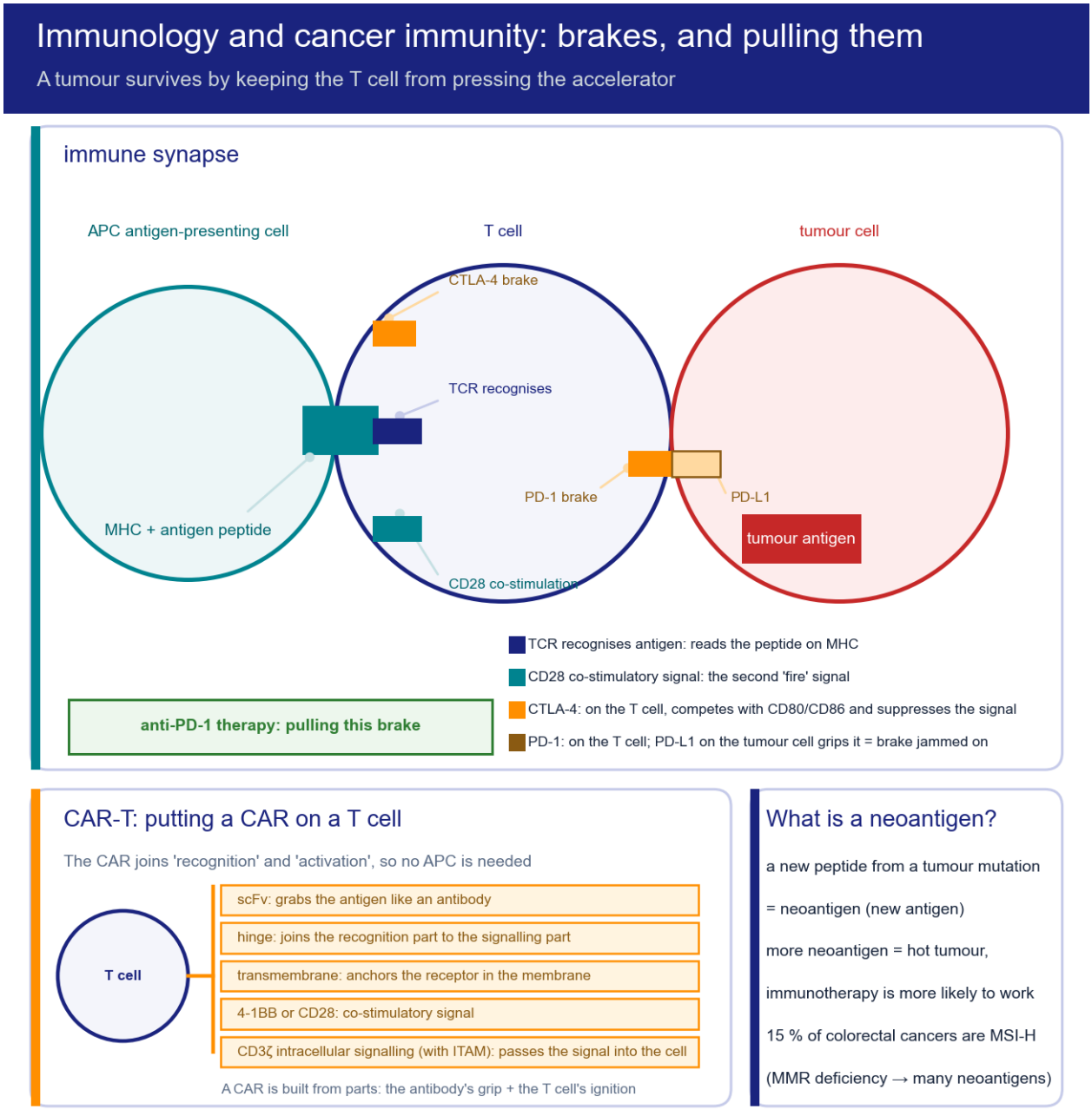


Figure 9: The brakes at the immune synapse (CTLA-4, PD-1), and how CAR-T puts a CAR on a T cell.

Key vocabulary for this chapter

Term	In one sentence	Source	Did he name it?
antigen	The target molecule the immune system recognises.	Ch 2, Ch 9, Lect 1, Lect 2	Examined (the thing an antibody recognises; he uses it again and again)
antibody	The Y-shaped recognition protein: Fab grabs the antigen, Fc summons the immune system.	Ch 2, Ch 9, Lect 1, Lect 2	Examined (the antibody structure is the basic framework of the whole immunity thread)
epitope	The small region of the antigen that the antibody actually touches.	Ch 2, Ch 9, Lect 2	Not examined (he never names it on its own)
isotype	The class of an antibody, which decides how the Fc end behaves.	Ch 2, Ch 9, Lect 2	Examined (one half of the "IgG1 vs IgG4" contrast question)
IgG1	The antibody class that binds the NK cell's Fc receptor and triggers ADCC.	Ch 2, Ch 9, Lect 1, Lect 2	Examined (one half of the "IgG1 vs IgG4" contrast question)
IgG4	The antibody class with no ADCC but a longer serum half-life.	Ch 2, Ch 9, Lect 1, Lect 2	Examined (one half of the "IgG1 vs IgG4" contrast question)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
ADCC	The antibody uses its Fc end to call a killer cell over to kill.	Ch 2, Ch 9, Lect 1, Lect 2	Examined (one half of the "ADCC vs CDC" contrast question)
CDC	Complement takes part directly in the killing; no cell and no antibody needed.	Ch 9, Lect 2	Examined (one half of the "ADCC vs CDC" contrast question)
Fab	The two arms at the top of the antibody that grab the antigen.	Ch 9, Lect 1	Not examined (background structure)
Fc	The stem at the bottom of the antibody that calls the immune system over.	Ch 9, Lect 1	Not examined (background structure)
flow cytometry	A counting and analysis instrument that sweeps a laser across cells one by one.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13, the basics of flow cytometry)
FSC	Forward scatter, which reflects cell size.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: FSC = size)
SSC	Side scatter, which reflects granularity and internal complexity.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: SSC = granularity)
gating	Drawing a gate on the scatter plot around the population you want.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: gating isolates the target population)
FACS	The sorter that actually puts the living cells into tubes or wells.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: FACS really does separate the live cells)
LCM	Using a laser to cut out a homogeneous piece of tissue under the microscope.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: LCM avoids contamination by stromal and immune cells)
IHC	A stain that detects protein using an enzyme substrate.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: IHC vs IF)
IF	A stain that detects protein using a fluorescent dye.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: IHC vs IF)
multiplex IF	Using several fluorochromes at once to look at several proteins.	Ch 9, Ch 11, Lect 2	Not examined (he shows it only as a tool)
DAPI	The blue fluorescent dye that stains the nucleus.	Ch 9, Ch 11, Lect 2	Not examined (background)
confocal microscopy	A microscope that uses a pinhole to block out-of-focus light, so it can make optical sections and 3D reconstructions.	Ch 9, Ch 11, Lect 2	Not examined (background)
pinhole	The pinhole in a confocal microscope that blocks out-of-focus light.	Ch 9, Ch 11, Lect 2	Not examined (background)
immune checkpoint	The molecular brake that makes an immune cell stop.	Ch 9, Lect 2	Examined (the core concept behind hot tumour and the dMMR-versus-pMMR figure)
PD-1	The brake receptor that sits on the T cell.	Ch 9, Lect 2, Lect 4	Examined (anti-PD-1 is a treatment he names)
PD-L1	The ligand on the tumour cell that shakes hands with PD-1.	Ch 9, Lect 2, Lect 3, Lect 4	Examined (the same mechanism chain as PD-1)
PD-L2	The other ligand of PD-1.	Ch 9, Lect 2	Not examined (it appears only on the CAR-T limitation slide)
CTLA-4	Another immune checkpoint.	Ch 9, Lect 2, Lect 3	Not examined (he showed it but never named it)
LAG3	An inhibitory receptor on the effector cell.	Ch 9, Lect 2	Not examined (same)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
TIM3	An inhibitory receptor on the effector cell.	Ch 9, Lect 2	Not examined (same)
VISTA	An inhibitory receptor on the effector cell.	Ch 9, Lect 2	Not examined (same)
anergy	The powerless state a T cell is switched into by the brake.	Ch 9, Lect 2	Not examined (mechanism background)
immune evasion	The state in which the tumour has successfully dodged the immune attack.	Ch 9, Lect 2	Not examined (mechanism background)
hot tumour	A hot tumour with many neoantigens that responds to immunotherapy.	Ch 9, Lect 2, Lect 4, Lect 5	Examined (he names it directly: "we call this is a hot tumor")
cold tumour	A cold tumour with few neoantigens that responds poorly to checkpoint inhibitors.	Ch 9, Lect 5	Examined (second-level §4.4, the hot-versus-cold contrast)
MSI-H	The high microsatellite instability caused by mismatch repair deficiency.	Ch 6, Ch 9, Lect 4	Examined (the key number: 15 % of colorectal cancers are MSI-H)
dMMR	Mismatch repair deficiency.	Ch 9, Lect 2, Lect 4	Examined (second-level §4.4 and the dMMR-versus-pMMR figure)
pMMR	Normal mismatch repair.	Ch 9, Lect 2, Lect 4	Examined (the group the dMMR one is compared against)
frameshift	A shift of the reading frame caused by an insertion or deletion that is not a multiple of three.	Ch 3, Ch 9, Ch 10, Lect 1, Lect 4, Lect 5	Examined (he says a frameshift creates neoantigens)
neoantigen	A new antigen the immune system has never seen.	Ch 9, Ch 10, Lect 1, Lect 4, Lect 5	Examined (the start of the hot tumour mechanism chain)
gene addiction	The "addicted" state created when an HDAC inhibitor induces PD-L1.	Ch 9, Lect 3, Lect 5	Not examined (it explains how a cold tumour can be converted)
tumour microenvironment (TME)	The cells, stroma and signalling environment around the tumour.	Ch 9, Lect 1, Lect 2	Not examined (conceptual background)
cancer-associated fibroblast (CAF)	The cancer-associated fibroblast in the tumour.	Ch 9, Lect 2	Not examined (background)
TAM	Tumour-associated macrophage.	Ch 9, Lect 2	Not examined (background)
Treg	Regulatory T cell.	Ch 9, Lect 2	Not examined (background)
MDSC	Myeloid-derived suppressor cell.	Ch 9, Lect 2	Not examined (background)
TIL	Tumour-infiltrating lymphocyte.	Ch 9, Lect 4	Not examined (background)
extravasation	The four-step process by which a white blood cell climbs out of the blood vessel into tissue.	Ch 9, Lect 2	Not examined (background)
chemotaxis	Moving along a chemical concentration gradient.	Ch 9, Lect 2	Not examined (background)
transmigration	Crossing the vascular endothelium.	Ch 9, Lect 2	Not examined (background)
P-selectin	The selectin that makes white blood cells roll along the vessel wall.	Ch 9, Lect 2	Not examined (background)
ICAM-1	The cell adhesion molecule involved in white blood cell adhesion.	Ch 9, Lect 2	Not examined (background)
VCAM-1	The cell adhesion molecule involved in white blood cell adhesion.	Ch 9, Lect 2	Not examined (background)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
PECAM-1	The molecule that lets white blood cells cross the endothelium.	Ch 9, Lect 2	Not examined (background)
MCP-1	The cytokine that guides white blood cell chemotaxis.	Ch 9, Lect 2	Not examined (background)
CAR	A chimeric antigen receptor that attaches the antibody's grip to the T cell.	Ch 9, Lect 2	Examined (the autologous / allogeneic / in vivo contrast question)
scFv	The single-chain variable fragment made by linking the antibody's VH and VL.	Ch 9, Lect 2	Not examined (structural background)
hinge	The hinge on the CAR.	Ch 9, Lect 2	Not examined (structural background)
transmembrane domain	The transmembrane segment that anchors the CAR in the T cell membrane.	Ch 9, Lect 2, Lect 5	Not examined (structural background)
CD28	One of the standard co-stimulation regions of a CAR.	Ch 9, Lect 2	Not examined (structural background)
4-1BB	One of the standard co-stimulation regions of a CAR.	Ch 9, Lect 2	Not examined (structural background)
CD3ζ	The intracellular signalling region of the CAR; it carries the ITAM.	Ch 9, Lect 2	Not examined (structural background)
ZAP70	The kinase that receives the ITAM signal from CD3ζ.	Ch 9, Lect 2	Not examined (structural background)
IL-2	The cytokine that activates and expands T cells.	Ch 9, Lect 2, Lect 5	Examined (the cytokine time series in third-level §5 contains IL-2 and IL-15)
IL-15	The cytokine that activates and expands T cells.	Ch 9, Lect 2, Lect 5	Examined (same)
lymphodepletion	The chemotherapy that clears the patient's own lymphocytes before the infusion.	Ch 9, Lect 2	Not examined (process background)
autologous	CAR-T made from the patient's own cells.	Ch 9, Lect 2	Examined (the confusable pair "autologous vs allogeneic")
allogeneic	Off-the-shelf CAR-T made from donor or iPSC-derived cells.	Ch 9, Lect 2	Examined (same)
in vivo CAR-T	Using an LNP to deliver mRNA and do the engineering inside the patient.	Ch 9, Lect 2	Examined (the confusable pair "in vivo CAR-T")
lipid nanoparticle	The lipid carrier that takes mRNA into cells.	Ch 9, Lect 2, Lect 5	Not examined (background)
FAP	Fibroblast activation protein; the target used by in vivo CAR-T.	Ch 9, Lect 2	Not examined (background)
liquid biopsy	Using a blood sample instead of a biopsy to find the tumour.	Ch 9, Lect 2	Not examined (he does not name it)
CTC	Circulating tumour cell.	Ch 9, Lect 2	Not examined (he does not name it)
EpCAM	The surface marker used to pull out epithelial-origin CTCs.	Ch 9, Lect 2	Not examined (background)
CD45	The white blood cell marker used to exclude the white blood cells.	Ch 9, Lect 2	Not examined (background)
EMT	Epithelial-mesenchymal transition: the cell leaves its neighbours and becomes able to move.	Ch 9, Lect 2	Not examined (background)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
V(D)J recombination	The rearrangement that assembles the V, D and J segments into an antibody or TCR recognition region.	Ch 6, Ch 9, Lect 2, Lect 4	Examined (first-level §3.3: the causal chain from an NHEJ defect to SCID)
NHEJ	The fast, error-prone DSB repair route that needs no template.	Ch 6, Ch 9, Lect 4	Examined (first-level §3.3 and the §4.6 key contrast)
SCID	The severe immune deficiency in which antibody and TCR diversity cannot be built.	Ch 6, Ch 9, Lect 2, Lect 4	Examined (first-level §3.3, the clinical result he names in so many words)
ultramutator	The state of a cell whose mutation rate explodes because proofreading fails.	Ch 9, Lect 4	Not examined (he uses it to explain rising neoantigen and TIL)

Chapter 10

Cancer biology and targeted therapy

Cancer starts when one cell gets a few letters of its "recipe" — its DNA — wrong. But not every wrong letter is useful. Some mistakes make that cell grow faster than the others; those are **driver**s (a typo in an important word). Some are just along for the ride, carried along by the chromosomal instability; those are **passenger**s. He explains it most clearly in Lecture 3: a **driver** is a change that brings a growth advantage, a **passenger** is only a bystander; to separate the two you have to combine the **CNV** (copy number variation) profile with the mRNA expression — the gene whose copy number goes up and whose transcription goes up with it is the one that can be a real driver, and the classroom example is **NPM1** in leukaemia. He also reminds you that he only gives you "a taste" of the molecular mechanism behind CNV: **Lecture 3:** "I'm just give you a taste… don't be scared with all these mechanism." So you need the concept of **amplification** (more copies) and **deletion** (missing copies); you do not need the mechanism.

The way in from Lecture 1: the Warburg effect

Lecture 1 opens with the metabolism of a cancer cell. A quiet normal cell makes about **36 mol ATP** by oxidative phosphorylation when oxygen is available; when oxygen runs out it switches to anaerobic glycolysis and gets only **2 mol ATP**. But a proliferating cancer cell — in his words, proliferating tissue or tumour — turns **85 % of its pyruvate into lactate** even when oxygen is plentiful, and sends only about **5 %** into the mitochondrion (because PDC is inhibited by the **PDK**s), ending up with about **4 mol ATP**. This "glycolysis in spite of oxygen" behaviour is the **Warburg effect**, and the pathway is called **aerobic glycolysis**.

Why do it that way? Because the few ATP that are wasted buy something other than energy: the cancer cell keeps the carbon skeleton and uses it to make nucleotides, lipids and amino acids so that it can grow. This is the way-in concept of Lecture 1, and the mitochondrion section of Lecture 2 and the chromatin of Lecture 3 both come back to it.

A workflow that finds cancer from blood

The whole set of small molecules is the **metabolome**; measuring the whole set and finding the discriminative part of it is **metabolomics**. On his Lecture 1 slide he draws a seven-step workflow: experimental design (a cohort plus an animal model) → sample collection (urine, blood and CSF, tissue) → sample preparation (homogenate, LLE or SPE, centrifuge) → instrumental analysis (**NMR**, **GC-MS**, **LC-MS**) → data mining (feature extraction, multivariate statistics, **PCA**, heatmap) → biomarker discovery → multi-center validation. His screening criteria are very practical: a volcano plot with **$p < 0.05$, FDR < 0.15**, intersected with **VIP > 1.0**, and whatever survives that is a differential metabolite.

The most eye-catching branch is **breathomics**: **VOC**s (volatile organic compounds) in the exhaled breath are collected with SPME fibres, and lung and gastric cancer can be detected non-invasively. His slide also marks where the VOCs are made — in the **tumour microenvironment**, where cancer cells, alveolar cells, T cells, neutrophils, macrophages and fibroblasts together with cytokines and metabolites make them.

K-Ras: a broken signal point downstream, where blocking upstream does nothing

Under normal circumstances **RAS** only fires when the **EGFR** above it is switched on. Once it mutates into **K-Ras** (**KRAS G12C**, for example) things are different: a mutation at G12, G13 or Q61 destroys the GAP-mediated hydrolysis of GTP, Ras sticks in its active Ras-GTP state, and the cell keeps dividing even when the upstream receptor is blocked. The pathway is **EGFR → RAS → RAF → MEK → ERK**; his slide marks EGFR, RAS, PI3K and RAF as the sites most often mutated in cancer. KRAS G12C appears in **13 % of lung adenocarcinoma** and **3 % of colorectal cancer**.

This molecule brings a clinical question that must be answered. **Cetuximab** is an anti-EGFR antibody, and it was made for colorectal cancer; but if the tumour's **K-Ras is mutant**, the downstream signal is still on, blocking upstream achieves nothing, and the drug fails. That is exactly the definition of a **predictive biomarker**: the marker predicts "will this patient respond to this drug". He also puts up a survival curve stratified by KRAS mutation status in Lecture

1: the mutant group does worse than the non-mutant group, **p = 0.016**.

K-Ras was long called **undruggable**. The end of Lecture 1 is a mock FDA press release: **Rasonque (daraxonrasib)** is approved for metastatic pancreatic cancer — a RAS inhibitor; the trial enrolled **500** adults with previously treated metastatic pancreatic adenocarcinoma, with a **median OS of 13.2 months**, against **6.7 months** for standard chemotherapy. He puts this in Lecture 1 because it turns "targeted therapy" from a theory into something you can fill in a form for.

HER2, and the Kaplan–Meier curve you must be able to read

HER2 (**ERBB2**) is a receptor tyrosine kinase. Its **amplification** drives a whole chain reaction: gene copy number goes up → mRNA goes up → receptors on the cell surface go up → release of the receptor extracellular domain goes up. Lecture 5 mentions the other side of the same fact: HER2 cannot homodimerise effectively but heterodimerises with HER1/HER3/HER4, and one of the actions of **Herceptin** (trastuzumab) is to block that dimerisation.

And the clinical meaning of HER2 has to be read off the **Kaplan–Meier curve** from Lecture 1 — a first-level question that he says comes up every time he teaches a KM curve and that only one student got right last year. There are only four steps to reading it: the curve steps down from **1.0**; a flat stretch means no events in that interval and a downward step is one event, while the little tick marks on the curve are the **censored** patients; a curve that is **steep and drops early** means the gene is involved **early** and the prognosis is poor; a curve that **drops late** means late involvement; and two curves that **separate** mean the gene can tell the two groups of patients apart, while curves that **cross or overlap** mean it cannot.

The trap in this question is that "the same target runs in opposite directions in different cancers". His classroom example is a receptor tyrosine kinase that predicts worse survival in pancreatic cancer but **better** survival in renal cell carcinoma. The **ERBB2** entry in the Human Protein Atlas is an example of the same kind: renal cancer favourable, endometrial cancer unfavourable and pancreatic cancer unfavourable, all three with **p < 0.001**. The conclusion is one sentence: **a drug target and a prognostic biomarker must always be tied to a cancer type; you cannot extrapolate.**

ADC: tying the poison to an antibody

The logic of an **ADC** (antibody-drug conjugate) is very direct: the antibody finds the tumour, the **cytotoxic payload** does the killing, and a linker joins the two. The payloads he uses are topoisomerase inhibitors — **SN-38** and **DXd** are both of that kind; the clinical examples include **trastuzumab-deruxtecan** against **HER2** and datopotamab-deruxtecan against **TROP2**. Why pick a topoisomerase inhibitor? Because scenario question 13 in the question bank asks exactly this: "why do you put a topoisomerase inhibitor inside the antibody?" The answer is selective delivery — the antibody carries the drug to the surface of the tumour cell and releases it there, so normal tissue is spared.

There is one scope boundary to hold on to: he has stated plainly that the **structure** of an ADC and **dual-payload ADC**s will not be examined — **Lecture 4:** "all this would not be included in the midterm. I'm just want you to know more". What you need is the logic of an ADC and the role of the payload, not the chemical structure.

Synthetic lethality: killing the tumour with its own gap

The definition of **synthetic lethality** is: losing either gene on its own is survivable, losing both at once is fatal. The clinical use is direct — the tumour is already missing the first one (a **BRCA1** or **BRCA2** mutation → HR deficiency), so you use a drug to attack the second one (a **PARP inhibitor**). Normal cells still have intact HR and survive; the tumour cells do not, so they die. The margin in between is the **therapeutic window**, and synthetic lethality is what widens it.

The contrast on the other side is between **tumour suppressor** genes and **oncogene**s: the normal job of a **tumour suppressor** (such as **p53**) is to press the brake, and losing one copy causes trouble; an **oncogene** is the accelerator. p53 is mutated in about **40 %** of cancers.

The last two clinical details

MSI is where the mismatch-repair thread from Lecture 4 ends up: MMR deficiency → microsatellite instability → frameshift → neoantigen → hot tumour. The more mutations there are, the higher the **TMB** (tumour mutational burden), and a high neoantigen burden is exactly the predictive condition for a response to immunotherapy. **CEA** (carcinoembryonic antigen) is a circulating **surrogate marker**, used to track tumour burden in colorectal cancer.

And there is **receptor occupancy (RO)**: how much of the receptor the drug has to occupy before it works. His numbers are about **1 %** for a typical antibody; **Herceptin needs 2 % in breast cancer** and **10 % in gastric cancer**. The same drug needs a different dose in different tissues because the required RO is tissue-specific. Similar is **tamoxifen** (**SERM**): an antagonist in the breast (anti-tumour) but a partial agonist in the endometrium, so endometrial cancer risk goes up slightly. He reminds you: **Lecture 5:** "This is the concept that you have to bear in mind."

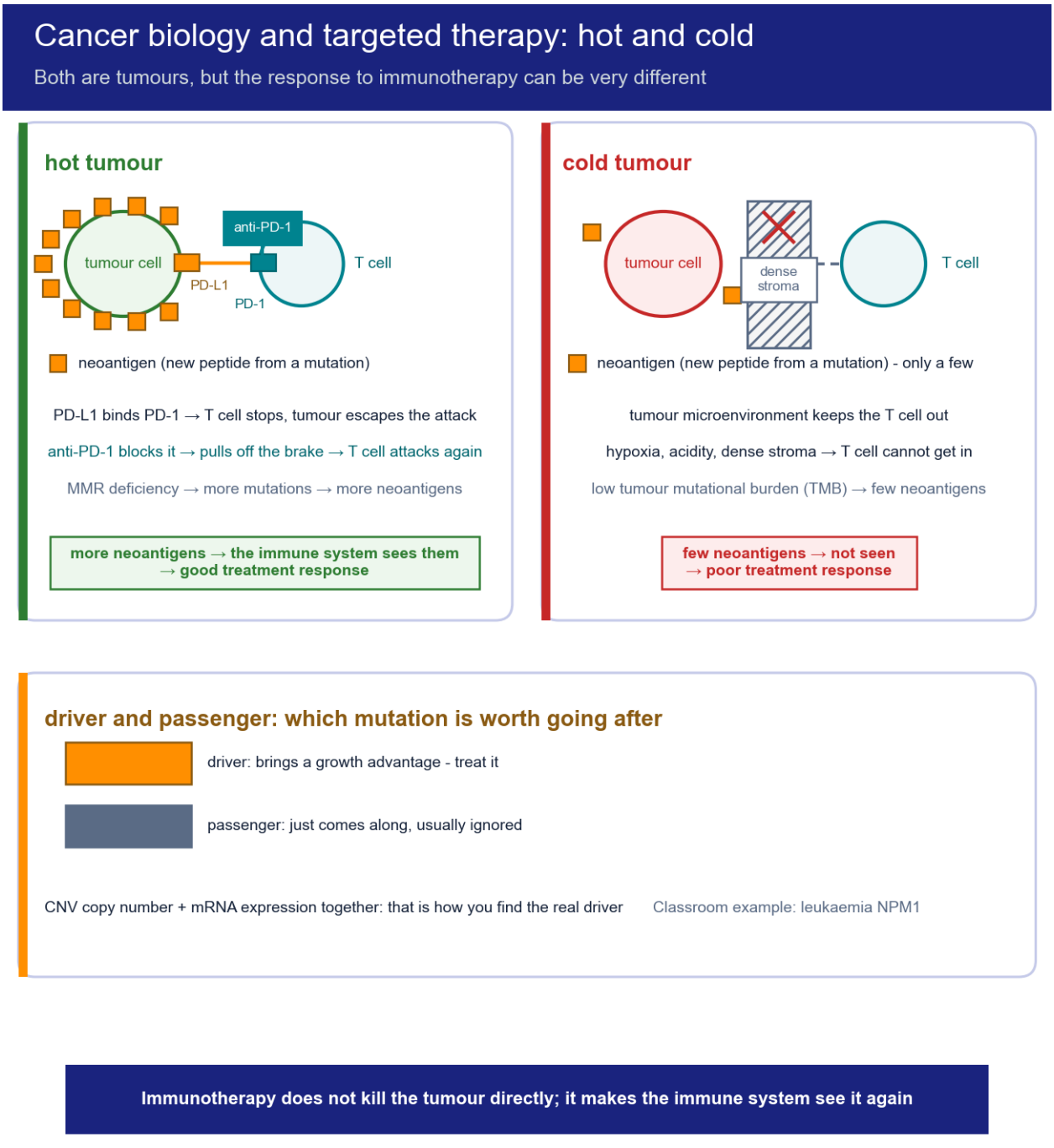


Figure 10: The difference between a hot and a cold tumour, and the difference between driver and passenger genes.

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
driver	A change that brings a growth advantage — a typo in an important word.	Ch 3, Ch 10, Lect 3	Examined (second-level §4.12, and named as one half of the confusable pair "driver vs passenger")
passenger	A bystander change that hitched a ride and was carried along by chromosomal instability.	Ch 3, Ch 10, Lect 3	Examined (same)
CNV	Copy number variation: a stretch of gene present in more or fewer copies.	Ch 10, Lect 3	Examined (second-level §4.12 requires combining the CNV profile with the mRNA; only the mechanism details are excluded)
amplification	An increase in the copy number of a stretch of gene.	Ch 10, Lect 2, Lect 3	Examined (second-level §4.13: FISH → HER2 amplification → the Herceptin candidate)
deletion	A missing copy number of a stretch of gene.	Ch 3, Ch 10, Lect 3	Not examined (background)
NPM1	The leukaemia gene used to show that "copy number and expression rising together" is what makes a driver.	Ch 10, Lect 3	Not examined (he uses it to demonstrate the integrative method; it is not a separate exam point)
Warburg effect	A cancer cell doing anaerobic glycolysis in spite of oxygen.	Ch 10, Lect 1, Lect 2	Examined (the way-in concept he names in Lecture 1)
aerobic glycolysis	Turning the products of glycolysis into lactate even when oxygen is present.	Ch 10, Lect 1	Examined (the mechanism name of the Warburg effect)
lactate	The reduced product of glycolysis and the evidence that the cancer cell has given up on oxidative phosphorylation.	Ch 10, Lect 1	Examined (he lists it as a metabolomics biomarker example)
PDK	The kinase that inhibits PDC and keeps pyruvate out of the mitochondrion.	Ch 10, Lect 1	Not examined (slide detail)
metabolome	The sum of all the small molecules in a sample.	Ch 10, Lect 1	Not examined (conceptual background)
metabolomics	Measuring the whole metabolome and finding the molecules that discriminate.	Ch 10, Ch 11, Lect 1	Examined (he drew the complete seven-step workflow and gave the FDR and VIP thresholds)
NMR	An analysis technique that measures small molecules by nuclear magnetic resonance.	Ch 10, Ch 11, Lect 1	Not examined (one instrument in the metabolomics workflow)
GC-MS	A technique that analyses small molecules by gas chromatography with mass spectrometry.	Ch 10, Ch 11, Lect 1	Not examined (same)
LC-MS	A technique that analyses small molecules by liquid chromatography with mass spectrometry.	Ch 10, Ch 11, Lect 1	Examined (also used for identifying the protein spots from 2D-PAGE)
PCA	The method that compresses thousands of measurements into a few axes and plots the samples on a plane.	Ch 10, Ch 11, Lect 1	Not examined (one step in the workflow)
FDR	False discovery rate; the control on the false positives produced by testing very many things at once.	Ch 10, Ch 11, Lect 1	Not examined (a statistical threshold in the workflow)
VIP	Variable importance in projection; the threshold used to screen differential metabolites.	Ch 10, Ch 11, Lect 1	Not examined (a statistical threshold in the workflow)
breathomics	Non-invasive cancer detection using the volatiles in exhaled breath.	Ch 10, Ch 11, Lect 1	Not examined (he showed it without naming it)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
VOC	Volatile organic compounds; the small molecules that can be measured in exhaled breath.	Ch 10, Ch 11, Lect 1	Not examined (same)
oncogene	The gene that presses the accelerator; its over-activity pushes a tumour forward.	Ch 10, Lect 1, Lect 3, Lect 4	Not examined (background)
tumour suppressor	The gene that presses the brake; losing its function lets the cell run away.	Ch 10, Lect 3, Lect 4	Not examined (background)
p53	The most important tumour suppressor; mutated in about 40 % of cancers.	Ch 6, Ch 10, Lect 2, Lect 3, Lect 4	Not examined (background)
RAS	The small GTPase sitting downstream of EGFR.	Ch 10, Lect 1	Examined (the RAS → RAF → MEK → ERK pathway has to be writable)
K-Ras	The signal point that, once mutated, sticks in its active state and ignores the upstream.	Ch 10, Lect 1	Examined (the core of the cetuximab question)
KRAS G12C	A K-Ras point mutant with one amino acid changed.	Ch 10, Lect 1	Examined (Lecture 1 has a dedicated KRAS G12C slide with the KM curve, p = 0.016)
EGFR	The surface receptor tyrosine kinase upstream of K-Ras.	Ch 10, Lect 1, Lect 4, Lect 5	Examined ("K-Ras mutant → cetuximab fails" is a question type he names)
RAF	The downstream of K-Ras in the MAPK pathway.	Ch 10, Lect 1	Not examined (pathway background)
MEK	The downstream of RAF in the MAPK pathway.	Ch 10, Lect 1	Not examined (pathway background)
ERK	The most downstream member of the MAPK pathway; it executes the growth signal.	Ch 10, Lect 1	Not examined (pathway background)
cetuximab	The anti-EGFR antibody that fails when K-Ras is mutant.	Ch 10, Lect 1	Examined (the §8 multiple-choice type: "K-Ras mutant → cetuximab fails")
daraxonrasib	The generic name of Rasonque, the RAS inhibitor approved for metastatic pancreatic cancer.	Ch 10, Lect 1	Examined (third-level §5: median OS 13.2 versus 6.7 months)
Rasonque	The trade name of daraxonrasib.	Ch 10, Lect 1	Examined (same)
undruggable	A target that was long thought impossible to attack with a drug.	Ch 10, Lect 1	Not examined (it sets up the breakthrough of Rasonque)
frameshift	A shift of the reading frame caused by an insertion or deletion that is not a multiple of three.	Ch 3, Ch 9, Ch 10, Lect 1, Lect 4, Lect 5	Examined (he says a frameshift creates neoantigens)
neoantigen	A new antigen the immune system has never seen.	Ch 9, Ch 10, Lect 1, Lect 4, Lect 5	Examined (the start of the hot tumour mechanism chain)
tumour mutational burden (TMB)	The total number of mutations accumulated in a tumour.	Ch 10, Lect 1, Lect 4	Examined (second-level §4.4 brings in the concept through "high mutation and neoantigen burden")
Kaplan–Meier curve	The step curve showing the proportion of patients still alive at each moment.	Ch 10, Lect 1, Lect 3	Examined (first-level §3.1: he says this question comes up every time he teaches a KM curve)
HER2	The receptor tyrosine kinase that is a breast cancer target.	Ch 7, Ch 10, Lect 2, Lect 4, Lect 5	Examined (second-level §4.13: FISH → the Herceptin candidate)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
ERBB2	The gene name of HER2.	Ch 10, Lect 1, Lect 5	Examined (third-level §5: renal favourable, endometrial and pancreatic unfavourable, all $p < 0.001$)
antibody-drug conjugate (ADC)	A design in which an antibody carries a cytotoxic drug to the tumour.	Ch 10, Lect 1, Lect 4	Examined (scenario question 13 tests the ADC logic and the payload)
cytotoxic payload	The poison part of an ADC that does the killing.	Ch 10, Lect 1, Lect 4	Examined (same)
SN-38	An ADC payload of the topoisomerase inhibitor class.	Ch 10, Lect 1, Lect 4	Not examined (he names it but does not ask you to memorise it)
DXd	Another ADC payload of the topoisomerase inhibitor class.	Ch 10, Lect 1, Lect 4	Not examined (same)
trastuzumab-deruxtecan	An ADC against HER2 that has reached the clinic.	Ch 10, Lect 1, Lect 4	Not examined (same)
TROP2	A cancer-cell surface marker used by an ADC.	Ch 10, Lect 1, Lect 4	Not examined (background)
dual-payload ADC	A design in which one ADC carries two payloads.	Ch 10, Lect 1, Lect 4	Excluded (he says "all this would not be included in the midterm")
PARP inhibitor	A drug that attacks PARP, used to kill HR-deficient tumours.	Ch 6, Ch 10, Lect 4	Examined (second-level §4.7, synthetic lethality and the therapeutic window)
synthetic lethality	A combination in which losing both genes is what kills.	Ch 6, Ch 10, Lect 4	Examined (second-level §4.7)
therapeutic window	The margin between the minimum effective dose and the toxic dose.	Ch 6, Ch 10, Lect 4	Examined (second-level §4.7)
MSI	Microsatellite instability; the tumour feature caused by MMR deficiency.	Ch 6, Ch 10, Lect 4	Examined (third-level §5: 15 % of colorectal cancers are MSI-H)
CEA	A fetal antigen that cancer cells re-express; it works as a circulating surrogate marker.	Ch 10, Lect 1, Lect 4	Examined (scenario question 11: CEA as a surrogate marker)
surrogate marker	A cheap stand-in for the thing you really want to measure, such as tumour size.	Ch 10, Ch 11, Ch 12, Lect 1, Lect 4	Examined (scenario question 11 and the passage on the cost of clinical trials)
receptor occupancy (RO)	How far the receptor has to be occupied by the drug before it works.	Ch 7, Ch 10, Lect 2, Lect 5	Examined (third-level §5: 1 % / breast 2 % / gastric 10 %)
Herceptin	The trade name of trastuzumab; it blocks HER2 dimerisation.	Ch 7, Ch 10, Lect 2, Lect 5	Examined (same)
tamoxifen	A SERM: an antagonist in the breast, a partial agonist in the endometrium.	Ch 7, Ch 10, Lect 5	Examined (second-level §4.8, where he says "you have to bear in mind")
SERM	Selective estrogen receptor modulator.	Ch 7, Ch 10, Lect 5	Examined (same)
predictive biomarker	A marker that predicts whether a patient will respond to a treatment.	Ch 10, Ch 11, Lect 1	Examined (the confusable pair "predictive vs prognostic biomarker")
prognostic biomarker	A marker that predicts how the disease itself will develop.	Ch 10, Ch 11, Lect 1	Examined (same)
OS	Overall survival.	Ch 10, Ch 12, Lect 1, Lect 4	Examined (third-level §5: Rasonque's median OS of 13.2 versus 6.7 months)

Chapter 11

Assay techniques and omics

Every assay method is really answering the same set of questions: **what are you measuring? what is the denominator? what can get in and contaminate it?** Once you have those three, you will not be afraid of any instrument in front of you. This chapter walks through the order "cells → tissue → DNA → RNA → protein → small molecules", and then comes back to one question: how do these numbers turn into a clinical judgement.

Measuring cells: sweeping a laser across them one at a time

The principle of **flow cytometry** is very simple: a suspension of single cells is wrapped by the sheath fluid into a thin stream and passes the laser one cell at a time. The light scattered and the light fluoresced by each cell are separated — **FSC** (forward scatter) reflects **size**, **SSC** (side scatter) reflects **granularity and internal complexity**. The real molecular information comes from the fluorescence: an antibody carrying a fluorescent label is a name tag.

Gating is drawing a circle on the scatter plot around the population you want, for example picking the CD3+CD4+ helper T cells out of PBMCs. To actually **get the living cells in your hands**, you need **FACS** (fluorescence-activated cell sorting): the instrument deflects the charged droplets into different collection tubes, and the whole thing is done inside a sterile hood.

To ask "which step of the cycle is this population of cells at", you stain with **PI** (propidium iodide) — PI binds the DNA and then fluoresces, and the fluorescence intensity is proportional to the DNA content, so the **DNA content histogram** you draw is the familiar curve: G0/G1 at 2N, S phase in the middle, G2/M at 4N. Two details also have to be read: **doublets** (two cells stuck together) on the scatter plot must be excluded using the relationship between FSC-A and DNA-A and DNA-W, or they get misread as G2/M; and the **SubG1** population is apoptotic cells with fragmented DNA. **Serum starvation** drops FBS to 0.5–1 % and can hold the cells in G1, giving a common starting point.

Measuring tissue: which protein is where, and how many copies of the gene

Where a protein sits is found by staining. **IHC** (immunohistochemistry) uses a **chromogenic enzyme substrate** — when the antigen binds the antibody, the enzyme deposits an insoluble dye right where you are looking; **IF** (immunofluorescence) uses a **fluorescent dye**, and the advantage is that you can look at several molecules at once, which gives you **multiplex IF**; **DAPI** stains the nucleus blue, so you know where every cell is. **Confocal microscopy** uses a **pinhole** to block out-of-focus light and collects the signal from one focal plane only, which is what lets it make optical sections and three-dimensional reconstructions.

The biggest enemy of a tissue specimen is mixing: a tumour sample contains cancer cells, stromal fibroblasts and infiltrating immune cells. **LCM** (laser capture microdissection) uses a laser to cut out a homogeneous piece of tissue under the microscope and send it for analysis, and that is why it is worth learning.

Gene copy number does not use antibodies at all; it uses **DNA probes** and **FISH** (fluorescence in situ hybridisation). His key sentence is: "FISH = DNA probes, not antibodies", so FISH counts how many copies of a gene there are, not how much protein there is. The clinical standard for reading it is a **HER2/CEP17** ratio of **≥ 2.0**, or a **HER2 copy number ≥ 6** (CEP17 is the centromere probe of chromosome 17, used as an internal control for the chromosome number).

There is one scope boundary to hold on to: the **FISH protocol parameters** are not examined. His words are: **Lecture 2:** "Definitely not include in the exam. I will not ask you about what's the denaturations, what's the temperature". In other words, the temperature at which the probe denatures and the temperature at which the hybridisation is done are not worth memorising; but what FISH can do and how to read HER2/CEP17 are.

Measuring DNA and RNA: amplifying the signal until you can measure it

PCR (polymerase chain reaction) uses a template, polymerase, dNTPs and a pair of primers to take a chosen DNA fragment repeatedly through denaturation, annealing and extension, amplifying it exponentially — theoretically **2^n** copies. **qPCR** adds real-time measurement on top, so it can give a quantitative result. **Sanger sequencing** reads out bands one at a time with the dideoxynucleotide chain-termination method and stacks them into a chromatogram; **NGS** (next-generation sequencing) reads millions of fragments in one go. Lecture 3 reminds you: **a point mutation and a single-nucleotide variant can only be found by sequencing** — a karyotype misses them.

Designing a primer runs into **GC content → melting temperature**: a G–C pair has **3 hydrogen bonds** and an A–T pair only **2**, so the higher the GC content, the harder it is to pull the DNA apart and the **higher the Tm**.

At the RNA level there are two routes. **RNA sequencing** goes the library way: input RNA → fragmentation → conversion to cDNA with sequencing adapters added → DNA library, and then the machine reads it. **Microarray** is the older route: the chip is covered with known gene sequences, the sample's RNA is converted into labelled cRNA and allowed to bind; the **oligonucleotide GeneChip** uses biotin labelling and a single colour, while the **cDNA microarray** labels normal tissue with **Cy5** and cancer tissue with **Cy3**, and the two compete to hybridise on the same chip, so whichever position leans to one colour tells you which gene is stronger in that group.

Measuring protein: bands, spots and mass spectra

Western blot is the most basic of them: electrophoresis separates the proteins, they are transferred to a membrane, an antibody is added, and you develop the image and look at "is this band at the right height and the right strength". **ELISA** is the method dedicated to measuring the concentration of one protein in a liquid sample (serum, plasma), and it is the tool that lands the liver cancer **Vimentin** case: in the control group (n=64) serum vimentin is **176.0 ± 61.3 ng/mL**; in **HCC ≤ 2 cm** (n=44) it is **271.5 ± 165.9**; in **HCC > 2 cm** (n=44) it is **231.3 ± 132.3**. There are 152 people in the whole dataset. Vimentin became a non-invasive biomarker precisely because it can be measured in serum.

Proteomics means looking at the whole set of proteins at once, and it relies on **2D-PAGE**: the first dimension is **isoelectric focusing**, separating by charge on a gel strip at pH **4.0–9.0**, and the second dimension is SDS-PAGE, separating by molecular weight from **220 kDa** all the way down to **10 kDa**; the spots are cut out and then identified by **LC-MS/MS**. Modern proteomics has mostly been taken over by mass spectrometry working directly, but his teaching material still uses this workflow to make the logic of "separate → cut the spot → identify" completely clear.

Measuring small molecules: reading the whole set of metabolites in one go

Chromatography is the separation step that comes before every mass spectrum. **LC-MS** pairs liquid chromatography with mass spectrometry and suits molecules that are heat-labile, unstable and highly polar; **GC-MS** vaporises the sample first and then separates it in the gas phase, which suits volatile and thermally stable molecules; **NMR** needs no separation at all and has no detection-limit problem, and it is a third route. Those three, plus the three data-processing gates of **PCA**, **FDR** and **VIP**, make up his **metabolomics** workflow. **Breathomics** is the most dramatic application of the whole workflow: **VOC**s (volatile organic compounds) in the exhaled breath are collected with SPME fibres and used for non-invasive cancer detection.

The other side of the omics: marking where the switches are

To ask "is this gene switched on or switched off", the sequence alone is not enough; you need the epigenetic and chromatin state. **Bisulfite sequencing** is the gold standard for measuring DNA methylation: **bisulfite** converts only the unmethylated C into U (which reads like T) and leaves methylated C alone, so you can read the methylation state at every single CpG position. The clinical markers from Lecture 3 — **MGMT** promoter methylation predicting sensitivity to alkylating chemo plus radiotherapy, and **CIMP** with the prognosis of colorectal cancer — all depend on it.

There are also two techniques you may meet in the literature but that never appear in his lecture material; knowing what each of them measures is enough: **ChIP** (chromatin immunoprecipitation) measures "which stretch of DNA a particular protein, or a particular histone mark, is bound to"; **ATAC-seq** measures "which regions of the DNA are open, where a transcription factor can get in". Both are background knowledge, not exam questions.

How the numbers become a clinical judgement

Back up to the top. All of these techniques exist to serve three kinds of **biomarker**: a **diagnostic biomarker** tells you whether the patient has the disease now; a **prognostic biomarker** tells you how the disease itself will go, whatever the treatment; and a **predictive biomarker** tells you whether the patient will respond to a particular treatment. The third one is the tightest — **cetuximab does not work in patients with mutant K-Ras**, and they are screened out in advance by the predictive biomarker.

And when a test has to be done before the drug is given and decides whether that patient will benefit, it is a **companion diagnostic**. Scenario question 6 in the question bank asks exactly this: given a specimen, can you say whether to use FISH, IHC, IF, ELISA or LCM, what it measures, and when to use it. As for the **surrogate marker** and the **surrogate endpoint**, those belong to the clinical-trial side: using the CEA that is easy to measure in blood in place of the tumour size on a CT or MRI scan, so that the trial does not have to rescan every patient over and over — which is precisely why biomarkers exist at all, given what he says a clinical trial costs, about **US$200,000 per patient**.

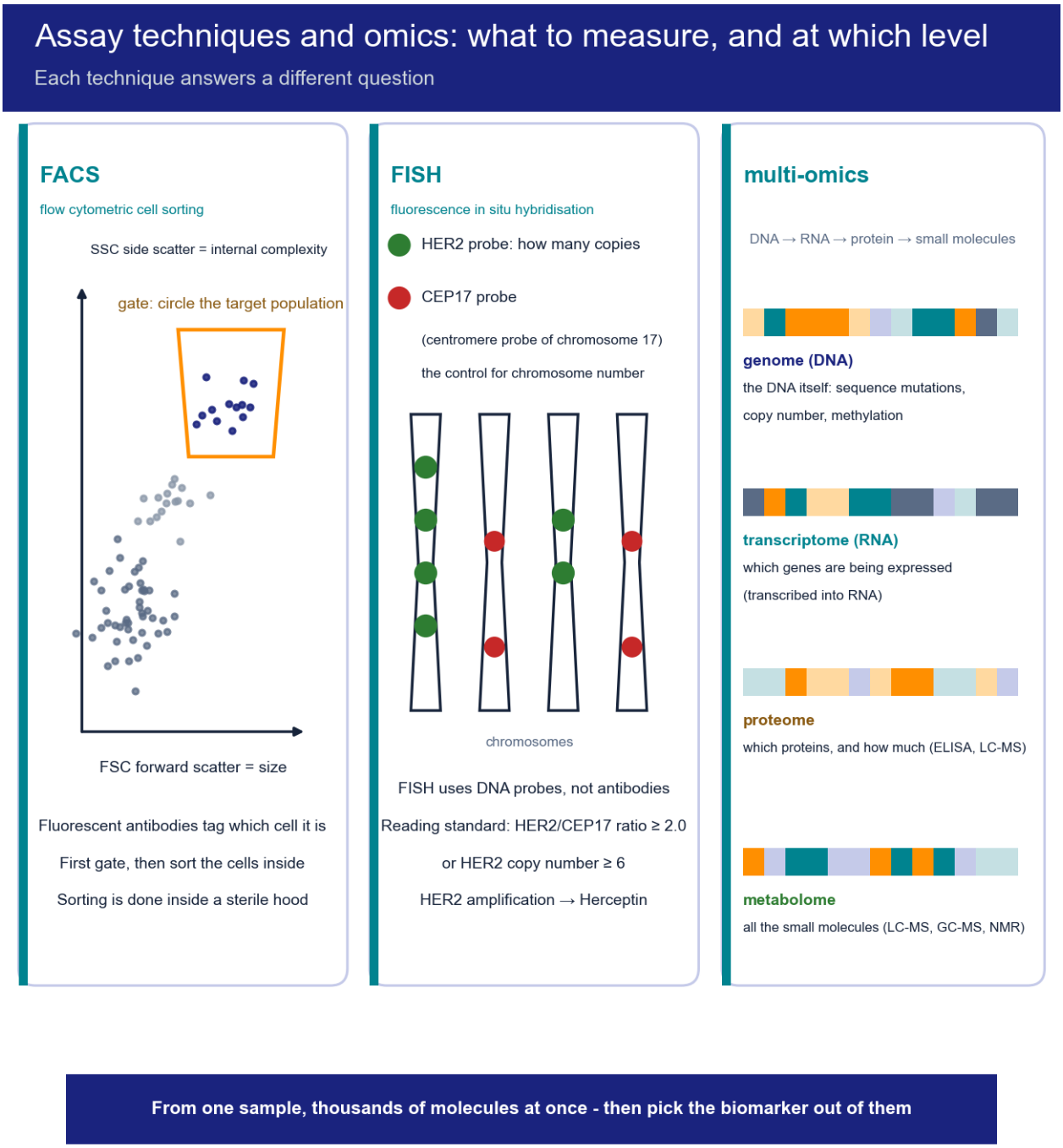


Figure 11: What FACS, FISH and the omics each measure.

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
flow cytometry	Sweeps a laser across cells one at a time and measures size, granularity and fluorescence.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13, the basics of flow cytometry)
FSC	Forward scatter, which reflects cell size.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: FSC = size)
SSC	Side scatter, which reflects granularity and internal complexity.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: SSC = granularity)
gating	Drawing a gate on the scatter plot around the target population.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: gating isolates the target population)
FACS	The sorter that actually puts the living cells into tubes or wells.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: FACS really does separate the live cells)
PI	Propidium iodide; once it stains the DNA its fluorescence is proportional to the DNA content.	Ch 11, Lect 2, Lect 4	Examined (second-level §4.13: PI staining measures DNA content)
DNA content histogram	The histogram on the DNA-content axis that separates G1 / S / G2-M.	Ch 11, Lect 2, Lect 4	Examined (second-level §4.13: G1 2N, S, G2-M 4N)
doublet	Two cells stuck together, an artefact that gets misread as G2/M.	Ch 11, Lect 2	Not examined (background)
SubG1	The below-G1 population of fragmented DNA, which represents apoptotic cells.	Ch 11, Lect 2	Not examined (background)
serum starvation	Dropping FBS to 0.5–1 % to hold the cells in G1.	Ch 8, Ch 11, Lect 2, Lect 4	Examined (third-level §5: the exact figure of 0.5–1 % FBS)
IHC	A stain that detects protein using a chromogenic enzyme substrate.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: IHC vs IF)
IF	A stain that detects protein using a fluorescent dye.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: IHC vs IF)
multiplex IF	Using several fluorochromes at once to look at several proteins.	Ch 9, Ch 11, Lect 2	Not examined (he shows it only as a tool)
DAPI	The blue fluorescent dye that stains the nucleus.	Ch 9, Ch 11, Lect 2	Not examined (background)
confocal microscopy	A microscope that uses a pinhole to block out-of-focus light, so it can make optical sections and 3D reconstructions.	Ch 9, Ch 11, Lect 2	Not examined (background)
pinhole	The pinhole in a confocal microscope that blocks out-of-focus light.	Ch 9, Ch 11, Lect 2	Not examined (background)
LCM	Using a laser to cut out a homogeneous piece of tissue under the microscope.	Ch 9, Ch 11, Lect 2	Examined (second-level §4.13: avoids contamination by stromal and immune cells)
FISH	Counting gene copy number with DNA probes rather than antibodies.	Ch 11, Lect 2	Examined (second-level §4.13: FISH = DNA probes, not antibodies)
HER2/CEP17	Reading the HER2 probe against the centromere probe by ratio and by copy number.	Ch 11, Lect 2	Examined (second-level §4.13: FISH → HER2 amplification → the Herceptin candidate)
FISH protocol parameters	The denaturation temperature and the hybridisation conditions of FISH.	Ch 11, Lect 2	Excluded (he says "Definitely not include in the exam. I will not ask you about what's the denaturations, what's the temperature")
PCR	The reaction that exponentially amplifies a chosen DNA fragment.	Ch 3, Ch 11, Lect 1, Lect 3, Lect 4	Examined (third-level §5: G–C 3 hydrogen bonds, A–T 2, and primer design)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
GC content	The proportion of G plus C in a stretch of DNA.	Ch 11, Lect 1, Lect 3	Examined (third-level §5: the more GC, the higher the Tm)
melting temperature	The temperature at which half of a double-stranded DNA has separated.	Ch 3, Ch 11, Lect 1, Lect 3	Examined (same)
qPCR	PCR with real-time measurement added, so it gives a quantitative result.	Ch 11, Lect 1, Lect 4	Not examined (he uses it only in the Vimentin validation workflow)
Sanger	Chain-termination sequencing that reads out a chromatogram.	Ch 11, Lect 2, Lect 3	Examined (second-level §4.11: point mutations need sequencing)
NGS	Sequencing technology that reads millions of fragments at once.	Ch 11, Lect 3	Examined (second-level §4.11, set beside Sanger)
RNA sequencing	The technique that turns RNA into a library and then sequences it.	Ch 11, Lect 1	Not examined (he puts it only on the platform comparison diagram)
microarray	The older method of measuring expression on a chip covered with known gene sequences.	Ch 11, Lect 1	Not examined (same)
oligonucleotide GeneChip	An oligonucleotide chip with a single biotin label.	Ch 11, Lect 1	Not examined (background)
cDNA microarray	A two-colour competitive chip with normal tissue labelled Cy5 and cancer tissue labelled Cy3.	Ch 11, Lect 1	Not examined (background)
Western blot	Electrophoresis, transfer and development: is this protein band at the right height?	Ch 11, Lect 1	Not examined (he uses it to validate Vimentin and codon optimisation)
ELISA	The standard method for measuring the concentration of a protein in a liquid sample.	Ch 11, Lect 1, Lect 2	Examined (scenario question 6 lists it as a candidate companion diagnostic)
proteomics	The study of measuring the whole set of proteins at once.	Ch 11, Lect 1	Examined (he walks the Vimentin case through the whole workflow)
2D-PAGE	Two-dimensional electrophoresis that separates proteins by charge and then by molecular weight.	Ch 2, Ch 11, Lect 1	Examined (the Vimentin case: IEF pH 4.0–9.0 × SDS-PAGE, 220 down to 10 kDa)
isoelectric focusing	Separating proteins on a pH-gradient strip according to their isoelectric point.	Ch 11, Lect 1	Examined (same)
LC-MS/MS	Liquid chromatography with mass spectrometry, used to identify the spots from 2D-PAGE.	Ch 2, Ch 11, Lect 1	Examined (the identification step of the Vimentin case)
chromatography	The step that separates the sample before it goes into the mass spectrometer.	Ch 11, Lect 1	Not examined (background)
LC-MS	Liquid chromatography with mass spectrometry, for measuring small molecules.	Ch 10, Ch 11, Lect 1	Examined (the instrument in the metabolomics workflow)
GC-MS	Gas chromatography with mass spectrometry, for volatile and thermally stable small molecules.	Ch 10, Ch 11, Lect 1	Not examined (the other instrument in the workflow)
NMR	Measuring small molecules by nuclear magnetic resonance, with no separation needed.	Ch 10, Ch 11, Lect 1	Not examined (same)
metabolomics	Measuring the whole set of metabolites and finding the molecules that discriminate.	Ch 10, Ch 11, Lect 1	Examined (he drew the seven-step workflow and gave the FDR and VIP thresholds)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
PCA	The method that compresses thousands of measurements into a few axes and plots the samples on a plane.	Ch 10, Ch 11, Lect 1	Not examined (one step in the workflow)
FDR	False discovery rate; the control on the false positives produced by testing very many things at once.	Ch 10, Ch 11, Lect 1	Not examined (a statistical threshold in the workflow)
VIP	Variable importance in projection; the threshold used to screen differential metabolites.	Ch 10, Ch 11, Lect 1	Not examined (same)
breathomics	Non-invasive cancer detection using the volatiles in exhaled breath.	Ch 10, Ch 11, Lect 1	Not examined (he showed it without naming it)
VOC	Volatile organic compounds; the small molecules that can be measured in exhaled breath.	Ch 10, Ch 11, Lect 1	Not examined (same)
bisulfite sequencing	Using bisulfite to turn unmethylated C into U and read the methylation at every site.	Ch 4, Ch 11, Lect 3	Examined (the gold standard of second-level §4.1)
ChIP	Measuring which stretch of DNA a particular protein or histone mark is bound to.	Ch 11	Not examined (it does not appear in his teaching material; knowing what it measures is enough)
ATAC-seq	Measuring which regions of DNA are open and accessible to transcription factors.	Ch 11	Not examined (same)
biomarker	A molecule measured from a sample to say something about the disease or the treatment response.	Ch 11, Lect 1, Lect 2, Lect 4	Examined (the framing concept of the whole course)
diagnostic biomarker	One that tells you whether the patient has the disease now.	Ch 11, Lect 1	Examined (the diagnostic / prognostic / predictive classification from Lecture 1)
prognostic biomarker	One that tells you how the disease itself will go.	Ch 10, Ch 11, Lect 1	Examined (same, and one half of the confusable pair)
predictive biomarker	One that predicts whether the patient will respond to a treatment.	Ch 10, Ch 11, Lect 1	Examined (same, and one half of the confusable pair)
companion diagnostic	A test that must be done before the drug and decides whether the patient benefits.	Ch 11, Ch 12, Lect 2	Examined (scenario question 6)
surrogate marker	A cheap stand-in for the thing you really want to measure.	Ch 10, Ch 11, Ch 12, Lect 1, Lect 4	Examined (scenario question 11: CEA as a surrogate marker)
surrogate endpoint	The measurement used in a clinical trial in place of the real clinical endpoint.	Ch 11, Ch 12, Lect 4	Examined (CheatSheet question 7: surrogate versus clinical endpoint)

Chapter 12

Drug development and clinical trials

Between a molecule and a pill you can sell, there are ten years, billions of dollars, and an entire regulatory process. This chapter is about how that road is travelled, and about the things on it that he specifically names as worth remembering.

Choosing the target first, but "important" is not the same as "makeable into a drug"

The first step is **target identification**: finding, in the mechanism of the disease, a molecule that might be worth attacking. Then comes **target nomination** — formally designating a particular molecule as a drug target. Lecture 4 has a distinction here that is very much worth remembering: **druggability** and **disease relevance** are two different things. The cytoskeleton is indispensable in metastasis, which makes it highly relevant to disease; but it is also indispensable to the cell staying alive, which makes it hard to hit with a selective small molecule. By contrast **telomerase** is genuinely druggable — both antisense oligonucleotides (ASO) and small molecules are already in clinical trials.

He also gives the complete criteria for this stage: a therapeutic target needs (a) **disease-specific overexpression** in patient data, (b) **structural tractability** that suits a small molecule, an antibody or a nucleic-acid drug, and (c) a competitive landscape that is not crowded.

There is one scope boundary to hold on to here: the whole pharmaceutical **target nomination and druggability process** is explicitly excluded — **Lecture 4:** "it's not included in exam, but I think it's really more interesting". So understanding those three criteria is enough; there is no need to memorise them, and no need to memorise the process details. What will be examined is the other end of this thread: the experimental data that makes **telomerase** a druggable target. His numbers are: three weeks of pre-treatment with an anti-TERT ASO brings the tumour take rate in mice down to **2/12**, against **9/12** in the untreated group, **p = 0.01**.

From candidate drug to IND

After that comes **medicinal chemistry** (turning a lead molecule into one that has both activity and pharmacokinetics), and then the **preclinical** stage: **MOA** (mechanism of action) studies and **toxicology**, with **IND-enabling studies** assembling the data package. The **IND** (Investigational New Drug) is the application that permits the drug to be studied in humans. Before submitting, a company may also hold **pre-IND**, or even pre-pre-IND, meetings with the FDA to settle the animal model in advance — a pharmacology-relevant non-human primate, or a humanised mouse.

Three phases, three questions

Phase 1A answers "is it safe": **dose escalation**, starting a small group of already-sick patients on a low dose and working upwards to find what dose is safe. **Phase 1B** is the **expansion cohort**: more patients, treated at the chosen dose, to characterise the side effects and look for a preliminary signal of efficacy. **Phase 2** and **Phase 3** answer "does it work, and is it better than what we have now" — that is, **efficacy**. The three phases are a relay, not three parallel things: the output of Phase 1A is a dose that is worth carrying forward; Phase 1B exchanges that dose for enough signal of efficacy and enough safety data; and only then are Phase 2 and Phase 3 entitled to answer "should this be approved, should the guidelines change".

To measure efficacy the trial uses three time endpoints: **OS** (overall survival), **PFS** (progression-free survival) and **ORR** (objective response rate). The first two have concrete cases in his teaching material: OS for Rasonque's **13.2 versus 6.7 months**, and PFS for that anti-PD-1 curve of dMMR against pMMR. As for the dose that is settled at the end of Phase 1 and carried into Phase 2, the standard name is **RP2D** (recommended Phase 2 dose); it is a general term he never covers, and knowing that it is the dose Phase 1A/1B converged on is enough.

One number to hold on to: about US$200,000 per patient

When Lecture 4 covers the structure of a clinical trial, he throws out the cost first: **about US$200,000 per patient**. A cohort of several hundred people is therefore tens of millions of dollars. This is precisely why biomarkers exist — a trial needs an endpoint that is **cheap and can be measured often**.

There is one place where this number is easily confused and the two have to be kept apart: **US$200,000 per patient is the cost of running one clinical trial for one person**; while the **USD 370,000 to 475,000** he gives in Lecture 2 for CAR-T is the cost of **one course of cell therapy**. They are not the same thing and cannot be interchanged.

How biomarkers decide who gets in

In a trial, biomarkers have two uses. The first is **biomarker-driven enrolment**: only patients whose biomarker meets the criteria are enrolled, which keeps the signal clean and allows a smaller sample. The second is the **surrogate marker** and the **surrogate endpoint**: what you really want to measure is the tumour size on a CT or MRI scan, but scanning people again and again is expensive and troublesome, so you use the easily measured **CEA** in the blood as the **surrogate endpoint**. The figure in his Lecture 4 is the evidence for this logic: after anti-PD-1, the tumour markers in **MMR-proficient** colorectal cancer mostly rose to above 0 % (some by more than **+200 %**), while **MMR-deficient** colorectal cancer and the non-colorectal cancers both fell to close to **−100 %**.

And when a test has to be done before the drug is given and decides whether that patient will benefit, it is a **companion diagnostic**. HER2, PD-L1 and EpCAM are all membrane proteins, all can be measured, and all are drug targets; when one of them is amplified, what you need is exactly a companion diagnostic, and **FISH** and **IHC** are what provide it.

A few fast tracks at the FDA

The Rasonque press release from Lecture 1 also lists three **FDA designations**. **Breakthrough Therapy** lets a drug skip Phase III; **Orphan Drug** designation covers rare diseases affecting fewer than **200,000** people in the United States and buys **7 years** of market exclusivity; and **Priority Review** speeds up the review. The three terms are not hard to understand — what is worth remembering is the number and the function attached to each.

The last piece: who owns the tool

The last two slides of Lecture 4 are about the **CRISPR** patent landscape, and there are two tracks: UC Berkeley's Doudna and Charpentier hold the invention of CRISPR-Cas9 itself, while the Broad Institute's Zhang holds the application of CRISPR to engineering eukaryotic cells; in between there is a **patent interference** and a string of dates, ending with the PTAB's second interference decision in March 2026. The Broad patent also has a key phrase in it: **first-to-invent**, that is, the "first to invent" principle rather than the "first to file" one.

He names none of this block: the CRISPR patent topic is not on the exam, so treat the timeline and the dates as reference material and do not memorise them. But the three ideas under the slide are worth keeping — the difference between **inventions** and **specific applications**, **non-obviousness** (an invention must not be an obvious extension of what is already known), and **Freedom to Operate (FTO)**: a review done before commercialisation that asks a very practical question — if I make this product, will I be sued? That is also why Broad has been pushing a **patent pool**, in which several holders licence to one another under agreed terms. The whole thing makes one point: **a discovery does not automatically become a product**, and ownership, licensing and cost stand in between — the same kind of practical constraint as a clinical trial costing US$200,000 per patient.

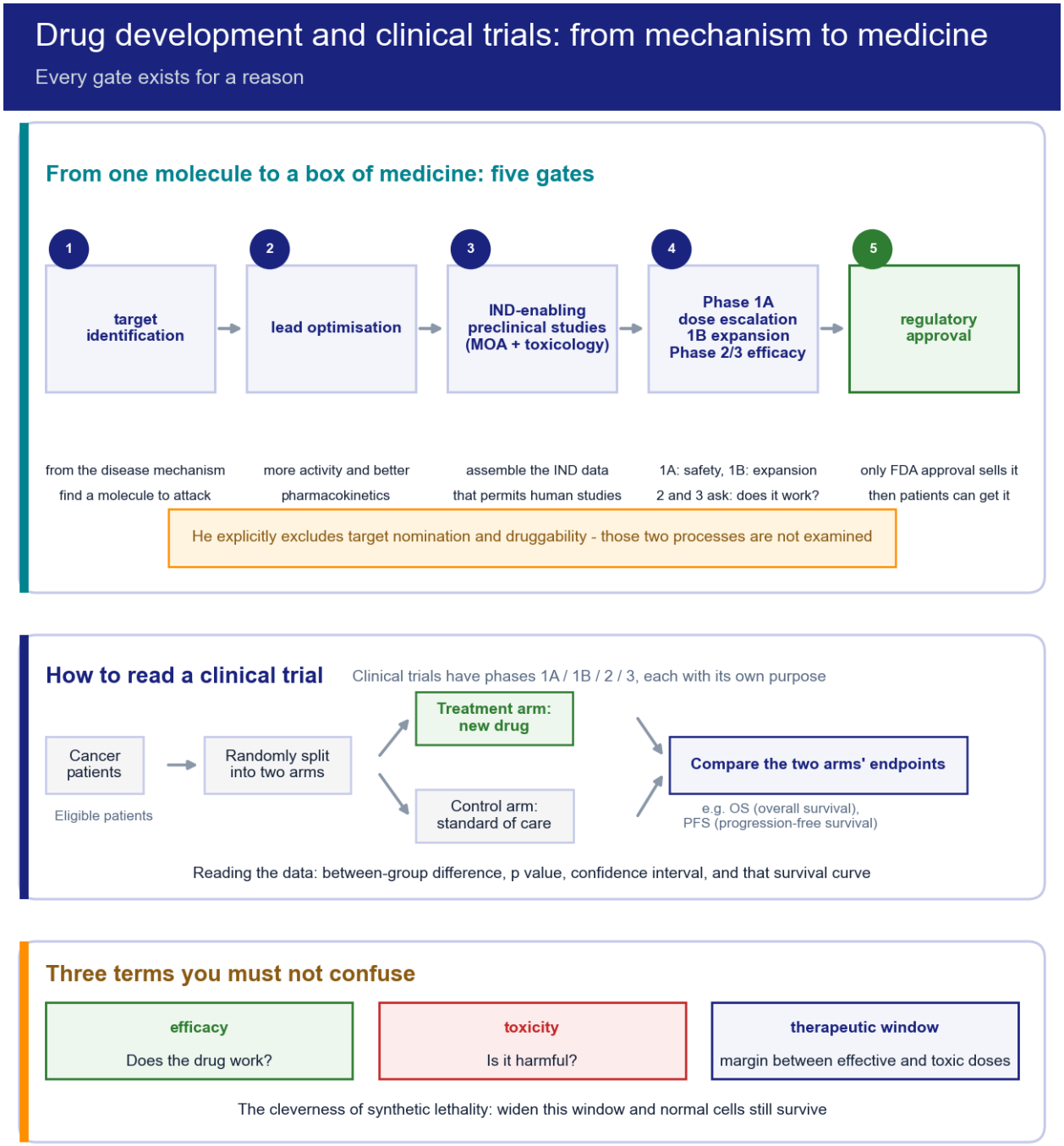


Figure 12: From finding a target to a box of medicine, and how to read the result of a clinical trial.

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
target identification	Finding, in the mechanism of the disease, a molecule that might be worth attacking.	Ch 12, Lect 2, Lect 4	Excluded (it belongs to the pharmaceutical target nomination process he excludes)
target nomination	Formally designating a particular molecule as a drug target.	Ch 12, Lect 1, Lect 4	Excluded (he says "it's not included in exam, but I think it's really more interesting")
druggability	Whether a molecule can be selectively attacked by a drug.	Ch 12, Lect 4	Excluded (same)
structural tractability	Whether a molecule has a structure a small molecule, an antibody or a nucleic-acid drug can bind.	Ch 12, Lect 4	Excluded (same)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
disease relevance	How much the molecule has to do with the disease.	Ch 12, Lect 4	Excluded (same; the concept must be understood: relevance is not the same as druggability)
telomerase	The reverse transcriptase that extends the end of a chromosome using its own RNA template.	Ch 5, Ch 12, Lect 3, Lect 4	Examined (third-level §5: three weeks of anti-TERT ASO pre-treatment, tumour take rate 2/12 against 9/12, p = 0.01)
antisense oligonucleotide (ASO)	A short nucleic-acid drug that binds the target RNA complementarily and shuts it down.	Ch 12, Lect 4	Examined (same)
medicinal chemistry	Turning a lead molecule into a drug with activity and pharmacokinetics.	Ch 12, Lect 2	Not examined (one step in the workflow)
MOA	Mechanism of action: how the drug actually works inside the cell.	Ch 12, Lect 2	Not examined (one step in the workflow)
preclinical	The stage of research before the drug goes into humans.	Ch 12, Lect 2	Not examined (one step in the workflow)
toxicology	The study of what the drug does to the body in the way of harm.	Ch 12, Lect 2	Not examined (one step in the workflow)
IND	Investigational New Drug: the application that permits research in humans.	Ch 12, Lect 2, Lect 4	Not examined (one step in the workflow)
pre-IND	The meeting with the FDA before the IND submission, to agree the animal model.	Ch 12, Lect 2	Not examined (one step in the workflow)
IND-enabling studies	The studies that support the IND application.	Ch 12, Lect 2	Not examined (one step in the workflow)
clinical trial	A regulated therapeutic study carried out in patients according to a protocol.	Ch 12, Lect 2, Lect 4	Examined (third-level §5: the phase structure and about US$200,000 per patient)
Phase 1A	The dose-escalation phase, which asks "what dose is safe".	Ch 12, Lect 2, Lect 4	Examined (third-level §5: Phase 1A dose escalation)
Phase 1B	The expansion-cohort phase, which studies more patients at the chosen dose.	Ch 12, Lect 2, Lect 4	Examined (third-level §5: 1B expansion)
Phase 2	The efficacy phase, which asks "does it work".	Ch 12, Lect 2, Lect 4	Examined (third-level §5: 2/3 efficacy)
Phase 3	The efficacy and comparison phase, which asks "is it better than the existing treatment".	Ch 12, Lect 2, Lect 4	Examined (same)
dose escalation	Raising the dose step by step to find the safe ceiling.	Ch 12, Lect 2, Lect 4	Examined (third-level §5)
expansion cohort	The cohort that enlarges the number of patients treated at the chosen dose.	Ch 12, Lect 2, Lect 4	Examined (third-level §5)
efficacy	Whether the treatment works.	Ch 12, Lect 2, Lect 4	Examined (third-level §5: 2/3 efficacy)
RP2D	Recommended Phase 2 dose: the dose Phase 1 converged on and carries into Phase 2.	Ch 12	Not examined (a general term he never covers)
OS	Overall survival.	Ch 10, Ch 12, Lect 1, Lect 4	Examined (third-level §5: Rasonque's median OS of 13.2 versus 6.7 months)

Key vocabulary for this chapter			
Term	In one sentence	Source	Did he name it?
PFS	Progression-free survival.	Ch 12, Lect 2, Lect 4	Examined (the dMMR-versus-pMMR curve he calls one of the most important figures in modern medicine)
ORR	Objective response rate.	Ch 12	Not examined (a general term he never covers)
biomarker-driven enrolment	Choosing who enters the trial according to their biomarker.	Ch 12, Lect 4	Not examined (what he connects to cost is the surrogate endpoint, not the enrolment rule)
surrogate marker	A cheap stand-in for the thing you really want to measure.	Ch 10, Ch 11, Ch 12, Lect 1, Lect 4	Examined (scenario question 11: CEA as a surrogate marker)
surrogate endpoint	The measurement used in a trial in place of the real clinical endpoint.	Ch 11, Ch 12, Lect 4	Examined (CheatSheet question 7: surrogate versus clinical endpoint)
companion diagnostic	A test that must be done before the drug and decides whether the patient benefits.	Ch 11, Ch 12, Lect 2	Examined (scenario question 6)
FDA designations	The accelerated-track designations the FDA gives a new drug.	Ch 12, Lect 1	Examined (all three are listed on the Rasonque press release)
Breakthrough Therapy	Breakthrough Therapy designation; it can skip Phase III.	Ch 12, Lect 1	Examined (he states its function plainly: "skips Phase III")
Orphan Drug	Orphan Drug designation: fewer than 200,000 people, and 7 years of market exclusivity.	Ch 12, Lect 1	Examined (he gives two exact figures)
Priority Review	Priority Review, which speeds up the FDA review.	Ch 12, Lect 1	Examined (listed on the Rasonque press release)
FTO	Freedom to Operate: the "will I be sued" review done before commercialisation.	Ch 12, Lect 2, Lect 4	Not examined (he shows it only on a slide and does not name it as an exam point)
non-obviousness	The rule that an invention must not be an obvious extension of what is already known.	Ch 12, Lect 4	Not examined (same)
first-to-invent	The principle the Broad CRISPR patent is judged under.	Ch 12, Lect 4	Not examined (same)
patent interference	The proceeding that decides whether two patents interfere with each other.	Ch 12, Lect 4	Not examined (same)
patent pool	A pool of patents in which several holders licence to one another under agreed terms.	Ch 12, Lect 4	Not examined (same)

PART 2

A–Z dictionary

(Every term, alphabetically)

905 entries, sorted by the English term. Every one of them carries a one-line summary, a detailed explanation, its place in the exam and where it appears.

▪ 14-3-3

In one sentence: A scaffold protein that pins phosphorylated YAP in the cytoplasm and lets the whole complex be degraded.

Detailed explanation: 14-3-3 is a family of scaffold proteins, and every molecule in the family carries a pocket that can recognise a phosphate group. When the Hippo pathway is running, kinase MST1 activates LATS1, and LATS1 then phosphorylates YAP.

YAP carrying that phosphate is grabbed straight away by 14-3-3, and the whole complex is degraded afterwards, so YAP never reaches the nucleus at all. That is exactly why this pathway can behave like a switch: press it and it stops dead. There is no half-on state to sit in and no slow slide back down; the YAP is taken off the board entirely, so the signal simply stops arriving.

The same family also takes part in a large number of other pathways, which is why it is better to think of it as a general-purpose shelter for phosphorylated proteins, a holding pen that many different signals end up passing through, rather than as a component that belongs to Hippo alone.

Why this course brings it up: Dr. Wong taught the Hippo–YAP pathway in both Lecture 1 and Lecture 5, but he never named 14-3-3 on its own as an examinable topic. It belongs to conceptual understanding. So the family name itself is background, and the chain is the part that gets examined: MST1 activates LATS1, and LATS1 phosphorylates YAP. **Appears in:** Ch 7,Lect 1,Lect 5

▪ 2D-PAGE

In one sentence: Two-dimensional protein electrophoresis, separating first by charge and then by molecular weight.

Detailed explanation: The first dimension of 2D-PAGE is isoelectric focusing, which spreads the proteins out along a strip according to their isoelectric point (net charge); the second dimension is SDS-PAGE, which takes that same strip and separates it vertically by molecular weight.

Together they make a two-dimensional map: charge on the horizontal axis, size on the vertical one, each spot standing for one protein. Several spots on the same horizontal line therefore mean several isoforms or modified versions of the same protein.

When Lecture 1 walks through the proteomics workflow, this is the first step. Two separations on two different physical properties is what makes the map two-dimensional rather than just a smear.

Why this course brings it up: He uses this workflow as the example in Lecture 1, but he does not name 2D-PAGE as an exam topic in its own right, so it is a concept to understand. What you need from it is the idea that proteins can be separated on more than one property at once, which is what makes a map possible. **Appears in:** Ch 2, Ch 11, Lect 1

▪ 3 prime untranslated region (3 prime UTR)

In one sentence: The stretch of an mRNA that comes after the stop codon and is never turned into protein, but that still carries all the regulatory sites for that message.

Detailed explanation: The 3' UTR is the part of the mRNA that runs from the stop codon all the way to the poly-A tail. It is never itself translated into protein, and yet it is where the regulation is packed most densely.

The seed sequence of miRNAs, the binding sites of RNA-binding proteins, and the structures that decide how long the mRNA survives all sit together in here. Take one and the same gene and attach its poly-A tail at a different position, and the 3' UTR comes out longer or shorter; the stability of the mRNA and its translatability change along with it. The most practical example on the clinical side is the circulating miRNA that you find in serum. One of the reasons it can be measured so stably in blood is that structures of this kind are standing guard over it.

Why this course brings it up: It is where Lecture 5 lands when he talks about mRNA stability and miRNA, and it is also the direct consequence of alternative polyadenylation changing the length of the 3' UTR. He never names this entry on its own, so it belongs to the pile of things you need to understand rather than the pile of labels you need to memorise. **Appears in:** Ch 7, Lect 5

▪ 3D reconstruction

In one sentence: Stacking optical sections back together to rebuild the three-dimensional shape of a cell.

Detailed explanation: A confocal microscope puts a pinhole in front of the detector to block out light that is out of focus, so each image records only one depth. Stack the optical sections back together in the right order and you get a reconstruction of the three-dimensional inside of the cell.

Without that step a thick cell under an ordinary fluorescence microscope simply glows as one sheet, the signals overlap each other, and you cannot tell whether the protein is on the membrane or inside the nucleus. So the pinhole is what buys the depth, and the reconstruction is what turns a stack of pictures into a picture of a cell.

Why this course brings it up: In Lecture 2 he uses confocal to explain the flexibility of immunofluorescence, and he does not name this entry in particular, so it is concept-only. The takeaway is that seeing where a protein is, not just whether it is there, needs the extra machinery. **Appears in:** Ch 9, Ch 11, Lect 2

▪ 4-1BB

In one sentence: One of the two co-stimulatory signals built into a CAR; the other one is CD28.

Detailed explanation: A CAR takes the scFv, the variable part of an antibody, and bolts it onto the intracellular signalling tail of a T cell. The point is that the T cell no longer has to use its own native TCR to recognise the tumour.

That intracellular tail normally carries two co-stimulatory domains at the same time: CD28 and 4-1BB. They supply the second signal, and what that signal says is "this is genuine antigen, not me". Without co-stimulation, a T cell can look straight at an

antigen and still fall into anergy instead of responding. So a CAR that can only see is no use on its own; you have to build the ignition key into it as well.
Why this course brings it up: It comes up in the CAR-T section of Lecture 2. Dr. Wong lists CD3ζ, 4-1BB and CD28 as the intracellular activation domains of a CAR, but he never singles 4-1BB out as an exam question on its own, so all you need here is to understand the concept. **Appears in:** Ch 9, Lect 2

▪ 53BP1, RIF1 and Shieldin

In one sentence: The three factors that push the cell towards NHEJ and hold HR back.
Detailed explanation: When the cell is deciding which repair shop to use, it looks first at whether these three factors are there. In G1, S and G2 they protect the broken ends and stop the cell from resecting them, so the repair system takes the NHEJ road: fast, but prone to error.
Only in S and G2 does long-range resection win, and only then does the cell take the HR road, which needs a sister chromatid to use as a template. Resection is the decision point: once the end has been chewed back, there is nothing left to butt together, and the cell has to use the template route instead. That is why these three proteins show up on the decision tree rather than in the enzyme list.
Why this course brings it up: HR versus NHEJ is the most likely multiple-choice pairing he lists (Tier 2); these three factors are what sit on the decision tree on his slide 42. He does not ask you to memorise the names, so it is a concept to understand.
Appears in: Ch 6, Lect 4

▪ accuracy

In one sentence: The rate at which a diagnostic test gives the right answer, the overall performance of sensitivity and specificity weighed together.
Detailed explanation: On the classifier for breast cancer patients against controls, Dr. Wong's numbers are AUC 0.981, sensitivity 98.25 %, specificity 91.53 %, accuracy 94.83 %. The reading is this: out of 100 people without HCC, about 91 to 92 are correctly called negative;
out of 100 people with HCC, 98 are correctly called positive. Those four numbers together are a complete description of how useful the classifier is, and if a question asks “how many false alarms would there be among 100 healthy people”, you calculate it with the specificity.
Why this course brings it up: These four numbers are Tier 3 precise numbers on his list, and you have to know them to at least one decimal place. So this is one of the entries where the figures matter more than the prose. The reason he wants all four is that they answer different questions: AUC describes how the classifier behaves across all thresholds, sensitivity and specificity each describe one side, and accuracy is the summary the lay reader quotes. **Appears in:** Ch 11, Lect 1

▪ Acetylation

In one sentence: Adding an acetyl group onto a histone loosens the chromatin, so the gene underneath it can finally be read.
Detailed explanation: The tail of a histone carries lysine amino acids, and acetylation is the chemical move of pushing an acetyl group into the positively charged end of lysine. Once that positive charge has been neutralised, the histone's grip on negatively charged DNA weakens, the nucleosome wraps the DNA more loosely, and only then can transcription factors and enhancers squeeze in far enough to reach the promoter.
Put another way, acetylation is opening the bookshelf so that other people can reach the books, while HDAC (the enzyme that takes the acetyl groups back off again) does the opposite and shuts the shelves again. This matters to you for two reasons. First, it is a switch you can throw in either direction, and which way you throw it decides whether the gene is read at all. Second, it is also a drug target, because a drug that keeps HDAC away from the histone is doing the same job as acetylating it would do.
Why this course brings it up: Acetylation is the core mechanism behind the "histone code" segment of Lecture 3, and behind the epigenetic material of Lecture 5 as well. The story of HDAC inhibitor creating PD-L1 "gene addiction" also sits on his Lecture 3 slide. But acetylation itself was never named as an exam topic — he teaches it as something you should understand rather than something you have to memorise, so you can be hazy about the detail but you should not be hazy about the mechanism.
Appears in: Ch 4, Lect 3, Lect 5

▪ Actin

In one sentence: One of the two proteins of the contractile ring, the one that pinches the cell in half; also the track along which cargo is moved inside the cell.
Detailed explanation: Actin is one of the most abundant proteins in the cell, and it polymerises into two completely different shapes. During division it joins hands with myosin to build the contractile ring, a rope that tightens from the middle of the cell and cuts the cytoplasm into two.
The rest of the time it is the track of the cytoskeleton: vesicles and organelles get hauled along actin filaments by motor proteins. It is also the scaffolding that gives the whole cell its shape, which is why cancer cell migration and invasion nearly always mean re-laying the actin network. That is why actin is both the "division machine" and the "road network" at the same time.
Why this course brings it up: "Inhibiting the contractile ring produces binucleated cells" is a Tier 1 question he named (Lecture 2). But he has also said you do not need to memorise the cyclin-CDK pairings, and actin itself is never singled out either, so it is concept-only. **Appears in:** Ch 8, Lect 2

▪ Activation energy

In one sentence: An energy threshold that every chemical reaction has to climb over, and enzymes exist to push it down.
Detailed explanation: For a reaction to happen, the molecules have to get over an energy barrier first, even when the total energy released at the end is larger than what you put in at the beginning.
That extra push you need at the very start is the activation energy. An enzyme does not change where the reaction ends up, and it does not manufacture energy. What it does is offer an alternative route whose shape fits the reaction, and that route lowers the barrier until the cell can cross it at ordinary temperature. A good analogy is protein folding: getting a chain into the right shape is hard, but a chaperone lets it get there in small steps, which is the same as cutting the barrier into pieces that are easy to clear one at a time. The useful way to picture it is a hill in the middle of the journey. The molecules start on one side and finish on the other, and no matter how much energy they release on the way down, they still have to reach the top of the hill first.
Why this course brings it up: This is a basic concept from the protein structure section of Lecture 1, and Dr. Wong did not name it as an examinable topic. But he explicitly said not to memorise chemical structures and metabolic pathway details, so this is background you need in order to understand mechanisms. Conceptual understanding, in other words. **Appears in:** Ch 2

▪ Activator

In one sentence: A protein that binds the enhancer and helps pull the transcription machinery over to the promoter.
Detailed explanation: An activator is the helper that switches a gene on — think of it as the thing that physically hands the transcription machine its instructions. Different kinds of cells carry different sets of transcription factors, which is why they express completely different lists of genes;
the difference comes down to which activators each cell happens to carry. The mechanism runs like this. The activator binds the enhancer first, and then, using proteins that bend DNA, loops the enhancer across to a promoter thousands of bases away, recruiting RNA polymerase and the general transcription factors over to it. Notice that the enhancer does not have to sit next to its own gene; the looping is what makes a distance of thousands of bases workable at all. So the same DNA behaves differently in a nerve cell and in a liver cell, purely because of which activators happen to be sitting there, and that is really what "expression is cell-specific" means in practice. Finally, there are seven ways a transcription factor can be switched on; every one of those seven is somewhere a drug can take aim, which is why activators are worth learning as a group rather than one at a time.
Why this course brings it up: Lecture 5 covers cis versus trans, enhancer looping, and the concept he "always emphasize" that expression is cell-specific and time-specific at the same time. The activator itself was never singled out by name as an exam topic, so you are expected to understand how it works rather than recite a definition of it — but the larger concept it belongs to is a named one, and that is where the marks are. **Appears in:** Ch 7, Lect 5

▪ Active site

In one sentence: The pocket on an enzyme where the substrate slots exactly into place, and where the chemical reaction happens.
Detailed explanation: The efficiency of an enzyme comes down to one very specific shape. The active site is a three-dimensional pocket whose shape and charge complement a particular substrate, so the reacting molecules can be posed properly and then handed to the reaction.
That is also why the pocket is the favourite target of drugs. Most inhibitors do nothing more sophisticated than jam themselves into the active site and block the substrate from getting in. Enzymes have a second control mechanism, though, and it is called allostery. A drug binds at some other site, away from the active site, and yet it changes the shape of the main pocket. Enzymes that control the rate of metabolism often work this way, letting the end product come back and switch off the enzyme that made it.
Why this course brings it up: It is a basic concept from the protein-and-enzyme section of Lecture 3, and allostery comes from the same lecture. He does not name active site on its own as an exam question, so this belongs to understanding the concept.
Appears in: Ch 2

▪ adaptive CNV

In one sentence: A difference in copy number that is the result of adaptation, not of a defect.
Detailed explanation: The number of copies of the amylase gene AMY1 differs between humans and other mammals: populations that eat more starch carry more copies, make more amylase, digest it more efficiently, and the copy number rises over generations.
His figures are that the high-starch population has n = 133 with a median of 7, while the low-starch population has n = 93 with a median of 5. So a CNV is not itself a disease. What you have to look at is where it falls and whether it changes the gene dosage, and only then does it mean anything for health or disease. Which is the whole point: copy number variation is not a synonym for mutation with bad consequences, it is simply a difference in how many copies a genome carries.
Why this course brings it up: He does not list adaptive CNV as an exam question on its own, so it is concept-only. He says outright that the molecular mechanisms of CNV are not examined, but the example of CNV not meaning disease is worth remembering. It is the example that stops you treating every copy number change as a diagnosis. **Appears in:** Ch 3, Ch 10, Lect 3

▪ adenine

In one sentence: One of the four main bases of DNA and RNA, written A.
Detailed explanation: A is a purine base, with an extra ring on the molecule. In DNA it pairs with thymine and in RNA it pairs with uracil, and when it pairs it forms two hydrogen bonds, which makes it the looser of the four pairs.
The difference between DNA and RNA comes down to the sugar and to this base: deoxyribose goes with thymine and ribose goes with uracil. Adenine and guanine are both of the larger, double-ringed pair among pyrimidines and purines, so the molecules are big and easily form several hydrogen bonds with other molecules. Two hydrogen bonds is a weaker grip than three, and that single fact is why the melting temperature of a stretch of DNA depends on how much GC it contains.
Why this course brings it up: Dr. Wong does not name adenine as an exam question on its own, so it is conceptual understanding, but the pairing of A-T through two hydrogen bonds and G-C through three is one of his Tier 3 precise numbers.
Appears in: Ch 3, Lect 1

▪ Adenosine

In one sentence: What is left of ATP once you cut off the phosphate tail: one base plus one sugar.
Detailed explanation: Adenosine is a nucleoside, meaning a nitrogenous base (adenine) joined to a five-carbon sugar (ribose). The only difference between ATP, ADP and AMP is how many phosphate tails happen to be hanging off that same sugar.
The breakdown is useful to keep in your head: the bond the cell really spends energy on is the one between the terminal phosphate and the phosphate sitting next to it. Cut the terminal phosphate off and you are left with ADP; put the phosphate back and you are at ATP again. So ATP, ADP and AMP are one molecule with a different number of phosphate tails. The cell does not stockpile ATP like precious currency. It spends ATP constantly, transferring phosphate groups onto proteins, and that is exactly the job of a kinase. A kinase-driven phosphorylation is the fastest switch the cell has.
Why this course brings it up: In Lecture 1 he lists ATP as one of the two uses of a biomolecule, energy source and signalling molecule, and the phosphorylation switch is built right on top of it. He does not ask you to draw ATP out from memory, so this stays concept-only. **Appears in:** Ch 9

▪ adenovirus

In one sentence: A viral vector for getting genes into cells.
Detailed explanation: The coding sequence of CRISPR, Cas9 plus gRNA, has to be delivered into the cell before it can do anything. Adenovirus is used as the vector because it can infect many kinds of mammalian cell;
the other route is the lipid nanoparticle, which delivers mRNA wrapped inside a fatty vesicle. The advantage of adenovirus is that it is stable and efficient; the disadvantage is that it is strongly immunogenic and can set off an immune response. The vector is just the delivery vehicle — the working parts still have to fit inside it.
Why this course brings it up: He does not name adenovirus as an exam topic in its own right, so it is a concept to understand; the point of that passage is that "Cas9 is short enough to fit inside a vector". **Appears in:** Lect 2

▪ adhesion and activation

In one sentence: The adhesion and activation step, after an immune cell has rolled along the vessel wall.
Detailed explanation: After rolling, the integrin binds ICAM-1 and VCAM-1, and the immune cell sticks to the vessel wall. At the same time these molecules pass a signal in, and the cell is activated.
ICAM-1 and VCAM-1 can also be shed and dissolve into the blood, where they become measurable markers. The binding of integrin to ICAM-1 and VCAM-1 does both jobs at once: it sends the signal into the cell, and this is where the cell is genuinely switched on rather than merely parked against the wall.
Why this course brings it up: Dr. Wong does not name it as a topic on its own, so it is conceptual understanding, and the point that lands is this: the four-step journey of an immune cell to a tumour offers a therapy a place to intervene at every step.
Appears in: Ch 9, Lect 2

▪ Adoptive cell therapy

In one sentence: Take cells out of the patient, modify or treat them, and put them back in to go after the tumour.
Detailed explanation: The logic here is plain. If the immune system can already recognise the tumour but cannot get the better of it, then rebuild the fighter by hand and let it loose on the tumour.
CAR-T is the most mature version of that idea: you take the patient's own T cells out of the body, fit a CAR onto them outside the body, expand the population out to millions of copies, and then transfuse that population back into the same patient. If the cells you use come from the patient herself, the product is called autologous. If they come from a donor or from an iPSC, it is called allogeneic, the off-the-shelf version. Dr. Wong's third lecture adds an alternative route to the same problem: CAR-T cannot burrow into solid tumours, so there are CAR-NK and CAR-macrophages, where a different effector cell is swapped in so that you can buy better infiltration in exchange for something else.
Why this course brings it up: The whole CAR-T manufacturing workflow gets a full stretch of Lecture 2, and Lecture 3 adds CAR-NK and CAR-M on top of it. Autologous versus allogeneic is one of the multiple-choice distractor pairs he listed, so that contrast is examinable and getting it wrong costs you the question. CAR-T itself, however, was not named as a standalone examinable topic. **Appears in:** Ch 9,Lect 2,Lect 3

▪ ADP (adenosine diphosphate)

In one sentence: What is left of ATP after one phosphate has been spent; put the phosphate back and it is recharged.
Detailed explanation: ATP is the only universal energy currency that a cell has, and the way it "spends" itself is by cutting off the terminal phosphate group. The phosphate that gets cut off is usually handed over to some protein, and that handover is phosphorylation, which is the fastest switch a cell owns: add one small negatively charged group and the shape of the protein gets pushed open.
The part that is left behind inside the cell is ADP. Turning it back into ATP takes oxidative phosphorylation in the mitochondria, and substrate-level phosphorylation at the bottom of glycolysis. In other words, every time the cell spends an ATP, it produces a fresh piece of recharging feedstock, ADP, at exactly the same moment.
Why this course brings it up: Lectures 1 and 6 both place ADP in the context of ATP and of the phosphorylation switch. He does not ask you to memorise the ATP cycle diagram, so this belongs to understanding the concept rather than to the list of numbers you have to hold in your head. **Appears in:** Ch 1

▪ Aerobic glycolysis

In one sentence: Turning sugar into lactate even when oxygen is plentiful — the pathway cancer cells prefer.
Detailed explanation: A normal cell with oxygen burns sugar all the way down to CO2 and earns roughly 36 ATP per glucose. A cancer cell often does something else: with oxygen available, it converts the pyruvate produced by glycolysis into lactate — that is the Warburg effect.
The numbers on his slide are like this. In a proliferating cancer cell, 85 % of the pyruvate flows to lactate and only 5 % enters the mitochondria, so aerobic glycolysis earns about 4 ATP. That sounds like a terrible waste of fuel, and yet it is good business: the intermediate products kept upstream can be reused to support further proliferation, so the cell is buying building material rather than energy. The oxygen is not needed for the glycolysis itself — it is only that the cell chooses this route even when the oxygen is there.
Why this course brings it up: He brings it up when Lecture 1 covers cancer cell metabolism, and again in Lecture 2 on organelles; but he explicitly said the details of the metabolic pathways do not have to be memorised — you are meant to understand the concept, and the concept is that cancer rewires its own metabolism for its own reasons. **Appears in:** Ch 10, Lect 1

▪ AF-1 and AF-2

In one sentence: The two activation regions on a nuclear receptor that recruit co-activators.
Detailed explanation: A nuclear receptor has three functional regions. The NTD is the variable part and does the regulating. The DBD is the conserved part and recognises the response sequence in DNA.
The LBD is the part that binds the ligand. The real switch is two areas of it. AF-1 sits at the NTD end, AF-2 at the LBD end, and these are the two places where co-activators get recruited. When a ligand binds, the shape of the LBD changes, and that is like swapping AF-2 for a brand new docking surface, so the co-activator gets invited in. Without a ligand, the very same receptor's LBD becomes the docking place for a co-repressor instead, and the gene stays shut. This is also the molecular basis for why tamoxifen behaves in opposite ways in breast tissue and in the endometrium: the same receptor, the same drug, a different co-activator or co-repressor decision in each tissue.
Why this course brings it up: In Lecture 5 the structure of the nuclear receptor, and the DBD against LBD contrast, is one of the MCQ distractor pairs he lists. Of tamoxifen's tissue difference he said, "the concept that you have to bear in mind". AF-1 and AF-2 themselves are never named on their own, so you only have to know what they do and where they sit. **Appears in:** Ch 7, Lect 5

▪ AGO

In one sentence: The protein that carries a microRNA inside the silencing complex.
Detailed explanation: AGO (Argonaute) is the scaffold of RISC, the machine that turns a microRNA into a working silencer. Once a microRNA has been cut out of its longer precursor, it is loaded into AGO — and in the process it is actually AGO itself that does the cutting, keeping only one strand of the duplex as the guide strand and discarding the other.
That guide strand finds its target mRNA through the seed sequence at its 3′ end (bases 2 to 8). When the pairing is complete, AGO cuts the mRNA on the spot; when it is only partial, AGO does not cut at all and simply calls a halt to translation on the ribosome, leaving the message alive but unread. Because recognition rests on just 7 bases, a single miRNA can shut down hundreds or thousands of target mRNAs at once, which is why one short message can silence so much of the cell at the same time.
Why this course brings it up: It is the miRNA biosynthesis segment of Lecture 5; he did not list AGO as an exam topic by itself, but miRNA and the seed sequence are part of what he teaches, so this belongs to the concepts you are expected to understand rather than to the list of facts you must recite. **Appears in:** Ch 7, Lect 5

▪ Alkylation

In one sentence: A kind of damage that glues a small chemical blob onto a DNA base, after which the base pairs with the wrong partner.
Detailed explanation: Alkylation is one of the most typical chemical forms of DNA damage that a cell has to deal with. A reactive electrophilic group is attached to a base, and what used to be a clean base is chemically modified in place, so the letter in the sequence is no longer the structure it was born with.

The trouble starts once the base has been modified, because the rules of pairing change along with it. O6-methylguanine, for example, will go on to pair with thymine instead of its usual cytosine, and so the mistake gets locked in as a fixed mutation the next time the DNA is copied. Dr. Wong's fourth lecture also points out that MGMT does the most elementary version of the repair job of all: it removes the alkyl group straight off the base, without cutting the base out, without resynthesising anything, and without sealing anything up afterwards.
Why this course brings it up: The DNA carcinogenesis chemistry section of Lecture 3. MGMT methylation as a predictive biomarker is a second-level concept on his list, but alkylation itself is conceptual understanding, and the mechanism is what you need rather than a list of chemicals. **Appears in:** Ch 6,Lect 3,Lect 4

▪ Allele

In one sentence: A different version of the same gene; you carry two of them, one from each parent.
Detailed explanation: The same gene, such as the one that controls blood group in the ABO system, can be a different version in different people, and those different versions are alleles of each other.
In a diploid cell there are two copies at the same position, one from the father and one from the mother, so at nearly every locus each of us is a combination of two alleles. The thing to keep in mind is that carrying a certain allele does not mean you will definitely fall ill. That is incomplete penetrance, and it is why he says the same BRCA mutation does not guarantee that you get sick.
Why this course brings it up: It comes from the pedigree and genetic counselling part of Lecture 3. Reading a pedigree is one of the two scenario questions he names, and the word allele itself was never named on its own. **Appears in:** Ch 3, Lect 3

▪ Allogeneic

In one sentence: Cells taken from a donor rather than from the patient themselves, so they are an off-the-shelf product.
Detailed explanation: An autologous CAR-T has to wait for the patient's own cells to be taken out, engineered, expanded and put back, and the price is time, money, and T cells that may not be in good enough shape to begin with.
Allogeneic sidesteps all of that. The source can be cord blood, a healthy donor, even iPSC: you build the standard product in advance, park it in the freezer, and ship it out whenever a patient needs it. The price on the other side is allogeneic immune reaction and rejection, and you have to deal with that with lymphodepletion therapy first.
Why this course brings it up: It comes from the CAR-T section of Lecture 2. Autologous and allogeneic are one of the multiple-choice distractors he lists, and picking the wrong one costs you the whole question. **Appears in:** Ch 9, Lect 2

▪ Allolactose

In one sentence: Allolactose is an isomer of lactose, and it is the molecule that actually knocks the Lac repressor off the operator and lets the lac operon run.
Detailed explanation: The common misunderstanding is that the presence of lactose is enough to switch the lac operon on. The reality is a little more interesting than that, because the system is one step more deliberate.
The β-galactosidase encoded by lacZ does two things: it breaks down lactose, and it also converts part of the lactose into a second isomer, allolactose. The molecule that actually binds to the Lac repressor, and makes the repressor let go of the operator, is allolactose and not lactose itself. So why would the cell be built that way? Because only allolactose genuinely carries the message that the bacterium has food. Lactose on its own is ambiguous; allolactose is not. In the laboratory, the reagent you normally use is IPTG: it looks similar enough to allolactose to switch the repressor off, but the cell has no enzyme that can metabolise it, so it stays put and keeps holding the switch down for as long as you want.
Why this course brings it up: The line about the lac operon needing two conditions rather than one is a second-level concept on Dr. Wong's list, and it is also one of the scoring points for case 16 in his case bank. Allolactose and IPTG are two of the four items he says you must be able to name. **Appears in:** Ch 7,Lect 5

▪ Allostery

In one sentence: A different molecule binds somewhere other than the main working site, yet changes the shape and the activity of that main site.
Detailed explanation: An enzyme is just a protein, so any mechanism that changes shape applies to it. Allostery is exactly this kind of indirect control: a molecule binds at a position other than the active site, causes a conformational change, and activity rises or falls as a result.
The most famous physiological example is feedback inhibition: the end product of a metabolic pathway binds the enzyme that catalyses the first step and switches it off, so the cell does not go on making something it already has plenty of.
Pharmacologically there is a second opening here: allosteric sites are usually flatter and easier for small molecules to bind than the active site itself, which makes them attractive drug targets.
Why this course brings it up: It is a basic concept in the protein and enzyme section of Lecture 3; he explicitly said not to memorise the chemistry of enzymes, so this one is a concept to understand rather than a fact to learn, and it will show up as reasoning rather than as a definition. **Appears in:** Ch 2, Lect 1

▪ Alpha helix (alpha-helix)

In one sentence: The drill-shaped helix a protein chain usually takes, held up by hydrogen bonds between neighbouring turns.

Detailed explanation: A polypeptide chain can travel through space in many ways, and the two most common are the α-helix and the β-sheet. The α-helix is a right-handed spiral, about 3.6 amino acids per turn, and it is held up by a hydrogen bond between the carbonyl oxygen of amino acid i and the amide hydrogen of amino acid i+4.

In other words, what stabilises it is the backbone itself, not the side chains, and each turn of the ladder is tied to the one above it. The side chains all point outward, like blades on a screw, so the chemical properties of a protein can be collected on its outer surface to build interfaces with other things.

Why this course brings it up: It is the second level of the four-level protein structure from Lecture 1, primary, then secondary, then tertiary, then quaternary. He does not ask you to memorise the table of amino acid R-groups, but the four levels of structure you do have to know, and you have to be able to say that the helix is held by backbone hydrogen bonds. **Appears in:** Ch 2

▪ alternative end joining (a-EJ)

In one sentence: The secondary pathway that can rejoin broken ends without needing much homologous sequence.

Detailed explanation: NHEJ is too casual, and HR needs a sister chromatid, so the cell has a third option. If the two ends have a few base pairs of homologous sequence they can be joined, and in the process a small amount of sequence is deleted or left in.

The opposite case is single-strand annealing, which needs roughly 20 base pairs or more of homology and wipes out everything in between. What these pathways leave behind is usually a short deletion or an insertion, in other words an indel, and it is exactly those indels that generate diversity in the immunoglobulin genes. So the minor repair pathways are not just a fallback, they are a deliberate source of variation in the one place where the cell needs variation on purpose.

Why this course brings it up: He does not name a-EJ as an exam question on its own, so it is concept-only. The Tier 2 pairing is the comparison of HR needing a template against NHEJ not needing one. That is the pair to hold, and the other two names are context for it. **Appears in:** Ch 6, Lect 4

▪ Alternative polyadenylation

In one sentence: Choosing to put the poly-A tail on at one position or at another, and in that way changing how long the 3' UTR ends up being.

Detailed explanation: The 3' end of a eukaryotic mRNA can pick between different polyadenylation signal sites, and the cell is free to choose which one it uses, transcript by transcript. If the cell picks the nearer one, the proximal PAS, the 3' UTR comes out short;

if it picks the further one, the distal PAS, the 3' UTR comes out long. The difference between the two is not only about the length of the tail. A longer 3' UTR carries extra binding sites for RBP, for miRNA and for lncRNA, and with those extra sites come changes in the stability of the mRNA, in its translation efficiency and in where it ends up inside the cell. His own example is BDNF, brain-derived neurotrophic factor: the short isoform stays around the cell body, while the long isoform gets carried out into the dendrite of the neuron.

Why this course brings it up: It comes from the part of Lecture 5 that deals with the structure of mRNA. He does not name this entry on its own, but when you need to understand why alternative splicing can expand the diversity of proteins, this is the concept that gets paired with it most often, and it is worth learning alongside the splicing itself. **Appears in:** Ch 7, Lect 5

▪ Alternative splicing

In one sentence: One gene uses different combinations of exons to build several different proteins.

Detailed explanation: After a eukaryotic gene is transcribed, the introns in the pre-mRNA are cut out by the spliceosome and the exons are joined together. But splicing is not limited to a single combination — one gene can produce several protein isoforms, so the same DNA behaves completely differently in a nerve cell and in a liver cell.

His Lecture 5 slides list a run of clinical examples. MCL1, an apoptosis regulator, is cut into a pro-apoptotic version and an anti-apoptotic version, and which version you get decides whether the cell dies or survives. And when a cryptic exon of AR gets spliced in, it produces the constitutively active AR-V7, which makes prostate cancer stop responding to enzalutamide. The drug still works on most of the tumour; the isoform is simply a way for part of it to escape. The drug still works on most of the tumour; the isoform is simply a way for part of it to escape, which is what makes it worth a lecture of its own.

Why this course brings it up: Splicing is a key mechanism of gene expression in Lecture 5, and AR-V7 is one of the case scenarios he listed (an isoform that predicts drug resistance); alternative splicing itself, however, was never named as an exam topic, so you should be able to explain the idea rather than recall a definition of it. **Appears in:** Ch 7, Lect 5

▪ Amino acid

In one sentence: The twenty small molecules that make up proteins, each carrying a side chain that decides how the chain folds.

Detailed explanation: A protein is really just a string of beads made of amino acids, and each bead's personality lives entirely in its side chain. A small, flexible side chain such as glycine gives that spot a bit more room to move.

A large, charged side chain such as tryptophan or arginine makes that spot stiff. To fold into the right three-dimensional shape, a chain relies on the attractions and the repulsions between its side chains working together. The phosphorylation switch he teaches is a perfect demonstration of that principle: add one charged phosphate group and the shape of the protein gets pushed

open.
Why this course brings it up: It comes up in the protein section of Lecture 1, but he explicitly said, "do not memorise the R-group table of amino acids, that will definitely not be examined", so all you need is the principle. **Appears in:** Ch 2

▪ Amp

In one sentence: The ampicillin resistance marker on a plasmid, used to select the bacteria that took the plasmid up.
Detailed explanation: In a molecular cloning experiment the donor plasmid carries the Amp resistance gene together with an insert. Once it has been put into the bacteria, only the cells that took the plasmid up and are expressing the resistance can grow on a plate containing ampicillin, and the ones that did not are killed by the antibiotic.
This step is called selection, and it lets you pick the few genuine recombinants out of several million colonies. So the ampicillin plate is doing the selection for you, and everything that grows on it has already passed the first test.
Why this course brings it up: Dr. Wong does not name it specially, so it is conceptual understanding. In the same workflow he talks about the cutting with EcoRI and NotI and the sealing with DNA ligase. **Appears in:** Ch 1, Ch 3, Lect 4

▪ AMP (adenosine monophosphate)

In one sentence: The level left over after ATP has spent two of its phosphate groups.
Detailed explanation: ATP → ADP → AMP is a chain in which the number of phosphates steps down one at a time. The place AMP shows up most often in energy metabolism is the cell's energy sensor.
A high AMP level usually means the cell is short of energy, which means ATP is low, and that is the signal that switches on the kinase AMPK. AMPK then shuts down the processes that consume ATP and opens up the processes that produce it, so the cell starts to pay its energy bills less and earn more. The Warburg numbers from Lecture 1 make this more interesting: cancer cells running aerobic glycolysis earn only about 4 ATP, while normal cells earn about 36, so the AMP/ATP ratio in a cancer cell is naturally pushed up, and it keeps the sensor pressed.
Why this course brings it up: Lecture 1 puts ADP, AMP and ATP on the same chain, so you should know where AMP sits in it and what it is telling the cell. Dr. Wong did not name AMP on its own, so this is conceptual understanding. **Appears in:** Ch 1,Ch 3,Lect 4

▪ Anabolism

In one sentence: The building half of metabolism: spend energy to assemble large molecules.
Detailed explanation: Metabolism has two halves. Catabolism is the demolition half, breaking big molecules apart and releasing energy. Anabolism is the construction half, assembling small molecules into proteins, lipids and nucleic acids, and it costs ATP.
Cancer cells depend on anabolism so badly precisely because they are dividing at high speed, and every daughter cell needs a fresh set of parts. The upstream intermediates also have to be diverted to make nucleotides, lipids and amino acids. Clinically, the surge in glucose uptake by a tumour, the thing a PET scan sees, is a direct readout of how strong the anabolic demand is.
Why this course brings it up: He teaches both directions together when he covers cancer cell metabolism in Lectures 1 and 2. But he says outright that the details of the metabolic pathways do not need to be memorised, so this is concept-only. **Appears in:** Ch 1

▪ Anaphase

In one sentence: The stage of mitosis in which the chromosomes split up from each other and travel in opposite directions towards the two poles.
Detailed explanation: The earlier part of the division cycle is prophase, where the spindle assembles and the chromosomes condense, and metaphase, where the chromosomes line up on the equatorial plate.
Then comes anaphase: the centromeres of the sister chromatids separate, and each side is pulled by the spindle fibres towards opposite poles, which stretches the cell as it goes. After anaphase comes telophase, and the nuclear envelope reforms around each new set of chromosomes. This stage matters because chromosome segregation and cytokinesis are two completely independent things. The chromosomes can separate perfectly well and the cell still fails to cut itself in half, and that is the binucleated cell question he names.
Why this course brings it up: It turns up in Lectures 3 and 5 alongside the mitosis comparison table he uses. What he actually names is the distinction between mitosis and cytokinesis, rather than anaphase itself, so it belongs to understanding the concept. **Appears in:** Ch 8, Lect 2, Lect 3

▪ AND-gate

In one sentence: A switch that only fires when two inputs are present at the same time; synNotch CAR uses it so it does not kill by mistake.
Detailed explanation: In logic gates, an AND-gate outputs "true" only when both of its inputs are "true". Applied at the level of cells, it handles the question "is this cell actually a tumour?".
Recognising a tumour antigen on its own is not enough, because normal tissue may occasionally express the same antigen, and the CAR-T would then kill it by mistake. The synNotch design strings two signals together: synNotch first binds tumour antigen A, and only the intracellular fragment released by that cleavage goes on to drive CAR expression; the CAR itself must additionally bind antigen B before it fires. Both conditions, not one. It is the same logic as a burglar alarm that needs two keys turned at once.

Why this course brings it up: In Lecture 5 he mentions this engineered design, and scenario case number 14 in his case bank is also "a compound that switches itself off"; but he did not name the AND-gate on its own as an exam topic, so it is something to understand rather than to memorise. **Appears in:** Ch 7, Lect 5

▪ Anergy

In one sentence: The state a T cell ends up in when a checkpoint signal switches it off: there is a signal, but no work gets done.

Detailed explanation: A T cell needs two signals to be fully activated. The first is the TCR recognising antigen; the second is co-stimulation, for example CD28 receiving B7. If only the first signal arrives and the second never does, the T cell does not attack.

It goes into anergy instead, and this is one of the immune system's protection mechanisms against autoimmunity. Tumours exploit it: PD-L1 on the surface of a tumour cell binds PD-1 on the T cell, and that fakes up the situation in which there is antigen but no co-stimulation, so the T cell is told to stand down. The most striking thing about anergy is that it is reversible, which is why the clinical logic of anti-PD-1 antibodies is so simple: release the brake and the T cells start working again.

Why this course brings it up: The tumour microenvironment and checkpoint section of Lecture 2, and the clinical use of anti-PD-1 is the core of the hot tumour story in Lecture 4. Dr. Wong did not list the word anergy as an exam topic on its own. Conceptual understanding, but you should be able to explain it. **Appears in:** Ch 9,Lect 2

▪ Aneuploidy

In one sentence: A cell whose chromosome number is wrong, usually because segregation failed during meiosis.

Detailed explanation: A normal human somatic cell has 23 pairs of chromosomes, 2n = 46. Aneuploidy means the number is not 2n: one extra, as in trisomy 21, Down syndrome, or one missing, as in Turner syndrome, 45,X.

The cause is failed segregation in meiosis, and he insists you tell M1 apart from M2. In an M1 failure the homologous chromosomes fail to separate, so one meiosis produces 2 normal gametes plus 2 abnormal gametes of the same type. In an M2 failure the sister chromatids fail to separate, producing 1 normal gamete plus 3 abnormal ones, and the M2 column also involves the absence of recombination. So the difference is not academic: M1 gives you two of one kind, M2 gives you two of one kind and one of another, and a pedigree you have to read back to the patient has to be built on the right one.

Why this course brings it up: Telling M1 apart from M2 is a Tier 1 question he named (Lecture 3), and a pedigree combined with aneuploidy is case number 8 in the scenario bank. He expects you to produce the two different gamete counts on demand, not just recognise the words. **Appears in:** Ch 3, Lect 3

▪ Angstrom

In one sentence: The unit for measuring atomic distances; a crystal structure at 2.8 Å counts as good, and anything below 2.0 Å counts as excellent.

Detailed explanation: One Ångström, written simply Å, is 10^-10 metres, which is about the scale of a single atom. What he cites in Lecture 3 for the nucleosome structure is exactly the 1997 paper by Luger and colleagues, "Crystal structure of nucleosome core particle at 2.8 Å".

The lower the resolution number, the more clearly you can see where every single atom sits. For an assembly as complicated as the nucleosome, 2.8 Å is already quite good; anything below 2.0 Å is usually described as excellent. The same yardstick applies to drug design, because a small molecule has to fit into the pocket of a receptor, and for that you need to be able to see every single turn of the side chain.

Why this course brings it up: Both numbers, 2.8 Å and 2.0 Å, sit on the list of precise figures he gives you, and they are standard raw material for a multiple-choice question. X-ray resolution itself was never named as a standalone exam topic, so it stays in the background column rather than in the exam column, even though he clearly expects you to know what the numbers mean. **Appears in:** Ch 2, Ch 4, Lect 3

▪ Annealing

In one sentence: The process by which two complementary single strands of DNA pair up again as the temperature comes down.

Detailed explanation: This is the second step of every PCR cycle: once the temperature comes down, the primers find the place on the template that matches their own sequence and stick there. The temperature of this step decides whether it works — too high and the primers will not stick, too low and they stick to each other or to the wrong place.

The rule he states explicitly is that the higher the GC content, the higher the temperature needed, which is the same thing as saying the higher the Tm. The primer is looking for its partner by shape as much as by sequence, and the temperature is what decides how fussy it is.

Why this course brings it up: The relationship between GC content and Tm is one of the Tier 3 numbers he lists, and it runs straight into PCR primer design; he also recaps the "three steps of PCR" in every lecture. But he does not name Annealing as an exam topic on its own. **Appears in:** Ch 3, Lect 4

▪ Anti-inflammatory mediator

In one sentence: The class of molecules that turns inflammation off once the danger has passed.
Detailed explanation: Inflammation is not only about starting; it has to be able to stop, otherwise the body stays on alert forever and the response outlasts the threat that provoked it. Dr. Wong's NF-κB story is a good example with a timing to it.
After LPS stimulation, IκB is degraded, NF-κB enters the nucleus and transcribes TNF-α; TNF-α peaks at 4–6 hours, and then IL-1β, IFN-γ, IL-2 and IL-15 follow one after another. The cytokines reinforce each other and form a positive-feedback loop: each new cytokine brings the next one in, and each one makes NF-κB work harder still. That is what makes the defence response swell up so quickly rather than crawling into place, and it is also why the response needs a planned ending rather than simply running out of steam. Once it has swollen far enough, the anti-inflammatory mediators arrive and push the loop back down again, and the whole thing settles.
Why this course brings it up: The closing of the NF-κB section in Lecture 5. But he explicitly said not to worry about the order of the cytokines, because that was only there to let you see that different responses can be observed at different times. Conceptual understanding. **Appears in:** Lect 5

▪ Anti-PD-1 antibody

In one sentence: A drug antibody that blocks the PD-1 checkpoint and puts the T cell back on duty.
Detailed explanation: Tumour cells display PD-L1 on their surface, and it binds PD-1 on the T cell, which is like stamping on the T cell's brakes, so the T cell enters anergy and the tumour escapes the immune attack.
An anti-PD-1 antibody does one thing: it occupies the PD-1 site, so PD-L1 has nothing to bind and the brakes come off. It only works on a "hot tumour", one with a high neoantigen load and MMR deficiency, MSI-H, because there the T cells are there but held down. A cold tumour simply does not have enough T cells to liberate, so there is no point using it. The KM curve he quotes in Lecture 4, dMMR versus pMMR treated with anti-PD-1, is what he calls one of the most important figures in modern medicine: the same drug, the same tumour type, and the whole separation comes from the mismatch status.
Why this course brings it up: It is the core clinical application in Lectures 2 and 4, and it is a scoring point in both the MCQ distractors and the hot versus cold tumour scenario case. The term "anti-PD-1 antibody" itself is never named on its own, but the logic behind it is examinable, because he built the whole immune checkpoint story around it. **Appears in:** Ch 4, Ch 6, Ch 9, Lect 2

▪ Anti-PD-L1 antibody

In one sentence: It blocks PD-L1 on the tumour cell, and the effect is the same as anti-PD-1.
Detailed explanation: This route and anti-PD-1 are two doors into the same thing. PD-1 sits on T cells, PD-L1 sits on tumour cells and on other cells; block either side and the signal cannot get through, so the T cell regains its killing power.
The difference is in where the drug ends up: an anti-PD-L1 antibody acts directly on tumour cells and on immune cells, so its concentration in tumour tissue is high — hence the example he named, atezolizumab for locally advanced or metastatic breast cancer. Its clinical position is comparable to anti-PD-1, and the multiple-choice question usually tests that both work by the same mechanism and merely block different partners in the handshake. Block either half of the handshake and the brake is off.
Why this course brings it up: It appears in Lectures 2 and 4; he did not list anti-PD-L1 on its own as a named exam topic, so it belongs to the concepts you are expected to understand, and the distinction to hold on to is the mechanism being the same rather than which drug name goes with which. **Appears in:** Ch 9, Lect 2

▪ Anti-TERT

In one sentence: An antisense oligonucleotide that degrades TERT mRNA, so the cell can no longer keep rebuilding the ends of its chromosomes.
Detailed explanation: TERT encodes the catalytic subunit of telomerase. Telomerase is a reverse transcriptase, an enzyme that reads RNA in order to build DNA, and in this case it carries its own internal RNA template so that it never has to go looking for one.
Its job is to make the ends of chromosomes longer. With it, a cell can break through the Hayflick limit and keep dividing. The anti-TERT antisense oligonucleotide (ASO) does not attack the enzyme directly; it degrades the messenger RNA that describes it, and telomerase then simply cannot be made. Dr. Wong's fourth lecture has the complete preclinical data: subcutaneous xenografts in immunodeficient nude mice, where the tumour take rate in the untreated group was 9/12, and after 3 weeks of anti-TERT pretreatment only 2/12 remained (Fisher $p = 0.01$). Read that comparison carefully, because the endpoint being measured is the tumour take rate, so that is the number to hold on to: nearly half the untreated animals grew a tumour, while the treated animals barely did at all.
Why this course brings it up: The TERT therapeutic target section of Lecture 4. The 2/12 against 9/12 and the $p = 0.01$ are on his list of precise numbers and are standard multiple-choice material. **Appears in:** Ch 5,Ch 7,Ch 12,Lect 4

▪ Antibody

In one sentence: A Y-shaped protein made by B cells that recognises one antigen and then either blocks it or tags it for removal.
Detailed explanation: The shape of an antibody is a Y. The two arms are the antigen-binding region; the stem is the Fc portion — the Fc portion hooks onto receptors on immune cells, while the arms do the recognising.
His Lecture 1 slide gives the molecular weights: Fv about 25 kDa (VH plus VL), Fab about 50 kDa, Fc about 50 kDa. The arms can recognise so many different antigens because three hypervariable loops, CDR1, CDR2 and CDR3, each contribute a

different contact surface; on top of that, V(D)J recombination and somatic hypermutation let the number of possible antibodies reach billions. A useful mental picture is a pair of tweezers on a stalk: the tweezers grip the target, and the stalk delivers the payload. Finally he reminds you of something easy to forget: an antibody only sticks to its antigen, and the actual killing is usually done by the effector cells that the Fc portion calls in. It is the molecule the whole immune half of the course is built on.
Why this course brings it up: Antibody diversity and ADC come from Lecture 1, ADCC and IgG1 / IgG4 from Lecture 2, and the V(D)J to SCID causal chain from Lecture 4; all of it leans on the antibody. But he never listed antibody structure itself as a named exam topic, so you need the shape and the parts straight rather than the numbers memorised word for word. **Appears in:** Ch 2, Ch 9, Lect 1, Lect 2

▪ Antibody-dependent cellular cytotoxicity (ADCC)

In one sentence: The Fc portion of an antibody grabs an NK cell and orders it to kill whatever the antibody is stuck to.
Detailed explanation: Antibodies have two ways of killing cells, and this is the more important one. The antibody first uses its two arms to stick to an antigen on the surface of a tumour cell, and then the Fc portion is recognised by FcγRIII on an NK cell. That is equivalent to dragging the killer to the scene, and the NK cell then releases perforin and granzyme to destroy the target. The key point is that sticking and killing are two separate events with an effector cell sitting in between, and that is exactly how ADCC differs from CDC, where the complement does the job by itself with no cellular help needed at all. Notice that the antibody itself never kills anything here. It is doing two jobs at once: it picks the target, and it summons the executioner. The direct clinical application is trastuzumab: it is an IgG1, so besides blocking HER2 dimerisation it also borrows ADCC on the way past.
Why this course brings it up: The antibody isotype section of Lecture 2 and the ADC passage of Lecture 4. ADCC versus CDC is one of the multiple-choice distractor pairs he listed, and getting it wrong costs you the whole question. **Appears in:** Ch 2,Ch 9,Lect 1,Lect 2

▪ Antibody-drug conjugate (ADC)

In one sentence: An antibody used as a delivery van, carrying a toxic drug only to the place where it recognises its target.
Detailed explanation: An ADC has three parts: the antibody that recognises the antigen, the linker that joins the two, and the payload that does the killing. The thinking is very practical. Chemotherapy on its own kills normal cells too, so you dress it up as "antibody plus toxin" and let the toxin be released only where the antibody finds its antigen.
The antibody is the delivery van and the payload is the cargo, and the linker is the hook that holds them together until they arrive. The example he gives is trastuzumab-deruxtecan, an HER2 ADC, whose payload is the topoisomerase inhibitor SN-38 together with DXd. In Lecture 4 he also mentions a dual-payload version: a branched linker carrying a topoisomerase inhibitor and a DDR inhibitor, such as an ATM blocker, to deal with the compensatory resistance a single drug produces.
Why this course brings it up: He explicitly puts the structure of an ADC and the dual-payload idea on the "will not be examined" list (Lecture 4). What you need to understand is the concept of why you would tie a toxin to an antibody in the first place, and the answer is selective delivery, so that the toxicity lands on the tumour instead of on the patient. He also asks you to remember that the ADC story keeps coming back, in Chapter 10 and in both Lecture 1 and Lecture 4. **Appears in:** Ch 10, Lect 1, Lect 4

▪ Antigen

In one sentence: Something that the immune system identifies as foreign and prepares to deal with.
Detailed explanation: An antigen is not simply one molecule; it is a shape that the immune system "recognises". It can be a protein, a polysaccharide or a lipid, and it can even be a whole bacterium or a whole virus.
The key point is that antibodies and TCRs recognise shape and not name, so molecules with similar structures cross-react with each other, and that is the root of both vaccine design and allergy. Tumour immunology has a special version of this, called a neoantigen: a new peptide fragment that normal cells do not have, and that can be used as a handle for treatment. The common source of neoantigens is exactly the frameshift mutation caused by MMR deficiency.
Why this course brings it up: It is the backbone of the immunology material in Lectures 1 and 2, and of the MSI to hot tumour mechanism in Lecture 4. But he never listed the word antigen on its own as an exam question. **Appears in:** Ch 2, Ch 9, Lect 1, Lect 2

▪ Antigen-presenting cell

In one sentence: A cell that lays out fragments of protein on its surface so T cells can inspect them.
Detailed explanation: The dendritic cell is the professional antigen-presenting cell. The sequence runs like this: it swallows a pathogen or a piece of tumour, cuts it into small pieces inside, loads them into MHC molecules, and carries them up to the cell surface;
when a T cell arrives, it reads this "menu" with its own TCR. The T cell only truly starts up if it reads both the antigen on the menu and a co-stimulatory signal (B7 against CD28) on the dendritic cell at the same time, so the antigen alone is not enough. That explains a clinical fact: cancer cells often escape immune recognition by refusing to express MHC I or by failing to put up B7, and immunotherapy is exactly the attempt to put those missing signals back.
Why this course brings it up: Dendritic cells and T cell activation come from Lecture 2, and antigen presentation plus T cell recognition of neoantigen from Lecture 4; this chain of presentation is itself upstream of the SCID causal chain in Lecture 4, so it belongs to the concepts you are expected to understand rather than to the facts you have to recall. **Appears in:** Ch 6, Lect 2, Lect 4

▪ Antiparallel

In one sentence: The two strands of DNA run in opposite directions: one from 5′ to 3′, and the other from 3′ to 5′.
Detailed explanation: The DNA double helix is like a zip, but the two sides of that zip are pulled in opposite directions. One strand runs from its 5′ phosphate end towards its 3′ hydroxyl end, and the other runs the other way round.
This is not decoration in the design; it is a chemical necessity. DNA polymerase can only add new nucleotides to a 3′ end, so replication can only travel along the template 5′ → 3′. Because the two template strands face opposite ways, two asymmetric replication products have to follow: one continuous leading strand, and one lagging strand stitched together out of Okazaki fragments. That same one-way rule then fixes the direction of proofreading, because the exonuclease activity that cuts a mismatch out of the chain runs 3′ → 5′. Every direction you have to remember in this part of the course comes back to this one fact: the polymerase can only build forwards, and the proofreading enzyme has to run backwards.
Why this course brings it up: In Lecture 4, elongation 5′→3′ and proofreading 3′→5′ is one of the fixed details he marks with the phrase "you need to understand", and it is one of the distractor pairs he lists for multiple choice. Antiparallel itself was never named on its own as an exam question, so it is a concept you pick up on the way rather than one you memorise. **Appears in:** Ch 3

▪ AP site

In one sentence: The gap left behind when a base is cut out and only the sugar backbone remains.
Detailed explanation: When DNA glycosylase cuts out a damaged base entirely, the sugar and phosphate backbone is still there but there is no letter at that position, and that is an AP site (abasic site).
The next step is AP endonuclease cutting at the gap, polymerase putting the right base back in, and ligase sealing it. Because only the sugar skeleton is left at the gap, replication is very prone to a second mismatch there, so not one of the steps in BER can be skipped. It is a small hole with no letter in it, and the machinery has to get there in time.
Why this course brings it up: Dr. Wong says explicitly that this part is not examined, in effect that the chemistry of DNA glycosylase and the AP site will not appear in the midterm; but the concept that BER repairs a single damaged base is listed separately as a multiple-choice pairing, and that still has to be mastered. **Appears in:** Ch 6, Lect 4

▪ APAF-1

In one sentence: The adaptor protein that assembles cytochrome c and procaspase-9 into the apoptosome.
Detailed explanation: Once the mitochondrion has released cytochrome c, these proteins meet in the cytoplasm. APAF-1 is the one that does the recruiting: it binds cytochrome c and changes shape, then pulls in procaspase-9, the precursor form of caspase-9, and the three of them assemble into a wheel-shaped apoptosome, which activates caspase-9 and drives the whole intrinsic apoptosis pathway.
Without APAF-1 the released cytochrome c simply drifts around the cytoplasm, the apoptosome never forms, and caspase-9 is never switched on. So the release of cytochrome c is not enough on its own, something has to gather the pieces, and that something is APAF-1. It is the scaffolding piece, and without scaffolding nothing is brought together in the right place. The apoptosome is also the point at which the mitochondrial signal becomes an enzymatic one, because that is where caspase-9 gets switched on.
Why this course brings it up: Dr. Wong uses these three steps in Lecture 2 to explain intrinsic apoptosis, and he does not name APAF-1 as an exam question on its own, so it is conceptual understanding, but the three-step version is the picture you should have in your head. **Appears in:** Ch 8, Lect 2

▪ APC

In one sentence: The first brake to fail on the multi-step road to colorectal cancer.
Detailed explanation: APC is an inhibitory protein in the Wnt signalling pathway; normally it holds down beta-catenin (a Wnt signalling protein) so that the cell does not keep dividing. His colorectal cancer model is a timeline.
APC fails first, and beta-catenin signalling runs out of control; next KRAS turns into a continuous growth signal; and last of all p53, the tumour suppressor, fails too, and a polyp grows all the way into an adenocarcinoma. Along the same line, the proportion of MSI rises as the sequence goes on. Each step is a separate brake going, and the tumour only arrives at cancer when enough of them have gone. Order matters as much as the list itself.
Why this course brings it up: This multi-step carcinogenesis model is on his slides, but he does not list APC as an exam topic on its own, so it is a concept to understand; the point he marks as Tier 2 is the difference between a driver and a passenger.
Appears in: Lect 4

▪ Apo-repressor

In one sentence: A repressor with no ligand bound to it, and therefore one that cannot hold on to the operator.
Detailed explanation: A repressor has two forms, and the contrast between them is clearest in the trp operon. When there is tryptophan around, tryptophan acts as a corepressor and binds to the Trp repressor;
the shape of the resulting complex changes, it can sit down on the operator, and transcription is blocked. That is the active repressor, and it is the loaded form. When there is no tryptophan, the repressor has no ligand, its shape is wrong, it cannot hold the operator, and it has to let go of the DNA so that RNA polymerase is free to transcribe. This free form is what gets called the apo-repressor, with apo meaning that the carrier is empty. The practical upshot is that the cell does not have to build two different repressors and keep them separate. It builds one protein, and the ligand decides which shape that protein is in, which is the same trick the lac repressor plays with allolactose.

Why this course brings it up: It is the vocabulary on the trp operon slide in Lecture 5, and it is part of the lac-versus-trp mirror-image contrast that runs through that lecture. Dr. Wong did not name apo-repressor as an exam topic on its own, so this is conceptual understanding. **Appears in:** Ch 7,Lect 5

▪ APOLO

In one sentence: A long non-coding RNA that forms an RNA–DNA hybrid with DNA.
Detailed explanation: An lncRNA is an RNA longer than 200 nucleotides that does not code for protein. APOLO is one of the archetypes: it pairs with DNA using its own sequence, forming an RNA–DNA hybrid, and it then delivers a chromatin remodelling mechanism to the target locus and switches the gene off.
The role of this class of lncRNA is like that of a guide molecule, or a delivery address. It does no work itself, but it takes the machine to the right address and lets the machine work there. In his Lecture 5 slides he sorts the functions of lncRNA into four archetypes: APOLO; PNCTR, which binds PTBP1, a splicing factor, and affects splicing; XIST, which binds SPEN, an RNA-binding protein, and PRC2, a chromatin silencing complex, to carry out epigenetic silencing; and TINCR, which pairs with mRNA and affects its stability.
Why this course brings it up: It is one of the archetypes in the lncRNA part of Lecture 5. He never names APOLO on its own as an exam question, so it belongs to understanding the concept rather than to the short list of labels you have to memorise before the exam, and it is still worth knowing what it does. **Appears in:** Ch 7, Lect 5

▪ Apoptosis

In one sentence: Programmed cell death: the cell takes itself apart neatly instead of bursting.
Detailed explanation: There are two ways a cell dies: apoptosis and necrosis. Apoptosis is regulated. The cell wraps its own contents into apoptotic bodies, cuts itself into fragments, and lets phagocytes clear it away;
the membrane stays broadly intact and the process causes no inflammation. Necrosis is the accidental rupture that follows an injury, and the leaking contents set off inflammation. Apoptosis has two starting paths. The intrinsic path is triggered by stress inside the cell, through the mitochondria, and the extrinsic path is triggered by a death receptor on the cell surface; both end up in caspase 3, which is where the cell actually gets taken apart. A tumour often survives precisely by switching this gate off, which is one of the cheapest ways for a cancer cell to stay alive.
Why this course brings it up: It is a core concept in Lectures 2 and 4, and it is what the bispecific antibody BI 905711, which hits TRAILR2 and CDH17 at the same time, is aiming at. He does not list apoptosis as a named question, but you are expected to be able to draw the two pathways and say where they meet. **Appears in:** Ch 8, Lect 2

▪ Apoptosome

In one sentence: The wheel-shaped structure assembled after cytochrome c is released, and it is what starts caspase 9.
Detailed explanation: The turning point of the intrinsic pathway sits in the mitochondria. When members of the Bcl-2 family judge that it is time to die, cytochrome c leaks out of the mitochondrial intermembrane space into the cytoplasm and binds APAF-1.
This complex assembles itself into a wheel-shaped heptamer — that is the apoptosome. Its job is to pull in procaspase-9 (the precursor form of caspase-9) and activate it; caspase 9 then cuts procaspase 3, and the execution phase begins. This "wheel" is worth remembering, because it is one of the very few giant structures that self-assembles out of cytoplasmic proteins, and because it is the point at which the cell has officially committed rather than merely prepared. Everything upstream of it can still be talked out of; everything downstream is on rails. Everything upstream of the wheel can still be argued with; everything downstream of it runs on rails.
Why this course brings it up: It is the structural node he covers in Lecture 2 when he teaches the intrinsic and extrinsic apoptosis pathways; but he never named the apoptosome as an exam topic on its own, so it is a concept to understand.
Appears in: Ch 8, Lect 2

▪ AR-V7

In one sentence: A shortened version of the androgen receptor produced by a splicing error, which is permanently active and which resists enzalutamide.
Detailed explanation: AR-V7 is an isoform produced when the AR gene is spliced in an abnormal way. In the normal transcript, AR skips over that stretch of cryptic exon CE3, whereas AR-V7 splices it in, and the protein that comes out is missing its ligand-binding domain, so it is already active without any ligand at all.
The clinical meaning of that is very direct: drugs that go after the androgen receptor, such as enzalutamide, stop working on it, and the patient relapses in spite of them. In his Lecture 5 slides he puts AR-V7 in the same list as MCL-1, PKM (the pyruvate kinase gene), Bcl-x (an apoptosis control gene), VEGF165 (vascular endothelial growth factor) and HIF3α (hypoxia-inducible factor), and together they make a complete roster of one gene, different splicing, and a completely different fate for each protein.
Why this course brings it up: It is the clinical example of alternative splicing in Lecture 5. "An isoform that can predict drug resistance" is case number 15 in his scenario bank, and he never names AR-V7 on its own as an exam question, so what he wants from you is that you recognise the pattern rather than the label. **Appears in:** Ch 7, Lect 5

▪ arginine

In one sentence: A positively charged basic amino acid.
Detailed explanation: Its R-group is a long, positively charged side chain. Apart from forming electrostatic interactions with the negative charges of phosphate groups and nucleic acids, it is often used inside the active site of an enzyme to grab a negatively charged substrate.
Like the bulky tryptophan, arginine makes that part of the protein chain stiff. Its charged side chain also often reaches into the active site of an enzyme to hold on to the negative substrate, and that is a good example of the idea that an R-group affects function as well as shape. So a side chain can be doing two jobs at once, pushing the fold one way and grabbing a molecule in the pocket.
Why this course brings it up: He does not name arginine as an exam question on its own, so it is concept-only, and the table of amino acid R-groups in the same section is explicitly not examined. What you take from it is the principle about side chains, not the list. **Appears in:** Ch 2, Lect 1

▪ aristolochic acid I (AA-I)

In one sentence: An aristolochic acid from a Chinese herbal medicine, and one of the four carcinogens Dr. Wong names.
Detailed explanation: AA-I is one of the four carcinogens named on the cancer chain of his slide 17 (the other three are 4-aminobiphenyl, benzo(a)pyrene and PhIP). They react with specific DNA sequences to form adducts, and if those adducts are not repaired the second round of replication fixes the mutation, and cancer begins there.
His chemical carcinogenesis experiment uses these four as demonstrations, and the point of it is that there is a very concrete causal chain between DNA damage and cancer. Aristolochic acid is the one that comes from a traditional Chinese herbal medicine, which makes it the most useful of the four for making the point that a substance you take for one reason can cause cancer by a completely different route.
Why this course brings it up: He does not ask you to memorise the list of carcinogens, so it belongs to conceptual understanding. What he does require you to remember is the chain: carcinogen → adduct → replication error → fixed as a mutation. **Appears in:** Lect 3

▪ Arylamination

In one sentence: The chemical step in which an activated carcinogen precursor attaches an aromatic amine group to a DNA base.
Detailed explanation: The nitrosamines in tobacco, NNK and NNN, are not directly carcinogenic by themselves. They have to be activated by P450 enzymes in the liver until they become electrophilic intermediates that can react with DNA.
The liver does the activation, and only the activated form can bond to a base. This intermediate then goes through arylation, gluing an aromatic amine group onto a base and forming a DNA adduct, like sticking a chemical sticker onto a base so that the pairing rules go wrong. If the adduct is never repaired, the second round of replication locks the error in as a mutation. The line he quotes from his Lecture 3 slides is "NNK induces a GC→AT mutation at K-ras gene in mice", and the arrows in that sentence are the point: a chemical event upstream becomes a specific base change downstream.
Why this course brings it up: It is the section on DNA carcinogenic chemistry in Lecture 3. How an unrepaired adduct becomes a fixed mutation and then connects to KRAS as a driver gene is concept-only. He never lists arylation as a question, and the chemistry of the adduct itself is not examinable either. **Appears in:** Lect 3

▪ asparagine

In one sentence: The amino acid that readily undergoes deamidation.
Detailed explanation: A protein left standing in a bottle will lose an ammonia group spontaneously from the amide next to an asparagine, going through a succinimide intermediate, and adding water back turns it into aspartic acid or iso-aspartic acid. Aspartic acid and iso-aspartic acid have similar structures but different stereochemistry, and that difference changes the behaviour of the protein, which is the common reason a therapeutic protein goes wrong within its shelf life. Because deamidation is a function of time, the storage conditions and the expiry date of a therapeutic protein are strictly specified, and that is a real pharmaceutics problem. The time element is the crucial one: nothing goes wrong on the day the protein is made, and something has gone wrong by the time it reaches the patient, so the specification has to cover the whole shelf life. It is a slow, boring chemical accident that turns into a regulatory requirement, which is a good example of how small a change can be.
Why this course brings it up: He does not name asparagine as an exam question on its own, so it is concept-only. The deamidation mechanism sits in the same passage as his mass numbers, Asn 114.04 Da, succinimide 97.02 Da and Asp/isoAsp 115.03 Da, and those numbers are worth having exactly. The numbers come from the same paragraph, so you learn them together, and note that the two acids are given the same mass of 115.03 Da, so the mass number on its own cannot separate them. **Appears in:** Ch 1, Ch 2, Lect 1

▪ aspartic acid

In one sentence: One of the two products you get when asparagine is deamidated.
Detailed explanation: It is a negatively charged acidic amino acid, with a carboxyl group at the end of its side chain. In deamidation, asparagine first becomes succinimide and then takes on water to become aspartic acid or iso-aspartic acid.
The two have the same mass (115.03 Da) but different bonding arrangements, so they affect protein folding and binding ability differently. Because the two products can be mixed together inside a protein, manufacturers of therapeutic proteins have to use methods such as high-performance liquid chromatography to separate the iso-Asp from the real Asp. Same mass, different

molecule — and that is exactly the problem, because mass is what the measuring instrument reports. So a protein can carry a subtle chemical change that no amount of mass measurement would reveal.
Why this course brings it up: He does not name aspartic acid as an exam topic in its own right, so it is a concept to understand. **Appears in:** Ch 1, Ch 2, Lect 1

▪ AT-rich

In one sentence: A stretch of DNA in which the proportion of A, adenine, and T, thymine, is high.
Detailed explanation: An AT-rich stretch pairs only through the two hydrogen bonds of A-T, while a GC-rich stretch uses three. The difference is visible on the chromosome: the dark bands of G-banding are the AT-rich regions and the light bands are the GC-rich ones, because the two kinds of region condense to different degrees during the cell cycle.
The higher the overall GC content of a DNA, the more stable the double helix and the higher the melting temperature, and that is the practical basis for designing PCR primers. So the same number that decides how hard a stretch of DNA is to pull apart is also the number you use when you write a primer.
Why this course brings it up: "The more GC, the higher the Tm" is one of the precise numbers he lists, at Tier 3, and primer design is the application he explicitly links it to. The two words AT-rich and GC-rich themselves he does not name, so they are conceptual understanding. **Appears in:** Ch 3, Ch 11, Lect 3

▪ Atezolizumab

In one sentence: Atezolizumab is an immune checkpoint inhibitor that goes after PD-L1 rather than after PD-1.
Detailed explanation: Atezolizumab is an anti-PD-L1 antibody, and it is the representative drug standing behind the dMMR / pMMR Kaplan–Meier plot on his Lecture 3 slide and in Lecture 4. What it blocks is the PD-L1 sitting on the surface of tumour cells and of immune cells.
Because PD-1 on the T cell can no longer receive that brake signal, the T cell recovers its cytotoxic activity and goes after the tumour cells that carry neoantigen. It runs on exactly the same logic as anti-PD-1 (pembrolizumab, nivolumab); the only difference is which side of the checkpoint the drug lands on. Block either half of that handshake and the result is the same: the brake is off and the T cell can kill again. Lectures 3 and 5 also mention a different way of combining the two ideas: first use an HDAC inhibitor to make the tumour express PD-L1 (he calls this "gene addiction"), and only then add the checkpoint inhibitor on top of it. The order matters, because the first step is what makes the second step work.
Why this course brings it up: Dr. Wong never named the drug atezolizumab as a standalone exam topic. What was named is the concept: hot / cold tumour, and the way MSI-H tumours respond to anti-PD-1. If you understand that concept, you can work out what atezolizumab is doing without having memorised the drug name. A multiple-choice question is far more likely to test that concept than the brand. **Appears in:** Lect 3

▪ ATM

In one sentence: The first sensor kinase to be switched on after DNA damage, and it mainly answers double-strand breaks.
Detailed explanation: ATM stands for ataxia telangiectasia mutated, and it is itself a kinase, which means putting a phosphate group onto other proteins, which is the same as turning their switches on.
When DNA takes a double-strand break, ATM is activated and passes the signal down to effector proteins such as p53, so that the cell stops to repair, or starts apoptosis if it cannot be fixed. ATM and ATR together are the two key signalling components of the DNA Damage Response, the two Dr. Wong calls out by name. So ATM is the sensor and not the repairer; it does not fix the break itself, it announces the break and sets the cell's response in motion. And the two possible answers are the same two answers as everywhere else in the cell: stop and repair, or die.
Why this course brings it up: In Lecture 4 he lists ATM as the sensor kinase of the DDR. He also names an “ATM blocker” as a possible second payload in a dual-payload ADC, though he says explicitly that the ADC structure is not examined. **Appears in:** Ch 6, Lect 4

▪ Atom

In one sentence: The smallest unit of an element; a molecule is what you get when atoms are bonded together.
Detailed explanation: Chemistry starts here. An atom has electrons orbiting a nucleus, and inside the nucleus sit protons and neutrons; the number of protons decides which element it is, and the number of neutrons decides which isotope it is.
A molecule is the result of atoms joining by covalent bonds. Carbon, with four bonds and therefore the ability to hook up to many other things at once, becomes the skeleton of organic chemistry, and that is why nearly all the molecules of life are built on carbon. The four common classes of molecule in a living thing (carbohydrates, lipids, proteins and nucleic acids) are all derivatives of that carbon skeleton.
Why this course brings it up: This is the chemical foundation in Lecture 1. Conceptual understanding: Dr. Wong explicitly said not to memorise chemical structures, so this entry is the foundation rather than an exam point. **Appears in:** Ch 1

▪ Atomic nucleus

In one sentence: The dense centre of an atom, and the place that holds the protons and neutrons.
Detailed explanation: Almost all the mass of an atom is concentrated in an extremely small nucleus, while the electron cloud takes up nearly all the volume and almost none of the mass. That is why a piece of metal can be dense and still conduct, because the electrons move freely, and why the nuclei of solid atoms sit far apart and can be packed loosely.

The number of protons in the nucleus is the atomic number of the element: hydrogen is 1, carbon is 6, oxygen is 8. This structure also explains the idea of isotopes. Add one neutron or take one away and the mass number changes but the chemistry does not, which is why carbon-14 can be used for dating.
Why this course brings it up: It is the chemical foundation of Lecture 1 and belongs to understanding the concept. He never names this entry. **Appears in:** Ch 1

▪ ATP (adenosine triphosphate)

In one sentence: The energy the cell can spend; cut the terminal phosphate off it and the power comes out.
Detailed explanation: ATP is made of adenine, ribose and three phosphates. What the cell actually spends is the bond between that terminal phosphate and the phosphate next to it. Cut the terminal phosphate off and you have ADP; put the phosphate back and you are recharged, so the cell is always paying out and taking back the same phosphate group. The key point is that ATP is not only fuel. It is also the currency of the fastest switch the cell has: a kinase takes a phosphate from ATP and sticks it onto a protein, the negatively charged phosphate pushes the protein apart, the shape changes and the switch changes state. A phosphatase uses water to take the phosphate off again, and the protein springs back to where it was. His whole Hippo-YAP example runs on this mechanism, and so does the phosphorylation switch in the protein section.
Why this course brings it up: In Lecture 1 he lists ATP as one of the two uses of a biomolecule, energy source and signalling molecule, and the Hippo example is built on it. He does not ask you to draw ATP, so it is concept-only, but the kinase-phosphatase pair that runs on it is examinable. **Appears in:** Ch 1

▪ ATR

In one sentence: The other sensing kinase, the one that answers replication stress and single-stranded DNA.
Detailed explanation: It divides the work with ATM: what ATR mainly watches for is the kind of signal that comes with replication stress — a replication fork that has stalled, and exposed single-stranded DNA.
Once it is switched on it passes the message on by phosphorylation, just like the other one, so that the cell pauses replication, stabilises the damaged DNA and then hands it to the repair system. When both kinases are open at once, the cell has detected something serious and has to stop everything. The two kinases are a division of labour, not a division of importance: different triggers, same downstream response.
Why this course brings it up: He does not list ATR as an exam topic on its own, so it is a concept to understand; the chain he has taught is the one where a DDR sensor kinase fires and the signal ends up at p53 and apoptosis. **Appears in:** Ch 6, Lect 4

▪ AUC (area under the curve)

In one sentence: A number between 0 and 1 that summarises how well a test can tell two groups of people apart, and higher is better.
Detailed explanation: AUC is the area under the ROC curve, and it squashes a whole curve into a single number: 0.5 means you are guessing, and 1.0 means you separated the two groups perfectly.
Dr. Wong used a circulating miRNA classifier as his example in Lecture 1. The AUC for miR-15b plus miR-130b was 0.981, paired with sensitivity 98.25 %, specificity 91.53 % and accuracy 94.83 %, and the result was confirmed in a validation cohort of 116 people. The CDKN2A methylation work in Lecture 3 was measured with the same ruler: AUC 0.847 for exon 1, 0.924 for exon 2, and 0.938 when the two were combined. Read it with care, though. The whole reason for reducing a curve to one number is convenience, but the number hides the shape of the curve, which is exactly why you cannot skip the question of where you set the cutoff. A test with a high AUC is not automatically good value in the clinic, because you still have to look at the trade-off between sensitivity and specificity at whatever cutoff you choose.
Why this course brings it up: The number 0.981 is on his list of precise numbers, which makes it standard multiple-choice material, so memorise it exactly as written. The concept behind the number, the receiver operating characteristic curve itself, is conceptual understanding rather than a memorisation item. **Appears in:** Ch 4,Lect 1

▪ Autologous

In one sentence: Taken from the patient themselves and given back to the same patient.
Detailed explanation: Autologous is the most intuitive version of cell therapy: the patient's own cells carry no risk of immune rejection, so although the manufacturing is slow and expensive, it is biologically the safest option available.
The drawbacks are very practical — after chemotherapy the patient's T cells are often in poor shape (he calls it poor T-cell fitness), and because each batch has to be made individually and shipped frozen, the cost ends up high. His Lecture 2 diagram of the CAR-T process spells it out: draw blood → engineer it → expand to millions of copies → infuse back → first give lymphodepleting chemotherapy. Every allogeneic version trades exactly those three pain points for the benefit of an off-the-shelf product, and that is the whole trade: convenience against safety.
Why this course brings it up: It is the CAR-T section of Lecture 2; autologous versus allogeneic is one of the distractor pairs he lists for multiple-choice questions, so the contrast is fair game even though neither word was separately named as an exam topic. **Appears in:** Ch 9, Lect 2

▪ Autophagy

In one sentence: The cell eats the parts of itself it does not need, and gets nutrients back in return.
Detailed explanation: When nutrients run short, for instance in the hypoxic core of a solid tumour, the cell switches on autophagy. It packages unwanted organelles and proteins into autophagosomes, ships them to the lysosome for breakdown, and then spends the recovered amino acids and energy on staying alive.
From the tumour's point of view this is a survival strategy, which is why he singles it out in Lecture 3, when he covers the progression of cancer: the promotional stage comes with a rise in autophagy, a rise in angiogenesis and a drop in immune homeostasis. From the treatment side it is a double-edged sword: blocking autophagy can starve a tumour that depends on it, but it can also let the remnants of a treated cell survive.
Why this course brings it up: It comes from the stress and disease part of Lecture 2, and from the cancer progression model in Lecture 3. He never names autophagy on its own as an exam question, so it belongs to understanding the concept. **Appears in:** Ch 8, Lect 2

▪ Avery, MacLeod and McCarty

In one sentence: The three who proved in 1943, using DNase and protease, that the transforming factor was DNA.
Detailed explanation: The story has two steps. In 1928 Griffith found that certain strains of Streptococcus pneumoniae could turn a harmless strain into a virulent one. He had found the transforming factor, but he did not know what it was, so the famous experiment was waiting for someone to run it on one substance at a time.
In 1943 Avery, MacLeod and McCarty took the DNA that carried the transforming ability and tested it on its own: add DNase, an enzyme that breaks down DNA, and the transforming ability disappears; add protease and the transforming ability is unchanged. The logic is clean. An enzyme that destroys the transforming factor must be acting on the transforming factor, so whatever survives the protease and dies with the DNase is the transforming factor. DNA was therefore the genetic material.
Why this course brings it up: It is the nucleic acid history section of Lecture 1, and the central premise of DNA as genetic material in Lecture 4. He does not list these three as a question, so it is concept-only, but you should be able to walk through the DNase versus protease logic if it comes up inside a longer question. **Appears in:** Ch 3, Lect 1

▪ Basal transcription

In one sentence: The baseline transcription that keeps running before any activator has arrived.
Detailed explanation: Transcription comes in two intensities. While a repressor is still sitting on the operator, RNA polymerase can only squeeze past, and very little mRNA is made — that is basal transcription, also called low-level expression.
For his teaching of the lac operon this is the crucial fourth state: when lactose is present but glucose is plentiful, the repressor has already been pulled away by allolactose, yet cAMP is still low and cAMP-CAP is absent, so the genes are "on, but slow". The clinical point is that pushing a gene from off to on and pushing it from slow to fast are two different things, and a drug that does the first has not necessarily done the second. A drug that removes a repressor gets you the basal level; a drug that recruits the machinery gets you the full level. Two intensities, not one.
Why this course brings it up: It is one of the four states of the lac operon in Lecture 5; the word basal itself was never named as an exam topic, so it is part of the framework you should be able to reason with rather than a term you need to define.
Appears in: Ch 7, Lect 5

▪ Base

In one sentence: One of the four letters of DNA: A, T, G and C; in RNA, U takes the place of T.
Detailed explanation: The four bases differ only in the functional group hanging off the side of the ring, and that difference is what decides the pairing rules: Adenine and Thymine form 2 hydrogen bonds, and Guanine and Cytosine form 3.
So GC-rich DNA needs more energy to pull the two strands apart, which shows up as a higher Tm, and that is the principle behind PCR primer design: you have to know roughly how hot you are going to make the tube before you can pick a primer.
There is one more difference that is easy to overlook: only cytosine gets methylated, so the whole stage on which epigenetic regulation performs is built on cytosine. Every one of those letters is a base, and every strand of DNA is just a long string of them read in order.
Why this course brings it up: The three hydrogen bonds of G–C, the two of A–T, and the effect of GC content on Tm are all precise numbers on his list. What he did not ask you to do is memorise the chemical structure of the bases. **Appears in:** Ch 1,Ch 3

▪ Base excision repair (BER)

In one sentence: The pathway that repairs a single damaged base: cut that base out and then put a new one in its place.
Detailed explanation: The cell keeps different repair shops for different kinds of damage, and BER is the one with the smallest job of them all. It handles chemical damage to a single base, such as the modified bases produced by alkylation or by oxidation. The first step is that DNA glycosylase cuts the damaged base clean out of the strand, which leaves behind an AP site, a position with no base at all. The second step is that AP endonuclease cuts there, on one side of the gap. The third step is that polymerase fills in the correct nucleotide to replace the one that was lost. And the last step is that ligase seals the gap shut. The core idea running through the whole pathway is a simple one: change a single letter of the text rather than rewriting the entire paragraph.
Why this course brings it up: BER against NER is the first distractor pair he lists for a multiple-choice question. But he explicitly said that the chemistry of DNA glycosylase and of the AP site will not be examined, so all you really need to carry away

is which pathway repairs what, and that every kind of lesion has its own shop. **Appears in:** Ch 6, Lect 4 **Video:** V06_BER_vs_NER.mp4

▪ Base pairing

In one sentence: The rule for pairing bases in DNA: A pairs with T, G pairs with C.
Detailed explanation: The pairing rule makes the two strands complementary, so one strand determines the other. That is the physical basis of both replication and sequencing. Pairing depends on hydrogen bonds, and the number of hydrogen bonds directly affects stability: G–C has 3 hydrogen bonds, A–T has 2, so a GC-rich region needs a higher temperature to denature, which means a higher Tm.
PCR primer design therefore leans directly on this rule, and the same rule is what lets you sequence a genome at all.
Why this course brings it up: The 3 hydrogen bonds of G–C and the 2 of A–T are precise numbers he lists, and the direction from GC content to Tm is slide content of his. Base pairing itself is a basic premise of Lectures 1 and 4. **Appears in:** Ch 3, Lect 1

▪ Bcl-2 family

In one sentence: The group of proteins that decide whether the mitochondria release the apoptotic signal.
Detailed explanation: The members of this family stand on two sides of the outer mitochondrial membrane. The anti-apoptotic members (Bcl-2 itself, for example) guard the outer membrane and stop cytochrome c from leaking out;
the pro-apoptotic members (bak, bax) puncture holes in the outer membrane, letting cytochrome c escape to assemble the apoptosome and start caspase 9. So the family is a balance, and the balance tips depending on which side is heavier. It is a tug of war played out on the same piece of membrane. The clinical application is the bispecific antibody BI 905711 — it hits TRAILR2 and CDH17 at the same time, so apoptosis only starts when "only a tumour has both conditions at once", and normal tissue stays safe.
Why this course brings it up: It is the key node in Lecture 2 when he teaches the intrinsic and extrinsic apoptosis pathways, and it is the mechanistic basis of his bispecific antibody story; he did not name the Bcl-2 family as an exam topic on its own, so the two-sides-of-the-membrane picture is what you need, not the membership list. **Appears in:** Ch 8, Lect 2

▪ Benzo(a)pyrene

In one sentence: A tobacco carcinogen that forms an adduct with DNA after the cell has activated it.
Detailed explanation: Polycyclic aromatic hydrocarbons like Benzo(a)pyrene are present in tobacco smoke, and they do not damage DNA directly as they are. They have to be activated first by the cell's P450 enzymes, and only then do they become compounds that can react electrophilically with DNA.
Dr. Wong's third lecture slide spells the steps out very clearly: the carcinogen reacts with a particular sequence in the DNA and forms a DNA adduct; if that adduct is never repaired, replication copies it wrongly; and the second round of replication fixes the error as a mutation. Exactly the same logic applies to 4-aminobiphenyl, aristolochic acid I and PhIP. Notice that the dangerous form is not the one on the cigarette, it is the one the cell manufactures out of it. And notice the order: carcinogen first, adduct second, copying error third, mutation last.
Why this course brings it up: The chemical carcinogenesis section of Lecture 3. The idea that an unrepaired adduct gets fixed into a mutation is conceptual understanding. He did not list Benzo(a)pyrene as an exam topic on its own. **Appears in:** Lect 3

▪ Beta sheet (beta-sheet)

In one sentence: Several protein chains pulled apart and lying side by side, a shape held in place by hydrogen bonds between them.
Detailed explanation: A β-sheet is the second most common secondary structure in proteins, and together with the α-helix it forms the skeleton. It looks like a paper boat that has been folded: the chain first zigzags back and forth into segments of parallel or antiparallel strands, and then those strands come together side by side.
What holds the sheet together is hydrogen bonds between the carbonyl oxygens of the main chain and the amide hydrogens. That is the same trick as the α-helix, so it is still the main chain doing the work rather than the side chains. Biologically it matters a great deal, because many fibrous proteins, such as keratin and collagen, are almost entirely made of β-sheet and get their tensile strength from it, and because the misfolding that causes disease, amyloid for instance, is often a stack of β-sheets as well.
Why this course brings it up: It is the second level of the four levels of protein structure in Lecture 1. He does not make you memorise the table of R-groups, but you are expected to understand the four levels and the secondary structures. **Appears in:** Ch 2

▪ BI 905711

In one sentence: A tetrahedral bispecific antibody that recognises TRAILR2 and CDH17 at the same time.
Detailed explanation: This is an example from his own research: one molecule that recognises two targets at once, the death receptor TRAILR2, which starts apoptosis, and CDH17, cadherin-17, which appears on the surface of cancer cells.
It only attacks when both of them are present, and that is the AND-gate design in the glossary, put there to solve the problem of killing normal cells. His own research has been built around CDH17 for twenty years, and the example ties the whole design logic of immunotherapy together in one go. Two targets, one molecule, and a rule about when it fires, and that is the entire design in a single sentence. The advantage is obvious once you have the rule, because the normal cells that carry only one of

the two are simply left alone.
Why this course brings it up: Dr. Wong does not make the drug name an exam question on its own, so it is conceptual understanding, and the point of the passage is that an AND gate avoids collateral damage. **Appears in:** Ch 8, Lect 2

▪ bid

In one sentence: The apoptosis signal protein that caspase-8 cuts into tbid.
Detailed explanation: The extrinsic and intrinsic apoptosis pathways start out as two separate roads, but when caspase-8 is switched on in the extrinsic pathway it cuts bid, and the tbid that comes out goes to the mitochondrial membrane and promotes the release of cytochrome c, which joins the two roads together.
In other words, a death signal from outside can be delivered into the mitochondrion and converted into the intrinsic execution programme. So the bridge is a single cut, and everything on the intrinsic side of the mitochondrion runs from there.
Why this course brings it up: He does not name bid as an exam question on its own, so it is concept-only. He does draw the meeting of the extrinsic and intrinsic pathways as a diagram. The diagram is the examinable part. **Appears in:** Ch 8, Lect 2

▪ Binucleated cell

In one sentence: A cell with two nuclei inside it and a DNA content of 4N.
Detailed explanation: This is the standard answer you get after inhibiting the contractile ring. The mechanism runs like this: chromosome segregation, which is mitosis, and division of the cytoplasm, which is cytokinesis, are independent of each other. Inhibiting the contractile ring blocks only the second one, and the first carries on regardless. So the genetic material has been copied and has separated, but it was never cut into two portions, and the result is one cell, two nuclei, and a DNA content of 4N. On the PI-stained plot from flow cytometry, what you see is not two peaks at 2N and 4N but a single 4N peak, because nothing was divided. He also adds the clinical consequence: organelles and mitochondria are handed out at random during cytokinesis, so siblings in the same family can differ a great deal in how severe their mitochondrial disease turns out to be.
Why this course brings it up: This is a Tier 1 question he named (Lecture 2): "this is the type of the question that I would try to include in the midterm". It is scenario case number 5 in the case bank. The whole point of the question is that mitosis and cytokinesis are separate processes, so blocking one of them does not block the other, and the answer to "what do you end up with" is a binucleated cell carrying 4N of DNA. **Appears in:** Ch 8, Lect 2

▪ Biomarker

In one sentence: A biomarker is a molecule you can measure in a sample and that tells you something about the disease or about how a treatment is working.
Detailed explanation: Biomarkers split into three uses, and that classification is one of the main threads running through his whole course, so it is worth learning early and getting straight.
A diagnostic biomarker answers the question "is there a disease right now" — SEPT9, a methylation marker gene used in colorectal cancer screening, is an example. A prognostic biomarker answers "how is this going to end up" — ERBB2, for instance, is favourable in kidney cancer and unfavourable in pancreatic cancer. A predictive biomarker answers "will this drug work for this particular patient" — cetuximab, for example, does not work in rectal cancer carrying a K-Ras mutation. Being able to tell prognostic apart from predictive is one of the distractor pairs he lists, and it is the distinction most likely to trip you up, because the two words sound interchangeable in English and mean quite different things in practice. One tells you the patient is sick now, one tells you how the illness will end, and one tells you which drug to reach for.
Why this course brings it up: In Lecture 1, when he set up the course, he said outright that the goal of the subject is to turn genomics into the three kinds of biomarker — diagnostic, prognostic and predictive. Every research case he runs through the course (vimentin, circulating miRNA, CEA) is a biomarker story, so the distinction runs through the whole thing, and these three words are the vocabulary of everything that follows. **Appears in:** Ch 11, Lect 1, Lect 2, Lect 4

▪ Biomolecule

In one sentence: Molecules that a living system makes: carbohydrates, lipids, proteins and nucleic acids.
Detailed explanation: Dr. Wong split the molecules of life into four families in Lecture 1, and he pointed out that their uses come in four flavours as well: disease biomarkers, energy sources, structural components and signalling molecules.
Every family has a clinical story attached to it. Carbohydrates are fuel, and they are also structural material such as cellulose. Lipids are the skeleton of membranes, and they are also the source of inflammatory signals such as LPS. Proteins are the machines, and they are also antibodies, enzymes and drug targets. Nucleic acids are the recipe and the photocopy. Cancer cell metabolism (the Warburg effect) and recombinant protein production (CHO cells plus codon optimisation) are both worked examples of him tying these four together. The categories are not decoration: every clinical problem later in the course attaches itself to one of these four lists, so this is the vocabulary you will keep reusing.
Why this course brings it up: This is the very first topic of Lecture 1 and also the frame for the whole course. Dr. Wong explicitly said not to memorise chemical structures and metabolic details, so you need to know the four categories and you do not need to memorise the details inside them. **Appears in:** Ch 1,Lect 1

▪ Bispecific antibody

In one sentence: An antibody with two different binding heads, so that it can hit two targets at the same time.
Detailed explanation: A single-specificity antibody has obvious limits: the target has to be expressed almost entirely by the tumour, otherwise the antibody damages normal tissue as well. A bispecific antibody puts two binding heads on the same molecule, so it can bind a tumour-specific antigen and one other molecule at the same time.
His own example is BI 905711, a tetrabivalent bispecific antibody that hits TRAILR2 and CDH17 at the same time, and it uses a logical gate so that apoptosis is only switched on when both conditions hold at once. The difference between the three tissues is therefore very clean: normal liver has no CDH17, normal gut is insensitive to TRAIL, and colorectal cancer has both of them.
Why this course brings it up: It comes from the bispecific antibody material in Lecture 2 and from the ADC section in Lecture 4. It is his own CDH17 research story landing in the clinic, so it belongs to understanding the concept rather than to the list of things to memorise. **Appears in:** Ch 8, Lect 2

▪ Bisulfite sequencing

In one sentence: The gold standard test for methylation: unmethylated C turns into U, methylated C stays as it is.
Detailed explanation: The principle is a chemical selectivity. Sodium bisulfite sulfonates the amine group of cytosine, and once cytosine has been methylated that reaction simply cannot happen.
So in DNA you have treated, an unmethylated site reads out as thymine, because it is really a U, while a methylated site still reads out as cytosine. You are reading methylation off the sequence without ever having to detect the methyl group itself. After sequencing, the difference between those two signals paints the methylation state across the whole map, with resolution down to a single base. The CDKN2A study is exactly what he used it for, measuring a tumour methylation rate of about 28 % in exon 1 and about 41 % in exon 2, so this is the method that turns a methylation call into a number you can put in a paper.
Why this course brings it up: It is the epigenetics testing section of Lecture 3, and he places it on his second-level (A) concept list as "Bisulfite sequencing = gold standard". The label matters, because it tells you that this, and not some other assay, is the method you are expected to name, and that the underlying chemistry is not what gets examined. **Appears in:** Ch 4, Ch 11, Lect 3

▪ Boundary elements

In one sentence: A chromatin boundary that fences off the active region from the silent one.
Detailed explanation: Chromatin marks spread. If a region gets marked with repressive methylation, that signal may keep travelling outward and shut down genes that were perfectly active. The boundary element is the wall that stops it, so that the open region and the closed region each mind their own business.
It is the same family of idea as the insulator in the glossary, only told from the point of view of chromatin. Without a wall, a silent patch of chromatin would quietly spread its silence across the genome.
Why this course brings it up: He mentions it on slide 27 in Lecture 3 without particularly naming it, so it is a concept to understand. **Appears in:** Ch 4, Lect 3

▪ BRAF

In one sentence: A kinase sitting in the MAPK signalling pathway, and an early event on the serrated colorectal cancer route.
Detailed explanation: BRAF is a kinase downstream in the MAPK pathway, normally under the control of the upstream RAS signal. Once it mutates it sends out a growth signal on its own, and the cell no longer needs the upstream switch turned on.
In Dr. Wong's colorectal cancer model, the adenoma route has APC early, KRAS in the middle and p53 late, while the serrated route has BRAF early. The two routes are different until the end, where both of them arrive at a high-MSI adenocarcinoma. Which is the multi-step story in miniature: several hits, several different genes, and one common destination.
Why this course brings it up: He does not list BRAF as a topic on its own, so it belongs to conceptual understanding. What is Tier 2 is “driver and passenger” and “multi-step carcinogenesis”. **Appears in:** Ch 10, Lect 4

▪ branch migration

In one sentence: The step that extends the stretch of DNA that has matched up.
Detailed explanation: After strand invasion, a region of heteroduplex DNA forms between the two molecules, and inside it there is still a loop that has not paired. Branch migration is what squeezes that loop out gradually, so the paired region gets longer and longer, and the process is driven by Rad54.
At the end there is still a resolution step to separate the intermediates, and only then is HR finished. What it achieves is the conversion of a match of a few bases into a match over a long stretch, which is what lets the polymerase copy the missing sequence from the template afterwards. It is like zipping a zipper: the teeth have to slide along before the two sides are properly closed.
Why this course brings it up: He does not name branch migration as an exam question on its own, so it is concept-only. The point of the entry is that HR needs a long enough stretch of homology before it can repair, and this is the step that provides it.
Appears in: Ch 6, Lect 4

▪ branched linker

In one sentence: A forked linker that carries two different drugs at the same time.
Detailed explanation: An ordinary ADC linker is linear, with the antibody at one end and a single drug at the other. Change it to a branched linker and the linker forks into two arms, each carrying a drug: usually a topoisomerase inhibitor plus an inhibitor of the DNA Damage Response, an ATM blocker for instance, put there to deal with the resistance that a single payload breeds.
The price of a branched linker is that the molecule gets bigger and the drug payload drops, so it is still a research-stage design rather than an established standard practice. Two drugs, two arms, one antibody, and the whole idea is to make it harder for the tumour to escape by becoming resistant to both at once.
Why this course brings it up: Dr. Wong is explicit that the structure of an ADC and the linker chemistry are not examined. All you need is that a dual-payload ADC exists and what the two payloads usually are. **Appears in:** Ch 10, Lect 4

▪ BRCA1 and BRCA2

In one sentence: Lose these two genes and the cell can no longer repair broken chromosomes by homologous recombination.
Detailed explanation: BRCA1 and BRCA2 are the key proteins of the HR pathway. The broken ends are first trimmed by a nuclease into 3′ single-stranded overhangs, RPA covers the exposed single strand so that nothing else can attack it, and RAD51 then winds itself into a filament to carry out strand invasion.
The job of BRCA1 and BRCA2 is to steer the whole complex towards the sister chromatid and to make sure RAD51 is loaded correctly. Both of them are recruiters and scaffolds rather than enzymes: they do not cut or copy DNA themselves, they make sure the right machinery arrives at the right place on the right chromosome. Without them, HR fails, the cell has to fall back on the error-prone NHEJ, and chromosome rearrangements pile up until the genome is visibly scrambled. The cleverness on the pharmacology side is synthetic lethality: the tumour has already lost HR, so press the second pathway down with a PARP inhibitor and it has nowhere left to go. Normal cells still have a working HR to fall back on, so they survive, and the therapeutic window ends up very wide. Drugs of this kind have been approved for BRCA1/2 mutant ovarian cancer and breast cancer.
Why this course brings it up: This is the core case for synthetic lethality and the PARP inhibitor in Lecture 4, and it is also case 10 in the case bank. Dr. Wong said that the PARP inhibitor and BRCA chain has to be something you can draw. **Appears in:** Ch 6,Ch 10,Lect 4

▪ BRD4

In one sentence: A protein that reads the acetylation mark and drags the transcription machinery over, and it is a current drug target.
Detailed explanation: BRD4 carries a bromodomain, which recognises acetylated lysine on histones; the moment it reads that mark, it recruits the transcription machinery to the spot and keeps the fact that "the genes here are being read" alive.
It is the most typical member of the bromodomain family and one of the hottest cancer drug targets in recent years. In the HDAC inhibitor section of Lecture 3, Dr. Wong gives BRD4 a specific role: HDAC3 inhibitor RGFP 966 raises PD-L1 expression through BRD4, so the cancer cell becomes more dependent on the checkpoint and more sensitive to PD-L1 blockade — the "gene addiction" he describes. Read the mechanism as three parts: the acetyl mark is the signal, the bromodomain is the reader, and BRD4 is the machine that does something about it. The logic is slightly counter-intuitive: you give a drug that turns a brake off, and the tumour becomes more eager to rely on that brake.
Why this course brings it up: BRD4 appears on his Lecture 3 slide on HDAC inhibitors as one link in that "gene addiction" mechanism; but he did not name BRD4 as an exam topic on its own, so you should be able to place it in the story rather than recite the mechanism from memory. The drug attached to the mechanism is not one you will be asked to list; what you may be asked is why the tumour becomes more dependent on its own brake. **Appears in:** Ch 4, Lect 3

▪ Breakthrough Therapy Designation

In one sentence: The FDA's fast track for drugs with breakthrough therapeutic potential.
Detailed explanation: It is one of the FDA's fast tracks, and it lets the developers talk to the review agency more often and move the approval along on earlier clinical data, so that a genuinely important drug reaches the market sooner.
On the same slide he also names Orphan Drug Designation (for rare diseases, drugs for fewer than 200,000 patients, and it gives seven years of market exclusivity) and Priority Review. The point of all three is the same: they change how quickly a drug is allowed through the gate, not whether it has to prove itself. Seven years of exclusivity is what makes the development of a drug for a very small population worth doing.
Why this course brings it up: He does not make those three FDA names exam topics, so they are concepts to understand; what is really examined is the logic of the phases of clinical trials. **Appears in:** Ch 12, Lect 1

▪ Breathomics

In one sentence: The study of the chemical content of exhaled breath, used to see inside the body.
Detailed explanation: Gas can be "sniffed" for disease, and that is exactly what this field does. His Lecture 1 slides describe the whole workflow from beginning to end. The tumour microenvironment, made up of cancer cells, alveolar cells, T cells, neutrophils, macrophages, fibroblasts, cytokines and metabolites, keeps giving off volatile organic compounds.
Those compounds are collected on solid phase microextraction fibres and then analysed by GC/MS. The clinical value of all this is that it is completely non-invasive: one breath is enough to screen for lung cancer and stomach cancer, with no needle and no radiation at all. On the risk side, you have to keep an eye on the false positive rate and on the consistency of the breathing

manoeuvre.
Why this course brings it up: It is an application branch of the metabolomics material in Lecture 1. He never names breathomics on its own as an exam question, so it belongs to understanding the concept rather than to the exam list, even though the application is the point he is making. **Appears in:** Ch 10, Ch 11, Lect 1

▪ C-terminus
In one sentence: The very last end of a protein chain, where the last amino acid carries a free carboxyl group sticking out.
Detailed explanation: Every polypeptide chain has two fixed ends: the N end carries a free amino group and the C end carries a free carboxyl group, and you read a protein from the N end towards the C end.
This is not just a label. It decides the working direction of a great many proteins. Here is a key example: the N-terminal domain (NTD) of a nuclear receptor is the regulatory region, the LBD sits on the C-terminal side and is the part that binds ligand, and putting the ligand-binding region at the C end is what makes the sequence of events, ligand binds, shape changes, co-activator recruited, make sense in three dimensions. Antibodies are the same: the Fab on the N-terminal side does the antigen recognition, and the Fc on the C-terminal side calls in the effector cells. Get the direction wrong and the protein stops making sense, in the same way that a sentence read backwards stops making sense.
Why this course brings it up: The protein structure section of Lecture 3. Dr. Wong did not name C-terminus as an exam topic on its own, so this is conceptual understanding, although the direction of the chain comes up everywhere. **Appears in:** Ch 2

▪ CA loop
In one sentence: The small loop left on the new strand when the polymerase slips while copying a repeated sequence.
Detailed explanation: When the polymerase is copying a repeat sequence such as a microsatellite it sometimes slips, and the newly synthesised strand ends up with a small unpaired loop, written out as 5′-CACAC[CA]ACACA-3′.
If mismatch repair is intact, the loop is picked up, cut out and redone, and the result is a stable microsatellite (MSS). If MMR has failed, nobody tidies up the loop, the repeat length keeps changing, and you get MSI-H. So MSI-H is not simply “more mutations”, it is repeat lengths starting to go wrong, and that length change can be measured directly with PCR and sequencing. The loop is the visible trace of a slip, and MMR is the cleaner that notices it. A mismatch in a copied base pair is a different shape of mistake, and it needs a different recogniser.
Why this course brings it up: MMR defect leading to MSI-H is Dr. Wong's Tier 2 point and one of the scenario questions he names. The term CA loop itself is not something he asks you to memorise, so it belongs to conceptual understanding. **Appears in:** Ch 6, Lect 4

▪ calcium ions
In one sentence: Calcium ions inside the cell, one of the second messengers.
Detailed explanation: IP3 travels from the cell membrane to the endoplasmic reticulum, binds the IP3 receptor in the ER membrane, and releases the stored calcium ions into the cytoplasm. The rise in calcium concentration is itself the signal, and it activates downstream proteins such as calmodulin, carrying the message from outside the cell right into the control room.
That is a textbook second messenger. The calcium signal is fast, because the concentration of calcium in the cytoplasm is extremely low to begin with, so any release immediately makes a big difference. The speed comes from the same place as the sensitivity: a low resting level means a small release is a large relative change.
Why this course brings it up: He does not name calcium ions as an exam question on its own, so it is concept-only. The skeleton he keeps coming back to is that the second messenger carries the signal inside the cell. So the mechanism matters more than the ion. **Appears in:** Ch 7, Lect 1

▪ cAMP (cyclic adenosine monophosphate)
In one sentence: A small second messenger; its concentration goes up when glucose is low, and it lets CAP pull RNA polymerase onto the promoter.
Detailed explanation: cAMP is cyclic AMP, a small molecule that can diffuse everywhere, a textbook second messenger. A second messenger is simply a small chemical that carries a signal from the membrane into the cell, and cAMP is the one everybody learns first.
In the bacterial lac operon, it plays the part of the energy-status reporter: when glucose is low, cAMP rises inside the cell, cAMP binds the CAP protein, the cAMP-CAP complex sits down on the promoter and drags RNA polymerase over, and transcription goes from slow to strong. The key thing to remember is that this is a second condition, independent of the lactose condition. When there is lactose but glucose is plentiful, the gene is only transcribed at a basal level, because one condition on its own is not enough to make it run. The cell is asking two questions at once, is there lactose to eat and is there glucose to burn, and only when both answers are right does the operon open wide.
Why this course brings it up: cAMP-CAP is the second condition itself, inside his second-level concept from Lecture 5, the "two conditions, not one" of the lac operon. He expects you to be able to say what happens when only one of the two conditions holds, to draw the promoter with the CAP-cAMP complex sitting on top of it, and to keep the two conditions apart when you are asked about basal versus strong transcription. **Appears in:** Ch 7, Lect 5

▪ Cancer-associated fibroblast (CAF)

In one sentence: Fibroblasts that the tumour recruits and reprograms, so that they spend their time building the stroma around it.

Detailed explanation: A tumour is not just a crowd of cancer cells. It is cancer cells plus immune cells, plus CAF plus extracellular matrix. A CAF used to be an ordinary fibroblast in normal tissue, one whose job was to repair damage and to make matrix;
once the tumour recruits it and its expression changes, it starts making matrix that suits the tumour, promoting angiogenesis, and building an environment that suppresses the immune response. The clinical meaning is that treatment gets harder: the drug kills the cancer cells, but the scaffold held up by the CAF is still there, so the tumour has somewhere to rebuild itself. His Lecture 2 slides also list the other reasons CAR-T fails: TGF-β, IL-10, Treg and MDSC are all getting in the way.

Why this course brings it up: It comes from the tumour microenvironment section of Lecture 2. The t-SNE plot he shows also marks CAF as one of the cell populations, but he never names it on its own as an exam question, so it stays in the understanding column. **Appears in:** Ch 9, Lect 2

▪ CAP

In one sentence: The protein that binds cAMP and helps RNA polymerase onto the lac promoter.

Detailed explanation: CAP is catabolite activator protein, the positive regulator in E. coli — the protein that switches on the use of catabolites. It has no activity on its own: it must first bind cAMP to become the cAMP–CAP complex before it can do any work. That complex sits directly on the promoter, recruits RNA polymerase, and so transcription of the lac genes goes from slow to strong. In the clinic and the lab it plays a double role: in bacterial recombinant systems, CAP–cAMP is a switch often used to crank expression up; and it also demonstrates another way a protein can be activated — binding a ligand and changing conformation, which is the same logic used to activate nuclear receptors. So the lac operon quietly teaches you two lessons at once: there are two ways to switch a gene up, remove the repressor and recruit the machinery, and a protein can be switched on by binding something.

Why this course brings it up: It is the second condition of the lac operon in Lecture 5; scenario case number 16 in his case bank is graded on exactly these links, allolactose and cAMP–CAP among them, so the pairing of CAP with cAMP is worth having solid and the two conditions should be walked through in order. **Appears in:** Ch 7, Lect 5

▪ Cap (5 prime cap)

In one sentence: The chemical cap on the front end of an mRNA: it protects the message and calls the ribosome over to collect it.

Detailed explanation: The 5′ end of a eukaryotic mRNA carries a 7-methylguanine (m7G) cap. It has three jobs: it protects the mRNA from being chewed up by nucleases at the 5′ end, it helps the mRNA get exported out of the nucleus into the cytoplasm, and it is recognised by the initiation factors of the ribosome, which is what starts translation going.
That last one is the part beginners find surprising, because the ribosome does not go hunting for the start of the message; it is handed the message. An mRNA without a cap works much less efficiently, and that is exactly the design point in the mRNA vaccine. The cancer mRNA vaccine described on Dr. Wong's Lecture 5 slide takes an mRNA encoding the antigen, delivers it into muscle cells inside lipid nanoparticles, and the cell translates it into protein, cuts that protein into peptides, and presents the peptides through MHC.

Why this course brings it up: The mRNA structure and mRNA vaccine section of Lecture 5. The 5′ cap was not named as an exam topic on its own, so this is conceptual understanding. **Appears in:** Ch 7,Lect 5

▪ CAP-cAMP (cAMP-CAP)

In one sentence: The CAP and cAMP pairing that sits on the lac promoter and turns transcription up.

Detailed explanation: This is the accelerator of the lac operon. When glucose concentration is low, cAMP rises inside the cell, cAMP binds CAP to form a complex, that complex sits directly on the promoter, and it pulls RNA polymerase over together with the general transcription factors.
That is the whole positive half of the story: when the cell is short of glucose, it says so by raising cAMP, and cAMP recruits CAP, and CAP makes the polymerase sit down more readily. But note the limits of its authority: CAP can only add to a system that the repressor has already released. It is an accelerator, never a master switch. Among the four states on his Lecture 5 slide, the last one is "cAMP-CAP present, but the repressor still sitting on the operator", and the result is still blocked, because the repressor is the master switch. That is also the most common way people get this question wrong.

Why this course brings it up: It is his second-level concept from Lecture 5, the "two conditions, not one" of the lac operon, and scenario question number 16 in the case bank names cAMP-CAP as a scoring point. He also uses it as the classic distractor in the four-state slide, where the trap is assuming that cAMP-CAP alone can overpower the repressor, and where the answer always depends on which of the two is still sitting there. **Appears in:** Ch 7, Lect 5

▪ CAR (Chimeric Antigen Receptor)

In one sentence: A synthetic receptor fitted onto a T cell, so that the T cell learns to recognise a tumour protein and then kill that cell.

Detailed explanation: The structure of a CAR, from the outside in, has five parts. There is the scFv, the variable region of an antibody, which does the actual recognising of antigen. Then the hinge.

Then the transmembrane segment, the piece that crosses the membrane. Then the co-stimulatory domain, such as CD28 or 4-1BB, which supplies the second signal. And at the very end there is CD3ζ, which carries the ITAM and is responsible for carrying the signal into the inside of the T cell. The whole receptor is called chimeric because it splices together part of an antibody and part of a T cell receptor. That way the T cell does not have to use its native TCR to recognise MHC at all. It recognises the protein on the surface of the tumour cell directly, which means it can kill even a tumour that has lost MHC.
Why this course brings it up: It comes from the CAR-T manufacturing workflow in Lecture 2 and from the engineering applications in Lecture 4, in the ADC and the CRISPR parts. CAR itself was not listed as a separately named exam question, but you do need to understand its structure and the reasons it fails, because that structure is what the whole therapy rests on.
Appears in: Ch 9, Lect 2

▪ CAR-NK

In one sentence: A CAR put on a natural killer cell instead of on a T cell.
Detailed explanation: His Lecture 3 points out one limitation: CAR-T cannot get into solid tumours. The Lecture 3 answer is to change the effector cell — CAR-NK and CAR-macrophages (CAR-M), whose infiltration profile is better than that of T cells. CAR-NK has a further attraction: NK cells already kill with perforin and granzymes, so they do not have to pass a T cell activation threshold first. But it is held back by the tumour microenvironment in the same way: hypoxia, adenosine, TGF-β and a lack of chemokine receptors (CCR2b and CCR4 are missing when CAR-T fails) all limit how far it gets in. Swapping the effector cell fixes the infiltration problem; it does not remove the siege.
Why this course brings it up: It is the CAR-NK and CAR-macrophages section of Lecture 3; it is the alternative he uses to show that "CAR-T is not the only answer", so it belongs to the concepts you are expected to understand rather than to the list you have to recite. His point is that the failure of CAR-T in solid tumours is a delivery problem before it is a design problem.
Appears in: Lect 3

▪ Carbohydrate

In one sentence: One of the four great biomolecules; mainly the fuel of the cell, and also structural material such as cellulose.
Detailed explanation: They come in three levels of complexity: monosaccharide (glucose, fructose), disaccharide (sucrose), and polysaccharide (glycogen, starch, cellulose). These three levels are simply a matter of how many sugar units are hooked together: one, two, or many.
Dr. Wong's slide points out that glucose and fructose have the same molecular formula, both $C_6H_{12}O_6$, but they are a six-membered ring (pyranose) and a five-membered ring (furanose) respectively. Functionally they split two ways: glycogen and starch are for storage, with glycogen built from α1,6 branches and α1,4 links, while cellulose is for structure, standing up inside the plant cell wall to resist tension. The difference between the two groups is not the sugar unit but the way the units are joined, and that is what decides whether a chain can be broken down for energy or has to hold its shape. In the human body, the liver and the kidney rely on glucose-6-phosphatase to turn the glucose-6-phosphate released from glycogen back into free glucose and let it out, and a defect in that enzyme produces an energy crisis for the whole body.
Why this course brings it up: One of the four classes of biomolecule in Lecture 1. Dr. Wong said plainly not to memorise metabolic pathway details and chemical structures, so you need the three levels of classification and the two functional routes, but not the chemistry inside them. **Appears in:** Ch 1,Lect 1

▪ Carbon

In one sentence: The central element of the organic chemistry of life.
Detailed explanation: Carbon has four electrons in its outermost shell, so it can form covalent bonds with four other atoms, and that four-valued nature lets it build long carbon chains, ring structures and double bonds.
The skeletons of sugars, lipids, amino acids and nucleotides all have carbon at the core. His description is blunt: carbon is the element that builds life.
Why this course brings it up: Dr. Wong spends a whole section on carbon in Lecture 1, but he does not make it an exam question. **Appears in:** Ch 1

▪ cardiotoxicity

In one sentence: The toxicity a drug does to the heart.
Detailed explanation: Early in drug development, testing cardiomyocytes made from iPSCs in vitro lets you see the cardiotoxicity before any human being has taken the drug. The same logic is used for the liver and the kidney: put human cells into organoids, run the drug screen on a multiwell plate, read out the activity first and then validate the target.
The advantage of this arrangement is that it is fast and repeatable: the cardiomyocytes made from one batch of iPSCs can be used to screen out the compounds that damage the heart and leave the time for the promising ones. You are using the patient's own cells, or cells close to them, instead of waiting for a trial to tell you the bad news.
Why this course brings it up: He does not name cardiotoxicity as an exam topic on its own, so it is a concept to understand; the point of Chapter 9 is that the iPSC platform catches the toxicity before the drug is ever given. **Appears in:** Ch 8, Lect 2

▪ Carrier

In one sentence: A person carrying one problematic copy of a gene who does not themselves get ill.
Detailed explanation: For a single-gene recessive disease, a carrier has one normal copy and one mutated copy, usually does not get ill, but has a 50 % chance of passing the mutation to the next generation with every pregnancy.
This line matters in the clinic, because the doctor has to separate three questions: will this person get the disease, will this person pass it on, and how large is this person's risk. Incomplete penetrance and differences in expression can make the answers come out differently, and genetic counselling exists precisely to sort those three questions apart.
Why this course brings it up: It is the pedigree and genetic counselling section of Lecture 3. Reading a pedigree and working out carrier probability is one of the scenario questions he named, case number 2 in the case bank, so the vocabulary itself is concept-only even though the pedigree reading is examinable. **Appears in:** Lect 3

▪ Caspase

In one sentence: Enzymes that cut up other proteins; the caspase family is what actually carries out apoptosis.
Detailed explanation: A caspase is a cysteine protease, and the name comes straight out of that: cysteine plus proteinase. It spends most of its life as an inactive procaspase, and when the time comes for it to be activated, an upstream initiator cuts it once.
That single cut exposes a new cleavage site, and the newly activated caspase then cuts the next caspase open in the same way, like a chain reaction, and that is how the cascade gets built. The final effector caspases, mainly caspase 3, go on to break up the cytoskeleton, the nuclear pores and the DNA repair proteins, and the cell falls apart into apoptotic bodies. The good thing about designing the whole thing as a cascade is that the signal can be amplified on the way down.
Why this course brings it up: It comes from the two apoptosis pathways in Lecture 2 and from the kill switch in the DNA damage response in Lecture 4. Caspase was never named on its own as an exam question, so it belongs to understanding the concept. **Appears in:** Ch 8, Lect 2

▪ Caspase 8 and caspase 9

In one sentence: The upstream starters of the two apoptosis pathways, in charge of the extrinsic and the intrinsic route respectively.
Detailed explanation: The extrinsic pathway begins outside the cell: a death receptor (DR5, for example) is bound by a ligand such as TRAIL, clusters, and cuts out caspase 8; caspase 8 not only cuts procaspase 3 directly, it also cuts bid to make tbid and passes the signal into the mitochondria.
The intrinsic pathway begins inside the cell: stress or DNA damage makes the mitochondria release cytochrome c, which assembles the apoptosome and activates caspase 9. Both routes converge on caspase 3, the executor that does the actual shredding. One route starts with a message arriving from outside the cell, the other with a decision the cell makes inside itself, and both end at the same executioner. The clinical point is that they give you two different ways in for a therapy — one you can trigger from outside the cell and one the cell triggers from inside itself.
Why this course brings it up: It is the starting point of the intrinsic and extrinsic apoptosis pathways in Lecture 2; he also mentions the crosstalk between caspase 8 and the mitochondria, so this is a concept to understand, and the two entry points are worth being able to draw from memory. **Appears in:** Ch 6, Ch 8, Lect 2

▪ Catabolism

In one sentence: The breakdown half of metabolism: it takes big molecules apart and releases energy.
Detailed explanation: A cell keeps itself alive on two things: taking apart, and building. Catabolism is the taking-apart half. Carbohydrates, lipids and proteins are broken down, the intermediates feed into the electron transport chain, and ATP comes out the other end.
The key point is that the carbon skeletons you get from breaking things down do not all have to be burnt for energy; they are equally the raw material for anabolism, the building of nucleotides, lipids and amino acids. Cancer cells push this to an extreme: running aerobic glycolysis they earn only about 4 ATP (a normal cell earns about 36), and yet they shunt a large share of the intermediates off to support proliferation instead. So catabolism is not only an energy statement; it is also a supply chain.
Why this course brings it up: One of the two directions he takes when he talks about cancer cell metabolism in Lectures 1 and 2. Dr. Wong said plainly that metabolic pathway details do not need to be memorised, so this is conceptual understanding.
Appears in: Ch 1

▪ cccDNA

In one sentence: The covalently closed circular DNA that HBV forms inside the nucleus.
Detailed explanation: After the DNA of HBV has entered the nucleus, the rcDNA is repaired into cccDNA, a circular molecule in which the two strands are fully paired and there is not a single gap in either of them.
It stays in the nucleus as an episome, it does not integrate into the host chromosome, and yet it keeps making viral RNA, which makes it a reservoir of virus and the hardest target for antiviral therapy to clear. The number of cccDNA molecules is large and they are stable, which is both the evidence that the virus persists in an infected hepatocyte and the reason that interfering with them needs a special strategy.
Why this course brings it up: Dr. Wong does not name cccDNA as an exam question on its own, so it is conceptual understanding, and the mechanism he has taught is that HBV cccDNA is a persistent episomal reservoir in the nucleus and is therefore hard to eradicate. **Appears in:** Lect 2

▪ CCR2b

In one sentence: The chemokine receptor that lets monocytes find a tumour by following CCL2, which is MCP-1.
Detailed explanation: For an immune cell to get from the blood to a tumour it needs a whole series of receptor to receptor conversations. CCR2b is one of them, and it recognises CCL2, which is the same thing as MCP-1.
The concentration of CCL2 around the tumour forms a chemical gradient, and T cells and monocytes move up that gradient, and that movement is chemotaxis. His list of the tumour microenvironment makes a particular point about this kind of chemokine receptor: if it is missing, the immune cells have no way of finding the road at all. So the receptor is what lets the cell read the gradient, and without it the cell is not lost so much as unable to perceive the direction to travel in.
Why this course brings it up: He does not name CCR2b as an exam question on its own, so it is concept-only. The backbone of this passage is the list of solid-tumour barriers. CCR2b is one line on that list. **Appears in:** Lect 2

▪ CCR4

In one sentence: The chemokine receptor that recognises CCL17, and one of the missing receptors on Dr. Wong's list.
Detailed explanation: The ligand for CCR4 is CCL17 (the chemokine TARC), and it mainly draws in lymphocytes. CCR2b and CCR4 are listed on his slide 63 as “missing chemokine receptors”, which means the tumour neighbourhood lacks these receptors, so immune cells can arrive nearby and still fail to gather.
This is one of the reasons CAR-T cannot get into a solid tumour, and the tumour can keep the door shut even when the immune cells arrive. The ligand it recognises is CCL17 (TARC), which mainly attracts lymphocytes, and with CCR2b it forms the two arrival routes, one for monocytes and one for lymphocytes.
Why this course brings it up: He does not name it specifically, so it belongs to conceptual understanding. The point it illustrates is the general one: the tumour can keep the door shut even when the immune cells arrive. **Appears in:** Lect 2

▪ CD28

In one sentence: The co-stimulatory signal in a CAR, telling the T cell that it has been legitimately switched on.
Detailed explanation: A T cell needs two signals to be fully activated. The binding of TCR to MHC is the first, and co-stimulation, with CD28 receiving B7 from the dendritic cell surface, is the second.
The CD28 signal turns on the PI3K-AKT-mTOR pathway, which pushes the cell's metabolism towards glycolysis, starts proliferation, and produces IL-2. So the rule for a T cell is this: antigen without co-stimulation gives you anergy, antigen plus co-stimulation means you start work. It is an AND, not an OR, and that is the whole point of the checkpoint system. CAR design writes CD28 straight into the receptor, which skips the step of having to go and hold a meeting with a dendritic cell first.
Why this course brings it up: It is the CAR structure and T cell activation material in Lecture 2. The fact that CD28 and CTLA-4 compete for the same signal is also one of the distractor pairs he lists, CD28 against CTLA-4, so you should be able to say which of the two lets the activation signal through and which of them blocks it. **Appears in:** Ch 9, Lect 2

▪ CD3 zeta (CD3z)

In one sentence: The signalling tail of the T cell receptor, used inside a CAR to switch the T cell on.
Detailed explanation: CD3ζ is the signalling subunit of the T cell receptor complex. It has no recognition function of its own, but it carries the ITAM, and once that is phosphorylated it recruits ZAP-70 and pulls the whole signalling chain into place below it.
A CAR uses it as its last segment, which is like borrowing the answer button of a telephone: the scFv is the microphone, CD3ζ is the loudspeaker, and when the antibody sees antigen the signal travels through the ITAM and into the nucleus, where it starts IL-2 transcription, proliferation and the cytotoxic response. His Lecture 2 slide draws CAR-T and full T cell activation side by side, and the whole point of that drawing is that a CAR bundles the first signal and the second signal together into the receptor in one go.
Why this course brings it up: It comes from the slide on CAR structure in Lecture 2. He never names CD3ζ on its own as an exam question, so it belongs to understanding the concept, and to the structure of the receptor rather than to any list of names.
Appears in: Lect 2

▪ CD45

In one sentence: A general surface marker for white blood cells, used to exclude immune cells from a sample.
Detailed explanation: When you do single-cell analysis on a tumour, what you are after are the cancer cells of epithelial origin. CD45 is the typical marker of white blood cells (lymphocytes above all), so the procedure runs like this: first grab the epithelial cells with EpCAM, then remove the CD45-positive immune cells, and keep only the EpCAM-positive/CD45-negative cancer cells for the sequencing that follows.
The EpCAM-positive/CD45-negative combination is the most commonly used condition for picking out epithelial cancer cells in clinical liquid biopsy and in single-cell sequencing. One marker says what you want; the other tells you what to throw away.
Why this course brings it up: He does not name CD45 as an exam topic in its own right, so it is a concept to understand; flow cytometry and gating themselves are the Tier 2 basic operation he lists. **Appears in:** Ch 9, Lect 2

▪ CD47

In one sentence: The "do not eat me" signal on the surface of a tumour, which makes the phagocytes stand down.
Detailed explanation: CD47 is a signalling molecule on the surface of a tumour cell, and normal blood cells read it. There is a matching SIRPα on the blood cell, and when the two meet the blood cell does not swallow the cell up.

A tumour expresses far more CD47 than normal tissue, which amounts to wearing a "do not eat me" disguise, so the macrophages let it pass and the tumour survives. Signals of this kind are therefore one of the new directions for immunotherapy targets: blocking CD47 with an antibody is the same as forcing the macrophages to bite again.
Why this course brings it up: Dr. Wong lists it among the hostile physical environment on slide 63, and he does not name it as an exam question on its own, so it is conceptual understanding. **Appears in:** Lect 2

▪ CD73

In one sentence: The enzyme that turns AMP into adenosine, and makes the area around a tumour immunosuppressive.
Detailed explanation: ATP and AMP outside the cell are converted by CD73 into adenosine, and adenosine is a strong immunosuppressive signal: it inhibits T cell activation and promotes Treg. So a tumour that pushes CD73 up has laid a layer of immune brake around itself.
He lists it together with CD47, IDO, hypoxia and low pH as part of the hostile environment of a solid tumour. So a rise in the adenosine concentration around the tumour is a local turning down of the immune response, and the list puts this kind of signal alongside hypoxia and low pH to make the point that more than one mechanism is at work. The enzyme is interesting because the substrate is everywhere, ATP and AMP are normal in the extracellular fluid, so the tumour does not have to make anything unusual, it only has to convert what is already there.
Why this course brings it up: He does not name it in particular, so it is concept-only. What it illustrates is that the tumour environment attacks the immune response from several directions at once, and this is one of the directions. **Appears in:** Lect 2

▪ CDH17 (Cadherin-17)

In one sentence: CDH17 is the cell surface protein Dr. Wong's laboratory has studied for twenty years, and it is the tumour-targeting end of his bispecific antibody.
Detailed explanation: CDH17 is a calcium-dependent cell adhesion molecule. It is expressed very highly in the intestinal epithelium and much more weakly in other tissues. Dr. Wong used it as his research object for twenty years, and eventually made it the tumour-targeting half of a bispecific antibody.
The tumour specificity is assembled out of three pieces: normal liver (which lacks CDH17) is safe; normal gut (which has plenty of CDH17 but is not sensitive to TRAIL) is also safe; only colorectal cancer, which has both, gets killed. The antibody therefore does not need to be clever about which tissue it hits — the cleverness is in demanding that both of its targets be present at once. The clinical reality is that cancer cells throw this molecule away.
Why this course brings it up: It is his own research story from Lecture 2, and the only complete case in the whole course that runs from the bench to the clinic; case numbers 6 and 13 in his case bank both sit around it. If you are asked how a bispecific antibody can spare normal tissue while still killing the tumour, this is the story to tell. **Appears in:** Ch 8, Lect 2

▪ CDK4/6 inhibitor

In one sentence: It blocks the kinase that lets a cell leave G1, so the cell cycle stops; palbociclib is one of them.
Detailed explanation: The mechanism runs like this. For a cell to move from G1 into S phase, Cyclin D first has to bind and activate CDK4/6, and the activated CDK4/6 then phosphorylates pRb. Once pRb has been phosphorylated it stops holding E2F down, E2F is released, and the genes that S phase needs start to be transcribed.
The Cip/Kip proteins (p21, p27, p57) are the brakes on this pathway in the first place. A CDK4/6 inhibitor seals the very first gate, and the cell stops in G1. That makes it a therapy which does not kill the cell but only freezes it. Compare that with a cytotoxic drug, which destroys the cell outright and takes the tumour with it: here the cell stays intact, it is simply parked, and the decision of whether to divide is taken away from it. Dr. Wong's slide also notes that p53 mutations turn up in about 40 % of cancers.
Why this course brings it up: The clinical application of the cell cycle and drugs in Lecture 2, and it is also the cell cycle arrest that Dr. Wong named. But he did say you do not have to memorise the cyclin and CDK pairings. **Appears in:** Ch 8,Lect 2

▪ CDKN2A

In one sentence: The gene for p16; when its promoter is methylated the cell loses one of its cycle brakes.
Detailed explanation: CDKN2A is a compact gene with a lot of job. Using two different reading frames, it encodes two proteins: p16INK4a, from the INK4 family, which inhibits only CDK4/6, and p14ARF, from another set of codons, which stabilises p53. So once its promoter is methylated, the cell loses two brakes at once, not one. Nothing in the DNA sequence has been damaged here; the gene is simply switched off because its promoter has been chemically sealed, and the cell carries on as if the brake were gone. His Lecture 3 uses this gene for a complete statistical demonstration, and the numbers are worth memorising: when CDKN2A methylation is detected, the odds ratio for loss of p16 is 12.7 (95 % CI 7.84–20.7, $p < 0.0001$); and in the multivariable regression the strongest predictor is CIMP-high (OR 39.6, 95 % CI 20.6–76.1, $p < 0.0001$). Watch the position too, because exon 2 has higher diagnostic value than exon 1, and that contrast is one of the pairs he likes to put in front of you.
Why this course brings it up: OR 12.7 and OR 39.6 are both on his list of precise numbers, and exon 1 against exon 2 is one of his distractor pairs. He expects you to quote the odds ratios together with their confidence intervals, not just to say that methylation matters. **Appears in:** Ch 4, Lect 3

▪ CDR1, CDR2 and CDR3

In one sentence: The three hypervariable loops in the variable region of an antibody, which together form the antigen-binding site.

Detailed explanation: The variable region of an antibody, VH and VL, is coded in the germline as a fixed sequence, but once V(D)J recombination has assembled it, unpredictable changes appear around the joints where the pieces were joined, and somatic hypermutation adds another layer of mutation on top of that.

Together these produce the three hypervariable loops CDR1, CDR2 and CDR3. Those three loops stick out on the surface of the antibody and press directly against the antigen, and this is where a lock and a key actually meet each other and decide whether they fit. The third loop is usually the most variable of the three, and precisely because it varies so much, these three loops are the hard part that all of chimeric antibody and humanisation work has to deal with.

Why this course brings it up: It comes from the antibody anatomy slide in Lecture 1 and from the antibody section of Lecture 2. Neither CDR nor antibody diversity was ever named on its own as an exam question. **Appears in:** Ch 2, Lect 1

▪ CEA

In one sentence: A fetal antigen and a blood marker that rises in colorectal cancer.

Detailed explanation: CEA stands for carcinoembryonic antigen. It is present at very low levels in normal adult tissue but is expressed heavily on the fetal gut and on colorectal cancer cells, so a rise in blood CEA is often read as a sign of tumour burden.

Dr. Wong positions CEA as a circulating surrogate marker, a stand-in that is far cheaper than a CT scan, so that a trial does not have to scan every participant every few weeks. The substitution is the whole point: the trial is really measuring the marker and using it as a proxy for the thing it cannot afford to measure. A surrogate saves money and time, and the price is that you have to have shown the surrogate tracks the real thing. And the reverse also holds: one CEA value on its own does not make a diagnosis of cancer.

Why this course brings it up: He explicitly presents it as a surrogate endpoint in clinical trials, and it is also the content of the eleventh scenario type (MMR/MSI result → CEA as the surrogate). But you cannot conclude from a CEA value that the patient has cancer. **Appears in:** Ch 10, Lect 1, Lect 4

▪ Cell adhesion molecule (CAM)

In one sentence: An adhesion protein on the cell surface that decides who is allowed through the tissue.

Detailed explanation: A white blood cell has to get out of the blood vessel and into tissue in four steps, and each step has its own molecular player; his Lecture 2 slide draws the whole sequence very clearly.

Step one is rolling, handled by P-Selectin, E-Selectin and integrin. Step two is firm adhesion and activation, handled by ICAM-1 and VCAM-1. Step three is crossing the endothelium, handled by PECAM-1. Step four is moving towards the chemical signal, handled by MCP-1, a chemokine. So a CAM is not just a glue; it is also a pass. Which CAM a cell expresses decides which tissue it can enter. And because the same set of molecules is reused everywhere — inflammation, infection, tumour infiltration — the list is worth learning as a chain rather than as four separate facts.

Why this course brings it up: It is the leukocyte infiltration and tumour microenvironment section of Lecture 2; he never named CAM as an exam topic on its own, so it belongs to the concepts you are expected to understand. The four-step version is the version he draws, so if you remember nothing else, remember the order. **Appears in:** Ch 9, Lect 2

▪ Cell cycle

In one sentence: The timetable of G1, S, G2, M plus cytokinesis, the schedule by which a cell copies itself.

Detailed explanation: The cell cycle is an ordered timetable. G1 is the growth and preparation phase. In S phase the DNA is copied, and the content goes from 2N to 4N. G2 is the check before division.

M phase is mitosis, where the chromosomes are dealt out, and finally cytokinesis cuts the cytoplasm into two halves. Sitting beside the timetable is G0, the quiet state a cell retreats into when it leaves the cycle. What drives the timetable is a switch made of two parts: the cyclin is the key, and its concentration rises and falls with the cycle, while the CDK is the machine, which is always on standby and has to be switched on by the cyclin before it will work. In the laboratory, dropping the serum down to 0.5–1 % (serum starvation) is enough to stop cells in G1 and synchronise them. The change from 2N to 4N in S phase is the single number you should carry out of this entry.

Why this course brings it up: It is the experimental design foundation in both Lecture 2 and Lecture 4. But Dr. Wong explicitly said you do not need to memorise the cyclin and CDK pairings. **Appears in:** Ch 8,Lect 2

▪ Cell cycle arrest

In one sentence: The cell stops at a checkpoint and does not divide; this is exactly what a CDK4/6 inhibitor produces.

Detailed explanation: The cell cycle is built with checkpoints, gates that let the cell through only once everything before has been finished. Arrest means holding the cell at one of those gates, keeping it alive but not dividing.

Nothing is damaged and nothing is killed; the cell simply sits at the gate with the brake on. That is a completely different therapeutic logic from killing the cell, and it works on fast-growing tumour cells while leaving normal cells less affected. A CDK4/6 inhibitor achieves arrest by shutting the gate between G1 and S. The other side of the same coin is that checkpoints are also the weak point of therapy: a tumour that can switch off p53, the last brake, finds it easier to keep proliferating.

Why this course brings it up: It is the cell cycle and drug section of Lecture 2, and it sits on the same thread as the "binucleated cell" question he named, namely that mitosis and cytokinesis are separate processes. Arrest itself is never singled

out by name, so you are expected to know it as a concept and to recognise it when a CDK4/6 inhibitor is described. **Appears in:** Ch 8, Lect 2

▪ Cell-free DNA (cfDNA)

In one sentence: DNA fragments drifting about in the blood; a tumour sheds some of itself into them.
Detailed explanation: Plasma already contains plenty of DNA, and it comes from dead white blood cells and from skin cells. When tumour cells die they release their own DNA into the blood as well, so the composition of cfDNA changes, and that is the physical basis of a liquid biopsy.
The key point is that those fragments still carry epigenetic marks on them: the methylation state can be read straight out of the blood, with no tissue section and no biopsy needed. That is why in Lecture 3 he lists methylation markers such as SEPT9, SFRP2 and SDC2, all of them methylation marker genes, as diagnostic biomarkers for early blood screening. The practical difficulty is that most cfDNA comes from normal cells, so the tumour signal is only a small fraction of the total that you measure.
Why this course brings it up: It comes from the non-invasive methylation testing section of Lecture 3. He never names cfDNA on its own as an exam question, so it belongs to understanding the concept, but it is the substrate that the methylation tests read from. **Appears in:** Ch 4, Lect 3

▪ Cell-type specificity

In one sentence: The same gene is switched on only in certain cell types, and switched off in all the others.
Detailed explanation: Every cell in your body carries the same DNA, but the list of genes a nerve cell expresses and the list a liver cell expresses are worlds apart. The source of that difference is the set of transcription factors each cell carries, so the same promoter is watched by different activators and repressors in different cells.
Two cells with identical genomes can therefore look and behave completely differently, and no amount of looking at the DNA alone will tell you which. The gene is not the difference; the switch is. In Lecture 5 he ties this to time-specificity as well: "the expression is usually cell-specific and at the same time, it is time-specific… This is the concept that I always emphasize." The clinical meaning is very practical — a gene that is only expressed in one cell type is the ideal therapeutic target, because hitting it spares everything else. In other words, the gene list a cell expresses is not the list of what its DNA allows; it is the list of what the switches in that particular cell happen to be set to allow.
Why this course brings it up: It is one of the concepts he says he "always emphasize" in Lecture 5, and it is the landing point of the cis / trans regulatory section; cell-type specificity itself was never named as an exam topic. That is why he keeps coming back to it: it is the concept that connects the whole of gene regulation to something you can actually use in the clinic. It is the concept he keeps returning to because it is the one that ties the whole of Lecture 5 together: cis elements, trans factors, enhancer looping and time-specific control all end up here. **Appears in:** Ch 7, Lect 5

▪ Cellular memory

In one sentence: The cell remembers the state of a switch long after the signal that set it has gone.
Detailed explanation: This is the most surprising corner of epigenetics. There is a positive-feedback loop between DNA methylation and histone modification: DNA that has been methylated recruits proteins that build compact chromatin, and compact chromatin in turn makes the methylation easier to maintain.
The state therefore maintains itself, which is why it can still be remembered after several rounds of cell division. Mechanically the trick is the maintenance methyltransferase, DNMT1, which copies the existing marks onto the daughter strand after replication. The signal that started all of it is long gone; the switch is still in the same position. That is what makes it memory rather than merely a response.
Why this course brings it up: The cross-talk section of Lecture 3, where Dr. Wong said plainly that DNA methylation and histone marks reinforce each other to form a positive feedback loop and produce cellular memory. He did not list the term as an exam topic on its own, but you have to be able to tell that causal chain. **Appears in:** Ch 4,Lect 3

▪ Cellulose

In one sentence: The structural polysaccharide that makes up plant cell walls.
Detailed explanation: Cellulose is the polysaccharide plants use for structure. It is a polysaccharide like starch and glycogen, but the jobs are different: starch and glycogen are for storage, cellulose is the wall.
The difference is in the bonding. Starch and glycogen are linked by α1,4 bonds, which human enzymes can take apart; cellulose is linked by β1,4 bonds, and we have no enzyme that can break that bond, so we can only swallow it as fibre and get nothing out of it. Because humans cannot digest cellulose, it counts as dietary fibre, and ruminants such as cows and sheep get the job done with the help of symbiotic microbes. The two look almost identical on paper, and in your gut they behave in completely opposite ways.
Why this course brings it up: It is the representative of the "structural material" class of carbohydrates in Lecture 1. He does not name it as a question, so it is concept-only, but it is the example he uses to separate a structural polysaccharide from a storage one. **Appears in:** Ch 1

▪ censored

In one sentence: A patient who drops out during follow-up, or who has not yet had the event.
Detailed explanation: The small tick marks on a Kaplan–Meier curve are the censored patients: they left the study, they were lost to follow-up, or they were still alive at the time of drawing but had not had the event.

A censored patient has not provided negative information, because at that moment he still "has no event", so the curve holds flat at the point where he leaves instead of dropping.
Why this course brings it up: Reading a KM curve is the first-grade, A+ question he asks in every lecture, and interpreting a censored observation is a compulsory part of it. **Appears in:** Ch 10, Lect 1

▪ Central dogma

In one sentence: The rule that genetic information flows one way: DNA, then RNA, then protein.
Detailed explanation: The central dogma says that a DNA sequence can serve as a template for making RNA, and that an RNA sequence can serve as a template for making protein. The full workflow as he presents it in class is this: DNA is transcribed into mRNA inside the nucleus, the mRNA is exported to the cytoplasm, the ribosome reads it and joins amino acids together into one polypeptide, and that polypeptide folds into a protein.
The difference between eukaryotes and prokaryotes lies exactly here. A prokaryote has no nucleus, so transcription and translation can happen in the same place at the same time. A eukaryote keeps them apart, and that is why the RNA has to be processed, with a cap, with a poly-A tail and by splicing, before it can leave the nucleus.
Why this course brings it up: It is the framework the whole course is built on, and it is the backbone of all his molecular biology storytelling. He never names it on its own as an exam question, so it belongs to understanding the concept. **Appears in:** Ch 7, Lect 1, Lect 5

▪ Centromere

In one sentence: The narrow waist in the middle of a chromosome, where the spindle fibres grab hold.
Detailed explanation: A chromosome has two arms: the short one is called the p-arm and the long one the q-arm, and the two meet at the centromere. Before cell division, spindle fibres (microtubules) reach out from the two poles and grab the chromosome through the kinetochore at the centromere, which lines the chromosomes up on the equatorial plate in metaphase. That narrow waist is therefore what decides the shape a chromosome takes under the microscope, and it is also what karyotyping (G-banding) uses to identify each chromosome. In his third lecture Dr. Wong used the ideogram of chromosome 8 to show the band numbers on the p and q arms, and he made a point of reminding you how to read them: “q-two-one-point-one”, not “q-twenty-one-point-one”. So the same number written as q21.1 is a position on the long arm, and saying it out loud is how you make sure you have put the digits in the right order.
Why this course brings it up: The chromosome anatomy section of Lecture 3. He singled out how to read q21.1 as a common trap, and the centromere is part of his G-banding resolution discussion, so you should be able to place it on the chromosome and say what the number means. **Appears in:** Ch 3,Lect 3

▪ Centrosome

In one sentence: The organelle that organises the microtubules and builds the spindle.
Detailed explanation: The centrosome is the microtubule organising centre of the cell. During mitosis it duplicates into two, the two copies move to opposite poles of the cell, and each one pulls out microtubules from there, forming the spindle, a traction framework.
The spindle fibres grab the centromere and pull the two sets of sister chromatids towards opposite poles, and it is this structure that does the work. If you have ever watched cells divide, the spindle is the thing pulling the chromosomes apart, and the centrosome is where the pulling starts. Cancer cells often have abnormal numbers of centrosomes, so chromosome distribution goes wrong and aneuploidy results — in Lecture 3 he connects aneuploidy with failed chromosome segregation. Plant cells have no typical centrosome and use another mechanism instead.
Why this course brings it up: It is the mitosis and organelle section of Lecture 2, and part of the cause of chromosomal abnormality in Lecture 3; it was never named as an exam topic. **Appears in:** Ch 8, Lect 2

▪ Cetuximab

In one sentence: An anti-EGFR antibody; it works in K-Ras wild-type rectal cancer and fails in patients whose K-Ras is mutated.
Detailed explanation: Cetuximab binds EGFR and blocks the growth signal it receives, which only helps if the tumour is still listening to EGFR. EGFR is a receptor tyrosine kinase sitting at the very top of that chain.
The drug strategy has two forms: anti-EGFR antibodies such as cetuximab, which block the ligand at the receptor, or small-molecule kinase inhibitors, which press directly on the kinase activity. But the strategy carries a condition, and the condition is that the downstream has to be clean. The signal runs EGFR → RAS → RAF → MEK → ERK, and a KRAS mutation (most often G12C, G13, Q61) locks RAS in its active state, so it keeps sending signal without any push from upstream. That is why K-Ras mutant rectal cancer does not respond to cetuximab while K-Ras wild-type does, and it is the cleanest example you will get of a predictive biomarker deciding whether you use the drug at all. He also points out that KRAS G12C is about 13 % of lung adenocarcinoma and about 3 % of colorectal cancer.
Why this course brings it up: One of the MCQ question types he lists is "which drug is a patient with a mutation in this gene most likely to respond to". K-Ras mutation leading to cetuximab failure is the standard answer, and it is also the reason you test K-Ras in the first place. **Appears in:** Ch 10, Lect 1

▪ Chaperone

In one sentence: Proteins that help another protein fold correctly and stop it from clumping into a mess.
Detailed explanation: Protein folding is a search problem. A chain has to find the right three-dimensional shape, and the energy landscape is complicated enough that it gets stuck half way. A chaperone is the helper that walks the landscape with it: it occupies the regions that fold badly and lets the protein take small steps instead.
Inside the cell, the rough endoplasmic reticulum is in charge of folding and glycosylation; a protein that cannot get out of the ER is left where it is and is broken down. The most direct clinical application is recombinant protein production: if you induce too early, the protein misfolds into insoluble inclusion bodies.
Why this course brings it up: It comes from the protein structure and quality control material in Lectures 3 and 6. He never names chaperone on its own, but the part on when to induce a recombinant protein is impossible without this concept. **Appears in:** Ch 2

▪ CHD

In one sentence: One of the four families of chromatin-remodelling complex, and it moves nucleosomes by reading histone marks.
Detailed explanation: The members of the CHD family carry a chromodomain, which can read a modification mark on a histone, H3K4me3 for example, and then the ATP-powered part of the complex pushes, drags or flings the nucleosome away. His point is that there are four families of remodelling complex, SWI/SNF, INO80, ISWI and CHD, and each one does a different kind of hauling job in different organisms and in different chromatin states. The families have their own preferences: some push the nucleosome aside to expose the promoter, and some shift the nucleosome to a different position, which rewrites the DNA. So the mark on the histone is not just a note for a reader, it is the address that tells the machine which piece of DNA to move.
Why this course brings it up: He does not ask you to memorise the membership of the families, so it is concept-only. The Tier 2 point is that nucleosomes slide along the DNA to expose or to cover the promoter. The families are the detail, the sliding is the idea, and the idea is what is examinable. **Appears in:** Ch 4, Lect 3

▪ Checkpoint

In one sentence: A checkpoint in the cell cycle that confirms the earlier work is finished before it lets the cell proceed.
Detailed explanation: The logic of a checkpoint is "check first, then continue". The G1/S checkpoint confirms that the cell is big enough, the DNA is intact and nutrients are available. The G2/M checkpoint confirms that DNA replication is complete.
After M there is another one confirming that the chromosomes were distributed correctly. The proteins that grant or withhold permission (p53, for example) pause the cycle when they find DNA damage and start the repair; if the damage is too severe to fix, they switch to apoptosis instead. The DDR cascade in Lecture 4 works exactly this way: ATM and ATR sense the damage, and the effector p53 decides whether to stop, repair or die. The clinical significance is that cancer cells very often switch this killer off. The pause is the point — a cell that keeps dividing with broken DNA is a cell that turns its own accident into a cancer.
Why this course brings it up: It is in Lecture 2 and Lecture 4; he never named checkpoint as an exam topic on its own, but the causal chain of damage beyond repair leading to apoptosis is something you have to understand. It is in Lecture 2 and Lecture 4; he never named checkpoint as an exam topic on its own, but the causal chain of damage beyond repair leading to apoptosis is something you have to understand. **Appears in:** Ch 8, Lect 2

▪ Chemokine

In one sentence: A signalling protein that makes white blood cells move towards one particular place, by following a concentration gradient.
Detailed explanation: A chemokine is a chemical signal that guides immune cells. The last step of a white blood cell drilling out of the blood vessel and into tissue, the part where it moves towards its target, is completed by the concentration gradient of a chemokine such as MCP-1.
The gradient is the point: a receptor on its own is no use if there is nothing on the other side of it pulling the cell in. A chemokine's other job is deciding whether immune cells can get into the tumour at all. Dr. Wong's second lecture slide lists the reasons CAR-T fails, and one of them is that the effector cells lack the chemokine receptors CCR2b and CCR4, so even when the killing power is enough they cannot get in. That is why the missing receptor sits on the list of reasons CAR-T fails, and not as a footnote. This is one of the most underrated obstacles in solid tumour immunotherapy, and it is also why CAR-NK and CAR-macrophages infiltrate better. Think of the chemokine as the postcode on the destination: a cell that knows how to kill still has to know where to go, and the gradient is the address.
Why this course brings it up: The white blood cell infiltration section of Lecture 2. He explicitly said not to memorise the details of the signal transduction cascade on the slide, so all that is required here is that you understand what a chemokine does and why the receptor matters. **Appears in:** Ch 9,Lect 2

▪ chemotaxis

In one sentence: Cells moving along a chemical concentration gradient.
Detailed explanation: Chemokines such as CCL2 and MCP-1 around a tumour form a gradient running from high to low concentration, and immune cells read out which side is denser and swim that way.
The fourth step, “following the chemical signal”, is chemotaxis, and it is the last step of the four-step journey of the immune cell reaching the tumour that Dr. Wong draws.
Why this course brings it up: He does not name chemotaxis as an exam topic on its own, so it belongs to conceptual

understanding. It is the last of the four steps, and knowing where it sits in the sequence is what makes the diagram readable.
Appears in: Ch 9, Lect 2

▪ CHO cell

In one sentence: A hamster cell line used to manufacture human therapeutic proteins, because it puts on the right sugars and puts them on in the right way.
Detailed explanation: A great many therapeutic proteins need glycosylation, which means decorating the protein with particular sugar chains, and those sugar chains decide the half-life, the stability and the potency of the drug you end up with.
Bacteria cannot make the correct sugar modifications at all, so you have to use a mammalian cell instead. The CHO cell, Chinese hamster ovary, is the standard workhorse of the industry: it grows fast, it expresses protein reliably, and it has been through regulatory approval more times than you can count, which is why the industry settled on it in the first place. The price is that it is still not quite identical to a human cell, so you need codon optimisation to make the gene read smoothly. His Lecture 2 slides also list the manufacturing details for CAR-T: beads coated with anti-CD3/CD28 antibody, plus IL-2 and IL-15, to activate the T cells and keep them happy while they grow.
Why this course brings it up: He uses it in Lectures 1 and 5 to talk about recombinant protein production, and it is his standing example of why the industry cannot simply use bacteria for every drug. He said out loud, "So remember that when you use the CHO cell, you need to do the codon optimization", and that is a link you are expected to remember, because the two of them always travel together in his story. **Appears in:** Ch 1, Ch 2, Ch 7, Lect 1, Lect 5

▪ Cholesterol

In one sentence: A lipid wedged into membranes that keeps them fluid at low temperature.
Detailed explanation: Cholesterol is a permanent member of animal cell membranes, sitting in between the phospholipids. It plays a double role: at low temperature it stops the phospholipids packing tightly, so the membrane does not solidify;
above room temperature it limits how far the phospholipids can move, so the membrane does not get too fluid. Think of it as a spacer that keeps the phospholipids from freezing into a solid slab when it gets cold, and as a weight that keeps them from sliding around too freely when it gets warm. Its presence therefore keeps membrane fluidity steady as the temperature changes, and that is the heart of the idea of the liquid crystalline state. It is also the precursor for steroid hormones and bile acids, so it is tied to hormone signalling, lipid metabolism and atherosclerosis all at once.
Why this course brings it up: This belongs to basic biochemical background. In Lecture 1, when he teaches lipids he teaches fatty acid, triglyceride and phospholipid, and does not expand on cholesterol. He says outright that the chemical details need not be memorised, so it is concept-only. **Appears in:** Ch 1

▪ Chromatin

In one sentence: DNA plus the proteins that package it; the manual together with its binding.
Detailed explanation: Chromatin is the general name for DNA together with histone — like a recipe plus its binding and its bookshelf. The packaging steps up in size as you go: 2 nm double helix → 11 nm nucleosome "beads on a string" → 30 nm chromatin fibre → 300 nm extended fibre → 700 nm condensed fibre → 1,400 nm mitotic chromosome.
Packaging is not only about making things fit; it directly decides whether a gene can be read. Loose euchromatin is active, tight heterochromatin is silent, and physically it blocks transcription factors and enhancers from reaching the promoter. So chromatin is not wrapping paper; it is itself a layer of regulation. And because the packaging can be shifted, the same genome can be read in different ways in different cells — that is the connection back to cell-type specificity. The manual is the same; how tightly it is bound decides which pages are open. The layers themselves are worth keeping as a ladder, because each step up is a further increase in how compactly the same information is stored.
Why this course brings it up: It is the core concept of Lecture 3, and it is also the second-level concept he lists, "chromatin accessibility"; chromatin accessibility and nucleosome sliding are things he explicitly says "you need to understand". The packaging levels themselves he does not expect you to recite. It is the core concept of Lecture 3, and it is also the second-level concept he lists, "chromatin accessibility"; chromatin accessibility and nucleosome sliding are things he explicitly says "you need to understand", while the packaging levels themselves he does not expect you to recite. **Appears in:** Ch 3, Lect 3

▪ Chromatin fibre

In one sentence: The 30 nanometre fibre you get when nucleosomes are packaged up again, one step above the beads on a string of DNA.
Detailed explanation: Nucleosomes (about 10 nm across) strung together make a string of beads, and folding that string further gives the 30 nm chromatin fibre. Above that come the higher order structures at 300 nm and 700 nm, and finally the tightest form of all, the 1,400 nm chromosome during division.
The historical four-level packaging model comes from the 2003 compilation by Felsenfeld and Groudine, and it is the slide he uses in Lecture 3. One thing to keep in mind: the 30 nm fibre, taken as an intermediate state that is generally present inside living cells, has been questioned by more recent microscopy work, but in his teaching material it is the standard framework, so when you are answering exam questions, go with his version. The numbers are worth holding on to, because they form a ladder rather than a list: 10, 30, 300, 700, 1,400.
Why this course brings it up: The hierarchy of chromatin section in Lecture 3. He said plainly that the electron microscopy images of chromatin do not need to be memorised, but the 30 nm figure and the relationships between the levels do have to be understood. **Appears in:** Ch 4,Lect 3

▪ Chromatin remodelling complex

In one sentence: An ATP-powered machine that pushes nucleosomes along DNA, exposing genes or covering them up.
Detailed explanation: Nucleosomes slide along DNA, and there are two driving forces behind that. The first is the intrinsic sequence affinity between DNA and histone: some positions in the sequence are simply easier for a histone to occupy than others, so there is a preferred position even before any machine acts.
The second is the ATP-dependent chromatin remodelling complex: it uses the energy of ATP to push nucleosomes away, move them along, or swap them out entirely, and that is how a promoter ends up exposed or covered. His Lecture 3 lists four major subfamilies, SWI/SNF, INO80, ISWI and CHD, each with members in yeast, flies, humans and mice. Each family has its own preference: some push nucleosomes aside to let the promoter come out, others shift a nucleosome to a different place so that the DNA is rewritten. Once nucleosome positions move, whether a transcription factor or an enhancer can reach the promoter changes with them.
Why this course brings it up: It is the nucleosome positioning and chromatin accessibility material in Lecture 3. He lists it as a second-level concept, "nucleosomes slide along DNA to expose or cover promoters", and he points out those two driving forces, the intrinsic DNA-histone affinity and the ATP-powered machine. You do not have to memorise the four subfamilies, but you do have to know that the sliding is what regulates access, and that access is what decides whether a transcription factor can reach the promoter at all. **Appears in:** Ch 4, Lect 3

▪ Chromosome

In one sentence: A well-printed manual of the genome, neatly bound.
Detailed explanation: A chromosome is what the DNA becomes once it has been packaged to the extreme, a printed manual. Humans have 23 pairs of chromosomes, and in each pair one comes from the father and one comes from the mother; during mitosis they copy themselves into sister chromatids and then separate at anaphase. Structurally each one has a p-arm and a q-arm, joined by the centromere, with a telomere at each end of both arms. Clinically, abnormalities in the number and in the structure of chromosomes are the origin of genetic disease and of cancer: the number being wrong is aneuploidy, while the structural changes include deletion, duplication, inversion and translocation. He also reminds you that karyotype analysis only reaches a resolution of 0.5–10 Mb, so a point mutation has to be found by sequencing instead of by looking.
Why this course brings it up: It comes from the chromosome anatomy and the karyotype resolution in Lecture 3. He lists the set of numbers that includes 23 pairs of chromosomes, Trisomy 21 and Turner 45,X, and he also reminds you how to read q21.1, so that you can go back and decode any karyogram he puts on the screen. **Appears in:** Ch 3, Lect 3

▪ CIMP

In one sentence: CpG island methylator phenotype: a tumour-level pattern in which methylation is spread widely across many promoters instead of being confined to a single gene, and which picks out one particular group of colorectal cancers.
Detailed explanation: What CIMP describes is a feature of the tumour as a whole: not that one single gene has been methylated, but that the promoters of many CpG islands have been methylated at the same time, in numbers large enough to form a recognisable pattern.
Dr. Wong quantified it in Lecture 3 with multivariate logistic regression. For CDKN2A methylation, the strongest predictor is CIMP-high relative to CIMP-0, with an OR of 39.6 (95 % CI 20.6–76.1, $p < 0.0001$), and CIMP-low also carries an OR of 5.30. An OR of 39.6 means the odds of that methylation being present are nearly forty times higher in a CIMP-high tumour than in a CIMP-0 one, which is why the classification earns its keep rather than being decorative. The clinical significance is that CIMP is associated with a worse prognosis. One thing to be careful about: do not run CIMP and MSI together. They are related, but they are different concepts, and they answer different questions about the same tumour. CIMP asks how much of the promoter landscape is switched off at once; MSI asks whether the mismatch repair machinery is working.
Why this course brings it up: The OR of 39.6 and its 95 % CI are both on his list of precise numbers. CIMP itself is also on his list of clinical methylation biomarkers. So you should be able to quote the figure exactly, including the confidence interval around it, and you should know which of the two methylation patterns you are being asked about. **Appears in:** Ch 4,Lect 3

▪ Cip/Kip

In one sentence: One of the two big families of CDK inhibitors; its members are p21, p27 and p57.
Detailed explanation: The cell cycle has two braking systems. The Cip/Kip family (p21, p27, p57) is the broad-spectrum type: they bind the whole Cyclin–CDK complex and shut it down entirely. The other family, INK4 (p16/INK4A, p14ARF, p15, p18, p19), only targets CDK4/6 and is the specific type.
The difference between the two families is the difference between pulling the plug and stepping on it: the broad inhibitors take out the whole machine, the specific ones take out one gear. Both families sit side by side on his Lecture 2 slide with the Cyclin D–CDK4/6 → pRb phosphorylation → E2F release pathway. His Lecture 3 data also explains why the p16 line matters: p16 loss caused by CDKN2A methylation is strongly associated with the p16 loss seen in tumours (OR 12.7). The two brakes are not interchangeable, and a cell that loses one of them is not the same as a cell that loses the other.
Why this course brings it up: It is labelled on the cell cycle slide in Lecture 2; but he explicitly said you do not need to memorise every cyclin and CDK pairing, so this belongs to the concepts you are expected to understand. The label on the slide is what you need; the membership of each family is not what he will test. **Appears in:** Ch 8, Lect 2

▪ Circular DNA

In one sentence: DNA that is a ring rather than a line; mitochondrial DNA is a circle of about 17 thousand bases.
Detailed explanation: Mitochondrial DNA is double-stranded and circular, about 17 kb, carrying 37 genes, and the circular gene map labels the D-loop, 12S, 16S, ND1, COX1, ATP8/6 and CYTB among them.
Being circular has three practical advantages: there is no end-of-line DNA end problem, so replication does not need telomeres; the two strands can serve as templates for each other, so permanent mutations accumulate less easily; and the whole genome can be transcribed in one go into one long polycistronic message, so one promoter can drive several genes. His Lecture 4 links the circular structure to telomerase: telomerase works by carrying its own RNA template and extending the 3′ end of a linear chromosome, and that is the same trick a circle simply does not need.
Why this course brings it up: It is the mitochondrial section of Lecture 2 and the telomere section of Lecture 4. The number of about 17 kb comes from his slides, so it is concept-only, but the contrast between circular and linear DNA is the frame he wants you to hold on to. **Appears in:** Ch 3, Ch 8, Lect 2

▪ circulating surrogate marker

In one sentence: A stand-in measurement that can be taken from the blood.
Detailed explanation: The reason it exists is very practical. If you want to know how big a tumour is, the standard method is CT or MRI, which are expensive and cannot be done often, whereas a marker in the blood is cheap, can be measured repeatedly, and can be used to follow the tumour burden.
CEA is the most typical example, and he uses it in colorectal cancer as a surrogate that replaces repeated scanning. The point is not that the blood test is better, it is that it can be repeated as often as you like, and a tumour that is responding will show it in the number.
Why this course brings it up: Dr. Wong explicitly positions CEA as a circulating surrogate marker, and it also appears in his trial design section in its role as a surrogate endpoint. But he also warns that you cannot conclude from one number alone that a patient has cancer. **Appears in:** Ch 10, Ch 11, Ch 12, Lect 4

▪ Circulating tumour cell (CTC)

In one sentence: Cancer cells that have broken off from a tumour and are now travelling around in the blood.
Detailed explanation: Solid tumours shed cells into the bloodstream, and these CTCs can be picked out with the combination EpCAM+/CD45-, which makes them the star of the liquid biopsy: instead of repeating a CT or an MRI, you take a blood sample and follow them.
The neatest clinical application sits on his Lecture 2 slides, where the Kaplan–Meier curve of breast cancer survival splits the patients into three groups according to the CTC count. In the CTC 0 group, survival at 100 weeks is still about 67 %; the CTC 1–4 group is about 50 %; and the CTC ≥ 5 group falls below 20 % after 80 weeks. But watch out for one trap: a CTC has been through EMT on its way into the blood, so the markers on its surface are not necessarily the same as the ones on the original tumour, which is a problem when you are trying to match a blood sample back to a tumour you have never biopsied.
Why this course brings it up: It comes from the liquid biopsy section of Lecture 2, and the Kaplan–Meier curve for CTC counts is on his slides too. CTC itself was never named on its own as an exam question, so it belongs to the pile of concepts you need to understand rather than the pile of terms you need to write down. **Appears in:** Ch 9, Lect 2

▪ Cis face

In one sentence: The receiving end of the Golgi apparatus, where the vesicles coming in drop anchor and start being worked on.
Detailed explanation: The Golgi is like a production line with a direction to it. It has a cis face (the side nearer the nucleus, facing the endoplasmic reticulum) and a trans face (the side pointing towards the cell membrane).
Dr. Wong's second lecture slide draws the whole route very clearly: the rough ER synthesises the protein while the chaperones fold it and glycosylate it, then a transport vesicle carries it, then the vesicle docks at the cis face, then it is worked on step by step as it moves through the cisternae, then it is sorted at the trans face, then it is packaged into a secretory vesicle, and then it goes to the cell membrane. The cisternae in between are the workshops where that step-by-step processing happens. Getting the direction right matters: a secretory protein comes in at the cis face and goes out at the trans face, so everything that can happen to it happens along that one-way journey, in that order. That is also why the two names sound like the entrance and the exit.
Why this course brings it up: The vocabulary on the endoplasmic reticulum and Golgi slide in Lecture 2. Dr. Wong did not name cis face as an exam topic on its own, so this is conceptual understanding rather than a memorisation item. **Appears in:** Ch 8,Lect 2

▪ Cis-regulatory element

In one sentence: A DNA sequence such as a promoter, enhancer or silencer, which only regulates the gene on the very DNA it sits on.
Detailed explanation: "Cis" means "on the same side" — these regulatory sequences can only act on the genes next to them on the same DNA molecule, even if the cell carries two copies of the same gene.
So a cell can hold the same gene twice and still behave differently in the two copies, because the regulatory DNA sitting beside each copy can be different. They usually fall into four kinds: promoter (where the transcription machinery lands), enhancer (pulls transcription up), silencer (pulls it down) and insulator (marks the boundary and stops an enhancer hooking up to the wrong gene). He particularly stresses one common misconception: not all enhancers lie upstream, some sit downstream of the target

gene, on the 3′ side. A cis element stays where it is put, and it works on the DNA it is physically attached to; that is the whole of its job.
Why this course brings it up: The cis versus trans contrast is a second-level concept he lists in Lecture 5; and he also emphasises the "not all enhancers are upstream" point as one of the ideas you are meant to remember. The cis versus trans contrast is a second-level concept he lists in Lecture 5, and he also insists on the "not all enhancers are upstream" point as one of the ideas you are meant to remember. **Appears in:** Ch 7, Lect 5 **Video:** V11_cis_vs_trans.mp4

▪ cisternae

In one sentence: The flat sacs stacked up in layers inside the Golgi.
Detailed explanation: After a protein is carried out of the endoplasmic reticulum it first arrives at the cis face of the Golgi, then passes through the stacked layers of cisternae, and finally leaves by the trans face, where it is packed into a secretory vesicle; the vesicle then fuses with the cell membrane and releases the protein outside. The stack of cisternae is like the stations on a production line. Transport vesicles relay between one layer of cisternae and the next, passing the protein forward stage by stage; the direction is from the cis face to the trans face and it does not run backwards. Each layer is a different set of enzymes, so the order of the layers is the order in which the modifications are made.
Why this course brings it up: He does not name cisternae as an exam topic in its own right, so it is a concept to understand; "in at the cis face, out at the trans face" is the point of Chapter 5. **Appears in:** Ch 8, Lect 2

▪ Co-activator (nuclear receptor)

In one sentence: The protein summoned by an activated nuclear receptor to help drive transcription.
Detailed explanation: A nuclear receptor has no ability of its own to start an RNA polymerase, it works as part of a team. When a ligand binds and changes the shape of the LBD, the AF-2 region becomes a new docking surface and the co-activator is invited in;
those co-activators carry enzymatic activity themselves, they can acetylate the histones to loosen the chromatin, and they pull RNA polymerase over. Without a ligand, what docks at that same site is a co-repressor, and the effect is exactly the opposite: the chromatin gets tightened and the gene is held shut. The clinical consequence is therefore "double pharmacology": some drugs, tamoxifen for example, stop the receptor binding a co-activator in some tissues while allowing it in others. Same drug, same receptor, different tissue, different answer, and the difference is decided by which of the two helpers is allowed to dock.
Why this course brings it up: It is the nuclear receptor section of Lecture 5, and it pairs with the co-repressor as one of his distractor pairs. Of tamoxifen's tissue difference he said, "the concept that you have to bear in mind". So the pair of words matters more than either word alone, and you should be able to say which one opens the gene and which one closes it, and that the switch is decided by which of them is allowed to dock on AF-2 rather than by whether the receptor is present at all. **Appears in:** Ch 7, Lect 5

▪ Co-localisation

In one sentence: Two proteins found in the same place in the same cell, proved by overlapping fluorescent staining.
Detailed explanation: Co-localisation is a spatial inference. Two different fluorescent dyes label two different proteins, and if the overlapping signal shows up in the same spot, that means the two of them are sitting close together, and it is a common way of working out the steps of a signalling pathway.
Because one image can carry several dyes at once, you can watch several components of the same pathway together instead of one at a time. His Lecture 2 gives a nice example: using multiplex immunofluorescence to stain MST1, LATS and YAP of the Hippo pathway at the same time, you can see at a glance whether YAP has been pinned outside the nucleus, because a protein sitting in the cytoplasm means the signal is off, while the same protein sitting in the nucleus means the signal is on. The value of the method is that the answer comes out of a single image instead of out of a hypothesis.
Why this course brings it up: It is the immunofluorescence and confocal section of Lecture 2, and it is the tool he demonstrates with. He never names co-localisation as an exam question, so it is concept-only. **Appears in:** Ch 9, Ch 11, Lect 2

▪ Co-repressor (nuclear receptor)

In one sentence: The protein summoned by a nuclear receptor to press transcription down, exactly the opposite of a co-activator.
Detailed explanation: Nuclear receptors are two-way switches, and the pair of words co-activator and co-repressor is what makes that clear. With no ligand, what docks on the LBD of the receptor is a co-repressor, which recruits deacetylating enzymes such as HDAC, tightens the chromatin and keeps the gene shut.
When the ligand arrives, the receptor changes shape, AF-2 becomes a new binding surface, the co-repressor is swapped out for a co-activator, the enzyme is swapped for an acetyltransferase, the chromatin loosens and the gene is on. It is a clean swap: one group of proteins out, another group in, and everything downstream follows from which group got there first. In other words, what decides the switch is not whether the receptor is there, it is which group of proteins is parked on it. That is also why the same receptor can be a brake in one tissue and an accelerator in another.
Why this course brings it up: It is the nuclear receptor section of Lecture 5, and it forms a pair of opposite terms with the co-activator. He never names the co-repressor as an exam question on its own, but you do have to be able to talk about the contrast, because the contrast is where tamoxifen and its tissue difference live. **Appears in:** Ch 7, Lect 5

▪ Co-stimulatory domain

In one sentence: The section inside the CAR that sends out the second signal, the one that switches the T cell fully on rather than merely nudging it.

Detailed explanation: The intracellular part of a CAR usually has two sections: the co-stimulatory domain (CD28, 4-1BB), which does the confirming, and CD3ζ, which does the transmitting. Why is the confirmation necessary?

Because the design principle of a T cell is that antigen on its own is not enough; it needs co-stimulation as well before it is allowed to work, and that is the rule which stops it attacking your own tissue. Tumours cheat this rule: PD-L1 fakes up an environment in which there is no co-stimulation, and the T cells stall. CAR writes the confirmation straight into the receptor, which means it bypasses the environmental check completely. Note that bypassing the check also bypasses the protection that came with it, because a T cell switched on by antigen alone will go after whatever carries that antigen. The price of all this is that CAR-T still cannot infiltrate solid tumours well, which is why there are CAR-NK and CAR-macrophages to try. The CAR is the immune system's own rule, written into a piece of hardware instead of being asked for each time.

Why this course brings it up: The CAR structure section of Lecture 2. The composition of the CAR signalling tail is the basis for understanding why CAR-T works and why it still fails, so this is conceptual understanding rather than a memorisation item.

Appears in: Ch 9,Lect 2

▪ Codon

In one sentence: Three letters on the mRNA that specify one amino acid.

Detailed explanation: Three nucleotides make up one codon, and each codon specifies either one amino acid or a stop. There are 64 codons in total, but there are only 20 amino acids, so most amino acids are specified by more than one codon, and that is what is called the degeneracy of the genetic code.

The start codon is AUG, which at the same time specifies methionine, and the stop codons are UAA, UAG and UGA. His slides give a frameshift example, where the sequence AUG AAG UUU GGC UAA is read as MET–LYS–PHE–GLY–STOP. Now delete one nucleotide from the middle of that sequence and the reading frame shifts completely. The ribosome is still reading three bases at a time, but it is now grouping what is left of the sequence differently, so the amino acids that come out are entirely different ones. The protein becomes a different shape altogether, and the ribosome may even run straight into a stop codon and stop early.

Why this course brings it up: It comes from the central dogma slides in Lecture 1 and from the frameshift story in Lectures 3 and 4. A frameshift producing MSI-H and a hot tumour is a concept he names explicitly, and it is the same frameshift that runs through both of those lectures, so if you understand the reading frame you already understand the cancer mechanism. **Appears in:** Ch 2, Lect 1

▪ Codon optimisation

In one sentence: Rewriting a gene in the codon spelling that the host cell prefers, so that the protein comes out of the host factory easily and in good shape.

Detailed explanation: For a gene to express properly in a new host, having the right amino acid sequence is not enough. The ribosome of the host cell reads some codons easily and finds others hard to come by, so the same sequence can produce very different amounts of protein in different species.

Codon optimisation means swapping the codons in the gene for the set that the new host prefers. Dr. Wong's Lecture 1 slide compares the codon preferences of human cells and CHO cells directly, and in Lecture 5 he goes further and turns it into a sentence to memorise: “So remember that when you use the CHO cell, you need to do the codon optimization.” Note that nothing about the protein itself changes. The amino acid sequence of the product is identical; what changes is how efficiently the factory reads the order. That is why it is called optimisation rather than rewriting: you are not changing the instructions, you are changing the handwriting. The reason it matters in practice is easy to see, because a protein that is made slowly and badly is no use, however correct the gene is. And the comparison he puts on the slide is done side by side, so you can see how far apart the two preference tables are.

Why this course brings it up: In Lecture 5 he named this rule in his own words, and it is the exam link for the recombinant protein production section. This is a connection you are expected to remember. **Appears in:** Ch 1,Ch 2,Ch 7,Lect 5

▪ Codon usage

In one sentence: Different species prefer different codons for the same amino acid.

Detailed explanation: The idea of a "preference" means that in a given species, certain amino acids are specified by one codon more often than others. The reasons include the number and distribution of tRNAs, translation speed, and the demands of efficient protein folding.

The point is that the genetic code is universal in what it means, but not in how efficiently it is written. In practice this has three consequences. First, a recombinant gene has to be codon-optimised before it expresses well. Second, the same mutation can produce different degrees of phenotype in different species. Third, codon preference is itself a research subject in evolution and in the regulation of gene expression. His Lecture 1 puts it on the central dogma slide.

Why this course brings it up: It is the other side of the same coin as codon optimisation; he did not list it as a separate exam topic, so it belongs to the concepts you are expected to understand. The practical consequence is the part worth keeping, because it explains why a human gene often will not make protein in bacteria until it has been rewritten. **Appears in:** Ch 1, Ch 2, Ch 7, Lect 5

▪ Codon-usage bias

In one sentence: Different species prefer different codons, so a gene that moves between species has to be adjusted.
Detailed explanation: This codon-usage bias and codon usage are two names for the same phenomenon; what matters here is the practical problem it creates. A gene that expresses well in human cells can drop off a cliff in expression when you move it into a CHO cell, because the codon composition of the gene does not match the preferences of the new host.
What you do then is not change the amino acid sequence, because that would change the protein. You "translate" the codons into the idiom that the new host prefers. His Lecture 1 slides also carry a set of experimental data, FALVAC-1 at 38.7 kd and PTP4A3 at 24 kd with GAPDH at 39 kd as the control, comparing Original with the best practice of several different companies, to show how much difference good or bad optimisation makes.
Why this course brings it up: He never lists it as an independent exam question. It is the same exam link as codon optimisation, told in a second way, so it belongs to understanding the concept, and you can treat the two as one exam point rather than two. **Appears in:** Ch 1, Ch 2, Ch 7, Lect 5

▪ Coenzyme

In one sentence: The small helper molecule an enzyme needs in order to work.
Detailed explanation: Some enzymes are proteins themselves and still need a small molecule to operate, and that small molecule is the coenzyme. It usually comes from a vitamin or a derivative of one, so a dietary shortage hits enzyme function directly: niacin corresponds to NAD+ and NADP+, and the B vitamins correspond to a range of coenzymes.
Mechanically, the coenzyme is normally the end of the enzyme that binds temporarily, carries electrons or a functional group, and hands it on. What separates it from a cofactor is that a cofactor includes inorganic components such as metal ions.
Why this course brings it up: It is basic biochemical background and he does not expand on it in Lecture 1. He also says outright that chemical structures and metabolic details need not be memorised, so it is concept-only. **Appears in:** Background supplement (not lecture content)

▪ Cofactor

In one sentence: A partner that an enzyme needs but which is not itself a protein, and it is very often a metal ion sitting in the middle of the reaction.
Detailed explanation: A cofactor is the non-protein part of an enzyme that it needs in order to work, and it comes in two kinds: the ones with metal ions in them (zinc, iron, copper, manganese and so on) and the organic ones, the coenzymes.
They usually take part in the catalysis directly. A metal ion can hold a negative charge steady, help break a bond, or line up the substrate correctly, while the organic coenzymes carry electrons and functional groups around. Either way the cofactor is doing real chemical work in the middle of the reaction; it is not a spectator, and without it the enzyme is only a shape. Clinically this is the mechanism of heavy metal toxicity: take the enzyme's cofactor away and the enzyme stops working, so the toxicity shows up as a failure of enzyme function. Turned round the other way, the existence of cofactors is also the opening for designing metal-chelating drugs.
Why this course brings it up: Basic biochemistry background, and he did not expand on it in Lecture 1. He did say plainly that chemical structures and metabolic details do not need to be memorised, so this is conceptual understanding. **Appears in:** Background supplement (not lecture content)

▪ Cold tumour

In one sentence: A tumour with few neoantigens, which responds badly to checkpoint inhibitors.
Detailed explanation: The difference between hot and cold is simply how many neoantigens there are. A hot tumour has a high mutational burden, which produces enough "foreign" peptides for T cells to recognise, so once you release the checkpoint brake there is work for the T cells to do — the classic example is colorectal cancer with MMR deficiency and MSI-H.
A cold tumour has no such premise; release the brake and there are no troops to use. He also gives a route for turning it hot: use an HDAC inhibitor to make the tumour express PD-L1, creating the state he calls "gene addiction", so that the tumour becomes dependent on the checkpoint. This is not a small difference in response rate; it is the difference between a strategy that works and one that does nothing at all, which is why the classification is used to pick patients rather than merely to describe them. Release the brake and see whether anybody is there to drive is the whole diagnostic question behind this classification.
Why this course brings it up: It is the topic he says "definitely remember" in Lecture 5: "we call this is a hot tumor… And not all the solid tumor is a hot tumor." — and the hot / cold contrast is a regular visitor to the scenario cases. The pairing of hot with cold is the form he wants, because that is how the question will be phrased. **Appears in:** Ch 9, Lect 5

▪ Companion diagnostic

In one sentence: A test you have to do before the drug, because it decides whether this patient can have this drug at all.
Detailed explanation: A companion diagnostic ties "which drug" and "which test" into one single decision, so that the two are never separated. His chain of cases is very clear. HER2 amplification is measured by FISH, which counts gene copies, and the reading rule is a HER2/CEP17 ratio of ≥ 2.0 or a HER2 copy number of ≥ 6, and that single result is what decides whether the patient is suitable for Herceptin.
Rectal cancer with a K-Ras mutation should not get cetuximab even when HER2 is perfectly normal, because the drug it would block is not the one doing the damage. In practice a companion diagnostic is the product of developing the drug licence and the test together, because without the paired test you simply cannot decide who should get the drug, and nobody will sell a drug without a way of choosing the patient. By contrast, a marker like CEA, which tracks tumour burden rather than telling you which

drug to use, is a surrogate marker.
Why this course brings it up: It comes from the liquid biopsy and clinical decision section of Lecture 2, and it is also case number 6 in his scenario bank. You have to be able to explain the difference between a companion diagnostic and a surrogate marker, because that distinction is what the case is testing. **Appears in:** Ch 11, Ch 12, Lect 2

▪ Complement

In one sentence: A group of blood proteins that can kill cells on their own, with no antibody needed to point them at a target.
Detailed explanation: Complement is a set of plasma proteins that can attack cells directly, with no antibody present at all, and that is what complement-dependent cytotoxicity (CDC) means. The difference from ADCC is clear: ADCC needs an antibody to act as a bridge, and then an effector cell such as an NK cell does the killing, whereas CDC is the complement assembling a membrane attack complex by itself and punching through the cell membrane.
It is worth remembering why the two routes came apart in the first place. One of them depends on an antibody having found the target and recruiting help, while the other is a standing system in the plasma that can be triggered with no antibody at all. Clinically, that difference is what decides how an antibody product is designed. Dr. Wong compares IgG1 with IgG4 in Lecture 2: the Fc of IgG1 binds Fc receptors well, so it can call up NK cells for ADCC, while the Fc of IgG4 binds poorly, gives no ADCC, and buys a longer serum half-life instead. Which of those you want depends on what the antibody was built to do. One is built to kill, the other is built to last, and both are perfectly good designs for different jobs. So the choice is not an accident of manufacturing; it is the clinical intent written into the molecule.
Why this course brings it up: The antibody isotype section of Lecture 2. ADCC and CDC are one of the multiple-choice distractor pairs he listed, and so is IgG1 against IgG4. The distinction between them is the kind of thing that turns up as a wrong option in a question, so it is worth having straight in your head. **Appears in:** Ch 9,Lect 2

▪ Compound

In one sentence: A chemical substance made by combining two or more elements.
Detailed explanation: Elements on their own make simple substances; only when two or more elements are combined in fixed proportions do you get a compound. Examples are everywhere in living things: water, carbon dioxide, ATP.
One consequence of that is very important for understanding life: once a chemical bond has formed it usually takes an enzyme to break it, because organisms live at ordinary temperature and pressure in water and cannot carry out such conversions spontaneously. So an "enzyme" is really the thing that makes a chemical reaction possible under physiological conditions. Back to his opening of the course: you do not have to memorise chemical structures, but the framework of how molecules are put together is something you do need.
Why this course brings it up: It is the chemistry foundation of Lecture 1; he explicitly said "do not memorize these chemical structures, they will not appear in my exam", so it belongs to the concepts you are expected to understand. **Appears in:** Ch 1

▪ Confidence interval

In one sentence: A range around a measurement, the range the true value is most likely to sit inside.
Detailed explanation: A number on its own means almost nothing. "OR 39.6" only tells you the association is strong, and to say how reliable it is you need a confidence interval. The standard way he writes it in his Lecture 3 data is: the OR for CIMP-high relative to CIMP-0 is 39.6, 95 % CI 20.6–76.1, $p < 0.0001$;
and the OR for detected CDKN2A methylation with loss of p16 is 12.7 (95 % CI 7.84–20.7, $p < 0.0001$). There is one detail you have to see clearly: the interval for 39.6 runs from a lower bound of 20.6 to an upper bound of 76.1, which says the effect size is very likely to lie somewhere in that range, but it is still a long way from 1, and 1 is what no association looks like, which is why the p value is so small. Think of the interval as a statement about the precision of the measurement rather than about the size of the effect: a wide interval with a large point estimate means you have probably got something real, but you are not yet sure how big it is, while a narrow interval means the estimate is tight. When you write a short-answer question, giving the point estimate together with the interval is what makes the answer complete, and quoting the numbers without the intervals invites the examiner to ask where the error bars are.
Why this course brings it up: OR 39.6 and OR 12.7, with their 95 % CI, are all on his list of precise numbers, and they are standard material for both MCQ and short-answer questions. He also expects the point estimate and the interval together, because his rule for short answers is that you write the numbers out and state the limitation. **Appears in:** Ch 4, Lect 3

▪ Confocal microscopy

In one sentence: A microscope that focuses on one thin optical section at a time, used to locate a labelled protein precisely.
Detailed explanation: The problem with a widefield microscope is that light from above and below the focal plane also hits the detector, so a thick cell lights up all at once and everything blurs together.
The confocal answer is to put a pinhole in front of the detector so that only light from the focal plane gets through and the rest is blocked — one slice at a time, and stacking the slices gives you a three-dimensional reconstruction of the inside of the cell. His Lecture 2 slide draws the light path very clearly: excitation laser → dichroic mirror → objective lens → focal plane → emission pinhole → detector. The practical value is that a question like the Hippo pathway one (is YAP in the cytoplasm or in the nucleus) is a question about space to begin with. It is a machine for answering where, as opposed to whether, and it is the tool you reach for whenever the answer to a biological question is a location.

Why this course brings it up: It is the microscopy section of Lecture 2; he never named confocal as an exam topic on its own, so it belongs to the concepts you are expected to understand. **Appears in:** Ch 9, Ch 11, Lect 2

▪ Consensus sequence

In one sentence: The DNA “letter” that turns up most often in a real genome, when you line up many sequences of the same family and look at each position in turn.

Detailed explanation: When we say a sequence is typical, or call it a consensus, we mean this: line up many homologous sequences, take the most common letter at each position, and assemble the result into one representative sequence.

Its use is to describe the features that members of a family share. The TATAAA box in a promoter, for instance, is really the consensus sequence of many eukaryotic promoters, and individual genes vary from it. The thing to be careful about is that a consensus sequence is a statistical representative. It does not mean every gene matches it exactly, and if you drop it straight in as a template it may not work at all. It is a picture of the crowd rather than a member of it, and recognising one in a sequence is still useful, while building a promoter around one on the assumption that it will definitely work is not.

Why this course brings it up: Background supplement, and not part of the content he listed in his five lectures. Understanding it is enough, and it counts as conceptual understanding. **Appears in:** Background supplement (not lecture content)

▪ Constitutively active

In one sentence: Staying switched on without needing an upstream signal.

Detailed explanation: Normally a small G protein is active only briefly, when an upstream receptor such as EGFR fires, and then it switches itself off again. Mutated K-Ras is different: the mutation site (G12, G13, Q61) has broken its own “off” switch, because the GTP hydrolysis promoted by GAP no longer works, so Ras stays locked in the GTP-bound active state and sends out growth and survival signals whether or not EGFR has anything to say.

That is what “constitutively active” actually means here: the signal is not amplified, it is not switched off, and no amount of quiet upstream will stop it. Compare it with a normal protein, which is doing nothing at all until the receptor fires.

Why this course brings it up: Dr. Wong uses this example to explain the difference between a “sticky oncogene” and a car with its foot permanently on the accelerator. He does not list the term as a topic on its own, so it belongs to conceptual understanding. **Appears in:** Ch 10, Lect 1

▪ Contact inhibition

In one sentence: Healthy cells stop dividing as soon as they bump into a neighbour.

Detailed explanation: Normal cells stop before they fill up the dish. They touch a neighbour and they stop dividing, and that is the control mechanism that stops cells piling into a heap. Cancer cells lose this property, so they keep stacking up on top of each other and form a mass.

More importantly, contact inhibition is exactly the upstream stimulus for the Hippo pathway: the point where two cells join sets off the MST–LATS kinase cascade, which phosphorylates YAP and pins YAP in the cytoplasm, and proliferation switches off. That turns "growing into your neighbour" into a feedback signal that tells the cell it is time to stop.

Why this course brings it up: It is the starting point of Hippo in Lecture 1 and of the second pass over Hippo in Lecture 5. He never names contact inhibition on its own as an exam question, but it is the doorway into the Hippo mechanism, and you need it to make sense of what the cascade is doing. **Appears in:** Ch 1, Ch 7, Lect 1

▪ Contractile ring

In one sentence: A ring built out of actin and myosin that pinches a cell which is in the middle of dividing into two, so that the cytoplasm ends up in two separate cells.

Detailed explanation: Mitosis is responsible for handing the chromosomes to the two daughter cells, and cytokinesis is responsible for cutting the cytoplasm apart. The two are independent of each other.

The machine that cuts the cytoplasm is the contractile ring: bundles of actin filaments are pulled by myosin, tightening like a rubber band from the middle of the cell, until a cleavage furrow forms and the cell is pinched into two. If you inhibit the contractile ring, either with a drug or by knocking down its expression, the chromosomes still separate perfectly well but the cell does not split, and the result is one cell with two nuclei and a DNA content of 4N. So the three things to hold together are: the chromosomes separate, the cytoplasm does not, and the DNA content ends up at 4N. Dr. Wong says he asks this question every single year, and that only half the students got it right last year.

Why this course brings it up: It is a first-level exam topic that Dr. Wong named himself, from Lecture 2: “this is the type of the question that I would try to include in the midterm”. It is case 5 in the case bank. The reason he keeps coming back to it is that the answer has two halves that students tend to mix up, and mixing them up is exactly what costs the marks. So you should be able to give the complete answer, naming what still happens and what does not, and not just the mechanism. **Appears in:** Ch 8,Lect 2 **Video:** V02_Contractile_ring_binucleated_cell.mp4

▪ Copy number variation (CNV)

In one sentence: A stretch of DNA that has more copies than normal, or fewer.

Detailed explanation: Between people, and between populations, some regions of the genome do not carry the same number of copies, and that is both evolutionary material and a source of drivers in tumours.

The classic evolutionary example is the amylase gene AMY1: populations that eat a lot of starch carry more copies, with the median in his slide data at 7 for the high-starch group. More copies means more enzyme, better starch digestion, and over generations the copy number can rise. On the tumour side, an increase in copy number that comes with a rise in mRNA expression may be a driver, while one that merely rides along is a passenger, and that distinction is the integration method he stresses in Lecture 3. There are four mechanisms that generate CNV: NAHR, FoSTeS, NHEJ and MEI, and the point of listing them is that they are different accidents of DNA repair rather than different kinds of disease.
Why this course brings it up: It is that section of Lecture 3. He says outright that the molecular mechanisms of CNV need not be studied in depth, but the driver versus passenger distinction is a second-level concept, and the adaptive CNV example is the one he uses to show that a copy number change is not automatically a disease. **Appears in:** Ch 3, Lect 3

▪ Core promoter

In one sentence: The stretch of DNA directly in front of a gene, where the transcription machinery assembles.
Detailed explanation: Eukaryotic genes are switched on in two layers. The core promoter is a short piece sitting immediately upstream of the transcription start site (+1), usually with a TATAAA box at around −30;
the transcription pre-initiation complex is assembled right there. TFIID (which contains TBP) recognises the TATA first, then TFIIA (a general transcription factor) and TFIIB (a general transcription factor) join one by one, and finally RNA polymerase II arrives with TFIIE (a general transcription factor), TFIIF (a general transcription factor) and TFIIH. An enhancer thousands of bases away from the promoter loops back to it, using proteins that bend DNA. The clinical use is very direct: the promoter is the most common place to swap in a replacement promoter. The naming helps here: "core" means the part common to the machinery, as opposed to the enhancer, which is the variable part that decides when and where this particular gene is used.
Why this course brings it up: It is the core promoter and the TATA box in Lecture 5; the promoter is also one of the cis elements he lists, and it is paired with the trans factors. He does not expect you to recite the order of every factor, but the logic of the two layers is the point. It is the core promoter and the TATA box in Lecture 5; the promoter is also one of the cis elements he lists, paired with the trans factors, and he does not expect you to recite the order of every factor but does expect the logic of the two layers. **Appears in:** Ch 7, Lect 5

▪ Corepressor

In one sentence: The molecule that does the work for a repressor; in the trp operon the corepressor is tryptophan.
Detailed explanation: A repressor on its own is not quite enough. It usually needs a partner before it can really shut transcription down, and that partner is called the corepressor, because it supplies the piece the repressor is missing.
The trp operon is the textbook example. The product of the trp genes is the enzyme that makes tryptophan, and tryptophan is itself the end product of that pathway, so the end product of the pathway can act as the brake on the pathway. When there is enough tryptophan in the cell, it acts as a corepressor and binds to the Trp repressor, which changes the shape of the complex and lets it sit down on the operator, and the genes for making tryptophan are switched off, because the cell will not spend effort making something it already has. Once you understand this negative feedback, the logic of the lac operon becomes clear as well: lac switches on only when there is an inducer, and trp switches off as soon as there is product.
Why this course brings it up: It comes from the trp operon section of Lecture 5, and it is one half of the contrast he draws between lac and trp as mirror images, the other half being the inducer on the lac side. The corepressor itself was never named on its own as an exam question, so you learn it as part of the trp story rather than as a term in its own right. **Appears in:** Ch 7, Lect 5

▪ Covalent bond

In one sentence: The kind of bond where two atoms share a pair of electrons and hold each other together.
Detailed explanation: Covalent bonds are the backbone that keeps a molecule in shape: each atom contributes electrons and shares them in pairs, forming a stable structure. The most important covalent bonds in biology are carbon–carbon bonds and carbon–hydrogen bonds;
carbon has four valencies, so it can grow long chains, branches and rings, and that is the skeleton of organic chemistry. The other category you have to keep separate is non-covalent interaction: hydrogen bonds, ionic interactions and van der Waals forces. The binding of DNA to histone inside the cell is non-covalent, which is why chromatin state can change dynamically and reversibly — and that reversibility is exactly what makes epigenetic regulation possible at all.
Why this course brings it up: It is the chemistry foundation of Lecture 1; he explicitly said not to memorise chemical structures, so it belongs to the concepts you are expected to understand. **Appears in:** Ch 1

▪ CpG island

In one sentence: A short stretch of DNA rich in "CG"; methylate it and the gene next to it usually falls silent.
Detailed explanation: "CpG" is a cytosine followed by a guanine, and the phosphate sitting in between is what supplies the P. Across the whole genome, CG pairs turn up far less often than chance would predict, and the reason is that a cytosine which has been methylated readily deaminates into thymine, which is a change that causes a harmful mutation.
Because of that, over the course of evolution the organism was able to keep large areas of the genome that are only lightly methylated. The exception is the CpG island that sits near a promoter: that region is usually not methylated at all, and so the gene next to it is switched on and available to be read. Here is the twist: cancer cells go out of their way to methylate those islands anyway, silencing the tumour-suppressor gene, and that is the epigenetic version of the brake pedal being stamped on and held down.

Why this course brings it up: It comes from the DNA methylation and promoter hypermethylation section of Lecture 3, which he lists as a second-level concept. CpG island is where MGMT and CIMP are read out, so if you can explain why a CpG island is normally unmethylated you have the whole picture in your hand. **Appears in:** Ch 4, Lect 3, Lect 5

▪ CpG site

In one sentence: A cytosine with a guanine right next to it on the DNA; methylating that cytosine is the switch that silences the gene, and it is the basic unit of epigenetics.

Detailed explanation: A CpG site is the basic unit of epigenetics. The chemical act of DNA methylation is to attach a methyl group to the fifth carbon of a cytosine, and SAMe is the supplier of that methyl group.

DNMT3A and DNMT3B are responsible for de novo methylation, which means establishing new marks, while DNMT1 works at the replication fork copying the existing marks onto the daughter strand, which means maintaining them. Note the division of labour: two enzymes write the labels and one copies them, and getting that third job wrong is what destroys the pattern. The effect of methylation is to hinder transcription factors from binding, or to attract inhibitory proteins, and the gene is switched off. Clinically, methylation is used as a biomarker: MGMT promoter methylation makes glioblastoma more sensitive to alkylating chemo and to radiotherapy. So the CpG site is where a chemical mark, a cell identity decision and a treatment decision all meet. And because only cytosine carries the mark, the whole epigenetic system has a very specific address in the sequence, which is why the same chemistry can turn a gene off in one place while leaving the rest of the genome alone.

Why this course brings it up: He listed it as a second-level concept: DNA methylation is “usually involved in the suppression of the gene expressions”, and the split between de novo and maintenance methylation is part of the same item. So the wording of that idea matters, and so does knowing which enzyme does which job, because the two methyltransferases are easy to confuse and the exam will not be gentle about it. The CpG site also shows up again later, in the biomarker material on methylation-based tests. And note that methylation does not delete the gene or change its sequence; it changes whether the gene can be read. **Appears in:** Ch 4,Ch 11,Lect 3,Lect 5

▪ CRISPR/Cas9

In one sentence: A gene editing system that cuts DNA at a chosen place and lets the cell's own repair machinery make the change.

Detailed explanation: Cas9 is a nuclease and it needs two things before it acts: a designed guide RNA that points at the target sequence, and a PAM sequence next to the target (the recognition point Cas9 must have before it cuts).

With both in place, the RuvC and HNH domains of Cas9 cut the DNA into a double-strand break. The important part comes next: what the change looks like is decided by the cell's own repair machinery. If it goes down the NHEJ route you get small insertions or deletions, which gives you a knockout; if it goes down the HR route and you supply a donor template, you get a precise knock-in. HR can only happen in S / G2, whereas NHEJ can happen in any phase. Think of Cas9 as scissors with an address written on them, and the cell's repair systems as two different tailors who then stitch the cut up in their own way: one is careless, one is precise, and only one of them needs a pattern to copy from. He also mentions that Cas9 won the 2020 Nobel Prize in Chemistry, and the patent fight between UC Berkeley and the Broad Institute.

Why this course brings it up: It comes up in Lecture 2 (where he says CRISPR has replaced the older siRNA / shRNA knockdown) and in Lecture 4; he never named CRISPR as an exam topic on its own, but you have to be able to draw the mapping from NHEJ to knock-out and from HR to knock-in. The Nobel Prize and the patent war are colour, not examinable content. **Appears in:** Ch 6, Lect 2, Lect 4

▪ Cristae

In one sentence: The folds that the inner mitochondrial membrane folds inward, where the ATP-making machine sits.

Detailed explanation: Cristae are folds of the inner mitochondrial membrane, and they multiply the membrane surface area many times over. Both the electron transport chain and ATP synthase need a lot of membrane area to work, so the denser the folds, the more ATP can be made.

Oxidative phosphorylation happens right here: electrons travel along the chain, and finally ATP synthase, driven by the proton gradient, adds a phosphate to ADP to make ATP. Run the numbers and you see how efficient the design is: one glucose in a normal aerobic cell earns about 36 ATP, while an aerobic cancer cell that relies on glycolysis earns only about 4. The inner membrane is also the gate for apoptosis: once it is punctured, cytochrome c leaks out. The same membrane that makes the cell's energy also holds the cell's death sentence.

Why this course brings it up: It is the organelle section of Lecture 5, and he never names cristae as an exam question, so it is concept-only. What you should take away is the link between surface area and ATP output, because that reasoning sits behind the whole cancer metabolism story. **Appears in:** Ch 8, Lect 2

▪ Crossing over

In one sentence: A deliberate swap of DNA between homologous chromosomes during meiosis, which reshuffles genetic variation.

Detailed explanation: Crossing over happens in the first meiotic division, while homologous chromosomes are paired, and the point where the swap happens is called a chiasma. After the swap, one chromosome carries a stretch of maternal genes and a stretch of paternal genes, so every gamete that comes out is different from the parent and from its siblings.

The sources of genetic diversity are therefore very clear: the random assortment of chromosomes, the crossing over during meiosis, and random fertilisation. His Lecture 4 makes a particular point of this, that the whole business runs on the very same homologous recombination machinery that you met in the repair chapter. The machine that normally repairs broken DNA is deliberately used in meiosis to rearrange genes instead, which is a neat trick if you think about it.
Why this course brings it up: It comes from the meiosis sections of Lectures 2 and 3. He never names crossing over on its own, but the M1 against M2 distinction is a named exam question, and the crossover is exactly the key difference in the M1 column of the answer sheet. **Appears in:** Ch 8, Lect 2, Lect 3

▪ Cryptic exon

In one sentence: A stretch of sequence that is normally not used, but becomes a real exon when splicing switches it on.
Detailed explanation: Splice site choice is not just on and off. Some sequences are never treated as exons under normal conditions, and they only get switched on when a mutation creates a new splice site, and that is what a cryptic exon is.
In other words the sequence was always there in the transcript, but it was being read as intronic sequence; one letter of change reassigns it, and the cell starts making a protein that contains it. His Lecture 5 slide uses AR to show it: normal AR skips that cryptic exon CE3, while AR-V7 instead includes it, and the isoform that results is missing the ligand-binding region, so it is constitutively active and resistant to enzalutamide. This example covers all three clinical consequences of alternative splicing in one go: making neoantigens, cutting off a drug-binding region and causing resistance, and producing an isoform that can actually be measured. It is a good illustration of how a splicing decision can be worth more to a patient than anything in the drug catalogue.
Why this course brings it up: It is one of the clinical list for alternative splicing in Lecture 5, and it links to AR-V7 and to scenario case number 15. It is never named as an exam question on its own, so the point is the concept, but you should be ready to draw the AR against AR-V7 comparison if it comes up. **Appears in:** Ch 7, Lect 5

▪ CSA and CSB

In one sentence: The two proteins that TC-NER uses to notice that the gene being read has got stuck.
Detailed explanation: Most of the DNA in a cell is being copied or transcribed, so the repair system cannot simply dig out all of it. TC-NER, transcription-coupled repair, handles specifically the gene that is being read by RNA polymerase II: if the polymerase jams at a site of damage and stops, CSA and CSB arrive first and tell the repair system that this stretch is being read, so repair it first.
They work like a repair call. They do not do the repair themselves, they call TFIIH and the cutting team to the place where the polymerase is stuck. The logic is triage: a lesion that has halted the machinery is a bigger problem than one sitting in a quiet stretch of DNA, so the cell prioritises it.
Why this course brings it up: He does not name CSA and CSB as exam questions on their own, so it is concept-only. He does teach GG-NER and TC-NER as two separate routes. So the two paths are the examinable part, and these two proteins are how the cell tells them apart. **Appears in:** Ch 6, Lect 4

▪ CTC0

In one sentence: The group of patients in whom no circulating tumour cells can be found in the blood at all.
Detailed explanation: He uses Kaplan–Meier curves to compare overall survival in breast cancer patients grouped by the number of CTCs in their blood: CTC0 is the best, with about 67 % still alive at 100 weeks;
CTC1–4 is in the middle, with about 50 % at around 80 weeks; and CTC≥5 is the worst, with under 20 % left after 80 weeks. That is a clean demonstration that curves which separate mean the marker can group the patients. The relationship between the CTC count and prognosis is the example given on his slide 73, and the three curves from top to bottom are CTC0, CTC1–4 and CTC≥5.
Why this course brings it up: Reading a KM curve is the first-grade, A+ question he asks in every lecture, and the CTC grouping is the practical example he uses to show it, and the numbers are worth remembering. **Appears in:** Ch 9, Lect 2

▪ CTGF

In one sentence: Connective tissue growth factor, one of the target genes switched on by YAP through TEAD when Hippo is off.
Detailed explanation: When the Hippo pathway is switched off, unphosphorylated YAP enters the nucleus, binds TEAD, and starts a set of growth-promoting genes — his Lecture 3 slide lists CTGF, JAG1 (a Notch ligand) and NOTCH2, and Lecture 5 lists them again when he repeats the material.
Each gene does its own job. CTGF promotes fibroblast proliferation and extracellular matrix deposition, and it is central to tissue fibrosis and to the tumour stromal response. JAG1 and NOTCH2 instead connect the Hippo and Notch pathways to each other. So CTGF is the output line of the Hippo pathway rather than part of its machinery, which is exactly why it makes a reasonable read-out of the pathway being stuck on. The clinical meaning is that if this whole growth switch jams in the on position, the tissue overgrows. Read it as the output line of the pathway rather than part of its machinery, which is exactly what makes it a reasonable read-out of the pathway being stuck on.
Why this course brings it up: It is the Hippo section of Lecture 1 and Lecture 5; he never named CTGF as an exam topic on its own, but the chain from YAP–TEAD to the target genes has to be something you can talk through. **Appears in:** Ch 7, Lect 1, Lect 5

▪ CTLA-4

In one sentence: The checkpoint receptor that competes with CD28 for the co-stimulatory signal to T cells.
Detailed explanation: CD28 on the surface of a T cell binds B7 on the surface of a dendritic cell, and that binding brings the activation signal, whereas CTLA-4 competes with CD28 for the very same B7 and binds it more strongly, so it can cut into the queue and take the co-stimulatory signal away.
There is not much CTLA-4 on a CD8 T cell to begin with, but once the cell is activated the amount goes up sharply, and the effect is to shut down the cell's own response, which is the brake that stops T cells from overreacting and from damaging the host while they are fighting. The tumour's calculation is to exploit this: if both the tumour cells and the dendritic cells express B7, then even a T cell that does see the antigen is held back by CTLA-4 and never fires properly. The counter-move is anti-CTLA-4, ipilimumab for instance.
Why this course brings it up: It comes from the immune checkpoint section of Lectures 2 and 3. His Lecture 3 slides put CTLA-4, anti-CTLA-4 and PD-1/PD-L1 on the same picture, so it belongs to understanding the concept rather than to any list of names, and the picture is worth drawing from memory. **Appears in:** Ch 9, Lect 2, Lect 3

▪ Cyclin

In one sentence: The protein that rises and falls through the cell cycle and switches on its matched CDK; it is the key, not the machine.
Detailed explanation: The CDK is the machine that does the work, but it sits on standby all the time and has to be switched on by a cyclin. The mechanism is simple: the concentration of each cyclin rises and falls with the cycle, so a particular cyclin is made in bulk at a particular moment, pairs up with its CDK, turns that CDK on, and the step gets pushed forward;
later the cyclin is degraded and the CDK goes dark again. The whole design is a wave: the cyclins come and go, the CDKs they carry stay put, and the timing of the wave decides which step of the cycle happens when. That is where the name comes from: the cycle is named after the cyclin, not after the cycle. The clinical importance shows up in drugs. A CDK4/6 inhibitor blocks the Cyclin D-CDK4/6 step and the cell stops in G1, while inhibitor proteins of the Cip/Kip type such as p21 and p27 are the cell's own brakes.
Why this course brings it up: It is the cell cycle material in Lectures 2 and 5, but he explicitly said "you do not need to memorise all the cyclin or cyclin-dependent kinase pairings", so it is concept-only. He wants you to hold on to the idea that the cyclin is the key and the CDK is the machine, not the other way round. **Appears in:** Ch 8, Lect 2

▪ Cyclin-dependent kinase (CDK)

In one sentence: An enzyme that pushes the cell cycle forward one step at a time, and it only works when the matching cyclin switches it on.
Detailed explanation: CDK is a serine/threonine kinase, and its activity needs two things: that it has been bound by a cyclin, and that another subunit has been phosphorylated. It is not a machine that repeats the same job;
it is lit up by a cyclin, does one step, and then is switched off again. Its output is phosphorylation. Cyclin D–CDK4/6 phosphorylates pRb, pRb lets go of E2F, and the genes that S phase needs start to be transcribed. This is why the cyclin and the kinase are two separate proteins rather than one. The cyclin supplies the timing, because its concentration rises and falls through the cycle, while the kinase supplies the machinery, because it is always there waiting. That way the cell can run the same enzyme through a different step each time simply by handing it a different key. Pharmacologically this is a two-pronged attack: on one side a CDK4/6 inhibitor seals the gate directly (palbociclib), and on the other side the tumour breaks the brakes itself, by wrecking p16 (INK4) or p21 (Cip/Kip) so that the CDK cannot be stopped. Either way the result is the same, a cell that keeps trying to divide.
Why this course brings it up: Lectures 2 and 5. But he explicitly said you do not need to memorise the cyclin and CDK pairings, so this is conceptual understanding rather than a memorisation item. **Appears in:** Ch 8,Lect 2

▪ cysteine

In one sentence: The amino acid with a thiol on it; two cysteines can be fastened together with a disulfide bond.
Detailed explanation: The side chain of cysteine ends in a thiol (–SH), and two thiols that are oxidised form a disulfide bond, like two little hooks hooking into each other and stitching the tertiary and quaternary structures of a protein together.
On a therapeutic protein the disulfide bond is a structural nail. And a free cysteine, with its thiol still hanging out, is reactive with other residues, which is something the formulation has to watch out for. The same chemistry that lets you build the protein also lets you lose it, because a cysteine that finds the wrong partner will bond to that instead. Which is why the number and the placement of the cysteines matter when you are designing a protein drug.
Why this course brings it up: He does not name cysteine as an exam topic in its own right, so it is a concept to understand; the disulfide bond is already in the glossary. **Appears in:** Ch 2, Lect 1

▪ Cytochrome c

In one sentence: A protein that normally lives in the mitochondria and, once released into the cytoplasm, starts the intrinsic apoptosis pathway.
Detailed explanation: The day job of cytochrome c is one link in the electron transport chain — it carries electrons between the inner mitochondrial membrane and complex III. But it has a second job: in the intrinsic apoptosis pathway it is the switch.
When the pro-apoptotic members of the Bcl-2 family, bak and bax, punch holes in the outer mitochondrial membrane, cytochrome c leaks into the cytoplasm, binds APAF-1 and assembles into the apoptosome, which activates caspase 9. The

clinical double life is interesting: blocking cytochrome c release can keep a cancer cell alive, but the same drug might also save heart muscle cells from apoptosis after an ischaemic stroke. One protein, two completely different jobs, and the second one is what most of the apoptosis material in this course is built on. One protein, two jobs, and the second job is the one this part of the course runs on.
Why this course brings it up: It is the key node in Lecture 2 when he teaches the intrinsic and extrinsic apoptosis pathways, and it is also the premise that makes his BI 905711 story possible; it belongs to the concepts you are expected to understand.
Appears in: Ch 8, Lect 2

▪ Cytokine

In one sentence: Small proteins that cells release to pass a message to the cells nearby.
Detailed explanation: A cytokine is a short-range message between cells, secreted by one cell and then acting on the cells around it. His Lecture 5 uses NF-κB to show the timing of these signals, and this is one of the clearest sequences in the whole course: TNF-α peaks at 4–6 hours, then IL-1β, then IFN-γ, and last of all IL-2 and IL-15.
The order matters, because the order of the peaks reflects cause and effect upstream and downstream rather than everything exploding at once. Even more important is the feedback: the cytokines that get secreted strengthen the activation of NF-κB again, which forms a positive feedback loop and lets the defence response swell up very fast. Clinically, this is where the cytokine storm comes from.
Why this course brings it up: It comes from the NF-κB section of Lecture 5. He explicitly said not to worry about the order of these cytokines, that it is only there so that you can see different reactions being observed at different times, so it belongs to understanding the concept. **Appears in:** Ch 7, Ch 9, Lect 5

▪ Cytokinesis

In one sentence: The step after mitosis that divides the cytoplasm into two.
Detailed explanation: Cytokinesis is the last step of the cell cycle, and it is a different process from mitosis, which separates the chromosomes, and it runs on a different schedule. The two can be uncoupled from each other, and that uncoupling is exactly what makes the classic exam answer possible.
Animal cells use a contractile ring made of actin and myosin to pinch inwards and form the cleavage furrow; plant cells instead grow a cell plate out from the middle and extend it in both directions. The key clinical lesson comes from a question he named: inhibit the contractile ring, the chromosomes still separate, the cell still cannot be cut in two, and what you get is a binucleated cell, one cell with two nuclei and a DNA content of 4N. The other detail is that organelles are distributed at random, so mitochondrial disease can differ wildly in severity between siblings.
Why this course brings it up: This is a Tier 1 question he named (Lecture 2), and the entire point of the question is that mitosis and cytokinesis are independent processes, so blocking one leaves the other running. **Appears in:** Ch 8, Lect 2

▪ Cytoplasm

In one sentence: The watery jelly inside the cell, and almost every chemical reaction in the cell happens inside it.
Detailed explanation: Cytoplasm is everything inside the cell membrane and outside the nucleus. It is roughly water plus dissolved ions, small molecules, proteins and a lot of floating organelles.
Functionally it is the factory floor: glycolysis, protein synthesis and most metabolic reactions all take place here. A prokaryote has no nucleus, so the whole cytoplasm is its workspace and transcription and translation can run at the same time. A eukaryote shuts transcription inside the nucleus and puts translation in the cytoplasm, with a processing step in between. So when you hear the word cytoplasmic, it usually means “in the part of the cell that is not the nucleus”, which is not quite the same as saying “outside the cell”.
Why this course brings it up: The organelles and signalling section of Lecture 5. Dr. Wong did not name cytoplasm as an exam topic on its own, so this is conceptual understanding. **Appears in:** Ch 7,Ch 8

▪ Cytosine

In one sentence: One of the four bases in DNA, and the one that gets methylated.
Detailed explanation: Cytosine is the only one of the four bases carrying an amino group (NH2) — thymine carries a methyl group instead. That small chemical difference carries a big evolutionary consequence: cytosine readily deaminates spontaneously into thymine, and if the error is not corrected it becomes a permanent mutation.
Cells use two sets of answers. One is the DNA repair system, which keeps checking and reading T back as C. The other is evolution, which produced large regions of the genome that are sparsely methylated, so methylation is concentrated in a few places — the CpG islands near promoters, which are the stage on which epigenetic regulation is performed. So one single base ties DNA stability and gene regulation together. That is the neat part of this base: it connects the stability of the genome to the control of what gets read, and neither half of the course can do without it.
Why this course brings it up: It is the DNA methylation section of Lecture 3; he never named cytosine as an exam topic on its own, but it and the CpG site are the same exam link. If you know why cytosine is the vulnerable one, most of the methylation material follows without further effort. **Appears in:** Ch 3, Ch 4, Lect 3

▪ Cytoskeleton

In one sentence: The network of fibres that gives a cell its shape, moves cargo, builds the spindle and pinches the cell in half.

Detailed explanation: The cytoskeleton is made of three kinds of fibre, and each has its own job: actin microfilaments, intermediate filaments and tubulin microtubules. Actin handles shape and movement and is also part of the contractile ring. Microtubules are the tracks for long-distance transport and the skeleton of the spindle during division, while intermediate filaments mainly give the cell mechanical strength. Clinically there are three consequences you can use immediately. First, drugs against microfilaments, such as cytochalasin and latrunculin, block the contractile ring and so produce binucleated cells. Second, drugs against microtubules, such as vinca alkaloids and taxanes, disturb the spindle so that chromosomes cannot be distributed properly to the two daughter cells. Third, the migration and the invasion of cancer cells depend on the actin network being laid down again and again in new places.

Why this course brings it up: It comes from the organelle section of Lecture 2, and from the contractile ring question he names. It belongs to understanding the concept, but it is the kind of thing you should be able to draw on a blank page. **Appears in:** Ch 8, Lect 2

▪ cytotoxic payload

In one sentence: The poisonous part of an ADC that does the killing.

Detailed explanation: The antibody is the delivery man and the payload is the cargo. The payload has to satisfy two conditions at once. It has to be very toxic, so that it kills once it gets inside the cell.

And it has to be transported safely to the target and only then released, because if it leaks out into the blood the toxicity misses the tumour and hits normal cells instead. Dr. Wong's examples are DNA-damaging payloads of the topoisomerase inhibitor type (SN-38, DXd). The delivery requirement is the hard part, because the same molecule that kills the tumour cell will also kill a healthy dividing cell if it gets loose.

Why this course brings it up: He says plainly that the structure of ADCs and the structures of the individual drugs are not examined. The reason he does give, though, is why the payload is chosen to be a DNA-damaging type, and it is about dealing with drug resistance. And the delivery problem is why an ADC carries a linker at all, because the payload has to stay quiet in the blood and only let go once it is inside the target cell. Get that wrong and the drug becomes a general poison rather than a targeted one. **Appears in:** Ch 10, Lect 1, Lect 4

▪ DAG

In one sentence: Diacylglycerol, the second messenger that stays on the cell membrane.

Detailed explanation: When a GPCR switches on phospholipase C, PI(4,5)P is cut into two pieces. One is the soluble IP3, which travels to the receptor on the endoplasmic reticulum and releases the calcium ions.

The other is DAG, which stays in the cell membrane and calls up and activates PKC. The two messengers divide the work: IP3 summons the calcium ions out, DAG stays where it is and lights up PKC. DAG and IP3 are a pair of products made by the same pair of scissors, phospholipase C, and they deliver the signal to the membrane and to the endoplasmic reticulum at the same time. Same reaction, two destinations, and that is why one enzyme can start two parts of the same signal at once.

Why this course brings it up: He does not name DAG as an exam question on its own, so it is concept-only. The skeleton he emphasises is that signal transduction works by a kinase adding a phosphate and a phosphatase taking it away. The individual messenger names are the detail, and the phosphorylation skeleton is the idea. **Appears in:** Ch 1, Lect 1

▪ DAPI

In one sentence: A fluorescent dye that glows blue and stains cell nuclei specifically.

Detailed explanation: DAPI binds DNA, so on any multiple immunofluorescence image it lights up every nucleus in blue, and at a glance you can see the cell density and the arrangement of the tissue.

With DAPI providing the coordinates, the signals from other proteins know which nucleus to be laid over. That also makes it a control: in the same image, a signal that overlaps the DAPI means the protein is in the nucleus, and a signal that stays separate means it is outside. One dye, and every other channel on the slide is suddenly interpretable.

Why this course brings it up: He does not particularly name it, so it is a concept to understand; the point of that passage is that immunofluorescence can look at several markers at once while IHC can only look at one. **Appears in:** Ch 9, Ch 11, Lect 2

▪ Daraxonrasib (Rasonque)

In one sentence: A RAS inhibitor: the median overall survival is 13.2 months, against 6.7 months for the chemotherapy it was compared with.

Detailed explanation: RAS has long been called “undruggable”. It is a small GTPase that switches itself on by binding GTP and off by hydrolysing it to GDP, and its surface has no obvious deep pocket into which a small molecule can fit.

Dr. Wong's first lecture quotes a simulated FDA press release from 26 August 2026: Rasonque (daraxonrasib) was approved for metastatic pancreatic cancer. The trial enrolled 500 adults with previously treated pancreatic adenocarcinoma, and the result was a median overall survival of 13.2 months, against 6.7 months for standard chemotherapy. The point of the story is that K-Ras was called undruggable because of a pharmacological difficulty, and that is not the same as being biologically undruggable. The binding site was always there. What was missing was a way of hitting it. Notice also how he frames the approval: it is a simulated press release, used to make a point about target nomination rather than to report a real regulatory decision. And the numbers are the kind of thing he expects you to quote exactly, because they make the argument better than

adjectives do.
Why this course brings it up: The 13.2 against 6.7 months figures are on his list of precise numbers. The idea that undruggable can be solved is also one of the points he makes when he talks about target nomination. So the drug name itself is not the point; the point is that a target which looked impossible now has an approved inhibitor and a measurable survival difference. **Appears in:** Ch 10,Lect 1

▪ datopotamab-deruxtecan

In one sentence: An ADC that goes after TROP2.
Detailed explanation: It comes as a pair with trastuzumab-deruxtecan, and the two are the examples he names among the ADCs that are already on the market: the first one recognises HER2 and the second one recognises TROP2, and in both the payload is from the topoisomerase inhibitor family.
The logic is clear. HER2 and TROP2 are both antigens expressed on epithelial tumours, the antibody picks out the tumour cells that carry them, and the poison only acts in that one place. The difference between the two drugs is only which antigen the antibody recognises, and the design logic is exactly the same: deliver the topoisomerase inhibitor to the door of the tumour. So the two drugs differ in which antigen the antibody binds and in nothing else that matters for the design. One is the antibody, one is the toxin, and the pairing is what makes the delivery selective.
Why this course brings it up: The structure of an ADC is not examined. He names these two drugs to make the point that the clinical targets of ADCs at the moment are HER2 and TROP2. **Appears in:** Ch 10, Lect 4

▪ De novo methyltransferases

In one sentence: The enzymes that build the methylation marks from scratch.
Detailed explanation: When a cell first appears, or when a stretch of new DNA needs switching on or off, it is DNMT3A and DNMT3B that write the first methyl group on a stretch of DNA that carries no mark at all.
They need no cue at all, only the ability to recognise a CpG site, and then they put the mark down in place. Nothing has to be switched on for them to work, which is why the de novo enzymes are the ones that build the map in the first place. The opposite of them is the maintenance methyltransferase, whose job is to put the existing marks back onto the new strand after replication. Without that first step the methylation map of embryonic development could never be laid down, and keeping the marks across each round of copying is DNMT1's job.
Why this course brings it up: “The difference between de novo and maintenance” is one of the 19 multiple-choice pairings Dr. Wong lists (Tier 2), and the division of labour between DNMT3A/DNMT3B and DNMT1 has to be remembered. It is the pairing that lets you answer what happens when a cell divides, and what happens when it does not. **Appears in:** Ch 4, Lect 3 **Video:** V12_de_novo_vs_maintenance_methylation.mp4

▪ Deacetylation

In one sentence: Take the acetyl group off, the chromatin tightens up again, and the gene is silenced.
Detailed explanation: Deacetylation is what HDAC, histone deacetylase, does, and it is exactly the opposite of acetylation. The mechanism is symmetrical: acetylation neutralises the positive charge on lysine and loosens the nucleosome; deacetylation gives that positive charge back, the nucleosome wraps tight again, transcription factors and enhancers can no longer squeeze in to the promoter, and the gene is shut. So the same chemical group, added or removed, behaves like a switch, and the cell has an enzyme for each direction. The clinical use is in cancer. In many cancers HDAC is overexpressed and holds a tumour-suppressor gene down; an HDAC inhibitor switches it off, and the gene starts expressing again. His Lecture 3 also points out that cancer cells commonly show a global loss of H4K16 acetylation, and that low H3K18ac predicts a worse outcome. It is worth holding both of those facts together, because the loss of a mark and the gain of the enzyme that removes it are the same disease seen from two sides, and the drug that blocks the enzyme is aimed at exactly that.
Why this course brings it up: It is the epigenetic regulation material in Lectures 3 and 5. The HDAC inhibitor and the "gene addiction" story about inducing PD-L1 are both on his slides, and the HDAC inhibitor is also one of the MCQ question formats he lists, "promoter hypermethylation leads to HDAC inhibitor". He also wants you to remember the direction: acetylation opens chromatin, deacetylation closes it, and an HDAC inhibitor opens it again. **Appears in:** Ch 4, Lect 3, Lect 5

▪ Deamidation

In one sentence: The slow chemical change in which asparagine in a protein becomes aspartate, and it changes how well the protein works.
Detailed explanation: Even when the sequence of a therapeutic protein is completely correct, the protein still ages slowly during storage and use, and the commonest kind of ageing is deamidation.
The route runs like this: asparagine loses ammonia at high pH and becomes succinimide, an intermediate; the intermediate then takes in a water molecule and becomes aspartate or its stereochemical twin, isoAsp. The mass change is given by the numbers on his slide — asparagine 114.04 Da, succinimide 97.02 Da, and both final products 115.03 Da. The clinical significance is that a tiny change in mass can change how strongly a drug binds its target, and therefore change its potency. The mechanism is slow and unavoidable, which is exactly why it becomes a manufacturing and shelf-life problem rather than a research problem: you cannot redesign around every asparagine in a protein, you can only measure how much iso-Asp has built up. A change you cannot design away, only measure, is the definition of a liability in this business. Slow, chemical and unavoidable — the three adjectives that make it a liability rather than a bug.
Why this course brings it up: It is the therapeutic protein liability example he gave himself in Lecture 1; he does not expect you

to memorise the dalton numbers, but the idea that asparagine turning into aspartate changes potency is something you do need. The same chain also explains why a protein made today can behave differently from a batch made a year ago. **Appears in:** Ch 2, Lect 1

▪ Death receptor

In one sentence: A receptor on the cell surface that starts apoptosis when an outside signal touches it.
Detailed explanation: The death receptor is the entrance to the extrinsic apoptosis pathway. It belongs to the TNF receptor superfamily, and the most typical member is DR5, the receptor for TRAIL.
When the ligand binds, the receptor trimerises, the death domains inside the cell gather together, caspase 8 is recruited and activated, and caspase 8 goes on to cut procaspase 3, so apoptosis begins. Sometimes it also cuts bid to make tbid and sends the signal on into the mitochondria. The clinical use of all this is very direct. His bispecific antibody BI 905711 hits TRAILR2 and CDH17 at the same time, and the logic of a tumour having both conditions at once confines the switching on of the death receptor to the tumour and away from the normal tissue, which is exactly the problem the antibody was designed to solve.
Why this course brings it up: It comes from the entrance to the extrinsic apoptosis pathway in Lecture 2, and it is the mechanism behind his bispecific antibody story. It belongs to understanding the concept rather than to the list of terms you have to write down. **Appears in:** Ch 8, Lect 2

▪ Deletion

In one sentence: A stretch of DNA that has gone missing, taking a bit of the sequence with it.
Detailed explanation: Deletion is the simplest of the structural chromosome abnormalities: a piece of the sequence is simply not there, so the gene dosage changes and the regulatory relationships between neighbouring genes may be broken as well. The scale can vary enormously. Losing only a few bases is an indel, and that may cause a frameshift, while losing a whole arm of a chromosome (the p arm or the q arm) is a very large event. Dr. Wong's third lecture lines deletion up with duplication, inversion, substitution and translocation as the five structural abnormalities, and he makes a particular point about one of them: an inversion neither adds nor removes genetic material and does not change gene dosage, yet it can change expression dramatically by moving a gene away from its enhancer. That counter-example is case 9 in the case bank. So the size of the loss and the size of the consequence are two different questions, and the second one is the one that catches people out. What makes a deletion dangerous is usually not the hole itself but the fact that the genes either side of it lose the arrangement they were reading each other in, so they get turned up or turned down when nobody asked them to be.
Why this course brings it up: The structural chromosome abnormalities section of Lecture 3. The contrast he wants you to remember is that some abnormalities cause a loss of genetic material and some do not. **Appears in:** Ch 3,Ch 10,Lect 3

▪ Delta ligand

In one sentence: The molecule on the surface of a neighbouring cell that Notch listens to.
Detailed explanation: Notch is a neighbour sensor. The Notch receptor on the receiving cell carries EGF repeats and is what makes contact with the delta ligand on the other cell's surface; after contact Notch is cleaved by a protease, releasing the intracellular NICD, NICD runs into the nucleus and switches on a set of genes linked to differentiation.
The key point is that delta sits only on the neighbour, so the Notch system comes with position information built in. It does not answer "how is this cell", it answers "who is my neighbour". Because the signal only appears on contact, it is also naturally a communication channel between cells rather than a message arriving from far away. In Lecture 5 he turns this into synNotch by designing a synthetic receptor that recognises only a tumour antigen.
Why this course brings it up: It is the Notch and synNotch section of Lecture 5, the source of the AND-gate CAR design, so it is concept-only. The design lesson is that borrowing a natural neighbour-sensing mechanism gives you a receptor with spatial specificity built in, which is exactly what a CAR needs and does not have. **Appears in:** Ch 7, Lect 5

▪ Denaturation

In one sentence: The process of opening the double helix into two single strands of DNA; also called melting.
Detailed explanation: To separate the two strands of double-stranded DNA is to take the hydrogen bonds between them apart one by one, and that is denaturation. It happens in the cell every day, because replication and transcription both have to open the double strand, and in the laboratory you do it with heat (the first step of PCR, and the pretreatment for a Southern blot).
The higher the GC content, the more hydrogen bonds there are and the harder it is to pull them apart, so the temperature denaturation needs (the melting temperature, Tm) is higher as well. The strands do not break; they simply stop being held to each other, and they can be put back together again.
Why this course brings it up: The direction of the GC content and Tm relationship is one of the Tier 3 numbers he lists; he also reminds you himself that the numbers in two places do not match up, so it is the direction and the reasoning that must be known, not the figures to be memorised. **Appears in:** Ch 3, Lect 1, Lect 3, Lect 4

▪ Dendritic cell

In one sentence: The specialised scavenger among immune cells, which picks up antigens and lays them out for T cells to look at.

Detailed explanation: A dendritic cell is the most professional antigen-presenting cell there is, and it is also the thing that starts the immune response off. Its behaviour is like that of a dedicated collection truck: it patrols the peripheral tissues, swallows a pathogen or a piece of tumour, cuts it into small pieces inside itself, loads those pieces into MHC molecules, and then migrates to the lymph node and spreads those "menus" out on its cell surface for T cells to inspect.

At the same time it supplies the second signal, B7 to CD28, because antigen without co-stimulation pushes a T cell into anergy instead of an attack. The clinically important role is in vaccine strategy: hand the tumour antigen to a dendritic cell and let it start the T cells off.

Why this course brings it up: It comes from the immune activation sections of Lectures 2 and 4. The role of the dendritic cell in providing co-stimulation is the premise for understanding both anergy and T cell activation, so you need it before either of those two topics makes sense. **Appears in:** Lect 2, Lect 4

▪ Deoxyribose

In one sentence: The sugar in DNA, which has one oxygen atom fewer than the sugar in RNA.

Detailed explanation: The sugar of RNA is ribose and the sugar of DNA is deoxyribose; the difference is that the 2′ carbon of ribose carries a hydroxyl group (–OH) while deoxyribose carries a hydrogen (–H).

That apparently tiny difference has two consequences. First, the 2′ hydroxyl takes part in hydrolysis, so RNA is easily broken in alkaline conditions and can only exist temporarily; DNA is much more stable and suited to long-term storage. Second, DNA replication does not have to recognise a 2′ hydroxyl at all, which makes the synthetic machinery's job considerably simpler. One oxygen atom, and it is the difference between a molecule that can be read once and a molecule that can be archived.

Why this course brings it up: It is the nucleic acid chemistry section of Lecture 1; he says plainly not to memorise chemical structures, so it belongs to the concepts you are expected to understand. **Appears in:** Ch 3

▪ deruxtecan

In one sentence: The family name for the DXd class of topoisomerase inhibitor payloads.

Detailed explanation: The molecules of the deruxtecan series are a new kind of topoisomerase inhibitor, and they serve as the poison part of an ADC. They cause DNA damage, so that the cell cannot replicate, so an ADC using one is a DNA-destroying tool delivered to the door of the tumour.

In his ADC examples the payloads fall into two groups, SN-38 and DXd. This is the core reasoning of ADC design: the antibody is responsible for delivering, the payload is responsible for killing, and the mechanism of the payload has to be mismatched to whatever resistance strategy the tumour has. The last part is the clever bit. If the tumour has learned to cope with one kind of damage, a payload that works by a completely different route still gets through, and that is why there are two payload families in his examples rather than one.

Why this course brings it up: He does not list the molecular structure of the payload as an exam question, and he says outright that the structure of the ADC is not examined. All you need is what the payload is and why the DNA-damaging type was chosen. The reasoning is the answer, and the drawing is not. **Appears in:** Ch 10, Lect 4

▪ detergents

In one sentence: Molecules that keep hard-to-soluble membrane proteins holding their shape in water.

Detailed explanation: The stretch of a protein that crosses the membrane is made entirely of hydrophobic amino acids that do not like water, and ordinary mass spectrometry works in aqueous solution, so a protein like that precipitates out before it even reaches the instrument.

Adding a detergent or a lipid to wrap the hydrophobic surface keeps it in solution, but the detergent then interferes with the measurement as well, and that is exactly why membrane protein proteomics is the hardest part of proteomics. It is a practical example of the trade-offs in technique that he uses to make the point.

Why this course brings it up: Dr. Wong does not name detergents as an exam question on its own, so it is conceptual understanding. **Appears in:** Lect 2

▪ Developmental specificity

In one sentence: The same gene being switched on and off at different stages of the body's development.

Detailed explanation: Gene expression changes not only with cell type but with time. Embryonic, childhood and adult stages each need their own genes, each with its own switching timetable, so the same protein can be essential at one point in life and irrelevant at another.

In Lecture 5 he places this alongside cell-type specificity and says both are concepts he always emphasises. Mechanistically, temporal specificity runs on signalling that delivers a transcription factor into the nucleus: an external signal triggers a phosphorylation cascade or a protease cut, and the transcription factor is let into the nucleus. The signal itself is often the thing that changes with time, so the same circuit produces different outcomes at different stages. There is another layer, epigenetics: methylation and histone marks lock a state in place so that it persists across cell divisions, which is how a cell remembers which stage of development it is in.

Why this course brings it up: It is one of the two things in Lecture 5 about which he said, "This is the concept that I always emphasize". The other is cell-type specificity. He pairs them deliberately, so an answer that talks about one without the other is

only half an answer. **Appears in:** Ch 7, Lect 5

▪ Diagnostic biomarker

In one sentence: Something that tells you whether the patient has the disease right now or not.
Detailed explanation: A diagnostic biomarker answers the question of “now”, and it is used for screening and for confirming a diagnosis. Dr. Wong's third lecture sorts the available tests by the sample site and by the use they are put to.
For stool-based early screening the list is ITG4, SFRP2 (methylation marker genes), Cologuard, miR-21, miR-92a and miR-135b. For blood-based early screening it is SEPT9 (a methylation marker gene), SFRP2 methylation, miR-21, miR-92a and miR-29a. On the clinical side there is always a trade-off between sensitivity and specificity: a false positive sends a person into unnecessary further tests, and a false negative misses a patient who could have been treated. The line between diagnostic on the one hand and prognostic and predictive on the other is one of the multiple-choice distractor pairs he listed. So a test is never simply “good” or “bad”. It is good at something and bad at something else, and the question is which of the two mistakes you can afford to make in your particular clinic.
Why this course brings it up: This is one of the three kinds of biomarker he set up at the start of the course. His third lecture gives a concrete list of markers, but the diagnostic biomarker itself was not named as an exam topic on its own. What is examinable is the set of names and the distinction between the diagnostic, prognostic and predictive kinds. **Appears in:** Ch 11,Lect 1

▪ Dicer

In one sentence: Dicer is the enzyme that cuts long RNA short; one piece of what it cuts becomes a microRNA.
Detailed explanation: The making of a microRNA happens in two steps. Upstream, the Microprocessor complex (DGCR8, Drosha's partner protein, plus Drosha itself) cuts pri-miRNA into pre-miRNA inside the nucleus;
pre-miRNA is then exported to the cytoplasm by Exportin-5, where DICER1 (the miRNA cutting enzyme) plus TRBP (an RNA-binding protein) cuts it into a double strand of about 22 nucleotides. One strand of that is then loaded into AGO to form RISC. The division of labour is clean: Drosha handles the hair, Dicer handles the length, AGO does the guiding. The order matters, because the two enzymes are not interchangeable: the nuclear cut makes the hairpin and the cytoplasmic cut sets the length that decides which targets the message can reach. The most practical clinical use is tumour subtyping — in Lecture 5 he analysed 97 pairs of hepatocellular carcinoma tissue and found that the miRNA cluster at 14q32.2 is over-expressed in the tumours and strongly correlated with serum AFP. Both enzymes are named for what they cut, and both are pieces of machinery you will meet again whenever RNA processing is discussed.
Why this course brings it up: It is the miRNA biosynthesis material in Lecture 1 and Lecture 5; he never named Dicer as an exam topic on its own, but the two-step splicing of miRNA and its relationship to AGO are things you do need. The two-step picture is the thing to carry away, because it explains why AGO cannot work until Dicer has been. And the Dicer product is what AGO loads, which closes the loop back to the earlier entry. **Appears in:** Ch 7, Lect 1, Lect 5

▪ Differentiation

In one sentence: A cell decides to become one particular kind of specialist cell.
Detailed explanation: A fertilised egg divides and produces nerve cells, muscle cells, liver cells, and every one of them carries the same DNA while doing a completely different job. What sends them down separate roads is differentiation: each cell switches on a different set of genes, and so makes different proteins and folds into a different shape, even though the DNA sitting in the nucleus is identical in every one of them.
In Lecture 2 he turns this into a reversible experiment in the iPSC section: the Yamanaka factors, Oct3/4, Sox2, c-Myc and Klf4, reprogramme a fibroblast back to the pluripotent state, and it is then differentiated into cardiomyocytes or neurons, for use as disease models and for drug screening, including the cardiac toxicity test carried out on iPSC-derived heart muscle cells.
Why this course brings it up: It comes from the iPSC and classification section of Lecture 2. Differentiation, and pre-differentiation against post-differentiation, are the variables on his slides, so it belongs to understanding the concept, and the iPSC experiment is the proof that the whole thing can be run backwards. **Appears in:** Ch 8, Lect 2

▪ Diffusion

In one sentence: Molecules spreading by themselves from where there are many to where there are few, at no energy cost.
Detailed explanation: Diffusion is passive transport caused purely by thermal motion. Molecules are always jiggling about, so more of them leave a crowded region than a quiet one, and the overall trend runs from high concentration to low concentration. The cell uses it for all sorts of things: oxygen and glucose in, carbon dioxide and lactate out, drugs across the membrane. Because it is a consequence of thermal motion, the rate of diffusion is inversely proportional to the square of molecular size and directly proportional to temperature. The clinical application is transdermal absorption and inhaled delivery, letting a drug cross the skin or the alveoli down its concentration gradient, which is a great deal cheaper than an injection.
Why this course brings it up: It is the chemistry and transport foundation in Lecture 1. He never names diffusion as an exam question, so it is concept-only. **Appears in:** Ch 1, Lect 1

▪ Dimerisation

In one sentence: Two identical or different proteins joining up as a pair, which often makes them far better at binding DNA.
Detailed explanation: A nuclear receptor binds a DNA response element very poorly on its own, and it takes two of them together to read the DNA accurately. That is why the DBD of a nuclear receptor carries a zinc finger structure, and why it almost always acts as a dimer.
A symmetric dimer (homodimer) works, and so does an asymmetric one (heterodimer). His example is HER2: it cannot dimerise efficiently with itself, and instead forms heterodimers with HER1, HER3 or HER4, and Herceptin (trastuzumab) works precisely by blocking that dimerisation. Think of a receptor that has to be switched on by finding a partner: block the partner and the switch can never be thrown, no matter how much ligand is floating around. The general pharmacological rule is that a drug does not have to block the ligand to stop a receptor being activated. Sometimes it is enough to stop the receptor assembling itself into the shape that works. And notice that this is a different strategy from competing at the ligand binding site, which is why the two approaches can be combined rather than simply substituted for one another.
Why this course brings it up: Nuclear receptors and dimerisation in Lecture 5. He listed the monomer against dimer pair among his distractors, and he ties HER2 dimerisation to Herceptin in the same breath. So the pairing you should remember is the one that connects the structural idea to a drug that is actually used. **Appears in:** Ch 7,Lect 5

▪ Diploid

In one sentence: Two full sets of chromosomes, one set from the father and one from the mother; human body cells have 23 pairs.
Detailed explanation: Diploid, written 2n, means that the body cells of a species carry two complete sets of chromosomes. For humans that is 23 pairs: 22 of them are autosomes and one is the pair of sex chromosomes.
Gametes are haploid instead, with n = 23, so fertilisation puts two sets back together and restores 2n. This "two copies" business explains why a recessive disease allele can hide inside a healthy copy, which is why a carrier can carry a mutation and never fall ill, and it is also the basis on which genetic counselling works out your risk. It also means that the two copies of a gene in one person can be different from each other, so the word "diploid" tells you the number of sets, not that everything in the sets is identical. Human 2n = 46 is one of the precise numbers he lists, and it turns up in the same group as Trisomy 21, at 2n+1 = 47, and Turner 45,X, at 2n−1 = 45.
Why this course brings it up: Both the 23 pairs of chromosomes and the two aneuploidy syndromes are on his list of precise figures, and reading a pedigree needs the idea of ploidy as well, because deciding whether a trait is dominant or recessive depends on how many copies of a gene you have to work with. **Appears in:** Ch 3, Ch 8, Lect 3

▪ Direct reversal

In one sentence: A repair pathway that fixes the damage exactly as it stands, without cutting out a single nucleotide.
Detailed explanation: Most repair pathways follow the pattern of cut it out, redo it and seal it up, but direct reversal does not even need the excision step. MGMT is the textbook example: it lifts the alkyl group straight off O6-methylguanine and the DNA is immediately as it was, with no resynthesis required.
It is the cheapest repair there is and the one with the least chance of introducing a mistake. No excision, no resynthesis, no chance of getting the base wrong the second time round. Dr. Wong's classification has a fourth line as well: cut and redo (BER and NER), mismatch repair (MMR), and the two double-strand break pathways (HR and NHEJ).
Why this course brings it up: He does not name direct reversal as a topic on its own, so it belongs to conceptual understanding. The MGMT methylation prognosis in the same section is what he marks as the Tier 2 point. **Appears in:** Ch 6, Lect 4

▪ Disaccharide

In one sentence: Two sugar units joined together, like sucrose.
Detailed explanation: Carbohydrates come in three levels of complexity: monosaccharides (glucose, fructose), disaccharides (sucrose) and polysaccharides (glycogen, starch, cellulose). Sucrose is a disaccharide of glucose plus fructose;
lactose is glucose plus galactose. A disaccharide has to be broken into monosaccharides in the gut before it can be absorbed, so a missing digestive enzyme shows up directly as malabsorption. Structurally, the bond between the two units is a glycosidic bond, and whether that bond is alpha or beta decides whether human enzymes can take it apart — which is exactly why cellulose uses beta1,4 linkages and humans are helpless against it. Same sugar, same chemistry, different linkage, completely different fate in your gut; this is one of the cleanest examples in the course of a single bond deciding everything. Three classes, one bond type, and one alpha or beta decision that decides whether you can digest it.
Why this course brings it up: It is one of the three classes of carbohydrate in Lecture 1; disaccharide itself was not named, but the three-level classification and the alpha / beta difference in the glycosidic bond are things you do need. So treat the alpha / beta distinction as examinable even though the word disaccharide is not. Which is also why the alpha / beta distinction reappears later in the course whenever a carbohydrate is discussed. **Appears in:** Ch 1, Lect 1

▪ Disease relevance

In one sentence: How much this molecule actually matters to this disease.
Detailed explanation: It is the first question when you are choosing a target: is this molecule really pushed up in the patients' samples? A molecule can be extremely important to the disease and still be impossible to make into a drug;

it can also be very easy to make into a drug and have nothing to do with the disease. The example he uses for contrast is the cytoskeleton: it is absolutely necessary for metastasis, and it is also too important and too everywhere to be attacked selectively. The question is not "can I hit it" but "does hitting it matter", and the two answers can come out in opposite directions.
Why this course brings it up: He explains very clearly that "druggability and disease relevance are an easily confused pair", but at the same time states outright that the whole target nomination workflow is not examined (he says it is only there so that you know more). So the distinction is worth understanding, but you are not being asked to run the process. **Appears in:** Ch 12, Lect 4

▪ dispersive

In one sentence: The third of the three models of DNA replication, in which the old strands are broken up and mixed into the new ones.
Detailed explanation: The Meselson and Stahl style isotope-labelling experiment set out three models against each other. Conservative says the old double helix is left completely intact and a new one is built alongside it.
Semi-conservative says each new double helix carries one old strand. Dispersive says the old strands are cut into fragments and scattered into the new ones. The experimental result supports semi-conservative, and dispersive is the one that is ruled out. The reason the three models have to be judged with isotope labelling is that the result on its own cannot tell you which old strand went where, so you have to follow a label. In other words, under the dispersive model the old DNA is not preserved in one piece, it is distributed in pieces among the new molecules, and the banding pattern after one generation tells you which model you are looking at.
Why this course brings it up: Dr. Wong does not name dispersive as an exam question on its own, so it is conceptual understanding, and semi-conservative replication is already in the glossary. It is one of three, and knowing that it lost is enough. **Appears in:** Ch 5, Lect 4

▪ Disulfide bond

In one sentence: A covalent link between two cysteines that helps hold a protein's shape in place.
Detailed explanation: The sulfhydryl groups on two cysteine side chains get oxidised, and an S–S covalent link forms between the sulfur atoms. It is one of the few genuinely covalent bonds inside a protein, so it behaves like a staple, pinning down the tertiary and quaternary structure that is otherwise held together by hydrogen bonds and hydrophobic interactions.
It is also a sensitive spot for redox: the reducing environment inside the cell can turn it back into two free sulfhydryl groups, the fold can loosen or misfold, and in bacteria in the endoplasmic reticulum the result piles up as an inclusion body; wrongly paired disulfide bonds get returned by the ER quality control. That is also why the bond matters for drug design and not only for structure, because break the staple or force the wrong pairing and the protein stops fitting its target.
Why this course brings it up: In Lecture 1 he said that disulfide bonds stabilise the tertiary and quaternary structure of a protein, in his own words. He does not list it as a separate question, so it is concept-only. But in the same lecture he describes deamidation as a risk for therapeutic proteins, because it changes drug binding and efficacy, so you have to be able to run the whole chain, chemical change leads to shape change leads to function change, and to apply it both to a covalent drug and to a protein that has sat on a shelf for two years. **Appears in:** Ch 2, Lect 1

▪ Diversity

In one sentence: The genetic differences between individuals of the same species.
Detailed explanation: He lists it as one of the two main axes of the whole course: why people look different from each other, where new gene combinations come from, and why a tumour becomes less and less like the cell it started from.
The second axis is how the science he tells becomes something usable. The sources of diversity are three things: the crossing over in meiosis, the random distribution of chromosomes, and random fertilisation. Three sources and three levels of answer, which is why the question about diversity is usually a question about mechanism.
Why this course brings it up: He does not name this entry in particular, so it is concept-only. The Tier 1 question he does name is the difference between meiosis I and meiosis II, M1 against M2. That is where the precision is. **Appears in:** Ch 6, Lect 2

▪ dMMR

In one sentence: Deficient mismatch repair, meaning the mismatch repair machinery is broken.
Detailed explanation: dMMR is the other way of writing MSI-H on a pathology report. It describes the cause, that the cell's mismatch repair machinery is broken, while MSI-H describes the result, that the microsatellites are unstable because of it. The same event in two vocabularies, and the pathology report writes both so that you can line the molecular mechanism up against the clinical behaviour. One is the cause and one is the effect, and for treatment decisions the cause is what tells you which drug to use. You do not treat MSI-H, you treat the broken repair machinery that produced it.
Why this course brings it up: The Kaplan–Meier curve of dMMR against pMMR is what Dr. Wong calls one of the most important figures in modern medicine, because MSI-H patients do clearly better on anti-PD-1. **Appears in:** Ch 9, Lect 2, Lect 4

▪ DNA adduct

In one sentence: A chemical molecule stuck onto a DNA base, which may then be mismatched when the DNA is copied.
Detailed explanation: When a chemical carcinogen such as the NNK in cigarettes, or benzo(a)pyrene, is activated by the cell's P450 metabolism, the active intermediate that is produced attacks a base and glues a bulky add-on onto the DNA.

That is a DNA adduct. The thing stuck on changes the shape of the base, so the pairing rules no longer recognise it: during replication it gets paired with the wrong letter, and once that mismatch is fixed in, it is a mutation. On his slide, NNK in mice induces the GC→AT mutation in the K-ras gene, and the same chemical pathway in people is exactly where the KRAS mutations in lung cancer come from. So the order of events is: chemical change of shape, then shape changes the pairing, then a pairing mistake gets locked in. The adduct itself is not a mutation. It is a lesion, sitting there waiting for the next round of replication to turn it into one.

Why this course brings it up: Dr. Wong did not specifically name the term DNA adduct, but it is the starting point of the whole causal chain in Lecture 3, chemical carcinogenesis leading to mutation leading to cancer, and it is also the background to his KRAS G12C story. That is conceptual understanding, but you have to be able to tell the complete mechanism as one piece: the chemical changes the shape, the shape changes the pairing, and the pairing error gets fixed in. If you cannot say it in that order, you have not got the chain. **Appears in:** Ch 6,Lect 3

▪ DNA-bending protein

In one sentence: A protein that bends DNA into a loop, so that a distant enhancer can reach its own promoter.

Detailed explanation: An enhancer can sit thousands of bases away from its gene and still control how much that gene is transcribed. The reason is that DNA itself is flexible: a class of proteins called DNA-bending proteins grabs the DNA between the enhancer and the promoter and bends it into a loop, so that the activator sitting on the enhancer can touch the machine sitting on the promoter.

This is the physical precondition that makes long-range control possible at all, and it also explains why cis and trans are different levels: the loop is pulled together by protein, but the action still happens on the same DNA molecule. Without the bending, distance would be a wall; with it, distance is only a shape the DNA is willing to hold. The loop is the physical form of the sentence "this enhancer talks to this promoter".

Why this course brings it up: He brings this picture in when he talks about enhancers in Lecture 5, but he does not list it as an exam topic, so it belongs to the concepts you are expected to understand. His real emphasis is on the point that not all enhancers are upstream, some sit at the 3′ end downstream of the gene — so the long-range mechanism itself you must understand, or you cannot answer that question. The folding is the mechanism, not an optional extra. So the picture, not the protein name, is what you need. **Appears in:** Ch 7, Lect 5

▪ DNA-binding domain (DBD)

In one sentence: The highly conserved part of a nuclear receptor that recognises its own word in the DNA.

Detailed explanation: Every nuclear receptor has three parts: the NTD, which is variable and does the regulating, the DBD, which is highly conserved and recognises DNA, and the LBD, which binds the hormone or the drug and therefore changes shape and recruits a co-activator or a co-repressor.

The DBD is highly conserved for a good reason: the word it has to recognise, the response element, for instance the ERE that the estrogen receptor recognises, has to be read without a single mistake, and a misread word means binding the wrong gene. It usually carries a zinc finger, and a zinc ion holds the DNA-recognition shape in place. The division of labour between DBD and LBD is simply which part reads the gene and which part takes the drug, and that is the point you lean on when you decide which class of drug to use.

Why this course brings it up: DBD against LBD is one of the distractor pairs he lists, and it is second-level (A). He is explicit that there are three parts, NTD, DBD and LBD, and he uses a three-dimensional structure picture of the estrogen receptor to show how the LBD changes shape after ligand binding. You have to be able to tell this set apart in a single sentence, and he says much the same about DBD against LBD and about signal peptide against NTD. **Appears in:** Ch 2, Ch 7, Lect 5

▪ DNA Damage Response (DDR)

In one sentence: The cell's alarm system for DNA damage. It starts repair, and if the damage is bad enough, it starts cell death instead.

Detailed explanation: When DNA is damaged, the cell does not quietly fix it and carry on as if nothing had happened. The two sensor kinases ATM and ATR pick up the abnormality first and pass the signal on to effectors, proteins such as p53, and then there are two possible consequences.

One is that the cell cycle is stopped and repair gets under way. The other is that when the damage is bigger than the repair capacity, the caspase cascade and the mitochondrial signals are switched on and the cell undergoes apoptosis. So the DDR is really a checkpoint: check first, and if you cannot make it in time, destroy yourself. Every branch of it is about the same question, which is whether the damage is repairable at all. Cancer cells very often get their way precisely by switching this kill-switch off, which is how they keep dividing with damaged DNA, and this is one of the easiest and most damaging things a tumour can do for itself.

Why this course brings it up: He never lists the DDR itself as an exam question, but it is the skeleton of the whole line of Lecture 4: damage, then sensing, then repair or apoptosis. It is also the setup for his first-grade, A+, question, because a defect in NHEJ leads to SCID and NHEJ is one of the DSB repair pathways. It belongs to understanding the concept, but you do have to be able to describe the two branches of the DDR, because that is the frame in which every later question about DNA damage is asked. **Appears in:** Ch 6, Lect 4

▪ DNA (deoxyribonucleic acid)

In one sentence: The master manual, written with four letters, kept in the nucleus.
Detailed explanation: DNA is the whole set of master manuals in the control room. It is built from the four letters A, T, G and C, each letter being a nucleotide (base plus sugar plus phosphate);
the phosphate and sugar string together into two sugar–phosphate backbones, and the bases pair in the middle like the rungs of a ladder. It is about a metre long and still has to fit inside a nucleus far smaller than a millimetre, so it is wound around histones into nucleosomes, then folded into chromatin and folded again into chromosomes. Its information flows in one direction only: DNA → RNA → protein, and that is the central dogma. It is also why the same gene behaves differently in different cell types. The four letters are the whole alphabet, and everything else in this course is a story about what those four letters are doing.
Why this course brings it up: This is the frame of the entire course; he did not name the word itself as an exam topic, but every scenario question assumes you have it. He keeps returning to the DNA metaphor — recipe, printed manual, typo — so you should be able to use it to explain gene, chromosome and mutation. You are not expected to recite the structure; you are expected to be able to reason from it. **Appears in:** Ch 1, Ch 3

▪ DNA fingerprint

In one sentence: A method for identifying a person by the differences in length of repeated sequences.
Detailed explanation: The genome is littered with short repeated sequences (microsatellites). In one given person the number of repeats at one particular locus is fixed, while the lengths differ from person to person.
Record the lengths at several such loci together and you have a barcode that is unique — just like a fingerprint, except that it is made of DNA itself. When the barcode of a suspect matches the sample from the scene, they are the same person. One locus is not enough on its own, because the repeats are shared between people; it is the combination across many loci that makes the code personal.
Why this course brings it up: This is the application on his slide 28, so it is a concept to understand; the microsatellite and MSI mechanism in the same passage is the Tier 2 part. The forensic use is the memorable part; the mechanism underneath it is what the question asks about. **Appears in:** Ch 3, Lect 4

▪ DNA glycosylase

In one sentence: The enzyme that cuts out a damaged base entirely, and the first step of BER.
Detailed explanation: When a base is damaged by oxidation or alkylation, the glycosylase recognises it first and cuts the base away from the sugar backbone, leaving an AP site. Then AP endonuclease cuts at the gap, the polymerase puts in the correct base, and the ligase seals it up.
This procedure turns one damaged letter into a short phrase that has to be rewritten, because a sugar backbone that has lost its base pairs very easily during replication. It is a clever way of handling the problem: the cell does not try to repair the damaged base in place, it removes it and resynthesises that stretch, which is much easier than knowing what the correct original base was.
Why this course brings it up: He says explicitly that this part is not examined, and his words are roughly that the chemistry of DNA glycosylase and the AP site will not appear in the midterm. But the pairing of BER repairing a single damaged base against NER repairing a bulky lesion still has to be mastered. So the enzyme names are out and the two-pathway contrast is in.
Appears in: Ch 6, Lect 4

▪ DNA ligase

In one sentence: The enzyme that seals the last gap in a newly synthesised DNA strand.
Detailed explanation: DNA ligase closes nicks by forming a phosphodiester bond, joining a 5′ phosphate to a 3′ hydroxyl.
During replication, once the Okazaki fragments on the lagging strand have each been made, there is still an RNA primer and a seam left between them;
after RNase H removes the RNA primer, the remaining gap is what ligase fills, and only then do the pieces become one continuous strand. Without the ligase step you would have a row of fragments that individually look fine but do not add up to a usable strand. The same principle is used in recombinant DNA and molecular cloning: the sticky ends left by restriction enzymes are joined by ligase, and that is how a piece of foreign DNA gets pasted into a plasmid. Note what makes ligase different from the polymerase: the polymerase adds nucleotides to a growing end, whereas ligase joins two ends that are already finished, and it does so with no template at all.
Why this course brings it up: He does not ask you to memorise enzymes, but the directions are first-level (A) material: elongation is 5′→3′, proofreading is 3′→5′, and ligase seals a 5′ phosphate to a 3′ hydroxyl. Those three lines are the content of his Lecture 4 summary table, and MCQs love to ask about "which direction". He is explicit that enzyme names and domains are not examinable, so learn the three directions and the idea that a nick is a nick, whatever enzyme closes it. **Appears in:** Ch 5, Lect 4

▪ DNA polymerase

In one sentence: The enzyme that builds a new DNA chain, and it can only add nucleotides in the direction from 5 prime to 3 prime.
Detailed explanation: DNA polymerase has only one “hand”, so it can only reach in one direction: it adds new nucleotides to the 3′ end of the strand that is being extended, and therefore DNA can only be synthesised from the 5′ end towards the 3′ end.

The same enzyme also carries 3′→5′ exonuclease proofreading activity. When it catches a mispairing such as A:G, it cuts off the wrong one and puts the right letter back in. That proofreading is not only quality control: it also pulls nucleoside analogue drugs off the DNA, which is how resistance arises. The other way round, a polymerase whose proofreading has failed manufactures large numbers of mutations and neoantigens, and the tumour then responds well to immune checkpoint inhibitors. So the enzyme carries three jobs at once: it builds, it checks, and it cleans up the mess that the drugs leave behind. Notice that both of the polymerase's activities point in opposite directions, and exam questions love to mix them up. The chain is made 5′→3′. The proofreading runs the other way, 3′→5′. So when you are asked which direction each activity runs in, the answer is: building goes one way, proofreading goes the other.

Why this course brings it up: His own words are the exam point: “this is important, you need to understand the elongation happens from 5 prime to 3 prime”, and he listed it as second-level (A). Equally important is the direction of the 3′→5′ proofreading, because that is the discriminator in the 5′→3′ versus 3′→5′ distractor pair from Lecture 4. So there are two separate things to keep straight: the direction in which the chain is built, and the direction in which the enzyme walks back to check its work. The first is the one he quotes; the second is the one that trips people up. Both appear in his material, and getting them the wrong way round is an easy way to lose marks on a question you otherwise know. **Appears in:** Ch 5,Lect 4

▪ DNA repair

In one sentence: The set of pathways that repairs the damage before it turns into a permanent mutation.

Detailed explanation: DNA is damaged every single day: replication makes mistakes, reactive oxygen species attack it, UV light creates pyrimidine dimers, and ionising radiation breaks both strands.

The cell uses a different pathway for each size of wound, and the choice is not arbitrary. BER repairs a single damaged base, NER repairs bulky lesions, MMR repairs copying mismatches and small insertions or deletions, and HR and NHEJ repair double-strand breaks. The logic they all share is the same: cut the bad part out first, let a polymerase put the right sequence back in, and finish with a ligase sealing the join. When this machinery fails, what is left behind is a permanent mutation, and a mutation that lands on an important word is how cancer starts.

Why this course brings it up: It is the main line of Lecture 4. Although he never treats the four words "DNA repair" as a standalone exam question, the concept he describes is something you have to understand: every kind of damage is matched to one repair pathway, and it has to be connected to the clinic, where MMR deficiency leads to MSI-high, leads to a hot tumour, leads to a good response to anti-PD-1. He said explicitly that you do not need to memorise the chemistry of DNA glycosylase, but you must know what BER repairs, and which pathway belongs to which kind of damage. **Appears in:** Ch 6, Lect 4 **Video:** V16_DNA_repair.mp4

▪ DNase

In one sentence: The enzyme that cuts DNA into pieces. It destroyed the transforming activity, and that is how DNA was identified.

Detailed explanation: In 1943 Avery, MacLeod and McCarty ran an enzyme experiment, and that experiment is the turning point of molecular biology. They took the “transforming factor” that Griffith had described in 1928, a substance that could turn harmless pneumococcus into a virulent one, split it into portions, added a different enzyme to each, and looked for the portion that made the transforming activity disappear.

DNase, the enzyme that cuts DNA, made it disappear; protease, the enzyme that cuts protein, did not. Since DNA was the thing that could be destroyed, the transforming factor had to be DNA. That single step established both that DNA is the genetic material and that enzyme experiments are a way of asking “what is this thing?”. Notice how clean the logic is. You are not measuring DNA directly; you are removing things one at a time and seeing which removal kills the effect. It also shows why Griffith's name keeps coming up: he found the substance and knew it was transferable, but he could not say what it was, and it took another four years and a different method to answer him.

Why this course brings it up: Dr. Wong did not treat DNase as an exam topic in its own right, but it is the key step in the DNA history passage of Lecture 4 and also the focus of the DNA history section of his first lecture. It is conceptual understanding, and all you have to be able to say is the causal sentence: DNase destroys it, protease does not, therefore DNA is the genetic material. That one line is worth memorising as a line, because it is exactly how he puts it. **Appears in:** Ch 3,Lect 1

▪ DNMT1

In one sentence: A maintenance methyltransferase: once DNA has been copied, it copies the methylation pattern onto the new strand.

Detailed explanation: DNMT1 is the maintenance type of DNA methyltransferase. The problem it solves comes from the fact that DNA replication is semi-conservative. When the old strand separates and new strands are built alongside it, each new CpG in the copy receives only half of the information that the old one carried.

If nobody stopped that from happening, the methylation marks would be diluted at every division, and eventually the whole methylation map of the cell would wash away, and the cell would quietly forget who it was. DNMT1's job is to recognise the freshly made hemi-methylated DNA at the replication fork and put the missing methyl group back onto the cytosine on the new strand. The same mark is then copied faithfully from one sister cell to the next, and that is exactly what epigenetic cell memory is.

Why this course brings it up: De novo against maintenance is one of the distractor pairs he lists, and it is a second-level, grade A, concept. He said that a de novo methyltransferase lays down new marks and a maintenance methyltransferase copies

the marks onto the daughter strand after replication. You have to be able to draw that action at the replication fork, and you have to be able to say which of the two types of methyltransferase is the one doing the copying. **Appears in:** Ch 4, Lect 3

▪ DNMT3A and DNMT3B

In one sentence: The de novo methyltransferases; they put the first methylation mark on a stretch of DNA.
Detailed explanation: DNMT3A and DNMT3B are responsible for establishing methylation from scratch. They take the methyl group from SAM (S-adenosyl methionine), the methyl donor, and attach it to the fifth carbon of the cytosine, forming 5-methylcytosine;
when that cytosine sits in a promoter inside a CpG island, the mark blocks transcription factors and is effectively the gene switch being turned off. The phrase de novo is the whole point: these enzymes do the first marking, not the copying of a pattern that already exists. One common driver of cancer is exactly this: the promoter of a tumour-suppressor gene locked shut by hypermethylation — the brake has been stomped on and held there. Nothing about the DNA sequence itself has changed; only a chemical tag has been added, and that is enough to switch a gene off. The opposite route runs the other way: hypomethylation of the repetitive parts of the genome causes mitotic recombination and genomic instability. Two enzymes, one job, and one of the most important switches in the whole book: whether a gene is on or off. And the striking part for a beginner is that no base pair is changed at all; a tag is simply added to one of the carbons.
Why this course brings it up: In Lecture 3 he says the main function of DNA methylation is to suppress gene expression, and that promoter hypermethylation is a common driver of cancer, which makes it a second-level (A) concept. So when he says methylation suppresses expression, this pair of enzymes is the machinery doing the suppressing. Also remember his clinical example: MGMT promoter methylation → sensitivity to alkylating chemo and radiotherapy and better survival. The MGMT example is worth more than the mechanism, because it is the one that turns a mark on DNA into a decision about chemotherapy. **Appears in:** Ch 4, Lect 3

▪ dNTPs

In one sentence: The free nucleotide raw material used when DNA is synthesised.
Detailed explanation: A dNTP is a deoxyribonucleotide triphosphate, one nucleotide carrying three phosphate groups. Each time, the polymerase takes a phosphate off the 5′ end, joins the remaining two onto the 3′-OH of the new strand, and releases pyrophosphate at the same time.
PCR can "amplify" because every round turns this raw material into new double strands. The concentration of the raw material is one of the experimental conditions as well, too little and the reaction is slow, too much and it easily produces non-specific products. So the same sentence describes both the chemistry and the recipe, and every round of PCR consumes another batch of the raw material.
Why this course brings it up: Dr. Wong does not name dNTPs as an exam question on its own, so it is conceptual understanding, and the rule that elongation always runs 5′→3′ is the Tier 2 item. **Appears in:** Ch 5, Lect 4

▪ donor plasmid

In one sentence: In molecular cloning, the plasmid that carries the DNA you want to put in.
Detailed explanation: The donor plasmid carries an Amp resistance marker, a NotI cutting site and the insert (the gene you want to express). It is amplified in bulk with PCR and then cut open with EcoRI and NotI to release the insert.
The recipient plasmid is the backbone that has been cut into a line and is ready to receive the insert. Once the two are joined they become a recombinant plasmid. The names themselves explain the roles: one pays for the goods and ships them, the other receives them and puts them into stock. The insert is the cargo, and the restriction sites are the packing cases that make the cargo and the skeleton grip each other.
Why this course brings it up: Dr. Wong does not name the donor plasmid as an exam topic on its own, so it belongs to conceptual understanding. But it is the first half of the cutting-and-joining picture, and the pairing with the recipient plasmid is worth having. **Appears in:** Ch 3, Lect 4

▪ donor template

In one sentence: The template DNA that supplies the correct sequence when you are doing a knock-in.
Detailed explanation: After CRISPR has cut the DNA, if you also provide a piece of DNA carrying the correct sequence as a donor template, the cell will use homology-directed repair to copy that sequence in — that is a gene knock-in.
But it only works in S and G2, because only then is there a sister chromatid to copy from. With no template and the cell going down the NHEJ route, the result is a knock-out instead. To make a knock-in succeed, the two ends of the donor template are usually designed to be homologous to the two sides of the break, so that the cell knows where to copy from. The template is the difference between the two outcomes: with it, you get the sequence you designed; without it, you get a scar. That single design decision — whether to offer a template — is what selects HR over NHEJ.
Why this course brings it up: "Knock-in uses HR (S/G2 only), knock-out uses NHEJ (in any phase)" is one of the Tier 2 pairings he lists, and it is also category 12 of the situation questions. So it is worth knowing cold: the choice of repair pathway is decided by what the cell has available and what you have handed it. **Appears in:** Ch 6, Lect 4

▪ Dose escalation

In one sentence: The first stage of a trial, in which the dose is stepped up to find the safe ceiling.
Detailed explanation: Phase 1A is called dose escalation. The method is that a group of patients gets a small dose first, you watch toxicity and pharmacokinetics, and then you go up according to a pre-set rule until you hit a dose-limiting toxicity, at which point you step back and find the maximum tolerated dose, the MTD.
It produces two things: a safety boundary, and the dose range that the later Phase 2 and Phase 3 studies will use. Phase 1B is the expansion cohort, where you treat more patients at a dose near the MTD and look for signs of efficacy. Phase 2 and Phase 3 are where efficacy is really tested. He also mentions that a trial costs about US$200,000 per patient.
Why this course brings it up: He does not list the details of Phase 1A as an exam question, but "how trials are phased" is common knowledge that can turn up in a scenario question, and he supplied the number himself: about US$200,000 per patient. So it is concept-only. **Appears in:** Ch 12, Lect 2, Lect 4

▪ Double helix

In one sentence: Two DNA strands twisted around each other, with the bases pairing up in the middle like the rungs of a ladder.
Detailed explanation: The double helix is the final shape of DNA. The two strands run in opposite directions (antiparallel): one goes from 5′ to 3′ and the other from 3′ to 5′. The sugar-phosphate backbone is on the outside and the bases pair up on the inside as the rungs.
Every rung is held by hydrogen bonds: A–T is 2 hydrogen bonds and G–C is 3. That shape explains three things at once. It explains why replication is accurate, because the strict pairing rules are what make one strand a template for the other. It explains why PCR works, because heating pulls the two strands apart. And it explains why GC-rich fragments need a higher temperature to denature: there are more hydrogen bonds to break, so the melting temperature is higher. It is a shape that is simultaneously the storage format, the copying instruction and the laboratory tool. So when you are asked why a particular technique works, the answer is often found by looking at the geometry of the helix rather than by looking at the enzymes.
Why this course brings it up: In Lecture 1 Dr. Wong uses the Watson, Crick and Franklin model to bring out this shape, and he says it underlies PCR. He did not name double helix as an exam topic on its own, so it is conceptual understanding, but G–C = 3 and A–T = 2 are numbers he explicitly lists. So the shape is background, and the two hydrogen bond counts are facts. That combination is typical of how he sets the course: the story is for understanding, and the small numbers are what you are actually asked to produce. **Appears in:** Ch 3

▪ Double-strand break (DSB)

In one sentence: The most dangerous kind of DNA damage: both strands cut at the same position.
Detailed explanation: When DNA is cut into two strands, neither side has a complete template to pair against, which makes this the hardest of all the kinds of damage to deal with. What makes it dangerous is that the repair can go down one of two roads. One is HR, which uses the sister chromatid as a template and copies the sequence back exactly as it was; that is only possible in the S and G2 phases, when a sister chromatid exists, and it almost never goes wrong. The other is NHEJ, which needs no template and simply glues the two ends together; it is fast but error-prone, and it often leaves a deletion or an insertion behind, which is a permanent change in the sequence rather than a repair. Ionising radiation and abortive topoisomerase breaks both produce DSBs, and CRISPR/Cas9 makes one deliberately, because that is how it edits.
Why this course brings it up: DSB is the most important classification axis in his Lecture 4: BER, NER and MMR deal with single strands or small wounds, while HR and NHEJ deal with DSBs. The HR against NHEJ comparison he lists is the single most likely multiple-choice contrast, and the points that separate them are whether a template is needed, how accurate the result is, and the cell cycle restriction, and those three are usually what the question asks you to pick out. **Appears in:** Ch 6, Lect 4

▪ Down syndrome

In one sentence: Trisomy 21. Chromosome 21 comes in three copies instead of two.
Detailed explanation: Down syndrome is one form of aneuploidy, caused by chromosomes failing to separate during meiosis. The key is to be clear about which division failed. If it is meiosis I, where the homologous chromosomes fail to separate, one meiosis produces 2 normal gametes plus 2 abnormal gametes, and the two abnormal ones are identical to each other.
If it is meiosis II, where the sister chromatids fail to separate, it produces 1 normal gamete plus 3 abnormal ones, of which 2 are one type and 1 is another. The two identical abnormal gametes are the signature of a first-division error, and the mixed set of three is the signature of a second-division error, and that difference is what a pedigree can sometimes let you infer. Human somatic cells have 23 pairs of chromosomes, and the extra copy of chromosome 21 is what gives you the clinical picture.
Why this course brings it up: He names the need to tell M1 and M2 non-disjunction apart, and says "I think you need to distinguish these two situations here", which makes it a first-level (A+) question. A pedigree plus aneuploidy is one of the archetypes of the scenario questions. Turner 45,X is another example from the same set, and if you can do the gamete counts for M1 and M2 you can do them for both. What is examinable is the distinction and the counting, not the clinical features of the syndromes. **Appears in:** Ch 3, Lect 3

▪ Driver

In one sentence: A mutation that gives a cell a growth advantage. One of the typos in the important words.
Detailed explanation: A driver is the typo that makes a tumour grow faster and escape control better — it changes an important word, so the cell keeps manufacturing growth signals from the wrong recipe.
Its opposite is a passenger: a typo that simply hitched a ride, carried along by chromosomal proximity or by chromosomal instability, and giving no growth advantage of its own. To find the real drivers you have to put the CNV profile together with the mRNA expression: the gene whose copy number has risen and whose expression has risen at the same time is the one that might be a genuine driver. His classroom example is NPM1 in leukaemia. Most typos in a long book are harmless, and most mutations in a genome are passengers; the whole job of a cancer biologist is to work out which of the two you are looking at.
Why this course brings it up: Driver versus passenger is one of the distractor pairs he lists, a second-level (A) item, and he taught the method of integrating CNV with expression data in order to find drivers. It is also the central metaphor of the whole course: a mutation is a typo, and a driver is a typo in an important word. **Appears in:** Ch 3, Ch 10, Lect 3 **Video:** V08_Driver_vs_passenger.mp4

▪ Drosha

In one sentence: The enzyme that cuts the primary microRNA transcript into a short hairpin inside the nucleus.
Detailed explanation: miRNA is made in two steps, and both steps happen inside the nucleus first. Drosha, together with its partner protein DGCR8, forms the Microprocessor (the pri-miRNA splicing complex), and it cuts the primary transcript (pri-miRNA) into a hairpin of about 70 nucleotides (pre-miRNA).
Exportin-5 then carries it out of the nucleus into the cytoplasm, and Dicer cuts it again into a double strand of about 22 nucleotides, which is finally loaded into RISC to suppress the target mRNA. Drosha is therefore the gatekeeper: it decides which stretch of RNA is eligible to become a regulatory signal. The order matters, and it is one of the few sequences in the course that you should be able to recite: Drosha, then Exportin-5, then Dicer, then RISC. Each step also changes the scale of the thing being handled, from a long primary transcript to a short hairpin, then to a short duplex, and finally to something that can be loaded into a silencing complex.
Why this course brings it up: He did not name Drosha on its own, but the miRNA production pathway appears in both Lecture 1 and Lecture 5, and miRNA is an attractive non-invasive biomarker for cancers such as HCC precisely because it is so stable in serum. The number he gives in Lecture 1 is the AUC of 0.981 for miR-15b plus miR-130b. That is conceptual understanding, but you still need to know the order Drosha → Exportin-5 → Dicer, because that is the part a question can be built on. And keep the numbers separate: the enzyme names belong to the mechanism, while the AUC belongs to the diagnostic example, and mixing the two up is how you end up trying to remember an AUC for Dicer. **Appears in:** Ch 7,Lect 1,Lect 5

▪ Drug resistance

In one sentence: A tumour stops responding to a drug that used to work. It can be caused by a new mutation, or by an isoform that has lost the binding site.
Detailed explanation: Tumour drug resistance comes from three directions. At the gene level, a new mutation changes the binding site of the drug, or lets the downstream signalling run around whatever position was blocked.
At the splicing level, alternative splicing produces a different isoform that has cut out the exon the drug used to bind to, so the same gene and the same drug are still in play but the binding site is no longer there; the AR-V7 on his slides is exactly the cryptic exon product of AR, and it sits in the active state and gives enzalutamide resistance. At the replication level, the 3′→5′ proofreading activity of the polymerase cuts nucleoside analogue drugs straight back out of the DNA, so the drug that was inserted into the chain gets removed again, and the tumour survives the treatment that should have killed it.
Why this course brings it up: He covers proofreading as a cause of resistance in Lecture 4 and isoforms as a cause in Lecture 5, and case 15 is the scenario question built around the "isoform that predicts drug resistance" idea. It sits between the second level, grade A, and the third level, grade B, as a concept, but you have to be able to state all three mechanisms, because a scenario question will usually let you pick whichever one you remember. **Appears in:** Ch 5, Ch 7, Lect 4, Lect 5

▪ Drug target

In one sentence: The molecule a drug is designed to act on. Membrane proteins and kinases are the good examples.
Detailed explanation: A molecule that can serve as a drug target has to satisfy three things at once. First, in patient data it must be specifically over-expressed in the disease. Second, it must be structurally tractable, meaning there is some way to reach it with a small molecule, an antibody or a nucleic acid drug.
Third, the competitive landscape must not be too crowded. And "causes disease" is not the same as "can be drugged": cytoskeletal components are crucial in metastasis yet very hard to attack selectively, whereas telomerase is a genuine druggable target, with antisense oligonucleotides and small molecules in clinical trials. The three tests are not equally weighted in practice, and a molecule that fails any one of them is not a target. It is also why the structural biology in the earlier lectures is not decoration: tractability is a question about shape. Three tests, and a molecule only becomes a target when it passes all three at once. Which is why the failures are so varied: some molecules are abundant in the disease and untouchable, others are touchable and irrelevant.
Why this course brings it up: He explicitly rules the pharmaceutical industry's target nomination and druggability workflow out of the exam: "it's not included in exam, but I think it's really more interesting". At the level of concepts he still expects you to judge — his counter-example of the cytoskeleton is there so that you can say that causing disease is not the same as being

druggable. The workflow itself is not examinable; the judgement behind it is. So expect questions that test the judgement rather than the vocabulary. **Appears in:** Ch 10, Ch 12, Lect 1, Lect 2

▪ Druggability

In one sentence: Whether this molecule can be turned into a drug that hits it without hitting anything else.
Detailed explanation: This is the second question. Can you find a small molecule that fits into its pocket, an antibody that recognises it, or a piece of nucleic acid that can target it? K-Ras used to be called undruggable, and then it stopped being called that, and he uses that example to show that targets which used to be undruggable are becoming druggable.
So important and druggable are two questions that have to be answered separately, and that is exactly what his two examples, cytoskeleton and telomerase, are there to make. The reason the answer can differ is practical rather than theoretical. A molecule that is everywhere in the cell, and does everything, cannot be attacked without attacking the whole cell with it.
Why this course brings it up: He says explicitly that the whole druggability and target nomination workflow is not examined, he says it is only so that you know more. But the difference between a target and a drug is a concept he really did teach. So the process is out and the distinction is in. **Appears in:** Ch 12, Lect 4

▪ Dual-payload ADC

In one sentence: An ADC with two different drugs hanging off the same antibody.
Detailed explanation: The way you do it is to replace the single linker that was there with a branched linker and hang two drugs side by side off the antibody: usually a topoisomerase inhibitor plus an inhibitor of the DNA Damage Response, an ATM blocker for instance.
The aim is to get round the resistance that a single payload breeds, because if the tumour resists only the first drug the second is still working. The price is that the molecule gets bigger and the drug loading goes down, so it is still a research-stage design rather than an established standard practice. The idea is ahead of the practice here, and he is careful to say so. The linker is the part that makes it work.
Why this course brings it up: Dr. Wong is explicit that the structure of an ADC and the details of a dual-payload ADC will not be examined. All you need is that it exists and what the two payloads usually are. **Appears in:** Ch 10, Lect 1, Lect 4

▪ Duplication

In one sentence: A DNA segment that has been copied one extra time.
Detailed explanation: A duplication is one kind of structural variation: a stretch of a chromosome is copied once, so that stretch now appears in two copies. For a coding gene, one extra copy usually means more mRNA and more protein, and HER2 is exactly that: gene copy number goes up, mRNA goes up, the number of receptors on the cell surface goes up, and finally even the release of soluble receptor goes up.
That ladder is drawn very clearly on his Lecture 2 slide. There are two mechanisms that produce it: NHEJ during DNA repair, and unequal crossing over at a repeat sequence, NAHR, which can produce either a duplication or a deletion depending on which way the misaligned loops point. That symmetry is worth remembering, because the same accident of DNA repair gives you opposite clinical outcomes depending on the direction it goes. When two repeat sequences line up the wrong way and exchange, the DNA between them can be copied twice or lost entirely.
Why this course brings it up: He does not list duplication as an exam question on its own, but it is the premise for explaining why HER2 can be treated with Herceptin, and "gene amplified in some breast and gastric cancers" is his own wording. It also connects to receptor occupancy and the question of clinical dosing. So it is concept-only, but he wants you to see the chain from copy number to drug choice, because amplification is what makes HER2 a target and what makes the receptor occupancy numbers for Herceptin matter. **Appears in:** Ch 3, Lect 3

▪ DXd

In one sentence: The drug payload that trastuzumab deruxtecan carries.
Detailed explanation: DXd is the code name for the topoisomerase I inhibitor payload in deruxtecan. Trastuzumab deruxtecan, T-DXd, hooks it onto an anti-HER2 antibody with a linker, and that arrangement is the classic structure of an antibody-drug conjugate, an ADC.
The logic of the ADC is to separate two jobs that a single molecule cannot do at once: finding the tumour and killing the cell. Neither half works on its own, and that is the whole point of the design. The antibody does the finding, carrying the whole complex to the surface of cells that express a lot of HER2, and the payload is only released inside the cell, where it does its DNA damage. Hang the same payload off an antibody against Trop2 instead and you get datopotamab deruxtecan. There are also dual-payload ADCs, which use a branched linker to carry a topoisomerase inhibitor and a DNA repair drug at the same time, so that a tumour which has learned to shrug off the first drug still has to deal with the second.
Why this course brings it up: He says explicitly that the structure of the ADC and the dual-payload idea will not be examined: "all this would not be included in the midterm. I'm just want you to know more". So you do not have to memorise the details of the payload, and you do not have to remember which linker is used where. What you do have to understand is that the logic of an ADC is selective delivery, and that is exactly the skeleton of the answer to scenario case 13. **Appears in:** Ch 10, Lect 1, Lect 4

▪ E-selectin

In one sentence: One of the selectins, the one that lets white blood cells roll along the vessel wall.
Detailed explanation: The first step in a white blood cell leaving the blood vessel is rolling, and the molecules that make rolling possible are the selectins. E-selectin is the selectin that endothelial cells express in large amounts only during inflammation; it grabs the sialyl-Lewis X selectin ligand on the white blood cell and makes the cell roll and slow down along the vessel wall. Then integrins take part in the firm adhesion, ICAM-1 and VCAM-1 provide the adhesion sites, PECAM-1 lets the white blood cell squeeze out between the endothelial cells, and finally MCP-1 supplies the chemotactic direction. Those four steps are his "leukocyte extravasation" slide, and every one of them needs a different molecule. The order matters too: roll first, then stick, then squeeze through, and only then walk towards the target.
Why this course brings it up: He does not test experimental parameters, but this pathway matters in the tumour microenvironment, because tumours often block white blood cell entry by changing the endothelium and the ECM. So it is concept-only, and it is usable when you answer "why can't T cells get into a tumour". It is also a useful concrete example to have ready whenever you need to explain a barrier. **Appears in:** Ch 9, Lect 2

▪ E2F

In one sentence: A transcription factor that is normally held down by pRb, and released when the cell is ready to divide.
Detailed explanation: E2F is a family of transcription factors that control the genes needed for entry into S phase, for instance the enzymes of DNA replication. The gate of the cell is pRb: during G1, pRb binds E2F and holds it, so the S phase genes stay shut;
when the cyclin D–CDK4/6 complex phosphorylates pRb, pRb lets go, E2F is released, and the whole battery of S phase genes starts up, and the cell enters its replication phase. That is also why CDK4/6 inhibitors such as palbociclib work, because they weld the gate shut and the cell stays in G1. So E2F is the goal at the end of the signalling, and pRb is the brake that holds it back until the cell has finished growing. Everything in the G1 checkpoint story is built around this one contact between a brake and a pedal. Which is why the question about where the cell stops after CDK4/6 inhibition has a clean answer: G1, with the DNA content unchanged at 2N.
Why this course brings it up: He does not ask you to memorise the individual cyclin and CDK pairings: “you don't need to memorize all this cyclin or cyclin-dependent kinase, CDK”. But the pRb–E2F control line and the mechanism of action of the CDK4/6 drugs are conceptual understanding, and a scenario question may well ask where the cell ends up after CDK4/6 inhibition and what the DNA content is at that point. Notice that the pairing he tells you not to memorise is the pairing, while the regulatory line he wants you to understand is the logic: gate, brake, release. That is a fair description of how the course works. There is a great deal of detail that you are allowed to skip, and a small number of relationships that you are expected to be able to draw. **Appears in:** Ch 8,Lect 2

▪ ECM (extracellular matrix)

In one sentence: The scaffold of proteins and sugars that cells sit in and have to dig through in order to move.
Detailed explanation: The ECM is the web outside the cell woven from proteins and sugars, and cancer-associated fibroblasts, CAFs, make a great deal of it. It is a scaffold and a barrier at the same time: for a cell to move it has to remodel the web and cut through it, so a tumour can invade and metastasise by changing the ECM.
The stroma it sits in is also where immune cells have to travel, so the same mesh that blocks invasion also shapes whether an immune cell can get in. The ECM is also one of the obstacles that CAR-T runs into against solid tumours. When he lists the limits of CAR-T, he puts "stroma-rich ECM" alongside hypoxia, low pH, CD47 and IDO, which is his explanation of why CAR-T works against blood cancers but so often cannot get into a solid tumour.
Why this course brings it up: He does not name ECM as an exam question, so it is concept-only. But it appears on his CAR-T limitations page, and when you write a scenario answer about why CAR-T performs badly against solid tumours, this is one of the points you can make. **Appears in:** Ch 9, Lect 2

▪ EcoRI

In one sentence: A restriction endonuclease that cuts DNA at a specific position.
Detailed explanation: EcoRI recognises the six-base sequence GAATTC and cuts between the G and the A, leaving a single-stranded AATT overhang, a sticky end. Dr. Wong's molecular cloning slide uses EcoRI and NotI to open both the donor plasmid and the recipient plasmid, so that the two overhangs are complementary and can stick, and then DNA ligase seals the joint.
The cut is placed where you want the seam, and the two ends that come away from it are the ones that have to match up. The sequence GAATTC reads the same backwards as forwards, which is why the same enzyme can make compatible ends on two different pieces of DNA. This is the point of restriction enzymes in molecular cloning: different enzymes cut out different complementary ends, so DNA from different sources can be joined together.
Why this course brings it up: He does not name EcoRI as an exam topic on its own, so it belongs to conceptual understanding. What is the skeleton of this section is “restriction enzymes cut out sticky ends, ligase seals them up”. **Appears in:** Ch 3, Lect 4

▪ Ectoderm

In one sentence: Ectoderm. It becomes skin and the nervous system.
Detailed explanation: The embryo has three germ layers, each with its own fate. Ectoderm becomes skin and hair, the nervous system (brain and spinal cord), and the epithelium of the sensory organs;
mesoderm becomes muscle, bone, blood and heart; endoderm becomes the gut lining and the lung. In the validation of an iPSC these three layers are the gold standard: inject the iPSC into an immunodeficient mouse, and only if what grows is a teratoma containing tissue from all three layers (cartilage, central nervous system, muscle, fat, epithelium) can you claim it is genuinely pluripotent. The test is deliberately brutal: rather than asking what the cells can do, it asks what the cells did on their own inside a mouse, and a teratoma missing one layer fails the test. Three layers, three fates, and a validation assay built on all three.
Why this course brings it up: He did not list the germ layers as an exam topic, so it is a concept to understand — but it appears inside his iPSC validation workflow, and he says explicitly that the workflow has to be remembered: "unique expression pattern and timing, and also the negative staining to exclude the contaminated cell. Don't forget it." So the germ layers are a detail you need because they are the scoring criterion, not because anyone will ask you to list them. The teratoma assay and the marker panel are the two halves of the iPSC validation chain he draws. **Appears in:** Ch 8, Lect 2

▪ Edman degradation

In one sentence: The standard method for sequencing a protein, reading it off one amino acid at a time starting from the N terminus.
Detailed explanation: Edman degradation uses chemistry to cut the single amino acid at the very end of a protein (the N end) off and identify it, then repeats the whole thing on what is left.
What it gives you is the sequence read from the N end, which is the primary structure. The limitation is that it needs a clean single chain, and a real protein has to be chopped into fragments before it can be read, so it is a classical method rather than a production method. Modern LC-MS/MS does the same job a different way: it cuts proteins into peptides, separates them by liquid chromatography, weighs them by mass spectrometry, and matches them against a database, telling you both which protein it is and how much of it there is. One protein, one chain, one letter at a time; that is the whole idea. And because each round costs a full cycle of chemistry, it does not scale to a whole proteome.
Why this course brings it up: In Lecture 1 Dr. Wong presents Edman degradation as the classical method and LC-MS/MS as the modern workhorse. The pairing of which method answers which question is a common form for multiple-choice questions: Edman gives the sequence, LC-MS/MS gives identity and quantity, SDS-PAGE gives size, and X-ray gives the atomic shape. That is conceptual understanding, and the useful way to revise it is as a question-and-answer table rather than as a list of techniques. Because in the exam it will never be “what is Edman degradation?” but “which technique would you use for this?”. So learn the question each one answers. **Appears in:** Ch 2

▪ Effectors

In one sentence: The last link on the DDR signalling chain, the one that actually carries out the work.
Detailed explanation: His slide 23 draws the DNA Damage Response as a four-part chain. Signals are the things that do the hurting (DNA damage, replication stress). Sensors sit on the damaged DNA and are the ones that discover it.
Transducers pass the message along. Effectors carry it out, and he names p53 and BRCA at this level. There are four possible outcomes: stop the cell cycle, repair, senesce, or start apoptosis. Within that, p53 decides whether the cell stops to repair, and BRCA handles the high-fidelity repair of double-strand breaks; when both of them fail, the cell accumulates mutations. Each of the four stages has a different job, and the chain only makes sense if you keep them apart: something bad happens, somebody notices, the news gets passed on, and then somebody does something about it.
Why this course brings it up: He does not name Effectors as an exam topic on its own, so it is a concept to understand; the four-part chain and the four outcomes of DDR are the framework he has actually demonstrated. Learn the shape of the chain once and every one of these terms drops into its place. **Appears in:** Ch 6, Lect 4

▪ efficacy

In one sentence: The measure of whether a drug really works, in the patients who should get it.
Detailed explanation: Beyond safety, efficacy asks whether the drug actually changes the outcome in the group of patients who are supposed to receive it. It is a different judgement from safety.
Phase 1A and 1B deal mainly with safety and with dose, and only Phase 2 and Phase 3 formally test efficacy. Phase 3 is the large randomised trial that compares the new drug with the current standard of care. Safety decides whether the drug can be used at all, efficacy decides whether it is worth using, and a drug has to pass both before it can reach the market. And the two are not the same question, because a drug can be perfectly safe and still do nothing, in which case there is no point in it at all.
Why this course brings it up: The phasing of clinical trials, 1A escalation, 1B expansion and 2-3 for efficacy, is Tier 3 precise content that he lists. So the phase numbers and what each phase is for are both worth having exactly. **Appears in:** Ch 12, Lect 2, Lect 4

▪ EGFR

In one sentence: The growth factor receptor at the top of the RAS chain. Blocking it is a standard anticancer strategy.
Detailed explanation: EGFR is a receptor tyrosine kinase, an RTK, sitting at the very top of the growth signalling chain EGFR → RAS → RAF → MEK → ERK. Normally it only starts up when there is a ligand outside the cell;

once it starts, it passes a phosphate group downwards, and RAS is pushed into its active Ras-GTP form. EGFR itself sits on the cell surface, which is one of the reasons it is such a popular drug target: an antibody can reach it from outside the cell without ever having to enter the cytoplasm. The receptor is only allowed to fire when a ligand binds it on the outside, and the kinase activity happens on the inside, across the membrane, which is why blocking the extracellular side and inhibiting the intracellular kinase are two genuinely different ways of doing the same job. There are two drug strategies: anti-EGFR antibodies such as cetuximab, which block the ligand at the receptor, or small-molecule kinase inhibitors that press directly on the kinase activity. But the strategy carries a condition, and the condition is that the downstream has to be clean. If K-Ras downstream has mutated itself into a permanently active form, KRAS G12C, G13, Q61, then blocking the receptor achieves nothing and the signal keeps going anyway. That is why cetuximab fails in K-Ras mutant colorectal cancer, and it is why you test K-Ras as a companion diagnostic before you treat.

Why this course brings it up: He teaches K-Ras together with cetuximab as the predictive biomarker theme of Lecture 1, and it is one of the questions where he supplied numbers: KRAS G12C is 13 % of lung adenocarcinoma and 3 % of colorectal cancer. He also mentions the FDA-approved RAS inhibitor Rasonque, daraxonrasib, with a median OS of 13.2 against 6.7 months. He treats this as the place where biomarker logic becomes unavoidable: if you know the downstream mutation, you know whether the drug can work, so the test comes before the treatment rather than after it. **Appears in:** Ch 10, Lect 1, Lect 4, Lect 5

▪ Electron

In one sentence: A negatively charged particle that moves around the nucleus and is responsible for every bond.

Detailed explanation: An electron is a particle with a negative charge, and it occupies shells of fixed energy around the nucleus. The whole of chemistry is the redistribution of electrons. An atom whose outer shell is not full will try to fill it, either by sharing a pair of electrons with another atom to make a covalent bond, or by taking the electrons over completely to make an ionic bond.

Carbon can build long chains, rings, branches and cage-like structures because it has four valence electrons and wants to get to eight. The oxidation and reduction reactions inside a body, which is simply the movement of electrons from one molecule to another, are the same game being played with a biological score.

Why this course brings it up: Chemistry is the part he excludes explicitly: "Please don't memorize all these chemical structure… They would not be included in my exam." So this is background, but it is the floor underneath the idea that a kinase adds a phosphate using ATP and a phosphatase takes it away again, the switch that flips a protein on and off simply by adding a charged group, and that idea is used again and again in the later lectures. **Appears in:** Ch 1

▪ Electron shell

In one sentence: A fixed energy level outside the atomic nucleus. The outermost one decides what an atom can bond to.

Detailed explanation: Electrons do not wander; they occupy fixed energy shells, and each shell holds a limited number (two in the first shell, eight in the second, eighteen in the third). Chemical behaviour is decided entirely by the outermost shell, the valence shell: a full outer shell means the atom is satisfied and will not react, while one or two electrons short means it will go and take or share.

The number of electrons in that outer shell is the valency, and numbers like 2, 3, 6 and 7 directly determine what kind of bond the atom can form. Sodium has a valency of 1, so it hands an electron to chlorine; chlorine has a valency of 7, so it needs one — and the two meeting gives the ionic bond of table salt, which is the simplest possible picture of a transfer.

Why this course brings it up: It is part of the chemistry he excludes, but it is the bottom layer of the chain of analogies he gives: atom → bond → molecule → macromolecule → cell. Understand this and you understand why pH changes protein shape, and why a kinase can act as a switch just by adding a charged phosphate. **Appears in:** Ch 1

▪ ELISA

In one sentence: The standard test for measuring how much of a particular protein is in a fluid sample.

Detailed explanation: ELISA is an enzyme-linked immunosorbent assay: a pair of specific antibodies sandwiches the target protein in the sample, one antibody is stuck to the plate and the other carries an enzyme label, and once the unbound material has been washed away you add a substrate and the enzyme makes the colour, and the depth of the colour is proportional to how much target protein there was.

In his HCC case this is the last step and also the most important one: the vimentin signal that was originally found on 2D-PAGE was eventually moved into serum and measured by ELISA, which turned it into a biomarker that a patient can be tested for with a blood sample. So the whole ELISA story is a story about specificity at the front end and quantitation at the back end, and those two ends are why it works. The result is a number in a tube, which is exactly what a clinician can act on.

Why this course brings it up: He teaches the complete workflow, which is separation, identification, quantification, validation, and then turning it into a test the patient can actually have, and he gives numbers: serum vimentin against control 176.0 ± 61.3 ng/mL, HCC ≤ 2 cm at 271.5 ± 165.9, and HCC > 2 cm at 231.3 ± 132.3. That is conceptual understanding, but the logic of this pipeline has to be something you can recite. Note the order of the steps, because a question can easily ask you to put them in the right sequence, and note that the numbers are attached to specific groups of patients rather than being general facts about vimentin. **Appears in:** Ch 11,Lect 1,Lect 2

▪ End-replication problem

In one sentence: DNA polymerase cannot copy the very end of a linear chromosome, so the telomere gets shorter every time the cell divides.

Detailed explanation: DNA can only be synthesised 5′→3′, and once the RNA primer on the last piece of the lagging strand has been removed there is no 3′ end left to extend from, so the end of the chromosome is left with a gap that can never be filled. Every somatic division shortens the telomere by about 50 to 200 base pairs; once it passes a threshold the cell enters senescence, and that is the Hayflick limit — the cell's division timer. Telomerase is a reverse transcriptase with its own internal RNA template, and it can rebuild the 3′ overhang; it is active in embryonic stem cells and germ cells, and in cancer it is frequently switched back on (around 90 % of cancers upregulate telomerase). So the chromosome end is a slightly imperfect copy of the one before it, and after enough rounds of copying there is nothing left to copy — a neat, non-magical way for a cell to count its own age. The consequence is a slow countdown written into the ends of every chromosome you own.

Why this course brings it up: He has said that "telomerase inactivation → senescence" is one of the forms a multiple-choice question takes ("What is the consequence of inactivating X?"), and both the telomere shortening numbers and the reactivation of telomerase are his own material. It is a concept to understand plus a third-level number. The therapeutic corollary is the other half of the story: if about 90 % of cancers turn telomerase back on to escape this limit, then telomerase is an obvious target, and several drugs are aimed at it. And the cancer version of the story, the tumour switching the timer back on, is the same mechanism with the sign reversed. **Appears in:** Ch 5, Lect 3, Lect 4

▪ Endoderm

In one sentence: The endoderm, the innermost of the three germ layers. It becomes the lining of the gut and the lungs.

Detailed explanation: The endoderm is the innermost of the three germ layers of the embryo. It forms the digestive epithelium, the lining of the gut, the epithelium of the lung and the trachea, the parenchyma of the liver and the pancreas, and the epithelium of the bladder.

What makes it interesting is the point it carries: one genome, and completely different fates depending on the germ layer. That is the framework for understanding cell fate decisions and the differentiation capacity of iPSC. To push an iPSC into cells of different germ layers is to use different differentiation signals to switch on different genes on one and the same DNA, and the genes that get switched on are the ones that decide the fate.

Why this course brings it up: It belongs to understanding the concept, and you need the net idea, not the arithmetic. His gold standard for validating iPSC is tissue containing all three germ layers in a teratoma, and the rule for excluding contamination is something he also told you to remember, because a teratoma that is missing a germ layer means the cells were not actually pluripotent. **Appears in:** Ch 8, Lect 2

▪ endogenous

In one sentence: A source of DNA damage that comes from inside the cell.

Detailed explanation: It includes the mistakes the polymerase makes while copying, the breaks caused by abortive topoisomerase, and the reactive oxygen species produced by cellular respiration attacking the telomeres.

His organising principle is the question "where does it come from?", and endogenous means the cell made it itself. The one worth singling out is the attack of the ROS on the telomeres, because the telomere is exactly where DNA damage and telomere biology meet. So the answer to the question depends on where you look, and the three examples are all things the cell does to itself in the ordinary course of work.

Why this course brings it up: Dr. Wong does not name endogenous as an exam question on its own, so it is conceptual understanding, and he has taught the endogenous against exogenous classification. **Appears in:** Ch 6, Lect 4

▪ Endometrium

In one sentence: The endometrium, the lining of the uterus. Tamoxifen behaves differently there than it does in breast tissue.

Detailed explanation: Tamoxifen is a SERM (a selective estrogen receptor modulator), and the same drug gives different results in different tissues. In breast it is an antagonist: estrogen is kept out of the receptor, co-activators cannot be recruited, the cells stop proliferating, and you get an anti-tumour effect.

In the endometrium it is a partial agonist: the tamoxifen–ER complex can recruit co-activators and encourage proliferation instead, so the risk of endometrial cancer goes up slightly. The conclusion is that the tissue context decides the outcome — a drug's effect is not a property of the drug itself but of the relationship between the drug and the tissue. Which is why the same molecule can be a life-saving drug in one organ and a carcinogenic risk in the next, and why receptor occupancy and the tissue context have to be read together. The drug is the same in both tissues; only the tissue is different, and that is the entire lesson.

Why this course brings it up: He says plainly "This is the concept that you have to bear in mind.", which makes it a second-level (A) item. The tamoxifen example sits together with HER2 dimerisation and receptor occupancy (RO) as the whole nuclear receptor block in Lecture 5. You should be able to say, in one sentence, why the same drug can be an antagonist in one tissue and an agonist in another. Which is why receptor occupancy on its own never tells you what a drug will do. **Appears in:** Ch 7, Ch 10, Lect 5

▪ Endoplasmic reticulum (ER)

In one sentence: The membrane network beside the nucleus, where new proteins are made, folded, checked and glycosylated.

Detailed explanation: The ER is the workshop next to the nucleus. A newly made protein that carries an N-terminal signal peptide is sent straight into the rough ER, where chaperones help it fold, enzymes check it, and a sugar group is often added, which is glycosylation.

A protein that fails the check is kept in the ER and sent back for disposal; only one that passes is packed into a transport vesicle and sent to the Golgi. The rough ER is rough because of the ribosomes studding its surface, which is where the protein is actually made, while the smooth ER is the other one, without them. The signal peptide is the postal address, and without it the protein is made in the cytosol and has no idea that it is supposed to be somewhere else. Without the signal peptide the protein never enters the route at all, and that is exactly why his postdoc wasted four months of work on it.

Why this course brings it up: He names the signal peptide as important: "So remember, signal peptide is really, really important for your study", which makes it second-level (A), and he attaches the story of the four months of wasted work. The route you have to remember is rough ER folding and glycosylation, then Golgi sorting, then the cell surface or secretion. The story about his postdoc is there for a reason, which is that a single missing signal peptide can cost months of work, and that is the sort of thing he expects you to remember. **Appears in:** Ch 2, Ch 8, Lect 2

▪ Endosymbiotic origin

In one sentence: The idea that mitochondria began as bacteria that an ancestral cell swallowed and kept.

Detailed explanation: Mitochondria have their own DNA, a circle of about 17 kb carrying 37 genes, their own ribosomes, and a double membrane, with the inner membrane folded into cristae while the outer membrane still looks much more like a bacterial membrane.

The reasonable explanation is that long ago a prokaryotic cell took in an aerobic bacterium and did not digest it, and the two of them lived together for a very long time. The one that was swallowed kept its capacity for respiratory metabolism, and the host got an extra power station, and the arrangement has been running ever since. That leaves three testable consequences: mitochondrial DNA is inherited from the mother, because the sperm's mitochondria are destroyed after fertilisation, and mitochondrial disease can be explained by a threshold effect.

Why this course brings it up: He ties three things together in one story: endosymbiotic origin, maternal inheritance and the threshold effect. And he makes the mitochondrial pedigree a first-grade, A+, exam question, so the origin itself is the premise of the whole story, and the pedigree pattern only makes sense once you accept it. **Appears in:** Ch 8, Lect 2

▪ Endotoxin

In one sentence: The poisonous part of the outer membrane of a Gram-negative bacterium. It is the same molecule as LPS.

Detailed explanation: LPS (lipopolysaccharide) is made of lipid A, a core polysaccharide and the O antigen; the part that actually does the damage is the lipid A section, and it sits in the outer layer of the Gram-negative outer membrane.

When the bacterium dies or lyses, LPS gets into the blood and sets off a whole-body inflammatory response. His slide puts it as “overwhelming level of endotoxin in blood – Septic shock”, and he points out that this is one of the common causes of death in surgical ICU patients. The signalling route is LPS → LBP (LPS binding protein) → CD14 (the LPS co-receptor) → MD-2 (the LPS binding subunit) → TLR4 dimerisation, and then it splits into two branches, MyD88 (signalling adaptor) dependent and TRIF (signalling adaptor) dependent, and both of them end by switching on NF-κB. Notice how much machinery sits between the LPS and the response: four separate proteins in a row before the receptor is even reached, and then two parallel branches that both land on the same transcription factor. That arrangement is not decoration. The purpose of a chain like this is that the bacterium cannot turn the inflammation on by itself; the host cell has to agree to pass the signal along. And because the whole chain ends on NF-κB, you have already met the enzyme that takes the LPS signal and turns it into a gene expression programme. LPS is a fragment of the bacterial cell wall, and what makes it dangerous is that the human immune system recognises it as foreign and reacts to it violently.

Why this course brings it up: He explicitly said that the chain LPS → TLR4 → NF-κB is a published MCQ format for him, and what he excludes is the detailed cascade table on the slide, not this main chain. So what you need is the order, not every molecule. That is a distinction worth holding onto, because it is the difference between being examined on something and being mentioned in passing. **Appears in:** Ch 1,Lect 1

▪ Energy

In one sentence: The capacity to do work. In a cell it comes from the net release of a metabolic reaction.

Detailed explanation: Energy is not a thing; it is the capacity to do work. A cell has no warm energy available to it, only chemical bonds. When the cell breaks glucose down into pyruvate and takes it into the mitochondria, the new bonds that form along the way are stronger than the old ones that were broken, and the difference is what gets stored as ATP.

So the energy budget has to be worked out as a net figure: what it cost to break the bonds, what comes back from forming them, and only the difference is energy the cell can spend. Energy is not stored in the molecule itself, it is stored in the difference between the molecules before and after. The same principle explains the kinase and phosphatase switch: adding a charged phosphate group changes the shape, and taking it away puts it back.

Why this course brings it up: Chemistry belongs to the part he excludes, but "ATP is the money the cell spends" and the

Warburg effect both have to be taught with this concept in mind. Cancer cells would rather be inefficient, about 4 ATP per glucose, if that lets them keep the half-broken sugar as building material. It belongs to understanding the concept. **Appears in:** Ch 1

▪ Enhanceosome

In one sentence: A cluster of proteins gathered on an enhancer and its promoter, which is how a distant enhancer gets wired to its own gene.

Detailed explanation: An enhanceosome is a complex formed by a group of transcription factors and co-activators working together. His Lecture 5 slide draws it as "bound to enhancer plus promoter leads to the gene to be controlled": an activator takes up position on the enhancer first, DNA-bending proteins pull the DNA into a loop, so that the complexes on the enhancer and on the promoter end up facing each other, and only then is RNA polymerase II recruited.

Picture it as a telephone cord: the enhancer at one end, the promoter at the other, and the complex spanning the loop. The DNA-bending proteins and the loop anchors exist to shorten the distance, because the enhancer and the promoter have to be brought within reach of each other before the enhanceosome can form across that gap. What the enhanceosome shows is that several proteins have to be in place at the same time, because one activator on its own is usually not enough to drive transcription.

Why this course brings it up: The structure is on his slide, but he does not list it as an exam question, so it is concept-only. What really gets examined is the distinction between cis and trans, and the concept he stated explicitly that not every enhancer sits upstream. He also uses the enhanceosome slide to make the point that regulation is combinatorial, which is why the cAMP against lactose contrast in the lac operon works the way it does. **Appears in:** Ch 7, Lect 5

▪ Enhancer

In one sentence: A stretch of DNA that increases transcription, no matter where it happens to sit relative to the gene.

Detailed explanation: An enhancer is a DNA sequence that raises transcription, and it acts in a positionally independent way: it can sit thousands of bases upstream of the gene, and it can also sit downstream of it, on the 3′ side.

The mechanism is that the bending of DNA brings the enhancer into contact with the promoter, so that the activator and the promoter complex can meet. An enhancer therefore needs two things to work: a matching activator nearby, and a gene that is actually able to start. Silencing an enhancer (an insulator) stops the enhancer reaching the wrong gene. So an enhancer is not a switch on the gene it acts upon. It is more like a loudspeaker that can be pointed anywhere, which is exactly why the cell needs a second mechanism to keep it pointing in the right direction. Position independence is the surprising part, and it is the part he insists on: people assume regulatory DNA sits next to the gene it controls, and the enhancer is the counter-example.

Why this course brings it up: He says it explicitly: “Not all the enhancer are located on the upstream of the sequence… This is one of the concepts that you need to understand.” It is second-level (A). The cis elements in the same group are promoter, silencer and insulator, and together they make up the cis versus trans comparison set that he lists. So you need two things: the idea that an enhancer can act from a distance, and the four names of the cis elements. Note that the downstream case is the one people forget, because “upstream and downstream” both sound like “nearby”. And remember that position independence applies to the silencer and the insulator in the same way, not just to the enhancer. If a question gives you a regulatory sequence on the 3′ side of a gene and asks you whether it can still work, the answer he wants is yes. **Appears in:** Ch 7,Lect 5

▪ Enzyme

In one sentence: A protein that is not used up itself but makes a chemical reaction go faster.

Detailed explanation: An enzyme is a protein with a pocket called the active site, and the substrate drops into that pocket. What the enzyme does is lower the activation energy that the reaction has to climb over, which is why the reaction can run at body temperature at all, because otherwise the cell would have to heat itself up to do chemistry.

One enzyme molecule can be used again and again, many times over, because it comes out of the reaction unchanged. Almost every tool in this course is an enzyme: helicase, topoisomerase, primase, polymerase, exonuclease, ligase, kinase, phosphatase, histone deacetylase, DNA glycosylase and the caspases. So whenever he names an enzyme you can assume you are looking at a protein with a shape, and blocking that shape or changing it changes the chemistry. This is the single most useful shortcut in the whole course, because it lets you predict the effect of a drug before you know anything about the pathway it sits in.

Why this course brings it up: He never makes enzyme kinetics or the chemistry of the active site an exam question, so it is background, but it is the skeleton under everything that comes later. At the very least you have to be able to say in one sentence that an enzyme lowers activation energy and works on the substrate inside its active site. **Appears in:** Ch 2

▪ EpCAM

In one sentence: A surface protein used to pick tumour cells out of a blood sample.

Detailed explanation: EpCAM (epithelial cell adhesion molecule) is the marker on the surface of tumour cells of epithelial origin. The liquid biopsy method depends on it: in the blood, the combination EpCAM+ / CD45- is used to pick the tumour cells (CTC) out of the millions of white blood cells, and then to analyse them further.

But the same slide also states the limitation: tumour cells that have gone through EMT stop expressing EpCAM and so slip through your screen; and in the other direction, marker expression on a CTC is not the same as marker expression on the primary tumour. So the same marker that makes the assay possible is also what limits it. The marker is also the limitation, and the two are the same fact.

Why this course brings it up: He framed "choosing the right cell population is half the battle" as the opening theme of Lecture 2, and EpCAM+ / CD45- is the most concrete example of that principle. It belongs to the concepts you are expected to understand, but you should know why EMT makes this method miss cases. **Appears in:** Ch 9, Lect 2

▪ episome

In one sentence: A DNA molecule that is not part of the host chromosome and can replicate on its own.
Detailed explanation: The cccDNA that HBV forms in the nucleus is the textbook example: it is not part of the host chromosome, yet it replicates along with cell division and stays in the nucleus for a long time, becoming a standing reserve for making viral RNA.
And because it does not integrate into the chromosome, the viral DNA has to take a separate route, rcDNA → nucleus → cccDNA. HBV is therefore the standard example of causing disease without integrating, with the virus keeping an episome in the nucleus and continuously manufacturing protein and RNA.
Why this course brings it up: Dr. Wong does not name episome as an exam topic on its own, so it belongs to conceptual understanding. The mechanism he does state is the one worth keeping: HBV cccDNA is a persistent episomal reservoir in the nucleus, which is why it is so hard to clear. **Appears in:** Lect 2

▪ Epithelial-to-mesenchymal transition (EMT)

In one sentence: The change in which a cell loses its grip on its neighbours and becomes free to move and invade.
Detailed explanation: EMT is the process in which epithelial cells turn into a mesenchymal cell type. Normal epithelial cells stick tightly to each other and have a clear apical-to-basal polarity;
after EMT they lose adhesion molecules such as E-cadherin, the polarity collapses, and they can slide, secrete ECM enzymes and invade. For a tumour, EMT lets it break free of its anchors in place, acquire the ability to invade and metastasise, and become more tolerant of certain treatments. The price is a change in markers: epithelial markers such as EpCAM fall after EMT, and that is exactly why screening with EpCAM for liquid biopsy misses a proportion of CTCs. The change is not all or nothing in practice, because the marker change that goes with it is precisely what a screening test relies on, and that is the link to the clinic: a test that depends on an epithelial marker will miss the cells that have already left the epithelium.
Why this course brings it up: He does not list EMT as a separate exam question, so it is concept-only. But it is the key to the clinical reasoning about "what does a screening method miss", and it is the background to his tumour microenvironment and CAR-T obstacle page. **Appears in:** Ch 9, Lect 2

▪ Epitope

In one sentence: The small patch on an antigen that the antibody actually touches.
Detailed explanation: An antigen can be very large (a whole virus, a whole bacterial surface protein), and an antibody only recognises a small piece of shape; that piece is the epitope. At the tip of the Y-shaped arms of an antibody sit three hypervariable loops in the variable region, CDR1, CDR2 and CDR3, and those three are the real gripping surface, and the epitope is whatever they line up with.
The idea is central to drug development: a therapeutic antibody is designed to recognise an epitope that is specific to cancer cells and absent from normal ones, so that it does not damage healthy tissue, and the payload of an ADC is only released once the antibody has delivered the whole complex to the epitope. So the drug design problem is really a targeting problem: find a shape that is present on the tumour and, as far as possible, nowhere else. And because the antibody is specific to a shape rather than to a whole molecule, the same antigen can be presented in different places and still be recognised, which is what lets one antibody work against many tumour cells. That is also why the CDRs get so much attention in the antibody engineering literature: they are the part that is doing the recognising.
Why this course brings it up: On his first lecture slide he marks CDR1/2/3 and the antigen binding site, so this is conceptual understanding. You have to be able to use it to explain why an antibody drug can be selective, and why an ADC can release its payload inside the tumour cell. Both of those explanations are worth rehearsing out loud, because they are the kind of answer a short scenario question expects. **Appears in:** Ch 2,Ch 9,Lect 2

▪ EPO

In one sentence: Erythropoietin, the red blood cell growth factor switched on by HIF-1α.
Detailed explanation: EPO is the growth factor for blood cell production, made mainly by the kidney, and it acts on red blood cell precursors in the bone marrow. When oxygen is low, HIF-1α is stabilised, enters the nucleus, binds HIF-1β, and then switches on EPO and a whole set of other genes through the HRE;
EPO rises, red blood cell production rises, and the blood's oxygen-carrying capacity comes back. This is the body's most direct physiological response to hypoxia, and it also explains why HIF-1α simultaneously switches on VEGF (which grows blood vessels) and GLUT1 (which lets cells work without oxygen) — in a hypoxic tumour core, the same switch helps the tumour survive longer. One hormone, one switch, and a whole survival programme. It is the clearest example in the course of one hypoxia switch producing several outputs at once.
Why this course brings it up: He never names EPO on its own, so it belongs to the concepts you are expected to understand — but it is one member of the HIF-1α target gene list (VEGF, GLUT1, EPO), and that list itself can be used to answer the question of what a cell does when it is short of oxygen. **Appears in:** Ch 7, Lect 5

▪ ERBB2

In one sentence: The gene name for HER2.

Detailed explanation: ERBB2 is the gene name for HER2, human epidermal growth factor receptor 2, so the two are the gene name and the protein name for one and the same thing, and they are used almost interchangeably even though they are not really the same word.

He uses the ERBB2 page of the Human Protein Atlas to make a crucial demonstration of exactly that point: the same target can point in completely opposite directions for prognosis in different cancers. Renal cancer is favourable, endometrial cancer and pancreatic cancer are both unfavourable, and all three come with $p < 0.001$. The lesson is that a drug target and a prognostic biomarker have to be cancer-specific. You cannot take what worked in one cancer type and assume it carries over to another type of cancer, because the same protein can be a good thing in one organ and a bad thing in another, and that is what makes ERBB2 such a useful teaching example.

Why this course brings it up: This is the parallel case to his first-grade, A+, Kaplan–Meier question, and it is on his list of precise figures as well: the ERBB2 prognosis in renal, endometrial and pancreatic cancer, at $p < 0.001$. It is compulsory material, and it is one of the three places where he wants you to be able to recall a number exactly. **Appears in:** Ch 10, Lect 1, Lect 5

▪ Estrogen receptor (ER)

In one sentence: The nuclear receptor for estrogen. Tamoxifen acts on it.

Detailed explanation: ER is a type I nuclear receptor. Without a ligand it is held in the cytoplasm by chaperones such as HSP90, with the folded receptor just sitting there waiting; once estrogen binds, HSP90 lets go, the receptor dimerises, moves into the nucleus, recruits a co-activator, and finally binds the ERE to start a gene.

The key to the whole mechanism is the change of receptor shape: a drug and a hormone can both do it, and that is why both can act as switches. Tamoxifen goes down the same road, but it binds the LBD and brings about the opposite recruitment result. So the receptor itself does not change identity. What changes is which proteins are available to it at that moment, and that is the whole story. And notice that the dimerisation step is shared with everything else in this lecture: the receptor has to pair up before it can go to work.

Why this course brings it up: He treats ER, HER2, receptor occupancy and HSP90 together in the nuclear receptor section of Lecture 5. It is second-level (A): you should be able to describe the three-part structure of ER (NTD, DBD, LBD) and the detail that it is held in the cytoplasm by HSP90 when there is no ligand. Those two facts together explain the whole agonist and antagonist story, so they are worth more than the rest of the paragraph put together. **Appears in:** Ch 7,Lect 5

▪ Estrogen response element (ERE)

In one sentence: The stretch of DNA that the estrogen receptor dimer binds to.

Detailed explanation: An ERE is a particular sequence of DNA, like a single word. The DBD of ER recognises that word as a dimer, and it recognises only that one — so which response element is present decides which gene the receptor works on.

For drug development, the ERE sequence can be used in a reporter assay: attach a candidate promoter fragment in front of luciferase, trim it down piece by piece, and measure the activity of the reporter gene, and you find which part of the promoter is really doing the work. This is the standard way he teaches promoter mapping in Lecture 5. Think of a response element as the address a receptor is looking for: the receptor is the same for many genes, and the DNA sequence it is hunting is what redirects it to one of them. A receptor without its address goes looking; the address decides the target.

Why this course brings it up: He does not list ERE as an exam topic, so it belongs to the concepts you are expected to understand, but the reporter experiment on a promoter and the matching of cis to trans falls inside second-level (A) content. And the serial deletion experiment is the technique he expects you to recognise even though the word ERE is not on the list.

Appears in: Ch 7, Lect 5

▪ Euchromatin

In one sentence: Loosely packed chromatin that is being transcribed.

Detailed explanation: Euchromatin is the loosely packaged chromatin that is being read, and the DNA inside it is easier to get at, so transcription factors and enhancers can sit close by and the gene can be switched on.

It is the working form of chromatin, and it is where the transcription machinery is gathered. Heterochromatin, in contrast, is packed tight, and it physically blocks the transcription factors, so the gene is silent. The important thing is that this state is dynamic and reversible: nucleosomes slide along the DNA and can expose or cover a promoter at any moment, and there are two forces that push the sliding. One is the inherent affinity between the DNA sequence and the histone, and the other is the ATP-driven chromatin remodelling complexes. Because the binding of histone to DNA is non-covalent, made of hydrogen bonds and electrostatic attraction, this switch can be flipped at any time, and a drug that changes the binding can flip it as well. If the binding were covalent, the chromatin state would be locked in and there would be no point talking about a switch.

Why this course brings it up: Euchromatin against heterochromatin is one of the distractor pairs he lists, and it is a second-level, grade A, concept. He also uses a very precise formulation of it: heterochromatin is physical blocking, not "less efficient", and he wants the distinction in that form. **Appears in:** Ch 3, Lect 3

▪ Eukaryote

In one sentence: A cell with a nucleus. Animals, plants, fungi and yeast are all eukaryotes.
Detailed explanation: The dividing line between a eukaryotic cell and a prokaryotic cell is a membrane-bound nucleus. Have a nucleus, and the DNA is shut in a control room, transcription and translation happen in different places, and an RNA processing step sits between them, splicing, capping and adding a poly-A tail.
That two-stage layout is exactly what makes alternative splicing and fine-tuned control possible. A prokaryotic cell has no nucleus, replication and production happen at the same time and in the same place, fast and simple, but it also lacks the machinery for folding and modifying complex proteins, which is why E. coli cannot make a human therapeutic protein. So the difference between the two cell types is not a matter of degree: one of them has a control room and a processing step, and the other has neither.
Why this course brings it up: In Lecture 5 he teaches prokaryotic and eukaryotic systems side by side, because that is the premise for operons and for eukaryotic transcriptional control. What he explicitly excludes is the detailed cascade table on his slides, not this division of labour between the two systems. So it is concept-only, and the contrast between the two systems is what you will be asked to use, not the table of details. **Appears in:** Ch 8, Lect 2, Lect 5

▪ Ex vivo

In one sentence: Outside the body. Cells are taken out, worked on, and put back.
Detailed explanation: The ex vivo route runs like this: you take cells out of the patient (blood or tissue), culture them outside the body and give them a genetic or functional treatment, and then put them back into the patient.
His example is ex vivo autologous gene therapy: take the patient's own cells out, repair the defective gene outside the body, and transfuse them back. CAR-T is the same pattern: the patient's T cells are extracted, fitted with a CAR outside the body, expanded to millions, and reinfused. The advantage is that you can verify and expand the product properly outside the body. The disadvantages are that the process is expensive, that it takes time, and that the patient has to have lymphodepletion conditioning first. So the word is really about where the work happens, not about what the work is. And the ex vivo route is what buys you the chance to check that the cells are what you want before you give them back.
Why this course brings it up: He did not list ex vivo as an exam topic on its own, so it is conceptual understanding, but it belongs to the same set of contrasts as allogeneic (off-the-shelf) and in vivo CAR-T, and a scenario question may ask why lymphodepletion is done or what the trade-off between allogeneic and autologous is. Those are the two places where this word does real work in an answer, so it is worth having the three routes lined up in your head: ex vivo, in vivo, and off-the-shelf. The conditioning question in particular is a favourite, because it links cell therapy to the immune system. **Appears in:** Ch 9,Lect 2

▪ EXO

In one sentence: The proofreading segment of DNA polymerase that cuts out mismatched nucleotides.
Detailed explanation: DNA polymerase does not only synthesise, it also carries exonuclease activity: when it puts in the wrong base it cuts that one out again itself, in the 3′→5′ direction, and then puts the correct one in its place.
His slide draws the polymerase as a machine with two segments, POL and EXO, and a proofreading-defective mutant strain is exactly a strain in which this segment has failed and which has therefore become an ultramutator. With proofreading, the replication error rate can be as low as one mistake per billion nucleotides; without it the mutation rate rises by several orders of magnitude. Which is why a proofreading-defective strain is called an ultramutator and why it makes a useful experimental model. That is the whole value of the EXO segment.
Why this course brings it up: Dr. Wong does not name EXO as an exam question on its own, so it is conceptual understanding, and he explains the chain of defective proofreading leading to more neoantigen and a better response to immunotherapy very clearly. **Appears in:** Ch 5, Lect 4

▪ exogenous

In one sentence: A source of DNA damage that comes from outside the cell.
Detailed explanation: Three main examples. UV causes pyrimidine dimers. Ionising radiation (X rays, gamma rays) causes double-strand breaks. And then there are mutagenic chemicals — for instance intercalating agents, which slot themselves in between the bases and twist the double helix out of shape — together with viruses like HBV that can integrate into the genome.
Telling endogenous and exogenous apart matters, because the two call for different prevention and treatment strategies: much of the exogenous damage can be reduced by avoiding the exposure, while the endogenous kind can only be held in check by the repair systems and the immune system. In other words, some of it is an accident of the world outside and some of it is simply the cost of copying DNA.
Why this course brings it up: He does not name exogenous as an exam topic on its own, so it is a concept to understand; the point is the classification of damage and which kind of damage goes with which repair pathway. That pairing is what makes a question about pyrimidine dimers answerable rather than a guess. **Appears in:** Ch 6, Lect 4

▪ Exon

In one sentence: The part of a gene that survives splicing and ends up in the mature mRNA.
Detailed explanation: The pre-mRNA that comes out of transcription is a rough thread with exons and introns threaded alternately through it. The spliceosome is made of five snRNP; it recognises the intron boundaries first, cuts the whole intron out and throws it away, then joins the exons on either side, so that the mature mRNA is one continuous run of exons.

The naming is the classic trap of this course: an exon is the part that is shown, and an intron is the part that is hidden, so the names are the wrong way round. The trap catches people out because the everyday meaning of "exon" sounds like the opposite of what it means here, and once you have been caught by it you will not be caught again. His mouse β-globin slide makes it clear: Exon 1, Intron 1, Exon 2, Intron 2, Exon 3, which only after splicing becomes a mature mRNA with a 5′ UTR, three exons and a 3′ UTR.

Why this course brings it up: It belongs to the concepts you are expected to understand, but splicing and alternative splicing are the spine of Lecture 5, and Case 15 (predicting drug resistance from an isoform) uses it directly. You have to be able to tell which parts are kept and which are removed. The exon and intron pairing is also the first thing you need in order to understand alternative splicing, because there the question is not which parts are removed but which combination of the kept parts is used. **Appears in:** Ch 7, Lect 5

▪ Exon inclusion and exon skipping

In one sentence: The two opposite choices the splicing machinery makes when it builds a mature message.

Detailed explanation: Splicing is not mechanical. The spliceosome makes a competitive choice near each splice site and decides whether a given stretch of exon is kept or not. Exon inclusion means it is packaged into the mature mRNA; exon skipping means the whole stretch is left out. The same gene can therefore produce several isoforms, and each isoform may carry different functional blocks: MCL-1S, the pro-apoptotic form, against MCL-1L, the anti-apoptotic form, in the apoptosis regulator MCL1; Bcl-x(L) and Bcl-x(S) in Bcl-x; and PKM1 against PKM2 in the pyruvate kinase gene PKM. The competition matters because the cell does not read the instructions for which exon to keep off a printed list; it reads them off the sequence itself, and a change in that sequence can move the decision. That is why a single-nucleotide variant deep inside an intron, far away from any exon, can still change the protein that comes out. Exon skipping can also be pathological. In his hepatocellular carcinoma case, a single-nucleotide variant inside an intron created a new splice site, so exon 7 was skipped in its entirety and a shorter isoform was produced, and patients carrying that variant had a significantly worse overall survival ($P < 0.001$). And because exon skipping can remove a whole functional block at once, the effect on the protein can be large rather than subtle, and it is that which gives you either a new neoantigen or a new drug target.

Why this course brings it up: It is the clinical version of alternative splicing in Lecture 5, and the hepatocellular carcinoma data are his own slides, n = 23 with no abnormal splicing against n = 19 with abnormal splicing. It is concept-only, but you have to be able to say the causal line, isoforms are the source of drug resistance and of neoantigens, and to connect the three clinical consequences: neoantigen creation, drug resistance through the loss of a binding site, and isoforms that can actually be measured. **Appears in:** Ch 7, Lect 5

▪ Exonuclease

In one sentence: The proofreading activity of DNA polymerase: the part of the enzyme that cuts out a base it inserted by mistake.

Detailed explanation: While DNA polymerase is adding nucleotides, it uses its 3′→5′ exonuclease activity to look back over its own work. If the base it has just put in cannot pair with the template, A sitting opposite G for instance, it excises that mismatched base and puts the correct one in its place.

It is a read-then-correct mechanism, and its direction is exactly opposite to synthesis, which runs 5′→3′, so the two are very easy to mix up in your head. Guard against that mix-up, because the whole point of proofreading is that it runs against the grain of the synthesis attached to it: the enzyme travels one way while it builds the chain and the other way while it checks the chain.

Exonuclease has a second job as well, removing the spent RNA primers from the lagging strand so that ligase can join the fragments together and the strand can be made continuous. But this proofreading ability has a clinical price. It also strips nucleoside analogue drugs back off the DNA, and that is one of the sources of resistance to antiviral and anticancer drugs. So the very activity that makes the polymerase a careful copy editor is also what gives a drug a way out.

Why this course brings it up: Dr. Wong lists "5′→3′ versus 3′→5′" as a distractor pair, and the discriminating point is exactly the difference in direction between elongation and proofreading, so it is worth pinning down before you meet it in a question. The other end of the entry reaches into the tumour immunology story of Lecture 4, where a disabled polymerase leads to a high neoantigen load, which leads to a good response to immune checkpoint inhibitors. **Appears in:** Ch 5

▪ Expansion cohorts

In one sentence: The group of patients in Phase 1B used to look at side effects and early signs of benefit.

Detailed explanation: The dose escalation of Phase 1A answers the question of what dose is safe, and it uses very few patients. Phase 1B switches to a fixed dose and brings in more patients, and these are the expansion cohorts.

The purpose is to see the side effects more clearly and, at the same time, to start looking for signs that somebody is responding. It is Phase 2 and Phase 3 that really test efficacy. So Phase 1B has more patients than 1A and a longer follow-up, and it is in effect the stage that collects safety and preliminary efficacy in the same go. The change in design between the two phases is what makes the difference: a handful of patients on a rising dose tells you about the dose, and a larger group on a fixed dose tells you about the drug.

Why this course brings it up: The phasing of clinical trials is one of the Tier 3 precise numbers he lists, 1A escalation, 1B expansion and 2/3 for efficacy, and it is worth learning by heart. So if you memorise one line from the clinical development section, make it this one. **Appears in:** Ch 12, Lect 2, Lect 4

▪ Exportin-5

In one sentence: The protein that carries the short hairpin RNA out of the nucleus and into the cytoplasm.
Detailed explanation: Exportin-5 (XPO5, a nuclear export receptor) works with Ran-GTP to move the pre-miRNA hairpin that Drosha has cut from the nucleus into the cytoplasm. The principle is a directional cargo label: XPO5 only grips RNA carrying a nuclear export signal while Ran-GTP is bound to it, and once the cargo arrives in the cytoplasm, RanGAP hydrolyses the GTP to GDP, the cargo is released, and XPO5 goes back into the nucleus carrying the GDP to be reloaded.
Because movement through the nuclear pore is directional, a protein has to be sent back before miRNA can make the trip from nucleus to cytoplasm and back again, over and over. That is the slightly counter-intuitive bit: the exporting protein is not free to leave on its own account, it has to be collected. And Ran is the timer. While the GTP is on, the door is open and the cargo moves; when the GTP is turned into GDP, the door shuts and the carrier has to go home.
Why this course brings it up: He did not name Exportin-5 on its own, but the complete miRNA production pathway (Pol II → pri-miRNA → Drosha/DGCR8 → pre-miRNA → XPO5 → Dicer → RISC) appears in both his first and his fifth lectures, and understanding it sits inside his second-level (A) range. So the pathway as a whole is examinable even though the individual carrier protein is not named. Write the arrow chain out once and it will stay straight in your head, because every arrow is a handoff, and the order of the handoffs is the point. **Appears in:** Ch 7,Lect 1,Lect 5

▪ Extension

In one sentence: The step where the polymerase adds new nucleotides along the template.
Detailed explanation: DNA polymerase can only add nucleotides to a 3′ end, so every extension runs in the 5′→3′ direction. It works a bit like a right hand: the fingers grip the template, the palm holds the growing strand, the thumb pushes each dNTP into place, and pyrophosphate is released with every addition.
Extension is the third step of PCR. Theoretically the PCR product doubles at the end of each cycle, so after n cycles there are about 2^n copies, and Dr. Wong has also pointed out that in practice there is a slight deviation from that. The deviation matters, because the last few cycles are the inefficient ones and they are where the numbers stop being exact. The deviation matters, because the last few cycles are the inefficient ones and they are where the numbers stop being exact.
Why this course brings it up: “Extension is always 5′→3′, proofreading is 3′→5′” is a Tier 2 contrast on his list, and it is one of the pairings he names. **Appears in:** Ch 3, Lect 1, Lect 4

▪ Extrinsic apoptosis pathway

In one sentence: The apoptosis pathway that starts outside the cell: a death receptor triggered by its ligand.
Detailed explanation: This pathway begins outside the cell: a ligand (TRAIL, for example) binds a death receptor on the cell surface (DR5, for example), and the intracellular part of the receptor immediately gathers procaspase-8 and starts the caspase cascade.
Caspase-8 does two things: it cuts procaspase-3 directly to make the executor caspase-3, and it cuts bid to make tbid and passes the signal into the mitochondria, so the intrinsic pathway starts as well. In other words the two pathways cross over, and the crosstalk is real. His clinical example is the bispecific antibody BI 905711, which recognises TRAILR2 and CDH17 at the same time: apoptosis is only induced when both are present, so normal liver (which lacks CDH17) and normal gastrointestinal tract (which is only weakly TRAIL-sensitive) survive, and only colorectal cancer dies. That is the whole design in one sentence: make the death signal depend on two things that only a tumour has at the same time, and the normal tissue never receives it. In practice this is the difference between a drug that kills the patient slowly and a drug that kills the tumour. Everything about this pathway is arranged to make the signal narrow and specific.
Why this course brings it up: He does not ask you to recite the caspase cascade (he explicitly rules out the detailed signal transduction cascade), but the strategy of restricting a death signal to the tumour with a bispecific antibody is a concept to understand, and it is the core of his own CDH17 story. He never asks you to memorise the order of the caspases; he asks you to understand why a normal cell is spared. So the bispecific antibody story is the place to see the principle in action. **Appears in:** Ch 8, Lect 2

▪ Fab fragment

In one sentence: The arm of an antibody, the part that grabs hold of the antigen.
Detailed explanation: An antibody is Y-shaped, made of two heavy chains and two light chains held to each other by disulfide bridges, so the whole molecule can be read as two Fabs sitting on top of one Fc.
Each arm of the Y is one Fab, an antigen-binding fragment: it contains one complete heavy chain plus half a light chain, and at its tip the variable region with the three gripping loops CDR1, CDR2 and CDR3, sitting right at the end of the arm where the antigen is met. The two arms are the same kind of part, so the antibody is symmetrical at the tips and lopsided at the stem. The Fc is what is left over, the tail ends of the two heavy chains, so the antibody divides neatly into two arms that do the binding and one stem that does the talking. Dr. Wong's Lecture 1 slide gives the masses: Fv about 25 kDa, Fab about 50 kDa, Fc about 50 kDa, so all three pieces can be placed on the same drawing. Fab is also where the CAR comes from. The extracellular recognition head of a CAR is the scFv of an antibody, which is VH plus VL joined by a linker, so the CAR takes the tip of the Fab and hangs it on a cell.
Why this course brings it up: The structure and the masses of an antibody are numbers from his Lecture 1 slides, so this is conceptual understanding plus a third-level number. CAR-T borrows exactly the same parts, the scFv is the tip of the Fab, and that link means you do not have to learn a second set of pieces when you explain CAR-T. **Appears in:** Ch 2, Lect 1

▪ FACS

In one sentence: The machine that sorts cells into different tubes according to their fluorescence.
Detailed explanation: FACS is fluorescence-activated cell sorting. It focuses a suspension of cells hydrodynamically into a single-file stream, one cell at a time, and passes each cell in turn through a laser and a detector;
from the scatter and the fluorescence signals the machine decides where that cell should go, and then gives it a charge, positive or negative, so that it is deflected and drops into a different collection tube. The charge is the clever part: the cell is deflected by an electric field while it is still in the stream, so analysis and sorting happen in the same machine, on the same cell, at the same time. And because you can stain as many markers as you like, the more markers you use the more precisely you can define the population you are after. On his slide he separates granulocytes (high SSC), monocytes (in between) and lymphocytes (low SSC) out of PBMCs, and then uses the four-quadrant plot of CD3-FITC against CD4-Alexa Fluor 647 to pick out the CD3+CD4+ helper T cells.
Why this course brings it up: He treats "isolating the right cell population is half the battle" as the opening theme of Lecture 2, and FACS is the most practical tool for that sentence. It is concept-only, but you have to keep straight the division of labour between gating and FACS: gating is drawing the circle, FACS is actually taking the cells out. He also wants you to be able to read a flow cytometry plot, not only to describe the machine. **Appears in:** Ch 9, Ch 11, Lect 2

▪ FAP

In one sentence: Fibroblast activation protein: the marker found on the surface of activated fibroblasts.
Detailed explanation: FAP is expressed in large amounts on the surface of activated fibroblasts in wound healing and in the tumour stroma, and hardly at all on normal tissue. Its clinical value is this: if you make it a target, you are aiming only at the reactive stromal cells around the tumour, and you leave the fibroblasts elsewhere in the body alone.
His in vivo CAR-T example is a FAP CAR doing exactly this kind of "hit the tumour only" recognition. The distribution is the point: a marker that is only there in one place is worth far more than a marker that is everywhere.
Why this course brings it up: He does not name FAP as an exam topic on its own, so it is a concept to understand; the point of that passage is "aiming a CAR at a molecule specific to the tumour microenvironment". That is the general design idea behind it. **Appears in:** Ch 9, Lect 2

▪ Fatty acid

In one sentence: A string of carbons with a small acidic end. Whether there are double bonds in the chain decides if it is saturated or unsaturated.
Detailed explanation: A fatty acid is made of a carboxyl group (COOH, the acidic end) joined to a chain of carbons. If the chain is all single bonds, it is a saturated fatty acid; if there is a double bond, it is unsaturated.
The difference is not decorative. A double bond makes the carbon chain bend, so unsaturated fatty acids cannot pack tightly, have lower melting points and are often liquid, while saturated fatty acids are straight, pack closely, have higher melting points and are often solid. Living things therefore tune membrane fluidity by adjusting the fatty acid composition. A fatty acid joined to glycerol is a triglyceride, and one joined to phosphate is a phospholipid, which is amphipathic and can assemble itself into a lipid bilayer. So the shape of the chain, and not just its length, is what a cell uses to control how fluid its membranes are.
Why this course brings it up: What he excludes is the chemical structures and the metabolic pathway details, but the lipid bilayer is the foundation of the whole of cell biology (membranes, receptors and signal transduction all depend on it), and the saturated versus unsaturated split is something he stated explicitly in Lecture 1. So this is conceptual understanding. You do not need the chemistry; you do need the reason the membrane behaves the way it does. **Appears in:** Ch 1,Lect 1

▪ Fatty-acid chain

In one sentence: The long carbon tail of a fatty acid, and whether it is saturated or not.
Detailed explanation: The length of the carbon chain decides the properties of a fatty acid: the longer the chain, the more hydrophobic it is, the higher its melting point and the larger its molecular weight.
If the chain contains a cis double bond, the chain bends at that point, and the kink stops neighbouring fatty acids from packing close together, so the membrane is more fluid and less likely to solidify; a trans double bond leaves the molecule nearly straight, with properties close to a saturated fatty acid. So on the same chain, with the same number of double bonds, cis and trans can have very different biological effects.
Why this course brings it up: It belongs to the concepts you are expected to understand. It is the demonstration of the running theme that structure determines property, applied to lipids, and it is the background to the "cancer cells reprogramme their metabolism" passage in the Warburg effect segment of Lecture 1. **Appears in:** Ch 1

▪ Fc fragment

In one sentence: The stem of the antibody, the part that recruits the rest of the immune system.
Detailed explanation: Fc is the stem of the Y-shaped antibody, built from the CH2 and CH3 domains of the two heavy chains, about 50 kDa. It does not recognise the antigen at all. It talks. The Fc receptors on NK cells and macrophages, he uses FcγRIII, recognise the Fc of IgG1, and so the antibody grabs the tumour on one side while the effector cell grabs the antibody on the other, and the tumour is killed.
That is ADCC, antibody-dependent cellular cytotoxicity, and the geometry is easy to picture: the Fab reaches out and holds the target, while the Fc is held by the cell that is going to do the killing. This talking ability of the Fc is also the physical source of the isotype difference. The Fc of IgG4 has one residue substituted at the joint, so it binds the Fc receptor poorly and does not trigger

ADCC, but its half-life in serum is longer. So the two isotypes differ by a single structural detail at the joint, and that one detail buys two different behaviours at once.
Why this course brings it up: Dr. Wong lists IgG1 versus IgG4 as a distractor pair, with ADCC and half-life as the discriminating points, so it is second-level, grade A. This is the division of labour where Fc decides the immune effect and Fab decides the specificity, and in a scenario answer you have to be able to say that trastuzumab kills the tumour through ADCC.
Appears in: Ch 2, Lect 1

▪ FDR

In one sentence: The false discovery rate, the proportion of false alarms among the results declared significant.
Detailed explanation: When you test several thousand features at once, some batch of them will turn out "significant" purely by luck. FDR accepts that: it estimates how many of the results you have marked as significant are expected to be false, and that is why the threshold commonly written on a volcano plot is $p < 0.05$ together with FDR < 0.15.
Drawing the p-value threshold and the FDR threshold on the same volcano plot and taking the intersection is a way of controlling the false positives while avoiding being too strict, and there is a balance to be struck between the two.
Why this course brings it up: Dr. Wong does not list FDR as an exam question on its own, so it is conceptual understanding, and he writes the threshold straight onto the volcano plot in Lecture 1. **Appears in:** Ch 10, Ch 11, Lect 1

▪ fetal bovine serum (FBS)

In one sentence: The cow serum used to culture cells, which supplies the growth factors.
Detailed explanation: FBS is the liquid part of calf blood, and it contains the factors that make cells grow and divide. Cell culture usually uses 10 % FBS. When you want to synchronise the cell cycle, though, you drop it to 0.5–1 %, and the cells stop in G1 because they are missing the growth factors.
He points out specifically that FBS is the main player in the serum starvation trick. And because the composition changes with the state of the animal, a rigorous experiment has to fix one batch of serum and record the batch number, otherwise the data carries a systematic bias. That last point is the part students usually miss, because it is not about the biology at all, it is about the bottle being different from the bottle you used last time.
Why this course brings it up: Dropping the serum to 0.5–1 % is a Tier 3 precise number that he lists, and cell cycle synchronisation is Tier 2 material. So both the number and the technique belong in your revision. **Appears in:** Ch 8, Ch 11, Lect 2, Lect 4

▪ first-to-invent

In one sentence: The filing principle used by the older American patent system.
Detailed explanation: Under first-to-file, whoever applies first gets it; first-to-invent looks instead at who genuinely made the invention first. On Dr. Wong's CRISPR patent timeline, the first CRISPR-Cas9 patent, covering plant and animal cells, was granted to Broad on 15 April 2015, and it was done under the first-to-invent principle rather than first-to-file.
That principle was gradually replaced by first-to-file in the United States after 2011, but he keeps it on the timeline because it explains why that 2015 patent was awarded to Broad in the first place. So the same invention, made in the same year, can end up with different owners depending on which rule is in force, and that is a question about paperwork as much as about science. He is not asking you to care about patent law, only to follow why the timeline looks the way it does.
Why this course brings it up: He does not list the patent principle as an exam topic, so it belongs to conceptual understanding. He says plainly that the whole intellectual property landscape is not examined. **Appears in:** Ch 12, Lect 4

▪ FISH

In one sentence: The test that uses DNA probes rather than antibodies to count gene copies inside a cell, for example HER2.
Detailed explanation: FISH uses fluorescently labelled DNA probes, which hybridise to complementary DNA sequences inside the cell, and then a fluorescence microscope shows you where the signal landed.
The reason FISH can count copies at all is that a probe is a piece of DNA with a fluorescent label attached, so if you make enough copies of it, each copy of the gene in the cell picks up one, and then you count the dots. Two-colour FISH makes that counting reliable: one probe for the gene of interest and one for the centromere of the same chromosome, so that the chromosome number can be used as an internal reference. The reading rule then becomes a ratio rather than a raw count, and that is what protects you from a cell that simply has a bad nucleus. The detection workflow he draws is this: nick translation works Dig-dUTP or Biotin-dUTP into the probe, the probe and the cellular DNA are denatured together (the probe at 75 °C and the cells at 42 °C with formamide), and after hybridisation the signal is detected with an anti-Dig or an avidin-fluorophore antibody. Nick translation uses a DNA polymerase to chew back one strand while copying the other, so the labelled nucleotide is built into the probe as it is made. The HER2 FISH uses two probes: the HER2/NEU probe labels 17q12 and the CEP17 centromere probe labels 17p11.1, and the reading rule is a HER2/CEP17 ratio of at least 2.0 or a HER2 copy number of at least 6. And the whole point of using DNA rather than an antibody is that you can count copies, which is a number an antibody stain cannot give you.
Why this course brings it up: He explicitly rules out the experimental parameters: "Definitely not include in the exam. I will not ask you about what's the denaturations, what's the temperature". But the contrast that FISH uses DNA probes and not antibodies, and can therefore count gene copies, together with its clinical use of finding candidates for Herceptin, is at the core of concept-only material. So the exclusion is narrow: the temperatures and the denaturation conditions are out, while the probe

design, the meaning of the ratio, and the clinical use stay in. **Appears in:** Ch 11, Lect 2

▪ Flow cytometry

In one sentence: A machine that counts cells with a laser and measures their size, their internal complexity and their fluorescence.

Detailed explanation: Cells pass through a laser beam one at a time, and the scattered light and the fluorescence are picked up by detectors pointing in different directions. Forward scatter (FSC) reflects cell size, side scatter (SSC) reflects internal complexity (how granular the cell is), and the fluorescence channel measures how much of a particular antibody-labelled target is present.

The same machine can also do DNA content analysis: stain with propidium iodide and the G1 cells are 2N, the S phase cells are somewhere in between and the G2/M cells are 4N, so if you inhibit the contractile ring you will see a 4N peak instead of separate 2N and 4N peaks. He has also taught using serum starvation (dropping the FBS from 10% to 0.5–1 %) to stop cells in G1 and synchronise them. Everything a flow cytometer tells you comes from two directions of light: how much goes straight through and how much gets knocked sideways. That is why the scatter plot is the natural first step, because it gives you broad populations before you have added any antibody at all.

Why this course brings it up: FSC equals size and SSC equals granularity are one of the exact pairings he lists, so that is second-level (A), and the flow plot is one of the scenario archetypes he uses (Case 4: FSC against SSC, which gate, which population, and what is the contaminant). In other words, you should be able to look at a plot and say what each axis means and what a sensible gate would look like. **Appears in:** Ch 9,Ch 11,Lect 2

▪ Fold (protein folding)

In one sentence: The process of a chain of amino acids twisting into a specific three-dimensional shape so that it can do its job.

Detailed explanation: Three kinds of force drive folding. First, the backbone collapses into the two regular local shapes of alpha-helix and beta-sheet, held in place by hydrogen bonds. Then the R-groups take their positions according to the rule that water-hating groups go inside and charged or water-loving groups stay out, folding the whole chain into one overall three-dimensional shape (the tertiary structure).

Finally, where needed, two or more folded chains come together into a quaternary structure. The process needs chaperones to help and it costs ATP. The outcome is not an aesthetic question; it is the function itself: change the shape and you change what it can bind, and whether a drug can bind it either. A protein that has not folded correctly is not a slightly worse protein; it is usually no protein at all. Three levels, one rule, and the rule is that shape decides what the molecule can do.

Why this course brings it up: What he excludes is the table of amino acid R-groups and the chemistry of folding: "don't memorize this… definitely not include in the exam". But "shape determines function" is a principle he demonstrates again and again, and it is the landing point for deamidation and the risks of therapeutic proteins. It belongs to the concepts you are expected to understand, so the level of detail to learn is the shape hierarchy, not the chemistry. And the hierarchy is examinable even though the chemistry is not. **Appears in:** Ch 2

▪ Forward scatter (FSC)

In one sentence: In flow cytometry, the signal that reflects cell size.

Detailed explanation: When the laser hits a cell, the light scattered straight forward is the light that was blocked head-on, and how much of it gets blocked depends roughly on the cross-section the cell presents to the beam.

So FSC mainly reports cell size. Side scatter, SSC, is the light that bounces off to the side instead, and it reports the granularity and internal complexity of the cell rather than its size. In his applications FSC does two jobs. The first is gating: separating lymphocyte, monocyte and granulocyte, because they differ in both size and granularity, so they can be told apart on the scatter plot before any antibody is added. The second is spotting doublets. Two cells stuck together look unusually large on FSC, which is the giveaway, so before you read DNA content you have to exclude doublets with FSC and SSC, otherwise the 4N peak is overestimated and every number downstream of it is wrong. A doublet does not merely add a stray event to the plot, it quietly inflates the tallest peak you are trying to measure, which is why the clean-up step has to come before the reading step and not after it.

Why this course brings it up: Dr. Wong lists FSC equals size and SSC equals granularity as a precise pairing, so it is second-level, grade A. The rule of excluding doublets before reading DNA content is a detail from his flow cytometry slide, and you can use it in a scenario answer to explain why the 4N peak needs care. **Appears in:** Ch 9, Ch 11, Lect 2

▪ FoSTeS

In one sentence: Fork stalling and template switching: a replication route that manufactures CNV.

Detailed explanation: When a replication fork gets stuck on a repeated sequence without actually breaking, it sometimes changes track: it leaves the template it was on, jumps to a similar sequence somewhere else and carries on copying from there, and then switches back afterwards.

This combination of stalling and template switching can produce both duplications and deletions, and it is one of the four CNV-generating mechanisms he lists side by side on his slide with NAHR, NHEJ and MEI. Note that nothing is broken; the fork simply loses its place and picks it up again somewhere that looks similar enough.

Why this course brings it up: He is explicit that the molecular mechanisms of CNV are not examined (he says "I'm just giving you a taste, don't be frightened by these mechanisms"); but what NAHR is, and the concept of adaptive CNV, are still within the

range of what you are expected to understand. **Appears in:** Lect 3

▪ Frameshift

In one sentence: A mutation in which the number of bases added or removed is not a multiple of three, so the reading frame shifts and the protein is usually truncated.

Detailed explanation: The ribosome reads three letters at a time, so if the number of bases added or removed is a multiple of three, the reading frame survives and the protein is merely shortened (in-frame).

If it is not a multiple of three, every codon after the change is regrouped, the whole sequence of the protein is rewritten, and it usually runs into an early stop signal and becomes a useless short protein. On his slide, a one-nucleotide deletion turns MET-LYS-PHE-GLY-STOP into MET-LYS-LEU-ALA and carries on; a one-nucleotide insertion terminates immediately after the AUG. A frameshift also manufactures neoantigens, and those are fragments the immune system has never seen before. So a single letter is enough to destroy a protein, which is why frameshifts are such a disproportionate piece of damage for such a small edit. And the neoantigen point is the clinically interesting one, because a broken protein makes new pieces, and new pieces are exactly what an immune system can lock on to.

Why this course brings it up: Frameshift is a core concept that appears in both Lecture 1 and Lecture 5, and its clinical consequence is precisely the start of the Lecture 4 chain from MSI-H to hot tumour to a good response to anti-PD-1. This is examinable content. So it earns its place in the dictionary twice over: once as a mechanism and once as the first step in a treatment story. **Appears in:** Ch 3,Ch 9,Ch 10,Lect 1,Lect 4,Lect 5

▪ Freedom to Operate (FTO)

In one sentence: The patent risk check you do before you commercialise something.

Detailed explanation: FTO answers a very practical question: if I make this product, will I be sued? That is a different question from whether I will be granted a patent. A company that did not invent the tool may still be barred from a particular application by somebody else's patent.

He uses the two CRISPR patent lines, Doudna and Charpentier of UC Berkeley against Feng Zhang of the Broad Institute, to show that the law can separate the invention from a particular application, but when it cannot separate them cleanly there is a dispute. The point of the example is that the science can be shared even when the commercial rights cannot.

Why this course brings it up: Dr. Wong does not list FTO as an exam question on its own, so it is conceptual understanding. He treats the whole intellectual property map as an extension, something for you to know more about. The CRISPR example is the one to remember, because the two lines are still being argued about. **Appears in:** Ch 12, Lect 2, Lect 4

▪ Functional group

In one sentence: A small group of atoms on a molecule that gives that molecule its chemical character and reactivity.

Detailed explanation: The carbon skeleton is only a skeleton. The real personality of a molecule comes from the functional groups hanging off it: a hydroxyl, -OH, makes it hydrophilic and able to take part in hydrogen bonds, a carboxyl, -COOH, makes it acidic, an amino group, -NH2, makes it basic, and a phosphate group takes part in the energy balance.

The most famous contrast among biomolecules is sugar against lipid. Sugars have several hydroxyls, so they are hydrophilic, they form hydrogen bonds, and enzymes can recognise them; the tail of a lipid is a pure hydrocarbon chain, hydrophobic, and that is what lets lipids assemble themselves into a membrane. Same carbon chain, opposite behaviour, all decided by the hydroxyls.

Why this course brings it up: What he excludes is the chemical structure itself, but the functional group is the shortest route to understanding the differences between the four classes of biomolecule, and it is the basis for clinical reasoning of the kind "sugars can be recognised by enzymes and lipids cannot". So it is concept-only. **Appears in:** Ch 1

▪ Fv fragment

In one sentence: The smallest part of an antibody, the variable region alone. About 25 kDa.

Detailed explanation: Fv is short for fragment of variable, and it is the smallest recognising unit of an antibody: it contains only the variable region of the heavy chain and of the light chain (VH and VL), and yet it already has the full antigen-binding ability, because CDR1/2/3 live inside it.

The mass on his Lecture 1 slide is about 25 kDa (against Fab about 50 kDa and Fc about 50 kDa). In CAR-T design the same idea is used: VH and VL are joined by a linker into an scFv (single-chain variable fragment), which is then joined to a transmembrane segment and an intracellular signalling region (CD3ζ, 4-1BB, CD28). The practical point of the design is that you can build a recognising molecule out of the smallest possible piece and then bolt on whatever signalling machinery you need. The smallest piece that still does the whole job of recognition.

Why this course brings it up: The 25 kDa figure is one of his precise numbers on a slide, which makes it a third-level (B) item. Fv and scFv are the entry point to understanding the structure of a CAR, which belongs to the concepts you are expected to understand. So the number and the design belong to two different tiers, and you should be clear about which is which. **Appears in:** Ch 2, Lect 1

▪ G-banding

In one sentence: The banded pattern of light and dark stripes a chromosome forms after staining, and it can expose large structural changes.

Detailed explanation: Stain a chromosome with Giemsa and it grows alternating dark and light bands. The dark bands mostly correspond to regions that are richer in AT, the light bands to regions richer in GC, so the stripe pattern is really a map of the sequence along the chromosome.

An experienced eye can read the chromosome number and the arm position straight off the stripes, which is why the technique is still the workhorse of karyotyping. What G-banding is for is catching abnormalities at the karyotype level: one chromosome too many or too few, large deletions, and translocations, where a piece of one chromosome has jumped somewhere it should not be. Its limit is resolution. You can only see gross structural changes of roughly 0.5 to 10 Mb, and point mutations or single nucleotide changes are completely invisible to it. For those you need sequencing, which reads the letters one at a time instead of looking at the shape of the whole chromosome.

Why this course brings it up: Karyotype analysis resolution is Dr. Wong's second-level (A) item, and he gives you two details to hold on to. The first is that q21.1 is said as q-two-one-point-one, not q-twenty-one-point-one. The second is that his own two statements about the resolution, 5–10 Mb and 0.5–1 Mb, do not actually agree with each other, so remember the direction and the limit instead of memorising one number. **Appears in:** Ch 3, Lect 3

▪ G0

In one sentence: The side branch off the cell cycle where a cell rests, alive but not dividing.

Detailed explanation: G0 is a side branch that leaves the cell cycle diagram at G1. A cell that has entered G0 is alive and metabolically normal, but it no longer copies its DNA and no longer divides.

It is also the state a cell falls back into when the environment tells it to stop. Highly differentiated cells such as neurons and muscle cells stay in G0 essentially for good, while stem cells can come back to G1 if they need to. Clinically G0 has two uses. First, serum starvation, dropping FBS to 0.5-1 %, parks cultured cells in G1 so you can release them together, and that is standard practice for DNA replication experiments. Second, "cellular quiescence" and "cellular death" are different things: a cell that has stopped cycling is not necessarily apoptotic.

Why this course brings it up: He teaches the four phases of the cell cycle plus this G0 side branch, so it is concept-only. He has also warned you that individual cyclin-CDK pairings need not be memorised, but "G1 arrest" and the serum starvation number, 0.5-1 % FBS, are precise numbers he lists himself, so the number to hold on to is 0.5-1 % FBS. **Appears in:** Ch 8, Lect 2

▪ G1

In one sentence: The growth phase of the cell cycle, before DNA replication.

Detailed explanation: G1 is the phase where the cell grows. It synthesises protein and organelles, does its own work, and gets ready to copy its DNA once. The DNA has not been replicated yet, so a G1 cell has a DNA content of 2N, and after PI staining it is the leftmost 2N peak on the flow cytometry plot.

The exit from G1 has two gates. One is the pRb–E2F gate, where cyclin D–CDK4/6 phosphorylates pRb, E2F is released, and the S phase genes are switched on. The other is the DNA damage checkpoint: when the DNA is damaged the cell stops in G1 and does not enter S, so a cell with a broken genome never gets its DNA copied. CDK4/6 inhibitors such as palbociclib jam the first gate, so the cell cannot get past the start line even when the signal to divide is there. The two gates are not redundant with each other. One asks whether the cell has permission to divide, the other asks whether the DNA is fit to be copied, and a cell has to get past both before it enters S phase.

Why this course brings it up: Conceptual understanding plus a third-level number. Dr. Wong has given the PI-stained readings of G1 at 2N and G2/M at 4N, and the step from blocking the contractile ring to the 4N peak is something the answer to his first-grade, A+ question has to contain. **Appears in:** Ch 8, Lect 2

▪ G2

In one sentence: The checkpoint stage after DNA replication is finished and before the cell divides.

Detailed explanation: G2 is the phase in which the DNA has already been copied (DNA content 4N) but the cell has not yet divided; its job is to check the copy. If the check fails, the cell stops here, and if it cannot be repaired it goes on to apoptosis; only if the check passes does it enter M phase. His flowchart goes into more detail: pRb is phosphorylated in G1, cyclin E–CDK2 takes over at the G1/S transition, cyclin A–CDK2 runs in S phase, cyclin A–CDK1 in G2, and cyclin B–CDK1 in M phase. The G2/M boundary is the most important gate in the cell cycle, which is why so many anticancer drugs are aimed at it. The pairing you want to carry away is the simple one: G2 is a checkpoint phase, and the DNA content in it is 4N.

Why this course brings it up: He explicitly excludes the individual cyclin–CDK pairings from memorisation, but "G2 is a checkpoint phase and the DNA content is 4N" is reading material he uses again and again. It belongs to the concepts you are expected to understand. **Appears in:** Ch 8, Lect 2

▪ GAP-mediated GTP hydrolysis

In one sentence: The GTP hydrolysis that GAP promotes, which is the small G protein's own off switch.

Detailed explanation: Normally a small G protein is on when it is holding GTP, but the GTPase activating protein, GAP, catalyses the hydrolysis of GTP into GDP, and the protein goes back to being off by itself.

The G12, G13 and Q61 mutations of K-Ras break exactly that catch, so Ras stays locked in the GTP-bound active state and keeps sending the growth signal down. The easiest way to remember GAP is as the handbrake: normally the handbrake bites, and when the mutation breaks it the car keeps going forward. Which is why a mutation in the brake mechanism and a mutation in the accelerator produce the same final picture, a signal that will not stop. And that is the whole difference between a normal small G protein and a mutated one: not more signal, but signal that cannot be switched off.
Why this course brings it up: Mutant K-Ras breaking GAP-mediated GTP hydrolysis is the mechanism written on his slide, and the clinical meaning of Ras mutations is Tier 2 background knowledge. The mechanism is on the slide and the clinical story is the background you build on top of it. **Appears in:** Ch 10, Lect 1

▪ Gating

In one sentence: Drawing a boundary on a flow cytometry plot so that one population can be picked out of many.
Detailed explanation: Gating is the first step of the analysis: on the FSC versus SSC scatter plot you circle the lymphocytes, and then within that circle you look at the four-quadrant plot of CD3-FITC against CD4-Alexa Fluor 647 and circle out the CD3+CD4+ helper T cells.
Every gate is a chain of filtering conditions, from coarse to fine: debris and doublets go first, and the target population is chosen last. Gating is the drawing; FACS is actually doing the separating. The same gate definition that is only a circle during analysis decides, during sorting, which cells end up in the culture flask. That is why a badly drawn gate ruins everything downstream: the wrong cells get analysed, and no later step can recover the mistake. Coarse to fine, always: that is how every gate you will ever draw is built.
Why this course brings it up: It belongs to the concepts you are expected to understand, but the flow plot is the prototype of the scenario questions he lists (Case 4: FSC versus SSC, which gate, which population, what is the contaminant), and the thing to be able to tell apart there is FSC = size and SSC = granularity. So the vocabulary, not the arithmetic, is what you need here. Case 4 is the one to practise. **Appears in:** Ch 9, Ch 11, Lect 2

▪ GC-MS

In one sentence: Gas chromatography mass spectrometry: it separates the chemicals in a sample first, and then identifies each of them as they come off.
Detailed explanation: GC-MS strings two instruments together. Gas chromatography (GC) does the separating: the sample is vaporised and injected into a column, the substances separate according to how volatile and how strongly adsorbed they are, and they come out one at a time.
Mass spectrometry (MS) does the identifying: each exiting molecule is hit with electrons and broken into charged fragments, and those fragments are laid out as a spectrum by their mass-to-charge ratio, and matching that against a database gives you a name. His metabolomics workflow is exactly this: sample collection → sample preparation → instrument analysis (LC-MS, GC-MS, NMR) → data mining (PCA, volcano plot, VIP > 1.0) → picking out differential metabolites → multi-centre validation. His breath VOC study also uses GC/MS to measure the exhaled air of lung cancer and gastric cancer patients. The order is the thing to take away: separate first, then identify, and only then quantify. You cannot identify what you have not separated, and that sequence is why the instrument comes as a pair rather than as one box. The two machines are partners rather than a single device, and each one is doing a job the other cannot do. A metabolomics run is therefore a pipeline, not a single measurement, and the quality of the answer depends on every stage of it.
Why this course brings it up: What he excludes is the chemical structures and the metabolic pathway details, but the logic of separate first, then identify, then quantify, and the reason GC-MS suits volatile substances while LC-MS suits polar ones, is the skeleton of the biomarker discovery workflow. Conceptual understanding. So the instrument choice is not a detail; it is a consequence of what the molecule is. And the same skeleton carries the metabolomics section later on. **Appears in:** Ch 10,Ch 11,Lect 1

▪ GC-rich

In one sentence: A stretch of DNA with a high proportion of G (guanine) and C (cytosine) in its base composition.
Detailed explanation: There are three hydrogen bonds between G and C and only two between A and T, so a GC-rich double helix is harder to pull apart and has a higher melting temperature. Dr. Wong uses the idea in two ways.
First, the pale bands on a G-banding karyotype are the GC-rich regions. Second, in PCR primer design, the higher the GC content of the primer the higher its Tm, and the annealing temperature has to be matched to it. So there are practical rules for primer design: too little GC and the primer is not stable enough, too much and it starts folding back on itself into a hairpin, and either way the reaction fails.
Why this course brings it up: “The more GC, the higher the Tm” is a Tier 3 directional fact on his list, and it connects straight into primer design. The term GC-rich itself is not one he names, so it belongs to conceptual understanding. **Appears in:** Ch 3, Ch 11, Lect 3

▪ Gene

In one sentence: A recipe: a stretch of DNA that specifies one product.
Detailed explanation: A gene is a stretch of DNA on a chromosome. It is transcribed into RNA, and that RNA is then used to make a product. For most genes the product is a protein; for a few it is a functional RNA, such as tRNA, rRNA or a small regulatory RNA.

The key point is that one gene does not equal one protein. Because of alternative splicing, the same gene can specify several isoforms, and whether it gets read at all depends on how tightly the DNA is packaged, which makes the packaging as much a part of the switch as the gene itself. Knowing the name of a gene is not enough. You also need to know what it does, where it is switched on, and what happens when it breaks. Knowing only the name K-Ras will not let you explain why cetuximab fails.
Why this course brings it up: This is the basic unit of the whole course, and Dr. Wong's metaphor for it is a recipe. What you should watch is his insistence that knowing the name is not enough, because that is exactly the part a scenario question asks you to fill in. **Appears in:** Ch 3

▪ Gene expression

In one sentence: The whole process of switching a gene on, copying out the message, and building the protein.
Detailed explanation: Gene expression is a production line made of three arrows: DNA → RNA (transcription), RNA → protein (translation), and then the protein goes off and does the work. Every arrow is a point where the output can be regulated, and therefore a point where a drug can be applied — this is the most useful single diagram in the whole course.
He comes back to it in Lecture 5: most clinical biomarkers (prognostic and predictive) act through the regulation of gene expression, and the same mutation (BRCA, for example) is not guaranteed to cause disease precisely because the regulatory layer decides. Mis-regulation also explains why some "cold tumours" can be remodelled by drugs back into "hot tumours". And the arrows are why an epigenetic mark, a transcription factor and a drug can all end up acting on the same gene from three completely different directions. Three arrows, and a drug can be placed on any one of them.
Why this course brings it up: The central dogma is the skeleton of his Lecture 1, and regulation of gene expression is the centre of Lecture 5, which makes this a concept to understand plus a second-level (A) item. What he names is the idea of multi-layer regulation, not the detail of any one arrow. So if you are asked to place a term, ask first which arrow it sits on.
Appears in: Ch 7, Lect 5

▪ Gene expression programme

In one sentence: The set of genes a particular cell type switches on and off, and what makes a T cell different from a dendritic cell.
Detailed explanation: The same DNA is read as a different combination of genes in different cell types, and that combination is called the expression programme. A T cell and a dendritic cell carry the same genome and still behave completely differently, because they read different parts of it.
The difference comes from each cell type expressing different transcription factors, and different transcription factors switch on different enhancers, so which genes are open is decided by cell identity rather than by the DNA itself. The genome is the hardware and the expression programme is the wiring that decides what the hardware does today. Dr. Wong's t-SNE clustering plot separates PBMC samples from melanoma and ascites samples into macrophage, cancer cell, T cell, NK cell, dendritic cell, CAF, endothelial cell, B cell and so on, and it does it by the expression profile of each group. Every one of those clusters is an expression programme in its own right, and a sample that contains all of them contains all of those programmes at once. He also stresses that the level of expression is cell-specific and time-specific at the same time, so the same cell does not read the same way on every day of its life. That is why an expression level is always a statement about a cell and a moment together, and never about the gene alone.
Why this course brings it up: Dr. Wong states it outright: "the expression is usually cell-specific and at the same time, it is time-specific… This is the concept that I always emphasize." That puts it at second level, grade A. It is also the premise underneath liquid biopsy and tumour microenvironment analysis, both of which begin from the question of what a given cell is currently reading. **Appears in:** Ch 7, Lect 5

▪ Gene knock-in

In one sentence: Putting a working copy of a gene into the genome, which needs a precise repair pathway to land in the right place.
Detailed explanation: knock-in is the exact opposite of knock-out. You are not trying to break a gene, you are trying to get it to carry a specified stretch of sequence and land in the genome accurately.
The method runs like this. Cas9 cuts a DSB, a double-strand break, at the target position. You then provide a donor template, a piece of DNA carrying the sequence you want, and the cell's own machinery copies it in using homology-directed repair (HR). The price is that HR is only available in S and G2, when there is a sister chromatid to work with, so knock-in efficiency is far lower than knock-out, and the operation often has to be timed to the right point in the cell cycle. That is the whole reason knock-in is the harder of the two strategies and the one you have to plan around. Dr. Wong's other detail is a reminder that tumour cells with suppressed HR, BRCA mutations for example, are actually easier knock-in targets.
Why this course brings it up: Choosing an editing strategy is one of the case question archetypes he lists, Case 12: knock-out uses NHEJ, knock-in uses HR and is limited to S/G2, and it is second-level (A). You have to be able to explain the mechanism and the limitation of both strategies clearly, in that order. **Appears in:** Ch 6, Lect 2, Lect 4

▪ Gene knock-out

In one sentence: Making a gene stop working altogether, usually by letting CRISPR cut it and letting the cell's own repair machinery break the reading frame.

Detailed explanation: Cas9 makes a double-strand break at the position specified by the guide RNA, and the cell repairs it with NHEJ. Because NHEJ is the quick-and-dirty repair, it often leaves an insertion or a deletion of a few nucleotides (an indel), and that breaks the reading frame, so the gene becomes non-functional.

That is a knock-out, and it is what he means when he says CRISPR has already replaced the older siRNA/shRNA knockdown techniques: a knockdown only pushes the expression level down, a knock-out switches it off completely. And because NHEJ works in every phase of the cell cycle, a knock-out does not need the cells to be synchronised, which makes it a great deal easier than a knock-in. Think of it as deleting a line from a book rather than covering it with correction fluid: one leaves the words missing for good, the other can be wiped away later. Which of the two you need depends entirely on what you want to see, a permanent loss of function or a temporary reduction you can wash out. And that is also why the technique is so widely used: it does not care what phase the cell is in, and it does not need a bespoke protein engineered for every target. The guide RNA is the part you design; the repair machinery does the rest.

Why this course brings it up: It sits between his second-level (A) and third-level (B) items. He said explicitly that "CRISPR/Cas9 is now standard for gene knock-out/knock-in; older siRNA/shRNA knockdown techniques are largely superseded." He also puts knock-out in the most likely multiple-choice contrast, the one between HR and NHEJ. So you should be able to say which repair pathway makes the indels, and why the whole technique is easier than making a targeted change. And note the word superseded: he is not saying the old methods are unknown, he is saying they are no longer what you would reach for. That is a small thing, but it is the kind of wording that tells you which answer the question wants. And notice that the knock-in is the harder half of that sentence, because that is the one that still needs the cell to be in the right phase of the cycle. **Appears in:** Ch 6,Lect 2,Lect 4

▪ General transcription factor

In one sentence: A protein needed to start transcription on any gene, rather than only on one particular gene.

Detailed explanation: Transcription does not work because RNA polymerase finds the gene by itself. The real way it starts is this: TFIID (which contains TBP) recognises the TATA box on the promoter (about 30 bases upstream of the transcription start site), then TFIIA (a general transcription factor) and TFIIB (a general transcription factor) join in order, and after that come RNA polymerase II together with TFIIE (a general transcription factor), TFIIF (a general transcription factor) and TFIIH, which together form the pre-initiation complex.

With that complex in place, RNA polymerase can pull the DNA open and start transcribing. The general transcription factors are the basic parts that assemble the machine and have nothing to do with which gene it is; what decides which gene gets switched on is the sequence-specific transcription factors and the enhancers. Think of the general transcription factors as the scaffolding and the crane: they are the same on every job, and none of them cares what is being built. The general transcription factors are therefore the parts of the machine you can swap around freely; the specific factors are the parts that decide the job.

Why this course brings it up: He teaches the assembly of the promoter and the pre-initiation complex, which is a concept to understand plus a second-level (A) item. You have to be able to tell apart the general factors that switch the machine on and the specific factors that choose the gene — that distinction is the core of any promoter question. If you get that division of labour wrong, every promoter question falls apart, because you will start looking for a gene-specific function in factors that are in fact universal. One more practical point: because the machinery is the same everywhere, the same seven ways of switching a transcription factor on apply to every gene in the cell. **Appears in:** Ch 7, Lect 5

▪ Genome

In one sentence: The complete set of DNA that one organism or one cell carries.

Detailed explanation: A genome is the full set of DNA in a cell or an organism, including the genes that code for proteins, and also including regulatory sequences and non-coding regions. The human genome has about 3 billion base pairs spread over 23 pairs of chromosomes.

Every somatic cell in the same person, sperm and eggs and a few exceptions aside, carries that same genome, and the only difference from cell to cell is which genes are switched on. A tumour's genome differs from a normal cell's, and that difference is exactly what CNVs, mutations and structural variants are. Telling which of them are drivers, the changes that push the tumour forward, and which are passengers, changes that come along for the ride, means you have to look at the CNV profile and the mRNA expression together. A copy number gain with no expression behind it is a passenger, not a driver.

Why this course brings it up: It is concept-only. It is the starting point of the "same DNA, different expression" thread, and it is the framework Dr. Wong uses in Lecture 3 to explain tumour genomics. **Appears in:** Ch 3

▪ genomic instability

In one sentence: The state of a genome that has become unstable overall.

Detailed explanation: When the repeated DNA of the pericentromeric heterochromatin loses its methylation and becomes hypomethylated, the cell no longer keeps it under control, and mitotic recombination can occur between the repeated sequences.

The chromosomes then start trading pieces with each other and rearranging, and the genome gets messier and messier. This is the same chemistry as "a promoter is methylated and the brake is switched off", but with the opposite result. Clinically, this

instability is also the engine of tumour evolution: every round of it produces a batch of new mutations, and now and then one of those turns out to be a driver that lets the cell grow better. The two directions of methylation are worth holding side by side: too much and a gene is shut, too little and the repeats that should be quiet start talking to each other.
Why this course brings it up: He does not name genomic instability as an exam topic on its own, so it is a concept to understand; the two-way comparison of "too much methylation shuts a gene down, too little methylation lets the repeats out" is one he has drawn. That comparison is the part that can actually turn up in a question. **Appears in:** Ch 3, Ch 4, Lect 3

▪ Genotype

In one sentence: The genetic make-up of a cell or a person: the version of each gene they carry.
Detailed explanation: Genotype is the composition at the level of genes, which version you carry, and phenotype is what is actually expressed. A whole chain of regulation sits between the two, so genotype is not phenotype, and you always need the phenotype beside the genotype before you can say anything about the person.
The same BRCA mutation gives one person disease and another person nothing. The same gene is a driver in some cancer types and anti-tumour in others. When you read a pedigree, the genotype is the symbol you put next to each person, which version they carry and inherited from which relative, and you then use it together with the phenotype to work out the inheritance pattern.
Why this course brings it up: Dr. Wong names the pedigree as one of the two scenario questions, and the core of a pedigree question is the pairing of genotype with phenotype. This is understanding within the second-level, grade A range, and you should be able to label a pedigree with both. **Appears in:** Ch 7, Lect 3

▪ Germ cell

In one sentence: A cell that will turn into a sperm or an egg, and that is how mutations get handed on.
Detailed explanation: A germ cell is the precursor of the germ line, and it ends up differentiating into a sperm or an egg. Its special standing in genetics is this: only it, and the gametes it produces, can pass a mutation on to the next generation.
A mutation in a somatic cell usually disappears when that cell dies, but a mutation in the germ line stays in the gene pool. During embryonic development the germ cells go through one large-scale DNA demethylation, which wipes out the epigenetic marks the parents left behind and resets them, and that is why genomic memory has a generational boundary. In other words, the body gets a fresh start each generation, but the germ line carries the shopping list forward. That reset is the reason epigenetic marks are not simply inherited forever. The cell is a courier, and the only address it ever delivers to is the next generation.
Why this course brings it up: One of his first-level (A+) exam topics is M1 and M2 non-disjunction, which is a question about gamete formation, and the germ cell is the other end of that line. Conceptual understanding, but you need it to explain why some mutations are inherited and some are not. It is a simple explanation, and it is the sort of thing a short-answer question can be built on. **Appears in:** Ch 4,Ch 8,Lect 3

▪ GG-NER

In one sentence: The repair route that searches for damage anywhere in the whole genome.
Detailed explanation: Global genome repair begins with UV-DDB, which is DDB1 and DDB2, also called XPE, recognising the distorted DNA, and then XPC recruits the rest of the team. Its opposite number is TC-NER, which only deals with the DNA that is currently being transcribed.
Both routes end up feeding into TFIIH, XPB and XPD, and then XPA, RPA, XPF and XPG cut the damaged piece out. Put simply, GG-NER is the patrol of the whole city and TC-NER is the emergency repair on the spot; the first one deals with most of the damage and the second one handles the crisis on the gene that is being read at that moment. The names are worth unpacking once, because global genome means the whole genome and transcription-coupled means the gene being read.
Why this course brings it up: Dr. Wong does teach GG-NER and TC-NER separately, and the difference between the two is conceptual understanding. That NER repairs the bulky lesion is one of the pairings he lists. The two are easiest to keep apart if you hold on to the picture of a patrol and a repair crew, because that is what the acronyms are really describing, and the two are easy to swap if you only half remember them. **Appears in:** Ch 6, Lect 4

▪ Glucose

In one sentence: The monosaccharide that cells burn as fuel.
Detailed explanation: The molecular formula of glucose is $C_6H_{12}O_6$, and it is the most universal fuel a cell has. The first stop in metabolism is glycolysis: without using oxygen it cuts glucose into pyruvate and nets 2 ATP;
with oxygen the pyruvate goes into the mitochondria for the TCA cycle and oxidative phosphorylation, and one glucose yields roughly 36 ATP. Cancer cells often prefer glycolysis even when oxygen is plentiful (the Warburg effect, aerobic glycolysis), because the half-broken pyruvate is precisely the raw material for making nucleotides, lipids and amino acids — what the cell wants is not fuel but feedstock. Blood glucose is also a classic clinical biomarker. Six carbons, six hydrogens and six oxygens, and almost every pathway in the cell either starts with it or ends with it. Glucose sits at the top of most metabolic charts, and the choice of what to do with it is one of the ways a cell decides what kind of cell to be.
Why this course brings it up: What he excludes is the detail of the metabolic pathways, but the Warburg effect is the opening story of his Lecture 1, and it is also his reason for using metabolomics to find biomarkers. It belongs to the concepts you are expected to understand. The single formula is worth remembering even if the pathways are not. **Appears in:** Ch 1, Lect 1

▪ GLUT1

In one sentence: The glucose transporter that HIF-1α switches on, letting a cell keep working when there is no oxygen.
Detailed explanation: GLUT1 is the membrane transporter that carries glucose into the cell. When oxygen is short, HIF-1α is stabilised, enters the nucleus, joins HIF-1β, and then uses HRE to switch on a set of genes: VEGF, GLUT1 and EPO. Switching on GLUT1 means that even without oxygen the cell can bring in more glucose and hold on with glycolysis, the pathway that makes ATP without oxygen. For a tumour core this is a survival mechanism, because the middle of a tumour is always short of oxygen. For the rest of the body, an ischaemic heart for example, it is a protective mechanism. The HIF-1α switch is therefore both the tumour's opportunity and the normal cell's rescue, and that is exactly why it is interesting to target.
Why this course brings it up: He does not name GLUT1 on its own, so it is concept-only, but it is one of the three target genes of HIF-1α, and that list is the answer to the question of what a cell does when it is hypoxic. **Appears in:** Ch 7, Lect 5

▪ Glycine

In one sentence: The amino acid with the smallest side chain, and the most flexible one.
Detailed explanation: Of the twenty amino acids, glycine has an R-group of only one hydrogen, so its backbone can rotate freely, and that makes it the most flexible and the least conformationally restricted position in a protein.
By contrast, a large or charged R-group such as tryptophan or arginine makes that section of the chain stiff. Part of the energy barrier of protein folding comes from the steric hindrance between R-groups, and because glycine has almost no side chain it often turns up at the bends. So a single hydrogen is what buys the flexibility, and that is a physical reason rather than a chemical rule.
Why this course brings it up: He uses this contrast to show that the R-group decides local rigidity, and in the same passage there is the table of amino acid R-groups that he says outright is not examined, he says do not memorise it, it will definitely not be tested. The comparison is in and the table is out. **Appears in:** Ch 2, Lect 1

▪ Glycogen

In one sentence: The form in which animals and fungi store glucose, a branched chain of sugar units.
Detailed explanation: Glycogen is a highly branched polymer of glucose, with α1,4 bonds making the main chain and α1,6 bonds forming the branch points. It is stored in the liver and in muscle, and it is the body's glucose warehouse.
Liver glycogen is broken down step by step by glycogen phosphorylase when blood glucose falls, becoming glucose-1-phosphate and then glucose-6-phosphate, and finally glucose-6-phosphatase, an enzyme the liver and kidney have and nothing else does, releases free glucose into the blood. That last step is why the liver can feed the rest of the body while muscle cannot, and the way out is a chain of conversions, each with its own enzyme named after it. Muscle glycogen is used only on its own account. The point of the branched structure is that many ends can be mobilised at once, and every branch is a separate exit point, so the breakdown is fast and can meet an emergency quickly. Dr. Wong's glycogenesis slide also reminds you that a lack of glucose-6-phosphatase leaves a person unable to release glucose from glycogen, which causes an energy crisis. The stored sugar is there and the machinery to use it is there, but the last step is missing, so the fuel stays locked in the warehouse.
Why this course brings it up: Conceptual understanding plus a third-level number. The α1,4 and α1,6 bonds and the clinical example of glucose-6-phosphatase deficiency are both from his Lecture 1 carbohydrate metabolism slide, where the whole glycogen story is drawn in one picture, the branches and the enzymes together. **Appears in:** Ch 1, Lect 1

▪ Glycolysis

In one sentence: The first step of making energy: glucose is broken down into pyruvate in a set of ten steps, and no oxygen is involved at any point in it.
Detailed explanation: Glycolysis is ten reactions that cut one six-carbon glucose into two three-carbon pyruvates. It happens entirely in the cytoplasm, it does not need oxygen, and the net result is 2 ATP and 2 NADH.
What happens to the pyruvate next is decided by oxygen: with oxygen available it goes into the mitochondria for the TCA cycle and oxidative phosphorylation, giving about 36 ATP in total; without oxygen, lactate dehydrogenase turns it into lactate and you keep only those 2 ATP. Cancer cells often take the second route (the Warburg effect), because a half-broken sugar is building material rather than fuel. The numbers on his slide are these: resting cells make about 36 ATP aerobically and 2 ATP anaerobically, and proliferating cancer cells send about 85 % of their pyruvate to lactate and only about 5 % into the mitochondria whether or not oxygen is present, netting about 4 ATP. So glycolysis itself is the same in a cancer cell and in a normal cell. What differs is what the cell decides to do with the pyruvate at the end of it. That is the whole point of the Warburg effect: it is not a broken pathway, it is a rerouting. And the economics are what he wants you to take away. A cancer cell chooses less ATP in exchange for keeping the half-built sugar, because the half-built sugar is what it uses to build new cells. Normal cells are doing the opposite and prefer the bigger return. You can also see the shape of the trade-off in the numbers themselves: about 36 becomes about 4, and the missing ATP is being spent on something else. And the cell that has decided to keep building is a cell that is about to divide, so the sugar gets spent on the components of the next cell. None of this requires a broken enzyme; it requires a different set of priorities.
Why this course brings it up: He explicitly excludes the metabolic pathway details: “Please don't memorize all these chemical structure… They would not be included in my exam.” But the direction of the Warburg effect and the numbers behind it (36 against 2 against 4) are third-level (B), and it is a theme he comes back to in both Lecture 1 and Lecture 2. So the figures are examinable while the ten reactions are not, and the sensible way to revise it is to remember what the cell is optimising rather

than the order of the steps. Also keep the two routes apart: glycolysis needs no oxygen at all, and the step where oxygen decides what happens to the pyruvate happens after glycolysis is finished. That is a question he likes. And if you are ever asked to draw the comparison, the three numbers are the whole drawing. **Appears in:** Ch 10,Lect 1

▪ Glycosidic bond

In one sentence: The chemical link that joins one sugar unit to the next in a sugar chain.
Detailed explanation: The glycosidic bond is the link between sugar units. It is formed when a hydroxyl group on one molecule reacts with a hydroxyl group on another and a molecule of water is lost, which is the same principle as peptide bond formation — the same dehydration reaction builds glycogen and cellulose in sugars and polypeptide in proteins.
The configuration of the bond decides the properties: glycogen uses alpha1,4 and alpha1,6 bonds and is something humans and animals can break down; cellulose uses beta1,4 bonds and humans have no cellulase, which is why herbivores need gut bacteria to make use of it. One reaction, two polymers.
Why this course brings it up: It belongs to the concepts you are expected to understand. The parallel between the glycosidic bond and the peptide bond (both are dehydration condensations) is part of the Lecture 1 thread about why water is so important inside cells; and the difference between alpha and beta bonds is what lets you understand why the configuration of one bond decides whether it can be broken down. **Appears in:** Ch 1

▪ Glycosylation

In one sentence: Attaching sugar groups to a protein, which usually happens in the endoplasmic reticulum.
Detailed explanation: Glycosylation is the attachment of sugar trees to a protein, usually finished in the rough ER and then modified in part by the Golgi. What matters is function: the sugars act like a label, they protect the protein from being broken down, help it fold, and decide where it ends up going.
The clinical consequence is direct, because cells and bacteria glycosylate differently, so a bacterial protein made inside a human is often attacked by the immune system as a foreigner, and the immune response wipes it out before it can do any work. That is why making therapeutic proteins uses mammalian cells, CHO cells, which decorate the protein the way a human cell would. In his ADC, the same logic applies one level down: the link between the payload and the antibody must also be cut by a specific enzyme inside the tumour, so the toxin is released only where it is wanted.
Why this course brings it up: It is concept-only. It is the reason behind choosing a platform for recombinant expression, and it is the background that goes with his CHO cells and codon optimisation material. **Appears in:** Ch 7, Ch 8

▪ Golgi apparatus

In one sentence: The packing room: it sorts finished proteins and sends each one to its destination.
Detailed explanation: The Golgi apparatus is a stack of flattened sacs, sitting downstream of the ER. Transport vesicles coming out of the ER first fuse with the cis face, the receiving side, and the proteins are modified and sorted layer by layer as they pass through the stack.
They then leave from the trans face, the dispatch side, and are packed into secretory vesicles, which fuse with the plasma membrane to deliver the protein to the cell surface or release it outside the cell. The same protein passing through the Golgi may pick up different modifying groups and end up in different destinations, and that is the working logic of sorting and delivery: the protein does not decide where it is going, the modification it picks up on the way through decides. Antibodies, hormones and many drugs all travel this way.
Why this course brings it up: Conceptual understanding. Dr. Wong teaches this route as a production line, rough ER folding and glycosylation, then Golgi sorting, then cell surface or secretion, and it also explains why the signal peptide matters. **Appears in:** Ch 8, Lect 2

▪ GPCR (G-protein-coupled receptor)

In one sentence: A large family of membrane receptors, and the best family of drug targets there is.
Detailed explanation: GPCRs are receptors that cross the membrane seven times, which is why the name is spelled out with the number in it. Seven crossings is a lot of membrane, but it lets the outside of the cell and the inside talk to each other through one molecule: the ligand binds on one side, and the G protein on the other is switched on.
The switch itself is the GTP exchange, Gα drops its GDP and picks up a GTP, and the switched-on G protein then goes looking for downstream effectors such as adenylyl cyclase. GPCRs are "the most successful family of drug targets" for two reasons: they sit on the cell surface, so a drug can bind from outside without entering the cell, and they are numerous and cover everything from vision and smell to hormone regulation. On his Hippo pathway diagram, GPCRs are one of the upstream inputs that turn Hippo OFF: activation of the G-protein-coupled receptor disables LATS, and unphosphorylated YAP walks into the nucleus and starts proliferation.
Why this course brings it up: It is concept-only. It appears in both the proposition that membrane proteins are the primary drug targets, in Lectures 1 and 2, and in Hippo signalling, in Lectures 1 and 5, and it is the entry point to his "receptor to kinase cascade" logic. **Appears in:** Lect 2, Lect 5

▪ granularity

In one sentence: How cluttered the inside of a cell is, and on a flow cytometer it is read off as side scatter.
Detailed explanation: Side scatter (SSC) measures how much a cell scatters light sideways, which reflects how complicated its internal structures are: a granular cell scatters a lot, while a lymphocyte is small and empty and scatters little.

Forward scatter (FSC) roughly corresponds to cell size. Used together, the two axes sort the cells into a few broad populations, which is the point of them: two numbers are enough to separate lymphocytes from granulocytes, and everything after that is fluorescence. So how well the gates are drawn decides how many reliable populations you can separate afterwards, and Dr. Wong's slide does exactly this, using the scatter plot to frame out the target cells and then using fluorescence as a second layer of filtering.
Why this course brings it up: FSC equals size and SSC equals granularity is the flow cytometry foundation he lists as Tier 2. It is the first thing to get right before any of the sophisticated gating makes sense. **Appears in:** Ch 9, Ch 11, Lect 2

▪ granulocytes

In one sentence: Granulocytes: the group of cells that sit high on side scatter in a flow plot.
Detailed explanation: Their cytoplasm is packed with granules, so they block and scatter a lot of light and they sit highest on the FSC–SSC plot; monocytes come in the middle, and lymphocytes, which are small and empty, sit lowest.
His slide uses these three groups to show how you draw the first gate and pick the cells you are interested in out of a mixed sample. The purpose of that gate is to separate the blood cells from the target cells; after that you use the fluorescence of CD45 and EpCAM to do a second layer of selection. The scatter is telling you something physical about the cell before any stain has been added — how much internal structure there is to scatter the light.
Why this course brings it up: He does not name granulocytes as an exam topic on its own, so it is a concept to understand; gating and the meaning of SSC are the Tier 2 parts. The gates are the examinable part, and the cells are just the example he uses to build them. **Appears in:** Lect 2

▪ Granzyme

In one sentence: The enzyme that kills a target cell once it has got inside it.
Detailed explanation: There are two routes by which cytotoxic T cells and NK cells kill a target cell: the Fas–FasL pathway and the granule pathway. In the granule pathway the T cell pushes its granules towards the target, perforin punches a hole in the membrane, and the granzymes (mainly granzyme B) travel through that hole into the target cell, where they activate the caspase cascade, shredding proteins and triggering apoptosis.
Granzymes are proteases, so they have to be delivered into the cytoplasm — the hole that perforin makes is the delivery route. So the sequence is: make a hole, squeeze the enzyme through, and let the enzyme start the cell's own death programme from the inside. Note that the killing machinery is brought by the attacker rather than asked of the target, which is why the same effector route serves T cells and NK cells alike.
Why this course brings it up: He drew the arrow "perforin → perforin pore → cell death" on the CAR-T slide, so this belongs to the concepts you are expected to understand. It also explains why CAR-T has to carry intracellular activation domains such as CD3ζ and 4-1BB: what they trigger is exactly this granzyme / perforin killing machinery. One mechanism, two contexts, and the second context is the one being engineered in the clinic. **Appears in:** Ch 9, Lect 2

▪ Griffith

In one sentence: The scientist whose 1928 bacterial experiment showed that material taken from dead pneumococcus could turn harmless live bacteria into virulent ones, which was the first hint that something transferable was at work.
Detailed explanation: Griffith worked with Streptococcus pneumoniae and split them into two kinds: the encapsulated S type (virulent) and the non-encapsulated R type (non-virulent). He added certain components of heat-killed S type bacteria to live R type bacteria, and the previously harmless live bacteria became virulent S type.
Griffith called this the “transforming principle”, and he knew he was looking at something that could be transferred, but he could not say what it was. Four years later Avery, MacLeod and McCarty answered the question with an enzyme experiment (DNase destroys it, protease does not, therefore it is DNA), and showed that DNA is the genetic material. Notice that Griffith's contribution is not a mechanism but a question. He identified the substance, he showed it could be passed from one bacterium to another, and he left the identity of the substance as an open problem. That is a common shape in the history of science: the first paper gives you the thing to work on, and the second paper tells you what it is made of. And the experimental design is the part worth learning. Griffith mixed two things and watched the phenotype change, and the phenotype change is what carried the whole argument. No microscope was needed for that, only the ability to tell a virulent colony from a harmless one, which was a matter of how the mice fared. The virulence of the S strain is what made the experiment legible in the first place.
Why this course brings it up: This is the first step of the three-part DNA history, so it is conceptual understanding. He does not make the years examinable, but conceptually you should be able to say that Griffith raised the question and Avery and his colleagues answered it with DNase. That division of labour is the examinable part. If a question asks you which experiment identified DNA, the answer is the enzyme one, not Griffith's. **Appears in:** Ch 3,Lect 1

▪ gRNA

In one sentence: The guide RNA that tells the CRISPR enzyme where to cut.
Detailed explanation: gRNA is the navigation for Cas9. A guiding sequence of about 20 nucleotides binds to Cas9, and Cas9 then goes looking in the genomic DNA for the complementary sequence, and it also requires a correct PAM nearby before it will act at all.
Cas9's two nuclease domains, RuvC and HNH, cut there, once, and that produces a double-strand break. That break is the hand-off point between the two decisions. Only after the gRNA does the choice come. If the break is repaired by NHEJ you get a knock-out; if it is repaired by HR with a donor template you get a knock-in. So the gRNA decides where the cut goes and the

repair pathway decides what the edit is, and that is the whole logic of his Lecture 4. In other words CRISPR hands you two separate decisions, where to cut and what the repair will do, and the usual mistake is to try to make them at the same time.
Why this course brings it up: Second level, grade A. Dr. Wong calls CRISPR/Cas9 "now standard for gene knock-out/knock-in" and draws the gRNA, the PAM and the RuvC and HNH domains as a diagram. In an editing strategy question you have to be able to explain that the choice of gRNA determines the efficiency and the off-target effects. **Appears in:** Ch 6, Lect 2, Lect 4

▪ GTP-bound

In one sentence: The active state of a small G protein, when it is holding GTP.
Detailed explanation: A small G protein is like a plug: plugged in with GTP, GTP-bound, and the current is on, and pulled out and turned into GDP, the current is off. Normally it is on for a few seconds and then switches itself off.
A mutated Ras is a plug that has got stuck and cannot be pulled out, so the signal keeps going. Once you see it that way you understand why a K-Ras mutation is the same thing as holding the accelerator down. And the point of the analogy is that the failure is mechanical rather than mysterious, a protein that cannot be switched off.
Why this course brings it up: He does not name GTP-bound as an exam question on its own, so it is concept-only. The K-Ras story is what falls in Tier 2 range. So the state is context and the mutation is the examinable story. **Appears in:** Ch 10, Lect 1

▪ guanine

In one sentence: One of the four main bases of DNA and RNA, written G.
Detailed explanation: G is a purine, and when it pairs with cytosine it forms three hydrogen bonds, which makes it the tighter of the four pairs. GC-rich regions are therefore harder to pull apart and have a higher melting temperature.
His chromosome lesson puts GC-rich in the light bands of G-banding and AT-rich in the dark bands. G and C can form three hydrogen bonds because the way their functional groups are arranged lets them make several at once, and that is also the reason GC-rich regions are more stable. So the single number to hold on to here is the three, and everything else in this entry follows from it. The two purines are the big ones with two rings and the two pyrimidines are the small ones with one, and that is also why the pairing works out.
Why this course brings it up: "Three hydrogen bonds for G–C, two for A–T" is one of his Tier 3 precise numbers. **Appears in:** Ch 3, Lect 1

▪ Guide (lncRNA)

In one sentence: The way a long non-coding RNA carries chromatin-remodelling proteins to the exact words of DNA they are supposed to act on.
Detailed explanation: An lncRNA is an RNA longer than 200 nucleotides that does not code for protein. One of its patterns of function is to act as a guide: it has no catalytic ability of its own, but it uses one stretch of sequence to recognise a particular position on the genome, and then it brings chromatin-modifying enzymes or transcription factors there so that they can do their work.
Think of it as a taxi that knows the address but does not do the job itself. His slides list a few examples. The repeated regions of XIST bind SPEN, an RNA-binding protein, and PRC2, a repressive chromatin complex, and cause epigenetic silencing; XIST is the extreme case, one RNA molecule coordinating the silencing of a whole chromosome. APOLO takes part in gene regulation in the form of an RNA–DNA hybrid. PNCTR uses a YYUCUY motif, a sequence recognised by splicing factors, to bind PTBP1, a splicing factor, and affect splicing. TINCR pairs with mRNA and binds Staufen 1, an RNA-binding protein, and affects mRNA stability. So an lncRNA is both a drug target and a new source of biomarkers.
Why this course brings it up: He does not list lncRNA as an exam question, so it is concept-only, but it demonstrates a general rule: RNA is not only a template for protein, it can also be a guide and a scaffold. **Appears in:** Ch 7, Lect 5

▪ H1

In one sentence: The linker histone, which sits on the DNA where it enters and leaves the nucleosome core like a bookmark.
Detailed explanation: The nucleosome core is an octamer made of two copies each of H2A, H2B, H3 and H4, with 147 base pairs of DNA wrapped around it. The fifth histone, H1, is not part of the core.
It sits at the doorway where the DNA enters and leaves the core, like a band on a spool, gripping the thread so that it cannot slip, and it influences the next level of folding of the chromatin fibre. How much H1 there is affects how tightly the chromatin fibre is packed, so its amount in the cell is not fixed but is adjusted to the state of the chromatin.
Why this course brings it up: Dr. Wong does not name H1 as an exam topic on its own, so it belongs to conceptual understanding. The nucleosome composition (the octamer and 147 bp, 10 nm) is his Tier 3 set of numbers. **Appears in:** Ch 4, Lect 3

▪ H2A K119ub

In one sentence: The mark left when lysine number 119 on histone H2A is ubiquitinated.
Detailed explanation: It is one way of writing down ubiquitination: a ubiquitin molecule is attached to the 119th lysine of H2A. In his cross-talk diagram, H4 K8ac and H4 K12ac influence this mark, and each other;
in other words, a mark is not an independent switch but one node in a web where everything pulls on everything else. He writes it as H2A K119ub in order to demonstrate that the same naming rule applies across different histones: the histone, the position

and the modification, and you need all three. Read it in pieces and it tells you the whole story at once — which histone, which residue, and what has been attached to it. And the web is the point, because it explains how a cell can keep a piece of chromatin switched off without anything being able to switch it back on.
Why this course brings it up: Cross-talk is one of the concepts he marks as Tier 2 (DNA methylation and histone marks reinforcing each other to form cellular memory); he does not ask you to memorise the names of the individual marks. The naming scheme is worth learning, because it is the same scheme in every case. **Appears in:** Ch 4, Lect 3

▪ H2B K123ub

In one sentence: The ubiquitination mark on lysine 123 of histone H2B.
Detailed explanation: In his slide 28, H2B K123ub is the start of one line: it pushes the state of H3 K79me and H3 K4me. A second line runs the other way, with H4 K8ac, H4 K12ac and H2A K119ub pulling on each other.
One mark sets the next, so the marks are not independent switches, they are a network. The notation is itself information: H2B is histone 2B, K123 is lysine number 123, ub is ubiquitination, and one mark is a coordinate you can look up in a database. So the practical value of the notation is that it turns a vague idea about histone modification into a specific, searchable address, and that is what makes the field reproducible at all. And the cross-talk runs in both directions, so the cell is not reading a list of independent settings but a small set of mutually reinforcing states.
Why this course brings it up: He does not ask you to memorise these mark names, so it is concept-only. The Tier 2 point he marks is that cross-talk and positive feedback produce cellular memory. So the network is the idea and the individual names are the detail. **Appears in:** Ch 4, Lect 3

▪ H3 K4me

In one sentence: The label that says lysine number 4 on histone H3 has been methylated.
Detailed explanation: The name breaks into three pieces: H3 is histone 3, K is lysine and me is methylation. H3K4me3, the triple-methylated form, generally marks the active chromatin near a promoter and is the usual clue that a gene is switched on, and his slide 28 uses H3K4me3 as the worked example in which he teaches the whole naming rule at once.
H3K4me3 usually marks the area near a promoter while H3K9me3 marks the silent regions, and the two are written on H3 in the same way and mean opposite things. The number after me is the number of methyl groups, so H3K4me3 is three of them and H3K9me3 is also three, written on a different residue.
Why this course brings it up: The naming rule of the histone code, which he gives when he talks about H3K9me3 as "Histone 3, lysine at position 9, trimethylation", is Tier 2 material. He does not ask you to memorise the individual mark names, but he does ask you to be able to read one, and to say what each part of the name is telling you. **Appears in:** Ch 4, Lect 3

▪ H3 K79me

In one sentence: The methylation mark on lysine 79 of histone H3.
Detailed explanation: It is one of several sites on H3 that can be methylated, and in Dr. Wong's cross-talk diagram it is driven by H2B K123ub. The number of methyl groups on the same lysine (me1/me2/me3) often stands for a different state, so which site, and how many times it has been methylated, is itself a code.
Because a different number of methyls on the same lysine changes which nearby proteins can recognise it, how many methyls are there is information, and the reading of it has to be learned site by site rather than guessed from the number alone. Me3 at one site and Me1 at another do not mean the same thing.
Why this course brings it up: He does not ask you to memorise it, so it belongs to conceptual understanding. The histone code and the cross-talk are what are Tier 2. **Appears in:** Ch 4, Lect 3

▪ H3K4me3

In one sentence: The mark that sits on active promoters.
Detailed explanation: The name of this mark is itself the decoding: H3 is Histone 3, K is lysine, 4 is the position it occupies counted from the N-terminus, me is methylation, and 3 is trimethylation.
H3K4me3 is the typical mark near a promoter and is associated with a gene being actively expressed. Its mechanism works in both directions: on one side H3K4me3 itself stops DNMT from binding the CpG, keeping the promoter in a permissive unmethylated state; on the other side, once the DNA has been methylated, methyltransferases and HDAC are recruited and push the chromatin back down. The mark and the DNA methylation therefore strengthen each other, forming a positive feedback loop — the cross-talk and cellular memory he talks about. Nothing about that loop is unusual in chemistry; what makes it interesting is that it means a cell which has switched a gene off can remember having done so. The naming is the smallest possible lesson in how histone marks are written: histone number, position, and modification, all three needed.
Why this course brings it up: The way he takes the naming apart (Histone 3, lysine at position 9, trimethylation) is his way of writing it, and the contrast between H3K9me3 and H3K4me3 is the skeleton of the epigenetic section. It is within second-level (A) understanding. And the naming is worth learning as a system, because you will be reading histone marks all through this part of the course. **Appears in:** Ch 4, Lect 3

▪ H3K9me3

In one sentence: A label that tells the cell a stretch of DNA ought to stay switched off.
Detailed explanation: The name unpacks like this: trimethylation of the ninth lysine of Histone 3. H3K9me3 is the typical marker of heterochromatin. His slide says that Suv39h plus HP1 will spread the methylation at K9 out into the neighbouring regions, and that is the mechanism behind silence being able to spread on its own.
The direction of causation runs the other way from what you might expect: H3K9me3 present → chromatin compacted → transcription factors cannot reach the promoter → the gene is silent. Read alongside H3K4me3, one mark is on and one mark is off, and that is the concrete content of the “histone code”. So the label is not a description of what is happening. It is one of the things that makes it happen. Which is why the two marks are better taught as a pair than as two separate entries. And the spreading is what gives the mark its power, because a single marked nucleosome cannot silence a whole region on its own. The mark recruits the machinery that reads it, and the reader recruits the writer, and the two keep each other going.
Why this course brings it up: When he teaches the histone code, the naming and the question of which mark means open and which means closed are the point, so this is second-level (A). And an HDAC inhibitor can open the chromatin up precisely because it blocks the opposite direction. So the pair to hold in your head is the mark that closes and the drug that undoes it.
Appears in: Ch 4,Lect 3

▪ H4 K12ac

In one sentence: The acetylation mark on lysine number 12 of histone H4.
Detailed explanation: Acetylation neutralises the positively charged end of the lysine, so the histone's attraction to negatively charged DNA weakens, the chromatin loosens, and the gene can be read.
His cross-talk diagram groups it with H4 K8ac and H2A K119ub and shows that acetylation is not an independent switch but something that interacts with the other marks. He draws those marks as a web of connected lines rather than a set of separate switches, and that diagram is the core of Lecture 3. The marks do not queue up one after another; they are all pressing on each other at once, and that is what makes the state of the chromatin a combination rather than a single decision. Which is why it is a network and not a list.
Why this course brings it up: He does not ask you to memorise the names of these marks, so it is a concept to understand; the line he has taught is "acetylation loosens the chromatin, an HDAC inhibitor releases the brake". That mechanism is the one to carry with you. **Appears in:** Ch 4, Lect 3

▪ H4 K8ac

In one sentence: The acetylation mark on lysine 8 of histone H4.
Detailed explanation: Together with H4 K12ac it is one of the acetylation sites on H4, and in his cross-talk diagram both of them connect to H2A K119ub. The consequence of acetylation is that the nucleosome is packed more loosely, so that transcription factors and enhancers can squeeze in to reach the promoter, and removing the acetyl group, deacetylation, tightens the chromatin up again.
The same acetylation can have different effects in different positions, so which site matters more than whether it is there, and that is exactly what the histone code means. Which is why an HDAC inhibitor has a place here: it keeps the acetyl group on, the nucleosome stays loose, and the gene can be read.
Why this course brings it up: He does not ask you to memorise it, so it is concept-only. The direction of acetylation and deacetylation is what he keeps stressing. Which way the switch points is the examinable part. **Appears in:** Ch 4, Lect 3

▪ Haploid

In one sentence: One set of chromosomes, which is what a gamete is. A human gamete has 23.
Detailed explanation: Haploid, written n, means one set of chromosomes, so a haploid cell carries one copy of each chromosome instead of the two copies a diploid cell carries. A human gamete has 23 chromosomes, 22 autosomes plus one sex chromosome, and after fertilisation it joins another gamete to become diploid, 2n = 46.
Meiosis is a machine for diversity because the first division separates homologous chromosomes and the second separates sister chromatids, so one diploid cell produces four haploid gametes that differ from one another. Each of those four is haploid, and each carries one set of chromosomes rather than two. The order of those two divisions is the part you have to hold on to. This also explains where abnormal chromosome numbers come from: if the chromosomes fail to separate, the gamete ends up with too few or too many. Failure to separate produces nullisomic or disomic gametes, and after fertilisation they become monosomy, 2n-1, or trisomy, 2n+1. The chromosome count is therefore decided in meiosis and read out afterwards in the offspring.
Why this course brings it up: Dr. Wong lists the 23 pairs of human chromosomes, trisomy 21 and Turner 45,X as precise numbers. And telling M1 failure from M2 failure, which is the first-grade question he names, uses exactly this concept: how many sets of chromosomes does the gamete have? Second level, grade A, through to third level, grade B. **Appears in:** Ch 8, Lect 3

▪ Hayflick limit

In one sentence: How many times a human cell can divide before it is forced to stop.
Detailed explanation: After every division the telomere gets shorter, by about 50 to 200 base pairs, because of the end-replication problem. Once the telomere reaches a critical length the cell stops dividing and enters senescence, and that ceiling is the Hayflick limit.

It is the cell's division timer. Who gets around it? Embryonic stem cells, some adult stem cells and germ cells keep their telomeres up with telomerase, while cancer cells simply switch telomerase back on (around 90 % of cancers upregulate it), which is how they divide almost without limit. In his HCC story, chronic injury causes hepatocyte turnover and telomere attrition, and then oncogenic MYC reactivates telomerase and the telomeres are lengthened again. And the twist is that the longest 10 % of telomeres is associated with worse recurrence-free survival (HR 1.66). So there are really two answers to the question of what sets the limit, and they are the same question asked from opposite ends. One is the shortening, which counts the divisions; the other is telomerase, which resets the counter. It also explains why the tumour story needs two steps. The liver damage comes first and wears the telomeres down, and only then does MYC come along and put the telomerase back. And it explains the result that sounds backwards at first. The cells with the most telomeres are the ones that kept dividing longest, and that is a bad sign rather than a reassuring one. So a long telomere here is not a healthy young cell, it is a tumour cell with a long runway. The counter is the point: the cell is counting its own divisions, and it counts them by spending the end of each chromosome. Telomerase is the mechanism that puts the counter back, which is why it sits at the centre of the senescence story rather than beside it. And the reason the limit exists at all is structural, not accidental. The enzyme that copies DNA cannot quite finish the end of a strand, so every round of copying leaves a little behind.

Why this course brings it up: The link from telomerase failure to senescence is one of the multiple-choice question formats he lists, and the numbers linking telomere length to HCC prognosis (the top 10 % of telomeres going with worse survival) are in his table of precise numbers. It is second-level (A) plus third-level (B). So both the mechanism and the figure are worth having, and the direction of the prognosis is the part people get wrong. Write it as “longer telomeres, worse survival” and you cannot get it backwards later. **Appears in:** Ch 5,Lect 3,Lect 4

▪ HBV

In one sentence: Hepatitis B virus. Its integration into liver DNA is a prognostic marker.

Detailed explanation: The life cycle of HBV leaves behind a store of cccDNA (episomal) in the nucleus, which is very hard to clear. HBV DNA sometimes integrates into the host genome, and the integration sites are often close to oncogenes: his slide lists TERT (chr 5), CCNE1 (a cell cycle gene, chr 19) and MLL4 (a chromatin remodelling gene, chr 19).

The breakpoints and the gene activation that integration brings are one of the mechanisms of liver cancer. He also gives a prognostic number you can use directly: patients with 3 or more HBV integration sites survive significantly worse (N_HBV < 3, n = 51; N_HBV ≥ 3, n = 32; P = 0.037), and the number of integration sites is inversely related to survival. The hard-to-clear nuclear store is the reason HBV is difficult to treat: even if you clear the virus in the blood, the reservoir in the nucleus stays. So the number of integration sites is both a record of what the virus has done and a forecast of how the patient will do.

Why this course brings it up: The rule "3 or more HBV integration sites → significantly worse survival" sits in his table of precise numbers, between third-level (B) and second-level (A). It is a concept to understand plus a number. He never names HBV as a standalone exam topic, so it is the concept plus the number that matters. **Appears in:** Ch 6, Lect 2

▪ HDAC inhibitor

In one sentence: A drug that blocks histone deacetylase, so acetylation piles up and the chromatin opens.

Detailed explanation: The job of HDAC is to take the acetyl group off a histone. The acetyl group carries a charge, and adding it makes neighbouring groups repel each other, which loosens the nucleosome and lets the DNA sit a little more openly.

So HDAC is the enzyme that tightens the chromatin back up, and an HDAC inhibitor blocks that action. The result is that acetylation accumulates, the chromatin loosens, and the genes are switched on again; his example is making a silenced tumour suppressor come back to life, because a tumour suppressor that has been silenced is a brake that has been welded on rather than removed. Notice that the drug never touches the DNA itself; it changes the packaging and the DNA follows. He also points out that HDAC does not only act on histones: p53, HSP90, NF-κB and tubulin are all its non-histone substrates, so the drug is doing more than loosening chromatin. The more interesting application is this: an HDAC inhibitor can induce tumour cells to express PD-L1, creating "gene addiction", and that turns what was a cold tumour into something checkpoint blockade can deal with. The inhibitor does not kill the tumour by itself; it makes the tumour visible to an antibody, which is the whole point of the combination.

Why this course brings it up: The HDAC inhibitor is one of the question formats he lists, "promoter hypermethylation to HDAC inhibitor". His own words are that a cold tumour can be reworked into a hot one by an HDAC inhibitor, and that is second-level (A). The thing to notice is that it is a drug for the cold to hot conversion, not a direct immunotherapy, so if the question asks you what to give a cold tumour you should reach for the HDAC inhibitor first and the checkpoint antibody second, in that order.

Appears in: Ch 4, Lect 3, Lect 5

▪ Heavy chain

In one sentence: The two longer chains of an antibody, which form the stem and part of the arms.

Detailed explanation: An antibody is built from two heavy chains and two light chains, and the four of them are held together by disulfide bridges into a Y. The variable region of the heavy chain, VH, and the variable region of the light chain, VL, together make the three gripping loops CDR1, CDR2 and CDR3.

The heavy chain is also cut up into domains, CH1, CH2, CH3 and the hinge: CH1 together with a light chain forms one Fab, and CH2 with CH3 forms the Fc. The source of antibody diversity sits in the heavy chain too. V(D)J recombination first joins V, D and J segments together, and for the heavy chain that is V 39, D 27 and J 6, while the light chain is V 36 and J 5. That is where the arithmetic of antibody diversity comes from, and the light chain does the same trick with fewer pieces because it has no D

segment to play with. The hinge region lies between CH1 and CH2, and it is the hinge that lets the two Fab arms open to different angles. Both chains are needed for the antibody to work: the heavy chain supplies the constant scaffold, the Fc and the hinge, while the light chain together with VH supplies the grip. The numbers above are not a random assortment, because a heavy chain is assembled from three kinds of segment and a light chain from only two, and the heavy chain is the one with the D segment in the middle.
Why this course brings it up: V ≈ 39, D = 27 and J = 6 are precise numbers he lists, so they are third-level, grade B, and they lead straight into the SCID question he names, because V(D)J recombination borrows the machinery of NHEJ. Second level, grade A, plus third level, grade B. **Appears in:** Ch 2, Ch 6, Lect 1

▪ Helicase

In one sentence: The enzyme that unwinds the two DNA strands so that they can be copied.
Detailed explanation: Helicase uses the energy of ATP to break the hydrogen bonds between the two DNA strands and separate the double helix into two template strands, so that polymerase has something to copy.
Its mechanical action is to walk along the DNA while peeling the helix apart, and it needs single-strand binding protein to hold the separated strands apart so that they do not tangle back together. Another enzyme has to work at the same time: topoisomerase. When the helix is unwound it generates torsional strain, and topoisomerase cuts and re-joins the DNA ahead of the replication fork to remove that strain. On his Lecture 4 slide, the supporting roles of SSB, topoisomerase and helicase are all on one picture. Think of helicase as the person walking ahead pulling a zipper open, and topoisomerase as the person behind untangling the twist that the zipper creates. None of it would work without the coordination, which is why the enzymes are taught as a chain rather than as separate items. Every one of these enzymes solves a problem created by the one before it.
Why this course brings it up: It belongs to the concepts you are expected to understand. He does not ask you to memorise a list of enzymes, but which enzyme does which step is the most basic set of pairings in Lecture 4: unwind (helicase), relieve strain (topoisomerase), primer (primase), extend and proofread (polymerase), remove primer (exonuclease), seal (ligase). So if you remember only the pairing list from this lecture, you will still be able to place every enzyme he mentions. And it is the enzyme list you should be able to reconstruct from the problem each one solves. **Appears in:** Ch 5, Lect 4

▪ Helper T cell

In one sentence: A T cell that releases cytokines, and in doing so tells the rest of the immune system what it ought to be doing next.
Detailed explanation: A helper T cell (a CD4+ T cell) is the commander of the immune system. Having recognised antigen, it activates into two states: one secretes cytokines, recruiting and activating other immune cells, and the other stays in the lymphoid organs and helps B cells carry out V(D)J recombination and differentiate into plasma cells.
In his CAR-T manufacturing process, the anti-CD3/CD28 antibody-coated beads together with IL-2 and IL-15 are what activate and expand the T cells outside the body, and that IL-2 is exactly the cytokine helper T cells secrete. The logic of immune checkpoints depends on the same division of labour: the PD-L1 on a tumour cell and the PD-1 on the T cell shake hands, and both the commander and its killers are switched off together. Notice the two jobs again, and notice that they are different jobs. One state talks to other cells through soluble signals; the other state helps B cells inside the lymphoid organ. Which is why the helper T cell is a good candidate for the cytokine story and for the antibody story at the same time. And in the CAR-T process the IL-2 in the culture is doing exactly what a helper T cell would have done in the body. So the helper T cell is the piece that connects the cytokine biology to the cell therapy and to the checkpoint story. And because the commander is silenced along with the killers, a tumour that can talk to the commander can stall the whole response rather than one soldier.
Why this course brings it up: Conceptual understanding. The flow cytometry example he gives is actually a separation of CD3+CD4+ helper T cells out of PBMC, so the name will appear on the figure he uses, and it is inside his second-level (A) range. It also appears in the PD-1 and PD-L1 story, which is where it matters most clinically. And in the CAR-T manufacturing section. **Appears in:** Ch 6,Ch 9,Lect 2

▪ Hepatocellular carcinoma (HCC)

In one sentence: The primary cancer of the liver.
Detailed explanation: HCC is a cancer that starts from a hepatocyte, the liver's own working cell, and his whole HCC story is the course's demonstration of how basic biology turns into clinical application.
The first case is vimentin, and it is worth walking along the pipeline slowly, because the pipeline is the lesson. You separate the proteins in a sample by 2D-PAGE, which sorts them first by charge and then by molecular weight. You identify the spot you are interested in by LC-MS/MS, which cuts the protein into pieces and reads them against a database. You then validate with Western blot and qPCR, two independent methods that both have to agree. And finally you build an ELISA so the marker can be read out of a blood sample rather than a tissue biopsy. The numbers he quotes are control 176.0 ± 61.3 ng/mL, HCC ≤ 2 cm 271.5 ± 165.9, HCC > 2 cm 231.3 ± 132.3 ng/mL. The second case is circulating miRNA, where miR-15b plus miR-130b gives an AUC of 0.981, sensitivity 98.25%, specificity 91.53% and accuracy 94.83%. The third case is HBV integration and telomere length used as prognostic markers. Three different kinds of molecule, three different kinds of clinical question, one workflow.
Why this course brings it up: This is the stage on which he demonstrates the whole biomarker development workflow, from separating the proteins to shipping a number out of a blood tube. The AUC for miRNA and the vimentin numbers are both on his list of precise numbers, third-level (B), while the workflow logic is second-level (A). The HCC material also keeps coming back later, in Chapter 5, in Chapter 11 and in both Lectures 1 and 2, so the same case is worth knowing from both directions: the

molecule and the workflow. **Appears in:** Ch 5, Ch 11, Lect 1, Lect 2

▪ Hepatocyte

In one sentence: The liver cell, and the main working cell of the liver.
Detailed explanation: The hepatocyte is the main functional cell of the liver: it metabolises, it makes albumin, it detoxifies and it makes bile. It has three roles in this course. First, it is the star of the Warburg effect.
His carbohydrate metabolism slide draws glycogen → glucose-1-P → glucose-6-P, and the last step is done by glucose-6-phosphatase, an enzyme the liver and kidney have and nothing else does, which puts free glucose back into the blood. Lose that enzyme and you get an energy crisis. Second, the hepatocyte is the cell that HBV can infect, and once HBV DNA integrates into nearby genes the hepatocyte starts on the road to cancer. Third, it is where telomeres get used up. Chronic injury keeps turning hepatocytes over, the telomeres get shorter and shorter, and then MYC switches telomerase back on and hepatocellular carcinoma appears. The hepatocyte is also the cell that decides when the rest of the body gets fed, because the liver and kidney are the only tissues that can release free glucose into the blood at all. In the first story it is the star of the Warburg effect, in the second it is the cell that HBV can infect, and in the third it is where the telomeres are consumed. Three different subjects, one cell type, and each of his HCC stories uses a different one of those roles. The chronic injury in the third story is what keeps the turnover going, and every round of division takes a little off the ends of the chromosomes.
Why this course brings it up: Conceptual understanding. All three of his HCC stories use the hepatocyte as a character, so you have to be able to say the chain out loud: chronic injury, then hepatocyte ageing and telomere shortening, then reactivation of telomerase, then HCC. **Appears in:** Ch 5, Lect 1

▪ HER1, HER3 and HER4

In one sentence: The other members of the HER receptor family, which pair with HER2.
Detailed explanation: The HER family has four members: HER1 (EGFR), HER2, HER3 and HER4. The key structural fact is this: HER2 is very poor at pairing with itself (homodimerisation), and it mainly forms heterodimers with HER1, HER3 or HER4 — and dimerisation is exactly how the downstream signalling is switched on.
That has a direct therapeutic consequence: HER2's antibody (trastuzumab) works precisely because it blocks HER2 from dimerising with HER3; his Lecture 2 slide also records that the downstream is PI3K (with two subunits, p85 and p110). Going the other way, HER1 (EGFR) and HER3 sit upstream of the K-Ras signalling chain, which is why EGFR inhibitors so often fail in K-Ras mutant tumours. This is the same dimer-versus-monomer contrast he flags elsewhere in the course, and the pairing partner is what decides which downstream pathway gets switched on. So the answer to why two receptors in the same family behave so differently is often simply which partner they end up with. Receptors in the same family are not interchangeable, and the difference is written in the pairing.
Why this course brings it up: It is second-level (A). Dimerisation versus monomer is one of his distractor pairs (a monomer binds DNA weakly, a dimer binds strongly), and "HER2 cannot efficiently homodimerise" is that contrast applied to receptors, which makes it examinable. And the drug consequence is the part worth remembering, because it explains a lot of resistance. So know which member pairs with which, and know that HER2 is the odd one out. **Appears in:** Ch 7, Ch 10, Lect 5

▪ HER2

In one sentence: A receptor whose gene gets amplified in certain breast and gastric cancers, and the receptor that trastuzumab is designed to block.
Detailed explanation: HER2 is a receptor tyrosine kinase of the EGFR family. Its disease mechanism is not a mutation but amplification: gene copy number goes up → mRNA goes up → the number of receptors on the cell surface goes up → downstream growth signalling is amplified, and the extracellular domain of the receptor is also shed into the blood.
His Lecture 2 slide draws those four steps as a staircase. Therapeutically, trastuzumab (Herceptin) blocks HER2–HER3 dimerisation, and at the same time its Fc is recognised by FcγRIII on an NK cell and sets off ADCC, while trastuzumab deruxtecan adds a topoisomerase I payload on top. Before you treat, you have to know whether the patient is HER2 positive, which is what a companion diagnostic is for: IHC or FISH, with the read-out being the HER2/CEP17 ratio (the centromere probe of chromosome 17) of ≥ 2.0, or a copy number of ≥ 6. Notice that every step of that staircase is an increase. Amplification is not a broken switch here, it is a volume knob turned up too far. And notice that one drug is doing two jobs at once, which is exactly what makes an IgG1 valuable in this setting. The reason the companion diagnostic is not optional is that the treatment and the biomarker are decided together: no measurement, no treatment. The ratio and the copy number are the two ways the same question gets asked, and either one being met is enough. And the receptor occupancy numbers are the last piece. A general antibody occupies 1 % of the receptors, Herceptin occupies 2 % in breast cancer and 10 % in gastric cancer, which is why the same drug needs a different dose in a different tissue. The HER2 story is also the cleanest example in the course of a genetic change in the tumour creating a target that the drug can aim at. That is the therapeutic logic in one sentence: find something the tumour has too much of, then make a drug that binds exactly that. And the same logic is why an ADC exists at all, since the amplification gives you both a target and a reason to believe the target matters.
Why this course brings it up: This is examinable content, and it pulls together three of the contrasts and number sets he lists: HER2 amplification and its treatment, the split of work between DxD and LBD, and receptor occupancy (1 % for a general antibody, 2 % for Herceptin in breast cancer and 10 % in gastric cancer, so the same drug needs a different dose in different tissue). Hold on to those occupancy percentages, because they are the kind of detail that turns up inside a case. And be ready to draw the chain: the amplification staircase is exactly that. It is a good diagram to have in your hand. **Appears in:** Ch 7,Ch

10,Lect 2,Lect 4,Lect 5

▪ Heterochromatin

In one sentence: Tightly packed, silent chromatin. It physically stops transcription factors from reaching the promoter.
Detailed explanation: Heterochromatin is tightly packaged chromatin, and the DNA inside it is not only hard to read, it is physically walled off. Dr. Wong's description is precise here, and the precision matters: it is not "less efficient", it is physical blocking, because the transcription factor and the enhancer simply cannot get near the promoter, so the gene is silent.
Not "the copying is slower", but "the machinery cannot get to the work". The state can be flipped, because nucleosomes sliding along DNA turn a region from tight to loose, and there are two forces that push the sliding. The first is the intrinsic affinity between the DNA sequence and the histone, the natural grip each has for the other. The second is the ATP-driven chromatin-remodelling complex, which is an active machine that shifts position deliberately. And because the binding between histone and DNA is non-covalent, hydrogen bonds plus electrostatic forces, none of this has to be rebuilt from scratch; the same bricks are just moved, and can be moved back again.
Why this course brings it up: euchromatin versus heterochromatin is one of his distractors, second-level (A), and he uses the precise phrase "physical blocking", so do not say "low efficiency". The distinction he wants you to hold is not open versus closed but accessible versus walled off, and the nucleosome sliding model is how you explain how a walled-off region can be opened.
Appears in: Ch 3, Lect 3

▪ Heterodimer

In one sentence: Two different proteins joined together, such as HER2 with HER3.
Detailed explanation: A heterodimer is built from two different subunits, as against a homodimer, which is two of the same thing. What matters is the function. When two subunits sit side by side, one can detect the signal and the other can do the work, so the same protein can be regulated by choosing which partners it pairs with.
A dimer of two identical subunits is a homodimer, and the difference between the two is not academic, because it decides whether the two halves of the protein can do different jobs from each other. Dimerisation of receptors and of transcription factors works this way. For nuclear receptors the monomer binds DNA badly and dimerisation raises the affinity dramatically, so dimerisation is itself the switch. For HER2 the heterodimer is almost the only way it works normally. Pairing with itself is inefficient, and it mostly starts its signalling by pairing with HER1, HER3 or HER4, and trastuzumab blocks exactly that pairing.
Why this course brings it up: Dr. Wong lists "dimerisation versus monomer" as a distractor pair, so it is second level, grade A. At the very least you should be able to use it to explain two things: why a receptor has to pair up, and why Herceptin works.
Appears in: Ch 7, Lect 5

▪ Heteroplasmy

In one sentence: The state in which normal and mutant mitochondrial DNA are mixed together inside one cell.
Detailed explanation: Every mitochondrion carries its own copy of DNA, so a single cell can hold both normal and mutant mtDNA at the same time; that mixed state is called heteroplasmy, while all-normal or all-mutant is homoplasmy.
The clinically crucial point is the threshold effect: carrying mutant mtDNA does not mean you will certainly be ill, because symptoms only appear once the mutant fraction passes the tolerance limit of a particular tissue. Tissues with high energy demand (skeletal muscle) have a low threshold and are affected first; low-demand tissues (hearing) have a high threshold and are affected later. He also explains why siblings can differ in severity: mitochondria are distributed randomly during cytokinesis. How much mutant DNA is in the cell, not whether there is any, is what decides the outcome.
Why this course brings it up: In his first-level (A+) questions the threshold effect is the turning point of the maternal inheritance question: skeletal muscle has a low threshold, hearing a high one. You have to be able to lay out the chain of proportion → over threshold → disease. It is one of the few places in the course where a number is less important than the shape of the curve.
Appears in: Ch 8, Lect 2

▪ Heterozygous

In one sentence: Two different versions of the same gene, one from each parent.
Detailed explanation: A pair of homologous chromosomes carries two copies of the same locus, which is a position on the chromosome where a particular gene sits. If the two copies are different versions, different alleles, then you are heterozygous; if they are the same, you are homozygous. The distinction has real clinical consequences. For a recessive disease a heterozygous carrier usually does not get ill, because one working copy is enough, but does pass the mutation on, so carriers have to be marked on a pedigree and the risk to a child depends on which partner they meet. For a dominant disease one heterozygous carrier may well get ill, because one faulty copy is enough to cause it. Humans have 23 pairs of chromosomes, so apart from the X and Y chromosomes, every locus has one copy from the father and one from the mother, and that is why the family tree is the first place to look when you are asked about a pattern of inheritance.
Why this course brings it up: The pedigree is one of the two case question types he names, Case 2: a genetic counselling report. Reading a pedigree means marking genotype and phenotype at the same time, so you have to use heterozygous and homozygous accurately and to say which of the two you are looking at. It falls within second-level (A) scope. **Appears in:** Ch 3, Lect 3

▪ HIF-1α

In one sentence: An oxygen sensor protein. When oxygen is normal it is made constantly and broken down just as constantly.

Detailed explanation: HIF-1α is the cell's oxygen meter. It is actually being made all the time, but at normal oxygen levels the two enzymes PHD and FIH hydroxylate its proline and asparagine residues, and once the hydroxyl groups are on, the VHL protein recognises it, tags it with polyubiquitin, and sends it to the proteasome to be destroyed.

When oxygen is scarce the hydroxylation step stops, HIF-1α survives, enters the nucleus, binds HIF-1β, recruits p300/CBP, binds the HRE, and switches on the group of genes including VEGF, GLUT1 and EPO. The heart of the mechanism is that oxygen is not detected directly. It is translated, through the activity of oxygen-dependent enzymes, into the stability of a protein. So the cell does not have an oxygen sensor. It has an oxygen-dependent chemical reaction, and the stability of the protein is the readout. Notice how indirect that is, and also how cheap it is: no special sensor molecule, just an enzyme that stops working. The three genes that come on at the end are all about dealing with a shortage of oxygen by getting more of it, or doing without it. And the logic of the circuit is a nice one: you do not need to know how much oxygen there is, you only need to know whether there is any.

Why this course brings it up: Conceptual understanding, but HIF-1α is one of the most important applications in Lecture 5: putting CAR expression under the control of an HRE, so that the T cell only arms itself in the hypoxic core of a tumour. He gives the figure that around 50 % of solid tumours contain hypoxic tissue. So the oxygen percentage and the HIF-1α behaviour are the two halves of the same answer. **Appears in:** Ch 7,Lect 5

▪ HIF-1β

In one sentence: The half that binds HIF-1α in the nucleus and makes up the working hypoxia factor.

Detailed explanation: HIF-1β, also called ARNT, is the fixed partner of HIF-1. It is far more stable than HIF-1α and is there even at normal oxygen levels; the half that oxygen actually regulates is the HIF-1α half.

The division of labour in this sensor is therefore very clean. HIF-1β supplies a constant scaffold, HIF-1α supplies a switch that rises and falls with the oxygen, and only the two together form the complete factor that can bind HRE and start transcription. The stable-and-moving design also explains why one oxygen signal can have several outputs, because HIF-1α can pair with other bHLH partners as well and form different factors such as HIF-2. Nothing about the oxygen reading is lost in the process, it is simply carried by a different half of the molecule.

Why this course brings it up: Conceptual understanding. You have to be able to say which half is under oxygen control, because that is where the mechanism behind the sentence "HIF-1α is degraded in normoxia and stabilised in hypoxia" comes from, and it is the two slides on degradation and assembly on his deck. **Appears in:** Ch 7, Lect 5

▪ HIF-CAR

In one sentence: A CAR whose expression is controlled by an HRE, so the T cell only arms itself inside a hypoxic tumour.

Detailed explanation: HIF-CAR is an application of synthetic biology: put the expression of the CAR gene under the control of an HRE (a hypoxia-responsive element), and the CAR protein is then only made in cells where HIF-1 is active.

Since HIF-1 is only active when oxygen is low, this version of CAR-T is switched on only in the hypoxic core of a tumour, and it steps around the oxygenated normal tissue around it. His slide shows green CAR expression curves for HIF-CAR1 and HIF-CAR2 that appear only under hypoxia; once normal oxygen levels return, the HIF-CAR signal falls below 0.2 within 2 to 3 hours, while the constitutive CAR stays between 0.8 and 1.0. It is a single-gate design that uses a molecular switch to confine a drug to the lesion. And it is reversible, which is what makes it a switch rather than a one-way decision. One gate, and the drug only works where the gate is open.

Why this course brings it up: It belongs to the concepts you are expected to understand — but it is the model answer for scenario case 14, "a compound that is only switched off in the correct cell", so you have to be able to state the causal chain between HRE, HIF-1α and hypoxia. The design principle, not the drug, is what the case is testing. **Appears in:** Ch 7, Lect 5

▪ High specific heat

In one sentence: The energy a substance needs to go up by one degree. Water has a high specific heat, so body temperature stays steady.

Detailed explanation: Specific heat is the number of calories needed to raise 1 g of a substance by 1 °C. Water's specific heat is unusually high, about 1 cal/g·°C, which means water has to absorb a lot of energy before it warms up noticeably, so your body temperature barely moves when you make a lot of heat or lose a lot;

dry tissue such as the protein in muscle has a much lower specific heat. Water's high specific heat also makes it a good heat carrier, with blood taking heat from the liver and the muscle out to the skin to be lost. The same principle at the molecular level explains how water can buffer pH, because its dissociation provides H+ and OH-.

Why this course brings it up: It is background, part of the chemistry he excludes. But "water's high specific heat keeps body temperature stable" is a structural reason the beginner's guide lists explicitly, and you have to be able to use it to explain stability and heat loss in biology. **Appears in:** Ch 1

▪ Hinge region

In one sentence: The flexible connecting piece between the DNA-binding domain and the ligand-binding domain of a nuclear receptor, the part that lets the two talk to each other.

Detailed explanation: On the structure diagram of a nuclear receptor, there is a flexible hinge between the DBD and the LBD (on his domain map it is region D). It behaves like a joint that can swivel, so that the shapes at the two ends can influence each other: when the LBD binds a ligand and changes shape, that change travels through the hinge to the DBD and affects how well it binds DNA and how well it recruits a co-activator.

This idea, that a connecting piece couples two functional modules into one machine, is everywhere in molecular biology: receptor and kinase, enzyme and regulatory subunit, the upstream and downstream of a signalling pathway. They are all hinged. So the hinge is not padding between two important parts. It is the part that decides whether the two parts can talk to each other at all. Which is why a drug that binds the ligand and changes the shape of the LBD can end up changing the DNA binding of a domain it never touches. And that is the same trick that the HER2 and ER examples are playing at a larger scale. Everything the receptor does is a two-handed operation, and the hinge is where the two hands meet.

Why this course brings it up: Conceptual understanding. It sits in the same piece of material as the DBD/LBD distractor pair, and the hinge is the concrete example of how a change of shape is carried across a module. So if you can explain the hinge, you can explain the rest of that section. **Appears in:** Ch 9,Lect 5

▪ Hippo pathway

In one sentence: The pathway that uses contact between cells to switch growth off. MST1 activates LATS1, and LATS1 phosphorylates YAP.

Detailed explanation: When a cell touches its neighbour and forms a junctional complex, a chain of proteins is recruited and phosphorylated. MST1/2, paired with SAV, is phosphorylated first, and that activates LATS1/2, and LATS then phosphorylates YAP.

The phosphorylated YAP binds 14-3-3, stays in the cytoplasm and is degraded, so YAP cannot enter the nucleus, TEAD cannot be activated, and the genes for proliferation and resistance to apoptosis are not switched on. The reverse also holds. GPCR signalling or actin stabilisation can disable LATS, and the unphosphorylated YAP goes into the nucleus, binds TEAD and opens CTGF, JAG1, which is a Notch ligand, NOTCH2 and other genes, and the result is proliferation, resistance to apoptosis, stemness and EMT. The whole pathway is therefore a contact sensor written as a phosphorylation cascade: touch your neighbour and the cascade runs and growth stops, lose the cascade and the cell grows whether or not anyone is touching it. Notice also that the two outputs are mirror images. Phosphorylated YAP sits in the cytoplasm and the growth genes stay shut; unphosphorylated YAP sits in the nucleus with TEAD and the growth genes open.

Why this course brings it up: Conceptual understanding plus second level, grade A. Dr. Wong says Hippo is a pharmacological target with druggable inhibitors in development, but he explicitly excludes the details of the signalling cascade. What you have to be able to recite is the three-step MST1 → LATS1 → YAP chain and the rule that phosphorylation means off while no phosphorylation means on. **Appears in:** Ch 1, Ch 7, Lect 1, Lect 5

▪ Histone

In one sentence: The protein that DNA is wound around to form a nucleosome.

Detailed explanation: Histones are proteins, and because they are rich in positively charged lysine and arginine they can bind negatively charged DNA. The core of a nucleosome is eight histones: two copies each of H2A, H2B, H3 and H4, with about 147 base pairs of DNA wrapped around them, the whole spool about 10 nm across, and the H1 linker histone clamping the point where the DNA enters and leaves.

Histones are highly conserved across species (almost identical from fruit fly to human), and that is exactly why model organisms can be used; and because they carry so much positive charge they are very hard to crystallise, so getting a 2.8 Å nucleosome crystal structure was a real achievement. The histone tails stick out, and they are the building sites for all the chemical marks. So a histone is at once a structural spool, a chemical surface and an evolutionary fossil, and all three uses are in this lecture. And the marks that do the regulating are all written on the tails that stick out.

Why this course brings it up: It is second-level (A). The 147 bp, 10 nm, 2.8 Å and "< 2.0 Å = very excellent" figures are all in his table of precise numbers. He also warned that you do not need to memorise the electron microscope images of chromatin. The numbers he wants you to have are the ones on the list, not the ones in your head from elsewhere. **Appears in:** Ch 4, Lect 3

▪ Histone code

In one sentence: The system of histone marks that tells a cell whether a stretch of DNA is open or shut.

Detailed explanation: The histone code means that on the same DNA sequence, different combinations of histone modification lead to different expression outcomes. The notation is simple. H3 stands for histone 3, K stands for lysine, the number is the position counted from the N terminus, me is methylation, and the number after it is how many methyl groups are there.

So H3K4me3 means histone 3, lysine 4, three methyl groups. The combinations multiply each other and also contradict each other. H3K4me3 marks an active promoter and blocks DNMT from binding CpG, which keeps the DNA in a demethylated, permissive state. H3K9me3 works with Suv39h and HP1 to spread the silence outward. The key point is that this system cross-talks with DNA methylation, reinforcing each other in a positive feedback loop, so the cell still remembers its state after it divides. That is cellular memory: the marks are copied onto the new DNA, and the new DNA then re-establishes the marks, so the pattern survives generation after generation.

Why this course brings it up: It is second-level (A). The notation and the cross-talk are the subject of his Lecture 3, and they connect directly to the clinical applications, the HDAC inhibitor and MGMT methylation. The other thing to take from it is that a mark is a switch with several settings, not an on and off button, and reading H3K9me3 out loud correctly is part of the job.
Appears in: Ch 4, Lect 3

▪ Histone modification

In one sentence: A chemical label on a histone, one that opens or closes the chromatin sitting around it, and that decides whether the gene underneath can be read.
Detailed explanation: The modifications happen mainly on the tails of the histones, and the four commonest are acetylation, methylation, ubiquitination and phosphorylation. The direction is fairly intuitive: an acetyl group carries a positive charge, adding it repels the negative charge of the DNA and pushes the nucleosome apart, so the chromatin opens up, while removing the acetyl group (the job of HDAC) tightens it again.
Methylation is more subtle and its effect depends on the position: H3K4me3 leans towards open, H3K9me3 leans towards closed. The modifications also influence each other, so ubiquitination of K123 on H2B drives H3K79me and H3K4me. So the simplest way to picture the whole system is as labels on the outside of a spool, each one pulling the DNA in or letting it out. And once you accept that the labels interact, the chromatin stops being a fixed structure and starts being a state that the cell keeps adjusting. That is also the point of the cross-talk he describes: the marks do not act alone, they act as a set. So the histone code is not a lookup table where one mark means one thing. It is more like a set of knobs that are wired to each other. And because the marks are editable, they are also something a cell can respond to rather than something fixed at the moment of birth.
Why this course brings it up: Second-level (A). The clinical application he can give is that the histone modification pattern itself can act as a prognostic marker (the KM curve in glioma, for example), and the HDAC inhibitor class of drugs. So the drugs and the marker are the same idea seen from two ends: change the marks, and you change the prognosis. Which is a good thing to be able to say in a short answer. **Appears in:** Ch 4,Lect 3,Lect 5

▪ Histone variants (H2A, H2B, H3 and H4)

In one sentence: The four proteins that make up the core of the nucleosome.
Detailed explanation: The octamer core of a nucleosome is built from two copies each of H2A, H2B, H3 and H4, and when you add the linker histone H1 that sits where the DNA enters and leaves, there are five kinds of histone in total.
The linker histone is the odd one out, because it sits at the entry and exit points of the DNA rather than on the core, and it is the part that has to let the DNA in and out. The core is about 5.7 nm high and 11 nm wide, with 147 base pairs of DNA wrapped around the outside. Besides the standard versions, cells also carry variant versions, such as macroH2A, H2A.Z and H3.3. They get placed at particular positions in the chromatin and they change how loose or how readable the neighbourhood is, which is the same as saying that they put a different label on the nucleosome. The variants of H3 matter most, because the positions of marks such as H3K4me3 and H3K9me3 are defined by amino acid numbering on H3. The variants are used in particular places rather than everywhere, and the H3 variants matter more than the others, because the marks are defined by a numbered position on H3, and that is exactly the numbering a variant of H3 would change.
Why this course brings it up: Conceptual understanding. The four core histones and the linker H1 are the structure from his Lecture 3 slides, and the numbers to hold on to are the octamer, the 147 bp and roughly 10 nm. **Appears in:** Ch 4, Lect 3

▪ HNH

In one sentence: One of the two nuclease domains of Cas9, and it cuts the target strand.
Detailed explanation: Cas9 is a pair of molecular scissors with an address label on them: the gRNA carries the sequence and goes looking for the target DNA near a PAM, and once it has found it the two nuclease domains make the cut, HNH and RuvC, and together they produce a double-strand break.
How the cell repairs that break is then what decides whether you end up with a knock-out or a knock-in. Each of the two nuclease domains cuts one strand, and together they make one clean double-strand break; if only one of them is working the cell is probably only nicked on one strand, and it will simply patch that straight away. And which strand each domain cuts is why the two names are worth keeping straight, even if you are not asked to recall them.
Why this course brings it up: Dr. Wong writes the names of the two domains, RuvC and HNH, on his slide, but he says in the same chapter that you should not memorise these names. What really gets examined is the fork after the cut: NHEJ goes to knock-out and HR goes to knock-in, and that fork is the same in both of the lectures in which he teaches it. **Appears in:** Ch 6, Lect 2, Lect 4

▪ Homodimer

In one sentence: Two identical proteins joined together. A dimer binds DNA far more strongly than a single protein.
Detailed explanation: A homodimer is a dimer made of two identical subunits, and it is the opposite of a heterodimer. It is extremely common in receptors and transcription factors: the DBD of a nuclear receptor usually binds a response element as a dimer, while a monomer binds DNA poorly, and dimerisation raises the affinity sharply — so "dimerisation" is itself a regulatory switch.
Cells also run mixtures of homodimer and heterodimer; HER2, for example, pairs with itself poorly and mainly pairs with HER1, HER3 or HER4, and that difference in composition is why every member of the same receptor family behaves differently. Which means that whether a receptor is switched on can depend on which partner happens to be present in the cell. So the

dimer-versus-monomer question is really a question about how much binding strength a receptor can bring to a DNA site, and therefore about how easily it can be switched on and off.
Why this course brings it up: Dimerisation versus monomer is one of his distractor pairs, a second-level (A) item. The test is one sentence: monomer weak, dimer strong, and that changes the activity of the receptor or transcription factor and decides which state a drug should bind. **Appears in:** Ch 7, Lect 5

▪ Homoeologous recombination

In one sentence: Recombination between similar but not identical chromosomes, the mechanism that makes new combinations in polyploid crops.
Detailed explanation: In a polyploid plant such as wheat, the sets of homologous chromosomes are not exactly the same as each other, so chromosomes that are only partly homologous can still exchange segments with one another.
That exchange is a crossover, and the result is that the offspring come out with new combinations of genes, and a breeder can pick out the traits they want from among them. That is the whole point of homoeologous recombination in crop breeding: it gives you new combinations out of a genome that already has everything it needs. It is different from the homologous recombination you meet in human disease genetics, where human homologous recombination happens between truly homologous chromosomes during meiosis, and the crossing over it produces is one of the sources of diversity. And the CNV you see in tumours comes from NAHR, non-allelic homologous recombination, where the exchange happens unequally at a repeated sequence and leaves one copy with extra and another with less. Different partners, different mechanism, different outcome, and all three get filed under the same broad word, recombination.
Why this course brings it up: This term is not in the content of his five lectures, so it is background knowledge: he neither names it nor excludes it. What you have to do is avoid confusing it with NAHR and with human homologous recombination. Homoeologous is partly homologous, homologous is completely homologous, and NAHR is the unequal exchange at a repeat. The three names are close and the mechanisms are not, so if a question gives you a mechanism, read off which one it is. The names being close is exactly why this one is worth separating cleanly. **Appears in:** Background supplement (not lecture content)

▪ Homologous chromosome

In one sentence: One of the two matching chromosomes in a pair, one from each parent, and the ones that separate at the first meiotic division.
Detailed explanation: A human somatic cell has 23 pairs of chromosomes. In each pair, one chromosome comes from the father and one comes from the mother. The two are broadly similar in shape and in which gene loci they carry, but the alleles sitting on them may be different, and that matched pair is what we call a homologous chromosome.
The important bit is when they come apart. Homologous chromosomes separate at meiosis I, and that is exactly the difference between meiosis I and meiosis II: the first division pulls the homologues apart, the second pulls the sister chromatids apart. And because the homologues are the thing that separates at the first division, an abnormal chromosome number (aneuploidy) shows up as a whole extra set or a whole missing set, rather than as one odd chromosome standing on its own. That is why the error is visible as a picture of the whole complement rather than as a detail inside it. After meiosis I the homologues are in two different cells, so meiosis II has no homologues left to pull apart, only the sister chromatids still joined at the centromere. So the pairing is not bookkeeping. It is the reason the arithmetic of inheritance comes out in neat halves.
Why this course brings it up: It is second-level (A). Dr. Wong's M1/M2 discrimination question rests entirely on that one sentence: are the things separating homologues or sister chromatids. And 23 pairs of chromosomes is one of the precise numbers he lists, so the count is worth having in your head as well as the pairing. So both halves of this entry are examinable, and the pairing is the half that gives you the answer. **Appears in:** Ch 3, Ch 8, Lect 3

▪ Homologous recombination (HR)

In one sentence: High-fidelity repair of a broken chromosome that uses the matching sister chromatid as a template. It only works in S and G2.
Detailed explanation: After a double-strand break, the cell first uses the MRN complex, Rad50, Mre11 and Nbs1, to trim the broken ends into 3′ overhangs. RPA then protects the single strand, Rad51 assembles into a filament and invades the homologous duplex, branch migration copies the information back, and finally a polymerase fills in the strand and a ligase seals it.
Because there is a template, the result is almost never wrong. The limitation is timing: only S and G2 have sister chromatids to work from, and after that the cell has to fall back on NHEJ. The physiological use is the crossing over of meiosis. The clinical use is the HR defect caused by BRCA1/2 mutation, and that is exactly the entry point for PARP inhibitor treatment. Note the order of the actors: MRN trims the broken end, RPA holds the single strand steady, Rad51 builds the filament, the branch migration brings the information back, and then a polymerase and a ligase finish the job. The clinical consequence sits at the other end of the same pathway, because a BRCA1/2 mutant cell has lost the template, and PARP inhibitors exploit that hole.
Why this course brings it up: Compulsory material. Dr. Wong lists "HR versus NHEJ" as the single most likely multiple-choice contrast on his list, and the discriminating points are precisely the template, the accuracy and the cell cycle phase. The synthetic lethality of BRCA with PARP inhibitors is one of the most important clinical stories of Lecture 4. **Appears in:** Ch 6, Lect 4

▪ homologous template

In one sentence: The stretch of DNA that HR copies from.
Detailed explanation: When HR repairs a double-strand break, the cell has to find a stretch of DNA that is almost exactly the same and copy from it. Under normal circumstances that template is the sister chromatid, the exact duplicate of the same chromosome that the cell made just before dividing.
So HR only happens in S and G2, the periods in which a sister chromatid is actually available to use. That is not a preference, it is a physical constraint: without a matching copy nearby there is nothing to copy the missing piece from.
Why this course brings it up: “HR needs a template, only works in S/G2, high fidelity” is a Tier 2 pairing on Dr. Wong's list, and the three parts go together. **Appears in:** Ch 6, Lect 4

▪ Homology-directed repair

In one sentence: The precise repair route that uses a matching template. CRISPR knock-in needs exactly this.
Detailed explanation: HDR repairs a break using a homologous template to copy from — and in gene editing that template is the donor template you supply, which is how a sequence can be inserted or changed precisely.
Its precondition is that a usable homologous template exists, and since the sister chromatid only exists in S and G2, HDR can only happen in those two phases, which makes it far less efficient than NHEJ. The medical reverse application is elegant: tumours with BRCA1/2 mutations have lost the ability to do HDR, and in that situation a PARP inhibitor can create double-strand breaks that HR cannot repair, while normal cells still rescue themselves — so the therapeutic window is wide. So the choice of repair pathway is not a detail of the experiment; it is the difference between a clean insertion and a random mess. Everything about the editing experiment depends on which of the two repair routes the cell happens to take.
Why this course brings it up: It is second-level (A). It is the answer to the "choose an editing strategy" scenario question (Case 12): knock-out uses NHEJ, knock-in uses HDR, and HDR is restricted to S / G2. And it is also the answer to the PARP inhibitor question, because the drug only works where the repair is missing. **Appears in:** Ch 6, Lect 2, Lect 4

▪ Homoplasmy

In one sentence: Every copy of the mitochondrial DNA inside one cell is identical, whether they are all normal or all mutated.
Detailed explanation: Homoplasmy is the mirror image of heteroplasmy: all the copies of mtDNA in a cell are the same. Either they are all normal (healthy homoplasmy) or they are all mutated (mutant homoplasmy).
Disease mostly turns up in the heteroplasmy state in between, and it only appears once the proportion of mutant mtDNA crosses the threshold set for that tissue, which is why the same patient can be very differently affected in different tissues. Clinically you estimate the mutant proportion from a muscle biopsy or from blood, but you have to remember that the blood threshold and the muscle threshold are not the same number, so a proportion measured in blood is not simply the proportion everywhere else in the body. Which also means you cannot read one tissue and assume the rest of the patient matches it.
Why this course brings it up: It is conceptual understanding, and it belongs to the same set as heteroplasmy and the threshold effect. In Dr. Wong's first-level (A+) mitochondrial pedigree question, the nuclear gene exception and the threshold effect are the two required points, and homoplasmy is the far end of the scale that threshold effect is measured along. So if you know where the middle sits, you know why the two ends behave differently. **Appears in:** Ch 8, Lect 2

▪ Homozygous

In one sentence: Two identical versions of the same gene, one from each parent, with no hidden difference between them.
Detailed explanation: Two copies that carry the same version of the gene at the locus are homozygous, which is the mirror image of heterozygous. In genetic disease reasoning this distinction decides the risk: a recessive disease needs the mutation on both copies, a homozygous affected individual, before it appears, while a heterozygous carrier usually does not get ill but passes the mutation on.
A dominant disease needs only one. The clinical application is that when you write up someone's genotype you have to state clearly whether it is het or hom, because that is what determines the carrier risk and the probability quoted in prenatal or preimplantation diagnosis.
Why this course brings it up: The pedigree case question, Case 2, uses it. What you have to be able to say is that the same mutation means different things clinically in a het and a hom background, and that is exactly why the presence of a mutation is not the same thing as the presence of disease. That falls inside second-level (A) scope. **Appears in:** Ch 3, Lect 3

▪ Hormone

In one sentence: A chemical messenger released by a gland into the blood, acting on cells far away.
Detailed explanation: A hormone is the messenger of the endocrine system: a gland secretes it into the blood, it travels with the blood all over the body, and it acts on distant cells that carry the matching receptor.
Estrogen works exactly this way — it enters the target cell, binds the estrogen receptor in the cytoplasm or the nucleus, changes the conformation of the receptor, and the receptor then enters the nucleus and binds the ERE. What separates a hormone from an ordinary signalling molecule is distance: local signals (paracrine, autocrine) act over a short range, whereas a hormone can act a long way away, and it is effective at a very low dose — which is also why he uses "receptor occupancy" to explain dosage. So the practical question with any hormone is always how much of the receptor it manages to occupy, and at what dose. Distance is the whole definition.
Why this course brings it up: It belongs to the concepts you are expected to understand. The whole nuclear receptor section

of Lecture 5 rests on the pairing of hormone with nuclear receptor, and both tamoxifen and receptor occupancy depend on this background. Which is a nice example of a concept you already need for one topic turning out to be the key to another. **Appears in:** Ch 7, Lect 5

▪ Hot tumour

In one sentence: A tumour carrying a large load of neoantigens, and therefore a good responder to immunotherapy.
Detailed explanation: A hot tumour is one with so many neoantigens that the immune system can recognise it and attack it.
The chain of causes is very clear. MMR deficiency means indels in the microsatellites are not corrected, frameshifts pile up, and those frameshifts produce a large number of peptides the immune system has never seen, which are the MANA.
Those peptides are presented by MHC to cytotoxic T cells. At that point the tumour cells turn up PD-L1 to fight back, and an anti-PD-1 antibody cuts off that counter-attack and restores the killing power of the T cells. His KM curve of dMMR against pMMR, where the MMR-deficient group is clearly better at $P < 0.001$, is data he calls one of the most important pictures in modern medicine. The reverse also holds: a cold tumour can be turned back into a hot one with drugs such as HDAC inhibitors.
The whole story is one of supply. A hot tumour is hot because it has handed the immune system something to aim at, and the more novel peptides it presents, the better the immune system can find it and the more of its T cells can be brought onto the tumour. The dMMR and pMMR curves are the measurement of that, because a dMMR tumour carries the MMR deficiency that starts the whole chain.
Why this course brings it up: Second level, grade A, up to first grade. Dr. Wong's own words are "you definitely remember that… we call this is a hot tumor", and he says the dMMR against pMMR KM plot is the kind of figure any presentation on immunotherapy will include. **Appears in:** Ch 9, Lect 2, Lect 4, Lect 5 **Video:** V07_Hot_tumour_vs_cold_tumour.mp4

▪ HP1

In one sentence: The protein that recognises H3K9me3 and spreads the methylation outwards in both directions.
Detailed explanation: HP1 is short for heterochromatin protein 1. It carries a chromodomain and it specifically recognises the trimethylation on the ninth position of H3. Having found that mark, it works with Suv39h to push the K9 methylation along the chromatin, and the silent region grows larger and larger — which is also why chromatin needs boundary elements, fences of that kind.
Put HP1 and Suv39h together and you can see that "spreading" and "boundaries" are a pair of concepts in epigenetics: without a fence, the signal keeps spreading. So the same protein is both the reader of the mark and the thing that extends it.
Why this course brings it up: He does not ask you to memorise it, so it is a concept to understand; the mechanism he has taught is that "marks spread and boundaries are needed". That mechanism is what makes the boundary element on slide 27 make sense. **Appears in:** Ch 4, Lect 3

▪ HSP90

In one sentence: A chaperone in the cytoplasm that keeps the estrogen receptor folded until the hormone arrives.
Detailed explanation: HSP90 is a molecular chaperone, which means its job is to help other proteins fold and stay folded. In type I receptors (estrogen, glucocorticoid and the like), the receptor is held by HSP90 in the cytoplasm when there is no ligand, sitting in a “loaded but standing by” shape, like a gun with the safety on.
When the ligand arrives, HSP90 lets go, and the receptor dimerises, moves into the nucleus, recruits a co-activator and binds DNA. And because HSP90 is also the chaperone for a great many oncoproteins (kinases, mutant p53), it is itself a hot drug target. The useful way to hold this is as a three-part story: a protein that cannot fold properly is not finished, so a chaperone holds it in shape until the signal arrives; the signal arrives and the protein changes shape; the change in shape is what switches the function on. HSP90 is one of the clearest demonstrations of that general rule in the whole course.
Why this course brings it up: It is conceptual understanding. It is a key box on the type I receptor diagram in Lecture 5, and it demonstrates the general rule “drug binds → shape changes → function changes”. It falls inside Dr. Wong's second-level (A) range, so you should be able to place it in the receptor story rather than treat it as a separate protein. **Appears in:** Ch 7, Lect 5

▪ HSPGs

In one sentence: Heparan sulfate proteoglycans, the large negatively charged molecules of the tumour matrix.
Detailed explanation: An HSPG is a proteoglycan carrying a long chain of heparan sulfate, sitting on the cell surface and in the matrix. In the tumour microenvironment they are two things at once: part of the scaffolding of the ECM, and also something that catches growth factors and signalling molecules and holds them, so that the signals pile up around the tumour.
That is one of the differences between a tumour and normal tissue at the level of chemical environment. So the matrix is not only a scaffold, it is a storage place that keeps a signal switched on by refusing to let it diffuse away.
Why this course brings it up: He does not name them in particular, so it is concept-only. The list of the TME is the real point of the passage. Know the list, and you can place this entry on it. **Appears in:** Lect 2

▪ Human Protein Atlas

In one sentence: The public database he uses to nominate drug targets and to find out where a protein is actually made.
Detailed explanation: The Human Protein Atlas is a public database of human protein expression. It records which tissues each gene is expressed in, how much, and what the prognostic link is in particular cancers.
He uses the ERBB2 page as a demonstration: the predicted location, inside the cell and at the membrane, eleven transcripts, an RNA category of low tissue specificity, a TCGA category of low cancer specificity, and the most important line of all, the

summary that reads "Prognostic marker in renal cancer (favorable), endometrial cancer (unfavorable) and pancreatic cancer (unfavorable)", with all three KM plots at $p < 0.001$. The low tissue specificity line is the one that matters commercially: if the protein is everywhere in the body there is no clean therapeutic window, and a drug that hits it will hit normal tissue too. The prognostic side answers a different question, whether patients whose tumours carry the feature do better or worse, and here the answer is not the same in the three cancers, which is why favourable and unfavourable sit side by side in the same sentence. In the biomarker workflow this is the checking and judging step: you confirm whether the protein is also expressed in normal tissue, which sets the therapeutic window, and you see whether it is linked to prognosis.
Why this course brings it up: It is concept-only plus third-level numbers, the three cancer types for ERBB2 and $p < 0.001$. What he explicitly excludes is the pharmaceutical workflow of target nomination and druggability, not the Human Protein Atlas as a tool, and the ERBB2 page is a highlighted item on his Lecture 1 slide. So you are expected to know that the database exists, what you would look up on it, and what a prognostic label on it tells you. The exclusion is of the process around it, not of the resource. **Appears in:** Ch 10, Lect 1

▪ humanised mouse

In one sentence: An animal model with a human gene put into a mouse.
Detailed explanation: Transplanting a human disease gene into a mouse lets you watch what that gene does inside a living body, which is far quicker and far cheaper than cell culture. But it is still not a person: the way a drug is metabolised in a mouse, the dose and the side effects can all differ from a human, so the model is the starting point and not the end point.

Why this course brings it up: Dr. Wong does not name the humanised mouse as an exam topic on its own, so it belongs to conceptual understanding. The real issue to remember is that choosing between a monkey and a humanised mouse is a genuine scientific debate to be discussed with the FDA at a pre-IND meeting. **Appears in:** Ch 12, Lect 2

▪ hydrodynamic focusing

In one sentence: The fluid trick that makes the cells pass through the laser beam one at a time in single file.
Detailed explanation: The instrument uses a fast-moving liquid called the sheath fluid to squash the sample stream from the middle down into an extremely thin thread, so the cells can only go through the laser one after another in single file.
That is exactly why flow cytometry can count accurately: if two cells pass at the same time their scatter signals overlap and cannot be told apart. If the focus spreads out the signals interfere with each other and the count becomes unreliable, so this is the physical basis of the counting power of flow cytometry. The cells are not being sorted out of the stream, they are simply being made to arrive one at a time.
Why this course brings it up: "Hydrodynamic focusing makes the cells pass in single file" is a mechanism he has demonstrated, and the basics of flow cytometry are Tier 2. **Appears in:** Ch 11, Lect 2

▪ Hydrogen bond

In one sentence: The weak attraction between a hydrogen attached to an electronegative atom and another electronegative atom nearby.
Detailed explanation: A hydrogen bond is the weak attraction between "a slightly positive hydrogen" and "a slightly negative atom", with O–H···O as the typical example. A single one is very weak, but dozens of them all pulling in the same direction are strong enough to fix a shape — the alpha-helix and the beta-sheet of a protein are held together this way, and the base pairing in DNA is also hydrogen bonding (three in G–C, two in A–T).
Between water molecules the same bonds are what give water a high specific heat, let it buffer temperature change and let it dissolve salts and sugars. The crucial property is reversibility: the cell can break and remake hydrogen bonds when it needs to, which is why folding is adjustable. Weak individually, strong in a crowd, and reversible — three properties, one bond.
Why this course brings it up: It belongs to the concepts you are expected to understand. The G–C = 3 and A–T = 2 counts are in his table of precise numbers, and the idea that many weak bonds can be strong and can be reversed is the common key to understanding alpha-helix, beta-sheet and DNA denaturation. **Appears in:** Ch 1

▪ hypermethylated

In one sentence: The state of being methylated too heavily.
Detailed explanation: Both directions cause trouble. If a promoter CpG island becomes hypermethylated, a tumour suppressor gene is repressed at the level of transcription, which is welding the brake on.
If pericentromeric heterochromatin becomes hypomethylated, the opposite happens and the repeated sequences are let loose, which allows mitotic recombination and genomic instability. Clinically, promoter hypermethylation of CDKN2A and of MGMT gives you a signal you can detect in a patient sample, and it is the entry point for predicting the response to treatment. So one word covers two opposite diseases, and which direction you are in decides whether the gene is silenced or the genome destabilised. The term you need is not the direction, it is the pair. And because the same measurement can report either direction, the laboratory test has to say which one it found, and that is why the pair of terms keeps appearing together.
Why this course brings it up: Promoter hypermethylation leading to brake failure is the Tier 2 core mechanism he lists, and it is the basis of both the MGMT story and the CDKN2A story. Both of those are clinical: one predicts a drug response and the other predicts a risk. **Appears in:** Ch 4, Lect 3

▪ hypomethylated

In one sentence: The state of being insufficiently methylated.
Detailed explanation: Repeated DNA such as pericentromeric heterochromatin is normally held down by heavy methylation, and when it becomes hypomethylated the cell loses the repression of those repeat sequences.
Repeats can then undergo mitotic recombination with one another, and genomic instability appears. The same chemistry, run in one direction, shuts a gene off, and run in the other, lets chaos out. Dr. Wong's slide in Lecture 3 puts promoter hypermethylation and repeat-region hypomethylation side by side on the same slide, to make the point that one chemical change can lead to two opposite outcomes. Too much mark on a promoter and the gene is silenced; too little on the repeats and the genome starts to come apart. Both are methylation problems, and both are epigenetic, which is the same word for a change with no change in the sequence.
Why this course brings it up: He does not name hypomethylated as an exam topic on its own, so it belongs to conceptual understanding. The contrast he does draw, two kinds of methylation abnormality with two kinds of consequence, is worth having.
Appears in: Ch 4, Lect 3

▪ Hypoxia

In one sentence: Too little oxygen. It is the signal that switches HIF-1α on.
Detailed explanation: Oxygen shortage sets off a whole set of adaptation reactions. The mechanism does not involve “sensing oxygen” directly. Instead the oxygen-dependent hydroxylating enzymes (PHD, FIH) stop working, so HIF-1α is no longer tagged for destruction, gets into the nucleus, binds HIF-1β, binds the HRE, and turns on VEGF (more blood vessels), GLUT1 (more glucose coming in) and EPO (more red blood cells).
Every one of those three is the same idea: deal with the shortage by getting more oxygen, or by doing without it. In a solid tumour the core is chronically hypoxic because the blood supply cannot keep up, and this reaction is what lets the tumour survive and sprout new vessels. Dr. Wong's figure is that around 50% of solid tumours contain hypoxic tissue. HIF-CAR is borrowing this switch, so the CAR only gets switched on in the part of the tumour that is short of oxygen.
Why this course brings it up: It is second-level (A). His own words connect hypoxia straight to HIF-1α and then to conditional CAR expression, and that chain is the answer to Case 14. So the chain itself is the examinable thing, and you should be able to walk along it without hesitating. **Appears in:** Ch 7, Ch 8, Ch 9, Lect 5

▪ Hypoxia-responsive element (HRE)

In one sentence: The piece of DNA that switches a gene on when HIF-1 is active.
Detailed explanation: An HRE is a particular stretch of DNA sequence, a word that HIF-1 can recognise. Once HIF-1α and HIF-1β have joined together in the nucleus and recruited co-activators such as p300/CBP, the complex goes looking for the HRE and starts transcription.
Every gene under an HRE is a list of things to do when oxygen is short: VEGF grows blood vessels in, GLUT1 lets glucose in, EPO makes more red cells. Every gene under an HRE is therefore a promise about what happens when oxygen runs short, and the list is always the same set of emergency responses: more blood vessels, more glucose, more red cells. The word in the DNA is only half of the story, since the complex also has to be there to read it. HRE has a second use that is purely engineered. Put the gene you want expressed, most typically a CAR, under an HRE and you have a switch that only works in hypoxia, and the cell that sits in the oxygen-poor part of a tumour is the only cell that turns it on.
Why this course brings it up: Conceptual understanding plus second level, grade A. It makes the idea of a molecular switch very concrete: the same HRE can be used to control a natural hypoxia response gene or to design a conditional therapy.
Appears in: Ch 7, Lect 5

▪ ICAM-1

In one sentence: An adhesion molecule that lets white blood cells grip the wall of the blood vessel.
Detailed explanation: Getting a white blood cell out of the blood and into tissue takes four steps, and Dr. Wong's Lecture 2 slide is exactly those four steps: (1) rolling, handled by P-selectin, E-selectin and integrins;
(2) adhesion and activation, with ICAM-1 and VCAM-1 providing the sites; (3) squeezing between the endothelial cells, handled by PECAM-1; (4) following the chemical gradient, guided by MCP-1. ICAM-1 is the adhesion molecule on the endothelial cell, and its matching receptor is the integrin (LFA-1, Mac-1) on the white blood cell. The two lock together, and that is the “gripping firmly” part. Rolling is what slows the cell down so that adhesion has time to happen, and adhesion is the step that both holds the cell and switches it on. Tumours often block this route by changing the endothelium and the ECM, so the T cells cannot get in, and that is a large part of why the immune system sometimes cannot reach a tumour.
Why this course brings it up: It is conceptual understanding. It appears on his four-step leukocyte extravasation diagram, and this pathway is ready-made material for explaining how the tumour microenvironment blocks immune cells. It falls inside his second-level (A) range. **Appears in:** Ch 9, Lect 2

▪ ideogram

In one sentence: A standardised diagram of one chromosome, with every band labelled.
Detailed explanation: His slide draws the complete ideogram of chromosome 8, labelling the bands of the p arm — p23.3, p23.2, p23.1, p22, p21.3, p21.2 and p21 among them — and the band names of the q arm as well.
The letter q comes after p, which is why the long arm is called q; "p is for petite", the French word for small, is the other little story about it. The ideogram is what gives laboratories a common language: when somebody reports a deletion at p21.3 on

chromosome 8, everyone in the world knows which stretch is meant. Without that shared map, the same words would not describe the same piece of DNA in two different labs. The centromere sits between the two arms, and it is the boundary that divides them.
Why this course brings it up: The relationship between the p arm, the q arm and the centromere is Tier 2; the word ideogram itself he does not ask you to memorise, so it is a concept to understand. What you need is the map, not the vocabulary. **Appears in:** Ch 3, Lect 3

▪ IDO

In one sentence: An enzyme that starves T cells of an amino acid they need and switches them off.
Detailed explanation: IDO, indoleamine 2,3-dioxygenase, breaks down tryptophan, and T cells need tryptophan in order to be activated and to proliferate. To protect itself the tumour cell raises the level of IDO, the tryptophan around the tumour is swept away, and the T cells cannot work any more, like a factory with no raw material.
The immune attack is defused that way. This is one of the mechanisms of immunosuppression in the tumour microenvironment, and it works together with the PD-1 and PD-L1 checkpoint brakes: one of them pulls the handbrake at the front, the other cuts the food supply at the back. IDO is therefore one of the combination targets in immunotherapy. Note that the enzyme does not attack the T cell. It attacks the tryptophan in the surroundings, and the T cell is the one that suffers for it, which is why this is a starvation strategy rather than a direct kill.
Why this course brings it up: Conceptual understanding. Dr. Wong lists IDO together with TGFβ, IL-10, CAF, hypoxia, CD47 and adenosine as the reasons CAR-T fails in solid tumours, and you can quote it directly when you answer the question of why CAR-T cannot get into a solid tumour. It sits inside the second-level, grade A range. **Appears in:** Ch 9, Lect 2

▪ IFN-γ

In one sentence: The third cytokine in the NF-κB response, after TNF-α and IL-1β.
Detailed explanation: IFN-γ (interferon gamma) sits third in his NF-κB time order: TNF-α peaks first, at 4 to 6 hours, then IL-1β, then IFN-γ, and finally IL-2 and IL-15. Biologically, IFN-γ is secreted by Th1 cells and NK cells, and its main actions are to activate macrophages, increase MHC expression and push cells towards Th1 polarisation.
It is also connected to checkpoint therapy: an HDAC6 inhibitor raises IFN-γR–STAT1 signalling and promotes M1 polarisation. The point of the ordering is not the list itself; it is that the same trigger produces different responses at different times, and that a drug which shifts one cytokine's timing changes the shape of the whole response. So the timeline is a picture of control rather than a list of names. And the cytokine that comes last in the sequence is the one that shifts the cell towards the type of response that fights infection inside cells rather than outside them.
Why this course brings it up: He explicitly excludes the complete ordering: "don't worry about all these cytokines. I'm just want to show you that different responses can be observed in different times." So a question like "what is the fourth cytokine" does not need memorising, but the TNF-α peak at 4–6 hours is in his table of precise numbers and is worth remembering. And the therapeutic hook is the part that matters clinically, because it is how an HDAC inhibitor is made to cooperate with a checkpoint blockade. So the cytokine list is optional, but the shape of the timeline and the clinical hook are not. **Appears in:** Ch 4, Ch 7, Lect 5

▪ IgG4

In one sentence: The antibody subclass that does not kill through ADCC and stays in the blood longer than IgG1.
Detailed explanation: IgG4 is one of the four subclasses of the IgG family, and the subclasses are not interchangeable. Compared with IgG1, its Fc region carries one key residue substitution, which drops its binding to the Fc receptor, CD16 or FcγRIII, sharply, so it cannot effectively pull an NK cell over to perform ADCC, which is the mechanism where the antibody hands the tumour over to a killer cell instead of killing it itself.
But there is an upside to the same change: the Fab arms of IgG4 can swap with each other, half-antibody exchange, so two different IgG4 molecules can form a bispecific structure. And its serum half-life is longer than IgG1's, so it stays in the blood longer. The clinical meaning is this: when the treatment you need is blocking rather than killing, an antibody that blocks signal transduction, you choose IgG4; when you need to kill the tumour, you choose IgG1.
Why this course brings it up: IgG1 against IgG4 is one of his distractors, and the discriminating points are ADCC and half-life, so it is second-level (A). It is also the clearest example of the larger idea that choosing an antibody subclass is choosing a function, because the subclass decides who does the killing and for how long the antibody stays around in the patient. **Appears in:** Ch 2, Ch 9, Lect 1, Lect 2

▪ IL-10

In one sentence: The cytokine that suppresses the immune response.
Detailed explanation: When the immune system is going too hard inside a tumour, or reacting to normal tissue too strongly, an anti-inflammatory signal such as IL-10 comes out to call a halt, pressing the activation of T cells and the inflammatory response back down.
In his tumour microenvironment list IL-10 is grouped with TGFβ as the signals that take the fire out of the immune system. Both of them are immunosuppressive, and if the tumour can turn up both of them it can keep the T cells silent even when they are looking straight at the tumour.
Why this course brings it up: Dr. Wong does not name it specially, so it is conceptual understanding. He has said explicitly that you do not have to memorise the order of the cytokines on that slide. **Appears in:** Ch 9, Lect 2

▪ IL-15

In one sentence: A cytokine that comes after IFN-γ in the NF-κB response and keeps NK cells alive.
Detailed explanation: In his time series of the NF-κB response IL-15 sits at the end, alongside IL-2. Biologically IL-15 is secreted by dendritic cells and macrophages, and it acts mainly on NK cells and memory CD8 T cells, supporting their survival and their proliferation.
It shares signalling components with IL-2, but at physiological concentrations it leans towards maintaining NK cells. It is also a key part of the CAR-T manufacturing process: when CAR-T cells are expanded in vitro, the beads coated with anti-CD3/CD28 antibodies are used together with IL-2 and IL-15 to activate the cells and keep them alive. So the same cytokine that is one line in the inflammatory sequence inside a patient is also a reagent on the bench, and the two contexts have to be kept apart.
Why this course brings it up: Conceptual understanding. He excludes the full order of the cytokines, but the role of IL-15 and IL-2 in the CAR-T manufacturing process is a different context, and you will need it when you answer how CAR-T cells are amplified outside the body. **Appears in:** Ch 9, Lect 2, Lect 5

▪ IL-1β

In one sentence: The second cytokine in the NF-κB response, coming after TNF-α.
Detailed explanation: In Dr. Wong's time ordering, IL-1β comes after TNF-α and before IFN-γ. It is a typical IL-1 family cytokine, matured by caspase-1 (the executioner of pyroptosis) cutting its precursor, and only after that cleavage is it active.
It is one of the endogenous pyrogens, acting on the hypothalamus to change the body's temperature set point, which is why fever is really the result of IL-1β and TNF-α acting together rather than a separate mechanism of its own. It is also part of the amplification loop: IL-1β goes back and activates NF-κB again, which makes even more inflammatory proteins. So in the NF-κB story it plays two roles at once, arriving in order and then feeding the loop that made it arrive. And that is why the timing matters even though you do not have to memorise it: the same molecule is an output of the cascade and an input back into it.
Why this course brings it up: He excludes the complete cytokine ordering, but these four cytokines form the backbone of his NF-κB section, and they come in handy when the question asks what is produced in sequence once NF-κB has been activated. It falls inside his second-level (A) range. **Appears in:** Ch 7, Lect 5

▪ IL-2

In one sentence: The cytokine that comes after IFN-γ in the NF-κB response and drives T cell proliferation.
Detailed explanation: IL-2 is the T cell growth factor. It was discovered and purified in the 1960s from the supernatant of cultured T cells, and only later was it found that it is the activated T cells themselves that secrete it.
Its job is to promote T cell proliferation and survival, and it is key to maintaining memory T cells. In Dr. Wong's time ordering, IL-2 and IL-15 come last. In the CAR-T manufacturing process, IL-2 is the growth factor added during in vitro culture, so the engineered cells expand to millions. So the same molecule is doing two quite different jobs across the course: in the NF-κB timeline it is one step in an inflammatory cascade, and in the CAR-T process it is simply the thing in the flask that makes the cells divide.
Why this course brings it up: It is conceptual understanding. He excludes the cytokine ordering, but IL-2 is a link you must remember in the CAR-T manufacturing process (beads + IL-2 + IL-15). It falls inside his second-level (A) range. **Appears in:** Ch 9, Lect 2, Lect 5

▪ Immune checkpoint

In one sentence: The molecular brakes that stop immune cells from attacking. PD-1 on the T cell is the most important one.
Detailed explanation: Checkpoints are brakes built into the immune system, and under normal circumstances they stop immune cells from over-attacking your own tissue, which is a real risk when the machinery is this powerful.
The most important set is PD-1 and PD-L1: PD-1 sits on the T cell, PD-L1 sits on the tumour cell, and when the two shake hands the T cell enters anergy, a state of doing nothing at all, and the attack stops. Tumours are clever about this, and they raise the amount of PD-L1 they display so that the brake stays pressed even when the T cells are there and ready, which is why the drugs work in the opposite direction. Anti-PD-1 antibodies, pembrolizumab and nivolumab, and anti-PD-L1 antibodies, atezolizumab among them, block the handshake and re-arm the T cell. The other checkpoint is CTLA-4, which competes with CD80, a co-stimulatory molecule, and CD86, another co-stimulatory molecule, early in T cell activation, so it stops the cell before it ever gets going. The anti-CTLA-4 drug ipilimumab releases that brake, and the drug you use therefore tells you where in the sequence the problem was.
Why this course brings it up: It is between second-level (A) and first-level. Every example he gives is built around checkpoints: MSI-H to hot tumour to anti-PD-1, and a cold tumour can be reworked into a hot one with an HDAC inhibitor. It is must-know material. The two brakes also work at different moments, CTLA-4 early in activation and PD-1 in the effector phase, so blocking them is not the same intervention and the drugs are not interchangeable. The point of the whole story is that the brake is the target: you do not need to make the T cell stronger, you need to take the brake off. **Appears in:** Ch 9, Lect 2

▪ Immune evasion

In one sentence: The state in which a tumour escapes an attack because the T cells have been switched off.
Detailed explanation: Immune evasion is the general name for a tumour escaping immune surveillance, and several mechanisms act at once behind it. The tumour cells raise PD-L1 expression and use the checkpoint to switch T cells off directly.

IDO clears tryptophan so the T cells run out of food. CAF build an ECM-rich barrier that keeps immune cells out. Hypoxia and low pH change the local metabolism. TGFβ and IL-10 push the immune response down. And the two suppressive cell groups, MDSC and TAM, sit alongside and weaken the T cells. On top of that there is the tumour's own heterogeneity: different regions of the same tumour express different antigens, so immunotherapy can only clear the part it can see. Every one of these is a place where a drug could be pushed, which is why the list is also a list of opportunities.
Why this course brings it up: It belongs to the concepts you are expected to understand, but it is the general answer to "why does immunotherapy fail in some patients", and it is the same list he uses to explain why CAR-T fails in solid tumours. It is within second-level (A) content. **Appears in:** Ch 9, Lect 2

▪ Immunofluorescence (IF)

In one sentence: Staining tissue with fluorescently labelled antibodies so you can see exactly where a protein is.
Detailed explanation: The principle of IF is a primary antibody plus a fluorescent secondary antibody. The primary antibody grips the target protein, and the secondary antibody, carrying a label such as FITC or Alexa Fluor, grips the primary, so the place that lights up is the place the target protein sits.
DAPI stains the nucleus blue and acts as the reference. His Lecture 2 demonstration uses multiplex IF to look at all three members of the Hippo pathway, MST1, LATS and YAP, in the same cell at once and to read their spatial positions. A confocal microscope, with optical sectioning and 3D reconstruction on top, can then read subcellular structure that wide-field IF cannot. The glow is the readout; everything else is plumbing, and the plumbing is what lets you see one protein, two proteins or six in the same tissue section.
Why this course brings it up: Dr. Wong lists IHC versus IF as a contrast, with IHC using a coloured enzyme substrate and IF using a fluorescent dye, so it is second level, grade A. You have to be able to answer which method gives you coloured localisation and which gives you multiple signals at once. **Appears in:** Ch 9, Ch 11, Lect 2

▪ Immunohistochemistry (IHC)

In one sentence: Staining tissue with an antibody plus an enzyme that leaves a coloured deposit, so that you can see where the protein is.
Detailed explanation: The difference between IHC and IF is only the form of the signal. IHC uses an enzyme (horseradish peroxidase, alkaline phosphatase) to turn a colourless substrate into a coloured deposit that stays at the position of the antigen, so the slide can go on an ordinary glass slide, be read under an optical microscope, even be kept for a long time and read by a pathologist.
IF uses fluorescence instead, which is more sensitive and allows multiple labelling. The clinical use is as a companion diagnostic: HER2 positivity can be judged by IHC or by FISH, and the clone, the scoring criteria and the cut-off value directly decide whether the patient can have Herceptin. That last part is the point to sit with. The test is not a yes/no from the sky; the same tumour, read with a different clone or a different threshold, can land on the other side of the line, and the treatment follows the line.
Why this course brings it up: IHC versus IF is one of the contrasts Dr. Wong lists, so it is second-level (A). And IHC is one of the tests most likely to be picked in his companion diagnostic scenario (Case 6), so you have to be able to say what it measures and when you would use it. **Appears in:** Ch 9, Ch 11, Lect 2

▪ Inclusion body

In one sentence: A hard, insoluble clump of misfolded protein. It appears when recombinant protein is made too fast or too early.
Detailed explanation: An inclusion body is an insoluble protein aggregate inside the cell, and it shows up most often in recombinant expression: the bacteria grow too quickly, the protein cannot fold correctly in time, and the molecules stick together into a precipitate in the cytoplasm, giving high yield and low activity.
The remedy is to slow down — delay induction until the exponential growth phase, because inducing too early loads the cell with synthesis pressure and both misfolding and inclusion bodies increase. The same principle shows up in disease: the core pathology of Alzheimer's and Parkinson's disease is misfolded protein collecting into amyloid plaques. Which is a lesson about production, not about biology: speed without control gives you junk. The precipitate is not a failed product so much as a product that arrived in the wrong place at the wrong time.
Why this course brings it up: It belongs to the concepts you are expected to understand. It demonstrates a general rule: the machine that makes protein should run as fast as possible, but the making has to be controlled, or what you get is rubbish rather than medicine — which is why he teaches codon optimisation and CHO cells together. **Appears in:** Ch 2, Ch 7, Lect 5

▪ IND-enabling studies

In one sentence: The whole set of non-clinical studies a company does before it applies for an IND.
Detailed explanation: It has two big blocks. The first is pharmacology and toxicology, meaning whether the compound does what it is supposed to do in cells and what it does to a whole animal.
The second is the quality data, of the kind called CMC. Only when that package is done is the company entitled to file an IND and apply to enter human trials. The point of the document set is that it turns we believe this drug works and is safe into a chain of evidence that a regulator can check item by item. Which is the whole philosophy of the thing: a belief is not enough, it has to be written down in a form somebody else can verify.
Why this course brings it up: He does not list IND-enabling studies as an exam question on its own, so it is concept-only. The

Tier 3 numbers are the phasing of clinical trials and the division of labour between 1A, 1B, 2 and 3. So the stages of the trial are the number to hold, and this is the paperwork around them. **Appears in:** Ch 12, Lect 2

▪ IND (Investigational New Drug)

In one sentence: The application a company has to file before it is allowed to test a drug in humans.
Detailed explanation: An IND is the application made to the regulator, the FDA in the United States, and it contains the CMC data, which is chemistry, manufacturing and quality control, the non-clinical reports on safety, pharmacokinetics and toxicology, and the clinical plan with its arrangements for protecting the volunteers.
An IND going into effect means the drug has been cleared to enter human trials, and the sequence runs pre-IND meeting, then non-clinical studies and IND-enabling studies, then Phase 1A with dose escalation, then Phase 1B for expansion, then Phase 2/3 for efficacy. Before the IND the company meets the FDA to agree which animal model to use, a pharmacology-relevant non-human primate or a humanised mouse. Dr. Wong also mentions that in January 2023 the FDA announced it would no longer require animal testing as a precondition for human trials. The IND is therefore a gate and not a formality: everything before it happens in the laboratory and in animals, everything after it happens in people, and the IND is the line between the two.
Why this course brings it up: Conceptual understanding. He does not put IND on the exam list, but the question of what gates a drug has to pass between being a candidate and reaching a patient is a common framework for a scenario question, and the dose escalation in Phase 1A together with the figure of US$200,000 per person is his own material. **Appears in:** Ch 12, Lect 2, Lect 4

▪ Indel

In one sentence: An insertion or a deletion in the DNA letters. If it is not a multiple of three, it causes a frameshift.
Detailed explanation: indel is the combined name for insertion and deletion, a short stretch of bases inserted into or deleted from a DNA sequence, and it is the most common kind of change after a point mutation.
It matters clinically because a microsatellite, a short repeat of two or three nucleotides, is the most likely thing to slip during replication and produce an indel, because a repeated run is exactly where the copying machinery is most likely to lose its place. If the cell's mismatch repair is working, those slips are corrected on the spot. When MMR is defective the correction fails, the length of the microsatellites starts to vary, and that is microsatellite instability, MSI. The arithmetic is simple, three bases per codon, so an indel of 3, 6 or 9 shifts nothing and an indel of 1, 2, 4 or 5 shifts everything after it. And when the length of the indel is not a multiple of three, the reading frame shifts, the protein is cut short and gains a new segment, and that new segment is a piece of protein the immune system has never seen, which is where the neoantigens of a hot tumour come from.
Why this course brings it up: It is must-know material. MMR deficiency to MSI-H to hot tumour to a good response to anti-PD-1 is one of his main threads, and the figure that 15 % of colorectal cancer is MSI-H is one of his precise numbers. The chain is worth memorising as a chain, because each step is a separate lecture topic and the exam question hands you the whole line at once. **Appears in:** Ch 6, Lect 1, Lect 3

▪ Inducible system

In one sentence: A system that stays switched off until the signal arrives.
Detailed explanation: The design of an inducible system is “get everything ready in advance, switch on on demand”. The gene for the protein is already wired up and the machine is already assembled, but the whole pathway stays shut until an inducer turns up, so nothing is wasted in manufacturing.
This matters especially for recombinant expression: let E. coli grow to size first, then switch on expression of the target protein, and both the efficiency and the folding quality are better. Dr. Wong's example is the lac operon: with no lactose, the repressor sits on the operator blocking RNA polymerase; with lactose present, allolactose binds the repressor and makes it move away, and only then does transcription begin. So the machine is always there and the switch is what waits, which is also why the timing of the switch matters so much. Switch on too early and the cell is still busy making its own machinery and has no spare capacity; switch on too late and you have wasted the growth phase. The design is deliberately lazy: build everything, then throw the switch.
Why this course brings it up: It is second-level (A). He explicitly says the lac operon is his demonstration example, and he stresses that it needs “two conditions” rather than one. The timing of induction and its relationship to inclusion bodies is what lets you apply the concept to the practice of recombinant expression. **Appears in:** Ch 7, Lect 5

▪ Induction

In one sentence: Opening a gene up by adding an inducer (lactose in the lac system).
Detailed explanation: Induction is the act of flipping the switch on. In the lac system the inducer is lactose: it is first converted into allolactose, and it is allolactose that actually binds the repressor.
Once it has bound, the repressor changes shape and can no longer grip the operator, so transcription starts — but that is only the first condition. The second condition is that glucose must be low, so that cAMP rises and CAP–cAMP can bind the promoter and recruit RNA polymerase; only when both conditions hold do you get strong transcription. That two-condition logic is the complete answer to his Case 16 scenario, and it is why he says "two conditions, not one". This is the part students get wrong, because removing the repressor feels like the whole story, and the lac system is carefully designed to be shut off as well as switched on. Two switches, and the cell needs both before it commits.
Why this course brings it up: It is second-level (A), and it comes with a note about what sits next to it on the exclusion list: IPTG and other artificial inducers are what recombinant systems use, whereas the lactose and glucose logic is what gets

examined. So if you are asked what has to be true for strong transcription, the answer has two halves and both are required.
Appears in: Ch 7, Lect 5

▪ Inhibitor of apoptosis (IAP)

In one sentence: A protein that blocks the caspase executioners.
Detailed explanation: IAP is a family of proteins that structurally contain repeated BIR domains, and those domains let them bind caspase-9 and caspase-3 directly and hold the executioners in place, so the signal for apoptosis is braked.
In a normal cell, mechanisms such as IAP and XIAP make sure that only the cells which really should die are removed. A tumour, by contrast, commonly escapes apoptosis by over-expressing IAP, survivin for instance, so IAP inhibitors are a direction for anticancer drugs under development, and they can also be paired with his bispecific antibody strategy. The brake and the accelerator are separate parts of the system, so a cell that wants to survive only has to push on the brake, and a tumour does exactly that with survivin. The brake is also the reason IAP inhibitors are interesting as drugs: take the brake off and the cell has to go back to being examined by the same machinery that would normally have killed it.
Why this course brings it up: Conceptual understanding, because it sits on the "inhibition" side of the apoptosis page. You have to be able to pair it with the opposite side: TRAIL and the death receptor are the signal that starts it, the caspases are the executioners, IAP is the brake, and the bcl-2 family together with IAP decides whether the mitochondrial route is taken at all.
Appears in: Ch 8, Lect 2

▪ INK4

In one sentence: A family of inhibitory proteins that block CDK4/6 kinase and so stop the cell cycle.
Detailed explanation: INK4 is short for Inhibitor of CDK4, and it is a family of CKIs, which is short for cyclin-dependent kinase inhibitor, a protein whose only purpose is to sit on top of a kinase and stop it working.
The family includes p16INK4A, p15INK4B, p18INK4C and p19INK4D. Their job is to bind CDK4 or CDK6 and stop it, so pRb is not phosphorylated, E2F is not released, and the cell stays in G1 and cannot enter S phase. The brake and the machinery it holds are both named in the same sentence: CDKN2A is the gene, p16INK4A is the protein it makes, and the protein is the one that sits on CDK4. The most relevant case clinically is the mutation or promoter methylation of CDKN2A, which is p16INK4A, and CDKN2A methylation is exactly what he quantifies in Lecture 3: an OR of about 12.7 when methylation is detected, and an OR of 39.6 for CIMP-high against CIMP-0, with a 95 % CI of 20.6–76.1 and $p < 0.0001$. Outside the INK4 family there is the other group, the Cip/Kip proteins, p21, p27 and p57.
Why this course brings it up: It is concept-only plus third-level numbers. He explicitly excludes the individual cyclin and CDK pairings from the list of things to memorise, but the OR values for CDKN2A methylation and the clinical meaning of p16 are on his list of precise numbers. The pairing you should hold is the one between the family and its members, because a question that gives you p16 wants the family, and a question that gives you INK4 wants p16. **Appears in:** Ch 8, Lect 2

▪ INO80

In one sentence: One of the four families of chromatin remodelling complex.
Detailed explanation: The INO80 family is one of the molecular machines that use ATP to move nucleosomes, pushing them, dragging them or flicking them off the DNA. Dr. Wong lines up four of them, SWI/SNF, INO80, ISWI and CHD, and puts their members in yeast, fruit fly, human and mouse in a table.
The point is not to memorise the list but to understand that opening a gene requires these machines to do active construction work.
Why this course brings it up: He does not ask you to memorise the membership lists, so it belongs to conceptual understanding. What is Tier 2 is the idea that nucleosomes slide and that chromatin is dynamic and reversible. **Appears in:** Ch 4, Lect 3

▪ insert

In one sentence: The piece of DNA you want, which is put into a plasmid.
Detailed explanation: In the molecular cloning workflow, the donor plasmid carries the insert (the gene you want to express), an Amp marker and a NotI cutting site. After cutting with EcoRI and NotI, the insert is cut off and then joined to the recipient plasmid, which has been cut open into a linear piece.
Terminologically, an insert is the segment that goes in, as against the vector, which is the carrier. Put in the same terms, the plasmid is the delivery van and the insert is the cargo; get the cargo right and the restriction sites right, and the experiment works. One is something you chose to carry, the other is the thing that does the carrying. Everything about the design flows from that one distinction.
Why this course brings it up: He does not name insert as an exam topic on its own, so it is a concept to understand; what matters is that you can tell the piece of DNA you are putting in from the piece of DNA that carries it. **Appears in:** Ch 3, Lect 4

▪ Insulator

In one sentence: A DNA sequence that stops an enhancer from working on the wrong gene.
Detailed explanation: An enhancer can act at a distance, and cell-type-specific and time-specific expression need precise control, so you need a wall. The insulator is that wall: it binds specific proteins and limits the range of an enhancer's action to before or after the correct promoter.

Mechanistically it can work in two directions, either blocking an enhancer from reaching across to another gene, or building a wall between two genes so that the neighbouring one is not switched on. Dr. Wong's list of cis elements has four: promoter, enhancer, silencer, insulator, and all four act on the single DNA molecule they sit on. That last point is what makes the cis and trans split so clean: a cis element is written into the DNA at the site, while a trans factor is a protein that has to travel there to do anything. So the insulator never leaves the DNA it is sitting on, and neither does the enhancer it is holding back.
Why this course brings it up: It is second-level (A). Together with promoter, enhancer and silencer it forms the cis side of the cis versus trans contrast he lists, and you need to know all of those words. The insulator is the one people forget, because it is a wall rather than a switch, but the contrast only works if you can name the four cis elements. **Appears in:** Ch 7, Lect 5

▪ integration

In one sentence: The permanent insertion of a piece of HBV DNA into the host genome.
Detailed explanation: In hepatocellular carcinoma cells the HBV DNA integrates into the human genome, and the integration sites are often close to oncogenes: TERT on chromosome 5, and CCNE1, a cell cycle gene, and MLL4, a chromatin remodelling gene, on chromosome 19.
The more integration sites there are, the worse the prognosis, and his data show that patients with more than 3 integration sites are significantly worse off, P = 0.037. The number of integration sites works as a prognostic marker because it reflects the infection history the tumour has been through: the more integration there is, the more trace the virus has left and the more the genes have been disturbed. So the count of integration sites is being used as a record of the past rather than as a description of the present tumour.
Why this course brings it up: "More than 3 integration sites means significantly worse survival" is one of his Tier 3 precise numbers. **Appears in:** Ch 6, Lect 2

▪ Integrin

In one sentence: A protein on the cell surface that grabs other cells and the matrix; it is how white blood cells hold on while they move.
Detailed explanation: Integrin is a family of transmembrane receptors made of an alpha and a beta subunit, so it is a heterodimer. Its ligands include proteins in the ECM (fibronectin, laminin) and molecules on other cell surfaces (ICAM-1).
In the four steps of leukocyte extravasation, integrin starts working at step one (rolling), and at step two the pairing of integrin with ICAM-1 / VCAM-1 provides firm adhesion. Integrin is also one of the popular targets for therapeutic antibodies: it sits on the cell surface, which makes it easy for an antibody to recognise and block. Two subunits, two names, one receptor. A receptor built out of two halves, which is why it can be switched on by two different kinds of contact.
Why this course brings it up: It belongs to the concepts you are expected to understand. It appears on his four-step diagram of leukocyte extravasation, and it is an example of the proposition that membrane proteins are the first choice for drug targets, which falls inside second-level (A) content. So the receptor, its two subunits and its two kinds of ligand are the useful level of detail. **Appears in:** Lect 2

▪ Intermembrane space

In one sentence: The gap between the two membranes of the mitochondrion, where protons pile up to drive ATP synthase.
Detailed explanation: A mitochondrion has an outer membrane and an inner membrane, and the space between them is the intermembrane space. As electrons pass along the respiratory chain from complexes I, III and IV, protons are pumped out of the matrix into that space, so the proton concentration there ends up far higher than in the matrix and an electrochemical gradient is built.
The protons then flow back into the matrix through ATP synthase, and the energy of that gradient is converted into ATP. That is oxidative phosphorylation, and it is why the mitochondrion is called the power station of the cell. The inner membrane is therefore highly impermeable to ions, so the gradient can hold, while the outer membrane is comparatively permeable and behaves like a sieve. The volume of the intermembrane space is small, which is what allows the proton concentration to be pushed so high. So the mitochondrion works like a dam holding back a reservoir: the inner membrane is the dam, the intermembrane space is the reservoir, and ATP synthase is the pipe through the dam that makes the flow useful.
Why this course brings it up: Conceptual understanding. It is the place-word in the oxidative phosphorylation model where the proton gradient drives ATP synthesis, and it is part of the beginner's guide to mitochondrial structure, with the outer membrane, the inner membrane, the cristae and the matrix. **Appears in:** Ch 8, Lect 2

▪ Intrinsic apoptosis pathway

In one sentence: The apoptosis pathway that starts inside the cell, usually at the mitochondrion.
Detailed explanation: This pathway is started by signals from inside the cell, and he lists three of them: DNA damage, hypoxia and a shortage of growth factors. Those push the balance of the Bcl-2 family, which in turn changes how permeable the outer mitochondrial membrane is.
When bak and bax win that balance, cytochrome c is released from the intermembrane space into the cytoplasm, joins APAF-1 and procaspase-9, the precursor form of caspase-9, to form the apoptosome, which switches on caspase-9, which then cuts procaspase-3 into the executioner caspase-3, and the cell is taken apart in an orderly way, which is what makes this apoptosis rather than an accident. If bcl-2 wins instead, the mitochondrion holds cytochrome c in and the cell survives, so the whole balance is decided by which family member is in charge. The sequence has three stages, the balance inside the mitochondrion,

the release of cytochrome c and the building of the apoptosome, and then the execution by the caspases. That is where the description of the mitochondrion as the control centre of apoptosis comes from: one organelle decides, and the rest of the cell obeys.
Why this course brings it up: It is concept-only. Intrinsic and extrinsic are the two pathways he gives, and the point where they cross is caspase-8 cutting bid to make tbid and send it into the mitochondrion. What you have to be able to do is tell which one is started by a death receptor and which one is started by the mitochondrion. **Appears in:** Ch 8, Lect 2

▪ Intron

In one sentence: The part of a gene that is cut out during splicing and never appears in the mature mRNA.
Detailed explanation: An intron is the part of the gene that gets removed, while the exon is the part that is kept. At transcription both are copied into the pre-mRNA, and then the spliceosome recognises the boundaries (5′ splice site, branch point, polypyrimidine tract, 3′ splice site), cuts out the intron, throws it away and joins the exons together.
Splicing is not done carelessly: a mutation in a boundary sequence can make a splice site be used wrongly, or make a new one appear, so an exon is wrongly kept or wrongly skipped. His liver cancer case is exactly that — a single-nucleotide variant inside an intron creates a new splice site, exon 7 is skipped, a shorter isoform is produced, and the patients carrying that variant have a markedly worse overall survival ($P < 0.001$). It is a nice illustration of how far a single letter can travel: a change that touches no protein-coding base at all can still decide whether a patient survives longer or shorter, because it changes which version of the protein gets made. And notice the direction of the risk: the variant sits in a region nobody thought was functional, which is why the discovery only became visible once a series of patients was compared. The boundaries themselves are a list worth knowing, because they are the landmarks the spliceosome reads to decide where to cut. Meanwhile intron is where microRNA and other non-coding RNAs hide.
Why this course brings it up: It belongs to the concepts you are expected to understand plus second-level (A). His seven types of alternative splicing (ME, RI, AD, AA, AP, AT, ES) and the liver cancer data (ALL $P < 2.2e\text{-}16$, ROC AUC 0.993) are both on his Lecture 5 slide, so the splicing vocabulary and the clinical data are the same slide, and a question can pull from either side of it. **Appears in:** Ch 7, Lect 5

▪ Inversion

In one sentence: A piece of chromosome is cut out and put back the other way round. No DNA is gained or lost, but the expression can change.
Detailed explanation: An inversion is a structural variant: a segment of the chromosome is cut out and reinserted in the opposite direction, so the total amount stays the same and the gene dosage does not change.
It matters because it can change the spatial relationship between a gene and its enhancer. A gene sitting near a certain enhancer can end up tens of thousands of base pairs away after the inversion, and its expression changes dramatically; in the other direction the inversion can pull an enhancer next to an unsuitable promoter. So the number of copies is untouched and yet the output of the gene can move a long way, which is the whole point of the example. Dr. Wong's own words are “some abnormalities cause loss of the genetic material, some do not”, and an inversion is exactly the “no loss but big functional change” kind. Think of it as a sentence cut out and put back the other way round: every letter is still there, and the meaning is different. Nothing is added, nothing is removed, and yet the gene behaves like a different gene.
Why this course brings it up: It is second-level (A). In the first-level (A+) M1/M2 section he singles out inversion: it adds nothing and removes nothing, yet it can change expression, and he says “you need to understand that some of these abnormalities can cause the loss of the genetic materials, but some does not.” That is exactly what the scenario question Case 9 is about, so the quote is worth keeping as it stands. **Appears in:** Ch 3, Lect 3

▪ ion channels

In one sentence: Channels set into the cell membrane that let ions move in and out down their concentration gradient.
Detailed explanation: They are excellent drug targets: a drug only has to sit outside the cell to block the channel, without entering the interior, and the membrane is full of good handles to grab, which is the whole appeal of a surface target over one buried inside the cell.
The three big classes of drug target on the membrane are GPCR, receptor tyrosine kinase and ion channels. It is also why so many channels have specific drugs: changing the ion permeability changes a cell's excitability, secretion or contraction, without having to go inside and alter its metabolism. You are editing the traffic through the gate rather than the machinery behind it. You are editing the traffic through the gate rather than the machinery behind it.
Why this course brings it up: Dr. Wong does not name ion channels as an exam topic on its own, so it belongs to conceptual understanding. The view that receptors on the membrane make better drug targets than intracellular ones is the Ch 6 point.
Appears in: Lect 2

▪ Ionic bond

In one sentence: The attraction between oppositely charged ions, like table salt.
Detailed explanation: An ionic bond forms after a complete transfer of electrons: sodium hands its electron to chlorine, both reach a stable outer shell of 8 electrons, and they become Na^+ and Cl^-, held in a crystal lattice by electrostatic attraction.
That lattice is table salt, and it is also why salt dissolves in water, because water molecules can surround the ions and prise the lattice apart. The division of labour against covalent bonding is clear: a covalent bond shares, an ionic bond takes. Inside a living cell the effect of ions is very direct. The Na^+ and K^+ gradient drives nerve signalling and muscle contraction, and Ca^{2+} is the

switch used in signal transduction.
Why this course brings it up: Background, and it belongs to the chemistry he excludes. But it is one of the three kinds of bond you have to tell apart, covalent, ionic and hydrogen, and that trio is the premise for understanding why a protein can fold dynamically through non-covalent interactions. **Appears in:** Ch 1

▪ IP

In one sentence: Intellectual property, the question of who owns the right to use a technology.
Detailed explanation: IP is a whole set of legal questions: who has the right to use a particular technology. In the case of CRISPR it becomes a real commercial conflict, because two patent lines each have a claim, and when the law can separate the invention itself from a particular application but cannot separate them cleanly, there is a grey area.
For a scientist doing business, IP is as much a reality to consider when choosing a project as druggability. So the biology can be identical in two laboratories and the commercial answer can still be different, because the answer is legal rather than biological.
Why this course brings it up: He does not name IP as an exam question on its own, so it is concept-only, and he says outright that this passage is so that you know more rather than because it will be tested. So you should be able to follow it, not prepare it. **Appears in:** Ch 12, Lect 4

▪ Ipilimumab

In one sentence: An anti-CTLA-4 checkpoint inhibitor; it works by releasing the T cell's brake.
Detailed explanation: CTLA-4 competes with CD80, a co-stimulatory molecule, and CD86, another co-stimulatory molecule, early in T cell activation, blocking the activating signal that comes through CD28, which functions like stamping on the brake very early.
ipilimumab is an antibody against CTLA-4. It sits on CTLA-4 and occupies it, so CD80 and CD86 are free to bind CD28 and give the T cell the full activation signal, with the brake unable to be applied because the thing that applies it is occupied. On his immune checkpoint diagram anti-CTLA-4 corresponds to ipilimumab, anti-PD-1 corresponds to pembrolizumab and nivolumab, and anti-PD-L1 corresponds to atezolizumab, durvalumab and avelumab, so the diagram is a drug list as well as a pathway. The difference between the two drugs is timing: the CTLA-4 brake is at the start, and the PD-1 brake is at the effector stage, which is why the setting you give them in and the order you give them in are not the same question.
Why this course brings it up: It is concept-only plus second-level (A). What you have to be able to say is where the two brakes sit, CTLA-4 and PD-1, and which drug goes with each. That comes into the answers for how a tumour escapes immunity and how immunotherapy breaks that escape, and both of those are things he asks in the case questions. The other thing to notice is the timing, because the two brakes are released at different moments and that is what separates the two drug classes. **Appears in:** Lect 3

▪ iPSC (induced pluripotent stem cell)

In one sentence: An adult cell reprogrammed back to an embryo-like state, so that it can become almost any cell type.
Detailed explanation: An iPSC is the product of taking a mature somatic cell, a fibroblast for instance, and reprogramming it with four factors back to a pluripotent state. Those are the Yamanaka factors: Oct3/4, Sox2, c-Myc and Klf4.
The advantage is that it sidesteps the ethical argument around embryonic stem cells, and because the cells come from the patient, in principle they can be made into an immunocompatible graft for that same person. Verification has three layers: morphology, then markers, meaning surface marker, PCR and flow cytometry, and then the gold standard, which is to inject the iPSC into an immunodeficient mouse and see whether a teratoma containing tissue from all three germ layers grows. That last one is the hardest of the three to fake, because a teratoma only grows if the injected cell really can build every kind of tissue. He also insists on excluding contamination, using a unique expression pattern and timing together with negative staining controls, and he repeats that line because it is the one students forget, so do not skip it when you revise. A cell that looks right and carries the right markers can still fail the teratoma test, and only the mouse tells you that, which is why the gold standard is still done.
Why this course brings it up: Second level, grade A. Dr. Wong states the sentence to remember: "unique expression pattern and timing, and also the negative staining to exclude the contaminated cell. Don't forget it." iPSC is also what makes allogeneic CAR-T possible, because an off-the-shelf product can be built from iPSC-derived cells, and that is what turns a bespoke CAR-T into something you can order off the shelf. **Appears in:** Ch 8, Lect 2

▪ IPTG

In one sentence: A synthetic inducer used to switch recombinant genes on in E. coli, and it is not metabolised.
Detailed explanation: IPTG is isopropyl-β-D-thiogalactopyranoside, a lookalike of lactose. It binds the Lac repressor and changes its shape so that it leaves the operator, but the bacteria have no enzyme that can break it down, so it is not consumed. Its concentration stays stable, and the timing and strength of induction are therefore completely under control, which is the practical value of a synthetic inducer: you decide when the switch is thrown and how hard. The standard approach for recombinant expression is to put the target gene under a T7 promoter and keep the lacO site, then add IPTG to release the repressor and let T7 RNA polymerase start expression. Dr. Wong's slide also carries a bar chart of IPTG concentration against protein yield, with the optimum at 100 μM. Going below that leaves the repressor partly in the way and the yield drops; going above it adds nothing, because the system is already fully induced and more inducer only means more cost in the culture. That is the point of a non-metabolisable inducer: you can set the dose once and it stays there, so the experiment is reproducible.
Why this course brings it up: It is second-level (A). He explicitly lists IPTG as part of the lac operon teaching material. But note the distinction: IPTG is a tool for the recombinant system, and the real examinable point is the two-condition logic of lactose and

glucose (Case 16). **Appears in:** Ch 7, Lect 5

▪ iso-aspartic acid

In one sentence: The stereoisomer produced when asparagine is deamidated.
Detailed explanation: When asparagine is deamidated it forms a succinimide intermediate, and when that takes on water again it can give you two products: aspartic acid and iso-aspartic acid. The two have the same molecular formula and the same mass (115.03 Da), but iso-Asp carries an unnatural stereochemical configuration, and that perturbs the folding and the binding ability of the protein and becomes a source of degradation for a therapeutic protein.
The clinical significance is the shelf life: a protein left in the vial slowly accumulates iso-Asp, the activity falls with it, and so both the dose and the expiry date have to be reconsidered. The molecule is the same weight but the wrong shape, and the shape is what the protein cares about. Nothing is lost, and yet the product is no longer the product.
Why this course brings it up: He does not name iso-aspartic acid as an exam topic on its own, so it is a concept to understand; the four mass numbers for deamidation he gives very precisely. Those numbers are worth having straight, because they are the only handle you have on the chemistry. **Appears in:** Ch 1, Ch 2, Lect 1

▪ Isoelectric focusing

In one sentence: A method that separates proteins according to the pH at which they carry no charge.
Detailed explanation: Every protein has a particular pH at which its net charge is zero, the isoelectric point or pI. Put the sample on a strip of pH gradient gel (4.0 to 9.0, for instance) and apply an electric field, and the proteins move towards the state where their net charge is zero, so they finally stop at their own pI.
Proteins with the same pI therefore pile up on the same line, which gives very high resolution. The principle is simple: in an electric field a protein travels until its own charge has neutralised it, and then it has no more reason to move. What it cannot do is separate proteins of similar molecular weight, which is why his 2D-PAGE runs IEF as the first dimension (by charge) and SDS-PAGE as the second (by molecular weight, 220 kDa down to 10 kDa), and then cuts the spots out for identification by LC-MS/MS. One dimension cannot do everything, which is exactly why the second one exists.
Why this course brings it up: It belongs to the concepts you are expected to understand plus third-level numbers (pH 4.0–9.0, 220→10 kDa). He uses this workflow to teach the discovery of the vimentin biomarker, so the chain of separate → identify → quantify → validate → turn into a test a patient can have done has to be learnable. Which is the difference between a method and a workflow. **Appears in:** Ch 11, Lect 1

▪ Isoform

In one sentence: A slightly different version of a protein, made by the same gene choosing different combinations of exons.
Detailed explanation: One gene producing several protein versions by alternative splicing gives you isoforms, and the splicing decision is made while the RNA is still being processed, so the cell can change which version it makes without changing the DNA.
On his slide, the five exons of the DNA are spliced into three mature mRNAs, exons 1-3-5, 2-3-4 and 1-2-5, and each one corresponds to a different protein. The clinical meaning has three layers. Different isoforms can do different things, for example MCL-1S promotes apoptosis while MCL-1L resists it, so which version a cell makes decides whether it dies. A particular isoform can lose its drug-binding site and create resistance, AR-V7 against enzalutamide, where the cancer keeps growing because the drug no longer has anything to grip. And abnormal splicing can manufacture neoantigens. There is a second way to make isoforms too: choosing a different polyadenylation site. BDNF, brain-derived neurotrophic factor, comes as a short version that stays near the cell body and a long version that is shipped into the dendrite, and each place needs its own copy. So isoform is not decoration: it is the same gene producing different tools for different jobs, and the difference can decide whether a drug works.
Why this course brings it up: It is second-level (A), and it is the subject of scenario question Case 15. What you have to be able to say is the concept of one gene and different isoforms, and to give an example where splicing changes function or causes drug resistance. **Appears in:** Ch 7, Lect 5

▪ Isotype

In one sentence: A version of an antibody class. IgG1 kills through ADCC; IgG4 does not.
Detailed explanation: The isotype is the class of the antibody constant region (IgG, IgA, IgM, IgE, IgD), and within IgG there are subclasses (IgG1, IgG2, IgG3, IgG4). What decides the function is the Fc constant region: the Fc of IgG1 binds strongly to FcγRIII on an NK cell, so it can trigger ADCC, while the Fc of IgG4 has one residue swapped out and binds weakly, so it does not trigger ADCC, but it has a longer serum half-life.
Developing an antibody drug is therefore a functional choice: pick the IgG1 framework if you want to kill cells, and pick IgG4 if you only want to block a signal and need it to stay around for a long time. One small change in the constant region decides which of those two jobs the antibody is good at. You are not choosing an antibody and then discovering what it does; you are choosing the function first and picking the framework that delivers it. Which is why isotype shows up as a design decision in the ADC story rather than as a piece of trivia.
Why this course brings it up: It is second-level (A). IgG1 versus IgG4 is one of the distractors Dr. Wong lists, and the discriminating points are ADCC and half-life. It also explains why almost every ADC payload uses an IgG1 framework. **Appears in:** Ch 2, Ch 9, Lect 2 **Video:** V09_IgG1_vs_IgG4.mp4

▪ ISWI

In one sentence: One of the four families of chromatin remodelling complexes.
Detailed explanation: The machines of the ISWI family are ATP-driven and their job is to reposition the nucleosomes along the DNA. Along with SWI/SNF, INO80 and CHD they are listed on his slide, and the members in yeast, in fruit fly, in human and in mouse are marked as well.
The existence of four families says one thing clearly: a gene does not become readable because the DNA loosens on its own, it becomes readable because a crowd of ATP-burning machines pushes it there.
Why this course brings it up: He does not ask you to memorise the list, so it is conceptual understanding. **Appears in:** Ch 4, Lect 3

▪ IκB (I-kappa-B)

In one sentence: The protein that holds NF-κB in the cytoplasm until some signal destroys it.
Detailed explanation: When the cell is at rest, NF-κB (p50/p65) is held in the cytoplasm by IκB, with its DNA recognition region covered, so that even in the nucleus it cannot read genes. When a stimulus such as LPS arrives, IKK is activated and phosphorylates IκBα;
the phosphorylated IκBα is ubiquitinated and degraded by the proteasome, and NF-κB can then enter the nucleus, bind the promoters of inflammatory genes and start transcribing. The complication is that the proteins NF-κB transcribes (including TNF-α and IL-1β) then go on to activate NF-κB again, forming a positive feedback loop — which is why the inflammatory response arrives fast and hard, and is also called an amplifying loop. The design is a good one: the cell keeps the switch locked until there is a reason to unlock it. So the chain has a built-in delay and a built-in amplifier, which is why inflammation is both slower to start and harder to stop than you would expect.
Why this course brings it up: He says explicitly that LPS → TLR4 → IKK → IκB → NF-κB → nucleus is one of his "announced MCQ formats", so this order has to be writable from memory. What he excludes is the fine detail of the cascade on the slide, not this main chain. The order is the examinable part; the detail of the cascade is not. **Appears in:** Ch 7, Lect 1, Lect 5

▪ IκB kinase (IKK)

In one sentence: The enzyme that puts a phosphate on IκB, which is how IκB is marked for destruction.
Detailed explanation: IKK is a complex enzyme whose one job is to phosphorylate IκBα. The phosphorylation puts a negatively charged phosphate group onto IκBα, and the N terminus of IκBα is then recognised by an E3 ubiquitin ligase, which is the enzyme that decides what gets thrown away.
It adds a polyubiquitin chain, a row of molecular tags saying destroy me, and the tagged protein is then taken apart by the proteasome, the cell's shredder. When IκB disappears, the mask over NF-κB is removed, and NF-κB goes into the nucleus and starts transcribing inflammatory genes. The design point is release rather than activation: NF-κB was in there all along, ready, and IκB was keeping it outside the door. What opens the door is IKK, and the way it opens it is by destroying the lock.
Why this course brings it up: It is second-level (A). One of the MCQ question types he lists is which enzyme or pathway is responsible for this reaction, and IKK's place in that order of events is the answer. IKK is also a popular drug target in signalling research. **Appears in:** Ch 1, Ch 7, Lect 5

▪ K-Ras

In one sentence: The gene whose stuck-on mutant drives many cancers, including colorectal and pancreatic cancer.
Detailed explanation: Normal K-Ras is only a messenger, and a well-behaved one. It does nothing at all until the upstream EGFR fires, and only then does it become the active form, and the two states are what the drug has to deal with.
A mutation stops GAP from hydrolysing the GTP, so the protein is stuck in the on position and keeps sending the two growth signals, RAF-MEK-ERK and PI3K-AKT-mTOR, whether or not there is any signal from upstream. That is why it has long been called undruggable, and only recently have small molecules against the mutant form been approved. In the normal protein the switch is reversible, because GAP can arrive and hydrolyse the GTP and put it back to sleep, and a mutation is exactly the removal of that possibility.
Why this course brings it up: Lecture 1 uses K-Ras to show how a predictive biomarker works. Cetuximab, an anti-EGFR, fails in colorectal cancer with a K-Ras mutation, because the receptor is blocked while the downstream is still lit up. He also gives the proportions: KRAS G12C in 13% of lung adenocarcinoma and 3% of colorectal cancer, and for the new drug Rasonque, daraxonrasib, the median overall survival in metastatic pancreatic cancer is 13.2 against 6.7 months. **Appears in:** Ch 10, Lect 1

▪ Kaplan-Meier curve

In one sentence: A step-shaped curve showing how the probability of a patient surviving changes over time.
Detailed explanation: It is a step function assembled from the time at which each patient has an event: if someone dies or the disease progresses, the curve drops one step, and a flat stretch means no events happened during that period.
The small tick marks on the curve are censored patients — people who were followed for a while, were lost to contact, and did not have an event. The point at which the curve crosses 0.5 is the median survival. Two curves are compared with a log-rank test, and if they separate cleanly then the two groups of patients can be told apart.
Why this course brings it up: It is one of the five first-level questions he names out loud, and he asks it every time he mentions KM: last year only one student got it right. What is tested is: steep and early means early involvement and a poor prognosis, a late drop means late involvement, two lines that separate means you can tell the groups apart, and crossing lines means you cannot. **Appears in:** Ch 10, Lect 1, Lect 3 **Video:** V01_Kaplan_Meier_curve.mp4

▪ Karyotype

In one sentence: A picture of all the chromosomes of a cell, used to look for large structural abnormalities.
Detailed explanation: A karyotype is a photograph in which every chromosome of the cell has been spread out and laid out in numbered order, and it is usually taken from a cell in metaphase. It answers questions of the kind "is the number right, and is a large chunk missing or extra", so trisomy 21 and Turner 45,X are obvious at a glance on a karyogram.
What it cannot see is a change in a single letter.
Why this course brings it up: Dr. Wong was quite explicit in Lecture 3 about the resolution of karyotype analysis, and it counts as a concept you must understand. The figure he gives is that G-banding can only detect gross structural changes of roughly 0.5–10 Mb, he has used both a 5–10 Mb version and a 0.5–1 Mb version in different passages and both are in the material, and point mutations have to be found by sequencing. **Appears in:** Ch 3, Lect 3

▪ Karyotyping

In one sentence: Using Giemsa staining to produce G-banding, so that large structural changes can be read off directly.
Detailed explanation: Stain metaphase chromosomes with the Giemsa dye and the alternating dark and light stripes (G-banding) mark the regions of the DNA that are rich in AT; the position of a stripe is the address of that part of the chromosome.
When something goes wrong you see missing segments, extra segments, or an abnormal number of whole chromosomes. The whole procedure is karyotyping: stain, squash, photograph, and then arrange every chromosome by its address. It is the classic cytogenetic picture: bands, addresses, and a list. Bands, addresses, and a count at the end: that is the whole output of the test.
Why this course brings it up: In Lecture 3 he draws the division of labour between karyotyping and sequencing clearly — structural changes are read off the karyotype, point mutations come from sequencing. It belongs to the concepts you are expected to understand, but the resolution figures are often used in questions. **Appears in:** Ch 3, Lect 3

▪ KIBRA, MER and WILLIN

In one sentence: Proteins at the cell junction that recruit and switch on the Hippo kinase.
Detailed explanation: When cells touch each other, the proteins KIBRA, MER and WILLIN gather at the junction, and their job is to switch on the Hippo kinase, which is the point of contact inhibition: if you are touching your neighbours there is no room to grow.
The signal then travels down: MST1 is phosphorylated, SAV passes the message on, LATS1 is phosphorylated, YAP gets a phosphate added to it, YAP binds 14-3-3 and stays in the cytoplasm and is degraded, and the cell stops proliferating. That is the molecular explanation of contact inhibition. And once Hippo is off, that is cancer: when the switch fails, unphosphorylated YAP goes into the nucleus and binds TEAD, switching on CTGF, JAG1, which is a Notch ligand, and NOTCH2, genes that promote proliferation.
Why this course brings it up: Reviewing Hippo in Lecture 5, he points out that the cell junction is the entry point by which Hippo is switched on, and that Hippo being off is cancer. These three proteins are the first stop on his slide. So the practical question is not the names of the junction proteins but the order of the cascade behind them, which is the same order he wants you to recall for Hippo in general. **Appears in:** Lect 5

▪ Kinase

In one sentence: An enzyme that puts a phosphate group onto another protein, usually switching it on.
Detailed explanation: A kinase moves one phosphate group from ATP onto its target protein. The phosphate carries a negative charge, which creates electrostatic repulsion nearby, changes the shape of the protein, and flips the switch to on.
Because the phosphate can be taken away again at any moment by a phosphatase, the same protein can be switched on, off and on again within seconds. Four examples in the course all depend on this mechanism: Hippo–YAP, NF-κB, the CDK cycle, and the signalling cascades. So one reaction does a great deal of the work in this course, and you will recognise it by the shape of the sentence rather than by memorising the list.
Why this course brings it up: Dr. Wong calls kinase and phosphatase “fast on/off switches”. They are the backbone of the signalling in Lecture 5, and they are one of the categories of distractor pairing he lists. **Appears in:** Ch 1, Ch 2, Lect 1

▪ Ku

In one sentence: The protein complex that is the first to grip the two broken ends of DNA in NHEJ.
Detailed explanation: Ku is a pair of proteins (Ku70/Ku80) shaped like a ring, which fits neatly over the end of a broken strand and holds it. DNA-PKcs is then called in, the polymerase and nuclease deal with the ends, and finally the ligase joins the two ends together.
What Ku does is claim the position first, so that the ends cannot be chewed up before they are joined. It is the difference between a hand shielding a fragile end and a hammer swinging at it. Ku binding is also why NHEJ can work in G1: even with no sister chromatid available, the broken ends have something guarding them and can be joined directly.
Why this course brings it up: Dr. Wong does not name Ku as an exam topic on its own, so it belongs to conceptual understanding. The characteristics of NHEJ (no template needed, available throughout the cycle, error-prone) are the Tier 2 pairing. **Appears in:** Ch 6, Lect 4

▪ KU and DNA-PKcs

In one sentence: The two members of NHEJ that protect the broken ends.
Detailed explanation: Ku sleeves onto the two broken ends first, and DNA-PKcs is the kinase that works with it, passing the signal on and controlling how the ends are processed. Together their effect is this: in G1, in S and in G2 they fit the ends with a protective cap, do the minimum tidying up, and let ligase join them up.
It is fast, and it often leaves a deletion or an insertion behind. They only occupy the ends for a limited time, and that window is exactly what decides whether the cell has time to switch to HR instead: take the factors away and the cell is pushed down the longer excision route. So the competition between the two pathways is decided by a clock, not by a vote.
Why this course brings it up: He does not list these two names as exam questions on their own, so it is concept-only. The comparison of HR against NHEJ is where he is most likely to put a multiple choice question. So learn the comparison and not the two names. **Appears in:** Ch 6, Lect 4

▪ Lac operon

In one sentence: The classic bacterial switch, where several genes for using lactose are opened by one signal.
Detailed explanation: The three genes lacZ, lacY and lacA share one promoter and one operator, and are transcribed together into a single polycistronic mRNA. For strong expression you must satisfy two conditions at once: there is lactose present (allolactose pulls the Lac repressor away), and glucose is low (cAMP rises, CAP–cAMP binds the promoter and recruits polymerase).
With one condition missing you are left with only a weak basal level. The design makes sense: the bacterium only builds a lot of sugar-digesting machinery when it is genuinely short of sugar. Three genes, one promoter, one operator, one message — and two conditions on the switch. The design is an AND gate in bacteria: two conditions, one output.
Why this course brings it up: In Lecture 5 he stresses the "two conditions, not one" point especially hard, and it is also an entry in the scenario case bank; he attaches IPTG, an artificial inducer, to this system for recombinant expression, and that artificial inducer is the giveaway that this system was built for the lab as much as for the bacteria. Which is why Case 16 exists, and why he keeps repeating the phrase. **Appears in:** Ch 7, Lect 5

▪ Lac repressor

In one sentence: A protein that sits on the operator and blocks the lac genes, until allolactose removes it.
Detailed explanation: While the Lac repressor is sitting on that stretch of DNA called the operator, polymerase simply cannot get through to the genes behind it. The real inducer is not lactose itself but the isomer it turns into on its own, allolactose: once allolactose binds the repressor, the repressor's shape changes and it leaves the operator.
In recombinant experiments the artificial inducer IPTG is used instead, and because it is not metabolised it lets you control the moment expression starts with real precision. So there are two separate things to keep apart here: the switch itself, and the thing that releases the switch. The repressor is the switch, and whether anything is bound to it is the question you ask yourself every time you set up the system. What makes the whole design economical is that nothing has to be built from scratch when the signal arrives.
Why this course brings it up: It is the classic case of a switch and the release of that switch being separate things. IPTG appears on Dr. Wong's pET system slide in Lecture 5, with 100 μM as the optimum point for expression. So if you can explain what the repressor is holding back and what lets it go, you have the whole mechanism. **Appears in:** Ch 7, Lect 5

▪ Lactate

In one sentence: The molecule a cancer cell makes when it ferments sugar instead of burning it.
Detailed explanation: With oxygen, a normal cell can make about 36 ATP from one glucose. A cancer cell does something different: it sends about 85 % of its pyruvate off to become lactate and lets only about 5 % into the mitochondrion, so the yield drops to about 4 ATP.
That looks like waste, but fermentation keeps the half-broken-down sugar inside the cell, and those intermediates are exactly the building material for nucleotides, lipids and amino acids. A cell that is about to divide tens of billions of times needs far more raw material than fuel, and that is what the trade is for. The number to remember is the contrast itself, 36 ATP against 4, with the rest of the carbon going into building blocks rather than into energy.
Why this course brings it up: He says the details of the metabolic pathways will not be examined, "Please don't memorize all these chemical structure", but lactate and the other metabolites are the entry point to biomarker discovery. His examples include the volatile organic compounds in exhaled breath, used to detect lung cancer or gastric cancer without taking a sample. So the pathway is not the point, the measurement is. **Appears in:** Ch 10, Lect 1

▪ Lactose

In one sentence: The milk sugar that E. coli can use as an energy source, and the reason the lac operon exists.
Detailed explanation: Lactose is a disaccharide made of one glucose and one galactose, and neither half can be used until the bond between them is cut. Before E. coli can use it, the sugar has to be cut in half, and the three genes of the lac operon are really three parts of the same livelihood: cut the sugar, transport the sugar in, and deal with the half that is left over.
The cell is only worth running this machinery when it is genuinely short of sugar, so the default state of the switch is off.
Why this course brings it up: It is the inducer of the lac operon and the substance behind his Lecture 5 scenario question about reading the logic of the lac operon. Dr. Wong also points out explicitly that IPTG is an artificial inducer that is not metabolised. **Appears in:** Ch 7, Lect 5

▪ LAG3

In one sentence: An inhibitory checkpoint receptor, and like PD-1 one of the brakes on a T cell.
Detailed explanation: LAG3, lymphocyte activation gene-3, sits on the surface of the effector T cell, and when it is bound by a ligand in the tumour environment the activity of the T cell goes down.
His slide 63 puts CTLA-4, PD-1, LAG3, TIM3 and VISTA side by side as the inhibitory checkpoints, and all of them are listed among the reasons CAR-T cannot get into a solid tumour. You only get the firepower of the T cell back by blocking these checkpoints, and the drugs that exist at the moment mostly target CTLA-4 and the PD-1 and PD-L1 line. The list is long, and the reason it matters is that each of these brakes can be pressed on the T cell at the same time, so a T cell in a tumour can be held down by several of them at once. The rest are on the list but not yet on the market.
Why this course brings it up: Dr. Wong does not name LAG3 as an exam question on its own, so it is conceptual understanding, and hot and cold tumour along with the checkpoint drugs is Tier 2, and the five names together are worth learning as a group rather than one at a time. **Appears in:** Ch 9, Lect 2

▪ Lagging strand

In one sentence: The strand that cannot be copied continuously and has to be built out of short Okazaki fragments.
Detailed explanation: DNA polymerase can only add nucleotides to a 3′ end, so synthesis always runs 5′→3′. But the two template strands run in opposite directions, so one of them is forced to work in pieces: for each piece, primase lays down a short RNA primer, the polymerase can then extend from it, and at the end ligase seals the gap between one fragment and the next. The lagging strand is therefore slower and more troublesome than the leading strand. Each fragment needs its own starting point, and each of those starting points has to be removed afterwards and the gap filled and sealed, so the lagging strand carries more steps, more enzymes and more chances to go wrong. Which is why the lagging strand is the one that shows up in the questions: it is the strand that explains why copying has to be done in pieces at all.
Why this course brings it up: He says it very directly in Lecture 4 — "I think one concept you'll need to understand is there's two different strands. So one is the leading strand, and the other one is the lagging strand". This is a Tier 2 concept, and it is also the basis of the 5′→3′ versus 3′→5′ distractor pair. It is also the strand that explains the end-replication problem, because the last fragment on it is the one that cannot be finished. **Appears in:** Ch 5, Lect 4

▪ Laser capture microdissection (LCM)

In one sentence: Cutting a clean region out of a tissue slide under the microscope, so that other cells cannot contaminate it.
Detailed explanation: Under the microscope you first circle out a small patch where the morphology is consistent (a pure nest of cancer cells, for example), and then a laser cuts that patch away and puts it onto another glass slide.
A tumour specimen contains not only cancer cells but also cancer-associated fibroblasts, immune cells and extracellular matrix, and all of those contribute protein signal. The purpose of LCM is therefore to physically separate the wanted from the unwanted before any measurement is done, rather than trying to subtract the noise afterwards.
Why this course brings it up: In Lecture 2 Dr. Wong lists it as one of the core messages of “separating the right cell population is half of success”. He also points out that the homogeneous regions cut out by LCM keep the tumour specimen from being polluted by stromal and immune cells. So it is a technique you should be able to name and place. **Appears in:** Ch 9, Ch 11, Lect 2

▪ LATS1

In one sentence: The kinase in the Hippo pathway that phosphorylates YAP and holds it in the cytoplasm.
Detailed explanation: LATS1 is the second stop on the Hippo signalling route. MST1 is phosphorylated and activates SAV, SAV activates LATS1, and LATS1 puts a phosphate onto YAP. The phosphorylated YAP then binds 14-3-3, stays in the cytoplasm and is degraded, so TEAD has no partner and does not start transcription, and the cell stops proliferating.
When LATS1 fails, unphosphorylated YAP slips into the nucleus, binds TEAD and opens CTGF, JAG1, which is a Notch ligand, NOTCH2 and other genes. LATS1 is the Hippo brake, YAP is the thing it holds down, and between them they keep the signal to keep dividing outside the nucleus, where it cannot do any work on the DNA. So the kinase and its substrate are a matched pair, and remembering which one is which is enough to reconstruct the whole pathway from the start, without needing the rest of the cascade.
Why this course brings it up: Hippo and YAP is a pathway Dr. Wong returns to again and again in Lectures 1 and 5. He also links Hippo failure with tissue overgrowth and with enlarged livers in mice. **Appears in:** Ch 7, Lect 1, Lect 5

▪ LC-MS/MS

In one sentence: A method that cuts proteins into fragments, weighs them and reads out their sequence.
Detailed explanation: The first stage of the mass spectrometer separates the molecules by their mass-to-charge ratio. The second stage takes the ions you selected, breaks them up, reads the pattern of the fragments, and compares that pattern against a database to work out which stretch of sequence it was.
His proteomics workflow is 2D-PAGE, which separates the proteins by charge and by molecular weight, then cutting out that spot, then digesting it into peptides, and then LC-MS/MS to identify them. It can handle a mixture of thousands of proteins at once, which makes it far better suited to screening for biomarkers than Edman degradation, where each run peels off one amino acid at a time.
Why this course brings it up: It is the identification step in his serum vimentin case in Lecture 1. The case questions have also

asked which workflow produced a given candidate list, and the answer is 2D-PAGE plus LC-MS/MS. So you need to know the order of the workflow, not the chemistry inside the instrument. **Appears in:** Ch 2, Ch 11, Lect 1

▪ Leading strand

In one sentence: The strand that can be copied smoothly and continuously.
Detailed explanation: When the replication fork is running in the direction of this strand, the 3′ end of the new strand keeps pointing towards the fork, so the polymerase only has to sit down once and write all the way to the end, needing a single RNA primer.
And because the polymerase can only add nucleotides to a 3′ end, the “continuous” part is really a consequence of which way the template runs; it is not a special talent of the enzyme. Meanwhile the fork still has to open up the duplex ahead of it: helicase does the unwinding, and topoisomerase releases the twisting tension that builds up in front.
Why this course brings it up: Together with lagging strand it forms the two concepts Dr. Wong asks you to tell apart in Lecture 4, and it is the pairing that multiple-choice questions use most often (5′→3′ extension, 3′→5′ proofreading). So the distinction is worth having as a reflex rather than as a fact to look up. **Appears in:** Ch 5, Lect 4

▪ Ligand

In one sentence: The molecule that binds a receptor. It is the message, not the receiver.
Detailed explanation: The relationship between ligand and receptor is like a key and a lock: the shape of the ligand is exactly what the binding site of the receptor can take, and once they are joined the receptor changes shape and only then does the signal travel onwards.
In this course the ligand can be a hormone, a drug, a growth factor, or even a foreign small molecule. His lectures split the idea into two positions: a ligand outside the cell binds a receptor on the cell membrane, and a ligand inside the cell binds a nuclear receptor in the cytoplasm. Either way the rule is the same, the ligand brings the information and the receptor does the work.
Why this course brings it up: In Lecture 5, when he teaches the nuclear receptor, what the LBD binds is the ligand, a hormone or a drug, and only after binding does the conformation change and recruit a co-activator or a co-repressor. **Appears in:** Ch 7, Lect 5

▪ Ligand-binding domain (LBD)

In one sentence: The part of a nuclear receptor that holds the hormone or the drug, and changes shape while doing it.
Detailed explanation: A nuclear receptor has three domains: the N-terminal NTD regulates, the middle DBD recognises the response element in the DNA, and the C-terminal LBD grips the ligand. As soon as a hormone or a drug enters the LBD, the receptor puts on a different face: the AF-2 that was tucked into a crevice becomes exposed, and it recruits a co-activator (switching it on) or a co-repressor (switching it off).
The same receptor can therefore do opposite things depending on which ligand it has taken. So the receptor is not a simple on/off switch; it is a shape that a ligand can set. So the same receptor can be a brake or an accelerator depending on what is sitting in its pocket.
Why this course brings it up: In Lecture 5 he sets NTD, DBD and LBD side by side and puts DBD versus LBD on his list of the contrasts most likely to appear in a multiple-choice question; tamoxifen being an antagonist in breast and a partial agonist in the endometrium works precisely because the LBD changes face. **Appears in:** Ch 2, Ch 7, Lect 5

▪ Light chain

In one sentence: The two shorter chains of an antibody, sitting at the tips of the arms.
Detailed explanation: Every antibody has four chains: two long heavy chains and two short light chains, and the two kinds together make the Fab, which is the end that actually catches the antigen, so the light chain is not a decoration but half of the business end of the molecule.
The light chain is the smaller of the two, and it contributes half of the antigen-binding tip. The antigen specificity of an antibody is hidden in the three hypervariable loops CDR1, CDR2 and CDR3, and CDR3 is the most variable of them. On mass, Fv is about 25 kDa, Fab about 50 kDa and Fc about 50 kDa. The other end, the Fc, is the part that calls up the immune cells: IgG1 triggers ADCC through the Fc receptor, and IgG4 barely binds the Fc receptor at all and instead has a longer half-life in the blood.
Why this course brings it up: This is the basic vocabulary of antibody structure. Dr. Wong also uses the light chain to explain CAR, because the scFv of a CAR is a single-chain variable region made by joining VH and VL with a linker. **Appears in:** Ch 2, Ch 6, Lect 1

▪ linker

In one sentence: A short connecting peptide that joins two pieces of protein together.
Detailed explanation: In an scFv, the linker joins the antibody heavy chain VH to the light chain VL, making it a single protein that can be expressed while keeping its original antigen-binding ability.
In an ADC, the linker joins the antibody to the payload; change it to a branched linker and you can hang two different drugs off it at once. The length and the flexibility of the linker also affect whether the whole scFv can fold properly, so it is not just glue — it has to be taken into account at the design stage. A linker that is too stiff can stop the two halves folding into the shape they need.
Why this course brings it up: He does not name linker as an exam topic on its own, so it is a concept to understand; the linker chemistry of ADCs he states plainly is not examined. What matters is the idea that the joint has properties of its own. **Appears**

in: Ch 9, Ch 10, Lect 2

▪ Lipid

In one sentence: A family of molecules that fear water: they build walls, store energy, and also carry signals.
Detailed explanation: A fatty acid is a long carbon chain with a carboxyl group at one end; with double bonds it is unsaturated, without them it is saturated. Three fatty acids plus glycerol make a triglyceride, which is pure energy storage, while two fatty acids plus a phosphate make a phospholipid, with a water-loving head and a water-fearing tail, so it is amphipathic.
The cell membrane can assemble itself into a bilayer precisely because of that amphipathic nature, because each molecule puts its head in the water and its tail away from it. LPS, a component of the bacterial outer membrane, is also a lipid, and it starts NF-κB through TLR4, setting off inflammation and in the worst case septic shock. So the lipids are not only walls and stores; the outermost layer of a bacterium is made of them, and that layer is exactly what the immune system reads as danger.
Why this course brings it up: Dr. Wong says the chemical structures and the metabolic pathway details are not examined, but the immune half of the lipid story is worth remembering: LPS → TLR4 → NF-κB → inflammation → septic shock is a causal chain that comes back in Lecture 5. **Appears in:** Ch 1, Lect 1

▪ Lipid bilayer

In one sentence: The two-molecule-thick sheet that makes up the cell membrane.
Detailed explanation: The hydrophilic heads of a phospholipid face outwards and the hydrophobic tails face inwards, and the membrane arranges itself into two layers on its own, with no machinery to build it, because the chemistry of the molecules makes that arrangement the cheapest option.
The hydrophobic region between the tails keeps water out on both sides. That wall is also a door: the proteins embedded in the membrane decide what is allowed in or out, and it is also a signal board, because receptors, G-protein-coupled receptors, PI(4,5)P and PLC all live in this layer.
Why this course brings it up: In Lecture 1, when he follows PLC cutting PI(4,5)P into DAG and IP3, the membrane is the stage, and DAG stays on the membrane and switches on PKC. That is the foundation of understanding signal transduction. **Appears in:** Ch 1

▪ Lipid nanoparticle (LNP)

In one sentence: A lipid bubble that carries mRNA into cells. Used for in-vivo CAR-T and for mRNA vaccines.
Detailed explanation: mRNA delivered on its own is broken down immediately and cannot get into cells at all. The LNP wraps the mRNA up, and because the cell membrane is made of lipids the cargo is delivered through it;
once inside the endosome the mRNA is let out and translated by the ribosome. In the design of in vivo CAR-T, the mRNA encodes a fragment of an anti-CD5 antibody, which recognises the T cell and delivers the CAR's mRNA into it, so the CAR-T cell is manufactured inside the patient. The vehicle is the same in both cases; only the cargo changes, and that is the whole trick.
The problem it solves is not the mRNA itself but the membrane that mRNA cannot cross on its own.
Why this course brings it up: He mentions this route in Lecture 2 and again in Lecture 5: in vivo CAR-T does away with the ex-vivo culture and the frozen shipment, and an mRNA vaccine uses the same vehicle, the only difference being that its mRNA encodes an antigen. So a question about mRNA vaccines is a question about this delivery system. **Appears in:** Ch 9, Lect 2, Lect 5

▪ Lipoprotein

In one sentence: A particle wrapped in protein that carries lipids around the blood.
Detailed explanation: Cholesterol and triglycerides do not dissolve in water in the blood, so they have to be carried inside particles made of a phospholipid monolayer plus apolipoprotein. The protein on the surface of the particle is at the same time the docking place for enzymes and the label that receptors recognise, so the wrapper is not just packaging, it is the part that makes the particle work.
That is why cholesterol always turns up in the same sentence as HDL and LDL, and it is also the basis of cardiovascular risk assessment.
Why this course brings it up: It is background vocabulary for Part 1 basic chemistry and for the biomolecules of Lecture 2. Dr. Wong does not name it specially, so it is conceptual understanding. **Appears in:** Ch 1

▪ Liquid biopsy

In one sentence: Looking for tumour material in a blood sample instead of cutting a piece out of the tumour.
Detailed explanation: A solid tumour keeps shedding cancer cells into the blood, and those are the circulating tumour cells (CTC). They can be separated from the white blood cells by being EpCAM positive and CD45 negative, and their DNA can then be sequenced.
So the doctor can take blood repeatedly and track the disease, instead of asking the patient to have another biopsy or another scan. One thing to be careful about: a CTC may already have gone through EMT, so its markers may not match those of the primary tumour. Dr. Wong's slide groups the patients by CTC count, and the patients with CTC ≥ 5 have the worst survival.
Why this course brings it up: In Lecture 2 he puts it inside the framework of tying cell biology to real clinical applications. He also flags a trap: a CTC has been through EMT, so its marker expression may not be the same as that of the primary tumour. That is the detail worth carrying, because it stops you treating the blood sample as a perfect mirror of the tumour. **Appears in:** Ch 9, Lect 2

▪ LncRNA (long non-coding RNA)

In one sentence: RNA longer than 200 nucleotides that does not make protein.
Detailed explanation: It is a message that will never be used as a message, and the 200-nucleotide line is where it separates from the small RNAs. It works in two ways. A scaffold ties two or more proteins together in the right place, and a guide carries proteins to the words of DNA they are supposed to act on.
His slides list four archetypes: APOLO forms an RNA-DNA hybrid, PNCTR uses the YYUCUY motif, a sequence recognised by splicing factors, to bind PTBP1, a splicing factor, and control splicing, and XIST silences an entire X chromosome. The two modes are complementary, because a scaffold needs a guide to tell it where to be, and a guide needs a scaffold to hold the machinery together when it gets there.
Why this course brings it up: Lecture 5 lists it as the modern frontier of gene expression regulation, and points out that it is an emerging drug-discovery target and a biomarker. The point to take away is that the interesting RNA functions are not about making more protein. They are about deciding what happens at one particular place on the DNA, and that is a completely different kind of drug target. **Appears in:** Ch 7, Lect 5

▪ Log-rank test

In one sentence: The statistical test that compares two survival curves and produces a p value.
Detailed explanation: It does not compare survival at one particular time point, it compares how far apart the two curves are overall, and that is the difference between a test that asks about one date and a test that uses the whole curve: the time at which each patient has an event, and whether that patient was censored, both go into the model, and the question is whether the difference between the two groups accumulates over time.
A small p value means that the two curves separating is unlikely to be random fluctuation. Dr. Wong writes the result of the log-rank test straight onto the graph, so the dMMR against pMMR plot is marked $P<0.001$.
Why this course brings it up: Dr. Wong teaches KM reading together with the log-rank test, and whether the two lines separate is exactly the core of the scenario question he names. **Appears in:** Ch 9, Ch 12, Lect 1, Lect 3

▪ LPS (lipopolysaccharide)

In one sentence: A molecule on the bacterial outer membrane that, once recognised by TLR4, starts the NF-κB inflammatory cascade.
Detailed explanation: The outer membrane of Gram-negative bacteria contains LPS. It is carried by LBP (an LPS binding protein), handed to CD14 (the LPS co-receptor) and MD-2 (an LPS binding subunit), and then presented to TLR4, which dimerises.
After that there are two routes: the MyD88-dependent one (MyD88 being an adaptor) switches on NF-κB and the pro-inflammatory cytokines; the MyD88-independent one (via TRAM, an adaptor, to TRIF, an adaptor) also switches on NF-κB and adds type I interferon on top. An excess of LPS in the blood causes septic shock. So the molecule on the outside of a bacterium is, from the immune system's point of view, an instant signal of foreignness. The whole cascade is built for speed and for amplification, because a bacterium that gets into the blood has to be met hard and fast.
Why this course brings it up: This is the starting point of the possible exam stem he lists, "LPS → TLR4 → IKK → IκB → NF-κB → nucleus"; he says not to memorise the order of the cytokines, but you should recognise this cascade. And it is the same cascade that the IκB entry ends with. So recognise the cascade, and do not worry about the order of the cytokines.
Appears in: Ch 1, Lect 1, Lect 5

▪ Luciferase

In one sentence: The reporter gene most often used in promoter experiments, because it gives off light.
Detailed explanation: The logic of a reporter assay goes like this: you put a stretch of the candidate promoter sequence in front of the luciferase gene, transfect it into cells, and then measure the light that comes out.
The light is a proxy for how active the promoter is. Then you do serial deletion, removing a small piece of the promoter at a time, and whichever deletion makes the light drop sharply tells you which stretch of sequence was necessary, for example which variant of the TATA box was actually doing the work. Nothing about this needs the real gene to be present; the light is doing the reporting. And the deletion series is the part that makes it an experiment rather than a measurement, because each round removes one candidate and asks the same question again.
Why this course brings it up: In Lecture 5 Dr. Wong calls this set of reporter experiments “the standard experimental logic for locating promoter elements”. He does not ask you to memorise any particular construct, but the way of thinking that goes with promoter mapping has to be understood. **Appears in:** Ch 7, Lect 5

▪ Lysine

In one sentence: The amino acid that carries most of the histone marks. Its side chain is the handle the mark hangs on.
Detailed explanation: The tails of the histones stick out beyond the nucleosome, and that amino acid side chain is the chemical hook. The epsilon-amino group at the end of the lysine side chain can accept an acetyl group, a methyl group or ubiquitin, which is why the histone code always looks like H3K9me3: H3 is histone 3, K is lysine, 9 is the position, me3 is trimethylation.
Its ionised, positively charged form also attracts the negative charge of the DNA, and that attraction is the basis of the packaging. So lysine is doing two jobs at once, it is the surface the modifications are written on, and it is one of the main reasons the DNA and the histones hold each other together in the first place.
Why this course brings it up: Without the lysine positions you cannot write a histone code. In Lecture 3 he treats the way you

read H3K9me3 as compulsory, and he connects the histone code with DNA methylation through the cross-talk that makes cellular memory. The two are not separate subjects, so if you can explain one you should be able to explain how they reinforce each other. **Appears in:** Ch 4, Lect 3

▪ Lysosome

In one sentence: The acidic recycling and disposal plant inside the cell.
Detailed explanation: It is a vesicle wrapped in a single membrane with an acidic interior full of digestive enzymes. Its job is to take material swallowed from outside the cell, worn-out organelles, which is autophagy, and old proteins, and break them down into amino acids and monosaccharides that go back into the cytoplasm to be used again.
The acidity is maintained by proton pumps in the membrane, and the digestive enzymes are only active in that acidic environment.
Why this course brings it up: It is part of the organelle list from Lecture 2. What appears alongside it is the destination of the signal peptide, because a protein with an N-terminal signal peptide can travel through the endoplasmic reticulum and the Golgi to the lysosome, to the cell membrane, or out of the cell. **Appears in:** Ch 8, Lect 2

▪ M phase

In one sentence: The part of the cell cycle in which the nucleus divides. Mitosis is the first half of it.
Detailed explanation: M phase contains mitosis (chromosome segregation) and cytokinesis (division of the cytoplasm). He emphasises particularly that these two are independent: inhibiting the contractile ring only leaves the cell binucleate, while the chromosomes still separate normally.
In a DNA content analysis, G2 and M cells both carry 4N, so flow cytometry can only report them together as a G2/M peak. Chromosome separation is done by the spindle fibres, while the contractile ring only divides the cytoplasm in two. The two processes are controlled by separate proteins. Which is a distinction worth making early, because the flow cytometry histogram you will meet later lumps them together and you have to know why. Two processes, one phase, two sets of proteins.
Why this course brings it up: Telling mitosis apart from cytokinesis is one of the five questions he names out loud (from Lecture 2); he says that last year only half the students got it right. Which is why the flow cytometry peak is called G2/M and not G2 and M. **Appears in:** Ch 8, Lect 2

▪ Macromolecule

In one sentence: A very large molecule, such as a protein or a nucleic acid, built out of many small units.
Detailed explanation: When we say macromolecule, the word "large" is relative. Proteins are built from amino acids, nucleic acids from nucleotides and polysaccharides from monosaccharides. What follows from building these four families out of repeating building blocks is that they can do three jobs at once: they can form structure, as cellulose does, they can store information, as the sequence of DNA does, and they can catalyse reactions, as the active site of a protein does.
And because the units repeat, the order in which they are arranged is where the biological information lives.
Why this course brings it up: Dr. Wong opens Lecture 1 with the four-family classification, and that is the skeleton he gives to the whole set of terms. He does not name this word on its own. **Appears in:** Ch 1

▪ Maintenance methyltransferases

In one sentence: The enzymes that copy the existing methylation marks onto the new strand.
Detailed explanation: DNA replication pulls the two strands apart, and each new strand comes with only half of the old marks. DNMT1 is what recognises, at the replication fork, that this side used to carry a methyl group, and then puts the same methyl group on at the matching position of the new strand.
That is the mechanism by which a mark can be handed down through cell division, and it is the basis of epigenetic inheritance and of cellular memory. Compare it with the de novo enzymes, which write a new pattern where there was none. One copies and one creates, and the cell needs both.
Why this course brings it up: The difference between de novo and maintenance is one of the 19 multiple choice pairings he lists, at Tier 2. One of nineteen, which means it is worth getting exactly right. **Appears in:** Ch 4, Lect 3

▪ manufacturing cycle

In one sentence: The five steps that carry CAR-T from the patient and back into them.
Detailed explanation: Dr. Wong's slide lists them: take the T cells out of the patient with a blood draw; introduce the CAR with a virus or with electroporation; culture up millions of CAR-T cells;
transfuse them back into the patient; and then wait a while to see the tumour response. The whole workflow is where the cost and the time of an autologous CAR-T come from, and it is where the US$370,000 to US$475,000 per course of treatment comes from. The more steps there are, and the more of them that have to be handled in a sterile environment, the higher the time and the cost, which is also why people have proposed making CAR-T inside the patient's own body, in vivo. The in vivo idea removes the middle of the list entirely.
Why this course brings it up: He does not name the manufacturing cycle as an exam topic on its own, so it belongs to conceptual understanding. He does name the figure, though: “CAR-T costs US$370,000 to US$475,000 per course of treatment”. **Appears in:** Ch 9, Lect 2

▪ Mass spectrometry

In one sentence: The technology that weighs molecules and then identifies them from the pattern of fragments they make.
Detailed explanation: The instrument first separates the molecules by mass-to-charge ratio and counts how many of each there are, then it breaks up the ones you selected and reads the composition of the fragments, and comparing that against a database gives you the identity.
Once you add chromatography, LC or GC, a single instrument can read out thousands of different molecules in the same sample. Both his metabolomics workflow and his proteomics workflow are built around it as the core measuring tool. The logic of the second step is what makes it an identification method and not just a weighing method: the pattern of fragments is a fingerprint that the database already knows.
Why this course brings it up: The second step of the metabolomics workflow in Lecture 1 is instrument analysis, NMR, GC-MS and LC-MS. He pairs the question of what you read small molecules with, and the question of what you read proteins with. That pairing is the thing to remember, because the same instrument is not the answer to both questions. **Appears in:** Ch 2, Ch 11, Lect 1

▪ Maternal inheritance

In one sentence: The rule by which a mitochondrial disease passes from an affected mother to all of her children and from an affected father to none.
Detailed explanation: The mitochondria brought in by the sperm are destroyed after fertilisation, so mtDNA travels almost entirely down the maternal line. There are three features to read off a pedigree: an affected mother passes the disease to all of her children;
an affected father passes it to none; and the disease appears in every generation. Clinically you also have to take the threshold effect into account, because the disease only appears once the proportion of mutant mtDNA exceeds what that tissue can tolerate. Skeletal muscle has a high energy demand and therefore a low threshold, so it is affected first, while the threshold for hearing is higher and the ear is affected later.
Why this course brings it up: It is one of the five questions Dr. Wong personally named, because it doubles as a pedigree scenario. The turn he emphasises is this: mitochondrial disease can also come from a mutation in a nuclear gene, and then the inheritance pattern is Mendelian, so you cannot look at the pedigree alone and conclude. **Appears in:** Ch 8, Lect 2

▪ Matrix (mitochondrial matrix)

In one sentence: The liquid interior of the mitochondrion, where the TCA cycle runs and where mitochondrial DNA lies.
Detailed explanation: A mitochondrion has three layers of structure: the outer membrane, the intermembrane space and the inner membrane, and the inner membrane folds inward into cristae, and that is where the power station of oxidative phosphorylation sits, with the ATP synthase particles embedded in the inner membrane.
The matrix is the liquid space enclosed by the inner membrane, and it is where the TCA cycle runs and where the circle of roughly 17 kb of mtDNA and its own ribosome live, almost completely separated from the metabolism of the cytoplasm. It is also the checkpoint for apoptosis, because once cytochrome c leaks out of the intermembrane space the cell starts its own execution programme. So the matrix is both the metabolic kitchen and the place where the mitochondrion keeps its own genetic material, and the two of them sit in the same liquid.
Why this course brings it up: The organelle list of Lecture 2 pairs each organelle's function with its clinical meaning. The matrix is the scene he uses when he explains that a mitochondrion has its own DNA and its own machinery. **Appears in:** Ch 8

▪ MCP-1

In one sentence: The chemokine that calls white blood cells to a particular place.
Detailed explanation: A chemokine is the dispatch signal between cells, and MCP-1 is the muster call. It is responsible for the last of the four steps by which a white blood cell leaves the blood vessel and enters tissue: first P-selectin, E-selectin and integrin make the leukocyte roll, then ICAM-1 and VCAM-1 provide adhesion and activation, then PECAM-1 gets it through the vessel wall, and finally the cell follows the concentration gradient of MCP-1 towards the site of inflammation.
Four steps, four molecules, one journey. And the gradient is what makes the journey directional, since a cell cannot know how far away the signal is.
Why this course brings it up: He puts these four molecules side by side on the four-step diagram of leukocyte extravasation in Lecture 2; the pathway explains how a tumour summons immune cells, and also how MDSC gets called in. **Appears in:** Ch 9, Lect 2

▪ MDSCs

In one sentence: Cells the tumour recruits to turn the immune response down.
Detailed explanation: Myeloid-derived suppressor cells are a group of immunosuppressive immune cells that come from the myeloid line. The tumour microenvironment recruits them, and they use TGF-β, IL-10 and the PD-L1 on their surface to press down on T cell activation, so the tumour escapes immune surveillance.
This is one of the reasons CAR-T therapy fails against solid tumours: his CAR-T limitations slide lists Treg, MDSC and TAM alongside hypoxia, low pH, IDO and adenosine as the obstacles. Note what they are doing: they are not attacking the T cell, they are simply removing the signal that would make it active.
Why this course brings it up: He lists MDSC as one of the mechanisms of immune escape in both Lecture 2 and Lecture 3. The HDAC inhibitor slide in Lecture 3 writes that MS-275 induces M1 polarisation through STAT1 and inhibits MDSC. The HDAC

slide is the same idea in the opposite direction: take away the suppressor and the immune response comes back. **Appears in:** Ch 4, Ch 9, Lect 2

▪ median overall survival

In one sentence: The median overall survival of a group of patients.
Detailed explanation: This is the endpoint reported most often in clinical trials, and the median means that half the patients lived longer than it and half lived less long. In the trial of Rasonque, daraxonrasib, there were n = 500 patients with previously treated metastatic pancreatic cancer, and the median overall survival on the new drug was 13.2 months against 6.7 months for standard chemotherapy, which is roughly twice as long.
The advantage of the median is that it is not thrown off by the extremes, while the mean can be pulled out of shape by one or two patients who lived an unusually long or unusually short time.
Why this course brings it up: "Rasonque 13.2 against 6.7 months" is one of his Tier 3 precise numbers. The reason he quotes the median rather than the mean is worth noticing, because the same trial reported in the other way round would not tell the same story. **Appears in:** Ch 10, Ch 12, Lect 1

▪ medicinal chemistry

In one sentence: The stage where a target is turned into a compound that is actually fit for the job.
Detailed explanation: A target that is biologically meaningful still has to be turned into a molecule with the right shape, the right solubility and the right potency before it can be a drug at all.
This is the stretch of work that comes after target identification and before preclinical; it belongs to the drug company's pipeline and is usually not what a biologist is doing. The failure rate at this stage is the highest of the whole pipeline, so the company runs many chemical series at the same time, and only a few molecules are ever pushed as far as the animal study stage. Most of what is tried here never leaves the building. Biology says what should be blocked; chemistry has to produce something that can actually be given to a person.
Why this course brings it up: He does not name medicinal chemistry as an exam topic on its own, so it is a concept to understand; what he does name is the order of the whole pipeline. Knowing the order is often what a question asks for. **Appears in:** Ch 12, Lect 2

▪ MEI

In one sentence: Mobile element insertion, the mechanism by which a transposon creates a CNV.
Detailed explanation: A transposon is first transcribed into RNA, and then reverse transcriptase turns it back into DNA and inserts it into a break in another chromosome. After the insertion that stretch has one extra copy, and that is how a duplication is formed.
This is one of four CNV-generating mechanisms listed alongside NAHR, FoSTeS and NHEJ. Unlike NAHR, MEI does not need sequence homology, so it can happen anywhere in the genome, and for that reason it can also bring about a larger change. And because the copy lands somewhere it did not belong, the gene there now has an extra copy sitting next to it, which is the situation that can matter for expression.
Why this course brings it up: He says outright that the molecular mechanisms of CNV are not examined, he says he is only giving you a little taste. But NAHR and adaptive CNV, the other two concepts, he did teach, so they are concept-only. The list of four is context and the two concepts are the teaching. **Appears in:** Lect 3

▪ Meiosis

In one sentence: The special division in a germ cell that makes four genetically different gametes.
Detailed explanation: Compared with mitosis, meiosis does two things: the first division separates the homologous chromosomes and the second separates the sister chromatids, so one diploid starting cell ends up producing four haploid gametes.
Inserted between the two divisions is a crossover, in which the maternal and paternal homologues exchange segments and reshuffle the alleles. Add random segregation on top of that, plus random fertilisation, and genetic diversity comes close to unlimited. So meiosis is not simply mitosis done twice; the reduction in number and the shuffling are both part of the job.
Why this course brings it up: In Lecture 2 Dr. Wong links meiosis directly to the main thread of diversity. Its two divisions and the question of non-disjunction are among the questions he names (M1 vs M2 in Lecture 3). So both halves of it, the halving and the shuffling, are worth being able to explain. **Appears in:** Ch 8, Lect 2, Lect 3

▪ Meiosis I and meiosis II

In one sentence: The first division separates homologous chromosomes; the second separates sister chromatids.
Detailed explanation: The difference between the two divisions is this: in meiosis I the homologous chromosomes (each carrying one chromatid from each parent) come apart, and in meiosis II the sister chromatids come apart, with no further crossover along the way.
The difference shows up very clearly in the gametes: a failure of M1 segregation gives 2 normal and 2 abnormal (the abnormal ones all of the same type), while a failure of M2 segregation gives 1 normal and 3 abnormal (two of one type and one of another). Counting the gametes is the fastest way to remember which is which. One division deals with pairs, the other with copies.
Why this course brings it up: It is one of the five questions he names out loud, and he says you have to be able to tell an M1

failure from an M2 failure; it is also what his Lecture 3 scenario question, a pedigree plus aneuploidy, is asking for. **Appears in:** Ch 3, Ch 8, Lect 3

▪ Melting temperature (Tm)

In one sentence: The temperature at which half of a DNA double helix has come apart.
Detailed explanation: As the temperature rises the two strands separate, and this process is called denaturation or melting. A G–C base pair has three hydrogen bonds and an A–T pair has only two, so the higher the GC content, the higher the temperature you need.
That single number decides the annealing temperature of a PCR and the design of the primers, and it is also used to judge how similar two sequences are, which makes it one of the most quoted numbers in molecular biology. The reason is the number of hydrogen bonds, and that is why the melting temperature is a statement about the sequence rather than about the length.
Why this course brings it up: In Lecture 3 Dr. Wong lists the direction "more GC, higher Tm" among his precise numbers, and it connects to the PCR primer design of Lectures 1 and 4. **Appears in:** Ch 3, Ch 11, Lect 1, Lect 3

▪ Membrane protein

In one sentence: A protein sitting in the plasma membrane, often a receptor. Because a drug can reach it from outside the cell, it is an excellent drug target.
Detailed explanation: The transmembrane part of a membrane protein is hydrophobic and very hard to dissolve for analysis by mass spectrometry, and that is the technical difficulty of membrane proteomics.
But that is exactly why they are worth chasing: GPCR, receptor tyrosine kinase and ion channels are all the interfaces between the cell and the outside world, and a drug that binds on the outside of the cell works without ever having to cross the membrane. Amplification of HER2 is precisely a way of increasing the number of receptors on the cell surface.
Why this course brings it up: In Lecture 2 Dr. Wong says explicitly that membrane proteins are prime drug targets and cell-surface biomarkers. It is also the basis underneath HER2/trastuzumab and receptor occupancy (2% in breast cancer, 10% in gastric cancer). So when he explains why an antibody can treat a cancer cell from the outside, this is the entry point. The membrane is both the reason the drug works and the reason it is hard to analyse, and knowing which side of a protein you are dealing with tells you which kind of drug you can write. **Appears in:** Ch 2, Ch 12, Lect 2

▪ Mesoderm

In one sentence: The middle germ layer. It becomes muscle, bone and blood.
Detailed explanation: Of the three germ layers, mesoderm sits in the middle, and the tissues it produces include muscle, bone, cartilage, blood and most of the cells of the lymphatic system, as well as the kidney and the reproductive system.
The layering matters experimentally: if you inject iPSC into an immunodeficient mouse and a teratoma forms, then finding tissue from all three germ layers inside the tumour proves that this batch of cells really does have pluripotency. One injection, one tumour, and the answer to the question is read off the insides of it.
Why this course brings it up: The teratoma assay is the gold-standard validation method for iPSC that he specifies in Lecture 2, and he says do not forget it. The three germ layers are the criterion you read it against, so the assay and the three layers are the same fact. **Appears in:** Ch 8, Lect 2

▪ Metabolism

In one sentence: All the chemistry of the cell: breaking things down to get energy, and building things up.
Detailed explanation: Metabolism has two opposing forces. Catabolism takes big molecules apart and extracts energy (glycolysis and the TCA cycle produce ATP), while anabolism spends energy to assemble small units into big molecules (nucleotides, lipids, amino acids, the raw materials a cell needs if it is going to divide).
What is most unusual about cancer is not its energy efficiency but the way the balance between the two has been rewritten: it gives up energy production in order to keep the intermediates as building material. Breaking down to build up, rather than breaking down to burn. Two directions, one budget.
Why this course brings it up: He uses the Warburg effect in Lecture 1 to explain how cancer rewrites its metabolism, and he keeps coming back to it in Lectures 2 and 5; metabolomics therefore became the main route for biomarker discovery. Which is why metabolomics became the natural way to look for biomarkers. **Appears in:** Ch 1, Lect 1

▪ Metabolite

In one sentence: A small molecule that the cell makes or uses up as it works.
Detailed explanation: A metabolite is an intermediate or an end product of metabolism: glucose, pyruvate, lactate and acetyl-CoA are all of them. There are a great many of them, they are small, and they move quickly in response to what the cell is doing at that moment.
It is an ideal target for a biomarker for exactly that reason, because it reflects the state of the cell now rather than its genomic potential. Two patients with exactly the same genes can have completely different metabolites, so the disease has a chemical fingerprint.
Why this course brings it up: In Lecture 1, when he teaches metabolomics, Dr. Wong gives lactate and the volatile organic compounds in exhaled breath as the examples, and he also stresses that a biomarker only counts once it has been validated. The validation point is the one beginners skip, and it is the difference between an interesting correlation and something you can act on. **Appears in:** Ch 10, Ch 11, Lect 1

▪ Metabolite fingerprint

In one sentence: The pattern of small molecules that makes one sample different from another.
Detailed explanation: Measure all the metabolites in a sample at once and you get a compositional fingerprint. Line up the fingerprints of many samples, compress them with PCA into a few axes, and you can see which samples cluster together and which are outliers.
PCA turns many measurements into a few axes, so that the picture can be looked at. To pick out the differential metabolites you use a volcano plot, with the cut-offs p < 0.05 and FDR < 0.15, and then cross it with VIP > 1.0, and the numbers shown on the slide are 10, 50 and 25. The logic is the standard one: statistical significance on one axis, effect size on the other, and you want the features that pass both.
Why this course brings it up: He does not use this name. What it corresponds to is step five of the seven-step metabolomics workflow in Lecture 1, feature extraction, then multivariate stats, then PCA, then heatmap. It is concept-only. The steps before it decide what goes into the fingerprint, and the steps after it decide what you do with the differences you find. The heatmap is the last step, and it is the picture you would put in a paper. **Appears in:** Ch 10, Ch 11, Lect 1

▪ Metabolomics

In one sentence: The study of all the small molecules in a sample. A great many biomarkers are found this way.
Detailed explanation: Dr. Wong's workflow has seven steps: experimental design (a cohort plus animal work), sample collection (urine, blood and cerebrospinal fluid, tissue), sample preparation (homogenisation, liquid-liquid or solid-phase extraction, centrifugation), instrument analysis (NMR, GC-MS, LC-MS), data mining (feature extraction, multivariate statistics, PCA, heat maps), biomarker discovery (volcano plot plus Venn diagram), and multi-centre validation.
The screening thresholds are p below 0.05 and FDR below 0.15, and then the Venn diagram is used to pick out the differential metabolites that also satisfy VIP above 1.0. The whole point of the workflow is to squeeze thousands of small molecules down to a few candidates that can actually be validated. Every step is a filter, and the last one is the filter that matters most: a candidate that only works in the laboratory it was found in is not a test, it is a curiosity. And the sorting is done in two dimensions at once, because a metabolite can be significantly different without being large enough to matter, and large without being reproducible.
Why this course brings it up: He does not mark the steps of the workflow as examinable, but the general rule that a biomarker has to be validated before it becomes a test runs through the whole course. Lactate and VOC are the examples he gives.
Appears in: Ch 10, Ch 11, Lect 1

▪ Methyl group

In one sentence: The tiny -CH3 label that DNA methylation puts onto cytosine.
Detailed explanation: One carbon carrying three hydrogens is a methyl group. Hang it on the fifth carbon of the cytosine and you have methylation. The chemistry that changes is very small, but the consequence is large: the cytosine now looks different in the pairing, so DNA-binding proteins have trouble recognising it and transcription factors cannot get to it.
The carbon source for the methyl is S-adenosyl methionine (SAMe). The smallest possible change, applied to the one atom that decides how the base reads. It is worth pausing on how small the change is, because the size of the consequence is the surprising part: a tag on one atom of one base can silence a gene for a lifetime of cell divisions.
Why this course brings it up: In Lecture 3 he places methylation as an important driver of cancer, and in the epigenetic lecture he calls it the main mechanism of suppressing gene expression. **Appears in:** Ch 4, Lect 3

▪ Methylation

In one sentence: Adding a methyl group onto DNA, usually in order to silence a gene.
Detailed explanation: Where the methylation happens is the whole story, and Dr. Wong is careful about the distinction. If the methylation lands on a CpG island in the promoter, it puts the brake on and the tumour-suppressor gene simply stops talking; that is one of the common ways a cancer silences a gene that was watching its back. If it lands in the pericentromeric repeat region, it is the demethylation there that causes the chromosomal instability, so the same chemistry produces the opposite outcome depending on where it is written. The enzymes divide the work too: de novo methyltransferase establishes new marks, while maintenance methyltransferase copies the marks onto the daughter strand after replication, which is how a dividing cell keeps the same pattern. The two enzyme types are the two halves of the job: one writes the mark and the other copies it, and a cell that has lost the copying enzyme loses the pattern at every division. His example is colorectal cancer, where patients with high CIMP have a worse prognosis, and the measurement can be taken down to single-base resolution by bisulfite sequencing, because the bisulfite chemistry reads the methylation straight off the base. So methylation is not one switch but two, and the location tells you which of them you are looking at. The practical consequence for a beginner is that methylation is the classic case of a gene being switched off without anything changing in the DNA sequence itself.
Why this course brings it up: In Lecture 3 Dr. Wong puts it in the Tier 2 concepts, and his words are: "The most important aspect of the DNA methylation is it has a very significant functions on these gene expressions, because DNA methylation is usually involved in the suppression of the gene expressions". He also gives two numbers, OR 12.7 and OR 39.6, and those are the figures that carry the idea into a multiple-choice question. **Appears in:** Ch 4, Ch 6, Lect 3, Lect 5

▪ MGMT

In one sentence: Its promoter methylation predicts that a brain tumour will respond well to alkylating chemotherapy.
Detailed explanation: The enzyme encoded by MGMT directly repairs the DNA damage done by alkylating drugs. Methylate the promoter and this enzyme is no longer expressed, so the tumour cannot repair the damage the drug causes, and the tumour becomes sensitive to alkylating chemotherapy, temozolomide being his example, and to radiotherapy.
He quotes the numbers from the original paper: methylation compared with unmethylated, prognostic HR = 0.41 and predictive HR = 0.49. MGMT is the simplest repair pathway there is: it takes the alkyl group straight off the base and does not have to cut out a stretch and rebuild it, so it is a direct reversal rather than an excision repair. The clinical logic is a clean two-step: the drug does the damage, and MGMT is the undo button, so if the undo button is gone the damage stays and the cell dies.
Why this course brings it up: This is one of the case questions in his Lecture 3 and in the case bank: an MGMT methylation result predicts sensitivity to alkylating chemotherapy plus radiotherapy. He also insists that you keep the expression of a repair gene and the position of the methylation separate in your head, because they are different facts and the question usually hinges on which one you are being asked about. The patient A and patient B diagram he draws makes the contrast concrete. The number to hold is that methylation switches the tumour from resistant to sensitive, and the test is usually asked about as the MGMT methylation result, with temozolomide as the drug. **Appears in:** Ch 4, Lect 3

▪ MHC class I

In one sentence: The display cabinet that carries fragments of proteins from inside the cell out to the surface, where killer T cells can look at them.
Detailed explanation: MHC class I is present on the surface of almost every nucleated cell. It loads up fragments of proteins that have been chopped up inside the cell and delivers them to the surface for the cytotoxic T lymphocyte to inspect.
This is the route by which neoantigens become visible to the immune system: the more fragments there are, the more neoantigens get displayed. And because every nucleated cell carries MHC class I, no infected or mutated cell can hide, so the immune system gets the chance to check them one by one. The system is a catalogue rather than a search, and the size of the catalogue is set by how many protein fragments the cell happens to have.
Why this course brings it up: Dr. Wong does not name MHC class I as an exam topic on its own, so it belongs to conceptual understanding. The chain “more neoantigens → hot tumour → responds to anti-PD-1” is the one he explicitly marks as Tier 2.
Appears in: Ch 6, Ch 9, Lect 4

▪ MHC class II

In one sentence: The display cabinet that shows protein fragments taken in from outside the cell to helper T cells.
Detailed explanation: The opposite of MHC class I, MHC class II is used mainly by the antigen-presenting cells, the dendritic cells among them, and it displays protein fragments swallowed in from outside the cell to CD4 helper T cells.
In his account of the TCR, the CD4 T cell reads MHC class II through the αβ heterodimer of its own TCR. In practice anti-tumour vaccines and some therapeutic antibodies make use of exactly this channel: put a tumour antigen into MHC class II and you raise CD4 T cells to help out.
Why this course brings it up: Dr. Wong does not name MHC class II as an exam question on its own, so it is conceptual understanding, but the pairing with MHC class I is the one to hold on to, because one shows what is happening inside the cell and the other shows what was swallowed from outside. **Appears in:** Ch 6, Lect 4

▪ MHC (major histocompatibility complex)

In one sentence: The surface proteins that display a small fragment of protein so that T cells can inspect a cell.
Detailed explanation: MHC I molecules display short fragments of proteins from inside the cell on the surface of every nucleated cell; MHC II is used by antigen-presenting cells to display fragments of foreign protein.
What the T cell receptor recognises is the combination of "fragment plus MHC", so a cell has to lay out its internal state for the immune system to inspect. So the immune system does not see a cell; it sees a cell's molecules displayed on a stand. A tumour that loses MHC I has effectively broken the lock, and so escapes the cytotoxic T cells. And because the display is selective, what ends up on the stand decides what the immune system reacts to.
Why this course brings it up: In Lecture 5, when he talks about mRNA vaccines, he points out that the last step is the antigen being presented on MHC and starting T and B cells; this is also the theoretical starting point of neoantigen immunotherapy.
Appears in: Ch 6, Ch 9, Lect 5

▪ Microarray

In one sentence: A chip carrying thousands of gene probes, used to measure which genes are switched on.
Detailed explanation: There are two ways of doing it. An oligonucleotide GeneChip converts the sample's mRNA into biotin-labelled cRNA and uses a single colour. A cDNA microarray instead labels normal and cancer tissue with two different fluorophores, Cy5 and Cy3, mixes them, and lets them hybridise competitively on the chip, so that whichever position shows more of one colour tells you which gene is stronger in which group.
The same slide also lays out the RNA-seq workflow, so that students can tell the difference between two-colour competitive hybridisation and single-colour labelling. The practical difference is the same as the visual one: in the two-colour version every sample has to be compared against a common reference, because the colour only tells you about the difference between two things on the same chip.
Why this course brings it up: Dr. Wong's miRNA biomarker workflow is a three-stage validation: tumour versus adjacent tissue

→ cell line supernatant → patient serum before and after surgery, and the first stage uses this kind of tool for the overall comparison. He also warns that adjacent tissue is not the same thing as normal tissue. **Appears in:** Ch 11, Lect 1

▪ MicroRNA (miRNA)

In one sentence: A short RNA about 22 nucleotides long that blocks the translation of its target mRNA.
Detailed explanation: Its production is a chain. RNA polymerase II makes the primary miRNA. DGCR8, the partner protein, and DROSHA form the microprocessor, the pri-miRNA splicing complex, which cuts it into a hairpin pre-miRNA.
exportin-5 with RanGTP carries it out of the nucleus. DICER1, the miRNA cutting enzyme, with TRBP, an RNA-binding protein, cuts it into a double strand, the two strands separate and the guide strand is the one kept. And then it is loaded onto AGO1 to AGO4, together with GW182, the RISC effector protein, to form RISC. It works by pairing with the seed sequence, positions 2 to 8, on the target 3′ UTR. The pairing is checked only over those seven nucleotides, which is why a single miRNA can have so many targets: all that has to match is the seed, anywhere in the 3′ UTR. Partial complementarity represses translation; complete complementarity leads to the message being cut up. miRNA is quite stable in serum, so it is often used as a non-invasive biomarker. His hepatocellular carcinoma example is miR-15b and miR-130b, with a combined classifier AUC of 0.981. Every step has a named protein, and the names matter because the whole chain is the standard way of describing miRNA action, which is how the questions are usually phrased.
Why this course brings it up: miRNA is unusually stable in serum, so Lecture 1 uses miR-15b plus miR-130b to build an HCC classifier with AUC 0.981, sensitivity 98.25%, specificity 91.53% and accuracy 94.83%. His Lecture 5 also lists it as one layer of message regulation, alongside the other RNA classes, and the point he makes there is that a small RNA can regulate many targets at once. Those four numbers are on his list of precise numbers, so they are worth having exactly, and the workflow that produced them is second-level (A) material. The direction of the effect is the other half of the story, because partial pairing represses translation while complete pairing destroys the message. And the production chain is worth knowing as a chain, because a question can drop you into the middle of it and ask which enzyme comes next. **Appears in:** Ch 7, Lect 1, Lect 5

▪ Microsatellite

In one sentence: Short repeating stretches of DNA. When mismatch repair fails their length changes, and that is how MSI is measured.
Detailed explanation: These are short tandem repeats of two or three nucleotides scattered around the genome, for example TATATATATATA or GTCGTCGTCGTC. They behave like a ruler: two people can carry a different number of repeats at the same position, which makes them usable as a DNA fingerprint.
During replication the polymerase tends to slip in a repeat region, inserting or deleting one repeat unit, and the job of MMR is to put it back. A ruler with a movable mark, and the error rate decides how far the mark drifts. So the repeat count is a number you can read, and MSI is nothing more than the rate at which those numbers have drifted.
Why this course brings it up: In Lecture 4 he explains why failure of MMR causes MSI: the changes in repeat length that cannot be repaired are the basis of the measurement, and the cause of MSI-H tumours. **Appears in:** Ch 6, Lect 4

▪ Microsatellite instability (MSI)

In one sentence: The change that short repeated DNA undergoes once mismatch repair has failed. An MSI-high tumour is a hot tumour.
Detailed explanation: MMR fails, so nobody repairs the insertions and deletions in the repeat regions, frameshifts result, the protein is either truncated or given an odd new stretch, the neoantigen load goes up sharply, the tumour becomes hot and it responds to anti-PD-1.
The proportion he gives is that about 15% of colorectal cancer is MSI-H. The progression-free survival curve of dMMR against pMMR in Lecture 4, P<0.001 by log-rank test, he calls one of the most important pictures in modern medicine. So the label describes the tumour rather than the patient, and it is a behaviour rather than a mutation: nothing was deliberately changed, the repair system simply stopped correcting a small error, and the errors accumulated over time. Because the shifts are in frames, the protein built from them is not the protein that was written, and the altered protein is what the immune system sees as new.
Why this course brings it up: This is the central mechanism behind the hot and cold tumour story of his Lecture 5, and it is also one of the entries in his scenario bank: MMR/MSI result, then MSI-H, then hot tumour, then anti-PD-1, with CEA as the surrogate marker. The chain is worth being able to recite in order, because each step is a different kind of failure and each one can be examined on its own, and one of them asks you to think about what you would use instead. **Appears in:** Ch 6, Ch 9, Ch 10, Lect 4

▪ Microtubule

In one sentence: A hollow tube made of protein. It forms the mitotic spindle, and it is also a track for cargo.
Detailed explanation: A hollow tube built from tubulin, with a GTP end at one side and a GDP end at the other, constantly assembling and disassembling inside the cell. The spindle microtubules grab the centromere of the chromosomes and pull the sister chromatids towards the two poles.
At the same time they act as tracks for organelle transport, with kinesin and dynein dragging cargo along them across the cell. Vimentin turns up in the HCC case in Lecture 1 because the intermediate filaments are often rearranged in tumours. So the same cell has two kinds of scaffolding in it, one built from tubulin and one built from intermediate filament proteins, and the spindle is the part that divides the cell.
Why this course brings it up: In Lecture 2 Dr. Wong uses the separation of mitosis from cytokinesis to explain what the spindle

and what the contractile ring each do, and that is exactly the skeleton of the answer to a question he names. So the spindle is the half that belongs to mitosis. **Appears in:** Ch 8, Lect 2

▪ Misfolding

In one sentence: A protein that has not folded into the right three-dimensional shape. It is one of the risks of a therapeutic protein.

Detailed explanation: During recombinant expression a protein may fold wrongly and aggregate into an inactive clump, an inclusion body, and that costs you both yield and activity: the machine produced it and it cannot start work.

His approach in the E. coli system is to delay induction until the exponential growth phase, to avoid the misfolding that heavy pre-expression brings; and at the sequencing stage he also checks for chemical changes such as deamidation. So there are two separate questions about a manufactured protein: does it get made, and is what gets made any good? And the sequence of the checks matters, because the cheapest thing to do first is simply not to make too much protein too early.

Why this course brings it up: In Lecture 1 he puts deamidation and inclusion body side by side as the risks of a therapeutic protein; he also mentions that CHO cells need codon optimisation before there is any yield. **Appears in:** Ch 2, Lect 1, Lect 5

▪ misincorporation

In one sentence: The polymerase putting the wrong base onto the new strand.

Detailed explanation: It is the first step of a copying error, and it is also where the nucleoside analogue drugs make their move: the drug is taken for a nucleotide, gets mixed in, and the polymerase hooks it onto the end of the new strand.

After that there are two roads. If the exonuclease activity of p53 cuts the analogue back out, the cell develops drug resistance; if the DNA synthesis that follows is blocked instead, the signal turns into cell death signalling. The same analogue in the same cell can give different results, which says that the effect of the drug depends not only on the drug but on the state the cell happens to be in. One drug, two outcomes, decided by what the cell does next. That is why the same molecule can be a therapy in one context and the reason for resistance in another.

Why this course brings it up: He does not name misincorporation as an exam topic on its own, so it is a concept to understand; what he has taught is the fork: excision → resistance, blocked synthesis → cell death. That fork is the examinable shape of it. **Appears in:** Ch 5, Lect 4

▪ Mismatch repair (MMR)

In one sentence: The repair pathway that puts right the mistakes left behind after replication.

Detailed explanation: During replication the polymerase occasionally pairs the wrong base, for example A against G. The MMR machinery recognises that mismatch, cuts out the wrong stretch of it (exonuclease), and then the polymerase puts the right base back in and ligase seals it up.

The machine works with MSH2–MSH6 or MSH2–MSH3 recognising the mismatch and MLH1–PMS2 doing the excision. Without it, the length of repeat regions accumulates errors, and that is MSI. Loss of MMR is also the cause of Lynch syndrome. About 15% of colorectal cancers are MSI, and because they carry many neoantigens they become hot tumours that respond well to checkpoint inhibitors. So the repair defect is not only a thing that goes wrong in the cell; it is the thing that makes the tumour visible to the immune system, because a genome that keeps shifting produces novel peptides for the T cells to recognise. The damage and the opportunity come from the same fault, which is one of the more satisfying points in the whole course: the same broken machine that makes the cancer dangerous also makes it visible.

Why this course brings it up: MMR is one of the four repair pathways in Lecture 4, and with BER, NER and HR it forms the set of distractor pairings Dr. Wong lists. MMR defect → MSI-H → hot tumour → anti-PD-1 is a complete causal chain, and it is the chain you should be able to write out in one line. **Appears in:** Ch 6, Ch 9, Lect 4

▪ Missense mutation

In one sentence: A single-letter change that swaps one amino acid for another.

Detailed explanation: Once a base is replaced, the codon changes, the amino acid that comes out of the translation changes, and the three-dimensional shape and the function of the protein can change with it.

K-Ras G12C is the standard example: the glycine at position 12 is replaced by cysteine, and Ras can no longer switch itself off. The change is called missense because the codon still makes sense, it just makes a different amino acid. Contrast the two neighbours in the family: a nonsense mutation manufactures a stop codon and truncates the protein, and a frameshift shifts the whole reading frame and produces nonsense after it. Missense is the quiet one, because the protein is still made and it still looks like a protein. How much damage that does depends on whether the position you replaced was a critical one, because some missense mutations do nothing at all.

Why this course brings it up: Dr. Wong puts missense alongside frameshift and nonsense as the kinds of mutation, and the KRAS G12C example is from Lecture 1, where it connects to the continuous growth signalling of a cancer cell. **Appears in:** Ch 3, Lect 3

▪ Mitochondrial DNA (mtDNA)

In one sentence: The small circular DNA inside the mitochondrion, about 17 kilobases with 37 genes. You inherit it from your mother.

Detailed explanation: A circle of double-stranded DNA, about 17 kb, carrying 37 genes, most of them tRNA and components of the respiratory chain, and it encodes its own rRNA and tRNA. What makes it special is that it is passed on from the mother only, in a single line, and that every cell carries many copies, so the same person can carry normal and mutant mtDNA side by side. That state is called heteroplasmy. How severe the disease is depends on whether the proportion of mutant mtDNA crosses a threshold that is specific to each tissue, and that is the threshold effect. So the number of copies is what turns a mutation into a diagnosis.

Why this course brings it up: It is at the core of his Tier 1 questions: mtDNA and nuclear genes follow different patterns of inheritance and have different threshold behaviour, so you cannot read a pedigree and draw a conclusion from the shape alone. This is one of the places where the shape of the family tree is genuinely misleading, so the rule is to check which kind of DNA you are looking at before you interpret it. **Appears in:** Ch 8, Lect 2

▪ Mitochondrial Eve

In one sentence: The one woman from whom all existing mitochondrial DNA is said to descend.

Detailed explanation: Because mtDNA travels almost entirely through the maternal line, all the mtDNA in living humans can be traced to one maternal lineage, and mitochondrial DNA went through a "bottleneck" somewhere between thousands and tens of thousands of years ago, leaving only a few women, after which no mtDNA from any other source is seen.

Her existence is inferred backwards from the differences in mtDNA; she is not a person found in a fossil. The name makes her sound like a person; the evidence is a family tree written in the mitochondrial DNA. It is an inference, not a discovery.

Why this course brings it up: In Lecture 2 he uses her to make the point that all human mitochondria come from a single maternal ancestor, which is where the inference about maternal inheritance comes from; his slide also mentions that rare cases of paternal mtDNA transmission really do exist. **Appears in:** Ch 8, Lect 2 **Video:** V04_Mitochondrial_pedigree.mp4

▪ Mitochondrion

In one sentence: The power station. It makes most of the ATP, controls apoptosis, and keeps its own DNA.

Detailed explanation: Three identities in one organelle. The ATP synthase on the inner membrane generates power from the proton gradient; when the cell has to execute itself, the mitochondrion releases cytochrome c, which triggers the apoptosome and caspase 9;

and it carries its own circular DNA and its own ribosome. Its origin is endosymbiosis: an ancestral eukaryotic cell swallowed a bacterium and the two of them ended up living together, which is the answer to why a bacterium should be inside one of our cells at all, and the circular DNA is the leftover of that arrangement. The number of mitochondria in a cell also rises with exercise training, and that increase is called biogenesis. One organelle, then: both the factory that makes ATP and the switch that turns apoptosis on.

Why this course brings it up: In Lecture 2 Dr. Wong teaches the origin of the mitochondrion, maternal inheritance and threshold-effect disease as one complete case, and his Tier 1 question turns on the point that mitochondrial disease can also come from nuclear genes. **Appears in:** Ch 8, Lect 2

▪ Mitosis

In one sentence: The division of the nucleus, in which the replicated chromosomes separate.

Detailed explanation: In prophase the chromatin condenses into visible chromosomes and the spindle forms; in metaphase the chromosomes line up on the equatorial plate; in anaphase the sister chromatids are pulled towards the two poles;

in telophase the nuclear membrane reforms. Mitosis only touches the chromosomes and the nucleus, and whether the cytoplasm splits is a separate matter, which is why the questions Dr. Wong sets can be split cleanly in two. The nuclear division and the cytoplasmic division are two independent jobs, even though in a normal cell they happen one after the other.

Why this course brings it up: The separation of mitosis from cytokinesis is one of the five questions he personally named (Lecture 2). He says that last year only half the students got it right. **Appears in:** Ch 8, Lect 2

▪ Mitotic clock

In one sentence: The counting mechanism that turns cell division into a ceiling on how many times one cell may divide.

Detailed explanation: The ends of the chromosomes carry telomeres made of 3 to 20 kb of the TTAGGG (telomere repeat) sequence. Because the polymerase cannot start from scratch at the very end (the end-replication problem), every division shortens the telomere by about 50 to 200 bp.

Once it reaches a critical value (the Hayflick limit), the cell stops dividing and enters senescence. Stem cells and germ cells make up the length with telomerase, and around 90 % of cancers reactivate telomerase. In hepatocellular carcinoma, chronic injury makes hepatocytes proliferate compensatorily and telomeres wear down; later, when MYC is upregulated the mechanism for lengthening telomeres is switched back on, telomeres lengthen again, and that turns out to be a prognostic marker. Which is a strange inversion worth noticing: in this tumour, the telomeres that grew longest belong to the patients who did worst. So the clock is not just a countdown to senescence; it is a thing a tumour can rewind, and the rewinding can itself be a bad sign.

Why this course brings it up: It ties ageing and cancer together in Lectures 3 and 4; he also uses telomere length in HCC as a prognostic analysis — the longest 10 % of telomeres actually have worse recurrence-free survival (HR 1.66, p = 0.029). So the numbers on this entry are worth having: 50 to 200 bp per division, 3 to 20 kb of repeat, and about 90 % of cancers switching

telomerase back on. **Appears in:** Ch 5, Lect 3, Lect 4

▪ mitotic recombination

In one sentence: The exchange of chromosome segments that happens during mitosis.
Detailed explanation: When the repeated regions of pericentromeric heterochromatin lose their methylation, so they become hypomethylated, the cell can allow an exchange between the repeats during mitosis, and the chromosomes are then reassorted with duplications and deletions appearing.
That is one of the sources of genomic instability, and also one of the mechanisms for generating CNV. Which is the same thing said the other way round: genomic instability and copy number variation are two faces of one event. The loss of methylation matters because the repeats are what make an exchange possible in the first place, and once they are free of the marks that keep them quiet they can line up with each other.
Why this course brings it up: He does not name mitotic recombination as an exam question on its own, so it is concept-only, and he says outright that the molecular mechanisms of CNV are not examined. So the link between instability and copy number is the idea, and the enzyme names are not. **Appears in:** Ch 8, Lect 3

▪ Mitotic spindle

In one sentence: A rope of microtubules that pulls the copied chromosomes apart.
Detailed explanation: The spindle is made of microtubules and associated proteins. In metaphase it lines the chromosomes' centromeres up on the equatorial plate, and in anaphase the microtubules shorten and pull the sister chromatids towards the two poles.
Its assembly and breakdown are dynamic rather than fixed, and for a chromosome to end up correctly distributed into two daughter cells this machine has to grab each sister chromatid in the right place. It is a pulling machine, not a pushing one, and everything about cell division depends on the pulling being aimed correctly.
Why this course brings it up: It is part of the mitosis and cytokinesis explanation in Lecture 2, and the exam question he names has the answer that the chromosomes separate normally but the cytoplasm fails to divide. The spindle is what separates the chromosomes, so it is the part that has to be working when the answer says the separation was fine. **Appears in:** Ch 8, Lect 2

▪ Mlh1

In one sentence: The mouse Mlh1 gene, used to build an animal model with MMR deficiency.
Detailed explanation: His experiment compares three kinds of mouse: wild-type, the ones carrying proofreading-defective Pole and Pold1, and the ones with Mlh1 knocked out. The third group is exactly the model of human MMR deficiency, and when the Pole e/e and Pold1 e/e double mutations are stacked on top of it, survival is shortest and tumour formation highest, which proves that proofreading and mismatch repair are two independent lines of defence.
So the experiment asks a simple question: if you break both defences, is the outcome worse than breaking either one alone, and the answer is yes. The knockout stands in for the human disease because the human MMR genes work the same way, and Lynch syndrome is what happens when one of them fails. When he compares the three groups of mice, this knockout model is what stands in for the MMR deficiency of human Lynch syndrome.
Why this course brings it up: Dr. Wong does not name Mlh1 as an exam question on its own, so it is conceptual understanding, and the counter-intuitive conclusion that a proofreading defect produces an ultramutator with a better prognosis is the point he singles out. **Appears in:** Ch 6, Lect 4

▪ MLH1/PMS2

In one sentence: The second safety net of mismatch repair, the one that does the cutting.
Detailed explanation: MSH2 and MSH6 form MutSα (the mismatch recognition complex), which finds the mismatch first, and MLH1 and PMS2 form MutLα (the mismatch excision complex), which takes over from there.
When the polymerase has mispaired a nucleotide on the lagging strand, or slipped out a small loop, these two sets of proteins cut out the wrong piece and have it resynthesised. MLH1 expression can itself be pushed down by promoter methylation, and that is the real cause in a great many MSI tumours. MutLα does not recognise the error itself; it is called in by MutSα to do the job, and this layering of “one recognises, one cuts, one refills” is the typical division of labour in a molecular machine. One protein cannot do everything, and splitting the job is what makes the machine fast and specific. It is the same three-step shape you meet everywhere in the course: something recognises, something acts, something restores. The cell trusts that division to this kind of layering throughout. It is the same three-step shape you meet everywhere in the course: something recognises, something acts, something restores.
Why this course brings it up: Dr. Wong does not list these four genes as topics on their own, so it belongs to conceptual understanding. The chain MMR defect → MSI-H → frameshift → neoantigen → hot tumour is his clearest one. **Appears in:** Ch 5, Ch 6, Lect 4

▪ MMR deficiency

In one sentence: The state in which the mismatch repair system has failed.
Detailed explanation: Copying DNA goes wrong now and then anyway: the polymerase occasionally puts in the wrong base, and on repeated sequences it occasionally slips. In the normal case MMR catches those errors and fixes them.

MMR deficiency is when that checkpoint is missed, and mismatches and small indels then pile up, especially in microsatellite regions, where the length of the repeats starts changing back and forth — and that is MSI. The system is not there to prevent errors, it is there to catch the ones that slip through. Without it, the errors that were rare become the normal state of the cell.
Why this course brings it up: It is a core Tier 2 concept and also the category he names as situation question 11; he also remembers that 15 % of colorectal cancers are MSI-H. That number is one to have ready. **Appears in:** Ch 6, Lect 4

▪ MOA

In one sentence: Mechanism of action, the evidence for what this compound actually does.
Detailed explanation: MOA and toxicology are the two cornerstones of the preclinical stage: the first shows that the drug really acts on the intended target in cells and in animals, and the second shows what side effects it has in a whole animal.
Together they are what gets handed to the regulator as the IND-enabling package. MOA has to come first because the same compound can act in more than one way, and if you do not know how it acts, every dose decision after that is guesswork. The order matters and it is not bureaucratic: you cannot design a dose schedule for a mechanism you have not identified.
Why this course brings it up: He does not name MOA as an exam question on its own, so it is concept-only. The phasing of clinical trials and the safety questions are the parts that will be examined. So the vocabulary is useful and the examination is elsewhere. **Appears in:** Ch 12, Lect 2

▪ Molecular biology

In one sentence: The study of life at the level of molecules, above all DNA, RNA and protein.
Detailed explanation: This course is the worked example. From the base sequence of DNA, through the splicing and the stability of RNA, to the folding of proteins and the drug targets they become, every layer is asking the same question: how does this molecular thread turn into a number that can be measured, a treatment that can be given, and a prediction that can be made about what happens to a patient.
That is exactly what Dr. Wong means by “turning genomics and molecular biology into clinical application”. The thread is the same at every level. What changes is only how far down you have travelled to find it.
Why this course brings it up: In Lecture 1 he frames the course objective as “turning genomics and molecular biology into clinical and real-world application”. It is a word for where the course sits, and it will not be asked on its own. But it is the frame that makes the rest of the content hang together. **Appears in:** Ch 1

▪ Molecular cloning

In one sentence: The technique of putting a chosen piece of DNA into a small circular DNA and copying it in large numbers.
Detailed explanation: The whole workflow runs like this. You amplify the fragment of interest (the insert) by PCR. You cut both the donor plasmid and the recipient plasmid with EcoRI and NotI, so that complementary sticky ends are exposed.
You join the two pieces together and seal them with DNA ligase, which gives a recombinant plasmid carrying the insert. You then put the plasmid into bacteria and use the Amp resistance to pick out the ones that succeeded. The bacterium is the copying machine, and a few hours buys you several million copies. Each step has one job, and the whole thing is a chain you can draw from memory.
Why this course brings it up: He does not name molecular cloning as an exam topic on its own, so it is a concept to understand; what it gives you is the picture of how a piece of DNA goes from a paper idea to a factory that can make millions of copies of it. **Appears in:** Ch 3, Lect 4

▪ Molecule

In one sentence: Two or more atoms joined together in a fixed arrangement.
Detailed explanation: Atoms are held together by chemical bonds made out of electrons, and the kind of bond and the distribution of the electrons decide the shape and the properties of the molecule.
What is special about carbon is that it can make four bonds, so it can build long chains, branches and rings, and that is why it can be the backbone of every molecule of life. It is also, for exactly that reason, why all four of the great families of molecules start with carbon.
Why this course brings it up: It is the starting point of Part 1 basic chemistry. When Dr. Wong opens Lecture 1 with the four great families, this is the background you need, and it is never examined on its own. **Appears in:** Ch 1

▪ monocytes

In one sentence: Monocytes, which sit between the granulocytes and the lymphocytes on a flow plot.
Detailed explanation: They are bigger than lymphocytes and more internally complicated than lymphocytes, but they do not have as many granules as the granulocytes, so on an FSC–SSC plot they fall in the middle band.
These three groups, granulocytes, monocytes and lymphocytes, are his standard example of first gating on the scatter and then adding the fluorescence. They are also the precursor cells of macrophages, so the monocytes around a tumour differentiate into TAM and become part of the tumour microenvironment.
Why this course brings it up: Dr. Wong does not name monocytes as an exam question on its own, so it is conceptual understanding. The reason they matter is the second half of the story, because the monocyte is where the macrophage comes from. They are the middle band on the plot. **Appears in:** Lect 2

▪ Monosaccharide

In one sentence: A single sugar unit, such as glucose or fructose.
Detailed explanation: The smallest sugar unit, and the two of them have the same molecular formula, C6H12O6. Structurally, glucose is a six-membered ring, a pyranose, while fructose is a five-membered ring, a furanose, so the formula is the same but the shape is not.
Two monosaccharides joined make a disaccharide, sucrose and lactose, and long chains of them make polysaccharides, glycogen, starch and cellulose. In the body it is the main energy currency and also the unit in which glycogen is stored. The branches of glycogen are made by α1,6 bonds and the straight parts by α1,4 bonds, and glucose-6-phosphatase in the liver and the kidney is what lets the free glucose out. So the same building block can be arranged as a storage polymer, a structural polymer or a fuel, depending on how it is linked and who is doing the linking.
Why this course brings it up: In Lecture 1 he divides the carbohydrates into three layers and uses the disease example of glucose-6-phosphatase deficiency to explain the release of monosaccharide. The details of the metabolic pathway are not examined. The example is the one to remember, because it shows what happens when the last step of a pathway cannot be completed. **Appears in:** Ch 1, Lect 1

▪ Monosomy

In one sentence: A chromosome that is present in one copy rather than two, written as 2n-1.
Detailed explanation: It is one of the abnormalities of chromosome number, and the opposite of trisomy, which has one extra. The most famous example is Turner syndrome, 45,X: there is only one X chromosome, with short stature, incomplete gonadal development and infertility.
On his Lecture 3 slide he puts this karyogram next to the trisomy 21 karyogram of Down syndrome and insists that you look at the number rather than at the structure. The cause is that the chromosomes failed to separate during meiosis, so the gamete came out with one chromosome missing, and after fertilisation the result is 2n-1. In a picture of chromosomes, monosomy is something you can see at a glance, because the count is wrong rather than the shape. That is the whole practical difference between this and a structural variant, where the count is right and the material is not.
Why this course brings it up: In Lecture 3 Dr. Wong uses non-disjunction to explain how aneuploidy arises, and he treats Turner and Down as the two results of M1 and M2 failure, which connects straight to the M1/M2 question he names. **Appears in:** Ch 3, Lect 3

▪ mRNA half-life

In one sentence: How long an mRNA survives before it is broken down. A short half-life means less protein.
Detailed explanation: The lifespan of a eukaryotic mRNA is decided by three features: the 5′ cap, the length of the 3′ poly-A tail, and the RNA-binding proteins and miRNAs that sit on the 3′ UTR.
When the tail is trimmed shorter, or when miRNA and RISC haul the message off to a P-body to be digested, the message retires early. So the same gene can make more or less protein purely because its tail is longer or shorter. And the rule Dr. Wong wants you to hold on to is this: a longer poly-A tail means more translation, not a longer life.
Why this course brings it up: In Lecture 5 he lists mRNA stability as one layer of the control of gene expression. He does not specifically name the term “half-life”, so it belongs to conceptual understanding. But the distinction between a long tail and a long life is exactly the kind of thing that trips people up, so keep the rule. **Appears in:** Ch 7, Lect 5

▪ mRNA (messenger RNA)

In one sentence: A photocopy of a recipe, carried from the nucleus to the factory floor.
Detailed explanation: A photocopy is not the original. The original is the DNA in the nucleus; the photocopy is spliced (introns removed, exons joined), given a 5′ cap and a poly-A tail, and only then sent out of the building.
What it carries is the instructions, not the recipe itself: the ribosome reads it to order the amino acids and folds the bead chain into a machine. The copy is a working document, not a sacred text, and it is edited before it is used. Because each message can be adjusted in length and in fate independently, the cell can regulate a whole set of genes by changing the amount of a few messages.
Why this course brings it up: The central dogma is the skeleton of the whole course; he also uses mRNA to teach codon optimisation, alternative splicing, and the in vivo CAR-T and mRNA vaccine that use LNP to deliver mRNA. So whenever a term about RNA comes up, ask where it sits in this journey. **Appears in:** Ch 2, Ch 3, Ch 7

▪ MSH2/MSH6

In one sentence: The first safety net of mismatch repair, the one that recognises copying errors.
Detailed explanation: MSH2 and MSH6 form MutSα (the mismatch recognition complex), and their job is to identify the base that will not pair up properly on the new strand. Once it has been found, it is handed to MutLα (the mismatch excision complex: MLH1 and PMS2), the exonuclease cuts out the wrong stretch and the polymerase puts the correct bases back.
The other variant, MSH2/MSH3 (MutSβ), handles the looped mismatches that occur in repeat sequences. Recognising a mispaired base and recognising a looped one need different recognition combinations, and it is precisely because there are two kinds that an MMR failure produces both an accumulation of mismatches and a change in microsatellite length at the same time. One broken recogniser can leave you with both symptoms at once, and that is what makes an MMR defect show up as a distinctive laboratory pattern rather than as scattered random damage.
Why this course brings it up: Dr. Wong does not list these genes as topics on their own, so it belongs to conceptual

understanding. The causal chain from MMR failure to MSI is the Tier 2 point. **Appears in:** Ch 5, Ch 6, Lect 4

▪ MSI-H

In one sentence: Microsatellite instability high, the label for a tumour full of unstable short repeats.
Detailed explanation: When mismatch repair fails, the length of the short repeated sequences in the genome, the microsatellites, starts to grow or shrink at random. If a substantial proportion of the microsatellites in a tumour are in that unstable state, it is recorded as MSI-H, and that is about 15 % of all colorectal cancers.
The consequence is a large number of frameshift mutations, and frameshifts make neoantigens, so the tumour becomes a hot tumour and responds to anti-PD-1. So the chain is: a repair enzyme fails, the short repeats start wobbling, the reading frame slips, new proteins appear, and the immune system now has something to see. And because the wobbling is random, every tumour is unstable in a different pattern, which is why the label is a category rather than a measurement of one particular gene.
Why this course brings it up: MSI-H to hot tumour to anti-PD-1 is the core chain he marks as Tier 2, and it is also the dMMR against pMMR survival curve he calls one of the most important figures in modern medicine. So the chain is examinable as a chain, and the curve is the picture that goes with it. **Appears in:** Ch 6, Ch 9, Lect 4

▪ MSS

In one sentence: Microsatellite stable, which means the microsatellites stay stable.
Detailed explanation: MSS is the opposite case: this tumour's mismatch repair is still intact, so even when the polymerase slips, the small loop is caught, cut out and redone, and the length of the repeat sequences stays stable.
MSS and MSI-H are the two ends of the same marker, and they are the pair of words he uses to set a multiple-choice question. Clinical reports often put MSS and pMMR together, because the two describe different layers of the same thing: the first is the phenotype and the second is the mechanism. So you may see either word in a report and be talking about the same tumour.
Why this course brings it up: Dr. Wong does not name MSS as an exam question on its own, so it is conceptual understanding, but the contrast with MSI-H and the contrast with dMMR and pMMR are the same line. **Appears in:** Ch 6, Lect 4

▪ MST1

In one sentence: The kinase at the start of the Hippo pathway. It activates LATS1.
Detailed explanation: MST1 is the first stop on the Hippo message. When cells touch, KIBRA, MER and WILLIN pull MST1 to the junction and phosphorylate it, SAV then passes the signal to LATS1, YAP is phosphorylated and stays in the cytoplasm, and the cell stops proliferating.
The other way round, mice with both Mst1 and Mst2 knocked out develop massive liver growth and tumours. MST1 is the first stop of this switch: only once it has been phosphorylated does it hand the signal to SAV and pass it on down to LATS1 and YAP. Everything about this pathway is a phosphorylation switch, which is why it is his favourite example of a reversible on and off.
Why this course brings it up: In both Lecture 1 and Lecture 5 he uses Hippo as the example of how a reversible phosphorylation switch does tumour suppression. He also mentions that the Hippo pathway is a target he is looking for drugs against. The order of the kinases is the examinable part, and MST1 is where that order begins. **Appears in:** Ch 7, Lect 1, Lect 5

▪ Multinucleated cell

In one sentence: A cell with more than one nucleus, because the cytoplasm never divided.
Detailed explanation: When you block the contractile ring (the ring of actin and myosin), mitosis goes ahead as normal and the chromosomes still separate, but the cell cannot be cut into two daughter cells.
The result is a cell with two or more nuclei and a DNA content of 4N. Flow cytometry then shows a single 4N peak instead of separate 2N and 4N peaks. There are two routes to destroying the ring: inhibit the expression of its proteins, or use an inhibitor directly. Either way the mitosis goes on, which is the proof that the two events are separate. And this is exactly why the flow cytometry plot is diagnostic: a cell with two 2N nuclei looks like one cell with 4N of DNA, so the normal 2N peak and the tumour-like 4N peak are replaced by a single peak sitting at 4N.
Why this course brings it up: It is one of the five questions Dr. Wong personally named (Lecture 2). Last year only half the students got it right, and “mitosis is not cytokinesis” is the complete answer to the question. **Appears in:** Ch 8, Lect 2

▪ Multiplex immunofluorescence

In one sentence: Staining several proteins at the same time, each one in a different fluorescent colour.
Detailed explanation: Each target gets its own antibody carrying a different fluorescent dye, so one staining round shows you several targets at once, for instance whether MST1, LATS and YAP, the three members of the Hippo pathway, are in the same cell.
DAPI stains the nucleus blue and acts as the coordinate system, and a confocal microscope then stacks the different focal planes into a 3D reconstruction, which wide-field imaging cannot do.
Why this course brings it up: In Lecture 2 Dr. Wong uses it to make the point that looking at several proteins in the same cell at once is the only way to tell where a signal came from. In the tumour microenvironment, where cancer cells, immune cells and fibroblasts are all mixed together, that discrimination is exactly what you need, because in a mixed sample the colour tells you which cell a signal came from. **Appears in:** Ch 9, Ch 11, Lect 2

▪ Multipotent

In one sentence: A cell that can become several cell types, but only within one family.
Detailed explanation: Multipotency is a step below pluripotency. A haematopoietic stem cell is the typical example: it can become red blood cells, white blood cells and platelets, but it will not become a neuron or a skin cell.
Adult stem cells have limited self-renewal capacity and usually divide only a limited number of times; ESCs and iPSCs are pluripotent and can differentiate into nearly every somatic cell, yet still cannot form a whole organism. The ladder is the whole point: each rung buys you more cell types and costs you more of your flexibility.
Why this course brings it up: In the iPSC section of Lecture 2 he uses a hierarchy diagram running from totipotent to unipotent to separate these words, and the teratoma assay with three germ layers is the test for pluripotency. **Appears in:** Ch 8, Lect 2

▪ Multivariate logistic regression

In one sentence: The test that tells you whether a marker still gives independent information once several other factors have been taken into account.
Detailed explanation: Univariate analysis looks at one factor at a time and is easily fooled by a confounder. A multivariate model puts all the factors in together and asks whether, once the other variables are controlled for, the marker can still predict the outcome on its own, and every coefficient comes with an odds ratio and a confidence interval.
The practical point is that a marker which looks strong on its own can turn out to be nothing but a shadow of another variable, and only the multivariate model tells you which of the two it is. The CDKN2A methylation model in his Lecture 3 is built exactly this way, so the model does two things at once: it estimates the size of the effect and it shows you how much of that estimate is solid.
Why this course brings it up: His example gives OR 39.6 for CIMP-high against CIMP-0, with a 95% CI of 20.6–76.1 and $p<0.0001$, and OR 12.7 for CDKN2A methylation against loss of p16, with 7.84–20.7 and $p<0.0001$, and both numbers are ready-made multiple-choice material. The confidence intervals are as important as the odds ratios here, because a wide interval is a warning that the estimate is not solid. **Appears in:** Ch 4, Lect 3

▪ Mutagen

In one sentence: Anything that causes a mutation, a chemical substance or a kind of radiation for instance.
Detailed explanation: Endogenous mutagens come from replication errors and ROS; exogenous ones include UV (which makes pyrimidine dimers), ionising radiation (which makes double-strand breaks) and intercalating agents.
A chemical carcinogen works by first forming a DNA adduct: a methyl adduct produced from the cigarette carcinogen NNK by P450 metabolism, if it is not repaired, is fixed as a mutation when the DNA is copied, for example the GC→AT transition in the mouse K-ras gene. So the sequence is always the same shape: damage first, then a copying mistake, then the mistake locked in. That third step is the one that matters, because a lesion that is repaired before it is copied leaves no trace, while a lesion that survives one round of copying is now a permanent change in the sequence.
Why this course brings it up: In Lecture 3 Dr. Wong explains carcinogenesis with the chain of adduct → replication error → fixed as a mutation. He lists examples such as NNK and 4-aminobiphenyl, but he does not ask you to memorise the list.
Appears in: Lect 3

▪ Mutant splice site

In one sentence: A new splice site created by a single-letter change, which produces a shorter isoform than the original.
Detailed explanation: A single nucleotide substitution can create a new splice site inside an intron, so the RNA decides to join up in the wrong place. His hepatocellular carcinoma example is exactly this: a single nucleotide polymorphism inside an intron creates a new splice site and produces a short isoform with exon 7 skipped, 250 bp, against a full length of 425 bp, and patients carrying that variant have significantly worse overall survival, $P<0.001$.
Notice how much damage one letter can do. The gene itself is not broken, most of the protein is still made, and the cell is handed a different version of it, and that version is enough to matter.
Why this course brings it up: In Lecture 5 he links it to neoantigen, because abnormal splicing can produce a protein the immune system has never seen, and it echoes the Lecture 4 thread of neoantigen as fuel for immunotherapy. It is case number 15 in the case bank. So the clinical relevance is not the splicing itself but the new antigen it produces. **Appears in:** Ch 7, Lect 5

▪ Mutation

In one sentence: A typo in the printed manual.
Detailed explanation: Typos get copied, so they are handed down generation by generation — that is the physical basis of an inherited disease. A typo can also be a good one: it supplies variety, and it lets viruses and bacteria survive under immune pressure.
But a typo that lands in an important word can take away the cell's brakes and turn it into a driver. Which is the whole argument of the course in three lines.
Why this course brings it up: He lists the driver versus passenger contrast as a common multiple-choice pairing; he uses NPM1 and the integration of CNV data to teach how to tell the two apart. **Appears in:** Ch 3, Lect 3

▪ Myosin

In one sentence: The motor protein that builds the contractile ring together with actin.
Detailed explanation: The head of myosin is the motor. It walks along actin fibres step by step, spending ATP as it goes, pulling the thin filament towards itself. A dividing cell assembles an actin–myosin contractile ring near the equatorial plate, and the ring contracting produces a furrow that pinches the cell into two.
Block the ring and the furrow does not form, so you get one cell with two nuclei. So inhibiting the ring only ever produces binucleate cells, with the chromosomes still separating perfectly, and that is exactly the evidence Dr. Wong uses for the independence of mitosis from cytokinesis. The two jobs use different machinery, they are triggered separately, and one of them can fail without touching the other.
Why this course brings it up: This is the mechanistic core of one of the five questions he personally named (Lecture 2): the contractile ring equals actin plus myosin, and it does cytokinesis, not mitosis. If you are asked why a blocked cell still separates its chromosomes, this is the answer. **Appears in:** Ch 8, Lect 2

▪ N-terminal domain (NTD)

In one sentence: The regulatory region at the start of a nuclear receptor. It is not the same thing as a signal peptide.
Detailed explanation: A nuclear receptor is divided into sections. The A and B region holds the NTD, which is variable and does most of the regulating, and the AF-1 activation region is also there.
The C region holds the DBD, which is highly conserved and uses a zinc finger to recognise the response elements in DNA. The D region is the hinge. The E region holds the LBD, which grabs the ligand and swaps in a co-activator or a co-repressor. The names are similar but the functions are completely different: the NTD is the receptor's own control region, while a signal peptide is the protein's postal address. One is a switch on the molecule, the other is a shipping label on the molecule, and confusing them is an easy mistake to make in a multiple choice question.
Why this course brings it up: He lists signal peptide against NTD as one of the contrasts most likely to turn up in a multiple choice question, and in Lecture 5 he says explicitly not to confuse them. DBD against LBD is on the list as well, so both of the pairs in that entry are worth learning as pairs. **Appears in:** Ch 2, Ch 7, Lect 5

▪ N-terminal signal peptide

In one sentence: The short postal address at the very front of a protein that routes it to the endoplasmic reticulum and so into the secretory pathway.
Detailed explanation: A newly made protein arrives carrying this sequence, and after SRP recognises it the protein is delivered to the translocation pore of the rough ER and passes into the lumen.
After that it is folded, glycosylated, and then sorted by the Golgi to the cell membrane, to a lysosome, or out of the cell in secretion. Without this address the protein never enters the route at all, and the whole expression of a secretory or membrane protein fails. It is a postal address with a postcode and no return label.
Why this course brings it up: He says it very movingly in Lecture 2 — his postdoc lost four months of work because of it ("signal peptide is really, really important for your study") — and he lists it as a Tier 2 concept you must understand. So the sequence, the receptor and the destination are the three things to hold. **Appears in:** Ch 2, Lect 2, Lect 5

▪ N-terminus

In one sentence: The first end of a protein chain, where the amino acid keeps a free amino group.
Detailed explanation: A peptide chain has two chemically different ends: the N end keeps a free amino group and the C end keeps a free carboxyl group. That difference is useful in two places.
Edman degradation has to peel the amino acids off step by step from the end with the free amino group. And a protein carrying an N-terminal signal peptide is directed to the endoplasmic reticulum by that position. If the signal peptide is cut away, the protein cannot find the way into the ER, and the whole experiment fails to produce anything. Which is why the position of the signal peptide on the chain matters: put it at the other end and the whole addressing system no longer works.
Why this course brings it up: It is the premise of Edman degradation in Lecture 3, and it is the reason the signal peptide and the NTD, which are not the same thing, are so easily confused with one another. **Appears in:** Ch 2

▪ NAHR (non-allelic homologous recombination)

In one sentence: Recombination between two similar but not identical stretches of DNA, which produces chromosomal rearrangements.
Detailed explanation: When two highly repeated sequences line up in the wrong place, the cell swaps segments between them by homologous recombination, and the result is one chromosome with an extra piece and another with a piece missing.
NAHR is one of the four mechanisms by which CNV is produced (the other three are FoSTeS, NHEJ and mobile element insertion). His evolutionary example is that populations which eat a lot of starch carry a higher copy number of AMY1 (the amylase gene). It is the misalignment of two near-identical stretches that makes it possible, which is why repeated sequences are where the rearrangements happen. Duplication and deletion, in other words, rather than a point change.
Why this course brings it up: In Lecture 3, when he covers the CNV mechanisms, he says explicitly that this part is "just a taste" and that nobody is expected to dig deeper; but duplication and deletion caused by NAHR remain the standard explanation of CNV. So the definition is examinable and the details are not. **Appears in:** Lect 3

▪ Natural killer cell (NK cell)

In one sentence: An immune cell that can kill tumour cells without being told which target to go for.
Detailed explanation: It belongs to innate immunity and does not need to be sensitised by an antigen beforehand, because it recognises cells that have lost the signal of self. Tumour cells often lower their MHC I, and an NK cell reads exactly that clue. It kills by releasing perforin and granzyme, and it also releases IFN-γ to activate other immune cells. In this course NK cells have two roles: they are the effector cells against tumours, and they are the executioners borrowed by IgG1 antibodies to do the killing. So an NK cell does not need to know what the tumour is made of, only that the tumour has stopped looking like a normal body cell.
Why this course brings it up: In Lecture 2, when he teaches antibody isotype, he says IgG1, Trastuzumab, recruits NK cells through FcγRIII to carry out ADCC. He also mentions CAR-NK as another answer for solid tumours alongside CAR-T. That recruitment is why the antibody subclass matters, because IgG4 cannot do it. **Appears in:** Ch 2, Ch 9, Lect 2

▪ Necrosis

In one sentence: Uncontrolled cell death after an injury, in which the contents leak out and set off inflammation in the surrounding tissue.
Detailed explanation: The difference from apoptosis is in two places. Necrosis is not under programme control: the cell membrane breaks down first, the contents spill out and an inflammatory reaction follows.
Apoptosis is the cell tidying itself up: the membrane stays intact, the contents are packaged into apoptotic bodies and swallowed, and very little inflammation results. In a tumour, necrosis often produces hypoxia and acidification, and hypoxia is the signal that makes HIF-1α accumulate. So the two deaths look different from the outside: one is tidy and quiet, the other is messy and inflammatory, and that difference is the reason the two words exist.
Why this course brings it up: Dr. Wong does not name the word specially. It is standard background set against apoptosis, in Lecture 4 and in the cell biology chapter, and it is the premise of what he says about HIF-1α and the hypoxic tumour microenvironment. **Appears in:** Ch 8, Lect 2

▪ Neoantigen

In one sentence: A fragment of protein the immune system has never seen, usually produced by a frameshift or by a new splice site.
Detailed explanation: Normal protein fragments were catalogued by the immune system long ago as its own, so they are not attacked. A neoantigen is a product distorted by a mutation: a frameshift moves the whole sequence out of register and bolts on an unfamiliar stretch, and a new splice site gives the protein an extra piece or takes a piece away and changes its appearance. Both routes end the same way: the protein that reaches the surface is not the protein that was written. Once these fragments are presented by MHC, cytotoxic T cells recognise them and attack, and a T cell that has never met the peptide before will react to it. The count matters more than the size of any single fragment, which is why a mutation that produces many of them beats a mutation that produces one large one.
Why this course brings it up: Dr. Wong comes back to this line in Lectures 1, 4 and 5 and calls it the ideal fuel for immunotherapy. MSI-H, then frameshift, then a high neoantigen load, then hot tumour, then anti-PD-1 is his most complete causal chain. **Appears in:** Ch 9, Ch 10, Lect 1, Lect 4, Lect 5

▪ Neutron

In one sentence: An uncharged particle that sits inside the nucleus. It adds mass without changing chemical properties.
Detailed explanation: A proton carries a positive charge and an electron a negative one, and the neutron has neither, because inside it the number of protons equals the number of neutrons and the net charge is zero.
The nucleus exists because the strong nuclear force inside it balances the repulsion between the protons. One more neutron means one more unit of mass and no change in chemical behaviour, which is why isotopes are almost identical chemically. So the isotope label on a molecule tells you about the mass and leaves the reactivity alone.
Why this course brings it up: It belongs to the basic chemistry of Part 1. Dr. Wong does not single it out in the five lectures, so it will not be examined on its own. **Appears in:** Ch 1

▪ NF-κB (nuclear factor kappa B)

In one sentence: A transcription factor held in the cytoplasm by I-kappa-B until a signal such as LPS moves it into the nucleus.
Detailed explanation: When the cell is quiet, the p50 and p65 halves of NF-κB are bound to the inhibitory protein IκBα, and the whole complex stays in the cytoplasm. When a signal arrives (LPS, PKC activation, virus, oxidative stress), IκB kinase phosphorylates IκBα, IκBα is degraded, and p50/p65 can enter the nucleus, bind the promoters of inflammatory genes and start transcribing.
The design lets the cell fire only when it really needs to. And once inside, NF-κB is a transcription factor doing a job: reading promoters and starting transcription. A closed door that opens on a signal.
Why this course brings it up: In Lecture 5 he uses NF-κB to demonstrate signal-induced, rapidly amplified transcription, and he gives the timing of the cytokines: TNF-α peaks at 4 to 6 hours, then IL-1β → IFN-γ → IL-2 / IL-15; he says plainly not to memorise that order but to recognise the cascade. So the shape of the cascade is the examinable part. **Appears in:** Ch 7, Lect 1, Lect 5

▪ NICD

In one sentence: The part of Notch inside the cell, which is cut loose, moves into the nucleus and acts as a transcription factor.

Detailed explanation: When the delta ligand on a neighbouring cell touches Notch on the receiving cell, Notch is cut by a protease, and the intracellular piece, NICD, is released and goes into the nucleus, where it displaces the histones and switches on downstream genes.

The elegance of the mechanism is that the signal only exists while the neighbour is talking, so it is cell-to-cell communication by design. synNotch takes this principle and makes it artificial. Nothing happens until two cells meet, and that is the whole point of the design.

Why this course brings it up: In Lecture 5 he introduces synNotch as the AND-gate CAR: it has to see the tumour antigen and a second environmental signal before it fires, and that raises specificity a great deal. One antigen alone is not enough, which is the fix for the problem that CAR-T hits normal tissue. **Appears in:** Ch 7, Lect 5

▪ Nick

In one sentence: A break in only one strand, with the other strand still intact.

Detailed explanation: On the lagging strand of DNA replication, removing the RNA primer leaves a gap, which the polymerase fills in, but the backbone still carries one unsealed nick. The job of DNA ligase is to join the 5′ phosphate to the 3′ hydroxyl, forming a phosphodiester bond and sealing that nick.

He gives the example that if you inhibit ligase, replication carries on perfectly well, but the lagging strand is left with a pile of unsealed nicks and the cell cannot divide properly in the end. So a nick is not a break in the double helix; it is a break in one of the two rails, and the other rail is still holding the whole thing together. That is also why the nick is a different problem from a double-strand break, even though the word sounds similar.

Why this course brings it up: His example of a causal chain is "ligase seals the nick and forms a phosphodiester bond"; ligase and phosphodiester bond are already in the glossary. So you do not have to learn anything new here, only connect two things you already know. **Appears in:** Ch 5, Lect 4

▪ Nivolumab

In one sentence: An anti-PD-1 checkpoint inhibitor.

Detailed explanation: It is a monoclonal antibody against PD-1, and what it does is release the brake on the T cell. PD-L1 on the surface of a tumour cell binds PD-1 on the T cell, and that puts the T cell into anergy;
nivolumab occupies PD-1, so PD-L1 cannot grip it, and the T cell gets its killing power back. It belongs to the same class as pembrolizumab, and the two appear together in Lectures 3 and 5. The same family also includes atezolizumab, durvalumab and avelumab, which target PD-L1 and so act at a different point. And MSI-H tumours, with their many neoantigens, are precisely the patients this class of drug works best on. The distinction between PD-1 and PD-L1 is worth holding onto, because it decides which antibody you use and, more importantly, where the drug has to reach. Blocking PD-1 means the drug sits on the T cell; blocking PD-L1 means it sits on the tumour cell or on the other cells in the tumour. And the choice between the two is not arbitrary, because a tumour that does not express PD-L1 at all can still be attacked through PD-1. Nivolumab and pembrolizumab are interchangeable in that setting, and both are antibodies, so both are given by the same kind of injection.

Why this course brings it up: On the checkpoint slide in Lecture 3 Dr. Wong lists anti-PD-1 as pembrolizumab and nivolumab, and anti-PD-L1 as atezolizumab, durvalumab and avelumab. That set of drug names paired with the MSI-H indication is a common combination in scenario questions. **Appears in:** Lect 3

▪ NMR

In one sentence: Nuclear magnetic resonance. It is used to work out the shape of molecules in solution.

Detailed explanation: Atomic nuclei line up in a magnetic field. Put the sample in the field, disturb it with a radiofrequency pulse, then listen to the signal the nuclei send back, and the different chemical environments give different shifts, from which the molecular structure can be worked backwards.

It does not require the sample to be separated or derivatised first, so it is non-destructive. On division of labour, GC-MS suits molecules that are volatile and thermally stable, LC-MS suits the ones that are less volatile and larger, and NMR suits very high concentrations.

Why this course brings it up: He lists it in Lecture 1 as one of the three main instruments for metabolomics, but he also says plainly that the details of the metabolic pathways will not be examined. So it belongs on the slide and not in the details. **Appears in:** Ch 10, Ch 11, Lect 1

▪ NNK

In one sentence: One of the tobacco-specific nitrosamines, which is activated by liver P450 and then binds to DNA.

Detailed explanation: NNK is a carcinogenic precursor in cigarette smoke. It does not damage DNA directly. It first has to be chemically modified by the cytochrome P450 enzymes of the liver, and Dr.
Wong singles out the arylation step in particular, which turns it into a form that can react with a base and produce a DNA adduct. If the adduct is not repaired, the second round of replication fixes it as a mutation. In mice, what NNK produces is a GC→AT mutation in the K-ras gene. The activation step is what turns an inert chemical into something the DNA can actually react with, and it is also why the liver, which carries the P450 enzymes, is where the damage happens rather than in the lung where the smoke first lands.

Why this course brings it up: He names NNK and NNN, and he writes “NNK induces a GC→AT mutation at K-ras gene in mice” on his slide. He does not ask you to memorise the detailed chemistry of the carcinogens. **Appears in:** Lect 3

▪ NNN

In one sentence: One of the two tobacco-specific nitrosamines.
Detailed explanation: NNN and NNK are both carcinogenic precursors in cigarette smoke, and they take the same route. They are activated by P450 enzymes in the liver, they then react with DNA bases to form an adduct, and if that adduct is never repaired the error is fixed as a mutation.
He lists NNK and NNN for a reason: so that you know roughly how a chemical carcinogen completes its journey inside a human body. And because the lung cancer rate among smokers is far higher than among non-smokers, this path from a chemical to a mutation is supported by very strong evidence in humans. The whole chain has three links, activation, adduct, and then repair or mutation.
Why this course brings it up: He does not ask you to memorise it, so it is concept-only. The landing point of this passage is that an adduct which is repaired causes no trouble, while an unrepaired one becomes a permanent mutation. That is the sentence to keep. **Appears in:** Lect 3

▪ non-covalent

In one sentence: A way of bonding that does not involve sharing a pair of electrons.
Detailed explanation: The binding between histone and DNA is non-covalent: there is no covalent bond, only hydrogen bonds plus the electrostatic attraction between opposite charges. The consequence of that kind of binding is that it can let go and can be re-formed, so the state of the chromatin is a dynamic and reversible switch rather than a permanent label.
Precisely because the binding is not covalent the histones can slide and the marks can be removed, and only then can the "on" and "off" of the chromatin be twisted round quickly by a drug. A covalent bond would be a commitment, and the whole point of the chromatin switch is that it is not committed.
Why this course brings it up: "The histone-DNA binding is non-covalent, so the chromatin state is dynamic and reversible" is one of the facts he lists as Tier 2. **Appears in:** Ch 4, Lect 3

▪ Non-disjunction

In one sentence: Chromosomes failing to separate properly during meiosis, which produces an abnormal chromosome number.
Detailed explanation: When M1 fails, the homologous chromosomes do not separate; when M2 fails, the sister chromatids do not separate. Either way the gamete ends up with one chromosome too few or one chromosome too many, and after it fertilises a normal gamete the result is aneuploidy: 2n-1 is monosomy, which is Turner 45,X, and 2n+1 is trisomy, which is Down syndrome.
You can also see the non-separation directly down the microscope, by marking the chromosomes and looking for the cell in which they failed to divide apart.
Why this course brings it up: In Lecture 3 Dr. Wong says you have to be able to tell an M1 failure from an M2 failure, and it is one of the five questions he personally names. A pedigree combined with aneuploidy is also one of the entries in his case bank. **Appears in:** Ch 3, Lect 3 **Video:** V05_Meiosis_I_vs_Meiosis_II_M1_vs_M2.mp4

▪ Non-homologous end joining (NHEJ)

In one sentence: The repair route that needs no template when it fixes a broken chromosome, and therefore makes mistakes. It is also what builds antibody diversity.
Detailed explanation: After a double-strand break, Ku protein binds the broken ends first, and then polymerase, nuclease and ligase take over to join the two sides — but there is no template to copy, so the sequence that goes back in often has pieces gained and pieces lost, which is how indels and frameshifts appear.
V(D)J recombination borrows exactly this machinery: RAG1 and RAG2 (recombination activating proteins) cut out the V, D and J segments, and NHEJ joins them. CRISPR/Cas9 is designed around the same fact: after Cas9 cuts a double-strand break, a repair by NHEJ leaves a small insertion or deletion and the gene is knocked out; a precise knock-in has to go down the HR route instead. Which is fine when the goal is diversity and a disaster when the goal is fidelity: the same sloppy repair that generates antibody repertoire is the same sloppy repair that lets a genome drift. And the frame matters: an indel that is not a multiple of three shifts the reading frame and can change the protein completely.
Why this course brings it up: It is one of the five questions he names out loud: NHEJ defect → no diversity can be built up from V + D + J → insufficient antibody and TCR diversity → SCID. The segment copy numbers he gives are V ≈ 39, D = 27, J = 6; NHEJ is also the route CRISPR uses for knock-out. So the chain, not the enzymes, is what you should be able to recite.
Appears in: Ch 6, Ch 9, Lect 4 **Video:** V03_NHEJ_defect_SCID.mp4

▪ Non-obviousness

In one sentence: The "non-obvious" requirement that a patent application has to get past.
Detailed explanation: An invention cannot simply be a straightforward extension of known technology, because then anybody could have done it, so there has to be a non-obvious technical contribution.

He uses the CRISPR patent war to show what is at stake: the dispute is exactly about whether this application is only an obvious extension of what was already known, and about whether the invention itself and the particular application can be separated. It is a legal test, not a scientific one, and that is worth saying out loud.
Why this course brings it up: He does not list non-obviousness as an exam topic on its own, so it is a concept to understand; he states plainly that the whole IP landscape is not examined. **Appears in:** Ch 12, Lect 4

▪ Nonsense mutation

In one sentence: A single-letter change that creates a stop codon early, and the protein is cut short.
Detailed explanation: The codon is replaced by a stop codon, UAA, UAG or UGA, the translation machine stops there, and the protein is shorter than it should be. Most of those truncated proteins are degraded straight away, which amounts to the function of that gene disappearing, so the effect of a nonsense mutation is often bigger than that of a missense that simply swaps one amino acid.
A frameshift that happens to land on a stop codon ends the same way, and if the truncated product looks new to the immune system you have gained a therapeutic target as well. The logic is arithmetic rather than chemical: the codon table has an entry that means stop, and if the reading frame moves, the codon that used to spell an amino acid now spells stop instead.
Why this course brings it up: In Lecture 3 he classifies mutations as missense, nonsense and frameshift. A frameshift causing an early stop is a source of neoantigen, and it is the example on his Lecture 1 diagram where a 1-nt insertion produces an immediate STOP. That diagram is worth remembering, because it ties the mutation classes to the thing the immune system sees. **Appears in:** Ch 3, Lect 3

▪ Notch

In one sentence: The receptor that lets one cell tell its neighbour to change its gene expression.
Detailed explanation: The ligand sits on the neighbouring cell and the receptor sits on the receiving cell, and the two have to be in contact before any signal can start, so Notch is built to be a conversation between neighbours.
That kind of communication suits development and tissue differentiation very well, because one group of cells can change its identity while the neighbours hold their own. synNotch takes this principle and moves it into CAR-T, using an AND gate so that the CAR is only expressed when two conditions are met at the same time.
Why this course brings it up: In Lecture 5 Dr. Wong treats synNotch as one of the answers to the lack of specificity in CAR-T, and case question five also asks what happens to the safety profile if you build an antigen-independent safety switch out of Notch. **Appears in:** Ch 7, Lect 5

▪ NOTCH2

In one sentence: One of the target genes switched on by YAP, through TEAD, when the Hippo pathway is off.
Detailed explanation: When Hippo is off, YAP is neither phosphorylated nor degraded, so it enters the nucleus and binds TEAD, becoming a transcription factor, and it switches on three target genes: CTGF, JAG1 (a Notch ligand) and NOTCH2. What these drive is proliferation, anti-apoptosis, stemness and EMT, which is what actually happens once Hippo fails. NOTCH2 is the piece that joins the Hippo and Notch signals together, which is why it sits inside this story rather than beside it. One of the targets, JAG1, is itself a Notch ligand, so the switch that Hippo releases feeds straight into the Notch pathway, and the three of them together turn a Hippo failure into proliferation, resistance to cell death, stemness and EMT. Which is why NOTCH2 is the one worth remembering from this slide rather than CTGF: it is the point where the Hippo story connects to a second pathway.
Why this course brings it up: On the Hippo slide in Lecture 1 Dr. Wong labels the result of Hippo being OFF as “Proliferation, anti-apoptosis, stemness, EMT”, and NOTCH2 is in the target gene box of that diagram. So if you can read that slide, you have the entry. **Appears in:** Ch 7, Lect 1, Lect 5

▪ NotI

In one sentence: Another restriction endonuclease used in molecular cloning.
Detailed explanation: NotI recognises its own eight-base sequence, and the overhang it leaves is complementary to the one EcoRI cuts, so the insert of the donor plasmid can stick to the backbone of the recipient plasmid.
The detail of the experiment is the double digest: cutting with EcoRI and NotI together both releases the insert and linearises the recipient plasmid. The point of the double cut is that it handles two jobs in the same tube, cutting the insert off the skeleton while cutting the skeleton itself open. One enzyme, two jobs, and the sticky ends it leaves are what hold the two halves together until the ligase seals them. Choose two enzymes whose ends are compatible and the pieces assemble; choose two that are not and nothing happens.
Why this course brings it up: Dr. Wong does not name NotI as an exam topic on its own, so it belongs to conceptual understanding. It is the second half of the donor and recipient plasmid picture. **Appears in:** Ch 3, Lect 4

▪ NPM1

In one sentence: His Lecture 3 worked example: you have to put CNV copy number together with mRNA expression before you can call it a driver.
Detailed explanation: The real use of this entry is to teach how you find a driver. His method is to integrate the CNV profile with the mRNA expression, and only a gene whose copy number has gone up and whose transcription has gone up with it can be a driver that genuinely brings a growth advantage.

Looking at the mutations alone is not enough, and looking at the copy number alone is not enough either. NPM1 appears in his Lecture 3 heat map and PPI network, and patients who are NPM1-high differ in overall survival and in cumulative relapse. The passenger is the gene that happens to ride along with the change without contributing to the growth advantage, and the whole point of the integration is to separate the two.
Why this course brings it up: In Lecture 3 Dr. Wong uses it as the worked example for telling a driver from a passenger, and he says explicitly that he is giving the molecular mechanism of CNV only as "a taste", so you should not be frightened of those mechanisms. **Appears in:** Ch 10, Lect 3

▪ Nuclear envelope
In one sentence: The double membrane that seals in the cell nucleus.
Detailed explanation: Two membranes enclose the RNA and the DNA, and nuclear pore complexes link them. The envelope separates transcription from translation, because a eukaryotic cell has to process the pre-mRNA and check it before sending it out of the nucleus, and that adds a checkpoint where errors can be caught.
The envelope is continuous with the outer membrane of the endoplasmic reticulum, which tells you that this membrane is homologous with the ER in evolutionary terms.
Why this course brings it up: It is the core of cell compartmentation in Lecture 2. A prokaryote has no nucleus, so transcription and translation can happen in the same place; a eukaryote separates them, and that is where the two extra layers of control, splicing and post-transcriptional regulation, come from. **Appears in:** Ch 8, Lect 3

▪ Nuclear pore
In one sentence: The doorway in the nuclear membrane, letting messages out and proteins in.
Detailed explanation: Each nuclear pore is a large protein complex, and the channel itself transports selectively: mRNA, tRNA and ribosomal subunits go out, while proteins the nucleus needs (transcription factors, histones) go in.
Small molecules can pass straight through the channel of the pore without needing a transport receptor at all. This threshold is what lets the cell control which messages get in and which get out. It is a checkpoint with a doorway in it.
Why this course brings it up: In Lecture 5, when he describes "outside signal → signal transduction → transcription factor moves into the nucleus → changes target gene expression", the nuclear pore is the switch; one of the seven ways a transcription factor can be activated is entering the nucleus through the pore once its inhibition is lifted. So the pore is where the signal-to-nucleus chain physically happens. **Appears in:** Ch 7, Ch 8

▪ Nuclear receptor
In one sentence: A protein that binds a hormone or a drug inside the cell and then switches genes on in the nucleus.
Detailed explanation: It is the opposite of a membrane receptor: the ligand, a hormone, a drug or an oxysterol, does not have to go outside the cell and simply passes through the membrane to bind it.
Once the receptor has changed shape it exposes the activation region that was hidden, recruits a co-activator or a co-repressor, and then goes into the nucleus to bind DNA response elements. Type I receptors sit in the cytoplasm bound to HSP, while Type II receptors are already in the nucleus, sitting on the DNA as a heterodimer. It is the same shape of story as the membrane receptors, the ligand arrives, the protein changes shape, and the consequence happens somewhere else in the cell.
Why this course brings it up: In Lecture 5 he teaches it together with the DBD and LBD, with dimerisation, and with tamoxifen and receptor occupancy. That is the main thread of the nuclear receptor section, and each of those is a separate question you could be asked. **Appears in:** Ch 7, Lect 5

▪ Nuclear transcription factor
In one sentence: A protein that enters the nucleus and switches genes on or off there.
Detailed explanation: A transcription factor is usually inactive in the cytoplasm, and it needs a “switch-on” before it can go into the nucleus. Dr. Wong lists seven ways this happens: protein synthesis, ligand binding, phosphorylation, the addition of a subunit, unmasking, de-repression followed by entry through the nuclear pore, and release by being cut off a membrane by a protease.
Once inside, it binds the response element on the DNA and recruits the polymerase and the general transcription factors. So there is a whole list of entry tickets, and the factor in front of you has to have used one of them.
Why this course brings it up: His slide of the seven activation routes is the skeleton of the transcription factor section in Lecture 5. The clearest example in the whole list is NF-κB, where the degradation of IκB lets it into the nucleus. **Appears in:** Ch 7, Lect 5

▪ Nucleic acid
In one sentence: The class of biological molecules that includes DNA and RNA. It carries the information.
Detailed explanation: Both are polymers of nucleotides, and they store information as a sequence through base pairing. DNA keeps a backup copy in two strands, so a damaged base can be read off the other one, and it can be stored chemically stably for decades.
RNA is single-stranded and can fold into all sorts of shapes, which is what lets it do catalytic and regulatory work. He uses the Griffith and Avery, MacLeod and McCarty pneumococcal experiments to show that the genetic material is DNA and not protein.
Why this course brings it up: It is one of the four great families of molecules in Lecture 1 and the skeleton of the whole course. Dr. Wong says the history matters, because it establishes a principle you can actually use, and the principle is that the

information lives in the sequence of bases. **Appears in:** Ch 1

▪ Nucleoid

In one sentence: The place where the DNA sits inside a bacterial cell, with no membrane around it.
Detailed explanation: A prokaryotic cell has no nuclear membrane, so the chromosome sits directly in the cytoplasm as a region of DNA with no boundary. That brings two consequences. Transcription and translation can happen at the same time, because a prokaryotic mRNA is picked up by a ribosome as soon as it emerges, and DNA replication and transcription can run side by side, which is the reason a bacterium can divide every half hour or so.
There is no compartment to keep the two apart, so nothing stops them overlapping.
Why this course brings it up: In Lecture 5 Dr. Wong uses the prokaryote versus eukaryote comparison to explain the differences in expression systems, splicing and protein modification. That comparison is also the background to the lac operon versus the eukaryotic promoter. So the contrast is doing more work than it looks. **Appears in:** Ch 8, Lect 5

▪ Nucleolus

In one sentence: The dense region inside the nucleus where the parts of the ribosome are made.
Detailed explanation: The nucleolus is a dense region inside the nucleus with no membrane around it; rRNA is transcribed there and assembled with proteins into ribosomal subunits, which are then carried out through the nuclear pores to the cytoplasm.
Its size changes with the cell's need for protein synthesis — a cell busy making protein has a particularly large nucleolus. So the nucleolus is a factory floor with no walls.
Why this course brings it up: He listed it in the organelle list in Lecture 2; it connects to the ribosome and protein synthesis material. It is on the list, but it is there as a pointer rather than as a topic. **Appears in:** Ch 8, Lect 2

▪ Nucleoside

In one sentence: A base plus a sugar, with no phosphate. The building block before the phosphate goes on.
Detailed explanation: The sugar is ribose or deoxyribose, and which one it is decides whether this is RNA or DNA; the base decides which base it is, DNA uses A, T, C and G, and RNA swaps T for U.
Add one phosphate and you have a nucleotide. A nucleoside itself is never folded into the DNA backbone, but it is the basic unit for nucleotide synthesis and for designing nucleoside analogue drugs. A nucleoside has no phosphate, so it cannot become DNA raw material directly: it has to be phosphorylated into a nucleotide first, and only then can it be joined onto the DNA or RNA backbone. So the order is sugar and base first, phosphate second, and the phosphate is what makes it reactive enough to be built into a chain.
Why this course brings it up: In Lecture 1 he uses the difference between nucleoside and nucleotide as the way into the chemistry of DNA and RNA. It is also the basis for understanding the segment that NER cuts out and the additions that polymerase makes. Three names in one chain: nucleoside, nucleotide, and the backbone. **Appears in:** Ch 1, Ch 3, Lect 1

▪ Nucleosome

In one sentence: DNA wrapped around a spool of histone protein. The basic unit of chromatin packaging.
Detailed explanation: The core particle is an octamer built from two copies each of H2A, H2B, H3 and H4, with 147 bp of DNA wrapped around the outside, about 5.7 nm high and 11 nm across, and H1 acting as the linker where the DNA enters and leaves.
A whole chromosome is this string of beads on a thread, and it then folds in steps into a 30 nm fibre, then 300 nm, then 700 nm, and finally becomes the 1400 nm mitotic chromosome. His crystal structure resolution is 2.8 Å. Wrapping the DNA round and round does two jobs at once: it shortens the length, and it gives the enzymes a way in. That is why heterochromatin, which is wound tight, is silent, while euchromatin, which is loose, is active, and it is why the packaging state is something a drug can be aimed at. Two details on the slide are worth holding on to: the 147 bp of DNA per nucleosome, and the resolution of the crystal structure, which is 2.8 Å.
Why this course brings it up: Nucleosome accessibility is the first gate on the question of whether a gene can be read. Heterochromatin physically blocks the transcription factor and the enhancer from reaching the promoter, and that is one of the concepts you must understand in Lecture 3. **Appears in:** Ch 3, Ch 4, Lect 3

▪ Nucleotide

In one sentence: One base plus one sugar plus one phosphate. The building block of DNA and RNA.
Detailed explanation: Phosphate and sugar alternate to make the backbone (every strand of DNA has two of these backbones), and the bases point inward and pair up complementarily to pull the two backbones together — three hydrogen bonds for G–C and two for A–T.
The phosphate at the 5′ end and the free hydroxyl at the 3′ end give the strand its direction, which is why polymerase can only synthesise 5′→3′. A nucleotide is both the building block of DNA and RNA and the structural unit of signalling molecules such as ATP and cAMP. Three parts, and each of them does a job: the base carries the information, the sugar and phosphate carry the shape and the energy.
Why this course brings it up: It is the common basis of DNA replication, PCR primer design and nucleoside analogue drugs (for example the nucleoside analogue that he describes being proofread out and causing drug resistance). **Appears in:** Ch 1, Ch 3

▪ Nucleotide excision repair (NER)

In one sentence: The pathway that repairs bulky damage such as a UV dimer, and it cuts out a whole stretch of the strand.
Detailed explanation: When UV creates a covalent bond between adjacent pyrimidines (a pyrimidine dimer), the whole base is distorted beyond recognition and cannot simply be swapped for another one.
NER therefore cuts out an entire segment, 22 to 32 nucleotides, and then the polymerase fills it in and ligase seals it. The division of labour is this: GG-NER scans the whole genome with DDB1 (DNA damage-binding protein 1), DDB2 (DNA damage-binding protein 2) and XPC, while TC-NER calls the repair in on the strand that is actually being transcribed, using CSA, CSB and RNAPII. So one of them is a genome-wide patrol and the other is a local repair summoned by the machinery of transcription itself. Both end up cutting out a segment of 22 to 32 nucleotides and handing the gap to the polymerase. And that size is not arbitrary either: it is long enough to clear the twisted bases and short enough to be resynthesised without much trouble.
Why this course brings it up: In Lecture 4 Dr. Wong pairs it with BER as the combination multiple-choice questions use most often (BER fixes a single bad base, NER fixes bulky damage). He also warns that the chemistry of DNA glycosylase and the AP site will not be examined, but you must know what each pathway repairs. **Appears in:** Ch 6, Lect 4

▪ Nucleus

In one sentence: The control room. It keeps the DNA and decides which genes get read.
Detailed explanation: There is more in the control room than a ledger, there is a selection mechanism as well. How tightly the chromatin is packed decides which door opens. Transcription factors come in and bind the promoter.
An enhancer reaches around, using a DNA-bending protein, to deliver an activator to the promoter. That is why the same DNA can produce completely different gene sets in a T cell and in a dendritic cell. His metaphor is the control room: the same set of DNA, different cells stationing different transcription factors on duty, and the doors that open are naturally different. The door is not the DNA itself but the packing around it, which is why chromatin studies and gene expression studies are the same subject.
Why this course brings it up: In Lecture 1 he runs the control room metaphor through the whole course, and he uses it again in the chromatin and epigenetics section: whether it can be read is everything. It is the framing that makes the epigenetics material make sense rather than look like a list of marks. **Appears in:** Ch 8, Lect 2

▪ octamer

In one sentence: The structure at the core of a nucleosome, made of eight histone molecules.
Detailed explanation: Octamer means two copies each of H2A, H2B, H3 and H4, eight molecules in all, forming a globular core with 147 base pairs of DNA wrapped around it. The whole nucleosome is about 10 nm across, and his slide also gives the core particle as 5.7 nm high by 11 nm wide.
The assembly of the octamer is dynamic, histones slide along the DNA, and precisely because the assembly can move, chromatin can be rearranged by the remodelling complexes. So the nucleosome is a stable shape with a moving population of parts, and that is what gives the remodellers something to grip.
Why this course brings it up: The composition and the dimensions of the nucleosome, the octamer, 147 bp, 10 nm and 2.8 Å, are his Tier 3 precise numbers. So those four numbers belong in the list you memorise. **Appears in:** Ch 4, Lect 3

▪ Odds ratio

In one sentence: A way of expressing how many times more likely something is in one group than in another.
Detailed explanation: The OR is calculated as (the proportion with the event in one group ÷ the proportion without it) ÷ (the proportion with the event in the control group ÷ the proportion without it).
An OR of 1 means there is no difference between the groups; the larger the OR, the stronger the association. Its strength is that you can still estimate the strength of an association in a case-control study, and its weakness is that it is not intuitive, so a confidence interval is almost always reported alongside. It is the standard way to say "times more likely".
Why this course brings it up: In Lecture 3 he gives two ready-made numbers: when methylation is detected, OR ≈ 12.7; and CIMP-high versus CIMP-0 is OR 39.6 (95% CI 20.6–76.1, p<0.0001). Those two sets of numbers are multiple-choice material.
Appears in: Ch 4, Lect 3

▪ Okazaki fragment

In one sentence: A short stretch of DNA made first on the lagging strand and joined up later.
Detailed explanation: Each piece is about 200 nucleotides long, and it starts with a short RNA primer of about 10 nucleotides. The polymerase extends it from its 3′ end until it reaches the front of the next primer, the RNA primer is then removed, and the gap left behind is sealed by ligase.
Because that has to be repeated thousands of times, the lagging strand is actually copied more slowly than the leading strand, and every single piece carries an extra three-step procedure: lay down a primer, take the primer away, seal the seam.
Eukaryotic cells are more extreme still, because a single chromosome carries thousands of origins of replication at the same time, each running out a bidirectional pair of replication forks. That is why the lagging strand is the complicated half of the story.
Why this course brings it up: In Lecture 4 Dr. Wong uses the Okazaki fragment to explain the 5′→3′ directionality limit and the role of DNA ligase, and the length of the primer and the spacing between fragments both have numbers on his slide, which are the figures you should be able to recall. Each of those three steps is a separate enzyme, and they have to happen in the right order for the fragment to be finished. **Appears in:** Ch 5, Lect 4

▪ Oncogene

In one sentence: A gene that pushes the cell to divide once it has been switched on too far. It is a driver after it is mutated.
Detailed explanation: A normal gene is either the brake or the accelerator, and an oncogene is an accelerator that has got stuck: a gene such as K-Ras or HER2, once it has been mutated, amplified or translocated into permanent activation, sends the cell the message “keep growing”.
Unlike a tumour-suppressor gene, the problem with an oncogene is not that it has stopped working but that it cannot be switched off, so a drug can go straight after this permanently-on switch. The brake and the accelerator are two different therapeutic problems, and only one of them can be fixed by giving the patient back the thing that is missing.
Why this course brings it up: In Lecture 1 Dr. Wong uses K-Ras to explain the “stuck switch”. He also lists driver and passenger as one of the common multiple-choice pairings, and explains that finding a driver means combining CNV and mRNA expression. **Appears in:** Ch 10, Lect 1, Lect 3, Lect 4

▪ Operator

In one sentence: The stretch of DNA where the repressor sits. While the repressor is on it, the gene behind it cannot be transcribed.
Detailed explanation: The operator is next to the promoter but it is not the same thing: the promoter is where the polymerase lands, the operator is where the repressor parks. Once the repressor sits down it is like a hand held over the keyboard, and the polymerase cannot turn the downstream structural genes.
This is the simplest and most effective regulatory design in prokaryotes, because you do not have to touch the gene itself; the same stretch of DNA can be switched on and off. The operator is part of the DNA, so the repressor is reading a sequence, and the promoter is also DNA, so the polymerase is reading a sequence; what differs is which protein each one is built to recognise.
The lac operon also needs cAMP-CAP: when sugar is low cAMP goes up, and only then does the CAP-cAMP complex pull the polymerase over and turn the output up.
Why this course brings it up: In Lecture 5 he draws the operator, the promoter, the repressor and the structural genes together on one operon diagram. The difference between the operator and the promoter is the core of what he says when he explains that diagram aloud. So if you only learn one thing from that slide, learn that those two are different places with different jobs. **Appears in:** Ch 7, Lect 5

▪ Operon

In one sentence: A group of neighbouring genes switched on and off together by one shared control region.
Detailed explanation: An operon contains one promoter, one operator and several structural genes, transcribed into a single polycistronic mRNA. The benefit of sharing control is that the whole machine opens and closes together;
lac and trp are two exact opposites — the first is off by default and comes on when lactose appears, the second is on by default and switches off when tryptophan appears. Recombinant expression borrows the same design: the lac operon with IPTG as an artificial inducer is used to make glycosylated protein in CHO cells. One switch, several genes, one message — that is the design in three phrases.
Why this course brings it up: In Lecture 5 he teaches lac and trp together, describing trp as the "mirror image": tryptophan binds the Trp repressor and switches off its own synthesis. He also says the logic of lac is one of the entries in the case bank. And the two examples are opposites on purpose. **Appears in:** Ch 7, Lect 5

▪ Organelle

In one sentence: A compartment inside the cell that has one job, like a room in a factory.
Detailed explanation: The mitochondrion is the power plant, the endoplasmic reticulum is the protein-making workshop, the Golgi apparatus is the packing and sorting department, the lysosome is the recycling plant, and the nucleus is the control room.
The design principle they share is “keeping incompatible reactions apart”: what needs an acidic environment, what needs oxidising conditions, and what needs to be kept away from DNA each get their own room. It is the same reason a house has a kitchen and a bathroom rather than one sink.
Why this course brings it up: In Lecture 2 Dr. Wong pairs each organelle with its clinical meaning (mitochondria also carry the maternal inheritance diseases), and this structure–function–clinical arrangement is the part beginners most easily muddle up.
Appears in: Ch 8, Lect 2

▪ Organoid

In one sentence: A small piece of human tissue grown from stem cells, used to test drugs.
Detailed explanation: If you push iPSC or patient-derived cells to differentiate into a small three-dimensional piece of tissue, you can reproduce some of the properties of the original tumour, including its response to drugs.
The value is that drug screening no longer needs an animal to be grown first, and no longer needs a human to be dosed directly, and because the organoid keeps its microenvironment it is closer to the real thing than a flat two-dimensional culture. The organoid is therefore a small, self-contained stand-in for the tumour, and a plate of them is a cheap way of asking what a drug does to a real piece of tissue.
Why this course brings it up: On his iPSC slide in Lecture 2 Dr. Wong draws a drug R&D route: differentiation into an organoid, then multiwell plate drug screening, then the activity readout, then target validation. His drug R&D and therapy product are two separate branches, and an organoid belongs to the first of them. **Appears in:** Lect 2

▪ origin of replication

In one sentence: The marked position on the DNA where replication starts.
Detailed explanation: A eukaryotic chromosome is very large and cannot be copied from one end, so the DNA is scattered with tens of thousands of origins. Each origin opens a replication bubble, and each bubble has two replication forks running in opposite directions.
Neighbouring forks eventually run into each other, and that is how replication is finished. Because the forks stop when they meet, the completion of replication does not require anybody to know how long the whole chromosome is, and that is the elegance of the multi-origin design. Each origin works on its own timetable, and no fork has to be told where the next one begins. That is also why the number of origins is a number worth remembering, because it is the number that explains why a drug can hit a dividing cell so hard.
Why this course brings it up: "Thousands of origins, each bubble with two forks running in opposite directions" is one of the facts he lists as Tier 2. **Appears in:** Ch 5, Lect 4

▪ orthology

In one sentence: Homologous genes in different species whose functions correspond.
Detailed explanation: When a signalling pathway is highly conserved between species, an experimental result in mice or fruit flies can be used to reason about a human disease — and that is the entire reason model organisms exist.
His Hippo passage ends exactly this way: these pathways are conserved across species, so mouse models are useful. It is precisely because the functions of genes can be matched up between different species that it is possible to model a human disease in fruit flies and mice at all, and that the experimental results have to be usable for reasoning about people. The word for it is orthology, and the whole of the modelling business rests on it.
Why this course brings it up: He does not name orthology as an exam topic on its own, so it is a concept to understand; the example he has given is that histones are highly conserved, so what is found in fruit flies is worth something for humans. The reasoning is the point, not the word. **Appears in:** Ch 4, Lect 1

▪ Oxidation

In one sentence: Losing electrons, or gaining oxygen. It releases energy.
Detailed explanation: Oxidation and reduction are a pair: when a molecule loses electrons it is oxidised, and the electrons it loses are picked up by another molecule, which is therefore reduced.
In respiration, glucose is oxidised while oxygen is reduced to water, and in the chemistry of iron, losing electrons is what rusting is. Oxygen is such a powerful electron acceptor because it has unpaired electrons. So nothing in the reaction disappears; the electron simply moves from one side to the other, and that movement is where the energy comes from.
Why this course brings it up: It is Part 1 basic chemistry. In Lecture 1, when he teaches energy and metabolism, he uses it to explain electron transfer, and he does not name it on its own. But it is the vocabulary underneath the whole energy story.
Appears in: Ch 1

▪ Oxidative phosphorylation

In one sentence: The process by which the mitochondria use oxygen to make most of the ATP.
Detailed explanation: Electrons travel down the respiratory chain (Complex I to IV), and the energy released pumps protons from the matrix into the intermembrane space, building an electrochemical gradient;
the protons then flow back into the matrix through ATP synthase, driving it to make ATP from ADP and phosphate. Oxygen is the final electron acceptor, and without it the whole chain stops. The inner mitochondrial membrane is highly impermeable to protons, which is what allows the gradient to hold. So oxygen is not the fuel; it is the thing at the end of the chain that accepts the electrons.
Why this course brings it up: In Lecture 1 he uses it to contrast with glycolysis: a resting normal cell with oxygen gets about 36 ATP and without oxygen only 2, while a cancer cell gets about 4 in both cases; he also says the hypoxic core of a tumour is where HIF-1α starts to accumulate. **Appears in:** Ch 8, Ch 10, Lect 2

▪ p-arm

In one sentence: The short arm of a chromosome, above the centromere.
Detailed explanation: A metaphase chromosome has two arms: the short one is p, for petit, and the long one is q; they meet at the centromere, and the centromere is where the spindle microtubules attach in metaphase and the anchor that lines the chromosome up on the equatorial plate.
Addresses are written in three parts, arm, region and band, for example p21.1 and q21.1, and each part of the address corresponds to one chromosome diagram on the slide. So the address system is a map, and once you can read the map you can find any band on any chromosome.
Why this course brings it up: In Lecture 3 he makes a point of how it is read: q21.1 is said as q-two-one-point-one, not q-twenty-one-point-one, and that is a pronunciation trap he has stated explicitly. The other half is remembering that p is the short arm and q is the long one. **Appears in:** Ch 3, Lect 3

▪ P-body

In one sentence: The place in the cytoplasm where silenced mRNA is gathered up and destroyed.
Detailed explanation: When mRNA is caught by miRNA and RISC it is not broken down on the spot. It is carried to the P-body, a condensate in the cytoplasm with no membrane around it, rich in RNA and protein.
There the mRNA is de-capped, stripped of its poly-A tail, and then degraded. The P-body is therefore where the cell disposes of messages it has judged unnecessary, a centralised station for clearing away the messages you do not need, and that makes it an important place for the mRNA stability lesson. The lack of a membrane is the interesting part: it is not an organelle with an import system, it is more like a droplet where molecules gather because they want to be near each other.
Why this course brings it up: On the miRNA production diagram in Lecture 5 he draws the action of RISC and GW182 inside the P-body. He also explains that miRNA is unusually stable in serum, which is what makes it a good non-invasive biomarker. Two different things on one slide, both about where the message ends up. **Appears in:** Lect 5

▪ P-selectin

In one sentence: One of the selectins that make white blood cells roll along the wall of a blood vessel.
Detailed explanation: A white blood cell travelling from the blood out into tissue has four steps to go through: first rolling, then adhesion, then transmigration across the vessel wall, and finally chemotaxis along the chemical gradient.
The rolling step is the job of the selectin family, with P-selectin and E-selectin both involved. They catch carbohydrate ligands on the white blood cell and make the cell roll slowly along the endothelium, which slows it down enough for the next step to happen. Adhesion is the job of ICAM-1 and VCAM-1, and transmigration is the job of PECAM-1. So each step has its own set of molecules, and a cell that cannot complete one of them never reaches the next, which is why the immune cell journey to a tumour is usually described as four separate obstacles rather than one long road.
Why this course brings it up: These four steps are the diagram on Slide 19 of Lecture 2, and Dr. Wong does not list them as an examinable topic, so it is conceptual understanding. But it has direct clinical meaning: a tumour that wants to hide changes the microenvironment so that immune cells cannot get in, and the list of obstacles to CAR-T entering a solid tumour, namely the immunosuppressive factors TGFβ and IL-10 and the matrix environment, are all different faces of the same story. **Appears in:** Ch 9, Lect 2

▪ P-value

In one sentence: How likely the data would look this extreme if the null hypothesis were true.
Detailed explanation: The p-value answers the question of how unusual the result would be if the original hypothesis were true, and it is not the probability that the result is a random coincidence.
The traditional cut-off is 0.05. The volcano plot in the metabolomics material of Lecture 1 uses $p < 0.05$ together with FDR < 0.15 to pick features, and then crosses that with VIP > 1.0. The methylation study in Lecture 3 reports OR 39.6 with a 95 % CI of 20.6–76.1 and $p < 0.0001$, and OR 12.7 with 7.84–20.7 and $p < 0.0001$. The p value tells you how unusual something is, and it does not tell you how big the effect is. A p value of 0.0001 can sit on top of a difference so small that nobody would notice it in a patient, which is why the number always has to travel with the size of the effect.
Why this course brings it up: He does not name the p-value itself as an exam question, but it is the reading standard for nearly every data slide he shows, and it is what a short-answer question wants you to write down. His requirement that the answer include the numbers and state the limitations depends on exactly this language. And the warning is worth keeping: $p < 0.05$ is a line on a graph, not a truth. So treat it as a convention you report, not a verdict you deliver. **Appears in:** Ch 4, Ch 9, Ch 10, Lect 1, Lect 3

▪ p16

In one sentence: The tumour-suppressor protein encoded by CDKN2A.
Detailed explanation: p16 (INK4a) inhibits CDK4/6; when CDK4/6 is inhibited, pRb is not phosphorylated, E2F is not released, and the cell stays in G1 instead of entering S phase. So p16 is the brake on the cell cycle.
The gene CDKN2A also encodes a second protein, p14ARF, and the two products push p53 to guard the G1 gate through two different routes, so if either route fails the cell can get away with dividing when it should not. Two brakes, one gate. And because the gene carries both instructions, losing it takes out both brakes at once, which is a good deal more dangerous than losing either on its own. The methylation story is a tidy one: hypermethylating the promoter is not a mutation, the sequence is untouched, but the gene simply stops being read.
Why this course brings it up: In Lecture 3 Dr. Wong uses CDKN2A methylation as his example: promoter hypermethylation makes p16 disappear and the brake fails. He also gives the numbers for when methylation is detected, OR 12.7, and CIMP-high versus CIMP-0 OR 39.6. **Appears in:** Ch 4, Lect 3

▪ p300/CBP

In one sentence: The proteins that HIF-1 recruits to help it drive transcription of its target genes.
Detailed explanation: p300 and CBP are co-activators. They do not bind DNA themselves, but they carry a region that can catch a transcription factor. In hypoxia HIF-1α joins HIF-1β in the nucleus to form a heterodimer, and once that heterodimer has bound the HRE it pulls p300/CBP in.
Their acetyltransferase activity acetylates the nearby histones, the chromatin loosens, and only then is the target gene actually transcribed. So they are not readers of DNA at all. They are the machinery that makes the DNA readable, by loosening the chromatin around the gene that is about to be transcribed. That is also why an HRE on its own is not enough, because the

complex has to recruit the co-activators before the gene will produce anything at all.
Why this course brings it up: In Lecture 5 Dr. Wong uses this chain to set out hypoxia, then HIF-1α stable, then into the nucleus, then p300/CBP recruited, then HRE, then VEGF, GLUT1 and EPO, and he extends it from there into the design of an HRE-controlled CAR-T. **Appears in:** Ch 7, Lect 5

▪ P450

In one sentence: Cytochrome P450, the family of liver enzymes that turn drugs and toxins into more active molecules.
Detailed explanation: P450 is a large group of oxidising enzymes found in the liver and in other tissues, and what they do is chemically modify drugs and carcinogenic precursors. Sometimes that is detoxification, and sometimes the opposite happens and the precursor is turned into something that can bind DNA.
The tobacco-specific nitrosamines are activated exactly this way. The same family of enzymes can both detoxify and cause cancer, and the difference lies entirely in whether the product it rewrites the molecule into is less reactive or more reactive with DNA. That is an uncomfortable idea to sit with, because the enzyme is not the villain, it is the chemist.
Why this course brings it up: He does not name P450 as an exam question on its own, so it is concept-only. The crucial point is the order: a toxin has to be metabolised before it can damage DNA. So the enzyme comes before the lesion in every one of these stories. **Appears in:** Lect 3

▪ p50 and p65

In one sentence: The pair of NF-κB subunits that go into the nucleus and read the promoter of an inflammatory gene.
Detailed explanation: NF-κB normally exists as a p50/p65/IκBα complex, held in the cytoplasm by IκBα. When the signal arrives, IκB kinase phosphorylates IκBα, IκBα is degraded, and only then do p50 and p65 go into the nucleus and bind the promoter of an inflammatory gene to start transcription.
p65 is the main activating subunit, while p50 can also form homodimers with other factors, and the two are responsible for not entirely the same set of genes. The order is the whole point: nothing moves until the inhibitor is phosphorylated and destroyed, and the destruction is irreversible, so the only way back is to build a new IκBα.
Why this course brings it up: In Lecture 5 he uses these two subunits to teach the order of signal transduction, and links them to the cascade LPS, TLR4, IKK, IκB, NF-κB, nucleus, which is a chain that could turn up as a question. That whole line is worth being able to recite in order. **Appears in:** Ch 7, Lect 5

▪ p53

In one sentence: The tumour suppressor protein that makes a cell carrying damaged DNA stop and not divide.
Detailed explanation: p53 is the gatekeeper of the genome. When DNA is damaged or replication errors pile up, p53 is activated and holds the cell cycle at G1 while repair takes place; if the damage cannot be fixed, it switches to apoptosis instead, and it also takes part in mismatch detection and proofreading.
TP53 is the most frequently mutated gene in cancer, and about 40 % of cancers carry a p53 mutation, so the brake fails and the cell carries on dividing with damaged DNA. So p53 answers the cell's question about its own DNA before it is allowed to copy it.
Why this course brings it up: He gives the roughly 40 % figure on the cell cycle slide in Lecture 2; in Lecture 4 he also links p53's 3′→5′ exonuclease activity to resistance to nucleoside analogue drugs (excising the drug → resistance; not excising → cell death signalling). **Appears in:** Ch 6, Ch 10, Lect 2, Lect 3, Lect 4

▪ Palbociclib

In one sentence: A CDK4/6 inhibitor that stops the cell cycle at G1.
Detailed explanation: For a cell to move from G1 into S phase, Cyclin D and CDK4/6 have to phosphorylate pRb so that E2F is released and can switch on the S phase genes. Palbociclib holds CDK4/6 down, so pRb stays insufficiently phosphorylated, E2F stays held back, and the cell remains in G1 instead of entering S phase.
It is a textbook example of using a drug to control the cell cycle, and it is the figure where Dr. Wong names palbociclib inhibiting CDK4/6 and causing cell cycle arrest. The drug is not attacking the cancer cell directly; it is removing a licence that the cell needed in order to copy its DNA. So the mechanism sits upstream of the damage and upstream of the mutation, which is why it can work on a tumour that has no new mutation to target.
Why this course brings it up: In the treatment list in Lecture 2 he puts it alongside bispecific antibodies as drugs that change the fate of a cell. But he has also said you do not need to memorise the whole table of cyclin and CDK pairings. The point is the clinical logic of CDK4/6 inhibition → G1 arrest. **Appears in:** Ch 8, Lect 2

▪ PAM

In one sentence: The short DNA sequence that the CRISPR enzyme requires to be present next to its target.
Detailed explanation: Cas9 does not cut DNA anywhere it likes. It first has to find a short sequence called the PAM, the common SpCas9 case being NGG, and only after it has confirmed that it is standing in the right place does it pair its target with the guiding sequence of the gRNA and cut out a double-strand break.
Without a suitable PAM the enzyme does nothing at all, and that is an ancient mechanism that bacteria use to tell their own DNA apart from foreign DNA, and it also limits the sites that CRISPR can edit. The PAM is a condition rather than a target: it is not what gets cut, it is what tells the enzyme that cutting here is allowed, and that is why it is written as sitting next to the target.
Why this course brings it up: On his CRISPR cartoons in Lectures 2 and 4 Dr. Wong labels the gRNA and the PAM, and he points out that the two nuclease domains of Cas9, RuvC and HNH, do the cutting. He also says that CRISPR has replaced the

older siRNA and shRNA knockdown methods. **Appears in:** Ch 6, Lect 2, Lect 4

▪ PARP inhibitor

In one sentence: A drug that blocks a DNA repair enzyme, so it kills BRCA-deficient tumour cells while letting normal cells survive.

Detailed explanation: This is the textbook example of synthetic lethality. A normal cell has a working HR repair pathway; a tumour cell with a BRCA1/2 mutation has lost it, and in that situation a PARP inhibitor both suppresses PARylation and pins PARP onto single-strand breaks, so the replication fork stalls and accumulates double-strand breaks — damage it cannot repair, and so it dies.

Normal cells have HR as a fallback and survive. That is why the therapeutic window is so wide. Two broken things at once, and the cell can only survive one of them.

Why this course brings it up: In Lecture 4 he lists it as a Tier 2 concept and explains the definition of the therapeutic window (the margin between the minimum effective dose and the toxic dose); one entry in the case bank is "the tumour becomes resistant to a drug that is obviously working". **Appears in:** Ch 6, Ch 10, Lect 4

▪ Pasha

In one sentence: The partner of the miRNA cutting machine, the one that lets Drosha recognise the hairpin to be cut.

Detailed explanation: The first step of miRNA happens inside the nucleus. Drosha is a pair of scissors that cuts the primary miRNA into the precursor miRNA hairpin, but Drosha cannot recognise on its own which part it should cut, and it needs Pasha to supply that recognition.

The precursor hairpin is then carried out of the nucleus by Exportin-5, and finally Dicer cuts it into short double-stranded RNA. Without Pasha, Drosha does not know where to cut and the first step of miRNA goes wrong, and that is a good example of a molecular machine needing a recognition partner. The scissor cannot find its own work, so it is given an address. And the order matters: the first cut is in the nucleus, and only then does the hairpin leave for the cytoplasm.

Why this course brings it up: Dr. Wong does not name Pasha as an exam topic on its own, so it belongs to conceptual understanding. The staged production of miRNA is content he laid out in Lecture 1. **Appears in:** Ch 7, Lect 1, Lect 5

▪ Passenger

In one sentence: A hitchhiker mutation that gives no growth advantage of its own.

Detailed explanation: When a large stretch of chromosome is copied or rearranged, every gene nearby is changed along with it, and most of those changes have nothing to do with tumour growth, like a typo on page 900 of a manual.

The way to tell a passenger from a driver is not to look at the mutation itself but to ask whether it brings a functional growth advantage. So a passenger is defined by what it does not do, and identifying one means you have to look at the whole picture rather than at the lesion.

Why this course brings it up: In Lecture 3 Dr. Wong lists driver and passenger as one of the most common multiple-choice pairings, and stresses that finding the real driver means combining the CNV profile with mRNA expression. His worked example is NPM1. **Appears in:** Ch 3, Ch 10, Lect 3

▪ patent pool

In one sentence: A league of patent holders who license from each other.

Detailed explanation: When neither of the two CRISPR patent lines can monopolise the whole field on its own, one solution is to build a patent pool: everyone puts their patents into the same pool, agrees the terms on which they license each other, and each of them is then free to work on their own application.

Broad has been pushing this direction. It is the most practical lesson intellectual property has to offer in biotechnology: being able to make something, and being legally able to sell it, are two different questions. Pooling does not give up the patents; it just stops them cancelling each other out.

Why this course brings it up: He does not name patent pool as an exam topic on its own, so it is a concept to understand; he states plainly that the IP landscape is not examined. **Appears in:** Ch 12, Lect 4

▪ paternal

In one sentence: On the father's side.

Detailed explanation: In the context of mitochondrial inheritance the word paternal is used to describe the father's mitochondria: after fertilisation the sperm's mitochondria are marked and destroyed, so all of the embryo's mitochondria come from the egg and therefore from the mother, and none of the father's are left at all.

That is also why this inheritance pattern is very easy to recognise on a pedigree: it appears in several consecutive generations, it appears in every generation, and it passes only from the mother.

Why this course brings it up: Dr. Wong does not name paternal as an exam question on its own, so it is conceptual understanding, and the three features of mother to all the children, father to none, every generation is the first-grade question he names. **Appears in:** Ch 8, Lect 2

▪ PCA (principal component analysis)

In one sentence: The method that squeezes thousands of measurements down onto a few axes, so that cells can be drawn as points.

Detailed explanation: When thousands of genes are measured at once, every sample is a point in thousands of dimensions, and you cannot draw that. PCA looks for the few directions that explain most of the variation and projects the points onto those axes, and then the clustering becomes visible: which group of points sits together, which might be one cell type, and which points are outliers, which might be an odd sample.

It is an unsupervised method, so it does not tell you the answer, it only shows you the structure. It is worth remembering that the axes are computed from the data, so the picture is a summary of what is in the sample rather than a list of what you asked for.

Why this course brings it up: In the metabolomics workflow of Lecture 1 Dr. Wong lists PCA as step five, data mining, alongside the heat map and the volcano plot, and he also uses t-SNE to show the clustering of different cell types. **Appears in:** Ch 10, Ch 11, Lect 1

▪ PCR

In one sentence: Polymerase chain reaction, which copies a chosen piece of DNA into millions of copies so it can be measured.

Detailed explanation: Template, primer pair, dNTP and DNA polymerase go into the same tube, and the cycle runs three steps again and again: denaturation (separate the two strands), annealing (the primers stick), extension (polymerase writes the new strand).

Each round doubles the yield, so after n rounds you have about 2^n copies. The hot-start version keeps the polymerase bound by an antibody and releases it only at the first denaturation, so it cannot run away at the set-up temperature. Three steps, repeated, doubling each time.

Why this course brings it up: In Lecture 1 he uses PCR to show the practical consequence of the double helix model, and in Lecture 4 he uses it for in vitro manipulation; the relationship between GC content and Tm is exactly what primer design is based on. **Appears in:** Ch 3, Ch 11, Lect 1, Lect 3, Lect 4

▪ PD-1

In one sentence: A checkpoint receptor on the T cell. When PD-L1 binds it, the T cell is switched off.

Detailed explanation: PD-1 is an inhibitory receptor on the surface of the T cell, and under normal conditions it is there to stop the T cell from over-activating. The tumour turns that to its advantage by expressing PD-L1 on its own surface: once PD-L1 binds PD-1, the T cell goes into anergy and its attack power is shut down.

This is the commonest trick tumour immune escape has. Block it and you have taken the brakes off. Dr. Wong also reminds you that PD-L1 is only one of the routes out, because a tumour with a high MSI load escapes by a different route, carrying a large number of neoantigens.

Why this course brings it up: He uses it again and again in Lectures 2 and 3. Pairing the checkpoint drugs with MSI-H/dMMR tumours is the content of the eleventh entry in his case bank, and it is also the subject of the dMMR versus pMMR KM curve that he calls one of the most important figures in modern medicine. **Appears in:** Ch 9, Lect 2, Lect 4

▪ PD-L1

In one sentence: The protein on the tumour cell that binds PD-1 and switches the T cell off.

Detailed explanation: It pairs up with PD-1 on the surface of the T cell, and once the two are bound the T cell's signal is suppressed and the tumour escapes immune surveillance. PD-L1 can sit on tumour cells, and it can also sit on immune cells and on MDSCs.

His HDAC inhibitor slide shows RGFP 966 acting through BRD4 to push PD-L1 up, which makes the tumour more sensitive to the blockade instead of less, and that is what he calls gene addiction. The same group of HDAC inhibitors contains other drugs: ricolinostat and citarinostat make the tumour easier to kill, while RGFP 966 raises the activity of the CTL cells. The target is therefore not the T cell but the tumour, which is why the drug is an antibody against the tumour's own protein. And the level of PD-L1 is the measurement that goes with the response, which is the link he keeps making between expression and outcome.

Why this course brings it up: On his checkpoint slide in Lecture 3 Dr. Wong puts the two groups of drugs side by side, and the relationship between the level of PD-L1 expression and the response to treatment is the main thread of the immunotherapy story he has taught. **Appears in:** Ch 9, Lect 2, Lect 3, Lect 4

▪ PD-L2

In one sentence: PD-L2, the second ligand for PD-1.

Detailed explanation: Both PD-L1 and PD-L2 can bind PD-1 on the T cell, and once PD-1 is bound the T cell enters anergy and loses its attacking power. PD-L2 is expressed mainly on dendritic cells and macrophages, and it is a second source of inhibitory signalling in the tumour microenvironment.

His list puts PD-L1 and PD-L2 side by side, to show that a tumour can use different molecules to reach the same brake. That is the practical reason the checkpoint drugs need to know about both: blocking only one of the two ligands leaves the other one free to pull the brake.

Why this course brings it up: He does not name PD-L2 as an exam question on its own, so it is concept-only. The Tier 2 point is that PD-1 and PD-L1 are the treatment targets of a hot tumour. PD-L2 is the footnote on that story. **Appears in:** Ch 9, Lect 2

▪ PECAM-1

In one sentence: Hold on to it and a white blood cell can squeeze out between two cells and leave the blood vessel.
Detailed explanation: White blood cell extravasation has four steps. P-selectin and E-selectin make it roll along. ICAM-1 and VCAM-1 make it stick and activate. PECAM-1, which is CD31, forms a gate at the junctions between the cells of the vessel wall, and a white blood cell grabs it and squeezes through.
And then it follows the MCP-1 gradient towards its target. Every step needs a different molecule and none of them can be skipped, and the order cannot be rearranged either: roll first, then stick, then squeeze through, and only then walk along the signal to the target. Think of it as four locks on four different doors, and the cell has to open them in order.
Why this course brings it up: On the four-step white blood cell extravasation diagram in Lecture 2 he labels PECAM-1 as step three. It is also the same gate that tumour cells squeeze through and that immune cells must go through to enter a tumour. Which is why blocking it is interesting for both sides of the fight. **Appears in:** Ch 9, Lect 2

▪ Pembrolizumab

In one sentence: An anti-PD-1 checkpoint inhibitor. It is the drug for MSI-high tumours of the hot kind.
Detailed explanation: It occupies PD-1 on the T cell so that the tumour's PD-L1 cannot grab hold, and the T cell regains its killing power. It belongs to Keytruda, and it was the first immune checkpoint inhibitor approved for MSI-H / dMMR metastatic colorectal cancer.
The tumours it works best on are the ones with enough mutations to produce plenty of neoantigen. MSI-H / dMMR tumours have many mutations and many neoantigens, so he calls them hot tumours, and they are exactly the group this class of drug suits. Which is the point: the drug does not attack the tumour directly, it only takes the brake off the T cell.
Why this course brings it up: In Lecture 4 he cites the work of Le DT and others (the progression-free survival curve for dMMR against pMMR, P<0.001 by log-rank test), and he calls this graph "one of the most important graphs in modern medicine"; the checkpoint slide in Lecture 3 puts it alongside nivolumab and atezolizumab. So the concept is examinable even when the drug name is not. **Appears in:** Lect 3

▪ Penetrance

In one sentence: What proportion of the people carrying a mutation actually develop the disease. It is never 100%.
Detailed explanation: Among people carrying the same disease-causing mutation, some develop the disease and some never have a symptom in their lives, and the reasons include other modifying effects of the gene, the tissue specificity of its expression, and the environment.
Incomplete penetrance and variable expressivity are two sides of the same fact, and together they explain why patients in one family line can be so different from one another in how badly they are affected. So the answer to “will this person get it” is always a probability rather than a certainty. And the same fact has to be reported the same way, so two people in one family carrying the same mutation can get two quite different answers, and both can be correct.
Why this course brings it up: It is the necessary concept for one of the two scenario questions Dr. Wong names (the pedigree and genetic counselling one): he says the answer has to be a probability and not a certainty, and he explicitly lists incomplete penetrance and the threshold effect as the two reasons. **Appears in:** Ch 8, Lect 3

▪ Peptide

In one sentence: A short chain of amino acids joined by peptide bonds.
Detailed explanation: Short enough that it is usually called a peptide, long enough to have a definite three-dimensional shape and then it is a protein. In drug design, peptides are the middle road: much easier to make than small molecules, and much larger.
His TCR generation, the V(D)J assembly of antibodies, and neoantigens are all recognised as peptides, because what the cell presents to a T cell is a fragment. That last point is the one that ties it to his material: the immune system reads short pieces, not whole molecules.
Why this course brings it up: In Lecture 1 he uses peptides and peptide bonds to explain the hierarchy of protein structure, and he also points out that the mechanism of an mRNA vaccine is precisely to make a cell produce a piece of antigen and present it to the immune system. Same word, two uses. **Appears in:** Ch 2

▪ Peptide bond

In one sentence: The chemical bond that joins one amino acid to the next along a protein chain.
Detailed explanation: The carboxyl group of one amino acid, -COOH, and the amino group of the next, -NH2, condense with the loss of water and form an amide bond, releasing one water molecule.
That bond has partial double-bond character, so it is rigid and usually takes the trans configuration, and that rigidity is exactly what allows a protein to form a regular alpha helix and a beta sheet along one axis. Breaking a peptide bond takes energy, which is also why digesting protein needs an enzyme.
Why this course brings it up: It is the chemical foundation of the four levels of protein structure. Dr. Wong uses the C=O···H–N hydrogen bond to explain the beta sheet; the hydrogen bond is what holds the secondary structure, and the peptide bond is what holds the primary structure. **Appears in:** Ch 2

▪ Perforin

In one sentence: A protein that punches holes in the target cell so that granzyme can get in.
Detailed explanation: The killing by killer T cells and NK cells has three steps. After recognising the target, the granules move towards it and fuse with the cell membrane, releasing perforin.
Perforin punches a hole in the membrane, and the granzymes go in through the hole. The granzymes then start the caspase cascade and the target cell undergoes apoptosis. Perforin only does the opening; the actual execution is done by granzyme.
CAR-T goes down the same road: once the receptor recognises the antigen, the T cell kills the target with this same open-the-door and carry-out-the-sentence machinery. It is worth separating the two, because when this pathway is discussed people often say the T cell punches holes, when the hole is only the delivery step.
Why this course brings it up: On the CAR-T slide in Lecture 2 he draws perforin and granzyme, then the perforin pore, then cell death, as the killing mechanism of CAR-T. So the diagram is the same story he tells about killer T cells, with a different way of choosing the target. **Appears in:** Ch 9, Lect 2

▪ pH

In one sentence: The scale of acidity and alkalinity. Below 7 is acidic, above 7 is alkaline, and 7 is neutral.
Detailed explanation: pH is the negative logarithm of the hydrogen ion concentration, so it is a logarithmic scale: when pH drops from 7 to 6, the hydrogen ion concentration becomes ten times higher.
Blood is held at roughly 7.35 to 7.45 by the lungs and the kidneys, while the inside of the lysosome is kept at an acidic pH of about 5, because the digestive enzymes there only work in that acidity. This is also why HIF-1α accumulates in the hypoxic core of a tumour, since lack of oxygen brings acidification with it.
Why this course brings it up: It belongs to the basic chemistry of Part 1. Dr. Wong does not single it out, but the acidification of the tumour microenvironment is one of the obstacles he mentions when he talks about CAR-T failing in solid tumours. **Appears in:** Ch 1

▪ PHD and FIH

In one sentence: The enzymes that add an oxygen group to HIF-1α, and that step is what marks it for destruction.
Detailed explanation: Under normal oxygen conditions, PHD (prolyl hydroxylase) and FIH (factor inhibiting HIF) hydroxylate the proline and the asparagine of HIF-1α respectively. The hydroxylated proline is recognised by VHL, which is the adaptor of an E3 ubiquitin ligase, so HIF-1α gets polyubiquitination, is hooked to the proteasome and is destroyed.
In hypoxia the hydroxylases have no oxygen, the reaction stops, and HIF-1α accumulates and enters the nucleus. HIF-1α then binds HIF-1β, p300/CBP opens up the chromatin, and genes such as VEGF, GLUT1 and EPO are switched on. He calls this mechanism an oxygen sensor: with oxygen available HIF-1α hardly has time to accumulate at all. So the cell measures oxygen by destroying the protein that would otherwise respond to its absence. It is a slightly strange design, but it is efficient: no oxygen, no marking, no destruction, no response.
Why this course brings it up: In Lecture 5 he connects this chain to the design of HIF-CAR — roughly 50 % of solid tumours contain hypoxic tissue, so putting CAR expression under HRE control lets it fire only there. Which is the link between the oxygen sensor and the CAR that only fires in a hypoxic tumour. **Appears in:** Ch 7, Lect 5

▪ Phenotype

In one sentence: The way a cell or a person actually turns out.
Detailed explanation: Phenotype is decided by genotype plus environment, plus the regulation of gene expression. The genotype is the starting point and the phenotype is the outcome, and everything in between is regulation, which is where most of the biology in this course lives.
The same genotype can produce different cell types because the expression control is different, since a T cell and a dendritic cell run different expression programmes. The same target can also produce opposite tumour behaviour in different cancer types, and his example is the receptor tyrosine kinase, which predicts a worse outcome in pancreatic cancer but a better one in renal cell carcinoma.
Why this course brings it up: In Lecture 1 Dr. Wong uses this example to show that a biomarker and a drug target have to be tied to a cancer type, and it is also the premise on which the two words prognostic and predictive can be separated at all.
Appears in: Ch 7, Lect 5

▪ Phosphatase

In one sentence: The enzyme that removes the phosphate group and switches the protein back off.
Detailed explanation: It is the exact opposite of the kinase: the kinase uses ATP to add a phosphate, and the phosphatase uses water to cut the phosphate away again. The phosphate carries a negative charge, and once it has been removed the electrostatic repulsion disappears, the protein folds back the way it was, and the switch is off.
Because the reaction takes only seconds, this is the fastest way the cell has to control anything, far faster than making a new protein would be. The on and off are both quick, they are run by two different enzymes, and they are the pair Dr. Wong uses whenever he wants to explain how a cell changes its mind within seconds rather than within hours.
Why this course brings it up: In Lecture 1 he uses the kinase and phosphatase cartoon to explain the “fast on/off switch”, with Hippo–YAP as the complete worked case. Lectures 2 and 5 keep calling on the same mechanism. **Appears in:** Ch 1, Ch 2, Lect 1

▪ Phosphodiester bond

In one sentence: The chemical bond that joins adjacent nucleotides and forms the backbone of a DNA strand.
Detailed explanation: The 3′ hydroxyl of one nucleotide and the 5′ phosphate of the next condense with the loss of water, forming a 3′→5′ phosphodiester bond. When DNA ligase seals the gap between Okazaki fragments, it is catalysing the formation of exactly this bond, joining a 5′ phosphate to a 3′ hydroxyl.
Once the bond has formed it cannot be undone, so DNA replication and repair can only proceed forwards, and the directionality of the strand comes from that. The consequence is a one-way street: each nucleotide can only be added to the 3′ end, which is why a lagging strand has to be built in fragments and a leading strand does not. And that is a different idea from the hydrogen bonds that hold the two strands to each other, which are the ones that come apart when the helix opens.
Why this course brings it up: In Lecture 4 he writes the ligase sealing action out explicitly as a phosphodiester bond between a 5′ phosphate and a 3′ hydroxyl. The same bond is also the explanation of why DNA has a direction. One chemistry, two questions. **Appears in:** Ch 5, Lect 4

▪ Phospholipid

In one sentence: It has a hydrophilic head and a hydrophobic tail, and that is exactly why a membrane can assemble itself.
Detailed explanation: Amphipathicity makes phospholipids line up into a bilayer by themselves in water: the heads face the water and the tails point inward, so the hydrophobic middle is sealed off.
The membrane can therefore close itself, repair itself, and become a controllable switchboard by having proteins embedded in it. His signalling cascade is built on top of this: PI is phosphorylated to PI(4,5)P, and then phospholipase C cuts it into DAG, which stays on the membrane, and IP3, which is soluble.
Why this course brings it up: Dr. Wong returns to this cascade again and again in Lectures 1 and 2. The PI(4,5)P, DAG, IP3 and PKC on the membrane are the complete pathway on his slide, but the metabolic pathway details themselves are not examined. **Appears in:** Ch 1

▪ Phosphorylation

In one sentence: Adding a phosphate group, and that usually means switching a protein on.
Detailed explanation: The phosphate group carries a negative charge, and adding it changes the distribution of charge on the protein and so affects folding, often changing its conformation. Where the phosphate lands matters: acting on an activation region can switch a protein on, acting on an inhibition region can switch it off;
it can also change where the protein is located, allowing it to move from the cytoplasm into the nucleus.
Why this course brings it up: In Lecture 1 he uses Hippo–YAP to show a reversible phosphorylation switch; in Lecture 5 he lists phosphorylation as one of the seven ways a transcription factor can be activated, and he uses it to explain why phosphorylated IκBα is degraded. **Appears in:** Ch 1, Ch 2, Ch 7, Lect 1

▪ PI3K

In one sentence: The signalling kinase downstream of the receptor, the main carrier of the HER2 message.
Detailed explanation: When HER2 is bound by Herceptin and can no longer pair up with HER3, the PI3K downstream is never switched on, and the cell's survival signal is cut. PI3K has two subunits, p85 and p110;
he uses it to explain to a beginner the chain "the receptor is blocked by a drug → the downstream signal goes out". Downstream of PI3K is a whole run of growth and survival messages, so blocking the receptor upstream is the same thing as turning the volume on that whole downstream chain back to zero. The receptor is the handle and PI3K is the pipe the water runs through, so a blockage at the handle silences everything downstream of it. Nothing has to happen at PI3K itself for the effect to be complete.
Why this course brings it up: He does not name PI3K as an exam topic on its own, so it is a concept to understand; the Tier 3 numbers he lists are the receptor occupancy figures for Herceptin in breast cancer and gastric cancer (2 % and 10 %). Those are the numbers to write down. **Appears in:** Ch 10, Lect 2

▪ pinhole

In one sentence: The small hole a confocal microscope uses to block out out-of-focus light.
Detailed explanation: An ordinary fluorescence microscope lights up the whole field, so the top and bottom of a thick cell both glow and are superimposed on each other. A confocal microscope puts a pinhole in front of the detector, and only the light from the focal plane gets through, so it can take an optical section layer by layer, and stacking the sections gives the 3D reconstruction.
The size of the pinhole decides how thin the sections are: the smaller the hole, the higher the axial resolution and the closer the image is to a true three-dimensional reconstruction.
Why this course brings it up: Dr. Wong does not name the pinhole as an exam question on its own, so it is conceptual understanding, but it is the piece of hardware that makes the difference between a flat picture and a three-dimensional one.
Appears in: Ch 9, Ch 11, Lect 2

▪ PKC (protein kinase C)

In one sentence: A signalling enzyme that can switch on inflammation and growth pathways.
Detailed explanation: When PI(4,5)P is cut by phospholipase C, the DAG left on the membrane activates PKC, and the activated PKC is the starting point of a whole series of signal transduction steps, opening pathways such as NF-κB and MAPK.

In Dr. Wong's NF-κB amplification loop, PKC activation is itself one of the inputs: pro-inflammatory cytokines, PKC activation, viruses and oxidative stress all start NF-κB, and the TNF-α and IL-1β that NF-κB then produces go back and activate NF-κB once more. So PKC sits at the entry of the cascade and also at the feedback that keeps it running. In the picture it is the meeting point between the membrane signal and the nucleus.
Why this course brings it up: He marks PKC both on the signalling cascade slide in Lecture 1 and on the NF-κB amplification loop in Lecture 5. His slide lists the “detailed signal transduction cascade” as content you do not need to memorise, but you should recognise the starting point of the cascade and the direction it runs in. **Appears in:** Lect 1, Lect 5

▪ Plasma membrane
In one sentence: The outer skin of the cell. A lipid bilayer that controls what gets in and out and carries the receptors.
Detailed explanation: A phospholipid bilayer with proteins and sugar chains embedded in it. The phospholipids on their own do not let charged or large molecules through, so the selective permeability comes from the channels and the transport proteins. The receptors in the membrane carry messages from outside the cell inwards, and the export pumps and transport proteins in the membrane send substances back out. The cell's "inside" and "outside" are held apart by the hydrophobic core of this lipid layer, and any protein that wants to cross it has to pay for the journey.
Why this course brings it up: In the organelle and signalling section of Lecture 2 Dr. Wong treats the membrane as the first checkpoint, and signal transduction, receptors, dimerisation and receptor occupancy all happen on it. **Appears in:** Ch 8, Lect 2

▪ plasmid
In one sentence: A small circular DNA in bacteria that can replicate on its own.
Detailed explanation: It is not part of the bacterial chromosome, but it can replicate itself and it can carry genes in with it. Researchers use it as a vector: put the gene you want expressed into the plasmid, deliver it into the bacterium, and the bacterium becomes a living factory producing millions of copies of the protein in a few hours.
That speed is the reason recombinant expression became a routine technique rather than a curiosity. In Dr. Wong's anti-TERT experiment the plasmid has three further uses: propagation (bulk copying), expression (making the protein) and sequencing (reading out the sequence). So the same circular piece of DNA is asked to do the copying, the making and the reading, in that order.
Why this course brings it up: He does not name plasmid as an exam topic on its own, so it belongs to conceptual understanding. Note that the cccDNA of HBV is circular DNA as well, but that one belongs to the virus. **Appears in:** Ch 3, Lect 4

▪ Pluripotent
In one sentence: A cell that can become almost any type of cell in the body.
Detailed explanation: Pluripotent sits above multipotent and below totipotent, and only totipotent can form a complete organism, placenta included. ESC and iPSC both count as pluripotent: they can self-renew, and they can multiply indefinitely in a dish.
The standard validation is the teratoma assay. Inject the cells into an immunodeficient mouse, and if a tumour forms containing tissue from all three germ layers, that proves the cells have pluripotency. It is a blunt test, but it is blunt in the right direction, because a tumour is a system that makes every kind of cell the body knows how to make, and if your cells can seed all three layers they must have all three potentials in them.
Why this course brings it up: In Lecture 2 he specifies this as the gold-standard validation for iPSC, and he says do not forget it. He also reminds you that quality control on stem cells relies on a unique expression pattern, on timing, and on negative staining to rule out contaminating cells. The assay is the headline, the QC is the footnote, and both belong to the same slide. **Appears in:** Ch 8, Lect 2

▪ PNCTR
In one sentence: A long non-coding RNA whose YYUCUY motif binds PTBP1 and so controls splicing.
Detailed explanation: PNCTR is one of the prototypes of lncRNA function, and it plays the part of a guide: it carries the YYUCUY motif (the sequence recognised by splicing factors), which lets those sequences bind PTBP1, an RNA-binding protein, and PTBP1 then adjusts where the pre-mRNA gets spliced.
Its method is not to change the DNA; it is to deliver the splicing machinery to the right target message, so that one gene can be cut into different isoforms. The scaffold is not the message; it is the address on it. And the outcome is that one gene can produce more than one protein.
Why this course brings it up: In Lecture 5 he lists it alongside APOLO, XIST and TINCR as the four lncRNA prototypes, and he points out that lncRNA is becoming an emerging target for drug discovery and a biomarker. So the mechanism, not the name, is what you need. **Appears in:** Ch 7, Lect 5

▪ POB-DNA
In one sentence: One of the DNA adducts formed by benzo(a)pyrene.
Detailed explanation: The benzo(a)pyrene in cigarette smoke is activated by P450 enzymes in the liver, and it then reacts with DNA bases to form what is called a POB-DNA adduct, and there are PHB-DNA and methylated adducts as well.
These adducts distort the double helix, so the polymerase easily miscopies during replication, and if the adduct is not repaired the second round of replication fixes the error as a mutation. The adducts can be detected with antibodies or with advanced mass spectrometry, so the traces a chemical carcinogen has left in the human body can actually be measured. Which is the

whole argument for molecular epidemiology: you do not have to wait for the cancer, you can look for the damage that came before it.
Why this course brings it up: He does not ask you to memorise the abbreviation, so it is concept-only. The chain of adduct, then replication error, then a fixed mutation is one he teaches very clearly. Keep the chain and drop the acronym. **Appears in:** Lect 3

▪ Point mutation

In one sentence: A change in a single letter of the DNA. It may be harmless, or it may be a driver.
Detailed explanation: What it does depends on where it lands and what kind it is. A point mutation at an important site may be a driver, while one at an unimportant site may be completely harmless.
And if the substitution of a single letter causes an early stop (nonsense) or shifts the reading frame, the consequences are far more serious than simply swapping one amino acid. Dr. Wong's classroom example is KRAS G12C, a point mutation that leaves Ras stuck in its active state. So one letter, one position, one codon, and the cell behaves as though it has been given a permanent instruction it cannot ignore. The size of the change is not a reliable guide to the size of the consequence.
Why this course brings it up: In Lectures 3 and 4 he sets point mutation against the larger structural abnormalities of the chromosome. He also points out that a point mutation can only be found by sequencing, and that karyotyping will miss it.
Appears in: Ch 3, Lect 3

▪ POL

In one sentence: The segment of DNA polymerase that does the synthesis.
Detailed explanation: POL is the structural part of the polymerase, and its job is to join dNTPs into a new chain; the same molecule carries another segment, EXO, which does the proofreading.
He draws the polymerase as a right hand: the fingers grip the template, the palm holds the new chain, and the thumb pushes the nucleotides into place. His slide puts the POL and EXO segments side by side to make the point that synthesis and proofreading are two functions on the same machine. The hand is doing the two jobs at once, which is the point of the drawing.
Why this course brings it up: Dr. Wong does not name POL as an exam question on its own, so it is conceptual understanding, and the contrast of proofreading 3′→5′ against elongation 5′→3′ is the Tier 2 item. **Appears in:** Ch 5, Lect 4

▪ Polarity

In one sentence: One end of a molecule carries a little positive charge and the other a little negative, like a small magnet.
Detailed explanation: An uneven distribution of charge creates polarity, and the water molecule is the classic example of an amphipathic molecule, which is why it can dissolve other polar substances and can get on with the hydrophilic head of a phospholipid.
The idea carries straight on up to the polarity of a cell: the cell knows which side is up and which side is down, and the polarity regulators of the Hippo pathway, together with NF2, AMOT, α-catenin and F-actin, are the upstream factors that carry the signal from the cell junction into Hippo. And the same word covers both levels, because a cell with a top and a bottom is polar in exactly the same sense that a water molecule is polar: charge, or orientation, is distributed unevenly.
Why this course brings it up: On his Hippo slide Dr. Wong lists the polarity regulators as the first group of upstream factors that turn Hippo on, and polarity itself belongs to Part 1 basic chemistry. **Appears in:** Ch 1

▪ Pold1

In one sentence: The gene for DNA polymerase delta, the one that copies the lagging strand.
Detailed explanation: Pol δ works on the lagging strand at the replication fork, taking over the extension of each Okazaki fragment, while Pol ε handles the leading strand. Pol δ carries 3′→5′ proofreading exonuclease activity, and once that activity is broken by a mutation (which Dr.
Wong calls proofreading-defective) the error rate of replication goes through the roof and the cell becomes an ultramutator. It is the fastest way to manufacture neoantigens, and the cell does it by accident. The proofreading step is what normally keeps the error rate low enough for the genome to stay usable. Pold1 and Pole are the guardians of the two strands respectively, and he puts these two mutations alongside the Mlh1 knockout to show that replication proofreading and MMR are separate lines of defence. Two different mistakes, two different correction machines, and a cell that has lost both is in real trouble.
Why this course brings it up: He does not name Pold1 as an exam topic on its own, so it belongs to conceptual understanding. The conclusion he draws is the one to hold on to: proofreading defect → high mutation rate → many neoantigens → and the prognosis is actually better. **Appears in:** Ch 5, Lect 4

▪ Pole

In one sentence: The gene for DNA polymerase epsilon, the enzyme that copies the leading strand.
Detailed explanation: Pol ε carries proofreading exonuclease activity, just as Pol δ does. His experiments use proofreading-defective Pole mutants to make ultramutator tumours, and compare them against Pold1 mutations, against Mlh1 knockouts, and against the double mutant Pole e/e Pold1 e/e, which shows that each of the two strands of replication has its own gatekeeper.
Pol ε handles the leading strand and Pol δ handles the lagging one, and the two keep separate watch; his comparison experiment exists precisely to show that the two lines of defence can fail independently of each other. One gatekeeper failing is bad enough; two failing at once is what turns a cell into an ultramutator. And the comparison is what separates the two stories

from a single one, because without it you cannot tell whether the leading strand has its own defences or whether both strands share one.
Why this course brings it up: He does not name Pole as an exam topic on its own, so it is a concept to understand; the counterintuitive conclusion he states very clearly is that ultra-mutator tumours have more neoantigen, more TIL and a better prognosis, and that is the part to hold on to. **Appears in:** Ch 5, Lect 4

▪ Pole e/e Pold1 e/e

In one sentence: A double-mutant mouse that has lost the proofreading ability of both polymerases at once.
Detailed explanation: The e/e allele means the exonuclease has lost function. This double mutation shuts off the proofreading ability of both Pol ε and Pol δ, which means two lines of defence fail at the same time, so in his tumour incidence table these mice have the shortest survival and the most tumours.
It is used to show that proofreading and MMR are two separate lines of defence: you need either one, and losing both is worse than losing one. The experiment is neat because it uses two mutations at once to ask what each one contributes on its own.
Why this course brings it up: He does not name this genotype as an exam question on its own, so it is concept-only. Its job is to support the inference that there are two lines of defence. So the reasoning is the point, not the notation. **Appears in:** Ch 5, Lect 4

▪ Poly-A tail

In one sentence: The run of adenosines at the end of an mRNA. The longer the tail, the more translation.
Detailed explanation: A string of A's is attached to the 3′ end. It helps the RNA get out of the nucleus, it protects the message from enzymes that break it down, and it lets the ribosome hold on to the message.
His rule is simple: the longer the poly-A tail, the more translation, and the shorter it is the faster the message decays. So changing the length of the tail changes how much protein that message can make, and that is the setup for his next topic, alternative polyadenylation. Think of it as both a ticket out of the station and a handle the ribosome holds while it works.
Why this course brings it up: In Lecture 5 he puts it alongside the 5′ cap and the 3′ UTR as the three features that determine mRNA stability, and it appears again in the alternative polyadenylation material. He does not name this term in particular, so it matters as part of a set of three, not as a free-standing fact. **Appears in:** Ch 7, Lect 5

▪ Polyadenylation

In one sentence: Adding the poly-A tail to the end of an mRNA. Which site you pick is itself a regulatory step.
Detailed explanation: When the polymerase reaches a termination signal, poly(A) polymerase attaches the tail. But the message often has several polyadenylation sites to choose from: pick the proximal one and the 3′ UTR is short;
pick the distal one and the 3′ UTR is long enough to bind RNA-binding proteins, microRNAs and lncRNAs. Which one you pick decides how stable the message is, how much it is translated and where it goes — so the destination of the message is itself a control point. So the end of the message is not the end of the story; it is one of the places where the story is decided.
Why this course brings it up: On his Lecture 5 slide he puts proximal and distal PAS side by side, and BDNF is his example: the short isoform stays near the cell body, while the long isoform is carried into the dendrite, so that the protein is made exactly where it can do its job. And the BDNF example is worth remembering because it is concrete. **Appears in:** Ch 7, Lect 5

▪ Polyadenylation site (PAS)

In one sentence: The place on the RNA where the poly-A tail gets added. Choosing an early one or a late one changes the message.
Detailed explanation: The same gene can have several polyadenylation sites. Choosing a proximal PAS gives you a short 3′ untranslated region; choosing a distal PAS gives you a long 3′ UTR, long enough for RNA-binding proteins, miRNAs and lncRNAs to have somewhere to bind, and so stability, translation, nuclear export, the position of the message and the position of the protein all change.
That is what turns the length of a message into a control switch. The information is all still in the sequence, but how much of it gets used is decided by where the transcript is cut, and the cell can move that decision from one site to another and get a different protein out of the same gene. So a single gene can produce two isoforms that differ only in how much tail they carry, and the tail is where the regulatory molecules sit. Nothing about the coding sequence has changed, and yet the protein that comes out can behave in a different way.
Why this course brings it up: In Lecture 5 Dr. Wong puts it in the alternative polyadenylation material alongside the BDNF isoform. Alternative polyadenylation is the last piece he adds to the puzzle of proteome diversity. **Appears in:** Ch 7, Lect 5

▪ Polycistronic mRNA

In one sentence: One RNA message that carries several genes at once, the way a bacterial operon does.
Detailed explanation: A single RNA transcribed from several structural genes at once, with its own start codon for each of them, so it can be translated into several different protein chains.
This is commonest in prokaryotes, with the lac and trp operons, and rare in eukaryotes, where a gene usually corresponds to one message, with a few exceptions. It is also one of the reasons the prokaryotic expression system is cheap: one transcription gives you several products, and you never have to switch a promoter on and off again. Because the genes share a promoter, the whole group is switched on and off as one, and the cell does not need a separate copy of the switch for each of them.
Why this course brings it up: In Lecture 5 Dr. Wong uses the operon diagram to show one promoter driving several structural

genes, and he also uses it as a concrete contrast between the prokaryotic and the eukaryotic expression systems. **Appears in:** Ch 7, Lect 5

▪ polymerase slippage

In one sentence: The polymerase slips when it copies a repeated sequence.
Detailed explanation: When the template is a repeated sequence such as TATATATATATA, the polymerase in the middle of copying can lose its place and leave a small unpaired loop on the new strand.
That is polymerase slippage. If MMR is intact the loop is noticed and corrected and the result is stable; if MMR has failed the length of the repeat changes again and again and it becomes MSI-H. The slippage on its own is not a big deal in a healthy cell, because MMR tidies it up; what is really dangerous is the time when MMR has failed at the same moment.
Why this course brings it up: The chain from MMR deficiency to MSI-H is the Tier 2 causal chain, and he does not ask you to memorise the phrase "polymerase slippage" itself. The sequence TATATATATATA is the kind of example worth being able to write out, because it makes the loop obvious. The name is worth remembering even though he does not ask for it. **Appears in:** Ch 6, Lect 4

▪ Polypeptide

In one sentence: A long chain of amino acids. Folded up, it is a protein.
Detailed explanation: Before it folds it is only a string of beads, and after it folds it becomes a machine with a specific shape that does a specific job. The atoms of the backbone are joined by peptide bonds, and the side chains decide how each stretch behaves and how flexible it is.
The smallest R-group, glycine, is the most flexible, and a bulky or charged R-group such as tryptophan or arginine is the stiffest. Folding itself is the hardest part to predict for a therapeutic protein. So the sequence decides everything, but only because the side chains decide what the folded shape will be.
Why this course brings it up: In Lecture 1 he builds this language with the four levels of structure, primary, then secondary, then tertiary, then quaternary. The risk of misfolding and of inclusion bodies also happens at this level. One of the two is the vocabulary, the other is the manufacturing risk. **Appears in:** Ch 2

▪ Polysaccharide

In one sentence: A long chain of sugar units, such as glycogen, starch or cellulose.
Detailed explanation: Monosaccharides are joined into chains by glycosidic bonds, with alpha1,6 bonds at the branch points and alpha1,4 bonds along the straight runs. The branching is not decoration: every branch end is a place an enzyme can attack, which is why glycogen can be released quickly without the whole chain having to be processed.
Plants store energy as starch and build cell walls out of cellulose; animals use glycogen.
Why this course brings it up: In Lecture 1 he teaches the structure of polysaccharides alongside their function, and uses the branching of glycogen to illustrate "how structure serves function"; the details of the metabolic pathways are not examined.
Appears in: Ch 1, Lect 1

▪ Polyubiquitination

In one sentence: Stringing a chain of ubiquitin molecules onto a protein, which in this course's context is the label that leads to degradation.
Detailed explanation: Ubiquitin is a small protein, and three enzymes hand it over to the target protein in turn, E1, E2 and E3. When HIF-1α is hydroxylated and recognised by VHL, VHL acts as the E3 ligase and puts the polyubiquitination tag on, the proteasome recognises that tag, and the whole protein is taken apart into short peptides.
The ubiquitin itself is recycled, so the same molecule can go on tagging a different protein again and again. So the tag is a message written in a language only the proteasome reads, and it is a label rather than a change of the structure underneath. It is a statement about the protein's future, not about its shape, and that is why the proteasome can recognise it at all.
Why this course brings it up: Dr. Wong puts it at a key position in the oxygen-sensing pathway of Lecture 5: hydroxylation, then VHL, then polyubiquitination, then proteasomal degradation. That is the mechanism behind HIF-1α being made and then immediately disappearing in normal oxygen. **Appears in:** Ch 7, Lect 5

▪ Positive-feedback loop

In one sentence: The product of a reaction switches the same reaction on again, so the reaction amplifies rapidly.
Detailed explanation: Dr. Wong gives two examples. NF-κB transcribes TNF-α and IL-1β, and those cytokines then go back and activate NF-κB, so the inflammatory response amplifies within minutes.
And DNA methylation and histone marks strengthen each other, forming a cellular memory that survives cell division. The advantage of positive feedback is speed; the disadvantage is that it is hard to stop by itself. Which is exactly why every amplified system in the body also has a brake built in, and why the anti-inflammatory mediators arriving at the end of the NF-κB sequence matter as much as the ones at the start. His NF-κB example also has a time order to it: TNF-α arrives first, then IL-1β, IFN-γ, IL-2 and IL-15, and only at the end do the anti-inflammatory mediators come out to close it down.
Why this course brings it up: In Lecture 3 he uses it to explain the cellular memory of epigenetics, and in Lecture 5 to explain why NF-κB is designed to be this “noisy”. He states plainly that you do not need to memorise the order of the cytokines.
Appears in: Ch 4, Ch 7, Lect 3, Lect 5

▪ Post-translational modification

In one sentence: A chemical change made to a protein after the chain has been made.
Detailed explanation: Once the chain is finished it can still have phosphates added, acetyl groups added, methyl groups added, ubiquitin attached, or a sugar chain hooked on. Those changes usually alter the activity, the stability, the location or the interaction partners of the protein, and they are the fastest layer of control the cell has, because you can switch a protein on and off without making a new one.
The marks on histones are the biggest group of these, and that is the histone code. The reason this layer is fast is simple: the protein is already built and sitting there, and all the cell has to do is add or remove a small group.
Why this course brings it up: In Lecture 3 he teaches the histone code, which is acetylation, methylation, ubiquitination and phosphorylation, and in Lecture 5 he teaches the hydroxylation and polyubiquitination of HIF-1α, all of which are post-translational modifications. He also reminds you that an HDAC inhibitor can restore histone acetylation and switch silenced tumour suppressor genes back on. The clinical application and the cell biology are the same list of marks. **Appears in:** Ch 2, Lect 1, Lect 5

▪ pRb

In one sentence: The protein that holds the cell cycle at G1 until the cell is ready to copy its DNA.
Detailed explanation: Unphosphorylated pRb holds on to E2F, and while E2F is held the transcription factor cannot start the S phase genes, so the cell stays in G1. When mitogen signalling arrives, Cyclin D and CDK4/6 phosphorylate pRb, E2F is released, the S phase genes are expressed and the cell enters S phase.
That is why he draws the chain from mitogen to pRb as a single line. His clinical extension is CDK4/6 inhibitors such as palbociclib: they do not kill the cell, they simply park it in G1 so it can never reach S phase. So the brake is not the cell cycle; it is one protein holding another one hostage.
Why this course brings it up: He draws this chain on the cell cycle slide in Lecture 2 and attaches the palbociclib CDK4/6 inhibitor story to it; he also says you do not have to memorise the whole table of cyclin and CDK pairings, but this clinical logic you do need. **Appears in:** Ch 8, Lect 2

▪ pre-IND

In one sentence: An informal meeting with the FDA before a company formally submits its IND.
Detailed explanation: The purpose of the meeting is not to obtain the regulator's blessing but to lay the plan out and talk it through: which animal model to choose, what data has to be ready first, and what the regulator expects.
The real scientific controversy Dr. Wong gives as an example is whether to use a non-human primate whose pharmacology is closer to ours, or a humanised mouse that is cheaper and faster but may not be close enough. By the end of the meeting the company usually has a written agreement in hand, and the experiments that follow are designed against that document.
Why this course brings it up: He does not name pre-IND as an exam topic on its own, so it belongs to conceptual understanding. The phases of clinical trials and the safety questions are what are examined. **Appears in:** Ch 12, Lect 2

▪ Pre-initiation complex

In one sentence: The whole machine, made of general transcription factors plus RNA polymerase, that is assembled on the promoter before transcription starts.
Detailed explanation: In eukaryotes, RNA polymerase II cannot recognise the promoter by itself, so the general transcription factors have to build the machine for it. TFIID (which contains TBP) recognises the TATA box first, then TFIIA and TFIIB join, and then polymerase II arrives with TFIIE, TFIIF and TFIIH to form the pre-initiation complex, with TFIIH doing the work of opening the DNA duplex.
Only once the assembly is complete does transcription really begin. Dr. Wong sums it up in one sentence: none of these factors can be spared, and if any one of them jams, polymerase never gets to work. Notice the naming as you go, because it tells you the cast: TFIID is the odd one out, since it is the factor that carries TBP and does the recognising, while TFIIA, TFIIB, TFIIE, TFIIF and TFIIH are all general transcription factors. Bacteria locate the promoter differently, using a sigma factor to find the -10 and -35 sequences. The point of the whole assembly is that the promoter sequence alone is not enough of a signal: it has to be read by proteins before any RNA can be made, and each protein in the complex arrives for its own reason and does one job. It is a relay team that only starts running when every member is in place, and the DNA is opened only at the last step.
Why this course brings it up: On his Lecture 5 slide he marks TATAAA at position -30 and labels the assembly order TFIID → TFIIA/TFIIB → Pol II with TFIIE/TFIIF/TFIIH. The promoter and the TATA box are also the way into the cis versus trans lesson, because a TATA box written into the DNA is a cis element while the factors that read it are trans. So the same slide is doing two jobs, and knowing the order of assembly is what lets you read it. **Appears in:** Ch 7, Lect 5

▪ Pre-mRNA

In one sentence: The first RNA copy of a gene, before it has been spliced.
Detailed explanation: The primary product that RNA polymerase transcribes, containing exons and introns together. The spliceosome, the machine built from snRNPs, cuts the introns out, joins the exons together, and once the 5′ cap and the poly-A tail are added it becomes mature mRNA and leaves the nucleus.
So pre-mRNA is the unedited draft, and splicing is the editing process applied to that draft. Splicing can also be selective, an exon kept or dropped, so that one gene produces different isoforms, and in cancer cells mis-splicing can even manufacture neoantigens. The order is fixed: cap and tail, then splice, then export. A message that has not been through all of it is not a

message the ribosome will accept, and because the editing happens before export, a mistake in the splice is baked into the message that leaves the cell.
Why this course brings it up: In Lecture 5 he makes the eukaryotic against prokaryotic contrast very clear: a eukaryotic pre-mRNA has to be finished inside the nucleus before it is exported, while a prokaryotic mRNA can be translated as soon as it comes out. Alternative splicing also happens at this step. One slide, two reasons to care about the nucleus, and the same point he makes when he contrasts the two cell types. **Appears in:** Ch 7, Lect 5

▪ pre-pre-IND

In one sentence: The first, very early sounding-out that comes before a pre-IND meeting.
Detailed explanation: At the first contact the company usually brings only the roughest idea: is this target worth doing, what model should it be assessed in, and what experiments should be prepared first.
Once the direction is settled the pre-IND meeting happens and the plan is discussed in more detail. Both are preparation steps before the real IND. At this stage the company is usually not bringing data, it is asking whether the regulator will listen to the same story about the same direction. So the early conversation is about whether the story is plausible, and the later one is about whether the plan is good enough.
Why this course brings it up: He does not name pre-pre-IND as an exam question on its own, so it is concept-only. It exists to show you that the regulatory path has stages before the stages you can be asked about. **Appears in:** Ch 12, Lect 2

▪ preclinical

In one sentence: The whole stretch of development work that comes before formal testing in humans.
Detailed explanation: Preclinical work has two parts. MOA, the mechanism of action, which is the proof that the compound really acts on the intended target. Toxicology, which is the proof of what it does to a whole animal.
There is also the quality data on the drug itself. Together these three are the content of the IND-enabling studies. If the conclusion at this stage is not encouraging the IND is not filed, and the safety data is more often the real endpoint than the efficacy data, because the ceiling on the dose is usually set by toxicity rather than by effect.
Why this course brings it up: Dr. Wong does not name preclinical as an exam question on its own, so it is conceptual understanding, and the phases of clinical trials together with the division of responsibilities are Tier 3. **Appears in:** Ch 12, Lect 2

▪ Predictive biomarker

In one sentence: A marker that predicts whether a particular patient will respond to a particular treatment.
Detailed explanation: The question it answers is which drug works for this patient. The clearest example is K-Ras: cetuximab, an anti-EGFR, fails in mutated colorectal cancer because the downstream signalling is still lit up.
So in this case the K-Ras status is a marker that predicts "this will not work", and not a marker that predicts "this will be worse". Being able to separate those two kinds of marker is the key to making clinical decisions. It is worth being careful with the wording here, because the marker is not saying the patient is sicker, it is saying that the drug will not work, and those are completely different claims. Either way you need to know the biology of the pathway before you can use the marker.
Why this course brings it up: In Lecture 1 Dr. Wong uses K-Ras to show how a predictive marker is used, and he also puts predictive and prognostic on the list of the most common contrasts in a multiple-choice question. Case question one is "should he be given cetuximab, explain using the signalling pathway". **Appears in:** Ch 10, Ch 11, Lect 1 **Video:** V10_Predictive_vs_prognostic_biomarker.mp4

▪ Primary structure

In one sentence: The amino acid sequence of a protein, read from one end to the other.
Detailed explanation: In other words, the order of the amino acids itself. Every level of protein structure is built on that order: local hydrogen bonds decide the alpha helix and the beta sheet, the way the whole chain folds decides the tertiary structure, and the assembly of several chains decides the quaternary structure.
Change one letter and you can change the shape and the function of the whole machine — which is exactly how a point mutation such as K-Ras G12C can cause cancer.
Why this course brings it up: In Lecture 1 he asks you to name the four levels of structure in order, with primary first; both Edman degradation and LC-MS/MS are sequencing methods that work at this level. **Appears in:** Ch 2

▪ Primase

In one sentence: The enzyme that lays down the short RNA primer that gives DNA polymerase a place to start.
Detailed explanation: DNA polymerase cannot start from nothing, it needs a free 3′ hydroxyl before it can add the first nucleotide. Primase makes an RNA primer of about 10 nucleotides that provides exactly that starting point, and the polymerase takes over from there and extends 5′→3′.
The RNA primer is removed later, and the gap is sealed by ligase. Every single segment of the lagging strand needs a fresh primer laid down. In a eukaryotic cell each origin of replication also needs a primer to begin with, so replication has to start at an origin, and the direction is decided there. A polymerase can never begin a strand, it can only extend one, and that is why every new strand in the cell starts with a short piece of RNA rather than DNA. So the very first step of every piece of new DNA is the laying down of a short RNA primer, and there is no way round it.
Why this course brings it up: In Lecture 4 Dr. Wong lists primase as one of the replication enzymes, and every Okazaki fragment on the lagging strand needs its own primer, which is the key to how he explains why the lagging strand is the more

troublesome half. It is also the answer to the question beginners always ask, which is where the very first nucleotide comes from. **Appears in:** Ch 5, Lect 4

▪ Primer

In one sentence: A short RNA starting point that gives DNA polymerase something to attach to.
Detailed explanation: DNA polymerase cannot start a new strand out of nothing; it needs a free 3′-OH end. Primase is in fact a small RNA polymerase, and it lays down a short stretch of RNA on the template as the primer, so that the polymerase can start adding dNTPs from there.
Bacterial DNA replication and PCR use exactly the same logic. In the cell the RNA primers are laid down by primase, and in the laboratory you use synthetic DNA primers; the role is the same either way, giving the polymerase a starting point to build from. In the laboratory you can order the primer by sequence, which is the whole difference between the two settings. In the cell you cannot, which is why primase exists. And note that the primer is built from ribonucleotides rather than deoxyribonucleotides, which is what lets RNase H find the junction later on. The polymerase itself cannot tell the difference, which is why the cleanup step cannot be skipped.
Why this course brings it up: Dr. Wong does not name primer as an exam topic on its own, so it belongs to conceptual understanding. The Tier 2 point is that the leading strand needs only one primer while every lagging strand fragment needs its own. **Appears in:** Ch 3, Lect 1, Lect 3, Lect 4

▪ Primer annealing

In one sentence: The step where the primer, once cooled, finds its spot on the template and sticks.
Detailed explanation: This is the second step of PCR. The sequence of the primer decides where it stops, and that is the principle of targeting. But if the primer sequence occurs too many times, or if the annealing temperature is too low, it will stick in the wrong place, and then PCR amplifies several products at once.
The rule he states explicitly is that the higher the GC content, the higher the Tm, and the annealing temperature in primer design has to be adjusted to match. So the specificity of the whole reaction sits on one thermodynamic property.
Why this course brings it up: The relation between GC content and Tm is a Tier 3 fact he lists, and it runs straight into primer design. He also tidies up the three steps of PCR in every lecture he gives. So the relationship is the number to keep and the three steps are the frame. **Appears in:** Ch 3, Ch 11, Lect 1

▪ Primer pair

In one sentence: The pair of primers PCR needs, one for each strand.
Detailed explanation: Because double-stranded DNA comes apart into two template strands, and each strand needs a starting point, PCR needs two primers: one complementary to one strand and one complementary to the other.
Where the two of them sit decides what gets amplified — the stretch between them, rather than the whole genome. Choose the positions badly and what you amplify may not be the piece you wanted, and you may even get several products running at once. The two primers also have to point towards each other, or they will not bracket anything at all.
Why this course brings it up: He does not name primer pair as an exam topic on its own, so it is a concept to understand; the two-strand logic behind it is what makes the design of the reaction make sense. **Appears in:** Ch 3, Ch 11, Lect 4

▪ Priority Review

In one sentence: The priority review status the FDA gives to an application that represents a major therapeutic breakthrough.
Detailed explanation: Along with Breakthrough Therapy Designation and Orphan Drug Designation, it is one of the three FDA mechanisms Dr. Wong names on his slide. The difference between them is this: Breakthrough Therapy is for a drug with the potential to be a breakthrough, Priority Review gives the privilege of an earlier review timetable, and Orphan Drug gives seven years of market exclusivity for a rare disease.
The three differ only in the treatment they get from the regulatory process, they make no difference to the science, and they belong to the institutional background of drug development rather than to the molecular mechanism.
Why this course brings it up: Dr. Wong does not list these three FDA names as exam questions, so it is conceptual understanding. The point to take away is that all three are about process and timetable rather than about how the drug works. None of them changes the biology of the molecule or the evidence behind it. **Appears in:** Ch 12, Lect 1

▪ Prognostic biomarker

In one sentence: A marker that tells you how the disease will go, whatever treatment you give.
Detailed explanation: It answers the natural history of the disease, with no reference to treatment. The same marker can be prognostic in some cancers and predictive in others: ERBB2 is favourable in kidney cancer and unfavourable in endometrial and pancreatic cancer, but only breast and gastric cancer show a clear survival benefit from anti-HER2 treatment.
In other words, high expression does not mean "this patient should get the drug". The two questions are separate, and a marker can answer one without answering the other.
Why this course brings it up: In Lecture 1 he uses the ERBB2 example to make the point that a biomarker has to be qualified by cancer type; he lists prognostic and predictive among the most common contrasts in multiple-choice questions, and case number 2 is exactly this point. **Appears in:** Ch 10, Ch 11, Lect 1

▪ progression-free survival

In one sentence: The time a patient survives without the disease getting worse.
Detailed explanation: Unlike median overall survival, PFS does not ask whether the patient died, only whether the tumour progressed. In a trial it often shows a difference between the groups earlier than overall survival does, which is why it is used for early reading.
On his MSI-H against pMMR figure the y axis is exactly progression-free survival. When you read a KM curve you have to check which survival measure is written on the axis: for the same drug, the OS and the PFS curves can have completely different shapes and completely different separation times. And that is why the same result can look like a modest effect on one axis and a dramatic one on the other, without anybody being wrong.
Why this course brings it up: The finding that the MSI-H, MMR-deficient group does clearly better on the PFS curve is the figure he calls one of the most important in modern medicine. He does not ask you to memorise the definition of PFS. So you should be able to read the axis, not recite the glossary. **Appears in:** Ch 9, Ch 12, Lect 2

▪ Prokaryote

In one sentence: A cell with no nucleus. Bacteria and archaea are both prokaryotes.
Detailed explanation: There is no nuclear membrane, the chromosome lies directly in the nucleoid in the cytoplasm, and there are no membrane-bound organelles. That brings three practical consequences.
Transcription and translation can happen in the same place at the same time, because a prokaryotic mRNA can be picked up by a ribosome as soon as it comes out. Proteins need no splicing, so there is no alternative splicing. And there is no nuclear layer of post-translational regulation. Prokaryotes therefore reproduce quickly and have simple genomes. So the difference is not sophistication, it is architecture, and the architecture is what changes the rules.
Why this course brings it up: In Lecture 5 he uses the prokaryote against eukaryote contrast to explain the difference between expression systems and regulatory layers, and the lac and trp operons sit on the other side of that contrast. The operons are the prokaryote's way of doing regulation, and they are the simplest version of it. **Appears in:** Ch 8, Lect 2, Lect 5

▪ Proliferative signal

In one sentence: A message that tells the cell to start copying its DNA.
Detailed explanation: It begins when a mitogen binds a receptor on the membrane, and then the signal travels downward: PI(4,5)P is cut by phospholipase C into DAG and IP3, and DAG activates PKC;
PI3K activates AKT and mTOR in turn; and RTK goes down the RAF–MEK–ERK route. The common end point is that Cyclin D–CDK4/6 phosphorylates pRb, E2F is released, and the cell cycle moves from G1 into S phase. A tumour's trick is to make this message for itself, either by switching on the receptor, by switching on the adaptor in the middle, or by cutting out the part of the pathway that would have turned the signal off. All three routes converge on the same destination: raising Cyclin D–CDK4/6, phosphorylating pRb, setting E2F free, and letting the cell step across into S phase. So although the three routes look completely different on the slide, they are all saying the same sentence at the end, and that is why a single drug at the shared end point, like a CDK4/6 inhibitor, can intercept all of them at once. The routes differ in how the signal gets in, and they agree completely on what the signal is for.
Why this course brings it up: This is the central logic of K-Ras driven carcinogenesis, because a stuck RAS is a proliferative signal that is permanently on. Dr. Wong's Lecture 1 and Lecture 4 both put the EGFR → RAS → RAF → MEK → ERK and PI3K → AKT → mTOR routes on the slides. **Appears in:** Ch 10, Lect 2

▪ Promoter

In one sentence: The word in the DNA that tells RNA polymerase where to start.
Detailed explanation: In eukaryotes the promoter is where polymerase II and the general transcription factors assemble, and there is usually a TATA box about 30 bases upstream of the transcription start point which is recognised by TFIID, which contains TBP.
In bacteria it is the −10 and −35 regions near the promoter that let the sigma factor position the polymerase. The promoter is a stretch of DNA rather than a single word, but it behaves like one instruction, and everything that has to happen before transcription can begin has to happen there. One thing to keep in mind is that Dr. Wong is explicit that some enhancers sit at the 3′ end of their target gene. He uses deletion series of the promoter joined to a luciferase reporter to locate the fragment that is actually doing the work, which is a way of finding the useful sequence without having to guess it, and the same point about enhancers at the 3′ end comes up again. A promoter is therefore not one thing but a region with several parts in it, and the parts can be separated experimentally.
Why this course brings it up: In Lecture 5 Dr. Wong sorts the promoter into the cis elements, which act on the DNA they sit on, as against trans factors, and he uses a reporter assay, the luciferase deletion series, to show how you find the sequence on the promoter that is actually useful, and that is the practical skill the experiment teaches, which is how you dissect a DNA element into the parts that matter. **Appears in:** Ch 7, Lect 5

▪ Proofreading

In one sentence: DNA polymerase checks its own work as it goes, cutting out the wrongly placed base and putting the right one in.

Detailed explanation: Base pairing on its own gives an error rate of about one in every hundred thousand bases. The 3′→5′ exonuclease activity sits on the polymerase and acts like a proofreader: it checks the newly added base against the template strand and, if it does not pair, cuts it out and puts the right one in;
with MMR on top, the overall error rate falls to about one in a billion. That extremely low error rate is the reason a human genome can exist at all. The same mechanism can also cut out nucleoside analogue drugs, which is a source of drug resistance. A polymerase that has lost proofreading lets mutations pile up, and such a tumour has more neoantigen and responds better to immune checkpoint inhibitors. So proofreading is not tidying up; it is the reason the whole enterprise is worth attempting. And notice the twist: the same machinery that makes copying near-perfect is also the machinery that decides whether a drug survives long enough to work.

Why this course brings it up: In Lecture 4 he places it in a direction contrast, in his own words: "So this is important, you need to understand the elongation happens from 5 prime to 3 prime." — elongation runs 5′→3′, proofreading and excision run 3′→5′, and that pair is one of the distractor contrasts he lists. He also points out that proofreading does more than catch mismatches; the same machinery removes drugs, which is how tumours and viruses become resistant. Ultramutator tumours caused by POLE proofreading mutations have more neoantigen and a better prognosis, and they belong to this same line. So the direction pair, not the enzyme, is what the question is about. **Appears in:** Ch 5, Lect 4

▪ Propidium iodide (PI)

In one sentence: A dye that stains DNA, so once it is in a flow cytometer you know how much DNA each cell has.

Detailed explanation: PI slots in between the bases of the DNA, so the fluorescence intensity is proportional to the DNA content. It cannot get through an intact cell membrane, so you have to fix the cells and make the membrane permeable before you can stain them.

The histogram that comes out has two peaks: G1 is 2N, S phase is the continuous rise between the two peaks, and G2/M is 4N. That is the fastest way of resolving the cell cycle, and it is the measuring tool he uses when he studies DNA replication, where the DNA content before replication is 1 and after replication it is 2. The shape of that curve is the thing to learn, because the three answers are all visible in one picture without any further experiment.

Why this course brings it up: He puts it on the second-level list of flow cytometry basics in Lectures 2 and 4: PI staining measures DNA content, giving G1 at 2N, then S, then G2-M at 4N. The same second-level list also holds one of his key numbers, dropping FBS to 0.5–1 % for serum starvation so that the cells park in G1 and can be synchronised. He does not ask you to memorise how to run the instrument, only to know how to read this DNA content curve. **Appears in:** Ch 11, Lect 2

▪ Protease

In one sentence: An enzyme that cuts proteins open using its active site.

Detailed explanation: A protease is like a pair of scissors working on the peptide bonds of the protein backbone. It is the opposite of the polymerase: the polymerase adds to the chain one unit at a time, while the protease cuts a finished chain apart. Its most important use in the classroom is a negative experiment. In 1943 Avery, MacLeod and McCarty showed that the “transforming factor” was DNA by adding DNase and protease separately and seeing which one destroyed the transforming activity. The answer is that DNase does and protease does not. The genetic material is DNA, not protein. It is a clean design: add the thing you suspect and see whether the activity survives, and the answer falls out of which tube went quiet. Note what the protease is actually doing in the experiment: it is chopping up everything in the tube that is not the transforming factor, and the transforming activity survives anyway. The negative result is the informative one here.

Why this course brings it up: Dr. Wong does not name the protease itself as an examinable topic, but this DNase versus protease control experiment is one of the two or three experiments he expects students to reason with, so it is worth remembering as a whole. He also says plainly that the fine details of chemical structures and metabolic pathways will not be examined: “Please don't memorize all these chemical structure… They would not be included in my exam.” What you need is the mechanism, not the molecular formula. **Appears in:** Ch 3

▪ Proteasome

In one sentence: The protease machine that takes apart a protein once it has been given the recycling label.

Detailed explanation: The proteasome is a barrel-shaped multi-protein complex, with receptors at the mouth of the barrel that recognise the tag and enzymes inside the barrel that cut the protein.

A protein marked with a chain of ubiquitin is brought to the mouth, taken apart into short peptides, and those peptides are then shipped to the ER to be fitted with MHC so that the cell can show them to the immune system as an entry pass. HIF-1α takes exactly this route in Lecture 5: in normal oxygen PHD adds a hydroxyl group, VHL recognises it and adds the ubiquitin chain and hands it to the proteasome, and in hypoxia that route is interrupted, which is what lets HIF-1α get into the nucleus and switch on VEGF.

Why this course brings it up: Dr. Wong does not name the proteasome on its own, but the chain VHL, then ubiquitination, then proteasomal degradation of HIF-1α is the skeleton of the HIF-1α and VEGF section of Lecture 5 and the beginning of the "tumour builds its own blood vessels" part. Understand it and you understand why the HIF-1α section of Lecture 5 has to begin with oxygen sensing. **Appears in:** Ch 7

▪ Protein

In one sentence: A machine made by folding a string of amino acid beads; its shape decides what it does.
Detailed explanation: The identity of a protein is a sequence of amino acids, but the function comes from the three-dimensional shape, so sequence and shape are two different things, and a folding error turns directly into disease.
The clue he gives in class is the hierarchy of shapes. First level is the sequence. Second level is the α-helix and the β-sheet. Third level is the 3D folding of the whole chain. Fourth level is several chains assembled into one working unit. An antibody is the textbook example of the fourth level, where the Fc half talks to the immune system and the two Fab halves grab the antigen. Each level is built on the one before it, which is why the hierarchy is the spine of the whole course rather than a piece of vocabulary. Two proteins with the same sequence and a different shape are different machines, which is exactly why a misfolded protein can be harmful even when every letter is correct.
Why this course brings it up: In Lecture 1 he teaches the primary, secondary, tertiary, quaternary hierarchy as the spine of the entire course. On the exclusion list are the twenty standard amino acids together with the R-group table, he says don't memorize this, definitely not include in the exam. But the hierarchy itself, the idea of a domain, and the notion that a small R-group means local flexibility all still have to be understood. So the table is out and the structure is in. **Appears in:** Ch 1

▪ Protein domain

In one sentence: A stretch of protein that folds on its own and does one identifiable job.
Detailed explanation: A domain is the modular building material of a protein: cut a big protein into pieces and each piece can fold into a stable shape in water on its own, taking on one recognisable job each.
That is what lets engineers bolt "grab the antigen" and "wake up the T cell" together — the scFv of a CAR is one domain, and CD3ζ and CD28 are several more. Domains can also evolve separately, which is why the same domain often turns up in quite different proteins. So a protein is not one lump but a set of parts that can be swapped.
Why this course brings it up: In Lecture 1 he defines a domain as an independently folding functional unit, which is the premise for explaining why one big protein can be taken apart and designed as pieces. He never lists domain as an exam item on its own, so it is a concept to understand: grasp it and you can see that the structural design of CAR and of ADC is the same logic. Which is the practical payoff of the whole idea. **Appears in:** Ch 2, Lect 1

▪ Protein kinase

In one sentence: The enzyme that puts a phosphate group onto another protein, and ATP is its fuel.
Detailed explanation: A kinase takes a phosphate group off ATP and attaches it to the target protein. The phosphate carries a negative charge, and attaching it pulls the charge distribution of the protein apart, which exposes a site that was previously hidden, and the protein has changed shape and changed state.
A phosphatase does the reverse and takes the phosphate off again using water. Together the two are the cell's fast switch, because phosphorylation and dephosphorylation can flip a signal in a few seconds. In Lecture 1 he uses the Hippo cascade as the example: MST1 phosphorylates LATS1, and LATS1 phosphorylates YAP. The phosphatase is the part that is easy to forget, because a kinase acting on its own would ratchet the cell into a permanent state, and it is the pair that makes the switch reversible.
Why this course brings it up: In Lecture 1 Dr. Wong describes the kinase and phosphatase pair as the reversible phosphorylation switch, and the Hippo-YAP pathway comes back in Lecture 5. He does not name this cascade as an exam question in Lecture 1, but the idea that phosphorylation keeps a co-activator out of the nucleus and that losing the kinase lets it in is one you have to keep. He also says explicitly that the upstream list of activating factors does not need to be memorised.
Appears in: Ch 1, Ch 2, Lect 1

▪ Proteolytic cleavage

In one sentence: Cutting a protein once with an enzyme, and the fragment that comes off changes its identity.
Detailed explanation: Proteolysis is not simply destruction. It is a switch: cutting away an inhibitory piece, or cutting out a piece that carries a signal. Notch is the textbook example. Once the delta ligand on a neighbouring cell attaches, Notch is cut to release its intracellular domain (NICD), and the NICD moves into the nucleus to switch on genes.
In Lecture 5 the “seven ways a transcription factor is switched on” Dr. Wong lists includes this one: release from membrane, achieved by proteolytic cleavage. A different cut gives a completely different downstream response, so the cut is not a detail of the mechanism, it is the mechanism. It is also why a protease can be such a precise tool: the same enzyme, cutting in a slightly different place, produces a slightly different protein with a slightly different job. Which is also why a cell can use one enzyme many times over and get completely different outputs each time.
Why this course brings it up: He does not name proteolytic cleavage as a topic on its own, but it appears in two subjects he really does teach: the Notch/synNotch material in Lecture 5, and one of the seven ways of activating a transcription factor. It belongs to conceptual understanding, and once you know that one cut equals one new identity, you can explain why synNotch can be built as an AND gate. **Appears in:** Ch 7, Lect 5

▪ Proteome

In one sentence: The whole set of proteins that a cell or a tissue has made at one moment.
Detailed explanation: The genome is a fixed list, but the proteome changes with time and with the state of the cell: the same DNA gives different proteins depending on the phase of the cell cycle, on what the cell has been stimulated by, and on how far it has differentiated.

So the proteome is a snapshot of one square of that grid, and describing the state of a cell is often better done with the protein list than with the gene list. In Lecture 1 he demonstrates this with hepatocellular carcinoma: he stratifies samples of different tumour sizes, cirrhotic livers and healthy transplant controls, runs 2D-PAGE, identifies with LC-MS/MS, and only then finds vimentin in the serum as a non-invasive marker. The key idea is that the genome is what the cell could make, and the proteome is what it is actually making, and the difference between the two lists is where the biology is.
Why this course brings it up: The vimentin line is the biomarker discovery workflow he walks through by hand in Lecture 1, and the blood numbers for vimentin, control 176.0 ± 61.3 ng/mL, HCC ≤ 2 cm 271.5 ± 165.9, HCC > 2 cm 231.3 ± 132.3, are on his list of key numbers, so they have to be exact. Proteomics itself is not on his exam list; target nomination and the druggability workflow are explicitly excluded. The exclusion is about the business process, not about the science. **Appears in:** Ch 2, Lect 1

▪ Proteomics

In one sentence: The study of the whole set of proteins made by a cell, used to find biomarkers and drug targets.
Detailed explanation: His workflow starts with 2D-PAGE. The first dimension is isoelectric focusing, which spreads the proteins out along a strip according to their charge; the second dimension is SDS-PAGE, which takes that same strip and separates the proteins a second time, this time by molecular weight.
What you end up holding is a map of spots, each spot standing for one protein. You then cut a spot out, digest it into peptides, and let LC-MS/MS read out those peptides, working out what the protein is by matching the peptides against a database. In Lecture 1 he flags an easy mistake at this point: the tissue sitting next to the tumour may look perfectly ordinary, but it is not biochemically the same as genuinely normal tissue, so you must not treat it as the "healthy control" group and compare against it. That one decision, taken early, is what decides whether the whole biomarker paper can be trusted.
Why this course brings it up: In Lecture 1 he uses vimentin as a complete case study in biomarker discovery, and it is a demonstration of how you walk from a single two-dimensional map all the way to a marker you could test in a blood sample. But he also says plainly that the drug company's target nomination and druggability workflow is not examined: "it's not included in exam, but I think it's really more interesting". So the pipeline itself is not to be memorised; the idea behind it is to be understood.
Appears in: Ch 11, Lect 1

▪ Proton

In one sentence: A positively charged particle, and the ones in the nucleus decide the identity of the element.
Detailed explanation: The nucleus holds protons, which are positively charged, and neutrons, which are not charged, and the electrons outside run around in fixed energy levels. The number of protons is the identity of the element: six is carbon, eight is oxygen, one is hydrogen.
The neutrons mainly provide the weight. The electrons decide how that element behaves, the neutrons decide how heavy it is, and the protons decide what it is. So the proton count is the one number you have to know about any atom, and everything else about the element follows from it. What is special about carbon is that it has four valence electrons and can make four bonds, so it can build long carbon chains and hold up the whole skeleton of organic chemistry.
Why this course brings it up: Dr. Wong does not name atomic structure as an exam question in Lecture 1, and chemical structure is on his exclusion list anyway: "Please don't memorize all these chemical structure… They would not be included in my exam." So this entry is conceptual understanding. You do not need to memorise the periodic table, but you do have to be able to use the idea that the proton number is the identity of the element to explain why the four bases of DNA each have their own pairing rules, which is the first biological fact in the course that the chemistry has to carry. **Appears in:** Ch 1

▪ pyrimidine dimers

In one sentence: Adjacent pyrimidine bases stuck together by UV light.
Detailed explanation: Two adjacent thymine bases that have been hit by UV become chemically bonded to each other, and the double helix is twisted out of shape so that the replication machinery cannot read past it and transcription stalls as well.
This is a typical bulky lesion: it has to be cut out whole by NER and rebuilt, rather than dealt with by BER, which only swaps a single base. UV is not the only source — the chemical carcinogens in cigarette smoke also produce bulky adducts — and both kinds of damage are NER's business. The size of the damage is what decides which repair pathway is called, and that is the whole question in this entry.
Why this course brings it up: "NER repairs bulky lesions and BER repairs single bases" is one of the 19 multiple-choice pairings he lists. So the pairing is what you are being tested on, not the chemistry of the dimer. **Appears in:** Ch 6, Lect 4

▪ Pyrogen

In one sentence: A substance that raises body temperature and brings on a fever.
Detailed explanation: Pyrogen is the umbrella term for anything that causes a fever. On Slide 21 of Lecture 1 he labels LPS as an endotoxin and as a pyrogen at the same time, and in the margin he writes that an excessive amount of endotoxin in the blood causes septic shock, which is a common cause of death among surgical intensive care patients.
A fever is not automatically a bad thing: it pushes the body to speed up its metabolism and raises immune efficiency. The trouble starts when it runs out of control — an excessive inflammatory response turns round and damages the body instead, and that is exactly what happens when the NF-κB cascade goes out of control.
Why this course brings it up: He does not list pyrogen as an exam topic on its own. What is examinable is the chain of logic you can build out from it: LPS → TLR4 → NF-κB → inflammatory cytokines → septic shock, and his exam notes list that chain as a possible multiple-choice stem. Pyrogen itself is background: it is there so that you can understand what role LPS is playing.

Appears in: Ch 1, Lect 1

▪ Pyruvate

In one sentence: The product at the bottom of glycolysis, and where it goes decides what the cell does.
Detailed explanation: Glycolysis breaks glucose down into pyruvate, and from there there are two roads. With oxygen and healthy mitochondria, the pyruvate enters the TCA cycle and oxidative phosphorylation, and one glucose yields about 36 ATP. Without oxygen the cell only does lactic fermentation, and that gives 2. A cancer cell, though, often takes the lactic road even when there is oxygen, sending 85 % of the pyruvate to lactate and only about 5 % into the mitochondria, for a net of about 4 ATP. It looks wasteful, but the point is to keep the half-broken sugar molecule as building material.
Why this course brings it up: The Warburg effect is his entry point in Lecture 1, and it is also one of the background numbers in the third tier: about 36 with oxygen, 2 without, about 4 in a cancer cell, per glucose molecule. He says explicitly that you are not expected to memorise the glycolysis pathway, because the metabolic pathway details are on the exclusion list. What you do have to understand is the trade-off, because a low-efficiency conversion is a growth strategy and not a mistake. **Appears in:** Ch 1, Ch 10, Lect 1

▪ q-arm

In one sentence: The long arm of a chromosome, the side below the centromere.
Detailed explanation: A chromosome has two arms, separated by the centromere (the point where the spindle fibre grabs hold at metaphase): the shorter one is the p-arm and the longer one is the q-arm.
The naming runs q1, q2 and so on down the arm, so q21.1 means the first sub-band of region 21 on the long arm. A gene location is written as the chromosome number, then the arm, then the region and band, so 17q12 is where HER2 sits, and that is exactly the notation Dr. Wong uses when he talks about HER2 FISH. The centromere is what makes the two arms different in the first place, and it is also the handle that the spindle uses, so the same landmark that names the arms is the one that moves the chromosomes. The p and q were simply the letters left over after P and Q had been taken for other uses, and the convention stuck.
Why this course brings it up: On Slide 13 of Lecture 3 he specifically warns about how you read it: q21.1 is “q-two-one-point-one”, not “q-twenty-one-point-one”. It is a spelling detail, but he puts it beside the chromosome structure diagram, which makes it the kind of thing that can be quietly taken away from you in a multiple-choice question. The resolution of karyotyping (0.5–10 Mb) and the q-arm notation are in the same section. **Appears in:** Ch 3, Lect 3

▪ qPCR

In one sentence: Quantitative PCR: it uses a fluorescent signal to measure how much of a particular stretch of DNA is in the sample.
Detailed explanation: Ordinary PCR only answers one question — "is it there or not". qPCR answers two: "is it there" and "how much of it". The product doubles with every cycle, and because the fluorescent signal is read off the accumulating product, the signal grows as the cycles go by.
The instrument then works backwards: it notes the cycle number at which the fluorescence crosses a threshold, and from that number it estimates how much template you started with at the beginning of the reaction. Which means the fewer target molecules the sample held, the more cycles it takes to get there. qPCR is far more sensitive than plain PCR, and that is why it became the validation tool on both of his biomarker work lines: the hepatocellular carcinoma line used qPCR to validate the vimentin that 2D-PAGE had turned up, and the Lecture 4 line used TERT qPCR to check whether the anti-TERT antisense oligonucleotide really had pushed TERT mRNA down.
Why this course brings it up: He does not list qPCR itself as an exam item, but it is the validation step in both of the biomarker pipelines, so it turns up in the material whether or not it gets named as a topic. The TERT result on his Tier 3 number list is one to remember: 3 weeks of anti-TERT pretreatment → tumour formation rate 2/12 (9/12 untreated), Fisher's p = 0.01. **Appears in:** Ch 11, Lect 1, Lect 4

▪ Quaternary structure

In one sentence: The way several separately folded chains come together to make one working whole.
Detailed explanation: Not every protein has a fourth level. A single-chain protein stops at the tertiary structure and works perfectly well. He gives two examples that do have one: haemoglobin, two alpha plus two beta, carrying an oxygen molecule, and the antibody, two heavy chains and two light chains.
The advantage of assembling several chains is division of labour and control: each chain does one job while the whole is one complete machine, and the division between the Fc and the Fab of an antibody is the clearest example of that. There is also a practical advantage, because each chain can be made and folded on its own and then put together, which is how large complexes are built in the cell.
Why this course brings it up: In Lecture 1 he teaches the fourth level as the last rung of the protein hierarchy, so it is concept-only. What he excludes is the amino acid R-group table and the chemical structures, not this ladder. What you need is the difference between the four steps: primary, secondary, tertiary, quaternary. If you can order the four, you can answer almost any question about protein structure. **Appears in:** Ch 2

▪ Quiescent

In one sentence: The state of being alive but not dividing, which is G0.
Detailed explanation: A quiescent cell is not dead. It has simply left the cell cycle and stopped dividing, while it keeps its metabolism running and keeps itself ready. G0 is not a station inside the cycle, it is a parking place outside it, and when an external signal arrives, a growth factor coming back into the serum for instance, the cell can re-enter G1 and start replicating its DNA.
Most differentiated cells sit in G0, so quiescence and differentiation are often two sides of the same coin.
Why this course brings it up: In both Lecture 2 and Lecture 4 Dr. Wong uses the same trick to park cells in G0: serum starvation, taking the FBS from 10 % down to 0.5–1 %. That number is on the key numbers list in both lectures, because it lets every cell start from the same point and you can then measure replication or drug response cleanly. He does not ask you to memorise the pairing table of cyclins and CDKs. **Appears in:** Ch 8, Lect 2

▪ R-group (side chain)

In one sentence: The side arm on an amino acid that can change, and it decides how the whole chain folds.
Detailed explanation: Every amino acid has the same backbone — an amino group, a hydrogen, a carboxyl group — and the only thing that varies from one to the next is the R-group hanging off it.
That is why the R-group sets the character of the amino acid, and why it sets the way the whole chain folds: the backbone is the constant frame, and the side arms are what decide whether a given part of the folded shape is tight and rigid or loose and floppy. The size of the R-group also decides how floppy that part of the chain is: glycine has an R-group of just one hydrogen, so it is by far the most flexible of the twenty; the R-groups of tryptophan and arginine are both large and charged, so they are comparatively stiff. The same idea explains how an antibody can tell apart millions of different antigens: what the antibody actually reads is the shape of the R-groups sitting on those CDR loops, and those shapes are different for different antigens.
Why this course brings it up: In Lecture 1 he links the size of the R-group to local flexibility, which is the way in to the idea that a small R-group means local flexibility. But he explicitly excludes the R-group table: "don't memorize this… definitely not include in the exam" — the table of the 20 standard amino acids with their R-groups is not to be memorised. It is a concept to understand rather than a table to learn. **Appears in:** Ch 2

▪ RAD51

In one sentence: The protein that winds itself around single-stranded DNA in HR and leads the way into the sister chromatid.
Detailed explanation: RPA first covers the exposed single strand, and then RAD51 assembles as a filament on that strand, preparing for strand invasion. This is where BRCA1 and BRCA2 come in, directing the whole complex towards the sister chromatid of the same chromosome.
The tighter RAD51 winds, the better it can pair up the homologous sequences, and afterwards the polymerase fills in the missing sequence using the sister chromatid as the source. So the repair is not a patch, it is a copy job with an accuracy check built in, which is why HR is the high-fidelity route and the one that only runs when a template exists.
Why this course brings it up: Dr. Wong does not name RAD51 as an exam topic on its own, so it belongs to conceptual understanding. The relationship between BRCA1/BRCA2 and the PARP inhibitor is the Tier 2 point. **Appears in:** Ch 6, Lect 4

▪ Rad51, Rad52 and RPA

In one sentence: The three players that carry out strand invasion.
Detailed explanation: His slide 33 breaks the steps of HR apart: first the Rad50, Mre11 and Nbs1 complex processes the broken ends into 3′ overhangs, and then Rad51, Rad52 and RPA work together to carry out strand invasion, with RPA holding the single strand steady, Rad51 twisting into a filament and Rad52 helping to draw the homologous duplex close and pair it up. Rad54 then does the branch migration and the resolution separates the intermediate. The division of labour between the three is what makes this pathway both accurate and fast: RPA holds it steady, Rad51 does the pairing and Rad52 brings the two together, and without any one of them the chain cannot continue. So the three names are worth knowing as a team even if you are not expected to know their individual functions, because they are the names that appear on the HR slide and the ones that let you follow the story of the mechanism.
Why this course brings it up: He does not ask you to memorise these protein names, so it is conceptual understanding, and it is the trio of high fidelity, a template required and the S and G2 restriction that makes the Tier 2 pairing. **Appears in:** Ch 6, Lect 4

▪ Radiation

In one sentence: An outside source of damage that directly breaks both strands of DNA.
Detailed explanation: Ionising radiation such as X rays and gamma rays breaks DNA into double-strand breaks, which is the most serious kind of damage there is. It belongs to the exogenous sources of harm, listed alongside the pyrimidine dimers caused by UV and the bulky adducts caused by chemical carcinogens, and it is one of the three outside causes on his damage classification table.
The ionising radiation he describes is the most severe damage, and the cell usually cannot repair it faithfully, so it either goes into apoptosis or is left with a mutation. Notice that the three sources damage the DNA in three different ways, and the repair machinery is built around those differences rather than around the source.
Why this course brings it up: He does not name radiation as an exam question on its own, so it is concept-only. The

classification of exogenous against endogenous sources of damage is a skeleton he does teach. So the table is the examinable thing and the single agent is not. **Appears in:** Ch 6, Lect 4

▪ rcDNA

In one sentence: Relaxed circular DNA: the loose circular DNA HBV carries before it enters the nucleus.
Detailed explanation: The infection cycle of HBV runs like this: the virus binds a receptor and enters, the capsid is uncoated, the rcDNA is carried into the nucleus, and there it is repaired into cccDNA;
at the same time the viral DNA may integrate into the host genome. rcDNA has to be "relaxed" because it still has a gap in it and has not been closed covalently. Only once it has been repaired into cccDNA in the nucleus can the viral DNA make RNA in a stable way, and that step is why HBV is so hard to treat and why the virus reservoir is hard to clear. So the difference between relaxed and covalently closed is the difference between a broken circle and a whole one.
Why this course brings it up: He does not name rcDNA as an exam topic on its own, so it is a concept to understand; he draws the complete life cycle of HBV on slide 17. The cycle is the part worth learning. **Appears in:** Lect 2

▪ Reactive oxygen species (ROS)

In one sentence: Highly reactive molecules produced by the cell's own metabolism, which put dents in DNA.
Detailed explanation: Metabolism keeps leaking active oxygen molecules, and a normal cell clears them with antioxidant systems. Whatever cannot be cleared hits the DNA and forms a DNA adduct, and when the DNA is copied that adduct gets fixed as a mutation.
In Lecture 3 Dr. Wong takes the same route with the tobacco-specific carcinogen NNK: after P450 metabolism it forms a DNA adduct, the adduct interferes with replication, and the second round of replication settles the mutation. His slide actually carries the line “NNK induces a GC→AT mutation at K-ras gene in mice”. The endogenous version of the same story is happening quietly in every cell that is alive. The difference is only the source of the oxidising molecule, and the result is the same sort of damage.
Why this course brings it up: In Lecture 3 he ties the DNA adduct to chemical carcinogenesis as one causal chain, but the depth of that section is conceptual understanding and you do not need to memorise the reaction steps. What is genuinely examinable is his general conclusion: DNA damage is spontaneous, and the cell has to keep a whole repair shop running to hold the genome stable, which is the starting point of the whole of Lecture 4. **Appears in:** Ch 6, Lect 3

▪ Receptor

In one sentence: A protein built to catch one specific signal molecule, and changed in shape by it.
Detailed explanation: A receptor is like a socket or a sensor set into the cell surface: the ligand (the signal molecule) binds onto it, the receptor changes shape, and the changed shape is what starts the next step inside the cell.
Most receptors sit on the membrane, but nuclear receptors wait in the cytoplasm for their ligand to arrive; some receptors carry enzyme activity themselves (RTK), while others pass the message on through downstream adaptor proteins. The seven ways a transcription factor can be switched on that he lists in Lecture 5 are precisely a catalogue of how receptors get a signal in from outside and deliver it to the nucleus.
Why this course brings it up: He does not list receptor as an exam topic on its own; it is a concept to understand. But in Lecture 5 it is the foundation of receptor pharmacology: receptor occupancy (1 % / 2 % / 10 %), HER2 dimerisation, and tamoxifen as a SERM all rest on the one idea of how much of the receptor is occupied. **Appears in:** Ch 7, Ch 8

▪ Receptor occupancy

In one sentence: How much of the receptor has to be occupied by the drug before you get a clinical effect.
Detailed explanation: A receptor does not have to be filled up to work, and occupying part of it may already be enough. That threshold is receptor occupancy, and it is tissue-specific: the same drug, the same target, needs a different occupancy in a different tissue.
His numbers are very concrete: about 1 % for a general antibody, 2 % for Herceptin in breast cancer and 10 % for Herceptin in gastric cancer. So the same drug is used at a different dose with a different schedule in two cancers, and that is not an arbitrary decision, it is decided by the receptor biology. The practical reading is simple: a low percentage is not a weak drug, it means the target is easy to fill, and the interesting question is always which tissue you are asking about.
Why this course brings it up: In Lecture 5 he puts receptor occupancy alongside the nuclear receptor domains and dimerisation as second-level items, and the three numbers 1 %, 2 % and 10 % are on the key numbers list. He does not give a theory with a higher number; what he wants you to take is the conclusion that the same target in a different tissue has a different threshold. Keep the conclusion, not a memorised ceiling. **Appears in:** Ch 7, Ch 10, Lect 2, Lect 5

▪ Receptor tyrosine kinase (RTK)

In one sentence: A membrane receptor that carries its own kinase activity, and HER2 is one.
Detailed explanation: The structure of an RTK is a receptor body that crosses the cell membrane plus a kinase region on the inside of the cell. Two receptors lock together through the extracellular ligand-binding parts, dimerisation, and each one's intracellular kinase then phosphorylates the other, which passes the signal downstream.
That route is a growth switch in a normal cell and a runaway growth engine in a tumour. The detail that matters is this: a monomer binds DNA badly, and dimerisation raises the affinity dramatically. HER2 cannot dimerise efficiently with itself, and it forms heterodimers with HER1, HER3 and HER4 instead. So the receptor has two jobs: it catches the signal outside the cell and

it fires the enzyme inside the cell, and the dimer is what connects the two. So the dimer is not a detail of the structure, it is the thing that turns the enzyme on, and everything a drug has to do is either to break the dimer apart or to occupy the pocket where the ligand would go.
Why this course brings it up: In Lecture 1 Dr. Wong uses the RTK to make a clinical point he repeats again and again, that the same target can be pro-cancer in one cancer type and anti-cancer in another. RTK predicts worse survival in pancreatic cancer and a better prognosis in renal cell carcinoma, and in the Human Protein Atlas ERBB2 is good in renal cancer, bad in endometrial cancer and bad in pancreatic cancer, all three at $p < 0.001$. That is the mechanism of information recognition, and it is the raw material for the kind of question he asks most often. **Appears in:** Ch 10, Lect 2, Lect 5

▪ recipient plasmid

In one sentence: In molecular cloning, the plasmid backbone that has been cut open and is ready to receive the insert.
Detailed explanation: The recipient plasmid carries Amp resistance and four cutting sites, XhoI, EcoRI, HindIII and NotI. Cutting it with EcoRI and NotI linearises it, and the overhangs at the two ends are exactly complementary to those on the insert cut out of the donor plasmid, so joining them with ligase gives a recombinant plasmid carrying the insert.
The Amp resistance marker is what lets you pick the bacteria that took the plasmid up, rather than the ones that did not. What makes molecular cloning clever is that once the backbone is linearised you do not need to know the whole sequence, only that those few bases at the two ends line up. The seam does the work of the sequence.
Why this course brings it up: Dr. Wong does not name the recipient plasmid as an exam topic on its own, so it belongs to conceptual understanding. It is the other half of the donor plasmid pair, and the two of them are the whole of the cloning story.
Appears in: Ch 3, Lect 4

▪ recombinant chromatids

In one sentence: The chromatids that carry a new combination after a crossover.
Detailed explanation: Label the two homologous chromosomes A, B, C and a, b, c, and after the crossover you get two recombinant chromatids, A-B-c and a-b-C, while the chromatids that took part in no exchange are still A-B-C and a-b-c.
The first division of meiosis separates these combinations and the second division hands them out to four gametes, and that is where diversity comes from. The proportions in which those four combinations are distributed are the reason meiosis can produce so many different gametes, and they are the cell-level source of diversity. So the shuffling happens once, in the first division, and the second division only sorts out what was already shuffled. That is why the recombinant chromatids are worth drawing on paper before you answer anything about meiosis.
Why this course brings it up: The difference between M1 and M2, two normal plus two abnormal against one normal plus three abnormal, is the Tier 1 question he names. So the recombination products lead directly to the answer he wants. **Appears in:** Ch 8, Lect 2, Lect 4

▪ recombinant plasmid

In one sentence: The recombinant plasmid formed once the gene of interest has been joined into the backbone.
Detailed explanation: It carries an Amp resistance marker and the insert at the same time, so once it is put into bacteria you can select on ampicillin plates: the colonies that grow are the ones that really did take up the recombinant plasmid.
The entire molecular cloning workflow — cut, join, transform, select — is built around it. Because it has both the Amp resistance and the insert, the two properties together let the researcher screen first and verify afterwards: pick out the colonies that have a plasmid, then sequence to confirm that the insert is correct. The antibiotic answers "is there a plasmid here"; the sequencing answers "is it the right one".
Why this course brings it up: He does not name recombinant plasmid as an exam topic on its own, so it is a concept to understand; what it is good for is seeing why the screening step of the workflow needs two different checks rather than one.
Appears in: Ch 3, Lect 4

▪ Recombinant protein

In one sentence: A protein made for you by a host cell, after you have handed it a human gene.
Detailed explanation: The method is to get the gene of interest, insert it into an expression vector, transfer it into a host cell, and let the cell make a lot of it. Choosing the host is the key decision: E.
coli is fast, cheap and needs no glycosylation, whereas CHO cells (mammalian cells) can produce the correct glycosylation, so a complicated protein needs them. There is one extra difficulty Dr. Wong singles out with CHO cells: codon optimisation. His own words are “So remember that when you use the CHO cell, you need to do the codon optimization.” So the choice of host is not a detail of the workflow, it is the decision that determines whether the protein you get back is the right shape, and the codon step is the extra bit of work that the mammalian host demands. The reason codon optimisation matters is that the two hosts read the same DNA differently, each cell having its own preferred tRNAs, so the gene may have to be rewritten in the words the new host uses most comfortably.
Why this course brings it up: The recombinant expression section of Lecture 5 strings three things together: E. coli with IPTG as the non-metabolisable inducer, the timing of induction (too early causes toxic pre-expression and inclusion bodies), and CHO with codon optimisation. He does not list recombinant expression itself as an exam topic, but the codon optimisation line he explicitly says to remember. **Appears in:** Ch 7, Lect 5

▪ recombination-signal sequences

In one sentence: The signal sequences at the ends of the V, D and J segments that tell RAG where to cut.
Detailed explanation: The heavy chain genes of the immunoglobulins exist in the genome as many alternative segments, V 39, D 27 and J 6, and beside every one of them sits a pair of RSS. The RAG1 and RAG2, recombination-activating proteins, complex recognises the RSS and cuts in the right place, and afterwards the machinery of NHEJ joins the segments together.
If NHEJ is broken the antibody diversity cannot be built at all, and the child is born with SCID. The RSS is therefore the address that the RAG complex reads, and everything else in the rearrangement follows from where it points. One pair of RSS, one cut, one join, and the choice of which segments to use is what creates the diversity.
Why this course brings it up: V ≈ 39, D = 27 and J = 6 are Tier 3 numbers on his list, and "NHEJ deficiency gives SCID" is the first-grade question he names. **Appears in:** Ch 6, Lect 4

▪ Reduction

In one sentence: Gaining electrons (or losing oxygen), which is how energy gets stored inside a molecule.
Detailed explanation: Oxidation and reduction always happen together: one side loses electrons and is oxidised, the other gains them and is reduced. Reduction is called reduction because metal is reduced from its ore back into the element.
The most important reduction in biology is the one that TCA cycle and oxidative phosphorylation perform when they push electrons into carriers such as NADH and FADH2; the electrons then flow along the electron transport chain to oxygen, and the energy released on the way is used to build ATP. Put that together and you see it: the cell's "power station" works by running a whole series of reduction reactions whose real product is energy banked in ATP.
Why this course brings it up: In Lecture 1 he does not teach oxidation and reduction as a named concept, and this framing does not appear on his exam checklist either; he uses basic chemistry only to explain the two fates of pyruvate. The chemical structures and the metabolic pathway details are both on the exclusion list. So this one is background: learn the plain-language definition now, and it only starts to mean something when you meet mitochondrial oxidative phosphorylation in Lecture 2.
Appears in: Ch 1

▪ Regulatory gene

In one sentence: A gene that makes a switch protein, for example lacI.
Detailed explanation: The structure of an operon has a regulatory gene standing off to one side, and the repressor protein it transcribes travels to the DNA and sits down on the operator, where it blocks the polymerase.
lacI has exactly that role: it does not make the lactose-metabolising enzymes, it makes the lock. The advantage of splitting it up like this is that the amount of the enzyme can be turned up while the switch itself stays sensitive, and the schematic he draws is the order regulatory gene, repressor, operator, promoter, structural genes. So the regulatory gene is not part of the production run at all. It makes a protein whose only job is to block the production run, and that separation is the whole design. You could change the output tenfold without touching the switch, and that is the engineering advantage he is pointing at.
Why this course brings it up: The lac operon is on his second-level list in Lecture 5, and his case bank names reading the logic of the lac operon as one of the scenario questions: you need lactose present and glucose low, and only when both conditions hold is there strong transcription. The three roles, structural gene, regulatory gene and repressor, all have to be kept apart. One condition is not enough, and the trap in the question is assuming that either one on its own will do it. **Appears in:** Ch 7, Lect 5

▪ Replication bubble

In one sentence: The bubble where the DNA is pulled open, with one replication fork running on each side.
Detailed explanation: Replication does not run from one end of a chromosome to the other. A eukaryotic chromosome is far too long for that, so thousands of origins are opened at the same time, each origin is a replication bubble, and each bubble has a replication fork at each end going in opposite directions.
The practical result is that thousands of replication forks are running on the same chromosome at once, which compresses the replication time into something tolerable. The multiple bubbles you can see in an electron micrograph are direct evidence of exactly this.
Why this course brings it up: In Lecture 4 Dr. Wong lists it as a second-level fact: thousands of origins, each with two bidirectional replication forks. The clinical meaning is very direct, because most DNA-damaging anticancer drugs and most mutagens go after the cells that are being replicated a thousand times over. He does not ask you to memorise the micrograph.
Appears in: Ch 5, Lect 4

▪ Replication fork

In one sentence: The Y-shaped opening where the double helix is unzipped and copied as it goes.
Detailed explanation: The replication fork has two arms, and the arms are the two template strands. As it advances, helicase opens up the double helix, topoisomerase relieves the twisting ahead of it, primase lays down the RNA primers, and DNA polymerase joins the new strand piece by piece.
One side is the continuous leading strand, and the other is the lagging strand assembled in pieces. So the fork is not a single operation, it is two different operations running side by side at the same time, and that is why the leading strand needs one primer while the lagging strand needs one for every fragment. The speed and the direction in which the fork advances is the rhythm of the whole chromosome being copied, and on Dr. Wong's slide he draws the leadin, the lagging strand and the position of the polymerase all together.

Why this course brings it up: He does not name the replication fork as an exam topic on its own, so it belongs to conceptual understanding. The Tier 2 point is “each origin opens a bubble, and each bubble has two forks running in opposite directions”.
Appears in: Ch 5, Lect 4

▪ replicons

In one sentence: A stretch of DNA that begins at a single origin and can be replicated on its own.
Detailed explanation: Replication of a eukaryotic chromosome is not done in one go; it is divided into many replicons, each with its own origin and two forks running in opposite directions. When neighbouring forks meet, replication is finished off there — they simply stop because they have run into a new strand somebody else already made.
The advantage of the design is that replication can run in parallel, and that is why a human chromosome can be copied in a relatively short time instead of being read from end to end. Every origin is another crew starting work on another section at the same moment.
Why this course brings it up: "Multiple replicons and thousands of origins" is one of the Tier 2 facts he lists. It is a simple fact, and it explains a great deal about how fast genomes get copied. **Appears in:** Ch 5, Lect 4

▪ Reporter assay

In one sentence: You use a gene that glows to measure how active a given promoter really is.
Detailed explanation: The logic is straightforward: put the candidate promoter in front of a reporter gene (for example luciferase), put the construct into cells, and then measure the light that comes out.
The brighter the light, the better the promoter is at recruiting polymerase, and the stronger the gene expression. The clever design is the serial deletion. You cut away the promoter piece by piece from one end, and after each cut you record the reading; the segment where the light suddenly falls off is the functional element that was doing the work, while the sequences at either end that make no difference are the parts you could have deleted without harm. His example is finding out which TATA box variant is the effective one.
Why this course brings it up: In Lecture 5 he uses it to demonstrate how you locate the functional elements of a promoter experimentally, so it belongs to the concepts you are expected to understand. He does not list it as an exam item, and he does not ask you to remember which reporter gene is most commonly used; but a situation question may well ask what the logic of the experiment is, so you need to be able to say "delete a segment, read the value, find the key sequence". **Appears in:** Ch 7, Lect 5

▪ Reporter gene

In one sentence: A substitute placed at the position of a real gene, used to measure how busy a promoter is.
Detailed explanation: A reporter gene is not a gene the cell already has. It is a measuring instrument that has been engineered into the cell, and the common ones are luciferase and GFP. Put it downstream of a candidate promoter and the light signal you measure stands in for how hard that promoter is driving, so you can work backwards from the strength of the promoter to how much of the real gene would be expressed in a cell.
It turns “transcription you cannot see” into “a number you can read”, which is why it is the standard first step in promoter analysis. The strength of the signal is what you can compare between candidates.
Why this course brings it up: It makes sense together with the reporter assay: the reporter gene is the instrument, and the reporter assay is the whole procedure that uses it. Dr. Wong does not list either term as an exam topic, so it is conceptual understanding, but they are the concrete method of his Lecture 5 promoter section and worth recognising. **Appears in:** Ch 7, Lect 5

▪ Repressible system

In one sentence: On all the time, until its own product switches it off.
Detailed explanation: A repressible system is the other logic for an operon, the opposite of the lac kind where something has to be present for it to open. The trp operon is the standard example: the genes for the enzymes that make tryptophan are on all the time, but when enough tryptophan has been made, the spare tryptophan binds the Trp repressor, and the bound repressor grips the operator better than the free one, so the whole operon is switched off.
The more product there is, the harder the switch is shut, and that stops the cell wasting effort making something it does not need. It is feedback with a negative sign: the end product of the pathway is what turns the pathway off.
Why this course brings it up: In Lecture 5 he calls the trp operon the mirror image of the lac operon, and both are on the second-level list. The distinction he gives is that the lac repressor is released by lactose while the trp repressor is switched on by tryptophan. Understanding that pair gives you the logic of both operons at once. The mnemonic is that in one case the presence of the sugar lets the engine run, and in the other case the abundance of the product stops it. **Appears in:** Ch 7, Lect 5

▪ Repressor

In one sentence: A protein that sits on the DNA and blocks transcription.
Detailed explanation: A repressor recognises the operator, a particular stretch of DNA, and by sitting on it blocks the path of the polymerase. Its switch usually works through a ligand: once the ligand binds, the repressor either becomes unable to hold the DNA, which is the lac case, or becomes better at holding on, which is the trp case.

As well as blocking the road directly, Slide 15 of Lecture 5 lists two other ways of repressing: a repressor pushes an activator off an enhancer, which is competition, and a repressor binds the activator's DNA-binding domain or its activation domain directly, which is quenching. The point of a repressor is that it stops transcription without having to damage anything, so the DNA can be read again as soon as the repressor moves away. Either way the repressor is doing something reversible and non-destructive, which is what makes it a switch rather than an accident. The two named mechanisms, competition and quenching, are worth knowing because they are the ones that let one protein stop another without touching the DNA at all.

Why this course brings it up: In Lecture 5 Dr. Wong teaches the lac and trp operons side by side, and the repressor is the key character in both. His case bank lists "read the logic of the lac operon correctly" as one of the scenario questions, and the key to reading it correctly is that lactose present and glucose low must both hold at the same time. **Appears in:** Ch 7, Lect 5

▪ resolution

In one sentence: The smallest change that a detection method can tell apart.

Detailed explanation: What G-banding can detect reliably is a gross structural change at the level of roughly 0.5 to 10 Mb, and he quotes 5–10 Mb in one place and 0.5–1 Mb in another, with both numbers in his material.

Anything smaller than that, a single base change for instance, karyotype analysis cannot see at all, and you need sequencing for that. Which is why a chromosome that looks normal is not the same thing as a gene with nothing wrong with it: the resolution of karyotyping has a ceiling, and the small changes simply are not visible to it. The lesson generalises. Every method has a floor, and a clean result only tells you that nothing above that floor was found.

Why this course brings it up: That the resolution of karyotype analysis is roughly 0.5–10 Mb, and that point mutations need sequencing, is a Tier 2 fact he lists. And it is the reason he warns you not to memorise a single number. **Appears in:** Ch 3, Ch 4, Lect 3, Lect 4

▪ Reverse transcriptase

In one sentence: An enzyme that makes DNA using RNA as the template.

Detailed explanation: Ordinary DNA replication goes DNA → DNA, and reverse transcriptase runs it the other way as RNA → DNA. It needs an RNA template, which is what makes the versions that carry their own template so useful: telomerase is a reverse transcriptase with a built-in RNA template, and it uses that template to extend the 3′ overhang at the end of a chromosome.

HBV also replicates by reverse transcription, and Slide 17 of Lecture 2 draws its cycle from cccDNA to the packaging of pgRNA. A retrovirus goes further and reverse transcribes its whole genome into DNA before integrating it into the host. The reason the enzyme needs a template at all is worth sitting with: DNA → DNA replication is something the cell does with a template it already has, whereas going from RNA to DNA means working backwards up the central dogma, and a template is what makes that direction physically possible. Which is why the version that carries its own template is the one that does the interesting work in this course. Think of the built-in template as a piece of tape stuck to the end of the chromosome with the telomere sequence written on it. The enzyme reads the tape, writes the matching DNA, and when it reaches the end of the tape it lets go. No tape, no extension, and that is the whole trick.

Why this course brings it up: In Lectures 3 and 4 Dr. Wong describes telomerase as “a reverse transcriptase with a built-in RNA template”, and that is the key link in the mechanism chain of Lecture 4: end replication problem → telomere shortening → Hayflick limit → senescence → the tumour switches telomerase back on. Lecture 3 also mentions that around 90% of cancers upregulate telomerase. **Appears in:** Ch 5, Lect 3, Lect 4

▪ Ribose

In one sentence: The five-carbon sugar in RNA; DNA uses its deoxy version.

Detailed explanation: Ribose is a five-carbon sugar and every nucleotide in RNA uses it. DNA uses deoxyribose, the same five-carbon ring with one oxygen missing on the 2′ carbon, so one less OH group.

That difference is not just chemical fun: the 2′-OH of RNA takes part in reactions, which makes RNA easy to hydrolyse in alkaline conditions, so RNA is short-lived, while DNA without it is far more stable and suited to storing recipes for a long time. RNA also uses uracil, U, in place of DNA's thymine, T, and that is the same design logic. So the two molecules are deliberately built with different lifetimes, and almost every practical difference between them follows from that.

Why this course brings it up: He does not list ribose as an exam question on its own, and the chemical structure is on the exclusion list. It is concept-only: you have to be able to say that one of the differences between DNA and RNA is whether the 2′ carbon of the sugar carries an oxygen, because an RNA vaccine can exist for a short time and then be degraded quickly precisely because of this property of RNA. That is a functional reason, not a structural one. **Appears in:** Ch 3

▪ Ribosome

In one sentence: The machine that reads the mRNA and joins amino acids together one by one.

Detailed explanation: The ribosome is the translation machine in the factory. It grabs a copy of the mRNA, a photocopy of the recipe, matches an amino acid to each three-letter codon on it in turn, and joins the amino acids together with peptide bonds. The chain it builds is the backbone of the protein, and only afterwards does it fold into a machine. The ribosome has a large and a small subunit, each made of rRNA and protein. It is itself neither an enzyme nor a template, and yet it is the single exit through which all proteins pass.

Why this course brings it up: It is the last station on the DNA, RNA, protein production line and the end of the central dogma. Dr. Wong does not list the ribosome as an exam question on its own, so it is conceptual understanding. Attached to it is one

detail that will be examined: the start codon AUG for methionine, and the stop codons UAA, UAG and UGA. He puts those three on a slide, so have them ready. **Appears in:** Ch 8, Lect 2

▪ RISC

In one sentence: The machine that uses a microRNA to switch off a target gene.
Detailed explanation: RISC is the RNA-induced silencing complex. The miRNA starts life as a double-stranded molecule; Dicer cuts it into pieces, and one of those pieces — the guide strand — is loaded into an AGO protein, which is the heart of RISC. From then on RISC patrols the cell, and what it is looking for is the miRNA's seed sequence sitting in the 3′ UTR of a target mRNA. Recognition rests on that short stretch of bases, not on the whole molecule. If the pairing is complete, AGO cuts the mRNA on the spot and the message is gone. If the pairing is only partial, the machinery does not cut at all: it blocks translation and speeds up the degradation of the mRNA instead — Slide 38 of Lecture 5 even draws RISC together with GW182 (the effector protein of RISC) delivering the mRNA into a P-body, where it is broken down. In other words, the two cases are decided by how well the two match: a good match destroys the message, a poor match only quiets it. And because only a short stretch has to pair at all, one single miRNA can push down many genes at the same time.
Why this course brings it up: He draws the miRNA generation and action pathway in both Lecture 1 and Lecture 5, so this one is a concept to understand rather than a list to memorise. What is really examinable is the clinical consequence downstream: miRNA is unusually stable in serum, so it can serve as a non-invasive biomarker — it can be measured in a blood sample without any tissue being removed from the patient. The numbers he gives are miR-15b plus miR-130b: AUC 0.981, sensitivity 98.25 %, specificity 91.53 %, accuracy 94.83 %. **Appears in:** Ch 7, Lect 1, Lect 5

▪ RNA-binding protein (RBP)

In one sentence: A protein that grabs hold of a stretch of RNA and changes how stable it is or how fast it gets read.
Detailed explanation: An RBP decides the fate of an mRNA. The same mRNA, held by one RBP, may become very stable and go on being translated many times over; grabbed by a different RBP, it is rapidly broken down instead.
The message is identical in both cases — what has changed is which protein has hold of it. The polyadenylation diagram on Slide 31 of Lecture 5 is a demonstration of this principle: a long 3′ UTR attracts RBPs, miRNA and lncRNA to bind onto it, and what they bind then decides where this message sits in the cell, how efficiently it gets translated, and how long it survives. That is why the long isoform of BDNF (brain-derived neurotrophic factor) is carried out to the dendrite instead of staying in the cell body. In other words, the message is the same; what differs is which proteins are holding it, and that is what decides whether it is read often and lasts, or cleared away.
Why this course brings it up: In Lecture 5 he teaches mRNA stability as one layer of gene expression control (5′ cap, poly-A tail length, binding of RBP and miRNA), so it is a concept to understand. The practical payoff is this: the same gene producing different isoforms gives them different RBP binding sites, which is one of the reasons alternative splicing can change the fate of a protein. Different isoform, different set of proteins holding it, different lifetime — and that is the whole point of making different isoforms in the first place. **Appears in:** Ch 7, Lect 5

▪ RNA interference (RNAi)

In one sentence: The older way of switching a gene off, using small RNA molecules.
Detailed explanation: RNAi runs on the cell's own machinery. You put a designed double-stranded small RNA into the cell, RISC takes it apart into single strands and uses them for recognition, and then the target mRNA is degraded or its translation is suppressed, so the protein output falls.
Traditionally there are two ways of doing it: siRNA, where the synthetic double-stranded RNA is delivered directly, and shRNA, where a viral vector expresses it so the cell keeps making it. The shared weakness is that this is a knockdown, you push the expression down rather than turning the gene off completely. The distinction matters because a knockdown leaves a residue of protein behind, and for some experiments that residue is the difference between a clean result and a muddy one.
Why this course brings it up: In Lecture 2 he puts it bluntly: CRISPR/Cas9 has become the standard way to do gene knock-out and knock-in, and the old siRNA and shRNA knockdown technology has essentially been replaced. So if the exam asks how to turn a gene off completely, the answer is the CRISPR knock-out and not RNAi. That contrast is one of the distractors he lists. The wording of the question tells you which one they want. **Appears in:** Ch 8, Lect 2

▪ RNA polymerase

In one sentence: The enzyme that copies a gene into RNA.
Detailed explanation: RNA polymerase is a specialised photocopier: it recognises the promoter stretch of DNA, opens the duplex, takes one strand as the template, and reads it while joining nucleotides into an RNA chain.
Unlike DNA polymerase it does not need a primer, it can start from nothing, and it has no 3′→5′ proofreading activity, so the error rate of transcription is higher than that of replication. The product has to be extended in the 5′→3′ direction, and that single rule governs both replication and transcription. So the two enzymes differ in where they start, how carefully they check themselves, and how fast they go, but they build the product in the same direction, which is why one rule covers both. And the direction is not a convention that could have been chosen the other way round: the chemistry of adding a nucleotide only works onto a free 3′ end.
Why this course brings it up: In Lecture 5 Dr. Wong uses it to build the whole scene of the transcription machinery: RNA polymerase and the general transcription factors assembling on the promoter, especially the whole set that gathers on the TATA box. The 5′→3′ direction rule is also on his distractor list, paired against the 3′→5′ proofreading. It belongs to conceptual

understanding. **Appears in:** Ch 6, Ch 7, Lect 5

▪ RNA polymerase II

In one sentence: The enzyme that copies protein-coding genes into precursor mRNA.
Detailed explanation: A eukaryotic cell has several RNA polymerases and they divide the work. RNA polymerase II handles the great majority of mRNA, which means it copies the protein-coding genes into pre-mRNA, and that then goes through splicing, capping and polyadenylation to become mature mRNA.
Polymerase I and polymerase III handle rRNA and tRNA and the rest. Pol II does not work on its own. It relies on a set of general transcription factors, TFIID and TFIIH among them, that assemble on the promoter first, and only then does transcription start, and that is the central picture of the promoter section of Lecture 5. The order matters: the factors arrive and build the platform, and the enzyme can only work once the platform is there, which is why a broken promoter stops transcription even when the gene itself is perfectly intact. Pol II is therefore not the machine on its own but the motor that is lowered into a rig built for it, and the promoter is the address of that rig. The other two polymerases are mentioned so that you can place them, and they are not examinable in detail: polymerase I makes the bulk of the rRNA and polymerase III makes tRNA and the other small RNAs.
Why this course brings it up: On the promoter mechanism diagram in Lecture 5 Dr. Wong draws RNA polymerase II together with the general transcription factors, and that is conceptual understanding. He does not ask you to memorise the job of each general transcription factor, but you do have to recognise the two steps of TFIID bringing TBP and TFIIH opening the DNA and firing Pol II, because those two steps are the working principle of the promoter, and they are the two names that appear on his slide. **Appears in:** Ch 6, Ch 7, Lect 5

▪ RNA (ribonucleic acid)

In one sentence: A whole family of molecules that read out and carry out the instructions in DNA.
Detailed explanation: RNA is a single-stranded molecule and its main members are mRNA, which copies the recipe into a portable copy, tRNA, which carries amino acids to the ribosome and reads the codon, rRNA, which is the ribosome itself, and miRNA and lncRNA, which regulate.
The prokaryote against eukaryote diagram in Lecture 2 makes it clear: a prokaryotic cell has no nucleus and its mRNA can be translated while it is still being transcribed, while a eukaryotic cell has to process the message into mature mRNA inside the nucleus and export it to the cytoplasm before it is translated. That separation is what decides the rules of expression systems and of splicing. So the same molecule behaves quite differently in the two cell types, and that is why the compartment, not the chemistry, is the headline.
Why this course brings it up: He lists the kinds of RNA on Slide 7 of Lecture 1 and does the prokaryote against eukaryote comparison on Slide 6 of Lecture 2, so it is concept-only. What is really examinable is the consequence of that separation: eukaryotic mRNA has to go through splicing, a 5′ cap and a poly-A tail, so there are more layers of stability and control than in a prokaryote. Keep the technique names in English and do not translate them. **Appears in:** Ch 1, Ch 3, Ch 7

▪ RNA sequencing

In one sentence: The technology that reads all the RNA in a sample as sequences.
Detailed explanation: The workflow runs like this: you pull out the RNA, break it into fragments, use reverse transcriptase to turn each of them into DNA, add adapter fragments, build a DNA library, and then the machine reads the sequences and aligns them to the genome.
Its value is that you see which genes are expressed and by how much at the same time, whereas a microarray can only ask whether the few genes you specified are there or not. It costs more and asks more analysis than a microarray, but because you do not have to name the genes in advance it cannot be replaced when you want to find a new transcript. That is the trade you accept for the extra sensitivity.
Why this course brings it up: Dr. Wong does not name RNA sequencing as an exam question on its own, so it is conceptual understanding, and the omics workflow of Lecture 1 is the part that matters. **Appears in:** Ch 11, Lect 1

▪ RNA vaccine

In one sentence: Deliver the mRNA and let the patient's own ribosomes build the antigen.
Detailed explanation: Slide 37 of Lecture 5 draws the whole road: injection into muscle → the mRNA, packed in a lipid nanoparticle, entering the cell → ribosomes translate the mRNA into the antigen protein → the proteasome cuts it into short peptides → they are loaded onto MHC through the ER/endosome route → the TCR recognises them → T cells and B cells are activated at the same time → the cancer cells are killed.
The key idea is that you are not delivering the protein but delivering the recipe for it, so the cell makes the antigen itself and the immune system is looking at something the body produced. The same central dogma that runs through the course, used as a delivery mechanism.
Why this course brings it up: In Lecture 5 Dr. Wong places the RNA vaccine at the modern frontier of “control of gene expression”, and it belongs to conceptual understanding. He does not list it as an exam topic and does not ask you to memorise the chemistry of the lipid nanoparticle. The point is that you can lay out the whole chain using the central dogma: an instruction in DNA can turn directly into a drug. **Appears in:** Lect 5

▪ RNase H

In one sentence: The exonuclease that shovels the RNA primer off the DNA.
Detailed explanation: Every Okazaki fragment on the lagging strand begins with an RNA primer. Once the polymerase has extended the fragment, the RNA primer at the front is a foreign body that has to be swapped for DNA.
RNase H is the enzyme that does this: it recognises the RNA–DNA junction and cuts away the RNA part, the gap is filled in by DNA polymerase, and the ligase seals it. The RNA primer has to be replaced by DNA, otherwise the new strand is left with a seam of RNA mixed in with DNA, which is bad for the replication that follows and bad for stability. It is the one part of the lagging strand that cannot be left as it was made. The polymerase has no way of removing a primer on its own, which is why a separate enzyme is needed for the job.
Why this course brings it up: Dr. Wong does not name RNase H as an exam topic on its own, so it belongs to conceptual understanding. The Tier 2 point is that the lagging strand is made in pieces and the pieces have to be joined up. **Appears in:** Ch 5, Lect 4

▪ rolling

In one sentence: The first step, in which an immune cell rolls along the inside of a blood vessel.
Detailed explanation: As the blood flows, white blood cells roll along the vessel wall, and they do it by repeatedly binding and letting go with P-selectin, E-selectin and integrin. The step is slow, and that is the point: it gives the cell a chance to stop in the right place and move on to the next step of adhesion and activation.
Rolling matters because it lets an immune cell try out the surfaces around it and decide whether to stop; plenty of the problems with the tumour microenvironment come from this step being too inefficient. Bind, roll a little, let go, bind again — the whole journey along the wall is made of that repeating motion.
Why this course brings it up: He does not name rolling as an exam topic on its own, so it is a concept to understand; the point is the four-step journey of an immune cell arriving at a tumour, and every step of it can be blocked by a treatment. **Appears in:** Ch 9, Lect 2

▪ Rough ER

In one sentence: The part of the endoplasmic reticulum that carries ribosomes, where new proteins are made and folded.
Detailed explanation: The rough ER is the protein factory: the ribosomes on its surface feed the growing peptide chain into the ER lumen, the chaperones inside help it fold, and the glycosylation modifications are added on the spot.
Only a protein that has folded correctly is let through, and one that has folded wrongly is kept there to be dealt with. The finished protein is then packed into a transport vesicle and sent to the Golgi. The difference between rough and smooth is only whether there are ribosomes on the surface: where there are, it is rough and it makes protein, and where there are not, it is smooth and it handles lipid. That single question, ribosomes or no ribosomes, is the whole of the distinction. The rough ER therefore does three things in one place: it receives the chain, it folds it, and it marks it, and all three happen while the ribosome is still feeding it in. The rule about folding is the important one, because a protein that is not folded correctly can be kept and dealt with rather than let out into the cell.
Why this course brings it up: Slide 22 of Lecture 2 draws the whole ER-to-Golgi route, so that part is conceptual understanding. But it connects to an exam point he has emphasised in person: the signal peptide. Miss the N-terminal signal peptide and the secretory and membrane proteins simply do not express, and his postdoc lost four months over exactly that. His words are "So remember, signal peptide is really, really important for your study". **Appears in:** Ch 8, Lect 2

▪ RPA

In one sentence: The protective protein complex that first covers the exposed single-stranded DNA.
Detailed explanation: When HR begins, the broken end of the DNA is cut back to leave a single-stranded overhang. Single-stranded DNA is easily tangled up or attacked by nucleases, so RPA, replication protein A, covers it first like a plastic sleeve, holding that strand stable and at the same time stopping it pairing back up with itself.
Only then can RAD51 wind a filament on it and carry out strand invasion. RPA is what allows the RAD51 filament to form on a protected strand, because otherwise the two single strands would simply pair with each other again and the replication would jam. So the order is protection first and assembly second, and skipping the first step makes the second one impossible.
Why this course brings it up: He does not name RPA as an exam question on its own, so it is concept-only. The principle it illustrates, that the single strand has to be held before it can be used, is the transferable part. **Appears in:** Ch 6, Lect 4

▪ rRNA

In one sentence: Ribosomal RNA, the bulk of the ribosome, and the part that lines the mRNA up.
Detailed explanation: rRNA is the most abundant component of the ribosome, and together with protein it forms the large and small subunits. Its function is not to act as a template but to act as the machine: the reaction centre that catalyses peptide bond formation is made mainly of rRNA, which means the ribosome is in essence a ribozyme.
At the same time it positions the mRNA correctly, so that the three-letter codons line up against the anticodons of the tRNA. The ribosomal subunits are 70S in bacteria and 80S in the eukaryotic cytoplasm.
Why this course brings it up: On Slide 14 of Lecture 3 Dr. Wong asks you to recognise the difference between rRNA and the 70S/80S figures, and it belongs to conceptual understanding. He does not list it as an exam topic, and you should not memorise the sedimentation coefficients or the subunit details. What you need is this: rRNA is the skeleton of the ribosome, and it is also the standard example of the RNA world idea that RNA can be its own catalyst. **Appears in:** Ch 8

▪ RuvC

In one sentence: One of the two nuclease domains of Cas9, and it cuts the non-target strand.
Detailed explanation: Once Cas9 has been led by the gRNA to the target sequence near a PAM, the two nuclease domains divide the work of the cut: RuvC and HNH each cut one strand, and together they produce a double-strand break in the DNA. How the cell then repairs that break is the variable the researcher is really controlling when designing a CRISPR experiment. The two domains cut close together, and only together do they make one clean double-strand break, which is what gets the cell to call up NHEJ or HDR to fix it. So the two domains are a matched pair, in the same way that the two ways of repairing the break are the matched pair you actually have to understand.
Why this course brings it up: He does not ask you to memorise the names of the two domains, so it is conceptual understanding, and what matters is the fork after the cut: NHEJ goes to knock-out and HR goes to knock-in. **Appears in:** Ch 6, Lect 2, Lect 4

▪ RXR

In one sentence: The partner receptor that pairs with Type II nuclear receptors to form a heterodimer.
Detailed explanation: RXR, the retinoid X receptor, is the busiest member of the nuclear receptor family, because it can pair with many different nuclear receptors and form the heterodimers of the Type II nuclear receptors: the thyroid hormone receptor, the oxysterol receptor and others all bring it along when they go looking for DNA.
The Type II diagram on Slide 19 of Lecture 5 is drawn exactly this way. With no ligand present, the RXR-NR heterodimer is already sitting on the DNA response element with a corepressor pressing the gene down. When the ligand arrives, the corepressor is swapped for a coactivator, and only then is the gene opened. So in the Type II case the receptor does not wait in the cytoplasm for its message, it is already in position, and the message changes who is sitting next to it.
Why this course brings it up: In Lecture 5 he teaches Type I and Type II nuclear receptors side by side, so it is concept-only and RXR is not named as a question on its own. But the difference between Type I and Type II is on his second-level list, and RXR is the representative mechanism of Type II: Type I waits for the ligand in the cytoplasm, Type II is already on the DNA waiting for it. One is a receptor that goes looking, the other is a receptor that is already there. **Appears in:** Ch 7, Lect 5

▪ S-adenosyl methionine (SAMe)

In one sentence: The molecule that carries the methyl group, and the source of the methyl that goes onto DNA.
Detailed explanation: Almost all of the methylation of DNA and of histones gets its methyl group from SAMe, which makes SAMe the donor that has to be present wherever a mark is going to be written, whether that mark is on the DNA or on a histone. The enzymes divide the work into two kinds. The de novo methyltransferases, DNMT3A and DNMT3B, establish methyl marks in new places, in effect writing the instructions afresh, and they are the ones that put a mark on a region that has never carried one. The maintenance methyltransferase, DNMT1, does something different and much more mechanical: after DNA replication it copies the marks on the old strand over onto the new strand, so that a cell which has just divided still knows who it is and does not have to rewrite the whole pattern from scratch. It is copying rather than writing, and that is the whole difference between the two. So the two enzymes are not two versions of the same thing; one writes the map and the other keeps the map in step with the copying of the book. Where the methylation lands is the critical part. Methylation of a CpG island in the promoter switches the gene off, and that is a common way for a cancer to keep its brakes on: the tumour suppressor is still there, it is simply never called.
Why this course brings it up: In Lecture 3 Dr. Wong teaches DNA methylation as a second-level item, and his words are "The most important aspect of the DNA methylation is it has a very significant functions on these gene expressions, because DNA methylation is usually involved in the suppression of the gene expressions." The de novo against maintenance contrast is also on his distractor list, and you have to be able to tell the two apart, because a question that mixes them up is easy to get wrong in exactly the same way every time. **Appears in:** Ch 4, Lect 3

▪ S phase

In one sentence: The stretch of the cell cycle in which DNA is copied.
Detailed explanation: S phase is the synthesis phase: DNA is replicated in full during it, and the DNA content climbs from 2N to 4N as a result. The replication itself does not proceed as one long copying job;
it opens out at the origin into thousands of replication bubbles all at once, with a replication fork running on each side of every bubble, and that is how thousands of separate copying jobs can start at the same moment. One leading strand is made continuously, while the other lagging strand is made in pieces called Okazaki fragments. S phase is also when DNA damage is most likely to interrupt you: a replication fork that runs into damaged DNA simply stalls, which is why the DNA repair pathways matter so much in S and G2.
Why this course brings it up: His flow cytometry plot splits the cells into three buckets — G1 (2N), S (in between) and G2/M (4N) — and what the PI stain measures is exactly those three. It is a concept to understand, but he draws it very clearly. He also says explicitly that you should not memorise the cyclin and CDK pairing table. The clinical link: HR can only be done in S/G2, while NHEJ can be done in any phase, and that rule is what decides how the two divide the labour in drug development.
Appears in: Ch 8, Lect 2

▪ Sanger sequencing

In one sentence: The gold standard method for reading DNA, where a series of coloured peaks tells you which base is where.

Detailed explanation: The principle of Sanger is chain termination: four differently fluorescently labelled ddNTPs are put into each reaction tube, and the moment one of them is added the DNA stops there.

Run four tubes (or separate the colours in one tube) on electrophoresis and the sequence of peaks you read out is the order of the bases in the DNA. It is slow but it is the gold standard, reading a few hundred to over a thousand bases in one run. Today bulk sequencing uses NGS, which reads tens of thousands to millions of bases at a time at a lower cost per base.

Why this course brings it up: In Lecture 3 Dr. Wong uses it for the key contrast: karyotyping cannot catch point mutations or single-base variants, and those have to be found by sequencing, Sanger or NGS. The resolution of karyotyping is only about 0.5–10 Mb, so a point mutation is only visible if you sequence. This pairing of "which question needs which tool" is the consistent logic he uses when he talks about the tests. **Appears in:** Ch 3, Ch 11, Lect 3

▪ SAV

In one sentence: The protein that carries the signal from MST to LATS in the Hippo pathway.

Detailed explanation: The Hippo pathway is one he comes back to again and again in Lectures 1 and 5. When the cells are dense and touching each other, the junction complex starts the phosphorylation of MST.

The phosphorylated MST then uses SAV, a scaffold protein, to hand the signal to LATS for the next phosphorylation, and the phosphorylated YAP is grabbed by 14-3-3, kept in the cytoplasm and degraded, and the cell stops proliferating. SAV's role here is that of a scaffold, the piece that holds two enzymes together. Scaffolds are a recurring idea in signalling: the enzyme that does the work is not the enzyme that brings the other one to the right place.

Why this course brings it up: He draws the Hippo diagram in both Lecture 1 and Lecture 5, but he does not list SAV as an exam question on its own, so it is concept-only. What he explicitly excludes is the list of upstream activating factors and the detailed cascade steps. What you have to keep is the skeleton: MST1, then LATS1 phosphorylates YAP, then YAP cannot get into the nucleus, then no proliferation, and when the kinase is lost YAP enters the nucleus and the tissue overgrows. **Appears in:** Ch 7, Lect 5

▪ Scaffold

In one sentence: A molecule that ties two or more proteins together in the right place so that they can work as a team.

Detailed explanation: A scaffold does not necessarily catalyse anything itself, but it is the platform that lines up several separate parts in the right positions so that they can work together.

In the cell it is a common strategy: in the Hippo pathway SAV lines up MST and LATS, and in signalling, scaffold proteins string a run of enzymes into something like a ladder, with one enzyme's product becoming the next enzyme's substrate. In the world of lncRNA it is also a leading role — Slide 29 of Lecture 5 explains that an lncRNA can act as a scaffold to hold proteins together, and as a guide to carry a protein over to the DNA it ought to be acting on.

Why this course brings it up: In Lecture 5 he sorts out how lncRNAs do their work into two types, scaffold and guide, so it is a concept to understand. But it also explains why lncRNAs are becoming new drug-discovery targets: they are natural assembly platforms, and drugs love nothing better than a design that borrows somebody else's platform. You do not have to build the scaffold from scratch if one is already sitting in the cell, holding the parts in the right places. **Appears in:** Ch 7, Lect 5

▪ scFv

In one sentence: The smallest usable piece of an antibody, one heavy-chain variable region plus one light-chain variable region.

Detailed explanation: The Y-shaped tip of an antibody is the antigen-binding site, made up of two variable domains, VH on the heavy chain and VL on the light chain. Join those two pieces with a short peptide and you have an scFv (single-chain variable fragment), which can be expressed on a single polypeptide without having to assemble two chains.

The rest of the antibody can then be thrown away: the Fc half is what talks to the immune system, and what a CAR wants is only the one function of grabbing antigen. So the CAR takes the tip of the antibody and hangs it on the outside of a T cell, which is why a CAR-T cell can still find and hold on to its target the way an antibody would.

Why this course brings it up: In Lecture 2 Dr. Wong describes the structure of a CAR as scFv (grabs the antigen on the outside) plus hinge plus transmembrane plus the intracellular signalling region (CD3ζ, 4-1BB, CD28), and points out that full T cell activation needs two signals, while the CAR wraps both of those into one receptor. It is conceptual understanding, but it is the entry concept for CAR-T and worth having firmly. **Appears in:** Ch 9, Lect 2

▪ SCID (severe combined immunodeficiency)

In one sentence: Severe combined immunodeficiency: the failure to build a usable library of antibodies and T cell receptors.

Detailed explanation: This is the hardest causal chain in the course and you have to be able to draw it from beginning to end. NHEJ is borrowed by V(D)J recombination. When B cells and T cells develop, they assemble their receptor genes out of V, D and J segments, and it is NHEJ that joins the cut pieces together.

So if NHEJ is broken you cannot assemble the receptor library. And if you cannot build antibodies and TCRs, the immune system has no ammunition for recognising foreign antigens, and the result is SCID. The numbers he gives for the segment copies are V ≈ 39, D = 27 and J = 6. Those numbers are how many choices the cell has at each of the three points, and the

library that can be built from them is enormous. Note the direction of the logic, because it is the direction the question is asked in: the defect is in a repair enzyme, the symptom is in the immune system. The enzyme is generic, the antibody is specific, and the link between them is the way the cell puts a gene together.
Why this course brings it up: This is the Tier 1, A+ question he marks with his own mouth. His words are: "Therefore, if I ask you, if the people showing a defective in this non-homologous end joining, what's the outcome? Definitely the outcome is the severe immune deficiency, right?" A short-answer question may ask exactly why V(D)J has to rely on NHEJ, so this chain has to be able to say backwards as well as forwards. It is the highest confidence question in the whole course. **Appears in:** Ch 6, Ch 9, Lect 2, Lect 4

▪ SDS-PAGE

In one sentence: The standard gel electrophoresis for proteins, which lines the proteins up as a ladder by size.
Detailed explanation: SDS is a detergent that wraps every protein in the same negative charge, so the differences in charge disappear and the only variable left is size. Small proteins travel further through the gel than large ones, so the gel looks like a set of steps.
But be careful: four bands mean four sizes, not four identities, because a band of the right size can be any protein of that molecular weight. To identify what the molecule is you need a method that reads the molecule itself, and that is LC-MS/MS. So the gel is a ruler, and a ruler can only tell you how big something is. The reason people get it wrong is that the band looks so convincing, four neat stripes of blue, that it is easy to believe the stripe has an identity as well as a size.
Why this course brings it up: In Lecture 1 Dr. Wong puts SDS-PAGE alongside Edman degradation, LC-MS/MS and X-ray crystallography as four methods and four different answers. It is conceptual understanding, but his self-test questions include this one: "is SDS-PAGE wrong?" The correct correction is not that the method is wrong, it is that you took a question about size to a method that answers a question about identity. The other three methods each answer their own question: Edman gives the sequence, LC-MS/MS gives identity and amount, and X-ray gives the atomic shape. **Appears in:** Ch 2, Lect 1

▪ Second messenger

In one sentence: A small molecule inside the cell that carries the signal further along.
Detailed explanation: Once a signal has reached a receptor on the cell membrane, it usually needs a small molecule that can move freely about inside the cell to carry the message deeper in, and that is the second messenger.
The receptor has done its job at the surface, but the message usually has a long way still to travel, and a small diffusible molecule carries it better than a protein can. The most famous is cAMP: it accumulates when glucose is low, and once it has bound CAP it goes on to recruit RNA polymerase at the promoter. Another pair is DAG and IP3 — once the receptor is activated, PLC cuts PI(4,5)P into two pieces, DAG stays behind on the membrane and activates PKC, while IP3 travels across to the ER and releases calcium ions from there. The two work in different places, which is how cutting one membrane lipid in two gives you two messengers at once.
Why this course brings it up: In the lac operon example of Lecture 5, cAMP is the second messenger: it is how the condition "glucose is low" gets read inside the cell. He does not list second messenger as an exam topic on its own, but the logic of "two conditions must both hold" is something you do need, and the cAMP–CAP branch is exactly where the answer to case study 16 lives. **Appears in:** Ch 7, Lect 5

▪ Secondary structure

In one sentence: The local regular shape of a chain: the alpha-helix and the beta-sheet.
Detailed explanation: Secondary structure is the first meaningful level of folding, held in place by hydrogen bonds that repeat along the backbone. An alpha-helix is a right-handed corkscrew of about 3.6 amino acids per turn, and a beta-sheet is a flat folded panel that forms when the chain folds back on itself and runs alongside an earlier part of itself, either parallel or antiparallel.
Secondary structure is a local rule, and the side chains of the amino acids (the R-groups) are still free to move about, which is exactly why it is more sensitive to the environment than the third level and also more useful for designing proteins. The helix and the sheet are the shapes that protein folds keep running into, because the backbone hydrogen bonds are satisfied by them.
Why this course brings it up: In Lecture 1 Dr. Wong lays out the four levels as the skeleton of the whole course, and secondary structure is one of them. It is conceptual understanding, and what is examined is “which level is held by what”: secondary by hydrogen bonds, tertiary by hydrophobic, ionic and hydrogen bond interactions between side chains, and quaternary by the interfaces between subunits. What he explicitly excludes is the table of amino acid R-groups and the chemical structures. **Appears in:** Ch 2

▪ Secretory pathway

In one sentence: The outbound route, rough ER, then Golgi, then vesicle, then cell membrane, then outside the cell.
Detailed explanation: A protein that is going to be secreted carries a signal peptide on its N-terminus. SRP picks it up and brings it to the rough ER, where it is inserted into the membrane and threaded through into the ER lumen, where it folds and is glycosylated, and then it travels in a transport vesicle to the cis face of the Golgi, the receiving end, to be processed and sorted, and finally it leaves from the trans face, the dispatch end, packed into a secretory vesicle that goes to the cell membrane and fuses with it and spits its contents out of the cell.
Membrane proteins travel the same road, they just stop at the cell membrane instead of going further. There is one entrance and one exit, and the entrance is at the front of the route.

Why this course brings it up: Dr. Wong teaches it once in Lecture 2 and once in Lecture 5, and it is conceptual understanding. But it connects to a point he flags explicitly: the signal peptide is the key that opens the door for the whole route, and without it the secretory and membrane proteins do not express at all, which cost his postdoc four months. He also wants you to keep the signal peptide apart from the N-terminal domain, NTD: one of them sends the protein into the ER, the other is the regulatory region of a nuclear receptor. **Appears in:** Ch 2, Ch 8, Lect 5

▪ Seed sequence

In one sentence: The stretch of nucleotides 2 to 8 on a microRNA, the part that does the target recognition.
Detailed explanation: A miRNA is only about 22 nucleotides long, and the part that actually does the matching is only a small piece of it, the seed sequence, positions 2 to 8. It recognises the 3′ UTR of the mRNA by complementary pairing, and it tolerates some mismatches through G:U wobble pairing.
That is why one miRNA can hold down hundreds of genes at once: as long as the 3′ UTR of those genes carries the same stretch of seed sequence, they are all on the list. So the specificity is set by seven nucleotides, and everything else in the miRNA is machinery. A short recognition site, a long reach.
Why this course brings it up: Slide 40 of Lecture 5 marks the seed sequence as positions 2–8, so it is concept-only. What you really have to take is the deduction of why one miRNA acts on many: pairing needs only a short stretch, so a single miRNA can switch off a whole group of genes at once. That is why it can change several pathways in a tumour at the same time, and why it has a more global effect than siRNA. **Appears in:** Ch 7, Lect 5

▪ Selectin

In one sentence: A cell-surface protein that makes white blood cells roll along the walls of blood vessels.
Detailed explanation: Selectin comes in three members. E-selectin is expressed by vascular endothelial cells after they have been stimulated by inflammation. P-selectin is expressed by both platelets and endothelium, and it is stored inside granules so that it can be moved up to the surface very quickly the moment the cell is activated.
L-selectin is expressed by the white blood cell itself. What all three do together is make the white blood cell roll along the endothelium: the first step has to be a slowing down, because only after the cell has slowed can the second step, adhesion, happen at all. After rolling comes adhesion by ICAM-1 and VCAM-1, then crossing the wall via PECAM-1, and finally chemotaxis via MCP-1.
Why this course brings it up: This is the four-step diagram on Slide 19 of Lecture 2, a concept to understand; he does not list it as an exam topic. But it has direct clinical meaning: a tumour that wants to hide away changes the microenvironment and keeps immune cells from getting in, and the list of obstacles that stops CAR-T from reaching a solid tumour includes the immunosuppressive factors TGFβ and IL-10 along with the stromal environment. All of those are different faces of the same story that this diagram tells. **Appears in:** Ch 9, Lect 2

▪ Selective Estrogen Receptor Modulator (SERM)

In one sentence: A drug like tamoxifen: it blocks estrogen in the breast and slightly activates it in the endometrium.
Detailed explanation: The key to a SERM is “same drug, different tissue, different result”. After tamoxifen binds the estrogen receptor, in breast tissue it keeps estrogen out and does not recruit a coactivator, so it acts as an antagonist and gives the antitumour effect.
But in the endometrium of the same body it can recruit a coactivator and becomes a partial agonist, which is why the risk of endometrial cancer goes up slightly. The same receptor, the same drug and two opposite results, and the difference comes from the mix of coactivators and corepressors lying around the receptor. The receptor does not change, the drug does not change, the only thing that differs is the neighbourhood the receptor finds itself in. So the drug is not working on the tissue, it is working on the receptor in that tissue, and the tissue gets its vote through the proteins it keeps nearby. This is also why the same drug can be a good idea in one organ and a bad idea in another, and why the tissue has to be named when you prescribe it.
Why this course brings it up: In Lecture 5 Dr. Wong lists it as a second-level item and says “This is the concept that you have to bear in mind.” The concept is that the same target may need a different pharmacological strategy in different tissues, and it will turn up in an exam question in the form of “the same drug acts in opposite ways in different tissues”. He does not ask you to memorise the pharmacological details. **Appears in:** Ch 7, Ch 10, Lect 5

▪ Semi-conservative replication

In one sentence: Every new DNA double helix is made of one old strand and one new strand.
Detailed explanation: This is the rule of DNA replication, and it is also the conclusion the Meselson–Stahl isotope-labelling experiment came to. The DNA separates into single strands, each strand acts as a template, and a new strand is built alongside each one, so that every daughter molecule ends up with exactly one parental strand and one newly synthesised strand.
The rule has a practical consequence: it guarantees that every generation can use the old strand as a template to proofread the new one, which is why replication can be almost perfect. And the word itself gives the game away — "semi-conservative" means "keeping", because the old half is kept all the way through.
Why this course brings it up: In Lecture 4 he puts three models side by side on Slide 4 (semi-conservative, conservative, dispersive), so it is a concept to understand. There is only one discriminating point, and it is the one to memorise: after the first round you get hybrid molecules. A conservative model would give you an untouched parent molecule plus a completely new one; a dispersive model would give you two strands each containing a mixture of old and new material. The answer is always

semi-conservative, so do not answer conservative. **Appears in:** Ch 5, Lect 4 **Video:** V13_DNA_replication.mp4

▪ Senescence

In one sentence: The state in which a cell stops dividing but does not die.
Detailed explanation: When the telomere shrinks to a critical length, the cell stops dividing and enters senescence, and that is the Hayflick limit. It is not the same as apoptosis: the cell is not cleared away, it stays in the tissue, but it no longer proliferates. The difference matters in cancer, because an old cell that has stopped dividing can still be pushed past the brake by a runaway signal. The most direct example is p53: about 40 % of cancers carry a p53 mutation, which is taking that brake off. So senescence is not the end of the line, it is a working arrest, and a cell in it is still alive and still sitting in the tissue.
Why this course brings it up: In both Lecture 3 and Lecture 4 he treats telomere shortening, then Hayflick limit, then senescence, then the tumour switching telomerase back on, as one causal chain, so it is concept-only, and it is the passage in Lecture 4 where he uses one continuous mechanism to show how the same thing runs from ageing all the way to cancer. What he explicitly excludes is the percentage of TERT promoter mutations, not this chain. So the chain is in and one number on it is out. **Appears in:** Ch 5, Lect 3

▪ Sensitivity

In one sentence: Of the patients who really have the disease, the proportion the test correctly picks out.
Detailed explanation: Sensitivity is how many patients the test finds. The denominator is everybody who really is ill: out of 100 real patients, if the test calls 98 of them positive, the sensitivity is 98 %.
The question it answers is how many patients this test will miss, and the ones it misses are the false negatives, which is the group whose treatment gets delayed. The other way round, the denominator of specificity is everybody who really is well, and it answers whether healthy people get called patients. His miRNA classifier gives exactly this pair of numbers: sensitivity 98.25 %, specificity 91.53 % and accuracy 94.83 %. Sensitivity is therefore a statement about the sick people, specificity is the mirror-image statement about the well ones, and you need both of them before any single number means anything. The way they trade off against each other is what the ROC curve draws.
Why this course brings it up: In the biomarker section of Lecture 1 Dr. Wong uses these four numbers to teach students how to read a ROC curve, and that is the language the short "data analysis" question is written in. He does not list sensitivity on its own as an exam question, but it is the skeleton of case study 11, the MMR and MSI result, and of every diagnostic question: the wording about missing patients has to be exact, and the distinction between a false negative and a false positive is where the marks are. **Appears in:** Ch 11, Lect 1

▪ Sensors

In one sentence: The level of the DDR signalling chain whose job is to discover the damage.
Detailed explanation: His slide 23 splits the DNA Damage Response into four parts. Signals are the things that hurt (DNA damage, replication stress). Sensors sit on the damaged DNA and are the ones that notice it.
Transducers pass the message on. Effectors carry it out (he names p53 and BRCA), and the chain ends in four possible outcomes: stop the cell cycle, repair, senesce, or start apoptosis. The four-part chain is easy to remember because it is the cell's general routine for dealing with any problem at all: notice, tell somebody, do something, and then decide whether to stop, repair, senesce or die. The sensor is the bit that makes the difference between a problem that is noticed and one that is not.
Why this course brings it up: He does not name Sensors as an exam topic on its own, so it is a concept to understand; the four-part chain of the DDR is the framework he has actually demonstrated. Keep the four parts apart and the individual terms are easy to place. **Appears in:** Ch 6, Lect 4

▪ Septic shock

In one sentence: The collapse that follows once sepsis overwhelms the body's own inflammatory response.
Detailed explanation: The pathological order runs like this. LPS from a Gram-negative bacterium enters the blood → it is carried by LBP (LPS-binding protein) and handed to CD14 (the LPS co-receptor) and MD-2 (the LPS-binding subunit), which present it to TLR4, and TLR4 dimerises → two branches, MyD88-dependent and MyD88-independent, both open NF-κB → a large amount of inflammatory cytokines (TNF-α first, peaking at 4–6 hours) → the blood vessels dilate everywhere, the blood pressure collapses and the organs are under-perfused.
TNF-α and IL-1β then go back and strengthen NF-κB, forming a positive-feedback loop that runs the response out of control. So the same mechanism that is meant to defend you against a bacterium ends up dilating your blood vessels until the organs are starved of blood. The name septic shock is doing honest work there, because it is a shock: the circulation fails, and what kills the patient is not the bacterium but the body's own response to it.
Why this course brings it up: On Slide 21 of Lecture 1 Dr. Wong points out that LPS is both an endotoxin and a pyrogen, and that an excess of endotoxin in the blood causes septic shock, a common cause of death in the surgical intensive care unit. What is examinable is the ordered chain you can reason through: LPS → TLR4 → IKK → IκB → NF-κB → into the nucleus → cytokines, and his revision notes list that cascade in order as a possible multiple-choice stem. **Appears in:** Ch 1, Lect 1

▪ Serial deletion

In one sentence: You delete the promoter piece by piece and watch which deletion makes the function collapse, which tells you which part was actually doing the work.

Detailed explanation: The method is very plain. You attach a reporter gene and take one reading. Then you delete another stretch from one end of the promoter and take another reading. You keep going, and out of this you get a curve showing how much can be deleted before the function goes.

The place where the curve suddenly collapses is where the key functional element sits; the sequences at either end, which make no difference when they are removed, are the "delete it and nothing happens" regions. That is exactly how he found out which TATA box variant is the effective one: he put the candidate promoter in front of luciferase, deleted pieces of the sequence away, and read off the number for each generation.

Why this course brings it up: In Lecture 5 he uses the reporter assay to demonstrate how you locate the functional elements of a promoter experimentally, so it is a concept to understand. He does not list it as an exam topic, but a short question may ask about the logic of the experiment, so you need to be able to say "delete a segment, read the value, find the key sequence" rather than memorise the step-by-step details. **Appears in:** Ch 7, Lect 5

▪ Serum starvation

In one sentence: Dropping the serum in the medium from 10 % to 0.5–1 % so that the cells stop in G1.

Detailed explanation: FBS, fetal bovine serum, is the nutrient source in the medium that feeds the cells, and it contains growth factors. Take those away and the cells lose the signal to keep dividing, and they settle quietly in G1.

Put the serum back and every cell leaves G1 at the same moment and enters S phase. The whole procedure is called cell-cycle synchronisation, and the effect is to give a replication experiment or a drug experiment a clean starting line: at time zero every cell is in the same place, so anything you see later must have been caused by the experiment. It is the same idea as starting a race with everybody on the same mark, rather than letting the fast ones set off early and having to guess afterwards who was ahead.

Why this course brings it up: He teaches this trick in both Lecture 2 and Lecture 4, and the 0.5–1 % FBS number is on the key numbers list in both, so it is a number you have to know. The reason it is worth knowing is exactly that it makes the experiment clean. He does not ask you to memorise the cyclin and CDK pairing table, he says you don't need to memorize all this cyclin or cyclin-dependent kinase, CDK. So the number is in and the table is out. **Appears in:** Ch 8, Ch 11, Lect 2, Lect 4

▪ sheath fluid

In one sentence: The fluid in a flow cytometer that squeezes the sample stream into a single thin line.

Detailed explanation: The instrument injects fast-moving sheath fluid from the side, pinching the central sample stream and flattening it to a very thin thread, so that the cells have to go through the laser one after another.

Without this trick two cells passing at the same time would give overlapping signals and the counting would immediately go wrong. The ratio of the sheath flow rate to the sample flow rate sets the degree of focusing, and it is the parameter most often adjusted on the instrument.

Why this course brings it up: Dr. Wong does not name sheath fluid as an exam topic on its own, so it belongs to conceptual understanding. Hydrodynamic focusing belongs to the same section, and it is what makes the single-cell analysis possible. **Appears in:** Ch 11, Lect 2

▪ side scatter (SSC)

In one sentence: The scatter channel on a flow cytometer that reflects how complicated the inside of a cell is.

Detailed explanation: When light hits a cell it scatters forward, which reports size, and it also scatters to the side, and the bigger the side-scatter angle the more cluttered the inside of the cell and the more granules it has.

His demonstration plot puts granulocytes at high SSC, monocytes in the middle and lymphocytes at low SSC. The two scatter axes together are like a table of weight and luggage for a cell: one dimension is size, the other is internal complexity, so you make a rough classification first and then go to fluorescence to make a finer one. It is a crude measurement, but it is cheap, it needs no staining, and it often separates the populations well enough to save you from running the whole panel.

Why this course brings it up: FSC equals size and SSC equals granularity is Tier 2 flow cytometry basics on his list. So the two letters and the two meanings are worth pairing up. **Appears in:** Ch 9, Ch 11, Lect 2

▪ Signal peptide

In one sentence: The address label on the N-terminus of a protein that gets it delivered to the ER.

Detailed explanation: When a new protein has only just emerged from the ribosome with its first few amino acids showing, the signal peptide is recognised by SRP, the whole ribosome is taken to the membrane of the rough ER, the signal peptide is fed into the channel, and the protein is dragged into the ER lumen to fold.

From that moment it is on the secretory pathway: ER, then Golgi, then the cell membrane or outside the cell. Without this label the protein is made perfectly well and goes nowhere, and the cell treats it as rubbish and gets rid of it. The tag also decides the route, because a protein without it has no place on the secretory pathway and no reason to go there. Which is why a protein that is made but never delivered is simply destroyed, and the experiment looks like it never worked at all rather than like it failed in an informative way.

Why this course brings it up: Dr. Wong emphasises it in person in Lecture 2, and his words are "So remember, signal peptide is really, really important for your study", because his postdoc left it out and lost four months over it. It is listed as a second-level

item. He also wants you to keep the signal peptide apart from the N-terminal domain, NTD: the first one delivers the protein into the ER and the second is the regulatory region of a nuclear receptor, and that pair is one of the distractor contrasts on his list.
Appears in: Ch 2, Lect 2

▪ Signal transduction

In one sentence: Carrying a signal from the cell surface all the way to somewhere inside the cell.
Detailed explanation: Signal transduction is a relay race, and there are four common ways it is run. The first is a phosphorylation cascade: once the receptor is activated, one kinase phosphorylates the next kinase, and the number of activated molecules grows at each step.
The second is a change of shape: the ligand binds the receptor, the receptor changes shape, and a binding site that was hidden is now exposed. The third is cleavage: like Notch, a protease cuts out an intracellular domain that then goes into the nucleus.
The fourth is release: a receptor on the membrane has its tail cut off, and the tail carries the transcription factor into the nucleus.
Why this course brings it up: In Lecture 5 Dr. Wong draws the seven routes side by side under “the seven ways a transcription factor is switched on”, and that is one of the few slides in the course that gathers mechanisms in one place, so it belongs to conceptual understanding. He also says plainly that the detailed signal transduction cascade on the slide does not need to be memorised: “I don't memorize all this thing, I just highlight here in case some student are really interested in molecular biology”. What has to be grasped is the classification of the concepts, not every step. Four routes, four different answers to the same question of how a message gets from outside the cell to a gene. **Appears in:** Ch 7, Lect 5

▪ Silencer

In one sentence: A stretch of DNA that turns transcription down, and it only works on the DNA molecule it happens to be sitting on.
Detailed explanation: A silencer is a cis element. It can sit upstream of the gene, downstream of it, or even inside an intron; it can be thousands of bases away from the promoter; and it works by using proteins that bend DNA to pull a repressive complex down onto the promoter.
In other words the silencer does not block anything by itself — it is a landing place, and what it does is bring the right proteins to the right stretch of DNA. The crucial point is the range of its action, which is limited to the one DNA molecule it is attached to — and that is exactly the difference between it and a trans-acting factor. One stays where it is and affects its neighbours; the other can diffuse away, even onto another chromosome. Slide 15 of Lecture 5 also lists a second mechanism for repression: a repressor binding directly to the DNA-binding domain or the activation domain of an activator, so that the activator can no longer do its job. Either way the result is the same: transcription goes down. So when you are asked whether a DNA sequence is cis or trans, the question you are really being asked is how far its influence can travel.
Why this course brings it up: Cis versus trans is one of the distractor contrasts he lists, and the discriminating point is that a cis element only acts on the DNA molecule it is on, whereas a trans factor can diffuse and can cross between chromosomes — his example is a repressor on chromosome 1 going to act on a target gene on chromosome 13. He also reminds you that not every enhancer is upstream, and the same goes for a silencer. **Appears in:** Ch 7, Lect 5

▪ Silent mutation

In one sentence: A mutation that changes a base without changing the protein.
Detailed explanation: Because the genetic code is degenerate, some base changes still specify the same amino acid, and the sequence and the function of the protein are both unchanged. It belongs with missense, where one amino acid is replaced, and nonsense, where a stop codon is manufactured, as the third member of the same family, and it is the set that beginners mix up most often.
The name says what happened, nothing happened to the protein, and that is the whole of the definition.
Why this course brings it up: Dr. Wong does not name silent mutation as an exam question on its own, so it is conceptual understanding, and it is the same family of classifications, point mutation, frameshift and in-frame, that is the focus of Chapter 4 and Lecture 4. **Appears in:** Ch 3, Lect 3

▪ siRNA

In one sentence: Small interfering RNA, a short double-stranded RNA used to switch off a specific gene.
Detailed explanation: siRNA is cut out of a longer precursor by the cell's own Dicer, and then it enters the RISC complex, where one of its two strands serves as the template for pairing with the target mRNA so that it gets degraded or cannot be translated.
The difference from miRNA is mainly in where it comes from and how specific it is: most siRNA is exogenous and works on a sequence chosen by hand. Its clinical use is exactly “switch off this gene”. In the past you did that with RNAi, and now most of the time you use a CRISPR knock-out instead. The tool changed and the aim did not, which is the general pattern for most of the tools in this course: the method is replaced and the question is not.
Why this course brings it up: Dr. Wong does not name siRNA as an exam topic on its own, so it belongs to conceptual understanding. The trend he mentions is that CRISPR has largely replaced the old siRNA and shRNA knockdown methods.
Appears in: Ch 7, Lect 1, Lect 5

▪ Sister chromatid

In one sentence: The two identical copies of one chromosome that exist after DNA replication.
Detailed explanation: Before S phase a chromosome is a single chromatid. After replication it becomes two identical sister chromatids held together by the centromere. They separate in M phase and each becomes a chromosome of a daughter cell.
The sister chromatid is the only template HR, homologous recombination, can use, because only it guarantees that the sequence is exactly the same, and that is what lets HR repair without error. It is also why HR can only be done in S and G2. Everything else about HR follows from that one point: the template has to exist, and the template only exists for a window of the cell cycle. The centromere is what holds the two together, and it is also where the spindle microtubules grab, which is why the two sister chromatids move as a pair until they are deliberately separated.
Why this course brings it up: In Lectures 2 and 3 he uses the sister chromatid for two points: the crossing over between homologous chromosomes in meiosis, on Slide 40 of Lecture 3, and the M phase separation after DNA replication, on Slide 17 of Lecture 4. It is the premise for understanding the contrast between HR needing a template and NHEJ not needing one, and that contrast is one of the most likely multiple choice pairings he lists. So the sister chromatid is the shared object that links two of his topics. **Appears in:** Ch 3, Ch 8, Lect 3

▪ Small molecule

In one sentence: A chemical molecule the size of a drug, usually made by chemical synthesis rather than by biology.
Detailed explanation: The definition of a small molecule is not as casual as "a small molecule", it is an operational line: usually a molecular weight below about 500 Da, well absorbed by mouth, and able to get into the pocket of a receptor or the active site of an enzyme.
The advantages are that it is cheap, that it can be made in large amounts, and that it is easy to design from the structure. The weakness is poor selectivity, so it tends to hit several targets at once. By contrast, an antibody, a biologic, is much larger and highly selective, but it usually has to be injected and cannot be taken by mouth. The 500 Da line is a practical line drawn by what the body will let through and what a drug design can actually reach, not a definition taken out of a textbook. So the two categories of drug have opposite virtues and opposite inconveniences, and the choice between them is a decision about the biology of the target.
Why this course brings it up: In Lecture 4 Dr. Wong uses it as the test of whether a target is druggable, and he also uses it to explain the logic of an ADC: you take a small molecule such as a topoisomerase inhibitor, attach it to an antibody, and it is delivered only to the tumour. Mind the proportions, though, because he explicitly rules the structure of an ADC and the dual-payload design out of the examination: "all this would not be included in the midterm. I'm just want you to know more". **Appears in:** Ch 10, Ch 12, Lect 4

▪ Smooth ER

In one sentence: The part of the endoplasmic reticulum with no ribosomes attached, and it makes lipids.
Detailed explanation: The smooth ER has no ribosomes and it synthesises lipids (phospholipids and steroids), metabolises drugs and does detoxification, and it also stores and releases calcium ions.
The division of labour with the rough ER is very clean: the rough ER makes proteins and the smooth ER makes lipids, and the cell membrane itself is a phospholipid bilayer, so when Dr. Wong says the cell builds its own membrane, what he means is mainly the smooth ER. The PI at the start of the PI(4,5)P signal transduction pathway in Lecture 5 also sits on the membrane.
Why this course brings it up: On Slide 22 of Lecture 2 he gives it a single sentence: “Smooth ER makes phospholipids.” It belongs to conceptual understanding. He does not examine the organelles one by one in separate chapters, and the cell-as-factory metaphor is the real point. What you need is to be able to work backwards from “the membrane is made of lipids” to the job of the smooth ER, and to keep the division of labour straight: proteins come off the rough ER, lipids come off the smooth ER, and the cell membrane is built out of the second of those. **Appears in:** Ch 8, Lect 2

▪ SN-38

In one sentence: A topoisomerase-I inhibitor used as the cytotoxic payload of an ADC.
Detailed explanation: SN-38 is a topoisomerase I inhibitor. It stops the knots that the DNA replication fork unwinds ahead of it from being tied up again, so double-strand breaks pile up in the DNA and the cell dies.
The advantage of using it in an ADC is selective delivery: on its own it would kill every dividing cell, while attached to an antibody it is delivered only to the target tumour. The clinical targets of ADCs are HER2 and TROP2, and the drugs he names are trastuzumab-deruxtecan and datopotamab-deruxtecan. One related fact: SN-38 is also the active metabolite of the small molecule drug irinotecan, which means the same molecule can appear in two completely different forms, a free small molecule and an antibody-linked payload. That part is background knowledge, because irinotecan does not appear in his five lectures. So the payload is doing exactly what a chemotherapy drug does, and the antibody is doing the work of deciding where that happens.
Why this course brings it up: He uses SN-38 in both Lecture 1 and Lecture 4 to explain the design rationale of an ADC, and why you would put a topoisomerase inhibitor inside an ADC is one of the short scenario questions in his case bank. The answer you should be ready with is selective delivery. What he explicitly excludes is the structure of the ADC and the chemistry of the payload, so do not memorise structures. The point of the question is the design principle, not the molecule, and the structures are not examinable. **Appears in:** Ch 10, Lect 1, Lect 4

▪ snRNP

In one sentence: A short piece of nuclear RNA plus its proteins; several of them together make up the splicing machine.
Detailed explanation: snRNP is small nuclear ribonucleoprotein: a complex formed when snRNA (small nuclear RNA) is bound together with protein, and its job is to recognise specific signals inside an intron.
One snRNP on its own is not much; it is the combination of several of them, and the spliceosome they build together, that carries out the cut. Slide 33 of Lecture 5 draws splicing in three steps: several snRNP bind onto the intron first → the intron coils into a loop that pulls the exons on either side close together → the intron is released and the snRNP are recycled and used again, so every round of splicing draws on the same pool of parts. The snRNP are the parts of the spliceosome, and the sites they recognise are what decide where splicing happens — which is precisely the molecular starting point of alternative splicing.
Why this course brings it up: He uses it when he introduces the spliceosome in Lecture 5, so it is a concept to understand and not an exam topic. He does not ask you to memorise the types and numbers of snRNP; but there is one thing to grasp: splicing is not a random cut. It is done by snRNP finding the site and the spliceosome assembling around it, so a tiny change in the sequence can create a new splice site — with a single base change in a liver cancer gene he produced a new isoform, and the patients survived significantly worse. **Appears in:** Ch 7, Lect 5

▪ Sodium bisulfite

In one sentence: A reagent that turns unmethylated cytosine into uracil and leaves everything else alone.
Detailed explanation: The chemistry of bisulfite is clever: it reacts with cytosine, but only on the condition that the cytosine has not been methylated. Unmethylated C therefore becomes uracil, which reads as thymine when the DNA is sequenced; methylated C is left completely untouched. Compare the two sequences position by position afterwards and you can draw the methylation map for that sample — and that is why bisulfite sequencing became the gold standard. The selectivity is the whole key to the method: if bisulfite changed methylated cytosine as well, you could no longer tell which positions were methylated to begin with. One reagent, one chemical question asked of every position in the genome, and the answer is read off by seeing what changed and what did not. It is a roundabout way of asking, but it is asking a question no other method asked so cleanly.
Why this course brings it up: "Bisulfite sequencing = gold standard" is one of the facts he marks as Tier 2; the chemical logic of "unmethylated C becomes U, methylated C survives" he explains very clearly. That logic is the part you should be able to redraw from memory, because it explains every step of the method. **Appears in:** Ch 4, Ch 11, Lect 3

▪ solid-tumour barriers

In one sentence: The collection of obstacles that make CAR-T fail in a solid tumour.
Detailed explanation: His slide 63 lists a whole sheet of them: inhibitory checkpoints, CTLA-4, PD-1, LAG3, TIM3 and VISTA on the effector cell side and PD-L1 and PD-L2 on the target cell side;
missing chemokine receptors, CCR2b and CCR4; suppressive cells, Treg, MDSC and TAM; signals, TGFβ and IL-10; an ECM rich in HSPG; and the unfriendly conditions of hypoxia, low pH, CD47, CD73, adenosine and IDO. The clinical meaning of the list is very direct: CAR-T works against blood cancers because the environment around the tumour cells is relatively simple, while a solid tumour presents a complicated environment, so several barriers have to be dealt with at once. And that is why the synNotch and HIF-CAR designs exist, because they are attempts to move part of the decision making into the tumour cell itself, where the information is. Every entry on that slide is a place where the immune response can be blocked, and each of them was worked out separately, which is why the list looks so long.
Why this course brings it up: The list itself is the teaching content of this passage, and he does not ask you to memorise the individual molecule names on it. So know the categories, not the roll call. **Appears in:** Lect 2

▪ Solvent

In one sentence: The liquid that other things dissolve in; for a cell, that liquid is water.
Detailed explanation: A solvent is the medium that something else dissolves in. The cell is an aqueous environment: proteins, sugars, nucleic acids and ions can only work because they are dissolved in water.
Three features make water especially suitable. It is a polar molecule, so it dissolves polar salts and molecules. Its network of hydrogen bonds gives it a high specific heat and a high boiling point, which is why body temperature stays steady. And its density means ice floats, so the bottom of a lake stays liquid through the winter. How well something dissolves is what decides whether the biochemical reaction can happen at all.
Why this course brings it up: Dr. Wong does not list solvent as an exam topic, but it sits alongside his four structural reasons for water (solvent, participant, the shaping role of hydrogen bonds, high specific heat). It is conceptual understanding: the solvent sets the rules of the chemistry inside the cell, and it connects directly to the DNA damage and repair mechanisms in Lecture 4. **Appears in:** Ch 1

▪ somatic recombination

In one sentence: Somatic recombination, which is the source of antibody diversity.
Detailed explanation: In a developing B cell the gene segments that make the antigen-binding part are shuffled at random and joined together, and every B cell draws a different combination. The whole set of segments, V, D and J, together with RAG1 and RAG2, the recombination-activating proteins, and the machinery of NHEJ, produces billions of different antibodies, but it happens in the somatic cells and it is not passed to the next generation.
That last point is the reason the mechanism works without any consequence for the species, because the rearrangement happens after the animal has already been conceived. Which is why the word somatic is doing real work in the name, and why

the mechanism is called somatic recombination rather than simply recombination.
Why this course brings it up: "Somatic recombination produces billions of antibodies" is a mechanism he has taught, and "NHEJ deficiency gives SCID" is the first-grade question he names. **Appears in:** Ch 6, Lect 1, Lect 4

▪ Specificity

In one sentence: Of the people who really are well, the proportion the test correctly clears.
Detailed explanation: The denominator of specificity is everybody who really is well. Out of 100 healthy people, if the test calls 92 of them negative, the specificity is 92 %, and the other 8 are called positive, which makes them false positives, and false positives bring unnecessary biopsies, unnecessary tests and unnecessary panic.
It and sensitivity are a pair of complementary mirrors: the first one handles "will it turn a healthy person into a patient", the second handles "will it miss a patient". The clinical trade-off is that screening uses a high sensitivity to protect against missed cases, while diagnosis uses a high specificity to reduce the damage done to healthy people. The figure he gives for the miR-15b plus miR-130b classifier is a specificity of 91.53 %. And the two mistakes are mirror images, so a test can be very good at one and poor at the other without anything being wrong with the test. That is why the cutoff you choose on the ROC curve decides the balance, and why the AUC on its own does not tell you which patients the test will get wrong.
Why this course brings it up: In Lecture 1 Dr. Wong uses this pair of numbers to teach students to read a ROC curve, and that is mandatory language for the short "data analysis" question. He does not list specificity on its own as an exam question, but it is the skeleton of every diagnostic question, and it sits in the same framework as his ROC and AUC work, where the AUC is 0.981. **Appears in:** Ch 11, Lect 1

▪ Spliceosome

In one sentence: The machine that cuts the introns out and joins the exons together.
Detailed explanation: The spliceosome is a nuclear complex built from snRNPs, and it is bigger than a ribosome. Its working sequence is this. The snRNPs recognise five key positions in the intron one after another and bind them.
The intron rolls up into a loop, which pulls the exons on either side of it together. The 2′-OH of one exon then attacks the branch-point phosphate of the intron, a transesterification reaction, the two exons are joined by a new phosphodiester bond, and the intron is released as a lariat and degraded. A spliceosome that assembles wrongly turns straight into disease, and the commonest example is DMD. So the reaction does not cut anything and rejoin it. It is one bond being made and another being broken, and that is why the exons end up chemically linked rather than merely held next to each other.
Why this course brings it up: In Lecture 5 he explains splicing with a three-step diagram, so it is concept-only, and he does not ask you to memorise the kinds of snRNP. But you do have to be able to derive the consequences of splicing: cutting out a drug-binding domain causes drug resistance, and a spliced isoform that is new to the immune system is a neoantigen. In his liver cancer example, patients with abnormal splicing had significantly worse overall survival, $P < 0.001$. **Appears in:** Ch 7, Lect 5

▪ Splicing

In one sentence: Cutting the introns out of pre-mRNA and joining the exons together to make mature mRNA.
Detailed explanation: A eukaryotic gene is not one continuous coding stretch. Exons (the parts that are kept) and introns (the parts that are cut out) alternate along it, and both are transcribed together into pre-mRNA, which is then processed by the spliceosome.
The key word is “alternative”. Alternative splicing lets the same gene keep different combinations of exons, so one gene can make several protein isoforms. The Lecture 5 cases include MCL1 (an apoptosis regulator: the short isoform promotes apoptosis, the long one prevents it), Bcl-x (an apoptosis regulator: L prevents apoptosis, S promotes it) and AR-V7 (constitutively active in structure, which drives resistance to enzalutamide). So in the first two cases the choice of isoform decides whether the cell dies, and in the third it decides whether the drug works. The gene does not change in any of them; what changes is which part of it gets used. That is the neat part of splicing as a mechanism: the cell gets variety without touching the DNA at all, and the variation is produced at the level of the RNA. So the same transcript can carry different instructions depending on which segments the spliceosome decides to keep, and the cell can shift the balance between isoforms when it needs to.
Why this course brings it up: In Lecture 5 Dr. Wong makes alternative splicing one of the main threads of cancer biology: aberrant splicing produces neoantigens that the immune system can see, while cutting away a drug-binding site produces drug resistance. His case library lists “a certain isoform predicts resistance” as one of the scenario short questions, so you have to be able to answer with the logic of “drug-binding domain cut out → resistance”. **Appears in:** Ch 7, Ch 9, Lect 5

▪ Splicing factor

In one sentence: A protein that helps decide which exons get joined together.
Detailed explanation: A splicing factor is a protein that recognises a sequence and regulates the choice of splice site. It can be promoting or inhibiting; it can also be a member of the RNA-binding protein family — and that is the role PNCTR plays on the lncRNA slide of Lecture 5: PNCTR uses its YYUCUY motif (the sequence that splicing factors recognise) to bind splicing factors of the PTBP1 kind, and thereby control splicing.
An lncRNA has thus changed the isoform of a protein indirectly, by pulling the splicing machinery around rather than by doing any cutting itself. The motif is the handle; the splicing factor is the hand that grips it, and change the handle and you change what gets gripped.

Why this course brings it up: In Lecture 5 he connects lncRNA with splicing control, so it is a concept to understand. He does not ask you to remember the name of any splicing factor; what you need is the mechanism: the choice of splice site is made by proteins reading the RNA sequence, so if you change the splicing factors or the RNA anchors that hold them, you change the isoform. **Appears in:** Ch 7, Lect 5

▪ Starch

In one sentence: The polysaccharide that plants use to store sugar, the stored form of glucose.
Detailed explanation: Starch is a large molecule built from glucose, and two structures exist side by side in it: amylose, which is a straight chain held by α1,4 bonds, and amylopectin, which is branched, with α1,6 branch points appearing at intervals along the α1,4 chain.
The two are not different molecules that happen to occur together, they are two kinds of glucose polymer inside the same granule, and the difference between them is the branching. Humans have amylase, an enzyme that breaks starch down, so starch is an important dietary source of carbohydrate for us, while the plant uses the same polymer to keep energy in reserve. The human storage version is glycogen, which structurally resembles the branched amylopectin even more closely and is branched more densely, so it can be mobilised faster.
Why this course brings it up: In Lecture 1 Dr. Wong divides the carbohydrates into monosaccharide, disaccharide and polysaccharide, and starch is one of the representatives of the polysaccharide group. It is conceptual understanding, and when he teaches the glycogen slide he draws the branching structure and the enzyme action together: glycogen phosphorylase, and glucose-6-phosphatase, which only the liver and the kidney have, and he links it to the pathological example of glucose-6-phosphatase deficiency causing an energy crisis in the liver cell. The three of them, the branching, the two enzymes and the disease, all come from the same slide, so learn them together. **Appears in:** Ch 1

▪ Start codon

In one sentence: AUG: the signal that starts translation, and at the same time the code for methionine.
Detailed explanation: The ribosome scans down the mRNA from the 5′ end, stops at the first AUG it meets, and begins hooking on amino acids there. Because AUG codes for methionine, the first amino acid of a new protein is usually methionine.
The start codon is recognised with the help of initiation factors rather than the ribosome simply guessing on its own, and that is also why bacteria are allowed a handful of non-AUG start codons. The ribosome is a machine that does not decide for itself where to begin; it is told where to begin, and then it goes. Everything after the start is ordinary elongation, and the codon boundaries are set from that first AUG.
Why this course brings it up: On the central dogma diagram of Slide 43 in Lecture 1 he marks out Start codon AUG (Methionine) and Stop codons UAA, UAG, UGA, and that is basic groundwork to have ready at any moment. He does not list it as a standalone exam topic, but the frameshift story depends on it: when an insertion or a deletion is not a multiple of 3, the reading frame shifts, and it usually runs into a stop codon quite quickly, producing a truncated protein or an early finish. **Appears in:** Ch 1, Ch 2, Lect 1

▪ STAT3

In one sentence: A transcription factor that, on his Lecture 3 slide, is what pushes PD-L1 expression up.
Detailed explanation: STAT3 is a transcription factor that sits downstream of cytokine signalling. But on the Slide 50 diagram in Lecture 3, the one with HDACi and the immune checkpoint inhibitors, its role is this: HDAC6 activates STAT3, STAT3 binds the promoter of PD-L1, PD-L1 expression goes up, and the tumour cell finds it easier to switch off the T cell with PD-L1.
The matching therapeutic logic is to use an HDAC6 inhibitor, Nexturastat A for example, to push STAT3 down, which reduces PD-L1, and the tumour becomes sensitive again to anti-PD-L1. The whole diagram is a chain of four steps, and each step is a place a drug could be inserted. The surprising part of the diagram is that the first two steps are not immunotherapy at all. The HDAC inhibitor does not attack the tumour directly; it changes what the tumour is willing to display, and the antibody then does the attacking.
Why this course brings it up: In Lecture 3 he calls this one of the most important figures in modern medicine, because any symposium on checkpoint inhibitors will put it up. He does not list STAT3 as a question on its own, but the concept of gene addiction has to be understood: an HDAC inhibitor induces PD-L1, and the tumour therefore becomes more sensitive to anti-PD-1 and anti-PD-L1. That is one of the routes by which a cold tumour is reworked into a hot one. The surprising part, and the part worth remembering, is that the sensitising drug is not the immunotherapy. The four-step chain is what you would draw in an answer. **Appears in:** Ch 4, Lect 3

▪ Stem cell

In one sentence: A cell that can copy itself and can also turn into a more specialised cell.
Detailed explanation: A stem cell has two abilities: self-renewal, and differentiation into more specialised cells. Dr. Wong's classification is ESC (embryonic stem cells, pluripotent and able to self-renew indefinitely), adult stem cell (haematopoietic stem cells, for example, which are multipotent but limited), and iPSC (pluripotent cells reprogrammed from a somatic cell).
The degree of differentiation is a continuum: totipotent → pluripotent → multipotent → unipotent. He stresses that the value of an iPSC is that it solves two problems at once, getting around the ethical argument over embryonic stem cells and allowing disease models to be made, so you take the patient's own cells, turn them into iPSCs, differentiate them into cardiomyocytes and test the heart toxicity there. It is worth being clear about what the pluripotency label means here, because it is the whole point: the cell

has been pushed back far enough along its development to be able to become almost any cell type, and that is a different thing from being able to become every cell type equally well. The continuum from totipotent down to unipotent is really a continuum of how many doors are still open, and each step down closes a few more of them. An unipotent cell has one door left, an ESC has almost all of them.
Why this course brings it up: In Lecture 2 he describes iPSC as one whole verification chain: morphology (fibroblasts becoming embryo-like cell clusters), surface markers, and PCR/flow cytometry, ending with the teratoma assay as the gold standard. It is conceptual understanding. There is also a stem-cell QC point he reminds you of himself: “unique expression pattern and timing, and also the negative staining to exclude the contaminated cell. Don't forget it.” **Appears in:** Ch 8, Lect 2

▪ sticky end

In one sentence: The single-stranded overhang left behind when a restriction enzyme cuts.
Detailed explanation: A restriction enzyme cuts DNA at a specific sequence and leaves behind not a blunt cut but a short single-stranded overhang, like the sticky tongue on a piece of tape. Because the sequence of that tongue is fixed, the tongue of one piece of DNA and the groove of another will stick together as long as they are complementary, and molecular cloning uses exactly this to splice an insert into a plasmid.

Why this course brings it up: Dr. Wong does not name sticky end as an exam topic on its own, so it belongs to conceptual understanding. It is the physical detail that makes the cloning story make sense. **Appears in:** Ch 3, Lect 4

▪ Stop codon

In one sentence: UAA, UAG and UGA, where translation stops.
Detailed explanation: A stop codon is a codon with no amino acid to match it, and when the ribosome meets one it stops extending and hands the finished protein over to a release factor to close the book.
The mechanism is worth spelling out, because it explains why the release factor matters: nothing about the codon itself stops the ribosome, the release factor is what recognises it. Remember them as UAA, UAG and UGA, all three of which contain U, uracil, while AUG is the start. That is also why the stop codons are recognisable as a family, three of them, and why AUG is worth keeping separate in your head, because it does the opposite job and is the only one that means methionine. Without a stop codon the protein simply keeps going. In the other direction, one of the common consequences of a frameshift is that the reading frame runs into a stop codon early and produces a truncated protein.
Why this course brings it up: On Slide 43 of Lecture 1 Dr. Wong puts it on the central dogma diagram together with Start codon AUG (Methionine), and that is the basic stuff you have to be ready for. What will really be examined is its clinical consequence: a frameshift mutation causes early termination, and the neoantigen that a frameshift produces is exactly the fuel of a hot tumour. MMR deficiency, then MSI, then frameshift, then more neoantigen, then a good response to anti-PD-1 is the chain that joins the two lectures together, and it is the kind of chain he wants you to be able to draw in one line. **Appears in:** Ch 1, Ch 2, Lect 1

▪ Strand

In one sentence: One of the two long chains that make up the DNA double helix.
Detailed explanation: The two strands run in opposite directions, and that is called antiparallel: one is 5′→3′ and the other is 3′→5′. The two strands are therefore not interchangeable; you always have to say which way you are reading.
Because polymerase can only extend in the 5′→3′ direction, the two strands can only be copied in two different ways: the leading strand is made continuously, and the lagging strand is made in pieces. The base-pairing rule is that G–C makes three hydrogen bonds and A–T makes two, so a region with a high GC content takes more energy to pull apart and has a higher melting temperature (Tm) — and that is one of the things you have to take into account when designing PCR primers. More bonds, more heat needed to break them, and therefore a higher temperature at which the two strands let go of each other. Everything that works on DNA has to keep track of which end it is at, and that direction is written into the chemistry.
Why this course brings it up: His own words are "I think one concept you'll need to understand is there's two different strands. So one is the leading strand, and the other one the lagging strand." And G–C = 3 hydrogen bonds, A–T = 2, and "the more GC, the higher the Tm" are all on his key numbers list. Antiparallel and the 5′→3′ direction rule are the most basic scaffolding in the whole course. **Appears in:** Ch 3, Ch 5

▪ Structural gene

In one sentence: A gene that makes the products the cell actually needs, like the enzyme that digests lactose.
Detailed explanation: In the structure of an operon, the structural genes are the ones that are transcribed into mRNA and then translated into functional proteins, such as the enzymes that metabolise lactose.
Against them stands the regulatory gene, which makes only the repressor protein and does no work of its own. What is transcribed is one polycistronic mRNA, which makes several different enzymes at once. The schematic on Slide 7 of Lecture 5 lays the whole chain out clearly: regulatory gene, then repressor protein, then operator, then promoter, then structural genes. The key word is polycistronic: one promoter, one message, several proteins, which is why the whole operon can be switched on and off as a single unit. And because the regulatory gene is separate, the switch can be tuned separately from the output, which is the same design trick as in the repressible system. The structural genes are the payload of the operon and the regulatory gene is the control panel.
Why this course brings it up: Keeping the three roles apart, regulatory gene, repressor, operator and structural gene, is the

precondition for reading the lac operon, and reading the logic of the lac operon is one of the short scenario questions in his case bank. The special point is that lactose present and glucose low have to hold at the same time, and satisfying only one of the two does not give strong transcription. That two-condition rule is where most people lose the marks. **Appears in:** Ch 7, Lect 5

▪ structural tractability

In one sentence: Whether this molecule can be attacked, given its shape.
Detailed explanation: To make a drug you have to be able to get near it: a pocket a small molecule can fit into, a surface an antibody can bind, or a sequence a piece of nucleic acid can pair with.
That is a different question from how important it is in disease, because a great many important molecules are important everywhere, and there is no way to attack only the copy that is causing the disease. So the two questions, important and attackable, can have completely different answers, and that is the practical point of the distinction.
Why this course brings it up: He does not name structural tractability as an exam question on its own, so it is concept-only, and he says outright that the whole target nomination workflow is not examined. The workflow is out and the distinction is in.
Appears in: Ch 12, Lect 4

▪ Structural variant

In one sentence: Large-scale rearrangements of the chromosome: deletion, duplication, inversion and translocation.
Detailed explanation: Unlike a point mutation, a structural variant moves large blocks of DNA around, and the size of the block is what separates it from the single-letter changes. On Slide 40 of Lecture 3 he quotes genome.gov and lists five kinds: deletion, duplication, inversion, substitution, where a block of one chromosome is inserted into another, and translocation, where two chromosomes exchange segments.
The most surprising of the five is the inversion. It does not change the total amount of genetic material, and it does not change the gene dosage, and yet it can move a gene away from its enhancer, or drag an enhancer over to a different promoter, so the level of expression changes dramatically and nothing else does. That is the case that surprises people, because the chromosome looks normal on a karyogram and the gene is still there, and the only thing that has moved is where it sits relative to the DNA that controls it. Which is why the inversion is the one to spend time on, because it is the case where the amount of DNA and the amount of gene product can both stay the same while the behaviour of the cell changes completely. His words are "you need to understand that some of these abnormalities can cause the loss of the genetic materials, but some does not."
Why this course brings it up: In Lecture 3 Dr. Wong teaches inversion and translocation as abnormalities of a different kind, and that is conceptual understanding, but it maps onto his scenario question nine: an inversion with no dosage change, then enhancer and promoter relocated, then expression changed. His words also say you have to be able to tell which abnormalities cause a loss of genetic material and which do not. The molecular mechanism of CNV he rules out explicitly, so you can leave the enzymology out and keep the picture. **Appears in:** Ch 3, Lect 3

▪ Substitution

In one sentence: One kind of structural variant: a piece of chromosome cut out and inserted into another chromosome.
Detailed explanation: The word has two uses and they have to be kept apart. First, the structural variant use. On Slide 40 of Lecture 3 Dr. Wong quotes the five structural abnormalities from genome.gov and defines substitution as “a piece of chromosome inserted into another chromosome”, listed alongside translocation (two chromosomes exchanging segments). Second, the point mutation use, where substitution means a single base replaced by another, and he uses it in the alternative splicing example in Lecture 5: “a single-nucleotide substitution creates a novel splice site”. The first moves large blocks, the second moves one letter. So the same word covers a rearrangement you could see down a microscope and a single letter you could only find by sequencing. If you are asked to define it in a scenario question, the sense that matters is the one in the paragraph you are actually looking at. Read the surrounding paragraph before you write the definition down, because the two senses point in different directions and only one of them fits the case in front of you.
Why this course brings it up: He does not list substitution as an exam topic on its own, so it is conceptual understanding. What to watch is that you read Slide 40 of Lecture 3 in the structural variant sense and the liver cancer splice site example in Lecture 5 in the point mutation sense. His reminder is worth keeping too: some structural abnormalities cause loss of the genetic material and some do not. **Appears in:** Ch 3, Lect 3

▪ Substrate

In one sentence: The thing an enzyme acts on.
Detailed explanation: The substrate is the side that gets changed by the enzyme; the enzyme is the side that does the catalysing. The two meet at a place called the active site, which is a pocket whose shape and chemical properties fit the substrate exactly.
Once the two have bound, the enzyme lowers the activation energy of the reaction — it lowers the energy barrier, so the reaction can happen without needing a high temperature. After the reaction, the substrate has become the product and the enzyme is released completely unchanged, ready to do it all again.
Why this course brings it up: In Lecture 3 he turns the mechanism of an enzyme into a self-test question in class, and the answer has to be given in terms of "the change from substrate to product", so it is a concept to understand. He does not ask you to memorise the details of any particular enzyme, and the metabolic pathway details are on the exclusion list too; what you really need is the vocabulary of the active site and the activation energy, because that is what explains why a target blocked by a drug stops the reaction dead. **Appears in:** Ch 2

▪ Subunit

In one sentence: One of the chains that go into a protein made of several chains.
Detailed explanation: A subunit is the unit that makes up a quaternary structure. An antibody is the easiest example: two heavy chains plus two light chains, four chains each with its own task, where the Fc end of the heavy chain talks to the immune system and the VH and VL ends of the two chains grab the antigen.
The ribosome is made of a large and a small subunit. The functional advantage is division of labour and control: different subunits can be regulated separately, and they can also evolve separately. So a protein can be built out of parts that are individually simpler than the whole.
Why this course brings it up: In Lecture 1 he teaches quaternary structure together with the structure of the antibody, so it is concept-only. The difference from a domain is worth keeping: a subunit is a different polypeptide chain, a domain is a stretch of the same chain, and you need that distinction when you read the structure of a CAR or of an antibody. **Appears in:** Ch 2

▪ succinimide

In one sentence: The short-lived intermediate formed when asparagine is deamidated.
Detailed explanation: The amide side chain of asparagine spontaneously loses an ammonia, becoming succinimide (Asu) first, and then takes up a water molecule to give aspartic acid or iso-aspartic acid.
His slide gives the masses: Asn 114.04 Da, succinimide 97.02 Da, while Asp and isoAsp are both 115.03 Da — same mass but different molecules, and that is precisely why deamidation is hard to tell apart on a mass spectrometer. So when you see one mass in the spectrum, both products may be sitting there together; separating them needs fragmentation analysis, not the mass on its own. The intermediate is short-lived, which is why you never catch it in the finished protein even though you know it was there. The whole pathway is a sequence of three masses, and each one tells you how far the reaction has got.
Why this course brings it up: He does not name succinimide as an exam topic on its own, so it is a concept to understand; what he does make a point of is the set of mass numbers, and succinimide is the middle one of the three. **Appears in:** Ch 1, Ch 2, Lect 1

▪ Sugar (carbohydrate)

In one sentence: One of the four big biological molecules, used mostly as fuel and also as building material.
Detailed explanation: Carbohydrates often have the formula CH2O, and they come as monosaccharides (glucose, fructose), disaccharides (sucrose) and polysaccharides (glycogen, starch, cellulose).
They have two main jobs: energy (one glucose gives about 36 ATP through glycolysis and oxidative phosphorylation) and structure (cellulose is the scaffolding of the plant cell wall). Carbohydrates are also the most important cell-surface recognition molecules, since blood groups are decided by carbohydrate chains on the cell surface.
Why this course brings it up: In Lecture 1 Dr. Wong uses the four families of biological molecules as the vocabulary frame for the whole course, and carbohydrates are the first family. Their clinical significance is not in the sugar itself but in the trade-off he describes with the Warburg effect, where cancer cells give up some ATP in order to keep half-digested sugar as building material. He says plainly that the metabolic pathway details are not examined, and what you need is this economics. The carbohydrate has to be seen as both fuel and raw material, and the cancer cell has chosen to be a builder rather than an athlete.
Appears in: Ch 1, Lect 1

▪ Sugar-phosphate backbone

In one sentence: The backbone of one strand of DNA, made of sugars and phosphate groups alternating along it.
Detailed explanation: Each nucleotide is one base plus one five-carbon sugar plus a phosphate group. The sugars and phosphates string themselves together alternately into the backbone, while the bases stick outwards, ready to pair with a base on the other strand.
Because the backbone is covalent and the bases point outward where they are free to pair, DNA is able to hold its two strands together with complementary bases. The phosphate group at the 5′ end and the hydroxyl at the 3′ end give the strand its direction, and that is where the names 5′→3′ and 3′→5′ come from. The backbone is the stiff part that holds everything in place; the bases are the part that carries the information. The bases are the letters; the backbone is the line they are written on.
Why this course brings it up: It is the physical source of the direction rule: elongation runs 5′→3′, proofreading runs 3′→5′, and ligase joins a 5′ phosphate to a 3′ hydroxyl. On Slide 6 in Lecture 3 he marks out antiparallel and the positions of the 5′ phosphate and the 3′ hydroxyl; this direction rule is one of the distractor contrasts he lists. **Appears in:** Ch 1

▪ surrogate endpoints

In one sentence: Using a cheap and easy measurement in place of the endpoint you really want.
Detailed explanation: The real endpoint of a trial might be how long people survive or how big the tumour is, but those need scanning, blood taking and waiting. A circulating biomarker such as CEA provides a substitute that can be measured often and that moves along with the tumour burden, so the trial can read out a response without scanning every participant over and over. For a surrogate to stand up it has to be shown first that it moves in step with the real endpoint, and CEA can be used because it broadly tracks the tumour load. Show that first, and then you are allowed to use the cheap measurement as a stand-in.
Why this course brings it up: Dr. Wong explicitly positions CEA as both a surrogate endpoint and a circulating surrogate marker, but he also warns that a marker is only a stand-in and you cannot make a diagnosis on it alone. **Appears in:** Ch 10, Ch 11, Ch 12, Lect 4

▪ Suv39h

In one sentence: The enzyme that spreads the H3K9 methylation along the chromatin.
Detailed explanation: Suv39h is a histone methyltransferase, and it adds a methyl group specifically at the ninth position of histone H3. It works together with HP1, which recognises the same mark, and the two of them form a “write it and recognise it” mechanism, so that the region of silence keeps growing.
Without a boundary, the K9 methylation would spread out until it reached genes that were supposed to be active, and Dr. Wong's slide 27 puts the boundary element and Suv39h on the same diagram precisely because that balance is the point. The spreading is what makes the silencing heritable from one cell division to the next, because a daughter cell inherits the marks that were already written.
Why this course brings it up: He does not ask you to memorise it, so it belongs to conceptual understanding. What is Tier 2 is the idea that the marks spread and that cellular memory comes out of that. **Appears in:** Ch 4, Lect 3

▪ synNotch

In one sentence: A synthetic Notch, so that the presence of a neighbouring cell switches on one of your own genes.
Detailed explanation: The mechanism of wild-type Notch runs like this: the delta ligand on the surface of a neighbouring cell meets the Notch on this cell, Notch is cut by proteolysis to release its intracellular domain, NICD, and the NICD moves into the nucleus to regulate genes.
synNotch replaces the part that recognises the ligand with a fragment of an antibody, one that recognises tumour antigen 1 for instance, so the question "does the neighbour have this antigen on it" is translated into "should I express the CAR". You can then use the same logic to test a second antigen, and the cell only kills when both conditions hold at once, and that is the AND gate. Because the readout is a gene, the whole thing is a switch you can wire into any gene you like, which is the point of building it that way.
Why this course brings it up: In Lecture 5 Dr. Wong presents synNotch as a representative application of synthetic biology, and that is conceptual understanding. It maps onto case fourteen of his case bank, where the compound switches itself off in the right cell, an HRE-controlled CAR-T, and that question is about a single gate. synNotch is two gates. He does not ask you to memorise the structure of synNotch, but the design idea of conditional expression is one you have to understand, because it is the idea the whole engineered CAR story rests on. **Appears in:** Ch 7, Lect 5

▪ Synthetic biology

In one sentence: Engineering new biological parts and switches.
Detailed explanation: Synthetic biology treats a biological system as an engineering problem you can design: you rewire the gene circuits so that the cell does things it would not otherwise do.
Both of his examples come from the CAR-T design work in Lecture 5. The HIF-1α-controlled CAR puts the expression of the CAR under a hypoxia response element, so only the hypoxic core of a tumour switches the CAR on, and his figures show that about 50 % of solid tumours contain hypoxic areas. The other is the synNotch AND-gate CAR. What they have in common is that the condition for when to fire is handed over to the cell to judge for itself. That is the shift in thinking: not a drug that decides where to act, but a cell that decides for itself.
Why this course brings it up: He puts it at the modern frontier of the whole lecture, so it is concept-only. He does not list it as a question, but his case number 14, a compound that switches itself off inside the correct cell, is asking exactly about this idea: a gate controlled by an environmental signal rather than by a doctor. So the case question is the exam for the topic. **Appears in:** Ch 7, Lect 5

▪ Synthetic lethality

In one sentence: Two defects that the cell can live with one at a time but not together.
Detailed explanation: Normal cells and cancer cells often each carry one weak point, but as long as one of them is still there the other can be supported. The drug works by removing both weak points at once.
A cancer cell with a BRCA mutation has already lost HR and can only rely on PARP for repair, and if you then give it a PARP inhibitor the replication forks stall and turn into double-strand breaks, while normal cells, whose HR is intact, still survive. That margin of safety is the therapeutic window. The elegance of it is that you do not have to find something the cancer cell has and the normal cell lacks. You find something the cancer cell is already missing, and you remove the backup as well.
Why this course brings it up: Synthetic lethality and the PARP inhibitor are the Tier 2 core he lists, and it is scenario question type 10. He explains it with a Gene A and Gene B diagram, so you have to be able to say the two together. So the pairing is the answer, not either term alone. **Appears in:** Ch 6, Ch 10, Lect 4

▪ T-cell

In one sentence: A lymphocyte of the immune system, split into helper, cytotoxic and regulatory kinds.
Detailed explanation: T cells are not one single thing. CD4+ helper T cells do not kill; they issue cytokines and direct the other immune cells. CD8+ cytotoxic T cells are the ones that actually go into action, recognising fragments displayed on MHC and killing infected or tumour cells with perforin and granzyme.
Tregs are the brakes, suppressing an excessive response. The TCR built by the V(D)J recombination Dr. Wong covers in Lectures 3 and 4 is the recognition tool that all T cells share, while CAR-T is a T cell engineered into a single-target hunter. So the same cell type is the commander, the soldier and the brake, depending on which set of instructions it was given. And the recognition tool is shared, which is what lets one body mount a specific response against one particular fragment. The difference

between the branches is which signals they send out and which molecules they carry, not which fragments they can see.
Why this course brings it up: He uses the T cell group repeatedly in Lectures 2 and 4, but what is actually examined is its branches and their consequences: NHEJ defect → no diversity can be built in the TCR and the antibodies → SCID, which he labels himself as a first-level topic. And PD-L1 binding PD-1 putting the T cell into anergy is the core mechanism of how a tumour hides. **Appears in:** Ch 6, Ch 9, Lect 2, Lect 4

▪ Tamoxifen

In one sentence: A drug that blocks estrogen in the breast and slightly activates it in the endometrium.
Detailed explanation: Tamoxifen is the prototype of a SERM. The estrogen receptor in the cell has two working binding sites: one is the ligand-binding domain (LBD), and the other is in the receptor's AF-2 region.
Once tamoxifen occupies the ligand-binding site, the receptor in the breast cannot recruit activation factors in the breast's coactivator environment, so estrogen's growth signal is blocked; but in the endometrium's coactivator environment it can recruit and activate instead. Same receptor, same drug, two opposite results.
Why this course brings it up: In Lecture 5 Dr. Wong lists it as a second-level item and says “This is the concept that you have to bear in mind.” The idea that the same target needs different strategies in different tissues is more important than the drug details. He does not ask you to memorise tamoxifen's pharmacological mechanism, but he does ask you to understand that a drug's effect is a conversation between the molecule and the tissue it lands in. The receptor has one shape and one site, and the tissue supplies the rest of the answer. Tamoxifen does not change between breast and endometrium, and neither does the receptor. Only the company it keeps is different, and that is enough to turn an antagonist in one tissue into a partial agonist in another. **Appears in:** Ch 7, Ch 10, Lect 5

▪ TAMs

In one sentence: Tumour-associated macrophages, the macrophages that gather round a tumour.
Detailed explanation: A TAM is a macrophage that the tumour has recruited and reprogrammed. It can be used by the tumour while at the same time playing a double role: it releases VEGF to encourage new blood vessels and it releases PD-L1 and cytokines to hold the T cells down.
In his tumour microenvironment list the TAMs sit alongside the Tregs and the MDSCs as the group that suppresses the immune system. They are listed as one of the obstacles.
Why this course brings it up: Dr. Wong does not name TAM as an exam question on its own, so it is conceptual understanding, and the whole TME list is the part that matters. **Appears in:** Ch 9, Lect 2

▪ target identification

In one sentence: Finding, at the very start of drug development, the molecule the disease actually depends on.
Detailed explanation: This is the first step of the whole pipeline, and what comes after it runs in a fixed order: medicinal chemistry (making the compounds), preclinical (MOA and toxicology), IND-enabling studies (filing the IND), and only then the human trials of Phase 1A, 1B, 2 and 3.
If the first step is wrong, everything after it is wasted effort. The most typical example he gives is NPM1, where you have to put the CNV copy-number data together with the mRNA expression level and only dare call it a driver when the two agree. The order is the point: each stage has its own job and its own evidence, and the evidence of one stage does not substitute for the next. Choosing early is cheap to change; choosing late is not.
Why this course brings it up: He does not name target identification as an exam topic on its own, so it is a concept to understand; what he does name is the order of this pipeline. For a question, the order and the names of the stages are what matter. **Appears in:** Ch 12, Lect 2, Lect 4

▪ Target nomination

In one sentence: The decision a pharmaceutical company makes about which molecule to go after.
Detailed explanation: Dr. Wong gives three conditions. First, the molecule has to be raised in the patient's own samples, which is the same thing as disease relevance. Second, it has to have structural tractability: a pocket into which a small molecule can fit, a surface that an antibody can bind, or something a nucleic acid drug can attack.
Third, the market must not be too crowded. Only after the disease relevance and druggability gates are passed is it worth putting money in. The order is not arbitrary: relevance first, because a drug against something irrelevant is worthless however elegant the chemistry. The third condition is the one the biologist has no say in, which is why it sits last on the list.
Why this course brings it up: He says explicitly that this whole workflow is not examined, adding that it is there so that you know a bit more and that he finds it interesting. But the contrast he does draw, telomerase as the good example and the cytoskeleton as the bad one, is worth having. **Appears in:** Ch 12, Lect 1, Lect 4

▪ TATA box

In one sentence: The key DNA word roughly 30 bases upstream of the promoter, where the transcription machine assembles itself.
Detailed explanation: Slide 13 of Lecture 5 draws the assembly order of the pre-initiation complex: TFIID (which contains TBP) binds first to the TATAAA sequence → TFIIA and TFIIB (both general transcription factors) join → RNA polymerase II arrives together with TFIIF and TFIIE (general transcription factors) → TFIIH takes on the job of pulling the DNA near the promoter open, and only then can Pol II start transcribing.

Each factor has a job and an order of arrival, and the order is not arbitrary: TFIID has to be there first, because it is what finds the TATAAA sequence in the first place. The TATA box sits at about -30 upstream of the transcription start site. Sequence variation in the TATA box itself can also affect how efficient transcription is, which is why his experiment compares different variants of it. The arrows in his diagram are not decoration: each step depends on the step before it, and the last one is the actual opening of the DNA. One thing to watch: it is only one kind of promoter among many, and the promoters of housekeeping genes often have no TATA box at all, so do not treat it as universal. The TATA box is the address; the assembly order is the delivery.

Why this course brings it up: In Lecture 5 he draws it as the starting point of the whole transcription mechanism on the promoter diagram, so it is a concept to understand. He also uses the reporter assay experiment to find out which TATA box variant is genuinely effective — that is his promoter experiment logic in practice, and it is the same serial deletion you have already met. What is worth remembering is the assembly order and the separate roles of TFIID and TFIIH, not the sequence detail. **Appears in:** Ch 7, Lect 5

▪ tbid

In one sentence: The fragment cut out of bid by caspase-8, which links the extrinsic pathway to the mitochondrion.

Detailed explanation: Once caspase-8 has been activated it does not only cut caspase-3, it cuts bid as well, and the tbid that comes out of that goes to the mitochondrion, promotes the release of cytochrome c and assembles the apoptosome.

The extrinsic signal, a death message from outside the cell, and the intrinsic signal, a death message started by the mitochondrion, therefore meet on the same execution programme. So extrinsic and intrinsic apoptosis are not two unrelated machines but one execution programme with two entrances, and both of them end at caspase-3.

Why this course brings it up: Dr. Wong does not name tbid as an exam question on its own, so it is conceptual understanding, but the picture of two entrances into one execution programme is the one to keep. **Appears in:** Ch 8, Lect 2

▪ TBP (TATA-box-binding protein)

In one sentence: The part of TFIID that recognises the TATA box and sits on it.

Detailed explanation: TBP is the TATA-box-binding protein, and it sits on the TATA box in the minor groove of the DNA with a saddle-like shape, which bends the DNA into roughly 90 degrees, and that bend is the geometric precondition for everything else to stack on top correctly.

TBP is the core of the TFIID complex, and TFIID is the first general transcription factor to arrive at the promoter as part of the pre-initiation complex. TBP is made of two structurally similar subunits that bind in a symmetrical arrangement, two similar halves gripping the DNA from both sides. So the first thing that happens is not transcription but geometry: the DNA has to be bent into a shape that the rest of the machinery can grip, and everything else in the pre-initiation complex is assembled on top of that bend.

Why this course brings it up: He puts it at step one of the assembly order on Slide 13 of Lecture 5, so it is concept-only. He does not ask you to memorise the structure of TBP or the number of its subunits; what you have to keep is the order, that TFIID arrives with TBP and TBP grips the TATA box, and how that divides the work with TFIID and TFIIH. The order is the examinable thing and the structure is not: TFIID, then TBP, then the rest. **Appears in:** Ch 7, Lect 5

▪ TC-NER

In one sentence: The repair route that fixes damage on the gene currently being transcribed.

Detailed explanation: The test for transcription-coupled repair is whether RNA polymerase II is stuck: if it is, then this gene is being read, and repairing it will immediately lift the stall.

CSA and CSB are the ones that recognise this situation and call in the team, and then, sharing the machinery with GG-NER, the repair uses TFIIH, XPA, RPA, XPF and XPG to cut out the damaged stretch of 22–32 nucleotides. The difference between the two routes is who finds the damage: GG-NER is found by a protein patrolling the DNA, and TC-NER is found by a stuck RNA polymerase sending out a distress signal. That is a neat division of labour. The cell does not scan everything all the time, it waits to be told by the machinery that has already run into trouble.

Why this course brings it up: He does teach GG-NER and TC-NER as two separate routes, so it is concept-only, and the pairing that NER repairs bulky lesions is on his list. So the two routes and the lesion type are the two things to hold. **Appears in:** Ch 6, Lect 4

▪ TCA cycle

In one sentence: The circular set of reactions in the mitochondrion that takes the fuel apart and loads the energy into NADH and FADH2.

Detailed explanation: The TCA cycle, the citric acid cycle, the Krebs cycle, turns round inside the mitochondrial matrix, oxidising the carbons of acetyl-CoA step by step into carbon dioxide and storing the electrons that come off in NADH and FADH2, which act as carriers.

The carriers then take the electrons into the electron transport chain and eventually hand them to oxygen, and the proton gradient built along the way drives ATP synthase to make ATP. For a cancer cell this whole line is the one that gets abandoned: sending 85 % of the pyruvate to lactate is exactly a way of stopping it from entering the TCA cycle. So the cycle is not really a circle in the sense of something that repeats unchanged, it is a set of steps that feed into each other, with the carbon leaving as carbon dioxide and the electrons leaving on the carriers.

Why this course brings it up: In Lectures 1 and 2 Dr. Wong uses the TCA cycle to set out the trade-off between efficiency and

building material: go the whole way round the TCA cycle and one glucose gives about 36 ATP, go by lactic fermentation and it gives 2, and a cancer cell doing aerobic glycolysis gets about 4. The metabolic pathway details are on the exclusion list, but the logic of the trade-off will be examined. **Appears in:** Ch 8, Lect 2

▪ TCR (T-cell receptor)

In one sentence: The recognition protein on the surface of a T cell, which reads a foreign fragment displayed by another cell.

Detailed explanation: The TCR is built from an alpha and a beta chain, with a minority of T cells using gamma and delta. What it recognises is not a whole antigen but a short peptide displayed on the cell surface by an MHC molecule, and that is the biggest difference between a T cell and an antibody.

The diversity of TCRs comes from the same V(D)J recombination as antibodies: the beta chain uses three pieces, V, D and J, while the alpha chain uses two, V and J. And when the TCR does recognise its antigen, it still needs the CD3 complex to carry the signal into the cell, and it still needs CD28-B7 to supply the second signal. So a TCR that recognises the antigen on its own is not enough, and the same rule applies as for a T cell meeting any antigen at all.

Why this course brings it up: On Slide 41 of Lecture 4 he draws the generation and the recognition of the TCR, so it is concept-only, but it is the other half of the SCID chain: if V(D)J recombination is broken, neither antibodies nor TCRs can be built. His own words name the clinical consequence of an NHEJ defect as SCID, and that is the highest confidence question in the course. So the two halves of that chain have to be learned together. **Appears in:** Ch 6, Ch 9, Lect 4

▪ TEAD

In one sentence: The DNA-binding protein that works alongside YAP in the nucleus.

Detailed explanation: When the Hippo pathway is off, YAP is not phosphorylated and stays in the nucleus, but YAP is not itself a DNA-binding protein, so it needs TEAD to do that part of the job.

TEAD is YAP's partner: the two of them sit on the DNA together and switch on a set of growth-promoting target genes, and his slide lists CTGF, JAG1, which is a Notch ligand, and NOTCH2, and that overlap with the Notch system is no accident of naming. The two of them are useless apart and only work as a pair, which is a good example of how the cell builds a switch out of two parts with different jobs. So YAP and TEAD are the two halves of one complex: YAP decides whether the complex gets into the nucleus, and TEAD decides which gene is read once it is there.

Why this course brings it up: Dr. Wong draws the Hippo pathway in both Lecture 1 and Lecture 5, and on Slide 24 of Lecture 5 he points out that YAP binds TEAD once it is in the nucleus and opens genes such as CTGF. That is conceptual understanding. He does not make YAP and TEAD an exam topic on their own, but Hippo and YAP is the regulatory switch he says will come back in Lecture 5, and you have to be able to draw the mechanism chain: MST1, then LATS1, then YAP phosphorylated, then it stays in the cytoplasm; lose the kinase, then YAP enters the nucleus, then it binds TEAD, then proliferation. **Appears in:** Ch 7, Lect 5

▪ Telomerase

In one sentence: A special enzyme that can rebuild the ends of chromosomes, and which most cancer cells switch back on.

Detailed explanation: Telomerase is a reverse transcriptase that carries its own RNA template inside it. It uses the RNA it brings with it — complementary to the 3′ overhang — as a template, and bolts on the GGTTAG repeat sequence piece by piece, then shifts itself forward and adds the next piece, and so on.

That is how the gap at the end of the lagging strand, the one that never quite gets finished, is finally filled in. Each time the enzyme adds one repeat, it moves its own template forward and repeats the trick, so the chromosome end is not merely patched but actively extended, one repeat at a time. The reason the cell needs this is that DNA copying has an end problem: the very last stretch of the lagging strand has no template to copy from, so without telomerase every division would shave a bit off the end of the chromosome. In a normal somatic cell telomerase is almost entirely switched off, so every time the cell divides the telomere gets a little shorter. ESCs, some adult stem cells and germ cells keep it active; cancer cells switch it back on, and that is how they can divide without limit.

Why this course brings it up: In Lectures 3 and 4 he treats telomerase as the end point of one mechanistic chain: the end-replication problem → telomere shortening → Hayflick limit → senescence → the tumour switching telomerase back on. He also says that about 90 % of cancers upregulate telomerase. And the TERT experiment numbers from Lecture 4 are worth remembering: 3 weeks of anti-TERT pretreatment → tumour formation rate 2/12 (9/12 untreated), Fisher's p = 0.01. **Appears in:** Ch 5, Ch 12, Lect 3, Lect 4

▪ Telomere

In one sentence: The repeated DNA tail at the end of a chromosome, which gets a little shorter with every division.

Detailed explanation: From the outside in, the structure is: a unique sequence, a subtelomeric repeat region of 100–300 kb, and the outermost 3–20 kb of TTAGGG (the telomeric repeat sequence).

It protects the end of the chromosome from being mistaken by the cell for a break, and it solves the end replication problem, but the protection has a price: in a somatic cell the telomere shortens by about 50–200 bp with every division. Once it reaches a critical length the Hayflick limit is triggered and the cell enters senescence. In the liver, the hepatocyte turnover caused by chronic HBV and fatty liver accelerates the loss of telomere.

Why this course brings it up: Dr. Wong covers telomere in both Lecture 3 and Lecture 4, and it belongs to conceptual

understanding, but the direction has to be remembered: the patients with the longest telomeres have the worst prognosis. In his HCC data the longest 10% of telomeres goes with worse recurrence-free survival (multivariate analysis HR 1.66, p = 0.029). The reason is that a long telomere lets cancer cells keep dividing, so it is a marker of how well the tumour is maintaining itself. Which is the opposite of the intuition that a long telomere means a healthy young cell. In this dataset the long telomere belongs to the cell that has been dividing the longest, and that is a bad sign rather than a reassuring one. It is the same counter as the Hayflick limit, read from the other end. **Appears in:** Ch 3, Lect 3, Lect 4

▪ TEM

In one sentence: Transmission electron microscope.
Detailed explanation: TEM fires a beam of electrons through an ultrathin sample, and because different structures scatter the electrons to different degrees, the image comes out at high resolution — in principle you can see a single protein.
His slide is a TEM photograph of a mouse L cell at 15,000-fold magnification, with the organelles labelled on it and each label matching one part of the cartoon cell drawn beside it. The sample has to be that thin because the electrons have to get through it and still carry information about what they passed.
Why this course brings it up: He does not name TEM as an exam topic on its own, so it is a concept to understand; in the same chapter he says plainly that the chromatin images from electron microscopy do not need to be memorised. **Appears in:** Ch 5, Lect 2

▪ Template strand

In one sentence: The strand of DNA that is being read during replication.
Detailed explanation: Both strands get read, but at any one moment only one of them is the template for that moment. The template strand is the antisense strand, and its sequence decides the sequence of the new strand.
The matching sense strand has the same sequence as the newly made RNA, with T swapped for U, and the reason is that RNA polymerase reads the template strand directly to make RNA rather than first making DNA and then transcribing it. The critical limitation is direction: RNA polymerase can only extend in the 5′→3′ direction, so it can only travel along the template strand in the 3′→5′ direction. So the enzyme's own chemistry fixes the direction of the whole reaction, and everything else about replication is arranged around that. The two strands of a double helix are therefore not equivalent in this respect: one is read to make RNA, and the other is the strand whose sequence matches the RNA you get.
Why this course brings it up: On the central dogma diagram of Slide 43 in Lecture 1 he marks the sense and the anti-sense, template, strands, so it is concept-only. What you have to keep is not the diagram but the direction rule, because that rule also decides how leading strand and lagging strand get done: elongation is always 5′→3′, so the two anti-parallel template strands can only be copied in two different ways. One rule, two strategies, and the direction rule is the thing rather than the label on the diagram. **Appears in:** Ch 5, Lect 1

▪ Teratoma assay

In one sentence: The gold standard for deciding whether a cell really is pluripotent: inject it into a mouse and see whether tissue from three germ layers grows.
Detailed explanation: You inject the cells under test into an immunodeficient mouse, and if they really have pluripotency they grow inside the animal into a tumour containing tissue from all three germ layers, ectoderm, mesoderm and endoderm.
His pathology slides list cartilage, CNS, muscle, adipose and epithelium, and another photograph is a mature cystic teratoma with hair, sebum and a tooth in it. That is a great deal stronger evidence than saying the cells look like embryonic cells, because it is a functional proof: it says this one cell still remembers how to make every kind of cell. The reading is done on the pathology slide, and that is why the assay takes months rather than days, because you are waiting for a tumour to grow and then for it to be sectioned and stained. The five tissues he lists, cartilage, CNS, muscle, adipose and epithelium, come from three different germ layers, and the point of the assay is that one cell has produced all of them. So the assay is not measuring how many cell types are present in the animal, it is measuring whether the single cell you injected still carries the instruction set for every one of them, and that is a much stronger claim than anything the morphology check can make.
Why this course brings it up: In Lecture 2 Dr. Wong explicitly calls the teratoma assay the gold standard of in vivo validation, and that is conceptual understanding. But the iPSC line has a more practical point attached to it: proving pluripotency and proving the culture is not contaminated are two different things, and his stem-cell QC reminder is "unique expression pattern and timing, and also the negative staining to exclude the contaminated cell. Don't forget it." **Appears in:** Ch 8, Lect 2

▪ TERC

In one sentence: The RNA template part of telomerase.
Detailed explanation: Telomerase is itself a reverse transcriptase, and because it makes DNA from RNA it has to bring its own template along. TERC is that template RNA (the human telomerase RNA component), and it pairs with the 3′ overhang at the end of the chromosome, allowing the enzyme to add the TTAGGG (the telomeric repeat sequence) onto the end.
TERT is the protein part that does the catalysis. If TERC and TERT are both working, the telomere is maintained; if telomerase fails, the chromosome gets shorter with every round of replication, until it hits the Hayflick limit and the cell senesces. So the template and the enzyme are both required, and losing either one ends the same way. The template is the part that says what to build, and the enzyme is the part that builds it.
Why this course brings it up: Dr. Wong does not name TERC as an exam topic on its own, so it belongs to conceptual understanding. What is the point is that telomerase is a reverse transcriptase with an RNA template, and the Hayflick limit.

Appears in: Ch 5, Lect 4

▪ Tertiary structure

In one sentence: The three-dimensional shape a whole polypeptide chain takes once it has folded up.
Detailed explanation: Tertiary structure is the second big level of folding, and it is decided by the interactions between the side chains (R-groups). The hydrophobic groups hide away inside the molecule and the charged groups stay out on the surface where they can touch water, and between them they make a stable core.
Once the shape is settled, the function is settled with it — the active site and the binding sites are all consequences of that shape. A change in the surroundings (pH, temperature, a denaturant) can unfold it, so protein folding is a dynamic balance rather than a fixed sculpture. The level above it, quaternary structure, is a further step again: several chains assembled together. So the tertiary structure is where the sequence finally becomes a machine.
Why this course brings it up: In Lecture 1 he teaches primary → secondary → tertiary → quaternary as the skeleton of the whole course, and tertiary structure is one level of it, so it is a concept to understand. What he explicitly excludes is the table of 20 amino acids with their R-groups and the chemical structure and metabolic pathway details — not this hierarchy. What you have to be able to do is say how the phrase "shape determines function" holds separately at the tertiary and the quaternary level. **Appears in:** Ch 2

▪ TET

In one sentence: A family of enzymes that oxidise the methyl mark off, the executors of DNA demethylation.
Detailed explanation: The TET enzymes work on already-methylated cytosine and oxidise it step by step: 5mC → 5hmC → 5fC → 5caC, and the intermediates get picked up by the cell's DNA repair machinery and converted back into unmethylated cytosine.
The overall effect is to erase the methyl mark from the DNA. This is the force working in the opposite direction to the methyltransferase, and the two together decide whether a CpG island is open or closed. TET is therefore tied to demethylation of tumour-suppressor promoters and to the DNA damage repair pathways. And the picture to hold is a balance: one hand writes the methyl mark on, the other rubs it off, and whether a gene is readable depends on which hand is winning at that spot. So a CpG island can be shut and opened without a single letter of the sequence changing. That is the whole epigenetic idea in one sentence: the information is in the chemistry on top of the DNA, not in the DNA itself.
Why this course brings it up: On the chromatin regulation diagram of Slide 30 of Lecture 3 Dr. Wong draws TET as the end that removes the methyl, and it belongs to conceptual understanding. He does not list TET as a topic on its own. But it has to be understood together with another pair: de novo methyltransferase (DNMT3A/3B) establishing the mark and maintenance methyltransferase (DNMT1) copying it, and that pair is on his distractor list. **Appears in:** Ch 4, Lect 3

▪ TFIID

In one sentence: The first general transcription factor to arrive at the promoter, and it brings TBP with it.
Detailed explanation: TFIID is the first complex to arrive at the pre-initiation complex, and it is made of TBP plus a group of TAFs, the TBP-associated factors. TBP is what recognises and grips the TATA box, while the TAFs recognise other sequence elements and recruit the co-activators.
Once TFIID is in place, TFIIA, TFIIB, RNA polymerase II, TFIIF and TFIIE stack on in that order, and finally TFIIH opens the DNA and Pol II starts transcription. You can think of TFIID as the advance party of the construction crew: it marks out where the foundation goes, and only then do the others know where to assemble. And it is the reason the direction of the whole assembly is fixed, because everything after it depends on the TATA box being held in the right place. The TATA box is the address and TFIID is the postman. The sequence of names on the slide is therefore an assembly order, and if you remember nothing else, remember that TFIID is the first one there.
Why this course brings it up: He puts it at step one of the assembly order on Slide 13 of Lecture 5, so it is concept-only. He does not ask you to memorise the number or the names of the TAFs, but you do have to recognise the pairing of TFIID with TBP, because it is the most commonly examined relationship in the promoter mechanism. One pairing, and it is the one to know, and the order of the factors after it is also a sequence you should be able to recite. **Appears in:** Ch 7, Lect 5

▪ TFIIH

In one sentence: The transcription factor that pulls the DNA near the promoter open so that RNA polymerase II can get going.
Detailed explanation: TFIIH is a multifunctional complex that carries helicase activity (XPB and XPD) inside it. At transcription initiation it opens up a short stretch of DNA so that RNA polymerase II can sit down and start synthesising.
The very same complex also takes part in NER (nucleotide excision repair): on Slide 25 in Lecture 4 he draws the GG-NER and TC-NER pathways both converging on TFIIH (XPB, XPD), which works with XPA, RPA, XPF and XPG to complete the excision. The excised product is about 22–32 nucleotides long, and after that a polymerase fills the gap and a ligase seals it. The same machine, doing two quite different jobs in two quite different lectures. That dual role is the interesting part: a complex that is needed to open DNA for transcription is also needed to cut damaged DNA out, and both activities sit in the same bag of proteins. The two are chemically the same act, peeling the two strands apart, applied to a healthy promoter in one case and to a damaged stretch in the other.
Why this course brings it up: In Lecture 5 he uses TFIIH to explain how DNA is opened; in Lecture 4 he puts it at the convergence point of NER, so it is a concept to understand. He does not ask you to memorise the structure of TFIIH; but at both

places you should be able to say in one sentence what its job is — in Lecture 5, "opening the promoter", and in Lecture 4, "stringing excision and fill-in together". **Appears in:** Ch 6, Lect 4, Lect 5

▪ TGFβ

In one sentence: The cytokine that a tumour uses to suppress the immune attack around it.
Detailed explanation: TGFβ is normally a signal that regulates cell growth, repair and immunity, but in the tumour microenvironment it is hijacked and turned into an immunosuppressant. The mechanism works on several levels at once: it suppresses the activity of the effector T cells, it promotes the expansion of Tregs, which are the immune brakes, and it works together with MDSCs, the myeloid-derived suppressor cells.
Slide 50 of his Lecture 3 shows MDSCs secreting TGF-β, IL-10 and PD-L1 to suppress CD8+ T cells. The tumour builds a defensive line this way, and it is a parallel route to the one that anti-PD-1 blocks. And the tumour benefits twice, because TGFβ also feeds the growth of the tumour cells themselves, so the same signal that pushes the immune system down is pushing the tumour up at the same time.
Why this course brings it up: On Slide 63 of Lecture 2 Dr. Wong lists TGFβ and IL-10 as one of the obstacles that stops CAR-T getting into a solid tumour, alongside Tregs, MDSC, TAM, hypoxia and the stromal environment. That is conceptual understanding, but the conclusion that a tumour does not hide from the immune system only with PD-L1 changes how you read why checkpoint blockade sometimes fails. **Appears in:** Ch 9, Lect 2

▪ Therapeutic window

In one sentence: The margin between the lowest effective dose and the highest tolerated dose.
Detailed explanation: For a drug to work, its concentration has to be above some threshold; for it not to harm the patient, it has to be below some other one. The distance between those two thresholds is the therapeutic window, and the wider it is the more useful the drug.
Dr. Wong's textbook example is BRCA1/2 mutation plus a PARP inhibitor: the tumour cells have lost HR repair, so blocking PARP kills them, while the normal cells, with intact HR, survive by taking the other route, and synthetic lethality widens the window. An ADC is a second version of the same idea, delivering the drug to the tumour so that the concentration piles up at the target. Both tricks buy the same thing, which is a bigger gap between the dose that works and the dose that harms. A narrow window makes a drug hard to use, whatever it does, because the dose that helps and the dose that harms sit too close together for comfort.
Why this course brings it up: In Lecture 4 he lists the therapeutic window as a second-level concept: the margin between the minimum effective dose (efficacy) and the toxic dose (toxicity), and synthetic lethality is precisely what widens that margin. A short question may ask “why is a PARP inhibitor especially effective in BRCA-mutant patients”, and the answer has to be given in the language of this window. **Appears in:** Ch 6, Ch 10, Lect 4

▪ Threshold effect

In one sentence: Disease appears only when the proportion of mutant mitochondrial DNA rises above the tolerance limit of a particular tissue.
Detailed explanation: A person can carry normal and mutant mtDNA at the same time, and that state is called heteroplasmy. Carrying the mutation is not the same as being ill: the symptoms only appear once the proportion of mutant mtDNA passes a threshold that belongs to that tissue.
Tissues with high energy demand, and he gives contracting skeletal muscle as the example, have a low threshold and are hit first, while low-demand tissues, hearing for example, have a high threshold and are hit later. His body map lists the affected organs from neurodegeneration, epilepsy and deafness through to cardiomyopathy, liver disease, kidney disease, diabetes and infertility, so the range is extremely wide. The point of the threshold is that the same person can have a different amount of mutant mtDNA in every tissue of the body, and in one tissue the mutation is already past the line while in another it is not. That is why the severity of the disease varies from organ to organ within one patient, and because the threshold is set by how much energy a tissue needs, the tissues that fail first are the ones that cannot afford to wait.
Why this course brings it up: This is the Tier 1, A+ question he marks himself. His words are: "So, this is you'll have to a 100% understand, right? Not only the mutation within the mitochondrial DNA can cause the mitochondrial disease. Some mitochondrial disease can actually result from the mutation within the nuclear DNA." The answer has to bring up both mtDNA and nuclear genes, and it has to explain the threshold effect. And do not forget that the random segregation of homologous chromosomes is what explains why siblings can be affected to different degrees. So the answer has three parts: the two genetic sources, the threshold, and the random segregation. **Appears in:** Ch 8, Lect 2

▪ thymine

In one sentence: One of the four main bases of DNA, written T.
Detailed explanation: T is a pyrimidine, a small molecule with a single ring, and in DNA it pairs with adenine and forms two hydrogen bonds. RNA has no thymine and uses uracil instead, and that is the one crucial difference between DNA and RNA in the base itself, and it is also the basis of how the bisulfite chemistry can tell methylation apart.
There is an evolutionary reason for DNA and RNA to use different bases: RNA carries an extra hydroxyl group and so hydrolyses more easily, while DNA is better suited to storing information for the long term. So the difference between the two molecules is not a detail of the chemistry, it is a decision about what each one is for.
Why this course brings it up: "Two hydrogen bonds for A–T, three for G–C" is one of his Tier 3 precise numbers. **Appears in:**

Ch 3, Lect 1

▪ TIM3

In one sentence: An inhibitory checkpoint receptor, listed alongside PD-1.
Detailed explanation: TIM3 appears on his slide 63 list of inhibitory checkpoints together with CTLA-4, PD-1, LAG3 and VISTA, and they all sit on the surface of the effector T cell. When the tumour environment provides its ligand, the activity of the T cell is pushed down — and his point is this: CAR-T works in blood cancers, but in solid tumours it is blocked by this whole row of brakes.
These checkpoints matter because one drug on its own is often not enough; the list is his way of saying that a solid tumour needs several approaches at once. One brake, and the cell still moves; several brakes at once, and it does not.
Why this course brings it up: He does not name TIM3 as an exam topic on its own, so it is a concept to understand; the slide it comes from is about why solid tumours resist CAR-T, and the list of brakes is the point of that slide. **Appears in:** Ch 9, Lect 2

▪ TINCR

In one sentence: A long non-coding RNA that controls the stability of a message by pairing with the mRNA and then grabbing hold of Staufen 1.
Detailed explanation: TINCR is one of the four lncRNA archetypes listed on Slide 29 of Lecture 5. It does not encode a protein; it acts as a guide and as a scaffold. One stretch of it pairs up complementarily with a particular mRNA, and another stretch binds an RNA-binding protein such as Staufen 1, and by putting those two together it changes the stability of that mRNA and the extent to which it gets translated.
The other three archetypes on the same slide are APOLO (which forms an RNA–DNA hybrid to regulate genes), PNCTR (which uses its YYUCUY motif, the sequence that splicing factors recognise, to bind splicing factors such as PTBP1 and control splicing), and XIST (which binds SPEN, an RNA-binding protein, and PRC2, a chromatin repressive complex, to silence an entire X chromosome). None of the four makes a protein, and all four of them work by holding something else in the right place, which is the whole point of the archetype idea. The lncRNA is the handle and the mRNA or the protein is what gets carried; in TINCR's case both jobs are done by the same molecule, using two different parts of itself.
Why this course brings it up: In Lecture 5 he treats lncRNA as the new frontier in gene expression control, so it is a concept to understand. He does not ask you to memorise the names of lncRNAs, but you do need to know the mechanistic classification: scaffold (binding proteins in the right place) and guide (carrying a protein over to the DNA or RNA where it ought to act). The clinical interest is that they are becoming new drug-discovery targets and biomarkers. **Appears in:** Ch 7, Lect 5

▪ TLR4

In one sentence: The cell-surface receptor that recognises LPS, and the first stop of the sepsis cascade.
Detailed explanation: The delivery and recognition of LPS after it enters the body happens in steps. LBP, the LPS-binding protein, carries it to the membrane and hands it to CD14, the co-receptor, and MD-2, the LPS-binding subunit, which present it to TLR4, and TLR4 then dimerises.
The recognition is deliberately built in steps with dedicated carriers, so that the receptor is only ever handed a properly presented molecule and never has to sort out the lipid by itself. The inside of TLR4 splits into two routes. The MyD88-dependent route, through TIRAP to MyD88, switches on NF-κB and the pro-inflammatory cytokines. The MyD88-independent route, through TRAM to TRIF, also switches on NF-κB and additionally produces the type 1 interferons. Two routes out of one receptor means the cell can escalate rather than simply switch on, because the MyD88-independent branch adds the interferons on top of the cytokines. When the amount of LPS in the blood gets too large, the whole reaction overwhelms the body and produces septic shock.
Why this course brings it up: What will be examined is the chain of order that you can reason out: LPS, then TLR4, then IKK, then IκB, then NF-κB, then into the nucleus, then the inflammatory cytokines, and his exam notes list it as a possible stem for a multiple-choice question. The NF-κB section of Lecture 5 is the lower half of the same chain. He also says explicitly that the detailed signalling cascade on the slide does not have to be memorised. **Appears in:** Ch 1, Ch 7, Lect 1, Lect 5

▪ TMZ

In one sentence: Temozolomide, an alkylating chemotherapy drug.
Detailed explanation: TMZ is an alkylating agent that can activate itself in the blood, and it puts a methyl group onto a DNA base, which stops the DNA replicating normally. The critical clinical logic is this: the enzyme MGMT is the one that takes the alkyl group off O6-methylguanine, which is the damage TMZ creates.
So if the MGMT promoter is methylated the patient expresses very little MGMT, cannot repair the damage, and is sensitive to TMZ; if it is not methylated the damage is repaired well and the tumour shows resistance. So the drug and the repair enzyme are matched against each other, and the laboratory measurement is what tells you which side of the fight the patient is on.
Why this course brings it up: MGMT methylation predicting the response to TMZ is a Tier 2 point he lists, and he draws the patient A and patient B comparison very clearly. That diagram is the whole clinical story on one slide. **Appears in:** Ch 4, Lect 3

▪ TNF-α

In one sentence: The first cytokine to appear in the NF-κB inflammatory response, peaking at 4 to 6 hours.
Detailed explanation: Once NF-κB has been switched on by LPS or another stimulus, the cell secretes inflammatory cytokines in time order: TNF-α comes first and peaks at 4–6 hours, then IL-1β, then IFN-γ, then IL-2 and IL-15.

The key is the positive feedback: TNF-α and IL-1β go back and activate NF-κB again, amplifying the response into something fast and fierce. Later on, the anti-inflammatory mediators take over and shut the response down. The overall shape is a curve that rises and then falls, not a steady output. And that curve is the reason the timing in the cytokines matters more than the names. If you take a blood sample at the wrong hour you will measure the wrong thing and conclude that the pathway is quiet when it is only between peaks. The response has a shape in time, and you have to sample it at the right point on that shape. That is the entire lesson of the ordering, whether or not you can list the cytokines.
Why this course brings it up: The line “TNF-α peaks at 4–6 hours” and the order that follows are on Dr. Wong's list of key numbers. But he also says plainly: “don't worry about all these cytokines. I'm just want to show you that different responses can be observed in different times.” What he wants is the concept of different responses at different time points, not a full list of cytokines memorised. **Appears in:** Ch 7, Lect 1, Lect 5

▪ tobacco-specific nitrosamines

In one sentence: Carcinogenic precursors found only in tobacco.
Detailed explanation: They are present in cigarette smoke and have to be activated by the liver's cytochrome P450 enzymes before they can react with DNA. NNK and NNN are the two Dr. Wong names;
once activated they form DNA adducts, and if those adducts are not repaired the second round of replication fixes them as mutations, in mice the GC→AT mutation in K-ras being what NNK produces. The signature is specific enough to be read back out of the tumour, which is how the chain gets its final confirmation. You can read the history of the tumour backwards from the mutations it carries. Clinically, smoking is the clearest example of chemical carcinogenesis: in cancer cells you can find the DNA adducts made by these carcinogens and the mutation signature they left behind. The compound is a poison, but the adduct is the thing that actually touches the DNA.
Why this course brings it up: He does not ask you to memorise the list of carcinogens, so it belongs to conceptual understanding. What he asks you to remember is the causal chain from adduct to mutation. **Appears in:** Lect 3

▪ tolerance

In one sentence: The tolerance the immune system learns for the body's own proteins.
Detailed explanation: During development, B and T cells that react too strongly to the body's own proteins are filtered out, and only the rest are allowed to survive, and that "do not attack yourself" constraint is what tolerance is.
Neoantigens can raise an immune response precisely because no tolerance to them has ever been established. One of the commonest ways a cancer cell escapes the immune system is to re-establish a kind of tolerance, so that the neoantigen becomes something the immune system has seen before and does not need to react to. The word is doing important work here.
Why this course brings it up: Dr. Wong does not name tolerance as an exam question on its own, so it is conceptual understanding, and he has taught the logic that a neoantigen is new and therefore has no tolerance against it. **Appears in:** Ch 6, Lect 1

▪ Topoisomerase

In one sentence: The enzyme that cuts DNA ahead of the replication fork, lets it swivel, and puts it back, to release the twisting strain.
Detailed explanation: DNA is a double helix, and pulling it open builds up overwound tension ahead of the fork.
Topoisomerase's answer is to cut the DNA backbone temporarily, let it swivel and relax, and then rejoin it, and the cut is in one strand only, with ligase sealing it afterwards.
It is the core of the ADC therapeutic strategy: inhibiting topoisomerase I stops the knots ahead of the replication fork from being resolved, double-strand breaks pile up, and the cell dies. It is the same physical problem as a coiled rope being pulled from both ends, except the cell solves it by cutting the rope rather than by loosening the grip.
Why this course brings it up: In Lecture 4 he lists it as one of the five core enzymes of replication, helicase, topoisomerase, primase, polymerase and ligase, so it is concept-only. He does not ask you to memorise the domains of the enzyme or a hand diagram. The clinical link is the ADC, and why you would put a topoisomerase inhibitor inside an ADC is one of the short scenario questions in his case bank, with the answer being selective delivery and the two named targets being HER2 and TROP2. The structure of the ADC and the chemistry of the payload are explicitly not examined. **Appears in:** Ch 5, Lect 4

▪ Toxic pre-expression

In one sentence: Making the recombinant protein before the host is ready for it, which can kill the culture.
Detailed explanation: When you do recombinant expression with the lac system of E. coli, adding IPTG starts the manufacturing of the target protein. If you induce too early, the protein accumulates in large amounts while the host still does not have enough folding and breakdown capacity, and it forms insoluble aggregates, inclusion bodies, which lowers the yield and can also hurt the culture.
The answer is to delay the induction until the log phase, so that the cells grow first and produce afterwards. His Slide 10 also shows how the IPTG concentration affects the level of expression and points to the concentration that gives the best expression. So there are two levers here, when you induce and how hard, and the second one is on the slide.
Why this course brings it up: In Lecture 5 Dr. Wong teaches recombinant expression together with IPTG, CHO cells and codon optimisation, and that is conceptual understanding. He does not make the timing of induction an exam question, but this passage is part of the same narrative as the story of a manufacturer choosing a platform on the balance of cost and quality. He lets E. coli be the cheap and fast option, CHO does the glycosylation, and codon optimisation is the sentence he explicitly says

to remember. **Appears in:** Ch 7, Lect 5

▪ toxicology

In one sentence: The study of what a compound does to a whole animal.
Detailed explanation: MOA answers "what does it do", and toxicology answers "what does it do to a living body". Only the two together make up the core of IND-enabling studies: the first is the evidence of effectiveness, the second is the evidence of safety, and it is often the second that actually decides the dose ceiling.
The most important figures are the maximum tolerated dose and the therapeutic window, and the gap between them is the range within which a patient can safely be treated. A drug can be beautifully effective and still useless if the window between the two doses is too narrow to hit reliably. The window is what turns a laboratory finding into a prescription.
Why this course brings it up: He does not name toxicology as an exam topic on its own, so it is a concept to understand; the safety questions of clinical trials and the therapeutic window are the Tier 2 parts. Those are the ones to know. **Appears in:** Ch 12, Lect 2

▪ TRAIL

In one sentence: The ligand that starts apoptosis from an outside signal.
Detailed explanation: When TRAIL binds a death receptor on the cell surface, caspase-8 is switched on and goes on to activate caspase-3, and at the same time caspase-8 cuts bid into tbid and sends the signal into the mitochondrion, where it joins the intrinsic pathway.
The pharmacological attraction of TRAIL is that it is far less toxic to normal cells than to tumour cells, which is why it is used as one end of a bispecific antibody. Precisely because it is so much less toxic to normal cells, TRAIL has the chance to be used as one end of a bispecific antibody that only really fires on cancer cells. So the whole point is selectivity: the same molecule that would be dangerous on its own becomes usable once you attach something that delivers it to the right cell.
Why this course brings it up: He does not name TRAIL as an exam question on its own, so it is concept-only. He draws diagrams of both the extrinsic and the intrinsic apoptosis pathways. So the diagram is the examinable part. **Appears in:** Ch 8, Lect 2

▪ TRAILR2

In one sentence: The receptor for TRAIL, which is also called DR5.
Detailed explanation: When TRAILR2 binds TRAIL it starts the death receptor pathway. Dr. Wong's own bispecific antibody BI 905711 ties TRAILR2 (the death receptor on the tumour) to CDH17 (the cancer cell surface protein he has studied for twenty years) on a single molecule, and it only attacks when both are present at once, which is an AND gate.
Because TRAILR2 is expressed on only a small number of cells, the bispecific antibody can use it as a threshold: a cancer cell is killed only if it also expresses CDH17. The second requirement is what stops the antibody from killing the healthy cells that happen to express the death receptor. Two marks on the same cell, and only the combination counts.
Why this course brings it up: He does not name TRAILR2 as an exam topic on its own, so it belongs to conceptual understanding. The AND-gate design is the point, and it is the same idea as the synNotch switch. **Appears in:** Ch 8, Lect 2

▪ Trans-acting factor

In one sentence: A diffusible protein that can act on a gene somewhere else, even on a different chromosome.
Detailed explanation: Cis elements (promoter, enhancer, silencer, insulator) act only on the DNA molecule they sit on, while a trans factor is a protein that diffuses freely and binds to cis elements somewhere else, and the typical examples are transcription factors and co-activators.
Dr. Wong's concrete example is a repressor encoded on chromosome 1 that acts on a target gene on chromosome 13. He also notes that trans regulation is “much more common at least in eukaryotic system”, and he emphasises that expression is both cell-specific and time-specific. So the same genome has to produce a different set of proteins in a nerve cell and a liver cell, and a different set again in the same cell at a different moment, and diffusible proteins are how that is arranged. The cis elements are the same in every cell, because the DNA is the same. What differs is which diffusible proteins are floating around in that particular cell at that particular time, and which of them happen to land on which cis element. Same DNA, different furniture in the room, different output. And because the trans factors are proteins, they can also be modified and controlled in their own right, which is a whole extra layer on top.
Why this course brings it up: Cis versus trans is one of the distractor contrasts he lists, and the test is whether it can diffuse and cross chromosomes. His chromosome 1 to chromosome 13 example is also on the key numbers list. He adds the reminder that not every enhancer is upstream: some regulatory sequences sit at the 3′ end of the gene. **Appears in:** Ch 7, Lect 5

▪ Trans face

In one sentence: The shipping end of the Golgi, where finished proteins are packed into vesicles.
Detailed explanation: The Golgi has a direction to it. The cis face is the receiving end — "cis" meaning the side that faces the nucleus — and transport vesicles bring the protein there, where the Golgi processes it step by step and sorts it.
The trans face is the shipping end, the side facing the cell membrane, and from there the finished protein is packed into a secretory vesicle and sent off to the membrane. The division of labour among the enzymes follows position too: the glycosylation steps are spread from cis to trans, with different glycosyltransferases sitting in different little sacs, like stations along a conveyor belt. The protein enters at one end and leaves at the other, and it is never sent back the way it came, so the order in which the

enzymes meet it is fixed. That is why the Golgi is drawn with an arrow on it rather than as a plain stack of sacs. Each station adds a different sugar, and the sequence of sugars it ends up with is decided by the route it took.
Why this course brings it up: In Lecture 2 he covers the whole route with one ER–Golgi diagram on Slide 22, so it is a concept to understand. But its clinical significance lies elsewhere: which route a protein takes and which face it is delivered to decides whether it ends up secreted, sitting in the membrane, or kept inside the cell — which is exactly what a signal peptide does, and it is also the basis of the antigen presentation route (ER → MHC). **Appears in:** Ch 8, Lect 2

▪ Transcription

In one sentence: Copying the recipe in the DNA into a working message that can be carried out of the nucleus.
Detailed explanation: Transcription is the first stage of the central dogma: RNA polymerase uses one strand of the DNA as a template and synthesises an RNA. A prokaryotic cell has no nucleus, so the mRNA can be translated by a ribosome while it is still being copied.
A eukaryotic cell first processes the pre-mRNA in the nucleus, with splicing, a 5′ cap and a poly-A tail, and only then exports it to the cytoplasm for translation. The real consequence of that separation is that a eukaryotic cell has an extra layer of regulation, because the half-life of the mRNA, the way it is spliced and the RBP it binds can all change the final output. So transcription is a copying step with a control surface laid over it, and each of those three handles is a different way of changing the output without changing the sequence.
Why this course brings it up: The whole of Lecture 5 is about how gene expression is regulated, and transcription is only one layer of it. That is conceptual understanding, but you have to be able to use it to explain two phenomena: why a methylated promoter produces no mRNA, because the transcription layer has been switched off, and why one gene can make several isoforms, which is the splicing layer. What he excludes explicitly is the chemical structure and the detailed signalling cascades, not this flow. **Appears in:** Ch 7, Lect 5 **Video:** V14_Transcription.mp4

▪ Transcription factor

In one sentence: A protein that recognises a particular word in the DNA and helps or blocks transcription there.
Detailed explanation: Transcription factors come in two kinds. The general transcription factors, TFIID and TFIIH among them, turn up at every promoter and are what recruit the RNA polymerase.
The specific transcription factors recognise particular DNA sequences and act only on particular genes. Their activity is usually hidden, and Slide 17 of Lecture 5 lists seven ways of switching one on: the protein is made new, a ligand binds it, it is phosphorylated, a second subunit joins, the covering is removed, it is moved into the nucleus, or it is released from the membrane. Each of those is a different signalling design. Seven ways of doing the same job, which is why the list is worth learning as a list rather than as a fact about any one of them. In every case the protein is already present in the cell and is simply held in a state where it cannot work, and that is what makes the response fast.
Why this course brings it up: The seven modes he lists in Lecture 5 are the most concentrated mechanism diagram in the course, so it is concept-only. He says plainly that the detailed cascades need not be memorised, but the classification of how a transcription factor gets switched on has to be understood, because it strings the proteolytic cleavage of Notch, the phosphorylation cascade and the ligand binding of the nuclear receptor into one framework. One framework, three examples, and the seven modes are the seven ways of getting from off to on. **Appears in:** Ch 3, Ch 7, Lect 5

▪ Transcription start site

In one sentence: The base at which the new RNA copy really begins.
Detailed explanation: On a promoter diagram, once TFIIH has pulled the DNA open and RNA polymerase II has taken up its position, the polymerase starts making the first nucleotide at one particular place on the template strand — and that place is the transcription start site, marked +1 on the diagram.
What it does downstream matters too: a change in the length and the structure of the transcript affects how stable the mRNA is; and a different start site produces a different 5′ UTR, which affects how efficiently the message is translated and which places are available to be spliced. So the site is not only a fact about where transcription begins; it is also a decision that changes what the finished message looks like and how long it lasts. The polymerase is already in the right place, so the only question left is which base it copies first, and that choice is written into the promoter sequence — and into the TATA box sitting 30 bases upstream. Everything upstream of +1 is therefore upstream of the gene, and everything downstream is part of the transcript.
Why this course brings it up: On the pre-initiation complex diagram on Slide 13 of Lecture 5 he marks the +1 transcription start site, so it is a concept to understand. He does not make the site number an exam item, but it is the origin of coordinates on the promoter diagram: the TATA box is at -30 (30 bases upstream), the transcription start site is at +1. Understand those two coordinates and you understand what "upstream" means. **Appears in:** Ch 7, Lect 5

▪ Transducer

In one sentence: A protein that converts one signal into another. RAS is one.
Detailed explanation: The receptor “receives” the signal from outside the cell, and the transducer is the one that “converts” it, turning what the receptor has picked up into another form of signal inside the cell and passing it on.
RAS is the textbook example: once EGFR has been activated by its ligand, RAS converts the signal into the two downstream routes, RAF–MEK–ERK and PI3K–AKT–mTOR. Tumours get their growth by keeping these transducers “stuck on”: mutated K-Ras activates itself, and the downstream signalling carries on whether or not the upstream EGFR has anything to say. So the message the cell sends itself no longer needs the outside world to send it, and the cell has taken the steering wheel out of the

receptor's hands. That is also why an anti-EGFR drug can fail against a K-Ras mutation: it blocks the steering wheel at the top of the hill, and the car is already driving. The drug works exactly as designed and still has no effect, because the target it holds is not the part that is stuck.
Why this course brings it up: Lectures 1 and 3 put RAS on the EGFR → RAS → RAF → MEK → ERK diagram, and Dr. Wong notes that EGFR, RAS, PI3K and RAF are all marked as common mutation sites in cancer. What is examinable is the clinical application of that chain: mutated K-Ras → downstream still on → cetuximab (anti-EGFR) fails, because the target is not downstream. His words are “So this is you'll have to a 100% understand...” and that sentence belongs in the pedigree territory of Lecture 4, but the same RAS idea is useful on both sides. **Appears in:** Ch 6, Ch 10, Lect 1

▪ Transforming principle

In one sentence: The name Griffith gave to the thing that turns dead bacteria into live ones, later shown to be DNA.
Detailed explanation: In 1928 Griffith found that dead pneumococcus bacteria could turn a live harmless strain into a virulent one, and he called the active substance the transforming principle, but he could not say what it was.
In 1943 Avery, MacLeod and McCarty answered with a simple enzyme experiment: adding DNase, which cuts DNA, destroyed the transforming activity, while adding protease, which cuts protein, did not. The DNA destroyed it and the protein enzyme could not, so the genetic material is DNA and not protein. It is a neat design, because the two enzymes are a control pair: the one that cuts DNA destroys the activity and the one that cuts protein does not, and the only conclusion left is that the active substance is the thing the first enzyme attacks. That is the whole of the proof, and it is the sort of experiment you can reason your way through without memorising anything.
Why this course brings it up: In Lecture 1 Dr. Wong treats this piece as chemistry and history, and he uses one inferable experiment to establish the foundation of the whole course, that DNA is the genetic material. His way of putting it is that this enzyme experiment is the proof. The years 1928 and 1943 are shown as a story, and he does not list them as numbers to memorise; what is examinable is the logic of that controlled experiment. **Appears in:** Ch 3, Lect 1

▪ Translation

In one sentence: Reading the mRNA and assembling the protein exactly as the description says.
Detailed explanation: The ribosome reads the codons on the mRNA, each codon specifies one amino acid, and the amino acids are joined with peptide bonds. The direction is 5′→3′, the start signal is AUG, which is methionine, and the stop signals are UAA, UAG and UGA.
Because the genetic code is degenerate, with several codons specifying the same amino acid, different organisms prefer different codons, so when you express a human protein in a mammalian cell you have to do codon optimisation, to give the host the codons it is used to. The degeneracy is not a flaw. It is slack in the system, and it is also what makes it possible to swap one codon for another without changing the protein at all. So the reading frame is the unit of meaning: three bases at a time, and if the frame moves, everything after it means something different.
Why this course brings it up: On Slide 43 of Lecture 1 he marks AUG and the three stop codons, so this is basic technique you have to be ready with at any time. What is examinable is the clinical deduction from it: a frameshift mutation changes the reading frame, so the ribosome hits a stop codon early, giving a truncated protein or a neoantigen, and the neoantigen is largely why a hot tumour can respond to anti-PD-1. He also says that codon optimisation has to be done in the CHO system. One codon table and several consequences, and the codon optimisation point is a manufacturing point rather than a discovery point.
Appears in: Ch 7, Lect 1, Lect 5 **Video:** V15_Translation.mp4

▪ Translocation

In one sentence: A piece of one chromosome trading places with a fragment of another.
Detailed explanation: On Slide 40 in Lecture 3 he quotes genome.gov and defines a translocation as a reciprocal exchange (chr 4 ↔ chr 20): each of the two chromosomes cuts off a piece and they are swapped over and joined back on.
Clinically there are two cases worth telling apart. A balanced translocation has no net gain or loss, so the total amount of genetic material is unchanged and, like the inversion he describes, it need not cause any change in dosage. An unbalanced one does have a net gain or loss, and that means part of the genome is missing or some of it is duplicated. Translocations also often produce fusion genes, in which two genes are joined together and make a new protein or a new pattern of expression. The swap itself is a mechanical accident of chromosomes breaking and rejoining badly; what matters clinically is whether the exchange was tidy or untidy. A tidy swap takes a piece away and gives an equal piece back; an untidy swap takes more than it gives, or gives more than it takes, and the cell ends up with the wrong amount of some genes.
Why this course brings it up: In Lecture 3 he teaches the five structural abnormalities side by side, so it is a concept to understand, but there are two traps in the matching questions to remember. First, his own words: "you need to understand that some of these abnormalities can cause the loss of the genetic materials, but some does not." — you have to be able to tell which ones lose genetic material and which do not. Second, translocations and inversions belong to the same functional class: both change expression without adding or subtracting anything. **Appears in:** Ch 3, Lect 3

▪ Transmembrane domain

In one sentence: The hydrophobic stretch of a protein that sits inside the membrane; take it away and the protein dissolves into the water and can be measured in blood.

Detailed explanation: The transmembrane region is an alpha-helix or a beta-barrel rich in hydrophobic amino acids and long enough to cross the lipid bilayer of the cell membrane. A receptor such as HER2 uses it to join the extracellular part to the intracellular kinase region.

The alternative splicing Dr. Wong covers in Lecture 5 has a direct application here: cut out the transmembrane region and the protein becomes a soluble isoform, so it turns up in blood or urine and can be measured as a biomarker, and that turns a "protein on the membrane" into a "testable marker".

Why this course brings it up: He covers it once in Lecture 2 and once in Lecture 5, and it belongs to conceptual understanding. The genuinely useful inference is this: when you want to turn a protein into a biomarker, the first question is whether it has a transmembrane domain, and if it does, splicing out a soluble version is a standard move. He does not ask you to memorise the length of any particular protein's transmembrane domain. The principle is what carries over to any new protein you meet. **Appears in:** Ch 9, Lect 2, Lect 5

▪ transmembrane segments

In one sentence: The stretch of a protein that is buried in the membrane bilayer.

Detailed explanation: The amino acids in that stretch are all hydrophobic and do not like water, and ordinary mass spectrometry has to measure the protein in aqueous solution, so a protein like this drops out of solution before it ever reaches the instrument.

The remedy is to add a detergent or a lipid, but those molecules interfere with the measurement itself, which is why membrane protein proteomics has become the hardest corner of proteomics. The transmembrane segment has to be kept in solution by detergent or lipid, and the reagent that keeps it there is the same reagent that spoils the reading. That is the trade-off in one sentence, and it is the reason the whole corner of the field is hard.

Why this course brings it up: Dr. Wong does not name transmembrane segments as an exam question on its own, so it is conceptual understanding. **Appears in:** Ch 9, Lect 2

▪ transmigration

In one sentence: The step in which an immune cell squeezes through the vessel wall into the tissue.

Detailed explanation: After rolling and adhesion, the immune cell has to find a place in the vessel wall to squeeze out through, and it does it with molecules such as PECAM-1. Only once it is through does it reach the fourth step, chemotaxis, walking along the chemical gradient towards the tumour.

Every step is a point where a treatment can get in the way — blocking a chemokine receptor, for instance, lets the cell arrive but not gather. It is a physical squeeze through a gap in the wall, and that is why it needs its own adhesion machinery.

Why this course brings it up: He does not name transmigration as an exam topic on its own, so it is a concept to understand. **Appears in:** Ch 9, Lect 2

▪ Transport vesicle

In one sentence: A small membrane bubble that carries cargo from one compartment to the next.

Detailed explanation: A vesicle is a small bubble wrapped in membrane, a little delivery cart inside the factory. It unloads by fusing with the membrane it is aiming at: the rough ER loads new protein into a transport vesicle and sends it to the cis face of the Golgi, and the Golgi processes and sorts it and sends a secretory vesicle out from the trans face to the cell membrane, where the fusion releases the protein outside the cell or embeds it in the membrane as a receptor.

The direction is controlled by Rab proteins and SNARE complexes, which make sure a bubble only goes where it is supposed to go. And because the fusion is what delivers the cargo, the vesicle and the membrane it is aiming at have to be compatible, which is what the Rab and SNARE machinery is there to arrange. So the vesicle is the delivery van, and the labels on it are the Rab and SNARE names.

Why this course brings it up: Slide 22 of Lecture 2 uses this diagram to cover the whole secretory pathway in one go, and that is conceptual understanding. It is also the premise for understanding antigen presentation, because a protein has to pass through the ER and the Golgi before it can be fitted with MHC, and that is the last step in how an mRNA vaccine can set off an immune response. His reminder is that without the signal peptide the whole road is blocked. **Appears in:** Ch 8

▪ trastuzumab-deruxtecan

In one sentence: An ADC against HER2.

Detailed explanation: It is the antibody trastuzumab joined to a topoisomerase inhibitor payload, DXd. The reason it uses a DNA-damaging payload is that if the tumour becomes resistant to a single drug, switching to a different mechanism of action can still work.

Its clinical targets are the two of HER2 and TROP2, one matched to trastuzumab-deruxtecan and one to datopotamab-deruxtecan. Put it in the frame of a cancer cell surface antigen plus a topoisomerase inhibitor, and you can see why it works against HER2-positive tumours. The two halves have separate jobs, and that separation is the design. The antibody decides where the drug goes and the payload decides what happens once it gets there, and the payload is chosen to do something the tumour has no ready defence against. And the fact that two different antibodies can carry the same payload is what gives a second shot at the same target if the first one runs out of road.

Why this course brings it up: He says plainly that the structure of the ADC is not examined. But why the payload was chosen to be DNA-damaging, and the two clinical targets HER2 and TROP2, he does teach. So the structure is out and the reasoning and the two targets are in. **Appears in:** Ch 10, Lect 1, Lect 4

▪ Trastuzumab (Herceptin)

In one sentence: An antibody drug against HER2, used in HER2-positive breast cancer and gastric cancer.
Detailed explanation: Trastuzumab is an IgG1 antibody and its mechanism has two layers. The first layer is blocking dimerisation: HER2 cannot usefully dimerise with itself, it needs to bind HER1, HER3 or HER4 to switch the signal on, and Herceptin stands in the way of that binding, so the signal never gets through.
The second layer is ADCC: the Fc end of an IgG1 is recognised by FcγRIII on NK cells, which brings the killer cell right up beside the tumour cell to kill it there. Before you use the drug you have to confirm the patient is HER2 positive, and HER2 FISH does that with the centromeric probe CEP 17 and the LSI HER-2/neu probe, with a positive call being a HER2/CEP17 ratio of at least 2.0 or a HER2 copy number of at least 6. The ratio and the copy number are the same decision made two ways: the ratio compares HER2 with the chromosome it sits on, and the copy number simply counts the dots. The centromere probe is there to be the internal reference, so that a cell with a broken nucleus cannot fake a high count. So the antibody does two different jobs with the same molecule. One is to stop a signal, and the other is to summon a killer, and the Fc region is what makes the second job possible.
Why this course brings it up: In both Lecture 2 and Lecture 5 he uses it as the demonstration of molecular biology becoming a class of drug. The number that will be examined is receptor occupancy, 2 % in breast cancer and 10 % in gastric cancer, together with the whole set of contrasts: a monomer binds DNA poorly, dimerisation raises the affinity sharply, HER2 cannot dimerise with itself but only heterodimerises, and Herceptin blocks that heterodimerisation. Doing the FISH before you give the drug is the standard companion diagnostic workflow. Two numbers, one ratio, one mechanism. **Appears in:** Ch 7, Ch 10, Lect 2, Lect 5

▪ Tregs

In one sentence: Regulatory T cells: the brakes on the immune system.
Detailed explanation: A Treg is a T cell whose job is specifically to suppress immune responses, and in normal life it stops the immune system from attacking the body's own tissue. In a tumour, though, the tumour makes use of it: the tumour microenvironment induces the Tregs to expand, and they push down the effector T cells that were aimed at the tumour, so the immune attack goes limp.
His slide on the obstacles to CAR-T lines Tregs up alongside MDSCs and TAMs and TGFβ and IL-10, and adds hypoxia, an acidic environment and a rich ECM, all of which together make up the fortifications of a solid tumour. CAR-NK and CAR-macrophage are the alternative effectors proposed to get round the infiltration problem, which is the whole point of them. The Treg is not doing anything wrong; it is doing exactly what it is designed to do, only now it is doing it on the wrong side. That is why it makes the list.
Why this course brings it up: On Slide 63 in Lecture 2 he lists Tregs among the obstacles that stop CAR-T from reaching a solid tumour, so it is a concept to understand. What you have to be able to say is the logic of the line: the tumour is not going un-attacked, it is being held down with the brakes on — and that is also why checkpoint blockade works while CAR-T on its own is not enough. **Appears in:** Ch 9, Lect 2

▪ Triglyceride

In one sentence: Three fatty acids joined to one glycerol, the way the body stores fuel.
Detailed explanation: Glycerol is a three-carbon skeleton, and attaching three fatty acid chains to it makes a triglyceride (triacylglycerol). It is neutral and uncharged, so it can be stacked in bulk in adipose tissue as an energy reserve, and lipase breaks it down again to release fatty acids when fuel is needed.
Its fortunes are the opposite of the phospholipid's: a phospholipid carries two fatty acids plus a charged phosphate group, which makes it amphipathic, which lets it assemble itself into a bilayer, and that bilayer is the cell membrane itself.
Why this course brings it up: In Lecture 1 Dr. Wong divides the lipids into two destinations, triglyceride for storage and phospholipid for building membranes, and uses the amphipathic nature of the phospholipid to explain how the cell membrane forms. It belongs to conceptual understanding. Chemical structures and metabolic pathway details are on his exclusion list, so you do not need to memorise the structures, only which road leads to storage and which leads to the membrane. **Appears in:** Ch 1, Lect 1

▪ Trisomy

In one sentence: A chromosome present in three copies rather than two, written as 2n+1.
Detailed explanation: It is one of the forms of aneuploidy, and it comes from chromosomes failing to separate normally during meiosis. What he wants you to be able to tell is which of the two divisions failed, because the two produce different gametes.
A Meiosis I failure is homologous chromosomes not separating: one meiosis then produces 2 normal and 2 abnormal gametes, with the abnormal ones all the same type. A Meiosis II failure is sister chromatids not separating, and that produces 1 normal and 3 abnormal gametes, of which 2 are one type and 1 is another. So the M1 and M2 gamete combinations are different, and that is exactly the discriminating point he asks you to get right. The arithmetic is worth doing once on paper, because the number of abnormal gametes tells you which division failed, and because in an M1 failure the two abnormal gametes are identical to

each other while in an M2 failure the two abnormal gametes are identical and the third is different. That difference is the thing the question is testing, and it comes straight out of which stage the chromosomes were stuck at. And the same arithmetic is what tells you whether a case in a pedigree can be explained by one non-disjunction event or needs two.
Why this course brings it up: This is the first-grade, A+ question he labels in person. His words: "And you might need to distinguish the M1 or the M2 failure, because if there's a M1 failure, there's a formation of the germ cell without a losing copy of the chromosome after the M1…" and "And I think you need to distinguish these two situations here, yeah." It turns up in the scenario questions that combine a pedigree with aneuploidy, so the two skills are practised together. **Appears in:** Ch 3, Lect 3

▪ Trisomy 21

In one sentence: Down syndrome: three copies of chromosome 21.
Detailed explanation: A human somatic cell has 23 pairs of chromosomes, and normally there are two copies of chromosome 21. In Down syndrome there are three, written as 2n + 1. The cause is non-disjunction of chromosome 21 during meiosis: depending on whether it is an M1 or an M2 failure you get different gamete combinations.
It can also come from a Robertsonian translocation in one of the parents. Those three copies carry an extra dose of genes, and that is where the whole set of features comes from: the craniofacial and skeletal pattern, the effect on development and on intelligence, and so on. The point to hold on to is that the abnormality is a number, not a mystery: three instead of two, one extra copy of everything on that chromosome. Every gene on chromosome 21 is now present in three copies rather than two, and that is the arithmetic behind the phenotype. Down syndrome is one of the few conditions where the extra material is large enough to produce a recognisable pattern of features rather than a single subtle defect.
Why this course brings it up: On Slide 37 in Lecture 3 he draws Trisomy 21 (Down syndrome) next to Turner syndrome (45,X), and both are on his key numbers list: 23 pairs of chromosomes, Trisomy 21, Turner 45,X. They are the concrete products of that first-tier exam topic, M1 and M2 non-disjunction, so you have to be able to match the chromosome number to the name of the disease. **Appears in:** Ch 3, Lect 3

▪ tRNA (transfer RNA)

In one sentence: The adaptor molecule that delivers amino acids to the ribosome and reads the codons.
Detailed explanation: A tRNA has a function at each of its two ends. One end carries the anticodon, three bases, and reads the codon on the mRNA by complementary pairing. The other end catches the matching amino acid, and the loading and matching is done by aminoacyl-tRNA synthetase, a step that needs ATP.
The decoder rule is set by the codon: the same codon always means the same amino acid, which is the degeneracy of the genetic code. Because a codon is decided by its first or second base, the same amino acid can be specified by several different codons. So the tRNA is the translator in the middle, and the two ends of it are the two halves of the job: read the message on one side, bring the right building block on the other. And the charging step is the one that can go wrong on its own. If the synthetase puts the wrong amino acid on a tRNA, the ribosome will still read the codon correctly and still insert the wrong building block, which is a different class of error from a mutation.
Why this course brings it up: On the central dogma diagram of Slide 43 in Lecture 1 he marks the codon table and the flow of the central dogma, so it is concept-only. What really is examined is codon optimisation: human and CHO cells prefer different codons, so when you express a human protein in CHO you have to optimise the codons. His words are: "So remember that when you use the CHO cell, you need to do the codon optimization." That sentence is the whole point of the entry. **Appears in:** Ch 7, Lect 1, Lect 2

▪ Trop2

In one sentence: A cell-surface protein that is an antibody target in lung cancer and breast cancer.
Detailed explanation: Trop2 is a protein on the surface of epithelial cells, and it is expressed at high levels on a range of solid cancers (breast cancer, lung cancer and others), which makes it a good antibody target.
The idea behind an ADC is that the antibody is responsible for delivering the cell to the neighbourhood of the target, while the toxic payload is responsible for killing the cell. The Trop2 example is datopotamab-deruxtecan (listed alongside trastuzumab-deruxtecan, which goes after HER2), and both carry a topoisomerase inhibitor as their payload. Whether the antibody is IgG1 or IgG4 directly changes how the immune system treats the target.
Why this course brings it up: In Lectures 1 and 4 Dr. Wong uses HER2 and TROP2 to show the clinical targets of ADCs, and it belongs to conceptual understanding. What is examinable is the answer to that scenario short question: why put a topoisomerase inhibitor in an ADC, given that on its own it kills every dividing cell, and attached to an antibody it goes only to the tumour. He also says plainly that the structure of ADCs and the chemistry of the payloads are not examined. What you take from the slide is the design logic, and it is the same logic in both examples. The antibody is the address and the payload is the delivery, and both parts have to be there for the package to work. **Appears in:** Ch 10, Lect 1, Lect 4

▪ Trp repressor

In one sentence: The protein that grabs the operator once it has bound tryptophan and switches off the tryptophan synthesis system.
Detailed explanation: The trp repressor is the switch of a repressible system. When there is no tryptophan it is the apo-repressor, the form without a ligand, and it cannot hold DNA, so RNA polymerase goes ahead and transcribes trpE, trpD, trpC, trpB and trpA.

When tryptophan becomes plentiful, the excess tryptophan binds, the repressor changes shape and becomes able to grip the operator, and the whole operon is blocked. The more product there is, the harder the switch is pushed shut, and that is the classic design of feedback inhibition. The two states of the protein are therefore the same protein with and without a partner, and the partner decides which state it is in. That is the opposite of the lac arrangement, where the inducer takes the repressor off the DNA instead of putting it on.

Why this course brings it up: In Lecture 5 Dr. Wong teaches the trp operon as the mirror image of the lac operon, and the lac against trp contrast is an item on his second-level list. His case bank lists "read the logic of the lac operon correctly" as a scenario question, and the discriminating points include the fact that the lac repressor is released by lactose while the trp repressor is switched on by tryptophan, and that the two run in opposite directions. That direction of travel is the whole of the contrast, and it is exactly what a multiple-choice question will ask you to get right. **Appears in:** Ch 7, Lect 5

▪ Trp (tryptophan) operon

In one sentence: The tryptophan synthesis system: it makes its own tryptophan, and it decides when to stop making it.

Detailed explanation: The Trp operon contains five structural genes — trpE, trpD, trpC, trpB and trpA — coding for the enzymes of the tryptophan synthesis pathway. Its regulatory logic runs the opposite way to lac: lac switches itself off when the substrate is absent, while trp switches itself off when the product is in surplus.

The Warburg slide in Lecture 1 carries the same economics: the cell would rather keep half-digested sugar as building material than burn all the fuel. In both cases the question being answered is the same one a factory asks — do you have enough of the thing, and if so, stop the production line. That is the whole difference between the two operons: one watches the input and the other watches the output.

Why this course brings it up: In Lecture 5 he teaches it alongside the lac operon, so it is a concept to understand. The logic of either operon could turn up as a question, but the "two conditions must both hold" business belongs to lac alone: lactose present (which lifts the repressor) and glucose low (which activates the promoter through cAMP–CAP), whereas trp is a single condition. He explicitly excludes the metabolic pathway details. **Appears in:** Ch 7, Lect 5

▪ Tumor microenvironment

In one sentence: The whole neighbourhood of a tumour: cancer cells, immune cells, fibroblasts and stroma all together.

Detailed explanation: A tumour never exists on its own. Dr. Wong's definition is precise: tumour equals cancer cells plus immune cells plus CAFs (cancer-associated fibroblasts) plus ECM. Each of the four has a role: the cancer cells proliferate, the immune cells can either help or be turned, the CAFs make the stroma, and the ECM forms the physical barrier.

And the tumour reshapes this environment to protect itself, by producing TGFβ and IL-10, by inducing Tregs and MDSCs, and by expressing PD-L1 to put effector T cells into anergy. This is also why CAR-T works in blood cancers and so often fails in solid tumours. A blood cancer gives the engineered T cells easy access, while a solid tumour is a walled-off mass that the cell has to fight its way into through the very stroma the tumour has built to keep it out. So the microenvironment is not background scenery, it is the second half of the CAR-T story. Which is why the four parts of his definition matter: change any one of them and you change what the tumour can do to protect itself. The stroma can be a wall, the immune cells can be switched off, and the fibroblasts can be conscripted into building more of it. A tumour that has done all of that is much harder to attack than one that has only grown.

Why this course brings it up: In Lecture 2 he uses it to tie immune escape to the limits of CAR-T, and it belongs to conceptual understanding. But it connects to a conclusion that is examinable, the hot tumour versus cold tumour one: a hot tumour with many neoantigens responds to anti-PD-1, while a cold tumour can be induced to express PD-L1 by an HDAC inhibitor and converted into a hot one, which is gene addiction. His words are “So guys, you definitely remember that for the tumor that present many neoantigen and can be treated with the immunotherapy, we call this is a hot tumor, yeah.” And the cold tumour half of that pair is the one that pays: a tumour that has been hiding can sometimes be switched back on, and then it is treated by exactly the drug it was resisting. **Appears in:** Ch 9, Lect 2

▪ Tumor suppressor

In one sentence: A gene whose normal job is to restrain division; lose it and the cell runs away.

Detailed explanation: A tumour suppressor is the cell's brake. p53 is the most famous one, and his slide says that about 40 % of cancers carry a p53 mutation, and RB, APC and PTEN are the same kind of thing.

The copy number view of this is very intuitive: driver genes are usually copy number gains, a gain of function, while tumour suppressors are usually copy number losses, a loss of function, because take away one brake and the car runs out of control. The epigenetic layer plays the same trick: if the promoter is hypermethylated the gene is switched off, which is stepping on the brake rather than removing it. That difference matters when you are reading a report. A deleted tumour suppressor gene is a hole in the car, while a methylated one is a brake that is welded on and can in principle be released, which is exactly why the methylated version can sometimes be undone with a drug.

Why this course brings it up: In Lecture 3 he describes promoter hypermethylation as a common driver of cancer, silencing the brake, and on Slide 41 of Lecture 2 he puts p53 mutations at about 40 %. It is concept-only. What will be examined is this pair of contrasts: a gene that is switched off and a gene that is damaged are two different things. And MGMT promoter methylation makes glioblastoma more sensitive to alkylating chemo and radiotherapy, which is the clinical version of exactly this mechanism. So the same idea shows up as a test result and as a treatment decision, and the two are not interchangeable in a report or in an answer. **Appears in:** Ch 4, Ch 10, Lect 3

▪ Turner syndrome (45,X)

In one sentence: A woman with one X chromosome instead of two.

Detailed explanation: A normal female is 46,XX; Turner syndrome is 45,X, one X chromosome short. His Slide 37 draws this karyotype next to Down syndrome. The cause is non-disjunction of the sex chromosomes during meiosis (it falls under M1 or M2 non-disjunction), so you can end up with a normal egg carrying only an X and an abnormal egg carrying no sex chromosome at all;

in a few cases the X is lost after a somatic division instead. The clinical features include short stature, a shield-shaped chest, gonadal dysgenesis and low fertility. So the same fault in the same cell division produces two mirror-image pictures: one condition with an extra chromosome and one with a missing one. The missing X is not a fragment, it is a whole chromosome — which is why the loss is large enough to show up all over the body. The karyotype is the answer in two numbers, and both numbers are on the list he wants you to remember.

Why this course brings it up: Turner 45,X and Trisomy 21 are both on his key numbers list, alongside 23 pairs of chromosomes. It is the other concrete product of that first-tier exam topic, M1 and M2 non-disjunction, so you have to be able to match "one missing" and "one extra" to the right way of writing the karyotype: 2n - 1 and 2n + 1. **Appears in:** Ch 3, Lect 3

▪ Type I nuclear receptor

In one sentence: Like the estrogen receptor, it is held in the cytoplasm by a chaperone until the ligand arrives and it can set off.

Detailed explanation: The sequence for a Type I nuclear receptor runs like this: the receptor sits in the cytoplasm bound to HSP, a heat-shock protein chaperone, folded correctly but doing nothing.

When the ligand, estrogen, arrives the HSP dissociates, the receptor changes shape, then it homodimerises, which is two receptors binding each other and which greatly increases the affinity for DNA, and then it moves into the nucleus, binds the ERE, the estrogen response element, recruits a coactivator and gene expression switches on. Every one of those steps is a checkpoint, and a receptor that skips one of them does not work, which is why the HSP hold is not merely a way of storing the receptor but the reason it cannot fire by accident. The HSP dissociation is the release, and everything after it, the shape change, the dimer and the move into the nucleus, is the receptor actually doing its job.

Why this course brings it up: In Lecture 5 Dr. Wong teaches Type I and Type II nuclear receptors side by side, and that is conceptual understanding. The contrast maps onto his distractor list: Type I is HSP held in the cytoplasm, Type II is already sitting on the DNA. And the dimerisation rule for Type I connects to another pair, a monomer binds DNA badly while dimerisation greatly raises the affinity. **Appears in:** Ch 7, Lect 5

▪ Type II nuclear receptor

In one sentence: Like the thyroid hormone receptor, already sitting on DNA with RXR, just waiting for a ligand to swap the team.

Detailed explanation: A Type II nuclear receptor, the thyroid hormone receptor or the oxysterol receptor for example, is already bound to DNA when there is no ligand present: it forms a heterodimer with RXR and sits with a corepressor on a response element such as TRE or LXRE, pressing the gene down.

When the ligand arrives, the corepressor is swapped for a coactivator, and only then is the gene opened. So whether the gene is on or off is decided by which set of helper proteins is sitting on the DNA, and not by whether the receptor happens to be in the nucleus. The receptor does not move and the gene does not move. What changes is the team sitting on the response element, and the ligand is what swaps the team.

Why this course brings it up: On Slide 19 of Lecture 5 he cites Sever and Glass and draws it side by side with Type I, so it is concept-only, but that comparison is itself examinable: Type I waits in the cytoplasm for its ligand while Type II is already on the DNA waiting for it, and when the ligand arrives Type I has to travel into the nucleus and dimerise whereas Type II simply swaps its co-repressor on the spot. The pharmacological difference, why two nuclear receptor drugs can differ so much in tissue selectivity, starts right here. **Appears in:** Ch 7, Lect 5

▪ TZAP

In one sentence: The protein that shortens over-long telomeres and cuts off the t-loop.

Detailed explanation: Once a telomere has been lengthened, if nothing is done about it the end of the chromosome grows structures that are hard to handle. TZAP acts like a pair of scissors, trimming off the excess t-loop and pulling the telomere length back into a manageable range.

An over-long telomere has clinical significance of its own: in liver cancer, the patients in the top 10% for telomere length have worse recurrence-free survival.

Why this course brings it up: Dr. Wong does not name TZAP as an exam topic on its own, so it belongs to conceptual understanding. Telomere length as a prognostic marker is one of his precise numbers. **Appears in:** Ch 5, Lect 4

▪ Ubiquitin

In one sentence: A small protein chain that gets hung on another protein as the label that says "send this for destruction".

Detailed explanation: Ubiquitin is a small protein made of 76 amino acids. When a protein is due to be retired, the cell strings several ubiquitin molecules into a chain and hangs it on that protein;

this polyubiquitination is itself the signal that says "call in the proteasome", and the proteasome recognises the tag and chews the substrate up into peptides. HIF-1α in Lecture 5 is handled exactly this way: under normal oxygen PHD adds a hydroxyl

group, VHL recognises the result, ubiquitin is added, and then the proteasome breaks it down. The point is that the tag is a physical handle — a chain of small proteins that the degradation machine can grip, and a chain of several rather than a single ubiquitin is what counts. Note the order of it: the protein is tagged before it is destroyed, and the tagging is the step the cell controls, which is why changing one link of the chain changes the whole outcome.
Why this course brings it up: He does not list ubiquitin as an exam topic on its own, so it is a concept to understand, but it is the key link in the mechanism by which HIF-1α is rapidly broken down under normal oxygen and yet stabilised under hypoxia — and that mechanism is where the Lecture 5 material on a tumour growing its own blood vessels and the HIF-CAR design begins. What the exam wants is the chain told clearly: oxygen → PHD → VHL → ubiquitin → proteasome → HIF-1α gone. **Appears in:** Ch 7, Lect 5

▪ ultramutator

In one sentence: A cell whose mutation rate has been pushed up to an extreme level.
Detailed explanation: When the proofreading exonuclease activity of the polymerase breaks down (a proofreading mutation in Pol ε), the error rate of replication climbs sharply and the cell becomes an ultramutator.
On the surface that looks worse, and in fact it turns out better: the neoantigen go up in large numbers, the tumour-infiltrating lymphocytes go up, and the patient does better — which is the most counterintuitive conclusion of Lecture 4. It is the most counterintuitive lesson in the whole course: the most mutated tumour is the one the immune system can see most clearly, and that is why it responds best. The extra mutations are not random noise; they are extra handles for the immune system to grip. Which is the opposite of what "more mutations" usually implies.
Why this course brings it up: He explicitly names and explains the phenomenon, and it is the content of the Kaplan–Meier curve he draws (the POLE mutant group above the wild-type group). If you remember one strange result from the course, this is a good candidate. **Appears in:** Ch 9, Lect 4

▪ Ultraviolet light

In one sentence: An external injury that glues adjacent pyrimidine bases together.
Detailed explanation: UV makes two neighbouring thymines bond chemically to each other, a pyrimidine dimer, and the double helix is twisted by it so that the replication machinery cannot read past, and the whole stretch has to be dug out by NER and redone.
Too much UV is one of the main causes of skin cancer. His table of kinds of damage lists the UV pyrimidine dimer as the most typical bulky lesion, which is also why he puts NER next to it. The dimer is the damage and NER is the repair, and the two go together on the same slide. So NER does not exist as an abstraction: it exists because the cell keeps making these dimers and has to be able to dig them out again.
Why this course brings it up: Dr. Wong does not name UV as an exam question on its own, so it is conceptual understanding, and the pairing of NER with the bulky lesion and BER with the single base is one he lists. **Appears in:** Ch 6, Lect 4

▪ Undraggable target

In one sentence: A target that no drug can attack, with K-Ras being the most famous example until very recently.
Detailed explanation: “Undruggable” covers three different obstacles. First, there is no pocket to bind into, as with K-Ras, whose surface is smooth where it meets GTP. Second, the protein is indispensable in normal cells too, so there is no selectivity to be had, as with the cytoskeleton, which is crucial in metastasis, which makes it hard to attack only the copy inside the tumour. Third, there is not enough structural information to design a molecule against. K-Ras was long treated as undruggable, until the example Dr. Wong mentions in Lecture 1: on 26 August 2026 the FDA approved Rasonque (daraxonrasib), a RAS inhibitor, for metastatic pancreatic cancer, with a median overall survival of 13.2 versus 6.7 months in the trial. The three obstacles are worth keeping apart, because they call for different kinds of fix. A missing pocket is a chemistry problem, an indispensable protein is a selectivity problem, and missing structural information is an information problem, and a drug that solves one of them does nothing about the other two.
Why this course brings it up: In Lecture 1 he takes K-Ras from “undruggable” to “there is an approved drug”, which belongs to conceptual understanding, and the numbers have to be precise. The pairing that is examinable is mutated K-Ras → cetuximab (anti-EGFR) fails, because the signal is already on downstream. He also says plainly that the FDA details and the drug structures are not examined. **Appears in:** Ch 10, Lect 1

▪ unmethylated

In one sentence: A cytosine that has not had a methyl group added to it.
Detailed explanation: The key to bisulfite is its selectivity: unmethylated cytosine is converted to uracil by sodium bisulfite and so reads as thymine when sequenced, while the methylated form is left completely untouched.
So you compare the sequencing result against the unmethylated reference sequence and you know the state of every CpG site. In practice you need completely unmethylated reference DNA as a control, otherwise you cannot tell whether a site is genuinely unmethylated or whether the signal has simply dropped out. Without that control every result is ambiguous, and the whole method depends on being able to read a positive as a positive. The methylation state of a CpG is being read out of a chemical conversion, so the conversion has to be known to be complete.
Why this course brings it up: “Bisulfite sequencing = gold standard, unmethylated C → U, methylated C retained” is the content Dr. Wong marks as Tier 2, and it is the method behind the methylation figures he quotes elsewhere in the course.
Appears in: Ch 4, Ch 11, Lect 3

▪ Unsaturated fatty acid

In one sentence: A fatty acid with one or more double bonds in the chain, which makes it kinked and unable to pack tightly.
Detailed explanation: The double bond bends the carbon chain, so the fatty acid tail becomes short and crooked, and the molecules cannot lie down together in a tight stack. That is why unsaturated fatty acids have a lower melting point and are mostly liquid at room temperature, and it is also why a membrane that contains them stays fluid even in the cold.
Saturated fatty acids have no double bonds, the chain is straight, the molecules pack closely, and they are mostly solid.
Mammalian cells cannot make some of the unsaturated fatty acids themselves and have to get them from food, and that is the link between diet and lipid biology.
Why this course brings it up: In Lecture 1 Dr. Wong divides the fatty acids into saturated and unsaturated and explains that the kink of the unsaturated ones is the reason a cell membrane stays fluid, and that is conceptual understanding. Chemical structure is on the exclusion list, "Please don't memorize all these chemical structure…", so do not memorise structures; what you should keep is the causal idea that a straight chain packs tightly and a kinked chain does not. **Appears in:** Ch 1

▪ Untranslated region (UTR)

In one sentence: The two stretches of an mRNA that are not turned into amino acids: the 5′ UTR before the start and the 3′ UTR after the end.
Detailed explanation: The UTR does not code for amino acids, and yet it carries the key regulatory information. The 5′ UTR affects how efficiently the ribosome starts, so the same ORF can be translated very differently depending on which 5′ UTR it has. The 3′ UTR is where the regulation is dense: miRNA seed sequences pair up here, RBPs bind here, and lncRNA often intervene here as well — Slide 31 of Lecture 5 shows that a long 3′ UTR attracts RBPs, miRNA and lncRNA, and that decides the stability of the mRNA, its translation, its export and its location. The long isoform of BDNF (brain-derived neurotrophic factor) is carried out to the dendrite precisely because of its long 3′ UTR. So the picture of an mRNA is not simply "start here, read to there, stop": the coding sequence is the part that becomes protein, and the two untranslated stretches on either side are the control panel. Everything that regulates how much of a message is made, and for how long, and where it goes, is largely happening in those two regions. An ORF is the same in both cases; what changes is the length of the tail it comes with, and because the tail can be made longer or shorter by alternative polyadenylation, the UTR length is itself a variable the cell can set.
Why this course brings it up: In Lecture 5 he teaches mRNA structure (5′ cap, poly-A tail, 3′ UTR) together with mRNA stability and alternative polyadenylation, so it is a concept to understand. What you need is the inference: miRNA acts on the 3′ UTR → translation is inhibited and degradation is sped up; a different 3′ UTR length → a different location and a different lifetime, and that is one of the mechanisms by which the same gene makes different isoforms. **Appears in:** Ch 7, Lect 5

▪ uracil

In one sentence: The base that replaces thymine in RNA, written as U.
Detailed explanation: U is a pyrimidine like thymine, and it pairs with adenine. The difference is the methyl group: the uracil in RNA does not have it. Clinically it has another identity as the product of the bisulfite reaction, where an unmethylated cytosine is converted into uracil and then read as thymine during sequencing.
And because RNA contains no thymine, the synthesis and breakdown of uracil also becomes a point of attack for antimetabolites: a drug can impersonate uracil and slip quietly into the new strand. So one molecule carries three separate jobs in this course, base pairing, methylation readout and drug design, and each of them depends on a single small difference from its thymine cousin.
Why this course brings it up: He does not name uracil as an exam question on its own, so it is concept-only. But it is worth knowing because the bisulfite chemistry is the same chemistry he uses to read methylation. **Appears in:** Ch 3, Ch 11, Lect 1, Lect 3

▪ V(D)J recombination

In one sentence: Rearranging the V, D and J gene segments to make the diversity of antibodies and TCRs.
Detailed explanation: The diversity of antibodies and TCRs comes from joining gene segments together at random. The heavy chain uses four segments, V + D + J + C: roughly 39 copies of V, 27 of D and 6 of J, joined D–J first and then V, producing V-D-J-C mRNA.
The light chain uses only V + J (roughly 36 V and 5 J). The process is started by the RAG1/RAG2 (recombination activating proteins) complex, and the joining itself is done by NHEJ. So it is an example of borrowing the DNA repair machinery for a normal physiological purpose, which is worth noticing in its own right. The cell is using an enzyme that normally fixes broken DNA to deliberately break and rejoin DNA on purpose, in order to make a new receptor. The joining itself is done by NHEJ, the same non-homologous end joining path that repairs a double-strand break, which is exactly why a defect in that path destroys the immune system. The randomness is what makes the repertoire huge, and the repair machinery is what makes it possible to build.
Why this course brings it up: This is the core of a first-level (A+) question Dr. Wong personally named. His words: “Therefore, if I ask you, if the people showing a defective in this non-homologous end joining, what's the outcome? Definitely the outcome is the severe immune deficiency, right?” Because the receptor cannot be assembled, there is no antibody and no TCR, and the result is SCID. The segment copy numbers 39 / 27 / 6 are also on his key numbers list. And the arithmetic is worth doing in your head: 39 copies of V multiplied by 27 of D multiplied by 6 of J is already more than six thousand combinations for the heavy chain alone, before the light chain and before any junctional diversity is added. **Appears in:** Ch 6, Ch 9, Lect 2, Lect 4

▪ Variable region

In one sentence: The tip of the Y-shaped arms of an antibody, different in every antibody, and responsible for grabbing the antigen.

Detailed explanation: An antibody splits into a constant region, which never changes, decides the isotype and how the antibody talks to the immune system, and a variable region, which is different in every antibody and is responsible for recognising the antigen.

The variable region splits again into VH and VL, and the two together build a beta-barrel style antigen-binding surface, inside which three hypervariable loops are called CDR1, CDR2 and CDR3. Those three loops are the part that actually touches the target, and they are also where an engineer goes when modifying an antibody. All of that diversity comes from the random combinations made by V(D)J recombination. So the part that does the recognising is a small part of a large molecule, and the large part is the part that the immune system reads.

Why this course brings it up: The structure of the antibody is core material in Lectures 3 and 4. His numbers are worth keeping: Fv about 25 kDa, Fab about 50 kDa, Fc about 50 kDa, and the set of segment copies V ≈ 39, D = 27 and J = 6, and those last ones are what support the Tier 1 SCID chain. Both halves of that entry are numbers, and both are examinable.

Appears in: Ch 2, Ch 9

▪ VCAM-1

In one sentence: The adhesion molecule that lets a white blood cell grip the wall of a blood vessel.

Detailed explanation: Extravasation of a white blood cell has four steps: rolling, which uses selectin, then adhesion and activation, which uses ICAM-1 and VCAM-1, then transmigration, which uses PECAM-1, and finally chemotaxis, which uses MCP-1.

VCAM-1 is expressed on the activated vascular endothelium and binds the integrin on the white blood cell, catching the rolling cell and flattening it in preparation for the crossing. Its soluble form is shed off and can be measured in the blood, so it is also one of the things you can quantify when you study the tumour microenvironment. And the order is worth learning as an order, because each molecule is named after the step it does, so the name tells you where in the journey you are. The whole sequence is a hand-off from one adhesion molecule to the next.

Why this course brings it up: It is one part of the four-step diagram on Slide 19 of Lecture 2, and that is conceptual understanding. Dr. Wong does not list white blood cell extravasation as an exam question, but the whole order of rolling, adhesion, crossing and chemotaxis helps you understand why immune cells in the tumour microenvironment cannot get in or get out, which is one of the obstacles that stops CAR-T reaching a solid tumour. **Appears in:** Ch 9, Lect 2

▪ vector

In one sentence: The vehicle that carries DNA into a cell.

Detailed explanation: In molecular cloning the vector is the plasmid: a small circular DNA that can copy itself and be expressed in bacteria, and its job is to carry the insert in and make a lot of it.

More broadly, anything that gets nucleic acid into a cell counts as a vector, including adenovirus and the lipid nanoparticle. He uses the same idea himself in the CRISPR passage: the coding sequence for Cas9 has to be delivered into the cell by a virus or a lipid nanoparticle before it has any chance of acting. The word is about the job, not the material.

Why this course brings it up: He does not name vector as an exam topic on its own, so it is a concept to understand; recognising that a plasmid and an adenovirus are doing the same job in different chapters is what the word is for. **Appears in:** Ch 3, Lect 4

▪ VEGF

In one sentence: Vascular endothelial growth factor, switched on by HIF-1α, which lets a tumour build its own blood vessels.

Detailed explanation: A growing tumour needs oxygen and nutrients, but a tumour starts out hypoxic, because the surrounding blood vessels have not grown into it yet. Hypoxia stabilises HIF-1α and moves it into the nucleus, it binds the hypoxia response element, HRE, and that opens VEGF and the other genes.

VEGF makes the endothelial cells of nearby vessels grow new vessels into the tumour, so the tumour gets its own blood supply, and that process is called angiogenesis, and it is why a tumour can keep growing. HIF-1α also switches on genes such as GLUT1 and EPO. So angiogenesis is not a service the host provides out of kindness, it is something the tumour asks for and the body supplies, because the body cannot tell that the requesting tissue is malignant.

Why this course brings it up: In Lecture 5 he uses VEGF to string up the whole hypoxia response, and it is also the target of his HIF-CAR design: put the expression of the CAR under the control of HRE and the CAR-T only fires in the hypoxic core of the tumour, and about 50 % of solid tumours contain hypoxic areas. What is examinable is the mechanism chain: hypoxia, then HIF-1α stabilised and no longer marked for degradation by VHL, then HRE, then VEGF, then angiogenesis. **Appears in:** Ch 7, Lect 5

▪ Vesicle

In one sentence: A small bubble wrapped in a membrane that moves cargo between the compartments of a cell.
Detailed explanation: Vesicle is the general name for all the transport tools inside a cell, and it divides into several kinds: the transport vesicle, which goes from the endoplasmic reticulum to the Golgi, the secretory vesicle, which goes from the Golgi to the plasma membrane and does the secreting, and the endosome and lysosome, which swallow foreign material and break it down.
Outside it wears a membrane that is homologous with the plasma membrane, and inside it carries the cargo. Its relationship with the cytoskeleton is that the vesicles travel along microfilaments and microtubules, with the cargo towed along by motor proteins. So the word vesicle is a category rather than a single object, and when you meet it in a diagram you have to ask which kind it is. The endosome and lysosome are the odd ones out, because they take material in rather than carrying it out.
Why this course brings it up: Dr. Wong uses the ER-to-Golgi pathway diagram for it in both Lecture 2 and Lecture 5, and that is conceptual understanding. What you have to remember is the direction: rough ER, then transport vesicle, then the cis face of the Golgi, then the trans face, then secretory vesicle, then the plasma membrane. The first key to that road is the signal peptide, and that is the item he says in person is really, really important, because leaving it out cost four months of work, and the same road is the one that carries an antigen to the MHC for presentation. **Appears in:** Ch 8, Lect 2

▪ VH

In one sentence: The variable region of the antibody heavy chain.
Detailed explanation: The antigen-binding site is at the top of the antibody Y, and it is made of the variable region of the heavy chain, VH, together with the variable region of the light chain, VL.
Between VH and VL there are three hypervariable loops, CDR1, CDR2 and CDR3, and those are the parts that actually meet the target. The scFv on the outside of a CAR is simply VH and VL strung together with a linker. Pulling VH and VL out and using them on their own is where the scFv comes from, and that design is the technical basis on which CAR-T borrows the recognition power of an antibody.
Why this course brings it up: Dr. Wong does not name VH and VL as exam questions on their own, so it is conceptual understanding, and the antibody structure, Fab, Fc, Fv and CDR1-3, is the content of Chapter 3. **Appears in:** Ch 2, Lect 1, Lect 2

▪ VHL

In one sentence: The protein that recognises a marked HIF-1α and hands it over to the proteasome for breakdown.
Detailed explanation: VHL (von Hippel–Lindau) is the gatekeeper of HIF-1α under normal oxygen. When oxygen is plentiful, the enzymes PHD and FIH add a hydroxyl group to HIF-1α; the hydroxylated HIF-1α is exactly the shape VHL is able to recognise, so VHL binds to it, ubiquitin chains get added, and the proteasome takes it apart.
That is why under normal conditions HIF-1α barely exists. When oxygen runs short the hydroxylating enzymes have no oxygen to work with, HIF-1α escapes, stabilises, goes into the nucleus and starts switching genes on. The whole thing is a sensor built out of chemistry: oxygen is not measured directly, it is measured by whether the hydroxylation can happen at all. And note that the same chain in reverse is what hypoxia exploits — the absence of a reactant is read as a signal. The gatekeeper is doing its job perfectly; the cell is simply reporting on how much oxygen there is.
Why this course brings it up: On Slide 23 in Lecture 5 he draws out the whole oxygen-sensing pathway, so it is a concept to understand, but it is really one of the backbones of the course: loss of VHL function causes VHL disease, and VHL mutations are a driver in kidney cancer — that turns a molecular detail into a clinical target. You have to be able to lay out the chain: oxygen → PHD/FIH hydroxylation → VHL recognition → ubiquitination → proteasome → HIF-1α gone; under hypoxia the chain is cut. **Appears in:** Ch 7, Lect 5

▪ Vimentin

In one sentence: A structural protein inside the cell that leaks into the blood in liver cancer, and Dr. Wong's hands-on biomarker example.
Detailed explanation: Vimentin is an intermediate filament protein, part of the cytoskeleton, and it gives the cell mechanical strength. Under normal conditions it stays inside the cell, but in hepatocellular carcinoma it is released into the blood.
What he built is a complete biomarker discovery pipeline: 2D-PAGE (separating first by charge, then by molecular weight) comparing normal liver, cirrhosis and HCC specimens of different sizes, which showed Vimentin rising already in early HCC; then validation with qPCR, IHC and Western blot, and finally measurement in serum with ELISA. That last step is the one that turns a protein observation into a test, because a marker nobody can measure in blood is not going to help anybody. Each step narrows the claim: first the protein is different, then the difference is real, and only then is it something you can put in a patient's arm. The numbers he quotes are the output of that last step.
Why this course brings it up: This is the biomarker case he demonstrates himself in Lecture 1, and the numbers have to be precise (n = 152): controls 176.0 ± 61.3 ng/mL (median 163.0), HCC ≤ 2 cm 271.5 ± 165.9 (median 202.5), HCC > 2 cm 231.3 ± 132.3 (median 179.0). The reading is that the smallest tumour shows the biggest rise, because that is the one hardest to catch early. He also warns that the tissue next to a tumour is not the same as normal tissue. **Appears in:** Ch 2, Ch 11, Lect 1

▪ VIP (vasoactive intestinal peptide)

In one sentence: A neuropeptide that can switch T cells off and help the tumour hide.
Detailed explanation: VIP is a polypeptide secreted by the nervous system, and it is background knowledge, because it does not appear in the material of the five lectures. What it shows is that immune escape does not rely on protein factors alone, because the nervous system is involved too: once VIP binds its receptor, VPAC, it can render T cells inactive, and the nerve fibres around a tumour therefore become one of the sources of immunosuppression.
And there is a particular trap here, an abbreviation collision: the VIP > 1.0 on his first lecture metabolomics slide is variable importance in projection, the score the machine gives a feature, and it has nothing to do with this peptide. So in this course VIP is two unrelated things, and the exam-relevant one is the number. The abbreviation is doing two different jobs in the same course, which is exactly the kind of thing that trips people up in a question.
Why this course brings it up: He does not name it in particular, so it is concept-only, and it does not actually appear in the five lectures. All you need for the exam is this: VIP > 1.0 is one of the three conditions for screening differential metabolites in metabolomics, together with $p < 0.05$ and FDR < 0.15, and the three numbers are on the same slide. Read the acronym, not the peptide, and you will get that question right. And while the peptide itself does not appear in the five lectures, it is a reasonable thing to hold for the tumour microenvironment discussion. **Appears in:** Background supplement (not lecture content)

▪ VISTA

In one sentence: An inhibitory checkpoint sitting at the interface between the immune cell and the tumour cell.
Detailed explanation: VISTA is one of the inhibitory checkpoints listed on Dr. Wong's slide 63 (CTLA-4, PD-1, LAG3, TIM3 and VISTA are all on the effector cell side, while PD-L1 and PD-L2 are on the target cell side).
What sets it apart from the others is that it is mainly expressed on myeloid cells and on tumour cells, and it has only recently been recognised as an important target, with no mature drugs against it yet. His list puts all five together to make the point that the immunosuppression of a solid tumour is layered, and that hitting one route on its own is often not enough to get CAR-T in. Several brakes, and the cell has to release all of them before it can drive.
Why this course brings it up: He does not name VISTA as an exam topic on its own, so it belongs to conceptual understanding. The layered-suppression point is the part worth keeping. **Appears in:** Ch 9, Lect 2

▪ VL

In one sentence: The variable region of the light chain of an antibody.
Detailed explanation: VL is a variable region just like VH, and it carries the same three hypervariable CDR loops, the only difference being that it sits on the light chain rather than the heavy chain.
The scFv of a CAR is VH plus VL, joined by a linker, and that is how the recognition ability of an antibody is bolted onto a T cell. The recognition actually comes from the three CDRs on both sides together, so when an engineer modifies a CAR, changing either side changes the binding. Which means that a CAR built from a single chain is still using two variable regions doing one job between them.
Why this course brings it up: He does not name VL as an exam question on its own, so it is concept-only. It matters as one half of the scFv and as one half of the antigen binding surface. **Appears in:** Ch 2, Lect 1, Lect 2

▪ Warburg effect

In one sentence: A cancer cell ferments glucose into lactate even when there is oxygen, because what it wants is raw material rather than fuel.
Detailed explanation: His Lecture 1 slide puts three situations side by side: an aerobic resting cell gets about 36 ATP per glucose, hypoxia gives only 2, and a proliferating cancer cell goes by aerobic glycolysis whichever way the oxygen runs, for about 4, because it sends 85 % of the pyruvate to lactate and keeps only about 5 % for the mitochondria, with PDK inhibiting PDC.
The inefficiency is a strategy. The half-broken sugar molecule is the ideal building material for nucleotides, lipids and amino acids, and a cell that is about to divide a billion times needs building material far more than it needs fuel. So the three numbers on the slide, 36, 2 and 4, are not three facts about three organisms, they are three answers to the same question asked of three different cells. The cancer cell is not broken, it has simply chosen to run the less efficient line on purpose.
Why this course brings it up: The Warburg effect is his entry point in Lecture 1, and it is also the reason metabolomics became a useful biomarker tool. It runs through three lectures: Lecture 1 puts forward the trade-off, Lecture 2 tells it again in the context of the mitochondrion and glycolysis, and Lecture 3 explains that it causes DNA damage and genomic instability, which raises the mutational burden of the cancer cell. He says plainly that the glycolysis pathway details are not examined. **Appears in:** Ch 10, Lect 1, Lect 2

▪ Water

In one sentence: The solvent of the cell, whose unusual shape gives it a high melting point and a great dissolving power.
Detailed explanation: The water molecule is bent, not straight, so one end carries a partial positive charge and the other a partial negative charge, and that is what lets it form hydrogen bonds with the water molecules next door.
This shape gives four biological reasons for water's importance: it is the solvent in which every reaction happens; it is a participant in the chemistry of proteins and ATP (the hydrogen bonds are what give macromolecules their shape); its hydrogen bonds give macromolecules their three-dimensional shape; and its high specific heat keeps body temperature steady. Its high melting point also comes from the hydrogen bond network — you have to break the hydrogen bonds before you can melt it. So

the same property, the polar shape, is doing all the work: it is why water can dissolve things, and it is why water is hard to boil. That is why you cannot treat water as just a passive background fluid; its shape is doing work.
Why this course brings it up: In Lecture 1 he lists water as one of the "four structural reasons life needs water", so it is a concept to understand. The chemical structure is on his exclusion list, so do not memorise the structure; what you should be able to derive is why a polar, hydrogen-bonding molecule is at the same time the best solvent and the most stable environment. That is also why the hydrophobic effect is such a fundamental idea in protein folding. **Appears in:** Ch 1

▪ Watson, Crick and Franklin

In one sentence: The three people who built the double helix model, using Rosalind Franklin's X-ray images.
Detailed explanation: The model rested on two clues. The first was Rosalind Franklin's X-ray diffraction images, with the X-shaped pattern that points to a helix. The second was the rules known at the time, that the amount of A equals the amount of T and the amount of G equals the amount of C.
Watson and Crick combined the two clues and proposed the double helix, and the rule of complementary base pairing (three hydrogen bonds for G–C, two for A–T) is its chemical explanation. The model was useful the moment it was published: the two strands can be pulled apart with heat, and pulling the strands apart is exactly what PCR does. So the history runs from the shape of the molecule to a machine that makes millions of copies of it, and the three of them are the people who supplied the shape. And the shape was the missing piece: without knowing that the strands separate, the base pairing rules looked like a coincidence rather than a mechanism.
Why this course brings it up: In Lecture 1 Dr. Wong treats this as “chemistry and history”: Griffith raised the question, Avery–MacLeod–McCarty answered “is it DNA” with enzyme experiments, and Watson–Crick–Franklin answered “what is the shape”. He has said that the double helix is the basis of PCR. The years 1928 and 1943 are shown as part of the story, not as numbers to memorise. **Appears in:** Ch 3, Lect 1

▪ Western blot

In one sentence: The standard protein test: does this sample contain this particular protein?
Detailed explanation: The workflow is this. SDS-PAGE separates the proteins by size, and then the proteins in the gel are transferred onto a membrane. An antibody recognises the target protein, that is the primary antibody, and a labelled secondary antibody does the detecting.
If you can see a band, the sample contains that protein, and the darkness of the band says how much. Its strength is the presence and the amount, and its weakness is that it cannot show you the whole proteome in one run. The band is the answer, and everything about the method is arranged so that one band can be read: separation by size, transfer, and then two antibodies in series so that the protein is identified and the signal is amplified.
Why this course brings it up: In the vimentin case of Lecture 1 he uses it as one of the validation tools, alongside qPCR, IHC and ELISA, and on the codon-optimisation slide he uses it to compare target protein expression between different treatment groups. It is concept-only. He does not list it as an exam question; all you have to do is say that it answers presence and amount, and keep the division of labour clear between it, SDS-PAGE, LC-MS/MS and X-ray. **Appears in:** Ch 11, Lect 1

▪ Wild type

In one sentence: The normal, unmutated version.
Detailed explanation: Wild type is the baseline you compare the mutant versions against. K-Ras is the best example: wild-type K-Ras only becomes active when the upstream EGFR sends a signal, whereas the mutant version switches itself on and keeps sending growth signals whether or not there is any signal from upstream.
That contrast is the entire logic of a predictive biomarker: a K-Ras mutant means the downstream is already on, so the anti-EGFR cetuximab does not work, and a K-Ras wild-type means it might. Wild type is not a judgement and it is not a better version, it is simply the version you find in a healthy cell, and every mutant is described as a difference from it. The comparison is what lets you ask whether a change has made a protein more active, less active or permanently active, and that question is the whole of the mutation section of the course.
Why this course brings it up: His words are: "So, this is you'll have to a 100% understand, right?" The same mutation behaving differently in different genetic backgrounds is exactly why a biomarker has to be typed before it is of any use. The figures he gives are that KRAS mutants survive less well, p = 0.016, and that KRAS G12C accounts for 13 % of lung adenocarcinoma and 3 % of colorectal cancer. So the baseline is not a footnote, it is the thing that makes the mutant readable. **Appears in:** Ch 5, Ch 10, Lect 1

▪ X-linked dominant

In one sentence: An inheritance pattern carried on the X chromosome, in which it appears in every generation.
Detailed explanation: The signature of X-linked dominant is that there is an affected person in every generation, and that an affected father passes it to all of his daughters and to none of his sons.
The reason is the number of X chromosomes: a father has only one, and he hands it to every daughter, while his sons get his Y. A carrier female is usually less severely affected than a male, because she has two X chromosomes and the normal one can partly compensate. Dr. Wong's Slide 29 draws this pattern alongside X-linked recessive and mitochondrial inheritance as three pedigrees, for students to practise reading. So the skill being trained is not memorising the definition but looking at a family tree and working out which of the three it is. And the fastest way to do that is to count: does an affected father have an affected son, and does an affected mother pass it to every child? The answer to those two questions tells you which of the three patterns you

are looking at.
Why this course brings it up: He discusses pedigree reading in both Lecture 2 and Lecture 4, and the pedigree is one of the two scenario question types he names (the other is the survival curve). He has said: “Just make sure you understand that different mutation can cause the mitochondrial diseases, and all this mutation can cause the different inheritance patterns when you analyze this pedigree tree, yeah.” These three patterns have to be tellable apart on paper. **Appears in:** Lect 2

▪ X-linked recessive

In one sentence: An inheritance pattern carried on the X chromosome that mainly affects males.
Detailed explanation: The typical pattern of X-linked recessive inheritance is more males than females, and it often appears in alternate generations: a carrier woman does not get ill herself, but her sons have a 50 % chance of getting the X chromosome with the mutation and falling ill.
A son always gets the Y from his father, so this trait is never passed from father to son, and that is the cleanest way of telling it apart from X-linked dominant inheritance, as well as the most useful clue in a pedigree question. The logic is simply that a woman has two X chromosomes and can carry the mutation on one of them, while a man has only one X, so a mutation on it has no second copy to be masked by.
Why this course brings it up: On Slide 29 of Lecture 2 he puts X-linked dominant, X-linked recessive and mitochondrial inheritance side by side, and reading pedigrees is one of the scenario questions he names. The point to keep, the one that turns on a hinge, is that a mitochondrial disease does not necessarily come from mtDNA: a mutation in a nuclear gene can produce the same phenotype with a completely different pattern of inheritance. So the answer has to mention both. **Appears in:** Lect 2

▪ X-ray crystallography

In one sentence: The method that solves the precise three-dimensional atomic shape of a protein.
Detailed explanation: The protein is induced into crystals and bombarded with X-rays; the scattering of the X-rays by the atoms forms a diffraction pattern, and a mathematical inversion of that pattern gives the position of every atom.
The result is a map of atomic coordinates, and that is exactly the thing that explains function, because shape determines function. The limitation is that the molecule has to be able to form a crystal at all, and a protein like histone, which carries a large positive charge and is extremely floppy, crystallises very badly, which is why Dr. Wong singles out the difficulty of crystallising histone in his Lecture 3. That is also why the method is so decisive when it works, because there is no interpretation involved, the atoms are simply where the picture says they are.
Why this course brings it up: In Lecture 1 Dr. Wong puts four methods for studying proteins side by side: Edman gives the sequence, LC-MS/MS gives identity and amount, SDS-PAGE gives size, and X-ray crystallography gives the atomic shape. That is conceptual understanding, but the lesson that using the wrong method gets you no answer at all is worth keeping. He uses the 2.8 Å nucleosome crystal structure on Slide 9 of Lecture 3 as his example, and he calls anything under 2.0 Å very good, and both numbers are on his key numbers list. **Appears in:** Ch 2

▪ XIST

In one sentence: A long non-coding RNA that silences an entire X chromosome; the classic example of an lncRNA.
Detailed explanation: A male mammal has a Y chromosome it has little use for, while the genes on the X chromosome would be expressed in double dose if both copies were left alone to work. XIST is the switch that solves this problem.
It is transcribed from the X chromosome itself, it carries repeated sequences, and it binds SPEN (an RNA-binding protein) and PRC2 (a chromatin repressive complex), wrapping the whole X chromosome in a condensed, silent state — and that is what dosage compensation means. Without XIST the X chromosome would simply be expressed normally and the gene dosage would be out of balance. The scale is what makes it remarkable: this is not one gene being switched off, it is a whole chromosome being shut. The RNA is coating the chromosome it came from, and the coat is what holds the silence in place. It is a piece of RNA doing the job of a chromatin complex, which is the whole point of the lncRNA idea.
Why this course brings it up: On Slide 29 in Lecture 5 he lists XIST as one of the four lncRNA archetypes, so it is a concept to understand. He does not make it an exam topic, but it is the best example of an lncRNA doing something large without coding any protein, and that is the reason lncRNAs are becoming new targets. The clinical extension is XIST in the treatment of X-linked disease — using lncRNA technology to switch a silenced X chromosome back on. **Appears in:** Ch 7, Lect 5

▪ XPA

In one sentence: The protein that positions the NER enzyme team on the damaged DNA.
Detailed explanation: TFIIH opens a short stretch of DNA first, and then XPA comes along like a sliding clamp, travelling along the DNA until it finds the place where the helix is twisted out of shape.
It stops there and assembles the RPA and the two endonucleases (XPF and XPG) into place. The job of XPA is purely to locate, so that the cut is made only where the damage really is. If XPA cannot find the damaged position, the cut happens in the wrong place; the specificity of NER rests on this locating protein. One protein does nothing chemically, and without it the whole operation is random.
Why this course brings it up: He does not name XPA as an exam topic on its own, so it is a concept to understand. **Appears in:** Ch 6, Lect 4

▪ XPC

In one sentence: The protein that finds the damaged positions across the whole genome in GG-NER.
Detailed explanation: The first step of global genome repair is a patrol of the whole chromosome, and XPC is the protein that recognises that this stretch of DNA does not look right, while at the same time not being blocked by a DNA replication fork.
Only once XPC has made the recognition are XPA, RPA and XPF and XPG called in to cut out the damaged piece. And it is precisely because XPC has to keep away from the replication fork that this pathway can cover the whole genome without cutting in the wrong place in the middle of a replication.
Why this course brings it up: Dr. Wong does not name XPC as an exam question on its own, so it is conceptual understanding. **Appears in:** Ch 6, Lect 4

▪ XPF and XPG

In one sentence: The two endonucleases that cut the damaged piece out in NER.
Detailed explanation: After TFIIH has opened the double helix, XPA has positioned the machinery and RPA has protected the single strand, XPF and XPG cut on the two sides of the damaged stretch and remove a piece of 22–32 nucleotides.
The gap is then filled by the polymerase and sealed by the ligase. The 22–32 nucleotides left behind after the cut are the evidence that a piece of DNA damage has been removed in one piece, and because that gap size is fixed it is also the clue he uses to tell NER apart from the other repair pathways. So the two enzymes are the scissors and everything else is the preparation and the clean-up.
Why this course brings it up: He does not name these two as exam questions on their own, so it is concept-only, and the pairing that NER repairs bulky lesions is one he lists. So the names are context for the pathway. **Appears in:** Ch 6, Lect 4

▪ Yamanaka factors

In one sentence: Oct3/4, Sox2, c-Myc and Klf4 — the four factors that reprogram a fibroblast into an iPSC, the induced pluripotent stem cell.
Detailed explanation: In 2006 Takahashi and Yamanaka showed that expressing these four factors together is enough to take an already differentiated somatic cell, a fibroblast, back to a pluripotent state.
His slide lists two more, Nanog and LIN-28, and gives the RT-PCR marker set for iPSCs: Nanog, ERas, Oct3/4, Sox2, Fgf4, Cripto, Dax1, Zfp296 and Nat1. The significance of reprogramming is enormous: it sidesteps the ethical argument about embryonic stem cells, and it lets you make an iPSC from a patient's own cells and then differentiate it into cardiomyocytes to test cardiotoxicity. The point is that the cell's identity is not a fixed thing written into the DNA; it is a state the cell is in, and four proteins can talk it back into a different state. And because the iPSC comes from the patient, the cardiomyocytes you make from it are a personal sample of what that patient's heart cells do under a drug. No embryo is involved anywhere in that route.
Why this course brings it up: In Lecture 2 he teaches the validation chain for iPSCs as three stages: morphology, markers, and the gold standard teratoma assay (injecting into an immunodeficient mouse and looking for tissue from all three germ layers). It is a concept to understand. He also reminds you that QC has to include "a distinctive expression pattern and timing profile, and using negative staining to rule out contaminated cells". **Appears in:** Ch 8, Lect 2

▪ YAP

In one sentence: The protein at the end of the Hippo pathway; when it is not phosphorylated it enters the nucleus and starts pushing growth.
Detailed explanation: When the Hippo pathway is off, MST1 and LATS1 phosphorylate YAP, and the phosphorylated YAP is grabbed by 14-3-3, kept in the cytoplasm and degraded, and the cell stops proliferating.
When the pathway fails, as it does in the MST1/2 double knockout mouse, YAP is not phosphorylated, runs into the nucleus and binds TEAD, and that switches on CTGF, JAG1, which is a Notch ligand, and NOTCH2, growth-related genes, and the tissue overgrows. His Slide 25 shows exactly that phenotype: the folded, overgrown tissue of Drosophila hpo mutant clones, and the grossly enlarged liver of the Mst1-/- Mst2-/- mouse. So YAP is the switch that decides whether the growth programme runs, and phosphorylation is what keeps it switched off.
Why this course brings it up: He teaches this pathway again and again in Lectures 1 and 5, so it is concept-only. What you have to keep is the causal chain and the experiment that goes with it: lose the kinase, YAP enters the nucleus, the tissue overgrows. And in Lecture 5 he mentions that druggable inhibitors are under development. He does not ask you to memorise the upstream list of activating factors. The chain and the drug, not the list. **Appears in:** Ch 7, Lect 1, Lect 5

▪ ZAP70

In one sentence: The kinase that carries the signal inside a CAR.
Detailed explanation: A CAR is built in three parts: the extracellular scFv (which recognises the antigen), the hinge and transmembrane region in the middle, and the signalling tail inside the cell.
The tail usually contains CD3ζ and ZAP70. When the scFv grabs hold of the antigen, the tail passes the signal into the T cell and sets off a chain of phosphorylation reactions, and the T cell switches on and goes to kill. With ZAP70 in place the CAR's signal can be translated by the T cell's own internal signalling network into the order to kill, and without that link the CAR is nothing but a light sensor.
Why this course brings it up: Dr. Wong does not name ZAP70 as an exam topic on its own, so it belongs to conceptual understanding. The skeleton for Ch 9 is the four parts of a CAR: scFv, hinge, transmembrane, signalling tail. **Appears in:** Ch 9, Lect 2

▪ ZFN

In one sentence: Zinc finger nuclease, the older protein-based method for cutting DNA at a chosen spot.
Detailed explanation: A ZFN is a pair of protein nucleases: each one carries a zinc finger domain that recognises a short stretch of DNA, and the two bind on either side of the target and cut across the middle.
Because the recognition is done by the protein's finger, designing a new one means re-engineering the protein, which is laborious. CRISPR/Cas9 replaces that whole protein recognition module with a piece of RNA (the gRNA), which is far cheaper to design, and it quickly displaced ZFN.
Why this course brings it up: On Slide 67 of Lecture 2 Dr. Wong uses ZFN to edit the TRAC locus in the allogeneic CAR-T workflow, so that CAR-T cells from a donor do not attack the recipient. It belongs to conceptual understanding. He has said that CRISPR/Cas9 has become the standard approach for knock-out and knock-in and that the old methods are essentially displaced, but ZFN still appears on his allogeneic CAR-T slide, so you should know where it sits: still in use, but superseded by CRISPR. The historical order is worth having: the protein-based method came first and was hard to redesign, and the RNA-based one came along and made the whole question of design much easier. Which is why the RNA guide can be typed out on a computer and the protein finger cannot. That difference in cost is what moved the field, and it is the same reason the allogeneic CAR-T workflow is much easier to plan now. **Appears in:** Ch 6, Lect 2, Lect 4

▪ Zinc finger

In one sentence: A small fold held open by one zinc ion, which lets a DNA-binding domain grip the DNA.
Detailed explanation: A zinc finger is a small module inside a protein: a run of amino acids uses the side chains of cysteine and histidine to chelate a single zinc ion, and the zinc ion holds the whole structure open in the shape of a stable little finger.
The zinc is not doing chemistry in the usual sense, it is doing structure, and it is doing it with one atom, which is why the module is so small and so easy to repeat. Each finger reads about three bases, so a DNA-binding domain that contains several fingers can recognise a reasonably long stretch of a particular sequence, and the number of fingers sets the length of the sequence that can be read. The DBD of a nuclear receptor is built from zinc fingers, and TZAP, telomeric zinc finger-associated protein, on Slide 12 of Lecture 4 is another example.
Why this course brings it up: On the nuclear receptor structure diagram on Slide 18 of Lecture 5 Dr. Wong marks that the DBD contains zinc fingers, and that is conceptual understanding. The principle behind it is the thing to master: a nuclear receptor recognises the DNA response element with its DBD, and a monomer binds DNA badly while dimerisation greatly raises the affinity, and that is the thread that leads straight to the HER2 and Herceptin mechanism question. The design principle of a ZFN rests on the same structure, and that is also why the specificity of a zinc-finger drug can be engineered rather than hoped for.
Appears in: Ch 7, Lect 5